

Econometrics

Personal Study Notes and Formal Derivations

Victor Alves

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Disclaimer

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Part I

Econometrics I

Chapter 1

The Classical Linear Regression Model (CLRM)

1.1 Motivation and Geometric Interpretation

The Classical Linear Regression Model (CLRM) addresses the problem of estimating the conditional mean of a dependent variable (y) given a set of independent variables ($x_1, \dots, x_k$), aiming to describe the statistical dependence between them.

Geometric Interpretation in $\mathbb{R}^n$:

Let $\mathbf{y} \in \mathbb{R}^n$ be the vector of observations of the dependent variable and $\mathbf{X} \in \mathbb{R}^{n \times K}$ the data matrix. Geometrically, the Ordinary Least Squares (OLS) estimator seeks a vector of fitted values $\hat{\mathbf{y}}$ that represents the **orthogonal projection** of $\mathbf{y}$ onto the subspace spanned by the columns of $\mathbf{X}$ (the column space). The residual vector $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ is the vector of smallest Euclidean norm connecting $\mathbf{y}$ to the subspace, ensuring that $\mathbf{e}$ is orthogonal to every column of $\mathbf{X}$.

1.2 CLRM Assumptions and Their Roles

Assumption 1.1: Linearity in Parameters

- **Role:** Defines the functional structure of the model, ensuring that the dependent variable is a weighted sum of the regressors and an additive disturbance.
- **Counterexample:** A production model where parameters are exponents, such as $Y = \beta_0 X_1^{\beta_1} \varepsilon$, requires logarithmic transformation to be linearized; if not linearizable, algebraic OLS methods fail in direct identification.

Assumption 1.2: No Perfect Multicollinearity (Full Rank)

- **Role:** Ensures that the explanatory variables vary in a linearly independent manner, allowing the inversion of the matrix $(\mathbf{X}'\mathbf{X})$ and the unique identification of parameters.
- **Counterexample:** Including the same variable in different units (e.g., meters and centimeters) or falling into the “dummy variable trap” (including dummies for all

categories and an intercept) renders the matrix singular, preventing the estimator from being computed.

Assumption 1.3: Strict Exogeneity (Zero Conditional Mean)

- **Role:** Guarantees that the error term does not contain information systematically related to the regressors, being the necessary condition for the unbiasedness of the estimator.
- **Counterexample:** Omitting a relevant variable correlated with the included regressors causes the error to absorb this effect, generating correlation between $\mathbf{X}$ and ε and resulting in systematically biased estimates (omitted variable bias).

Assumption 1.4: Spherical Errors (Homoskedasticity and No Autocorrelation)

- **Role:** Establishes that the error variance is constant across observations and that errors are independent of each other, simplifying the covariance structure to a scalar matrix ($\sigma^2 \mathbf{I}$).
- **Counterexample:** In time series data, shocks in one period may persist into the next (autocorrelation), or in cross-sectional data, variance may increase with income level (heteroskedasticity), invalidating standard errors and OLS efficiency.

1.3 Formal Statement of the CLRM

Let a measure and probability space be associated with the data generating process. The Classical Linear Regression Model (CLRM) is defined by the following structure:

1. **Population Linearity:** The response vector $\mathbf{y} \in \mathbb{R}^n$ satisfies the equation:

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon \quad (1.1)$$

where $\mathbf{X}$ is an $n \times K$ matrix of regressors, $\beta \in \mathbb{R}^K$ is the vector of fixed parameters, and $\varepsilon \in \mathbb{R}^n$ is the vector of stochastic disturbances.

2. **Identifiability:** The matrix $\mathbf{X}$ has full column rank, such that $\text{rank}(\mathbf{X}) = K < n$ with probability 1.
3. **Exogeneity:** $E[\varepsilon|\mathbf{X}] = \mathbf{0}_{n \times 1}$.
4. **Sphericity:** $E[\varepsilon\varepsilon'|\mathbf{X}] = \sigma^2 \mathbf{I}_n$, where $\mathbf{I}_n$ is the identity matrix of order n and $\sigma^2 \in (0, \infty)$.

1.4 Derivation of the OLS Estimator

Theorem 1.4.1 (Ordinary Least Squares Estimator). *Under Assumptions 1.1 to 1.4, the OLS estimator is given by:*

$$\boxed{\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}} \quad (1.2)$$

Proof. We define the objective function $S(\mathbf{b})$ as the squared Euclidean norm of the residual vector:

$$S(\mathbf{b}) = (\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b}) \quad (1.3)$$

Expanding the quadratic form:

$$S(\mathbf{b}) = \mathbf{y}'\mathbf{y} - \mathbf{b}'\mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\mathbf{b} + \mathbf{b}'\mathbf{X}'\mathbf{X}\mathbf{b} \quad (1.4)$$

Since $\mathbf{b}'\mathbf{X}'\mathbf{y}$ is a scalar (dimension 1×1), it is identical to its transpose $(\mathbf{y}'\mathbf{X}\mathbf{b})$. Hence:

$$S(\mathbf{b}) = \mathbf{y}'\mathbf{y} - 2\mathbf{b}'\mathbf{X}'\mathbf{y} + \mathbf{b}'\mathbf{X}'\mathbf{X}\mathbf{b} \quad (1.5)$$

To find the minimum, we differentiate $S(\mathbf{b})$ with respect to $\mathbf{b}$ and set it to zero (First Order Condition):

$$\frac{\partial S(\mathbf{b})}{\partial \mathbf{b}} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\mathbf{b} = \mathbf{0} \quad (1.6)$$

Isolating $\mathbf{b}$:

$$\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y} \quad (1.7)$$

By Assumption 1.2 (Full Rank), $(\mathbf{X}'\mathbf{X})$ is positive definite and thus invertible:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \quad (1.8)$$

□

1.5 Proof of Unbiasedness of OLS

Theorem 1.5.1 (Unbiasedness of OLS). *Under Assumption 1.3 (Strict Exogeneity), the OLS estimator is unbiased:*

$$E[\hat{\beta}|\mathbf{X}] = \beta \quad (1.9)$$

Proof. Substituting the structural equation (1.1) into (1.2):

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \varepsilon) \quad (1.10)$$

Distributing:

$$\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon \quad (1.11)$$

Applying the conditional expectation given $\mathbf{X}$:

$$E[\hat{\beta}|\mathbf{X}] = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\varepsilon|\mathbf{X}] \quad (1.12)$$

By Assumption 1.3, $E[\varepsilon|\mathbf{X}] = \mathbf{0}$:

$$E[\hat{\beta}|\mathbf{X}] = \beta \quad (1.13)$$

□

1.6 Implications of Strict Exogeneity

1.6.1 Motivation

Exogeneity is the assumption that enables the transition from mere **statistical association** to **causal or structural inference**. In practical terms, it solves the problem of isolating the effect of a specific explanatory variable on the dependent variable, ensuring that the error term does not contain factors that vary systematically with the regressors.

Geometric Interpretation in $\mathbb{R}^n$:

Let $\mathbf{X}$ be the matrix of regressors. Exogeneity implies that, in the sample space, the vector of population errors ε is “orthogonal” (in expectation) to the subspace spanned by the columns of $\mathbf{X}$. This means that the projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$ recovers exactly the structural component $\mathbf{X}\beta$, without contamination from unobserved factors that could “pull” the projection in wrong directions.

1.6.2 Assumptions and Their Roles

Assumption 1.5: Strict Exogeneity (Zero Conditional Mean)

- **Role:** Guarantees that the error is not predictable from any information contained in the regressors, establishing mean independence.
- **Counterexample:** If we study the effect of education (x) on wages (y), and omit “innate ability” (a), the latter will reside in the error ε . Since more able individuals tend to pursue more education, $\text{Cov}(x, \varepsilon) \neq 0$. The estimator will attribute to education the wage gain that actually comes from ability, generating **omitted variable bias**.

1.6.3 Implications of Strict Exogeneity

Implication I: Zero Unconditional Mean ($E[\varepsilon] = 0$)

By the Law of Iterated Expectations:

$$E[\varepsilon] = E[E[\varepsilon|\mathbf{X}]] = E[\mathbf{0}] = \mathbf{0} \quad (1.14)$$

Implication II: Orthogonality and Zero Covariance

For any function $h(\mathbf{X})$ of the regressors:

$$E[h(\mathbf{X})\varepsilon] = E[E[h(\mathbf{X})\varepsilon|\mathbf{X}]] = E[h(\mathbf{X})E[\varepsilon|\mathbf{X}]] = \mathbf{0} \quad (1.15)$$

In particular, $\text{Cov}(\mathbf{X}, \varepsilon) = \mathbf{0}$.

Implication III: Identification of the Parameter β

Taking the conditional expectation of model (1.1):

$$E[\mathbf{y}|\mathbf{X}] = \mathbf{X}\beta + E[\varepsilon|\mathbf{X}] = \mathbf{X}\beta \quad (1.16)$$

Premultiplying by $\mathbf{X}'$ and taking the unconditional expectation:

$$E[\mathbf{X}'\mathbf{y}] = E[\mathbf{X}'\mathbf{X}]\beta \quad (1.17)$$

Isolating β :

$$\beta = (E[\mathbf{X}'\mathbf{X}])^{-1}E[\mathbf{X}'\mathbf{y}] \quad (1.18)$$

Equation (1.18) shows that the population parameter β is uniquely determined by the observable moments of $\mathbf{y}$ and $\mathbf{X}$. Without exogeneity, the OLS estimator would converge to a value $\beta^* \neq \beta$, resulting in **inconsistency**.

1.7 Ordinary Least Squares (OLS) as Orthogonal Projection

1.7.1 Motivation

The OLS process solves the problem of **approximation in subspaces**. When we have a system of linear equations $\mathbf{y} = \mathbf{X}\beta$ that is overidentified (more equations than unknowns, $n > K$), there is generally no exact solution due to noise in the data. OLS seeks the parameter vector that produces the “best fit,” minimizing the Euclidean distance between the observed vector and the predicted vector.

1.7.2 Assumptions and Their Roles

Assumption 1.6: Full Column Rank

- **Role:** Ensures that the columns of $\mathbf{X}$ are linearly independent, guaranteeing that the Gram matrix $(\mathbf{X}'\mathbf{X})$ is invertible.
- **Counterexample:** If one variable is a copy of another, the subspace S will have dimension less than K . This creates a “line” or “plane” of possible solutions for β , making the estimator indeterminate.

Assumption 1.7: $n > K$ (Positive Degrees of Freedom)

- **Role:** Necessary for the residual to not be trivially zero and to allow estimation of the noise variance.
- **Counterexample:** If $n = K$, the “approximation” becomes an exact interpolation; if $n < K$, the system is underdetermined and there are infinitely many solutions that zero the error, preventing the identification of individual effects.

1.7.3 Formal Statement and Derivation

The OLS estimator, denoted by $\hat{\beta}$, is defined as:

$$\hat{\beta} = \arg \min_{\mathbf{b} \in \mathbb{R}^K} S(\mathbf{b}) \quad (1.19)$$

where $S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = (\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b})$.

Theorem 1.7.1 (Uniqueness of OLS). *Under Assumption 1.6, the OLS estimator is unique and given by (1.2).*

Proof. The Hessian matrix is $\mathbf{H} = 2\mathbf{X}'\mathbf{X}$. By Assumption 1.6, $\mathbf{X}$ has linearly independent columns. If $\mathbf{v} \neq \mathbf{0}$, then $\mathbf{X}\mathbf{v} \neq \mathbf{0}$, which implies $\mathbf{v}'(\mathbf{X}'\mathbf{X})\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 > 0$. Hence, $2\mathbf{X}'\mathbf{X}$ is **positive definite**, guaranteeing that the critical point is a **strict and unique minimum**. $\square$

1.8 Gauss-Markov Theorem

1.8.1 Motivation

The Gauss-Markov Theorem addresses the problem of **estimator selection**. Given that there are infinitely many ways to combine the data to estimate a parameter β in an unbiased manner, the theorem identifies which of these combinations produces the smallest uncertainty (variance).

Geometric Interpretation in $\mathbb{R}^n$:

In the sample space, the OLS estimator $\hat{\beta}$ corresponds to the orthogonal projection of the observation vector $\mathbf{y}$ onto the subspace spanned by the columns of $\mathbf{X}$. Any other

linear unbiased estimator can be seen as an “oblique” projection or a transformation that does not minimize the Euclidean distance. The theorem guarantees that the “ellipse of uncertainty” (covariance matrix) of any other linear unbiased estimator will contain the ellipse of uncertainty of OLS.

1.8.2 Formal Statement

Theorem 1.8.1 (Gauss-Markov). *Under Assumptions 1.1 to 1.4 (Linearity, Full Rank, Strict Exogeneity, and Sphericity), the OLS estimator $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ has the smallest covariance matrix (in the positive definite matrix sense) within the class of all linear unbiased estimators of β .*

1.8.3 Granular Proof

Definition 1.8.1 (OLS Estimator).

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{A}\mathbf{y}, \quad \mathbf{A} \equiv (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \quad (1.20)$$

Definition 1.8.2 (Alternative Linear Estimator). Let $\tilde{\beta}$ be any linear estimator defined by $\tilde{\beta} = \mathbf{C}\mathbf{y}$, where $\mathbf{C}$ is a $K \times n$ matrix that may depend on $\mathbf{X}$, but not on $\mathbf{y}$. We define $\mathbf{D}$ such that $\mathbf{C} = \mathbf{A} + \mathbf{D}$.

Lemma 1.8.1 (Unbiasedness Condition). *For $\tilde{\beta}$ to be unbiased, we must have $\mathbf{D}\mathbf{X} = \mathbf{0}$.*

Proof.

$$E[\tilde{\beta}|\mathbf{X}] = E[(\mathbf{A} + \mathbf{D})\mathbf{y}|\mathbf{X}] = (\mathbf{A} + \mathbf{D})\mathbf{X}\beta + (\mathbf{A} + \mathbf{D})E[\varepsilon|\mathbf{X}] \quad (1.21)$$

$$E[\tilde{\beta}|\mathbf{X}] = \mathbf{A}\mathbf{X}\beta + \mathbf{D}\mathbf{X}\beta = \beta + \mathbf{D}\mathbf{X}\beta \quad (1.22)$$

For $E[\tilde{\beta}|\mathbf{X}] = \beta$ for any β , it is necessary that $\mathbf{D}\mathbf{X} = \mathbf{0}$. □

Step 1: Variance of OLS:

$$\text{Var}(\hat{\beta}|\mathbf{X}) = \mathbf{A} \text{Var}(\mathbf{y}|\mathbf{X})\mathbf{A}' = \sigma^2\mathbf{A}\mathbf{A}' = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} \quad (1.23)$$

Step 2: Variance of the Alternative Estimator:

$$\text{Var}(\tilde{\beta}|\mathbf{X}) = \sigma^2\mathbf{C}\mathbf{C}' = \sigma^2(\mathbf{A} + \mathbf{D})(\mathbf{A} + \mathbf{D})' \quad (1.24)$$

$$\text{Var}(\tilde{\beta}|\mathbf{X}) = \sigma^2(\mathbf{A}\mathbf{A}' + \mathbf{A}\mathbf{D}' + \mathbf{D}\mathbf{A}' + \mathbf{D}\mathbf{D}') \quad (1.25)$$

Step 3: Simplification of Cross Terms:

$$\mathbf{AD}' = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}' = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{DX})' = \mathbf{0} \quad (1.26)$$

Similarly, $\mathbf{DA}' = \mathbf{0}$.

Step 4: Efficiency Comparison:

$$\text{Var}(\tilde{\beta}|\mathbf{X}) = \sigma^2\mathbf{AA}' + \sigma^2\mathbf{DD}' = \text{Var}(\hat{\beta}|\mathbf{X}) + \sigma^2\mathbf{DD}' \quad (1.27)$$

Since $\mathbf{DD}'$ is an outer product matrix, it is **positive semidefinite (PSD)**. Hence:

$$\text{Var}(\tilde{\beta}|\mathbf{X}) - \text{Var}(\hat{\beta}|\mathbf{X}) = \sigma^2\mathbf{DD}' \succeq \mathbf{0} \quad (1.28)$$

Conclusion: OLS is **BLUE** (Best Linear Unbiased Estimator). $\square$

1.9 Projection and Annihilator Matrices

1.9.1 Motivation

The central problem of linear regression is to decompose a vector of observed data $\mathbf{y}$ into two mutually exclusive components: one that belongs to the subspace spanned by our explanatory data (the “explained” part) and another that is orthogonal to that subspace (the noise or “error”). The projection and annihilator matrices are the linear operators that perform this separation.

Geometric Interpretation in $\mathbb{R}^n$:

Let S be the subspace spanned by the columns of the matrix $\mathbf{X}$ (the column space of $\mathbf{X}$).

1. The **Projection Matrix ($\mathbf{P}$)** acts as a “map” that orthogonally projects any vector $\mathbf{y}$ onto S , resulting in the vector of fitted values $\hat{\mathbf{y}}$.
2. The **Annihilator Matrix ($\mathbf{M}$)** projects the vector $\mathbf{y}$ onto the orthogonal complement of S , resulting in the residual vector $\mathbf{e}$.

Geometrically, the vector $\mathbf{y}$ is the hypotenuse of a right triangle whose legs are $\hat{\mathbf{y}}$ and $\mathbf{e}$, ensuring the orthogonal decomposition $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{e}$.

1.9.2 Formal Statement

Definition 1.9.1 (Projection and Annihilator Matrices). Let $\mathbf{X} \in \mathbb{R}^{n \times K}$ with $\text{rank}(\mathbf{X}) = K$. We define:

1. **Projection Matrix (or “Hat Matrix”):**

$$\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \quad (1.29)$$

2. Annihilator Matrix (or “Residual Maker”):

$$\mathbf{M} = \mathbf{I}_n - \mathbf{P} \quad (1.30)$$

1.9.3 Properties of Projection Matrices

Property I: Symmetry

$$\mathbf{P}' = \mathbf{P}, \quad \mathbf{M}' = \mathbf{M} \quad (1.31)$$

Property II: Idempotence

$$\mathbf{P}\mathbf{P} = \mathbf{P}, \quad \mathbf{M}\mathbf{M} = \mathbf{M} \quad (1.32)$$

Property III: Orthogonality and Conservation

$$\mathbf{P}\mathbf{X} = \mathbf{X}, \quad \mathbf{M}\mathbf{X} = \mathbf{0}, \quad \mathbf{P}\mathbf{M} = \mathbf{0} \quad (1.33)$$

Property IV: Trace and Rank

$$\text{tr}(\mathbf{P}) = K, \quad \text{tr}(\mathbf{M}) = n - K \quad (1.34)$$

Since for idempotent matrices $\text{rank}(\mathbf{P}) = \text{tr}(\mathbf{P})$, the rank of the projection matrix is exactly the number of regressors K .

1.9.4 Orthogonal Decomposition of $\mathbf{y}$

Using the matrices $\mathbf{P}$ and $\mathbf{M}$:

$$\mathbf{y} = \mathbf{P}\mathbf{y} + \mathbf{M}\mathbf{y} = \hat{\mathbf{y}} + \mathbf{e} \quad (1.35)$$

where $\hat{\mathbf{y}} = \mathbf{P}\mathbf{y}$ are the fitted values and $\mathbf{e} = \mathbf{M}\mathbf{y}$ are the residuals.

Verification of Orthogonality:

$$\hat{\mathbf{y}}'\mathbf{e} = (\mathbf{P}\mathbf{y})'(\mathbf{M}\mathbf{y}) = \mathbf{y}'\mathbf{P}'\mathbf{M}\mathbf{y} = \mathbf{y}'\mathbf{P}\mathbf{M}\mathbf{y} = \mathbf{0} \quad (1.36)$$

1.10 Partitioned Regression and the Frisch-Waugh-Lovell Theorem

1.10.1 Motivation

The concrete problem solved by partitioned regression is the isolation of the marginal effect of a subset of variables, “cleaning” the influence of other regressors (controls). Researchers often face the dilemma of the “Short Regression” (estimated with available data) versus the “Long Regression” (the ideal population model), which generates systematically biased estimates when relevant variables are omitted.

Geometric Interpretation in $\mathbb{R}^n$:

Let S be the subspace spanned by the columns of $\mathbf{X} = [\mathbf{X}_1 \mid \mathbf{X}_2]$. The long regression estimator projects the vector $\mathbf{y}$ onto the total space S . The short regression projects $\mathbf{y}$ only onto the subspace $S_1 \subset S$ spanned by $\mathbf{X}_1$. If $\mathbf{X}_1$ and $\mathbf{X}_2$ are not orthogonal, the projection onto S_1 captures part of the variation that geometrically belongs to the direction of $\mathbf{X}_2$, “contaminating” the coefficient of $\mathbf{X}_1$. The Frisch-Waugh-Lovell (FWL) Theorem shows that the long regression coefficient can be obtained by projecting the residuals of one short regression onto the residuals of another, performing what we call *partialling out*.

1.10.2 Assumptions and Their Roles

Assumption 1.8: Relevance of the Omitted Variable ($\beta_2 \neq 0$)

- **Role:** Ensures that the excluded variable has a structural effect on the dependent variable in the population model.
- **Counterexample:** If we include irrelevant variables (whose true parameter is zero), the short regression will be unbiased, though less efficient.

Assumption 1.9: Correlation between Regressors ($\mathbf{X}_1' \mathbf{X}_2 \neq 0$)

- **Role:** Establishes that the included and omitted variables are linearly dependent in the sample, allowing the included variable to act as a partial *proxy* for the omitted one.
 - **Counterexample:** If the regressors are orthogonal ($\mathbf{X}_1' \mathbf{X}_2 = 0$), omitting a relevant variable does not bias the coefficient of the other, only increases the residual variance.
-

1.10.3 Formal Statement

Consider the Partitioned Linear Regression Model:

$$\mathbf{y} = \mathbf{X}_1 \beta_1 + \mathbf{X}_2 \beta_2 + \varepsilon \quad (1.37)$$

where $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X}_1 \in \mathbb{R}^{n \times k_1}$, $\mathbf{X}_2 \in \mathbb{R}^{n \times k_2}$, and $\varepsilon \in \mathbb{R}^n$. We assume strict exogeneity and full rank for $\mathbf{X} = [\mathbf{X}_1 \mid \mathbf{X}_2]$.

We define:

1. **Short Regression Estimator:** $\tilde{\beta}_1 = (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{y}$.
2. **Annihilator Matrix of $\mathbf{X}_1$:** $\mathbf{M}_1 = \mathbf{I} - \mathbf{X}_1 (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1$.

Theorem 1.10.1 (Omitted Variable Bias). *Under Assumptions 1.8 and 1.9:*

$$\boxed{E[\tilde{\beta}_1 | \mathbf{X}] = \beta_1 + (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{X}_2 \beta_2} \quad (1.38)$$

Proof.

$$E[\tilde{\beta}_1 | \mathbf{X}] = (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 E[\mathbf{y} | \mathbf{X}] = (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 (\mathbf{X}_1 \beta_1 + \mathbf{X}_2 \beta_2) \quad (1.39)$$

$$E[\tilde{\beta}_1 | \mathbf{X}] = \beta_1 + (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{X}_2 \beta_2 \quad (1.40)$$

□

1.10.4 Frisch-Waugh-Lovell (FWL) Theorem

Theorem 1.10.2 (Frisch-Waugh-Lovell). *The OLS estimator for β_2 in model (1.37) is identical to the estimator obtained by regressing the residuals of $\mathbf{y}$ (with respect to $\mathbf{X}_1$) on the residuals of $\mathbf{X}_2$ (with respect to $\mathbf{X}_1$):*

$$\boxed{\hat{\beta}_2 = (\mathbf{X}'_2 \mathbf{M}_1 \mathbf{X}_2)^{-1} \mathbf{X}'_2 \mathbf{M}_1 \mathbf{y}} \quad (1.41)$$

Proof (sketch). The partitioned normal equations are:

$$\begin{bmatrix} \mathbf{X}'_1 \mathbf{X}_1 & \mathbf{X}'_1 \mathbf{X}_2 \\ \mathbf{X}'_2 \mathbf{X}_1 & \mathbf{X}'_2 \mathbf{X}_2 \end{bmatrix} \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{X}'_1 \mathbf{y} \\ \mathbf{X}'_2 \mathbf{y} \end{bmatrix} \quad (1.42)$$

From the first row: $\hat{\beta}_1 = (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 (\mathbf{y} - \mathbf{X}_2 \hat{\beta}_2)$. Substituting into the second row and rearranging:

$$\mathbf{X}'_2 \mathbf{M}_1 \mathbf{X}_2 \hat{\beta}_2 = \mathbf{X}'_2 \mathbf{M}_1 \mathbf{y} \quad (1.43)$$

Isolating $\hat{\beta}_2$, we obtain (1.41). □

1.11 Relative Efficiency: Short vs. Long Regression

1.11.1 Motivation

The concrete problem here is the **bias-variance trade-off** in model selection. When deciding whether to include a set of controls ($\mathbf{X}_2$) to estimate the effect of interest ($\mathbf{X}_1$), we face a dilemma: the **Long Regression** (with $\mathbf{X}_2$) eliminates omitted variable bias, but

the **Short Regression** (without $\mathbf{X}_2$) may offer more “stable” estimates (lower variance), especially if there is collinearity among the regressors.

Geometric Interpretation in $\mathbb{R}^n$:

Think of the variance of an estimator as the “volume” of its confidence ellipsoid.

1. In the **Short Regression**, we project $\mathbf{y}$ directly onto $\text{Col}(\mathbf{X}_1)$. Precision depends only on the dispersion of $\mathbf{X}_1$.
2. In the **Long Regression**, the FWL Theorem tells us that we estimate the coefficient of $\mathbf{X}_1$ by projecting $\mathbf{y}$ onto the part of $\mathbf{X}_1$ that is **orthogonal** to $\mathbf{X}_2$ (i.e., $\mathbf{M}_2\mathbf{X}_1$).

If $\mathbf{X}_1$ and $\mathbf{X}_2$ are highly correlated, the residual vector $\mathbf{M}_2\mathbf{X}_1$ will have a very small norm. Geometrically, we are trying to extract information from a very narrow “shadow” of $\mathbf{X}_1$, which amplifies uncertainty and increases the variance of the estimator.

1.11.2 Formal Statement

Let the DGP be given by the Long Regression (1.37). Under homoskedasticity, the variances of the estimators for β_1 are:

1. **Long Regression:** $\text{Var}(\hat{\beta}_1|\mathbf{X}) = \sigma^2(\mathbf{X}'_1\mathbf{M}_2\mathbf{X}_1)^{-1}$.
2. **Short Regression:** $\text{Var}(\tilde{\beta}_1|\mathbf{X}) = \sigma^2(\mathbf{X}'_1\mathbf{X}_1)^{-1}$.

Theorem 1.11.1 (Efficiency of the Short Regression). *Under homoskedasticity:*

$$\boxed{\text{Var}(\hat{\beta}_1|\mathbf{X}) \succeq \text{Var}(\tilde{\beta}_1|\mathbf{X})} \quad (1.44)$$

Proof.

$$\mathbf{X}'_1\mathbf{M}_2\mathbf{X}_1 = \mathbf{X}'_1\mathbf{X}_1 - \mathbf{X}'_1\mathbf{P}_2\mathbf{X}_1 \preceq \mathbf{X}'_1\mathbf{X}_1 \quad (1.45)$$

By the Matrix Inversion Lemma (if $\mathbf{A} \preceq \mathbf{B}$, then $\mathbf{A}^{-1} \succeq \mathbf{B}^{-1}$):

$$\sigma^2(\mathbf{X}'_1\mathbf{M}_2\mathbf{X}_1)^{-1} \succeq \sigma^2(\mathbf{X}'_1\mathbf{X}_1)^{-1} \quad (1.46)$$

□

1.11.3 Practical Implications

The short regression is always **more efficient** (has lower variance) than the long regression, except in the case of orthogonality ($\mathbf{X}'_1\mathbf{X}_2 = \mathbf{0}$), where the variances are equal.

Indispensable Assumption: Homoskedasticity ($\sigma^2\mathbf{I}$) is indispensable for the mathematical proof. In the presence of heteroskedasticity, the short variance matrix becomes a “sandwich” and it is theoretically possible for the long regression to be more efficient if $\mathbf{X}_2$ helps explain the variance structure of the error, although this rarely occurs with standard OLS.

1.12 Estimator of the Residual Variance (s^2) and Distribution of Residuals

1.12.1 Motivation

The concrete problem here is **measuring uncertainty**. The Ordinary Least Squares (OLS) estimator provides the “best fit,” but does not reveal the magnitude of the random error (σ^2). Without an estimate of σ^2 , it is impossible to compute standard errors or perform hypothesis tests. The estimator s^2 solves this by correcting the downward bias that would occur if we used the simple average of the squared residuals.

Geometric Interpretation in $\mathbb{R}^n$:

The vector of observations $\mathbf{y}$ inhabits a space of dimension n . The model projects $\mathbf{y}$ onto a subspace of dimension K (spanned by the regressors). The residual vector $\mathbf{e}$ is the projection of $\mathbf{y}$ onto the **orthogonal complement** of that subspace. Since this complement has dimension $n - K$, the residual vector has only $n - K$ independent directions in which to vary. Dividing the sum of squared residuals by $n - K$ (degrees of freedom) is equivalent to calculating the average variance per available dimension in the residual space.

1.12.2 Additional Assumptions

Assumption 1.10: Normality of Errors ($\varepsilon|\mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$)

- **Role:** Necessary to derive the exact distribution of residuals and for finite-sample inference.
- **Counterexample:** Without normality, the distribution of residuals is not exactly known, although asymptotically it approaches a normal distribution.

1.12.3 Formal Statement

Let the model be $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, with $\varepsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ and $\text{rank}(\mathbf{X}) = K$. We define:

1. **Projection Matrix (P):** $\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.
2. **Annihilator Matrix (M):** $\mathbf{M} = \mathbf{I}_n - \mathbf{P}$.
3. **Residual Vector (e):** $\mathbf{e} = \mathbf{M}\mathbf{y}$.
4. **Variance Estimator (s^2):** $s^2 = \frac{\mathbf{e}'\mathbf{e}}{n-K}$.

1.12.4 Proof of Unbiasedness of s^2

Lemma 1.12.1 (Properties of Matrix $\mathbf{M}$). $\mathbf{M}$ is symmetric ($\mathbf{M}' = \mathbf{M}$), idempotent ($\mathbf{M}\mathbf{M} = \mathbf{M}$), and orthogonal to $\mathbf{X}$ ($\mathbf{M}\mathbf{X} = \mathbf{0}$).

Lemma 1.12.2 (Expectation of Quadratic Forms). For a random vector $\mathbf{z}$ with mean μ and covariance matrix Σ , $E[\mathbf{z}'\mathbf{A}\mathbf{z}] = \text{tr}(\mathbf{A}\Sigma) + \mu'\mathbf{A}\mu$.

Theorem 1.12.1 (Unbiasedness of s^2). *Under Assumptions 1.1 to 1.4:*

$$\boxed{E[s^2|\mathbf{X}] = \sigma^2} \quad (1.47)$$

Proof. Step 1: Express the Sum of Squared Residuals (SSR) in terms of the population error:

$$\mathbf{e} = \mathbf{M}\mathbf{y} = \mathbf{M}(\mathbf{X}\beta + \varepsilon) = \mathbf{M}\mathbf{X}\beta + \mathbf{M}\varepsilon = \mathbf{M}\varepsilon \quad (1.48)$$

Hence, the SSR is:

$$\mathbf{e}'\mathbf{e} = \varepsilon'\mathbf{M}'\mathbf{M}\varepsilon = \varepsilon'\mathbf{M}\varepsilon \quad (1.49)$$

Step 2: Apply the Expectation Operator:

$$E[\mathbf{e}'\mathbf{e}|\mathbf{X}] = E[\varepsilon'\mathbf{M}\varepsilon|\mathbf{X}] \quad (1.50)$$

Applying Lemma 1.12.2 (with $\mu = \mathbf{0}$ and $\Sigma = \sigma^2\mathbf{I}$):

$$E[\mathbf{e}'\mathbf{e}|\mathbf{X}] = \text{tr}(\mathbf{M} \cdot \sigma^2\mathbf{I}) + \mathbf{0}'\mathbf{M}\mathbf{0} = \sigma^2 \text{tr}(\mathbf{M}) \quad (1.51)$$

Step 3: Compute the Trace of $\mathbf{M}$:

$$\text{tr}(\mathbf{M}) = \text{tr}(\mathbf{I}_n - \mathbf{P}) = n - \text{tr}(\mathbf{P}) = n - \text{tr}(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') \quad (1.52)$$

By the cyclic property of the trace ($\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$):

$$\text{tr}(\mathbf{M}) = n - \text{tr}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}) = n - \text{tr}(\mathbf{I}_K) = n - K \quad (1.53)$$

Step 4: Finalize the Proof:

$$E[\mathbf{e}'\mathbf{e}|\mathbf{X}] = \sigma^2(n - K) \quad (1.54)$$

Taking the expectation of s^2 :

$$E[s^2|\mathbf{X}] = E\left[\frac{\mathbf{e}'\mathbf{e}}{n - K} \middle| \mathbf{X}\right] = \frac{\sigma^2(n - K)}{n - K} = \sigma^2 \quad (1.55)$$

□

1.12.5 Proof of the Distribution of Residuals

Lemma 1.12.3 (Transformation of Normal Vectors). *If $\mathbf{z} \sim N(\mu, \Sigma)$, then for a constant matrix $\mathbf{A}$, the vector $\mathbf{Az} \sim N(\mathbf{A}\mu, \mathbf{A}\Sigma\mathbf{A}')$.*

Theorem 1.12.2 (Distribution of Residuals). *Under Assumption 1.10 (normality):*

$$\boxed{\mathbf{e}|\mathbf{X} \sim N(\mathbf{0}, \sigma^2\mathbf{M})} \quad (1.56)$$

Proof. **Step 1: Identify the linear form:**

From equation (1.48), $\mathbf{e} = \mathbf{M}\varepsilon$. Since $\varepsilon \sim N(\mathbf{0}, \sigma^2\mathbf{I}_n)$ by assumption, the residuals are a linear transformation of a normal vector.

Step 2: Compute the First Moment:

$$E[\mathbf{e}|\mathbf{X}] = \mathbf{M}E[\varepsilon|\mathbf{X}] = \mathbf{M} \cdot \mathbf{0} = \mathbf{0} \quad (1.57)$$

Step 3: Compute the Second Moment (Covariance Matrix):

$$\text{Var}(\mathbf{e}|\mathbf{X}) = \mathbf{M} \text{Var}(\varepsilon|\mathbf{X})\mathbf{M}' = \mathbf{M}(\sigma^2\mathbf{I}_n)\mathbf{M} = \sigma^2(\mathbf{M}\mathbf{M}) = \sigma^2\mathbf{M} \quad (1.58)$$

Conclusion: The residual vector follows a multivariate normal distribution:

$$\mathbf{e}|\mathbf{X} \sim N(\mathbf{0}, \sigma^2\mathbf{M}) \quad (1.59)$$

□

1.12.6 Independence between $\hat{\beta}$ and $\mathbf{e}$

Theorem 1.12.3 (Independence between $\hat{\beta}$ and $\mathbf{e}$). *Under Assumption 1.10 (normality):*

$$\boxed{\hat{\beta} \perp \mathbf{e} \mid \mathbf{X}} \quad (1.60)$$

Proof. $\hat{\beta} = \beta + \mathbf{A}\varepsilon$, where $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. $\mathbf{e} = \mathbf{M}\varepsilon$.

We compute the covariance between the random terms:

$$\text{Cov}(\mathbf{A}\varepsilon, \mathbf{M}\varepsilon|\mathbf{X}) = \mathbf{A} \text{Var}(\varepsilon|\mathbf{X})\mathbf{M}' = \sigma^2\mathbf{A}\mathbf{M} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} = \mathbf{0} \quad (1.61)$$

Since $\hat{\beta}$ and $\mathbf{e}$ are linear transformations of a normal vector, zero covariance implies **stochastic independence**. □

1.13 Summary of CLRM Properties

Property	Expression	Condition
OLS Estimator	$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$	Full Rank
Unbiasedness	$E[\hat{\beta} \mathbf{X}] = \beta$	Strict Exogeneity
Variance of OLS	$\text{Var}(\hat{\beta} \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$	Sphericity
BLUE	$\text{Var}(\hat{\beta}) \preceq \text{Var}(\tilde{\beta})$	Gauss-Markov
Estimator of σ^2	$s^2 = \frac{\mathbf{e}'\mathbf{e}}{n-K}$	Sphericity
Unbiasedness of s^2	$E[s^2 \mathbf{X}] = \sigma^2$	Sphericity
Distribution of Residuals	$\mathbf{e} \sim N(\mathbf{0}, \sigma^2\mathbf{M})$	Normality
Independence	$\hat{\beta} \perp \mathbf{e} \mid \mathbf{X}$	Normality

Chapter 2

Hypothesis Testing in the Classical Normal Linear Regression Model

This chapter establishes the theoretical foundations for statistical inference in the Classical Normal Linear Regression Model (CNLRM), covering from the derivation of the t -statistic for hypotheses about a single parameter to the F -statistic for multiple linear restrictions, including confidence intervals and the trinity of asymptotic tests. The material is organized in progressive sections, moving from geometric intuition to rigorous algebraic formalism.

2.1 The Classical Normal Linear Regression Model (CNLRM)

2.1.1 Motivation

The concrete problem here is the **validity of inference**. Until now, the Ordinary Least Squares (OLS) method allowed us to obtain point estimates ($\hat{\beta}$), but did not tell us how likely it would be to obtain these values if the true relationship were null. By adding the normality assumption, we transform the model from an algebraic approximation tool into a complete probabilistic model. This allows us to determine the exact distribution of the estimators and construct tests that quantify uncertainty under unknown variance.

Geometric Interpretation in $\mathbb{R}^n$:

Under the normality assumption, the vector of population errors ε is not only orthogonal to the column space of $\mathbf{X}$ in expectation, but its probability density is spherically symmetric in $\mathbb{R}^n$. When we project $\mathbf{y}$ onto the subspace spanned by $\mathbf{X}$, the distance between the observed value and the projection (the residual) follows a distribution related to χ^2 , allowing us to “measure” the error in a standardized way.

2.1.2 CNLRM Assumptions

For the construction of exact tests, we add the following assumption to the CLRM:

Assumption 2.1: Normality of Errors ($\varepsilon|\mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$)

- **Role:** Ensures that any linear combination of y (such as the estimators $\hat{\beta}$) also has a normal distribution. This is the basis for the ratio between the estimation error and the standard error to follow a Student's t distribution.
 - **Counterexample:** If the errors follow a heavy-tailed distribution (such as Cauchy), the OLS estimator can still be computed, but the test statistic will not follow a t distribution in small samples, leading to incorrect conclusions about the significance of variables.
-

2.2 Inference about a Single Parameter: The t -Statistic

2.2.1 Motivation

The concrete problem solved by the t -statistic is hypothesis testing about a single regression coefficient when the population variance σ^2 is unknown. Instead of substituting σ^2 with an ad hoc estimator, the t -statistic incorporates the additional uncertainty from estimating the variance, resulting in a distribution with heavier tails than the Normal.

Geometric Interpretation in $\mathbb{R}^2$:

For a simple regression, we can visualize the estimator $\hat{\beta}_2$ as a random variable centered at β_2 . The standard error $se(\hat{\beta}_2)$ measures the dispersion of this variable. The t -statistic standardizes the distance between the estimate and the tested value, dividing it by the estimated standard error. The denominator s (root of the estimated variance) introduces additional variability, making the resulting distribution have heavier tails than the Normal.

2.2.2 Formal Statement

Consider the Simple Linear Regression Model defined for n observations:

$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i, \quad i = 1, \dots, n \quad (2.1)$$

where y_i is the dependent variable, x_i is the non-stochastic regressor, β_1 is the intercept, β_2 is the slope, and ε_i is the stochastic error.

We assume the following conditions of the Classical Normal Model:

1. **Exogeneity and Normality:** $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$, where σ^2 is constant (homoskedastic) and unknown.
2. **Identifiability:** There is variation in x , such that $\sum_{i=1}^n (x_i - \bar{x})^2 > 0$.

Theorem 2.2.1 (Distribution of the t -Statistic). *Under the CNLRM assumptions, the statistic:*

$$\boxed{T = \frac{\hat{\beta}_2 - \beta_2^0}{se(\hat{\beta}_2)} \sim t_{n-2}} \quad (2.2)$$

where $se(\hat{\beta}_2) = \frac{s}{\sqrt{\sum (x_i - \bar{x})^2}}$ and $s^2 = \frac{\sum_{i=1}^n \hat{e}_i^2}{n-2}$, follows a Student's t distribution with $n - 2$ degrees of freedom.

2.2.3 Formal Derivation

Step 1: Derivation of the OLS Estimators $(\hat{\beta}_1, \hat{\beta}_2)$

We minimize the sum of squared residuals $S(b_1, b_2) = \sum_{i=1}^n (y_i - b_1 - b_2 x_i)^2$.

The First Order Conditions (FOC) are:

$$\frac{\partial S}{\partial b_1} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) = 0 \quad (2.3)$$

$$\frac{\partial S}{\partial b_2} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) x_i = 0 \quad (2.4)$$

From (2.3):

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x} \quad (2.5)$$

Substituting (2.5) into (2.4):

$$\hat{\beta}_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \quad (2.6)$$

Step 2: Variance Estimator (s^2)

We define the residual as $\hat{e}_i = y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i$. The unbiased estimator of the variance is:

$$s^2 = \frac{\sum_{i=1}^n \hat{e}_i^2}{n-2} \quad (2.7)$$

(Justification: We lose 2 degrees of freedom when estimating β_1 and β_2 .)

Step 3: Sampling Distribution and Construction of the Test Statistic

Under the normality assumption, since $\hat{\beta}_2$ is a linear function of y (which is normal):

$$\hat{\beta}_2 \sim N\left(\beta_2, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right) \quad (2.8)$$

To test $H_0 : \beta_2 = \beta_2^0$, we standardize the estimator:

$$Z = \frac{\hat{\beta}_2 - \beta_2^0}{\sigma / \sqrt{\sum (x_i - \bar{x})^2}} \sim N(0, 1) \quad (2.9)$$

Since σ is unknown, we use s . The associated Chi-square variable is:

$$V = \frac{(n-2)s^2}{\sigma^2} \sim \chi_{n-2}^2 \quad (2.10)$$

Lemma 2.2.1 (Definition of the Student's t Distribution). *If $Z \sim N(0, 1)$ and $V \sim \chi_\nu^2$ are independent, then $T = \frac{Z}{\sqrt{V/\nu}} \sim t_\nu$.*

Applying Lemma 2.2.1 with $\nu = n - 2$:

$$T = \frac{\frac{\hat{\beta}_2 - \beta_2^0}{\sigma / \sqrt{\sum (x_i - \bar{x})^2}}}{\sqrt{\frac{(n-2)s^2}{\sigma^2} / (n-2)}} = \frac{\hat{\beta}_2 - \beta_2^0}{s / \sqrt{\sum (x_i - \bar{x})^2}} \quad (2.11)$$

Defining the estimated Standard Error as $se(\hat{\beta}_2) = \frac{s}{\sqrt{\sum (x_i - \bar{x})^2}}$:

$$T = \frac{\hat{\beta}_2 - \beta_2^0}{se(\hat{\beta}_2)} \sim t_{n-2} \quad (2.12)$$

□

2.3 Distribution of Linear Combinations of OLS

2.3.1 Motivation

The concrete problem here is **inference about multiple simultaneous hypotheses**. Frequently, economic or scientific theory does not predict only the value of a single parameter, but a relationship between them (e.g., constant returns to scale, where the sum of coefficients must be unity).

Geometric Interpretation in $\mathbb{R}^K$:

Let $\beta \in \mathbb{R}^K$ be the vector of population parameters. A single linear restriction defines a hyperplane of dimension $K - 1$ in the parameter space. A set of J linearly independent restrictions defines the intersection of J hyperplanes, resulting in an affine subspace of dimension $K - J$. Testing whether the restriction is valid is equivalent to measuring the probabilistic distance between the unrestricted estimator $\hat{\beta}$ (the point that minimizes the sum of squares in the total space) and the restricted subspace.

2.3.2 Additional Assumptions

Assumption 2.2: Full Column Rank ($\text{rank}(\mathbf{X}) = K$)

- **Role:** Guarantees the identifiability of $\hat{\beta}$ and the invertibility of $(\mathbf{X}'\mathbf{X})$.
- **Counterexample:** If the columns of $\mathbf{X}$ are linearly dependent, there is no unique point in the parameter space to compare with the restrictions.

Assumption 2.3: Normality of Errors ($\varepsilon|\mathbf{X} \sim N(\mathbf{0}, \sigma^2\mathbf{I}_n)$)

- **Role:** This is the generating assumption. Since the estimator is a linear transformation of the errors, the normality of ε is inherited by $\hat{\beta}$ in any finite sample.
- **Counterexample:** Without normality, $\hat{\beta}$ follows a normal distribution only asymptotically ($n \rightarrow \infty$). In small samples, the distribution would be unknown, invalidating exact tests.

Assumption 2.4: Independence and Rank of Restrictions ($\text{rank}(\mathbf{R}) = J < K$)

- **Role:** Ensures that the restrictions are not redundant or contradictory, guaranteeing that the variance-covariance matrix of the restrictions is invertible.
- **Counterexample:** If $\mathbf{R}$ does not have full rank, the “ellipse of uncertainty” of the restrictions will be collapsed in some dimension, preventing the calculation of the test statistic.

2.3.3 Formal Statement

Consider the model $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, with $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times K}$, $\text{rank}(\mathbf{X}) = K$, and $\varepsilon \sim N(\mathbf{0}, \sigma^2\mathbf{I}_n)$.

Consider a set of J linear restrictions on β of the form:

$$\mathbf{R}\beta = \mathbf{q} \quad (2.13)$$

where $\mathbf{R}$ is a $J \times K$ matrix of constants of rank J , and $\mathbf{q}$ is a $J \times 1$ vector of constants.

The unrestricted OLS estimator is $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.

Theorem 2.3.1 (Distribution of the Restriction Vector). *Under the CNLRM assumptions, the vector of sample restrictions $\mathbf{m} = \mathbf{R}\hat{\beta} - \mathbf{q}$ follows a multivariate normal distribution:*

$$\boxed{\mathbf{m}|\mathbf{X} \sim N(\mathbf{R}\beta - \mathbf{q}, \sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}')} \quad (2.14)$$

2.3.4 Formal Derivation

Lemma 2.3.1 (Affine Transformation of Normal Vectors). *If $\mathbf{z} \sim N(\mu, \Sigma)$ and $\mathbf{w} = \mathbf{a} + \mathbf{Bz}$ (where $\mathbf{a}$ and $\mathbf{B}$ are constants), then $\mathbf{w} \sim N(\mathbf{a} + \mathbf{B}\mu, \mathbf{B}\Sigma\mathbf{B}')$.*

Step 1: Distribution of $\mathbf{y}$ conditional on $\mathbf{X}$

Since $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, and $\varepsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$:

$$E[\mathbf{y}|\mathbf{X}] = \mathbf{X}\beta, \quad \text{Var}(\mathbf{y}|\mathbf{X}) = \sigma^2 \mathbf{I}_n \quad (2.15)$$

Hence:

$$\mathbf{y}|\mathbf{X} \sim N(\mathbf{X}\beta, \sigma^2 \mathbf{I}_n) \quad (2.16)$$

Step 2: Distribution of the Estimator $\hat{\beta}$

The OLS estimator is $\hat{\beta} = \mathbf{B}\mathbf{y}$, where $\mathbf{B} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

Applying Lemma 2.3.1:

$$E[\hat{\beta}|\mathbf{X}] = \beta, \quad \text{Var}(\hat{\beta}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} \quad (2.17)$$

Hence:

$$\hat{\beta}|\mathbf{X} \sim N\left(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\right) \quad (2.18)$$

Step 3: Distribution of the Restriction Matrix $\mathbf{m}$

We define the discrepancy vector $\mathbf{m} = \mathbf{R}\hat{\beta} - \mathbf{q}$.

Since $\mathbf{m}$ is an affine function of $\hat{\beta}$, we apply Lemma 2.3.1 again:

$$E[\mathbf{m}|\mathbf{X}] = \mathbf{R}\beta - \mathbf{q}, \quad \text{Var}(\mathbf{m}|\mathbf{X}) = \sigma^2 \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{R}' \quad (2.19)$$

Therefore:

$$\mathbf{m}|\mathbf{X} \sim N\left(\mathbf{R}\beta - \mathbf{q}, \sigma^2 \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{R}'\right) \quad (2.20)$$

□

2.4 Cholesky Decomposition

2.4.1 Motivation

The concrete problem solved by the Cholesky decomposition is the “**standardization**” of **multivariate linear systems**. Just as in scalar statistics we divide a variable by its standard deviation to obtain unit variance, in matrices we seek an operator $\mathbf{L}$ such that, when applied to a system with variance-covariance matrix Σ , it results in a system with identity $\mathbf{I}$.

Geometric Interpretation in $\mathbb{R}^n$:

A covariance matrix Σ defines the geometry of an ellipsoid of uncertainty. The Cholesky decomposition extracts a lower triangular matrix $\mathbf{L}$ that functions as a coordinate system.

Multiplying a white noise vector (spherical) by $\mathbf{L}$ “stretches” and “rotates” this vector so that it assumes the shape of the ellipsoid defined by $\mathbf{\Sigma}$. Conversely, multiplying correlated data by $\mathbf{L}^{-1}$ (the *whitening* process) collapses the ellipsoid back to a unit sphere.

2.4.2 Formal Statement

Theorem 2.4.1 (Cholesky Decomposition). *Let $\mathbf{A} \in \mathbb{R}^{K \times K}$ be a square, symmetric, and positive definite matrix. There exists a unique lower triangular matrix $\mathbf{L} \in \mathbb{R}^{K \times K}$, with strictly positive diagonal elements ($L_{ii} > 0$), such that:*

$$\boxed{\mathbf{A} = \mathbf{L}\mathbf{L}'} \quad (2.21)$$

where $\mathbf{L}$ is referred to as the *Cholesky square root* of $\mathbf{A}$.

2.4.3 Algorithmic Derivation

Notation Definition:

- A_{ij} : element in row i and column j of $\mathbf{A}$.
- L_{ij} : element in row i and column j of $\mathbf{L}$.
- $\mathbf{L}$ is lower triangular $\implies L_{ij} = 0$ for $j > i$.

Step 1: Expansion of the Matrix Product

By the definition of matrix multiplication:

$$A_{ij} = \sum_{k=1}^K L_{ik} L_{jk} \quad (2.22)$$

Since $\mathbf{L}$ is lower triangular:

$$A_{ij} = \sum_{k=1}^j L_{ik} L_{jk} \quad (2.23)$$

Step 2: Diagonal Elements ($i = j$)

For a generic diagonal element L_{ii} :

$$A_{ii} = \sum_{k=1}^{i-1} L_{ik}^2 + L_{ii}^2 \quad (2.24)$$

Isolating:

$$L_{ii} = \sqrt{A_{ii} - \sum_{k=1}^{i-1} L_{ik}^2} \quad (2.25)$$

Step 3: Off-Diagonal Elements ($i > j$)

From (2.23) and isolating L_{ij} :

$$L_{ij} = \frac{1}{L_{jj}} \left(A_{ij} - \sum_{k=1}^{j-1} L_{ik} L_{jk} \right) \quad (2.26)$$

Lemma 2.4.1 (Uniqueness of Cholesky Decomposition). *Suppose $\mathbf{A} = \mathbf{L}_1 \mathbf{L}_1' = \mathbf{L}_2 \mathbf{L}_2'$. Since $\mathbf{A}$ is positive definite, it is invertible. This implies $\mathbf{L}_1^{-1} \mathbf{L}_2 = \mathbf{L}_1' (\mathbf{L}_2')^{-1}$. The left-hand side is lower triangular and the right-hand side is upper triangular. A matrix that is simultaneously lower and upper triangular must be diagonal. Since the diagonals are positive, the only solution is the identity, hence $\mathbf{L}_1 = \mathbf{L}_2$.*

2.5 Multivariate Standardization and the χ^2 Statistic

2.5.1 Motivation

The concrete problem solved here is the **comparison of distances in correlated spaces**. When we test multiple simultaneous hypotheses (e.g., $\mathbf{R}\beta = \mathbf{q}$), the estimation error in one parameter may be correlated with the error in another. If we tried to sum the squares of these errors directly, we would be ignoring the geometry of the system.

Geometric Interpretation in $\mathbb{R}^J$:

Imagine that the uncertainty of the restrictions forms an ellipsoid of probability density. The Cholesky Decomposition extracts the dependence structure of this ellipsoid.

1. **Normal to Standard:** The Cholesky factor acts as a “whitening” operator, rotating and collapsing the uncertainty ellipsoid into a perfect unit-radius sphere.
 2. **Standard to Chi-square:** Once we have a unit sphere, the sum of the squares of the distances from the center to the surface follows, by definition, a χ^2 distribution.
-

2.5.2 Formal Statement

Let $\mathbf{m} = \mathbf{R}\hat{\beta} - \mathbf{q}$ be the discrepancy vector of the restrictions of dimension $J \times 1$. Under the null hypothesis $H_0 : \mathbf{R}\beta = \mathbf{q}$, we assume:

1. **Distribution:** $\mathbf{m} \sim N(\mathbf{0}, \mathbf{V}_m)$, where $\mathbf{V}_m = \sigma^2 \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{R}'$.
2. **Variance Structure:** $\mathbf{V}_m$ is a $J \times J$ symmetric and positive definite matrix.
3. **Decomposition:** There exists a lower triangular matrix $\mathbf{L}$ such that $\mathbf{V}_m = \mathbf{L}\mathbf{L}'$.

Theorem 2.5.1 (Wald Statistic). *The Wald statistic for testing $H_0 : \mathbf{R}\beta = \mathbf{q}$ is:*

$$\boxed{W = \mathbf{m}' \mathbf{V}_m^{-1} \mathbf{m} \sim \chi_J^2} \quad (2.27)$$

2.5.3 Formal Derivation

Step 1: Transformation to Standard Normal (Whitening)

We define the transformed vector $\mathbf{z} = \mathbf{L}^{-1}\mathbf{m}$.

By the Affine Transformation Lemma 2.3.1:

$$E[\mathbf{z}|\mathbf{X}] = \mathbf{L}^{-1}E[\mathbf{m}|\mathbf{X}] = \mathbf{0} \quad (2.28)$$

$$\text{Var}(\mathbf{z}|\mathbf{X}) = \mathbf{L}^{-1}\mathbf{V}_m(\mathbf{L}^{-1})' = \mathbf{L}^{-1}(\mathbf{L}\mathbf{L}')(\mathbf{L}^{-1})' = \mathbf{I}_J \quad (2.29)$$

Hence:

$$\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_J) \quad (2.30)$$

Step 2: Transformation to Chi-square (χ^2)

We define the test statistic W as the inner product (sum of squares) of the standardized vector:

$$W = \mathbf{z}'\mathbf{z} = \mathbf{m}'(\mathbf{L}^{-1})'\mathbf{L}^{-1}\mathbf{m} = \mathbf{m}'\mathbf{V}_m^{-1}\mathbf{m} \quad (2.31)$$

Lemma 2.5.1 (Distribution of Quadratic Forms). *If $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_J)$, then $\mathbf{z}'\mathbf{z} = \sum_{j=1}^J z_j^2 \sim \chi_J^2$.*

Applying Lemma 2.5.1:

$$W \sim \chi_J^2 \quad (2.32)$$

□

2.6 The Snedecor F Statistic

2.6.1 Motivation

Until now, we have derived test statistics that depended on the population parameter σ^2 . However, in real applications, this parameter is invariably unknown. The problem solved by the F statistic is to allow simultaneous inference about J linear restrictions using an estimate of the variance based on the model residuals (s^2).

Geometric Interpretation in $\mathbb{R}^n$:

The F statistic can be viewed as the ratio between two normalized orthogonal distances in $\mathbb{R}^n$. The numerator measures the (squared) length of the projection of $\mathbf{y}$ that is “lost” when we impose the restrictions of H_0 . The denominator measures the (squared) length of the residual vector of the unrestricted model, which represents the inherent variability

of the error. The independence between these two measures ensures that the ratio follows an F distribution.

2.6.2 Formal Statement

Consider the model $\mathbf{y} = \mathbf{X}\beta + \varepsilon$ with $\varepsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ and $\text{rank}(\mathbf{X}) = K$. Consider the null hypothesis $H_0 : \mathbf{R}\beta = \mathbf{q}$, where $\mathbf{R}$ is $J \times K$ with $\text{rank}(\mathbf{R}) = J$.

We define:

1. **Variance Estimator (s^2):** $s^2 = \frac{\mathbf{e}'\mathbf{e}}{n-K}$, where $\mathbf{e} = \mathbf{M}\mathbf{y}$.
2. **Discrepancy Vector ($\mathbf{m}$):** $\mathbf{m} = \mathbf{R}\hat{\beta} - \mathbf{q}$.

Theorem 2.6.1 (F Statistic). *Under the CNLRM assumptions, the statistic:*

$$F = \frac{\mathbf{m}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}\mathbf{m}/J}{s^2} \sim F_{J, n-K} \quad (2.33)$$

follows a Snedecor F distribution with J and $n - K$ degrees of freedom.

2.6.3 Formal Derivation

Step 1: Distribution of s^2

We start from the definition of residuals: $\mathbf{e} = \mathbf{M}\mathbf{y}$, where $\mathbf{M} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

Substituting the model structure:

$$\mathbf{e} = \mathbf{M}(\mathbf{X}\beta + \varepsilon) = \mathbf{M}\varepsilon \quad (2.34)$$

The sum of squared residuals (SSR) is:

$$\mathbf{e}'\mathbf{e} = \varepsilon'\mathbf{M}\varepsilon \quad (2.35)$$

Lemma 2.6.1 (Distribution of Quadratic Forms with Idempotent Matrices). *If $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$ and $\mathbf{A}$ is idempotent with $\text{rank}(\mathbf{A}) = \nu$, then $\mathbf{z}'\mathbf{A}\mathbf{z} \sim \chi_\nu^2$.*

Applying Lemma 2.6.1:

$$\frac{(n-K)s^2}{\sigma^2} \sim \chi_{n-K}^2 \quad (2.36)$$

Step 2: Independence between Numerator and Denominator

The Wald statistic W depends on $\hat{\beta}$, while s^2 depends on $\mathbf{e}$.

$$\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon, \quad \mathbf{e} = \mathbf{M}\varepsilon \quad (2.37)$$

We compute the covariance:

$$\text{Cov}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon, \mathbf{M}\varepsilon|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} = \mathbf{0} \quad (2.38)$$

Since $\hat{\beta}$ and $\mathbf{e}$ are linear transformations of a normal vector, zero covariance implies **stochastic independence**.

Step 3: Derivation of the F Statistic

We define $W^* = \mathbf{m}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}\mathbf{m}$. Note that $W = W^*/\sigma^2 \sim \chi_J^2$.

By the definition of the F distribution (ratio of two independent χ^2 variables divided by their respective degrees of freedom):

$$F = \frac{W/J}{\frac{(n-K)s^2}{\sigma^2}/(n-K)} \quad (2.39)$$

Canceling σ^2 :

$$F = \frac{\mathbf{m}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}\mathbf{m}/J}{s^2} \quad (2.40)$$

Applying the derived distributions:

$$F \sim \frac{\chi_J^2/J}{\chi_{n-K}^2/(n-K)} = F_{J,n-K} \quad (2.41)$$

□

2.7 The Feasible Wald Test

2.7.1 Motivation

The concrete problem solved here is the **operationalization of multiple hypothesis tests** when the noise variance (σ^2) is unknown. While the theoretical Wald statistic uses the true variance, the feasible test substitutes this parameter with its sample estimate (s^2), allowing the researcher to perform tests of linear restrictions using only the sample data.

Geometric Interpretation in $\mathbb{R}^J$:

Geometrically, the Wald statistic is a measure of the **weighted Euclidean distance** between the vector of sample restrictions ($\mathbf{R}\hat{\beta}$) and the vector of restrictions under the null hypothesis ($\mathbf{q}$). The use of s^2 in the denominator acts as an “adaptive ruler”: if the estimated noise in the data is high, the measured distance between the coefficients and the hypothesis must be significantly larger to be considered statistically relevant. The F statistic is nothing more than this distance normalized by the number of dimensions (restrictions) being tested simultaneously.

2.7.2 Formal Statement

Consider the model $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, where $\varepsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ and $\text{rank}(\mathbf{X}) = K$. Consider the null hypothesis $H_0 : \mathbf{R}\beta = \mathbf{q}$, where $\mathbf{R}$ is a $J \times K$ matrix of constants of rank J and $\mathbf{q}$ is a $J \times 1$ vector.

We define:

1. **Discrepancy Vector:** $\mathbf{m} = \mathbf{R}\hat{\beta} - \mathbf{q}$.
2. **Residual Variance Estimator:** $s^2 = \frac{\mathbf{e}'\mathbf{e}}{n-K}$, where $\mathbf{e} = \mathbf{y} - \mathbf{X}\hat{\beta}$.
3. **Feasible Wald Statistic (W):**

$$W = \frac{\mathbf{m}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}\mathbf{m}}{s^2} \quad (2.42)$$

Theorem 2.7.1 (Wald-F Relationship). *Under H_0 and normality, the statistic $F = W/J$ follows a Snedecor F distribution with J and $n - K$ degrees of freedom:*

$$F = \frac{W}{J} \sim F_{J, n-K}$$

(2.43)

2.7.3 Formal Derivation

Lemma 2.7.1 (Distribution of $\mathbf{m}$ under H_0). *Under H_0 and normality, $\mathbf{m}|\mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}')$.*

Step 1: Construction of the χ^2 Quadratic Form for the Numerator

Consider the theoretical Wald statistic (W_σ) using the population variance σ^2 :

$$W_\sigma = \mathbf{m}'[\text{Var}(\mathbf{m}|\mathbf{X})]^{-1}\mathbf{m} \quad (2.44)$$

Substituting the variance from Lemma 2.7.1:

$$W_\sigma = \frac{\mathbf{m}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}\mathbf{m}}{\sigma^2} \quad (2.45)$$

By the theory of quadratic forms of normal vectors, since $\mathbf{m}$ has dimension J :

$$W_\sigma \sim \chi_J^2 \quad (2.46)$$

Step 2: Construction of the χ^2 Quadratic Form for the Denominator

We define the chi-square variable based on the sum of squared residuals (SSR):

$$V = \frac{(n-K)s^2}{\sigma^2} \sim \chi_{n-K}^2 \quad (2.47)$$

Step 3: Stochastic Independence

As demonstrated in Section 1.6.3, $\hat{\beta}$ (and hence $\mathbf{m}$) is independent of the residual vector $\mathbf{e}$ (and hence s^2) under the normality assumption. Therefore, W_σ and V are independent.

Step 4: Definition of the F Ratio

By definition, an F variable is the ratio of two independent χ^2 variables, each divided by its respective degrees of freedom:

$$F = \frac{W_\sigma/J}{V/(n-K)} \quad (2.48)$$

Substituting (2.45) and (2.47) into (2.48):

$$F = \frac{\frac{\mathbf{m}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}\mathbf{m}}{\sigma^2 \cdot J}}{\frac{s^2}{\sigma^2}} \quad (2.49)$$

Canceling the unknown parameter σ^2 :

$$F = \frac{\mathbf{m}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}\mathbf{m}}{s^2 \cdot J} \quad (2.50)$$

Recalling the definition of W in (2.42):

$$F = \frac{W}{J} \quad (2.51)$$

□

2.8 Confidence Intervals and Ellipsoidal Regions

2.8.1 Motivation

Point estimators, although unbiased, do not quantify the intrinsic uncertainty of the sampling process. The Confidence Interval solves the problem of providing a **set of plausible values** for the population parameter, based on the precision of the estimator.

Geometric and Spherical Interpretation:

Consider the parameter space $\mathbb{R}^K$.

1. **Univariate Case:** The CI is a line segment centered at the estimate.
2. **Multivariate Case (Spherical Interpretation):** Under the assumption that the parameter estimates are independent and have the same variance (spherical errors), the joint confidence region in $\mathbb{R}^2$ is a **circle** centered at $(\hat{\beta}_1, \hat{\beta}_2)$.

3. **Covariance and Slope:** When there is covariance between the estimators ($\text{Cov}(\hat{\beta}_1, \hat{\beta}_2) \neq 0$), the circle deforms into an **ellipse**. The slope of the principal axis of this ellipse is governed by the covariance structure: if the correlation is positive, the ellipse tilts to the right; if negative, to the left.

2.8.2 Formal Statement

Let $(X_1, \dots, X_n)$ be a random sample from a population with parameter $\theta \in \Theta$. We define the Confidence Interval of level $1 - \alpha$ as the random set $\hat{C} = [\hat{L}, \hat{U}]$ such that:

$$P(\theta \in [\hat{L}, \hat{U}]) = 1 - \alpha \quad (2.52)$$

where $\hat{L}$ and $\hat{U}$ are statistics (functions of the data).

Theorem 2.8.1 (Confidence Interval for β_j). *Under the CNLRM assumptions:*

$$CI_{1-\alpha}(\beta_j) = \hat{\beta}_j \pm t_{n-K, \alpha/2} \cdot se(\hat{\beta}_j) \quad (2.53)$$

Theorem 2.8.2 (Confidence Interval for Linear Combination). *For $\mathbf{c}'\beta$:*

$$CI_{1-\alpha}(\mathbf{c}'\beta) = \mathbf{c}'\hat{\beta} \pm t_{n-K, \alpha/2} \cdot \sqrt{s^2 \mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}} \quad (2.54)$$

Theorem 2.8.3 (Joint Confidence Region). *The joint confidence region of level $1 - \alpha$ for β is:*

$$\hat{C}_{joint} = \left\{ \beta : (\hat{\beta} - \beta)'(\mathbf{X}'\mathbf{X})(\hat{\beta} - \beta) \leq K \cdot s^2 \cdot F_{K, n-K, 1-\alpha} \right\} \quad (2.55)$$

2.8.3 Formal Derivation (Univariate Case)

Lemma 2.8.1 (Distribution of the Sample Mean). *If $X_i \sim N(\mu, \sigma^2)$, then $\bar{X} \sim N(\mu, \sigma^2/n)$.*

Definition of the Pivot Variable:

We define the standardized variable Z :

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \quad (2.56)$$

Note that the distribution of Z does not depend on μ .

Step 1: Establishing the Central Probability

Given a significance level α , we select the quantile $z_{\alpha/2}$ such that:

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha \quad (2.57)$$

Step 2: Substitution of the Pivot Variable

Substituting (2.56) into (2.57):

$$P\left(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha \quad (2.58)$$

Step 3: Isolating the Parameter μ

Multiplying all terms by the standard error $\sigma/\sqrt{n}$:

$$P\left(-z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha \quad (2.59)$$

Subtracting $\bar{X}$ and rearranging:

$$P\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha \quad (2.60)$$

Conclusion: The interval $\hat{C} = \left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right]$ is the $1 - \alpha$ CI for μ .

When σ^2 is unknown, we substitute $z_{\alpha/2}$ with $t_{n-1, \alpha/2}$, resulting in (2.53).

2.8.4 Generalization to Ellipsoidal Regions

When dealing with a vector β of dimension K , the probabilistic distance is measured via the Mahalanobis quadratic form. Under the null hypothesis that β is the true value, the Wald statistic W is:

$$W(\beta) = (\hat{\beta} - \beta)' [\text{Var}(\hat{\beta})]^{-1} (\hat{\beta} - \beta) \sim \chi_K^2 \quad (2.61)$$

The joint confidence region is the set:

$$\hat{C}_{\text{joint}} = \{\beta : W(\beta) \leq \chi_{K, 1-\alpha}^2\} \quad (2.62)$$

When σ^2 is unknown, we use s^2 and the F distribution, resulting in (2.55).

2.9 Joint Significance Test and the Algebra of Sums of Squares

2.9.1 Motivation

The concrete problem solved by the joint significance test is the **collective validity of the regressors**. Frequently, individual variables may not be statistically significant in isolation (due to multicollinearity, for example), but together they explain a relevant portion of the variance of the phenomenon. The test seeks to answer: “Is the proposed model better than just using the sample mean to predict the dependent variable?”

Geometric Interpretation in $\mathbb{R}^n$:

In the sample space, the Total Sum of Squares (SST) represents the squared length of the vector of observations centered at their mean. OLS decomposes this vector into two orthogonal components: the Explained Sum of Squares (SSE), which is the projection of the vector onto the subspace spanned by the regressors, and the Residual Sum of Squares (SSR), which is the orthogonal distance of the vector to the subspace. The R^2 is the squared cosine of the angle between the observed vector and its projection, indicating the angular “proximity” between the data and the model.

2.9.2 Formal Statement

Consider the linear regression model $\mathbf{y} = \mathbf{X}\beta + \varepsilon$ with n observations and K parameters (including the intercept). We define:

1. **Total Sum of Squares (SST):** $SST = \sum_{i=1}^n (y_i - \bar{y})^2$.
2. **Explained Sum of Squares (SSE):** $SSE = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$.
3. **Residual Sum of Squares (SSR):** $SSR = \sum_{i=1}^n \hat{\varepsilon}_i^2$, where $\hat{\varepsilon}_i = y_i - \hat{y}_i$.
4. **Coefficient of Determination (R^2):** $R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}$.

Theorem 2.9.1 (Decomposition of Sums of Squares). *If the model contains an intercept, then $SST = SSE + SSR$.*

2.9.3 Derivation of the F Statistic via R^2

Step 1: Comparison between Restricted and Unrestricted Models

Let an unrestricted model (ur) with K parameters and a restricted model (r) with $K - J$ parameters, where J is the number of exclusion restrictions imposed (e.g., $\beta_j = 0$).

Proposition 2.9.1. $R_{ur}^2 \geq R_r^2$ and $SSR_{ur} \leq SSR_r$.

Proof. The unrestricted OLS estimator minimizes the sum of squared residuals over the space $\mathbb{R}^K$. The restricted model minimizes the same function over a lower-dimensional subspace ($K - J$). Since the set of solutions of the restricted model is contained in the set of the unrestricted model, the minimum value attained by the unrestricted model will necessarily be less than or equal to that of the restricted model. $\square$

Step 2: Construction of the F Statistic via SSR

To test the joint validity of J linear restrictions:

$$F = \frac{(SSR_r - SSR_{ur})/J}{SSR_{ur}/(n - K)} \quad (2.63)$$

Step 3: Construction of the F Statistic via R^2

From the definition of R^2 : $SSR_{ur} = SST(1 - R_{ur}^2)$ and $SSR_r = SST(1 - R_r^2)$.

Substituting into (2.63):

$$F = \frac{(R_{ur}^2 - R_r^2)/J}{(1 - R_{ur}^2)/(n - K)} \sim F_{J, n-K} \quad (2.64)$$

Step 4: Special Case of Global Significance

In the test where $H_0 : \beta_2 = \beta_3 = \dots = \beta_K = 0$ (all slopes are zero), the restricted model is $y = \beta_1 + \varepsilon$. In this case, $R_r^2 = 0$. Substituting into (2.64):

$$F = \frac{R^2/(K - 1)}{(1 - R^2)/(n - K)} \sim F_{K-1, n-K} \quad (2.65)$$

2.10 Summary of Test Statistics

Hypothesis	Statistic	Distribution	Degrees of Freedom
$H_0 : \beta_j = \beta_j^0$	$t = \frac{\hat{\beta}_j - \beta_j^0}{se(\hat{\beta}_j)}$	t_{n-K}	$n - K$
$H_0 : \mathbf{R}\beta = \mathbf{q}$ (with σ^2 known)	$W = \mathbf{m}'\mathbf{V}_m^{-1}\mathbf{m}$	χ_J^2	J
$H_0 : \mathbf{R}\beta = \mathbf{q}$ (with σ^2 unknown)	$F = \frac{W^*/J}{s^2}$	$F_{J, n-K}$	$J, n - K$
$H_0 : \beta_2 = \dots = \beta_K = 0$	$F = \frac{R^2/(K-1)}{(1-R^2)/(n-K)}$	$F_{K-1, n-K}$	$K - 1, n - K$

2.11 Role of Assumptions

1. **Normality of Errors (Assumption 2.1):** Indispensable to guarantee that test statistics follow exact distributions in finite samples. Without normality, t , F , and W have only asymptotic validity.
2. **Full Rank (Assumption 2.2):** Indispensable to guarantee the identifiability of parameters and the invertibility of $(\mathbf{X}'\mathbf{X})$, necessary for computing variances.

3. **Rank of Restrictions (Assumption 2.4):** Indispensable to guarantee that the matrix $\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$ is invertible, allowing the calculation of the Wald and F statistics.
4. **Sphericity** ($E[\varepsilon\varepsilon'] = \sigma^2\mathbf{I}$): Simplifies the variance structure, allowing closed-form statistics. Under heteroskedasticity, formulas require robust adjustments.
5. **Homoskedasticity:** Indispensable for the relationship $F = W/J$. If the variance were heteroskedastic, the term $\text{Var}(\mathbf{m}|\mathbf{X})$ would involve the matrix $\mathbf{\Omega}$, preventing the isolation of the scalar s^2 and breaking the identity.
6. **Presence of Intercept:** Necessary for the decomposition $SST = SSE + SSR$. If the intercept is omitted, SST is not comparable, making the statistic based on R^2 mathematically invalid.

Chapter 3

Heteroskedasticity

3.1 The Heteroskedasticity Problem

3.1.1 Motivation and Geometric Interpretation

The Ordinary Least Squares (**OLS**) estimator is built upon the premise that errors possess constant variance — the so-called **homoskedasticity**. **Heteroskedasticity** arises when this variance is not uniform but varies systematically with the explanatory variables.

In empirical contexts, this situation is frequent. When modeling savings as a function of income, for example, higher-income families exhibit much greater dispersion in their saving habits than low-income families — the uncertainty about behavior increases with the level of the variable itself.

Geometric Interpretation in $\mathbb{R}^2$:

In a scatter plot (X, Y) :

- Under **homoskedasticity**, the cloud of points exhibits uniform thickness around the regression line.
- Under **heteroskedasticity**, this cloud assumes a cone or fan shape: the vertical dispersion of residuals expands as we move along the X axis.

The OLS estimator fits the line through the center of this cloud but assigns equal weight to all observations — ignoring that points in regions of high dispersion are less precise informants about the location of the true line than those in regions of low dispersion.

3.1.2 Classical Assumptions and Their Roles

Assumption 3.1: Linearity in Parameters

- **Role:** Ensures that the population relationship is expressed as a linear combination of fixed parameters and an additive error, enabling the OLS estimator.
- **Counterexample:** If the model is intrinsically non-linear (e.g., $Y = e^{\beta X} \cdot u$), OLS fails to capture the curvature, generating systematically biased residuals.

Assumption 3.2: No Perfect Multicollinearity

- **Role:** Ensures that $\mathbf{X}$ has full rank, enabling the inversion of $(\mathbf{X}'\mathbf{X})$ and the unique identification of parameters.
- **Counterexample:** If $X_2 = 2X_1$, there are infinitely many lines that minimize the sum of squares, making β non-identifiable.

Assumption 3.3: Strict Exogeneity ($E[\mathbf{u}|\mathbf{X}] = \mathbf{0}$)

- **Role:** Ensures that the error does not carry systematic information correlated with the regressors — a fundamental condition for the unbiasedness of OLS.
- **Counterexample:** With endogeneity ($E[\mathbf{u}|\mathbf{X}] \neq \mathbf{0}$), the estimator absorbs into β effects belonging to the error, generating bias even in infinite samples.

Assumption 3.4: Heteroskedasticity ($E[\mathbf{u}\mathbf{u}'|\mathbf{X}] = \mathbf{\Omega}$)

- **Role:** Replaces the homoskedasticity assumption. The matrix $\mathbf{\Omega}$ is diagonal with elements $\text{diag}(\mathbf{\Omega}) = \{\sigma_1^2, \dots, \sigma_n^2\}$ not necessarily equal.
- **Counterexample:** Erroneously assuming $E[\mathbf{u}\mathbf{u}'|\mathbf{X}] = \sigma^2\mathbf{I}$ produces incorrect standard error formulas, invalidating all significance tests (t and F).

3.1.3 Formal Statement of the Problem

Consider the linear population model in matrix notation:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{u} \quad (3.1)$$

where $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times k}$ with $\text{rank}(\mathbf{X}) = k < n$, $\beta \in \mathbb{R}^k$, and $\mathbf{u} \in \mathbb{R}^n$. Conditionally on $\mathbf{X}$, we assume:

1. $E[\mathbf{u}|\mathbf{X}] = \mathbf{0}$ (strict exogeneity);
2. $E[\mathbf{u}\mathbf{u}'|\mathbf{X}] = \mathbf{\Omega} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$ (heteroskedasticity).

The OLS estimator is given by:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \quad (3.2)$$

Objective of this section: To demonstrate that, under heteroskedasticity, $\hat{\beta}$ remains unbiased and consistent, but:

- the classical formula $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ becomes invalid for the variance;
- the estimator ceases to be the Best Linear Unbiased Estimator (**BLUE**).

3.1.4 Formal Derivation of Consequences

Step 1: Decomposition of the Estimator

Substituting (3.1) into (3.2) and using $(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X}) = \mathbf{I}_k$:

$$\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} \quad (3.3)$$

Step 2: Unbiasedness

Applying the conditional expectation in (3.3) and using Assumption 3.3:

$$E[\hat{\beta}|\mathbf{X}] = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' E[\mathbf{u}|\mathbf{X}] = \beta \quad (3.4)$$

Partial conclusion: OLS is unbiased under heteroskedasticity — the variance of the errors does not affect the mean of the estimator.

Step 3: True Variance — Sandwich Form

For an unbiased estimator, $\text{Var}[\hat{\beta}|\mathbf{X}] = E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'|\mathbf{X}]$. From (3.3):

$$\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} \quad (3.5)$$

Computing the outer product and extracting the deterministic terms:

$$\text{Var}[\hat{\beta}|\mathbf{X}] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' E[\mathbf{u}\mathbf{u}'|\mathbf{X}] \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$$

Substituting Assumption 3.4:

$$\boxed{\text{Var}[\hat{\beta}|\mathbf{X}] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}} \quad (3.6)$$

This is the **sandwich form** of the true variance of the OLS estimator.

Step 4: Invalidity of Classical Inference

Classical inference assumes $E[\mathbf{u}\mathbf{u}'|\mathbf{X}] = \sigma^2\mathbf{I}$. Substituting into (3.6):

$$\text{Var}_{\text{classical}}[\hat{\beta}|\mathbf{X}] = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

If $\boldsymbol{\Omega} \neq \sigma^2\mathbf{I}$, this expression is **not** the correct variance estimator. Since the t and F statistics depend on the standard errors in the denominator, using the classical formula under heteroskedasticity produces biased tests, leading to incorrect conclusions about significance.

3.1.5 Synthesis of the Role of Assumptions

- **Strict exogeneity** ($E[\mathbf{u}|\mathbf{X}] = \mathbf{0}$) was indispensable for guaranteeing unbiasedness in (3.4).
- **Homoskedasticity** is the assumption that “fails” in this context: without it, OLS loses efficiency within the BLUE class (the Gauss-Markov Theorem no longer applies), making it necessary to use:
 - **robust standard errors** (EHW estimator) for valid inference; or
 - the **Generalized Least Squares (GLS) estimator** to recover efficiency.

3.2 The Eicker-Huber-White (EHW) Estimator

3.2.1 Motivation and Geometric Interpretation

Although $\hat{\beta}$ remains unbiased and consistent under heteroskedasticity, the classical variance formula $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ fails to capture the true uncertainty — typically underestimating standard errors and inflating t -statistics, leading to false positives in significance tests.

The **Eicker-Huber-White (EHW)** estimator solves this problem by allowing each observation to have its own variance, providing a robust basis for inference **without requiring knowledge of the functional form of heteroskedasticity**.

Geometric Interpretation in $\mathbb{R}^n$:

The EHW estimator can be interpreted as a non-parametric reconstruction of the **ellipsoid of uncertainty** around $\hat{\beta}$. Instead of assuming that this ellipsoid is a uniform hypersphere (homoskedasticity), the EHW estimator estimates its irregular shape using the squared OLS residuals as a local proxy for the variance of each observation.

3.2.2 Specific Assumptions of the EHW Estimator

Assumption 3.5: Contemporary Orthogonality ($E[\mathbf{x}_i u_i] = \mathbf{0}$)

- **Role:** Minimum exogeneity condition for OLS consistency. Ensures that the regressors do not carry systematic information about the error.
- **Counterexample:** With endogeneity ($E[\mathbf{x}_i u_i] \neq \mathbf{0}$), the estimator converges to a value distinct from β — no variance correction recovers the validity of inference.

Assumption 3.6: Independence between Observations ($E[u_i u_j | \mathbf{X}] = 0$ for $i \neq j$)

- **Role:** Ensures that $\mathbf{\Omega}$ is diagonal. The EHW estimator only needs to estimate the n individual variances σ_i^2 .
- **Counterexample:** With autocorrelation or spatial dependence, the off-diagonal elements of $\mathbf{\Omega}$ are non-zero. The EHW estimator, by ignoring them, becomes inconsistent — requiring clustered standard errors (Arellano) or Newey-West.

Assumption 3.7: Finite Fourth-Order Moments ($E[u_i^4 \mathbf{x}_i \mathbf{x}_i'] < \infty$)

- **Role:** Technical condition for the Law of Large Numbers and the CLT to apply to the quadratic product $u_i^2 \mathbf{x}_i \mathbf{x}_i'$.
- **Counterexample:** With excessively heavy-tailed distributions (e.g., Cauchy), the variance of the variance estimator is not well-defined, preventing convergence of the robust estimator.

3.2.3 Formal Statement

Consider the linear model $\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$ with $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. Under $E[\mathbf{u}|\mathbf{X}] = \mathbf{0}$ and $E[\mathbf{u}\mathbf{u}'|\mathbf{X}] = \mathbf{\Omega} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$, we wish to construct an estimator $\widehat{\text{Var}}[\hat{\beta}]$ such that:

$$\text{plim}\left(n \widehat{\text{Var}}[\hat{\beta}]\right) = \text{Avar}\left[\sqrt{n}(\hat{\beta} - \beta)\right]$$

3.2.4 Formal Derivation of the EHW Estimator

Step 1: Decomposition of the Estimation Error

From the definition of OLS:

$$\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} \quad (3.7)$$

Step 2: Asymptotic Scaling

Premultiplying (3.7) by $\sqrt{n}$:

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1} \left(\frac{\mathbf{X}'\mathbf{u}}{\sqrt{n}}\right) \quad (3.8)$$

Step 3: Central Limit Theorem for the Score

Let $\mathbf{w}_i = \mathbf{x}_i u_i$ with $E[\mathbf{w}_i] = \mathbf{0}$ (Assumption 3.5). By the multivariate CLT (under Assumption 3.7):

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{x}_i u_i \xrightarrow{d} N(\mathbf{0}, \Sigma) \quad (3.9)$$

where the variance matrix of the score is:

$$\Sigma = \text{plim}\left(\frac{1}{n} \sum_{i=1}^n u_i^2 \mathbf{x}_i \mathbf{x}_i'\right) = E[u_i^2 \mathbf{x}_i \mathbf{x}_i'] \quad (3.10)$$

Step 4: Asymptotic Variance — Sandwich Form

Let $\mathbf{Q} = \text{plim}(\mathbf{X}'\mathbf{X}/n) = E[\mathbf{x}_i \mathbf{x}_i']$. By Slutsky's Lemma applied to (3.8):

$$\boxed{\text{Avar}\left[\sqrt{n}(\hat{\beta} - \beta)\right] = \mathbf{Q}^{-1}\Sigma\mathbf{Q}^{-1}} \quad (3.11)$$

Σ is the “filling” and $\mathbf{Q}^{-1}$ are the “bread slices” of the asymptotic sandwich structure.

Step 5: Estimation of Σ by White's Estimator

Since u_i is unobservable, White (1980) proposed replacing it with the OLS residual $\hat{u}_i = y_i - \mathbf{x}_i'\hat{\beta}$:

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n \hat{u}_i^2 \mathbf{x}_i \mathbf{x}_i' \quad (3.12)$$

The consistency of $\hat{\Sigma}$ follows from:

- $\hat{\beta} \xrightarrow{p} \beta$;
- expansion $\hat{u}_i = u_i - \mathbf{x}_i'(\hat{\beta} - \beta)$;
- the second- and third-order terms involve $(\hat{\beta} - \beta) \xrightarrow{p} \mathbf{0}$, and the adjacent sums are $O_p(1)$ by the finiteness of fourth-order moments (Assumption 3.7).

Therefore:

$$\text{plim}(\hat{\Sigma}) = \Sigma \quad (3.13)$$

Step 6: Final Robust Estimator

Substituting the sample analogs into (3.11) and undoing the scaling by n :

$$\widehat{\text{Var}}_{\text{EHW}}[\hat{\beta}] = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n \hat{u}_i^2 \mathbf{x}_i \mathbf{x}_i' \right) (\mathbf{X}'\mathbf{X})^{-1} \quad (3.14)$$

This is the expression for the robust variance-covariance estimator **HCO** of White (1980).

3.2.5 Role of Assumptions in the EHW Estimator

Independence between observations (Assumption 3.6) was indispensable for ensuring that Σ depended only on the contemporaneous products $u_i^2 \mathbf{x}_i \mathbf{x}_i'$. With autocorrelation or spatial dependence, the covariance terms $E[u_i u_j \mathbf{x}_i \mathbf{x}_j']$ ($i \neq j$) would be non-zero, and the estimator (3.12) would be inconsistent for the true variance — requiring alternatives such as clustered standard errors (Arellano) or Newey-West.

3.3 Asymptotic Properties of the EHW Estimator

3.3.1 Motivation

In large samples, it is not necessary for the errors to be normally distributed to perform valid inference. The asymptotic challenge that the EHW estimator solves is the construction of correct hypothesis tests when the error variance is unknown and potentially heteroskedastic.

Geometric Interpretation:

The sampling distribution of $\hat{\beta}$ is a cloud of points that contracts as n grows (consistency). When scaled by $\sqrt{n}$, this cloud converges to a Normal distribution. The EHW estimator allows us to accurately calculate the “width” (variance) of this Normal even when the dispersion of the original data is irregular — ensuring that confidence intervals and t -tests do not distort the true precision of the estimate.

3.3.2 Asymptotic Assumptions

Assumption 3.8: i.i.d. Sampling of $(\mathbf{x}_i, y_i)$

- **Role:** Enables the direct application of the Weak Law of Large Numbers (WLLN) and the CLT in their classical forms.
- **Counterexample:** With temporal dependence (autocorrelation), the sums do not converge to the simple expectations, requiring Newey-West corrections or time series approaches.

Assumption 3.9: Finite Fourth-Order Moments ($E[\|\mathbf{x}_i\|^4] < \infty$ and $E[u_i^4] < \infty$)

- **Role:** Ensures that the variance of $u_i^2 \mathbf{x}_i \mathbf{x}_i'$ is finite, allowing the sandwich filling to converge in probability to the population value.
- **Counterexample:** With heavy-tailed distributions (e.g., Cauchy), the mean squared error does not converge, making the robust standard error unstable.

Assumption 3.10: Non-Singular Moment Matrix ($\mathbf{Q} = E[\mathbf{x}_i \mathbf{x}_i']$ positive definite)

- **Role:** Ensures that the regressors are not asymptotically collinear, guaranteeing that $\mathbf{Q}^{-1}$ exists and that β remains identified in the limit.
- **Counterexample:** If a regressor becomes constant or asymptotically collinear, the variance of the estimator explodes to infinity.

3.3.3 Formal Statement

Theorem 3.3.1 (Asymptotic Properties of OLS with EHW). *Under i.i.d. sampling, finite fourth-order moments, and contemporaneous exogeneity ($E[\mathbf{x}_i u_i] = \mathbf{0}$), the OLS estimator $\hat{\beta}_n = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ and the robust estimator $\widehat{\text{Var}}_{\text{EHW}} = (\mathbf{X}'\mathbf{X})^{-1}(\sum_i \hat{u}_i^2 \mathbf{x}_i \mathbf{x}_i')(\mathbf{X}'\mathbf{X})^{-1}$ satisfy:*

1. **Consistency:** $\hat{\beta}_n \xrightarrow{p} \beta$
2. **Asymptotic Normality:** $\sqrt{n}(\hat{\beta}_n - \beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{V}_\beta)$, where $\mathbf{V}_\beta = \mathbf{Q}^{-1}\Sigma\mathbf{Q}^{-1}$
3. **Consistency of the Robust Estimator:** $\text{plim}(n \widehat{\text{Var}}_{\text{EHW}}) = \mathbf{V}_\beta$

3.3.4 Formal Derivation

Step 1: Consistency of $\hat{\beta}_n$

We write the estimation deviation in scaled form:

$$\hat{\beta}_n - \beta = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i u_i \right) \quad (3.15)$$

By Khinchin's WLLN (under Assumption 3.8):

$$\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \xrightarrow{p} \mathbf{Q}, \quad \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i u_i \xrightarrow{p} E[\mathbf{x}_i u_i] = \mathbf{0}$$

By the **Continuous Mapping Theorem** applied to (3.15):

$$\text{plim}(\hat{\beta}_n - \beta) = \mathbf{Q}^{-1} \cdot \mathbf{0} = \mathbf{0} \implies \hat{\beta}_n \xrightarrow{p} \beta$$

Step 2: Asymptotic Distribution

Premultiplying (3.15) by $\sqrt{n}$:

$$\sqrt{n}(\hat{\beta}_n - \beta) = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{x}_i u_i \right) \quad (3.16)$$

By the **Multivariate CLT** (under Assumptions 3.8 and 3.9):

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{x}_i u_i \xrightarrow{d} N(\mathbf{0}, \Sigma), \quad \Sigma = E[u_i^2 \mathbf{x}_i \mathbf{x}_i']$$

By **Slutsky's Lemma**, combining $\hat{\mathbf{Q}}_n \xrightarrow{p} \mathbf{Q}$ and the convergence in distribution of the score:

$$\boxed{\sqrt{n}(\hat{\beta}_n - \beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{Q}^{-1} \Sigma \mathbf{Q}^{-1})} \quad (3.17)$$

Step 3: Consistency of the Filling ($\hat{\Sigma} \xrightarrow{p} \Sigma$)

White's estimator for the filling is $\hat{\Sigma} = \frac{1}{n} \sum_i \hat{u}_i^2 \mathbf{x}_i \mathbf{x}_i'$. Expanding $\hat{u}_i = u_i - \mathbf{x}_i'(\hat{\beta}_n - \beta)$:

$$\hat{u}_i^2 = u_i^2 - 2u_i \mathbf{x}_i'(\hat{\beta}_n - \beta) + (\hat{\beta}_n - \beta)' \mathbf{x}_i \mathbf{x}_i' (\hat{\beta}_n - \beta)$$

Upon substituting into the sample sum:

- The first term converges by the WLLN: $\frac{1}{n} \sum_i u_i^2 \mathbf{x}_i \mathbf{x}_i' \xrightarrow{p} \Sigma$.
- The second- and third-order terms involve $(\hat{\beta}_n - \beta) \xrightarrow{p} \mathbf{0}$, and the adjacent sums are $O_p(1)$ by the finiteness of fourth-order moments (Assumption 3.9).

Therefore:

$$\text{plim}(\hat{\Sigma}) = \Sigma \quad (3.18)$$

Step 4: Consistency of the Robust Variance

Rewriting $n \widehat{\text{Var}}_{\text{EHW}}$ in scaled form:

$$n \widehat{\text{Var}}_{\text{EHW}} = \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1} \hat{\Sigma} \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1}$$

By the Continuous Mapping Theorem, combining $\hat{\mathbf{Q}}_n \xrightarrow{p} \mathbf{Q}$ and $\hat{\Sigma} \xrightarrow{p} \Sigma$:

$$\boxed{\text{plim}(n \widehat{\text{Var}}_{\text{EHW}}) = \mathbf{Q}^{-1} \Sigma \mathbf{Q}^{-1} = \mathbf{V}_\beta} \quad (3.19)$$

3.3.5 Role of Assumptions in Asymptotic Properties

The **finiteness of fourth-order moments** (Assumption 3.9) was indispensable in Step 3. Without it, the WLLN cannot be applied to the product $u_i^2 \mathbf{x}_i \mathbf{x}_i'$, preventing the convergence of $\hat{\Sigma}$ to the population value and invalidating all asymptotic inference.

i.i.d. sampling (Assumption 3.8) ensures that sample means converge to population expectations. With dependence, the covariance structure of the score becomes more complex, requiring robust methods for time series or panel data.

3.4 Hypothesis Testing with the EHW Estimator

3.4.1 Motivation and Geometric Interpretation

The Wald test addresses the problem of simultaneously testing multiple linear hypotheses about β (e.g., $\beta_1 = 0$ and $\beta_2 = 1$). Instead of evaluating each parameter in isolation, it measures the **statistical distance** between the unrestricted estimator $\hat{\beta}$ and the constraint space imposed by the null hypothesis.

Geometric Interpretation in $\mathbb{R}^k$:

$\hat{\beta}$ is a point in the parameter space; the null hypothesis defines a linear subspace (hyperplane) of dimension $k - q$. If H_0 is true, $\hat{\beta}$ should be “close” to this hyperplane. The distance, however, is not Euclidean — it is weighted by the uncertainty of the estimator. Under heteroskedasticity, the confidence ellipsoid around $\hat{\beta}$ has irregular axes; the EHW estimator correctly estimates the curvature of this ellipsoid, ensuring that the Wald distance reflects the true precision of the data.

3.4.2 Assumptions of the Test

Assumption 3.11: Independent Linear Restrictions ($H_0 : \mathbf{R}\beta = \mathbf{r}$)

- **Role:** Defines q simultaneous linear conditions with $\mathbf{R} \in \mathbb{R}^{q \times k}$ of full row rank ($\text{rank}(\mathbf{R}) = q$).
- **Counterexample:** If $\text{rank}(\mathbf{R}) < q$, the restrictions are redundant or contradictory, making $(\mathbf{R}\mathbf{V}_\beta\mathbf{R}')$ singular and the statistic undefined.

Assumption 3.12: Consistency of the EHW Estimator

- **Role:** Ensures that the weighting matrix of the Wald quadratic form converges to the true asymptotic variance $\mathbf{V}_\beta$.
- **Counterexample:** With autocorrelation or violation of the EHW assumptions, the estimator converges to an incorrect matrix, causing the statistic W not to follow the χ_q^2 distribution, invalidating the test.

Assumption 3.13: Asymptotic Normality of $\sqrt{n}(\hat{\beta} - \beta)$

- **Role:** Allows the quadratic form of the Wald statistic to converge to a χ^2 distribution in large samples.
 - **Counterexample:** Without finite fourth-order moments, the CLT fails and the Wald statistic may exhibit heavy-tailed distributions, inflating false rejection rates.
-

3.4.3 Formal Statement

Under heteroskedasticity $E[\mathbf{u}\mathbf{u}'|\mathbf{X}] = \mathbf{\Omega} = \text{diag}(\sigma_i^2)$, we wish to test:

$$H_0 : \mathbf{R}\beta - \mathbf{r} = \mathbf{0} \quad (3.20)$$

where $\mathbf{R} \in \mathbb{R}^{q \times k}$ with $\text{rank}(\mathbf{R}) = q$ and $\mathbf{r} \in \mathbb{R}^q$. The **robust Wald statistic** is:

$$W = (\mathbf{R}\hat{\beta} - \mathbf{r})' [\mathbf{R} \widehat{\text{Var}}_{\text{EHW}}[\hat{\beta}] \mathbf{R}']^{-1} (\mathbf{R}\hat{\beta} - \mathbf{r}) \quad (3.21)$$

Objective: To prove that, under H_0 , $W \xrightarrow{d} \chi_q^2$ as $n \rightarrow \infty$.

3.4.4 Formal Derivation

Lemma 3.4.1 (Quadratic Forms of Gaussian Vectors). *If $\mathbf{z} \sim N(\mathbf{0}, \mathbf{\Sigma})$ with $\mathbf{\Sigma}$ non-singular, then $\mathbf{z}'\mathbf{\Sigma}^{-1}\mathbf{z} \sim \chi_{\dim(\mathbf{z})}^2$.*

Step 1: Asymptotic Behavior of the Discrepancy Vector

Under H_0 , substituting $\mathbf{r} = \mathbf{R}\beta$:

$$\mathbf{R}\hat{\beta} - \mathbf{r} = \mathbf{R}(\hat{\beta} - \beta) \quad (3.22)$$

Premultiplying by $\sqrt{n}$:

$$\sqrt{n}(\mathbf{R}\hat{\beta} - \mathbf{r}) = \mathbf{R} [\sqrt{n}(\hat{\beta} - \beta)] \quad (3.23)$$

Step 2: Limiting Distribution of the Discrepancy Vector

By the asymptotic normality of $\sqrt{n}(\hat{\beta} - \beta)$ proved in (3.17), and by the property of linear transformations of Gaussian vectors:

$$\sqrt{n}(\mathbf{R}\hat{\beta} - \mathbf{r}) \xrightarrow{d} N(\mathbf{0}, \mathbf{R}\mathbf{V}_\beta \mathbf{R}') \quad (3.24)$$

Step 3: Consistency of the Weighting Matrix

Defining $\hat{\mathbf{V}}_n = n \widehat{\text{Var}}_{\text{EHW}}[\hat{\beta}]$, by the consistency proved in (3.19) and the Continuous Mapping Theorem:

$$\text{plim}(\mathbf{R}\hat{\mathbf{V}}_n\mathbf{R}') = \mathbf{R}\mathbf{V}_\beta\mathbf{R}' \quad (3.25)$$

Step 4: Asymptotic Formulation of the Wald Statistic

Rewriting (3.21) with the scaling factor n balanced:

$$W = \left[\sqrt{n}(\mathbf{R}\hat{\beta} - \mathbf{r}) \right]' \left[\mathbf{R}\hat{\mathbf{V}}_n\mathbf{R}' \right]^{-1} \left[\sqrt{n}(\mathbf{R}\hat{\beta} - \mathbf{r}) \right] \quad (3.26)$$

By **Slutsky's Lemma**, combining the convergence in distribution of (3.24) and the convergence in probability of (3.25):

$$W \xrightarrow{d} \mathbf{z}' [\mathbf{R}\mathbf{V}_\beta\mathbf{R}']^{-1} \mathbf{z}, \quad \mathbf{z} \sim N(\mathbf{0}, \mathbf{R}\mathbf{V}_\beta\mathbf{R}') \quad (3.27)$$

Step 5: Convergence to χ_q^2

Applying Lemma 3.4.1 directly to (3.27), since $\mathbf{R}\mathbf{V}_\beta\mathbf{R}'$ has full rank q :

$$\boxed{W \xrightarrow{d} \chi_q^2} \quad (3.28)$$

3.4.5 Role of Assumptions

The **full rank of $\mathbf{R}$** ($\text{rank}(\mathbf{R}) = q$) is indispensable. Without it, the matrix $\mathbf{R}\mathbf{V}_\beta\mathbf{R}'$ would be singular, its inversion in (3.26) would be undefined, and the Wald statistic would not be mathematically well-posed.

3.5 The Inference Dilemma: χ^2 versus F Distribution

The choice between critical values from the χ^2 distribution or the F distribution reflects the balance between pure asymptotic theory and finite-sample performance.

Algebraic Relationship between W and F :

The two statistics maintain the direct identity:

$$F = \frac{W}{q}$$

where q is the number of restrictions. In the classical framework (homoskedasticity and exact normality), W/q follows strictly $F_{q,n-k}$. Under heteroskedasticity, the exact distribution of W/q in finite samples is unknown, but its asymptotic limit converges to χ_q^2/q .

Why does asymptotic theory point to χ^2 ?

There is no theorem guaranteeing that the Wald statistic computed with robust standard errors follows an F distribution in small samples with heteroskedasticity. As $n - k \rightarrow \infty$, the $F_{q,n-k}$ distribution multiplied by q converges exactly to χ_q^2 — in large samples, the two approaches produce numerically identical conclusions.

Why does practice use the F distribution?

The F distribution has heavier tails than the χ^2 . For the same significance level α , its critical values are larger, generating more conservative p-values. This offers practical protection against false positives when n is not sufficiently large to guarantee complete asymptotic convergence of White's estimator. In other words: if the choice between F and χ^2 materially alters the rejection decision, this is a sign that the asymptotic approximation of the EHW estimator is fragile in that sample — making the more cautious stance provided by the F distribution the prudent recommendation.

3.6 Generalized Least Squares (GLS) Estimator

3.6.1 Motivation and Geometric Interpretation

The Ordinary Least Squares (**OLS**) estimator is built under the assumption of spherical errors (constant variance and no serial correlation). In real data, however, observations often exhibit heterogeneous variances (heteroskedasticity) or stochastic dependence (auto-correlation). In these scenarios, OLS loses its efficiency property within the class of linear unbiased estimators, although it remains unbiased and consistent.

The Generalized Least Squares (**GLS**) estimator resolves this inefficiency. Instead of assigning the same mathematical weight to all observations, GLS incorporates the structure of the population covariance matrix of the errors to transform the original system, applying weights inversely proportional to the variability and codependence of each observation.

Geometric Interpretation in $\mathbb{R}^n$:

Geometrically, OLS orthogonally projects the data vector $\mathbf{y}$ onto the column space of $\mathbf{X}$, denoted $\mathcal{C}(\mathbf{X})$, under the standard Euclidean metric. Under non-spherical errors, the uncertainty region around $\mathbf{y}$ ceases to be a hypersphere and assumes the shape of an **ellipsoid** in $\mathbb{R}^n$.

GLS acts by applying a linear transformation through a weighting matrix that rotates and rescales the axes of the data space. This operation deforms the error ellipsoid, converting it into a unit hypersphere compatible with the classical Gauss-Markov assumptions. The standard orthogonal projection is then performed in this transformed space.

3.6.2 Assumptions of GLS

Assumption 3.14: Linearity and Full Rank

- **Role:** Defines the population structural relationship $\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$ and ensures that the transformed sample moment matrix is invertible, guaranteeing the unique identification and computability of β .

- **Counterexample:** If $\text{rank}(\mathbf{X}) < k$, there is perfect multicollinearity. The column space $\mathcal{C}(\mathbf{X})$ exhibits dimensional redundancy, generating infinitely many solutions to the minimization problem and rendering the estimator undefined.

Assumption 3.15: Strict Exogeneity ($E[\mathbf{u}|\mathbf{X}] = \mathbf{0}$)

- **Role:** Ensures that the conditional error has zero mean for any realization of $\mathbf{X}$, being the fundamental pillar for unbiasedness in finite samples.
- **Counterexample:** If $E[\mathbf{u}|\mathbf{X}] \neq \mathbf{0}$, the estimator will absorb into the vector β systematic effects belonging to the noise, generating bias in finite samples and asymptotic inconsistency.

Assumption 3.16: Generalized Covariance Matrix ($E[\mathbf{u}\mathbf{u}'|\mathbf{X}] = \sigma^2\mathbf{\Omega}$)

- **Role:** Determines the stochastic structure of the errors, where $\mathbf{\Omega}$ is an $n \times n$ symmetric positive definite (SPD) matrix.
- **Counterexample:** If $\mathbf{\Omega}$ is not positive definite, it will be singular (having at least one zero eigenvalue), implying the existence of a linear combination of errors with zero variance — a deterministic error. This precludes the existence of $\mathbf{\Omega}^{-1}$, necessary for GLS computation.

3.6.3 Formal Statement and Derivation

3.6.3.1 Formal Statement

Consider the linear population model in matrix notation:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{u} \quad (3.29)$$

where $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times k}$ with $\text{rank}(\mathbf{X}) = k$, $\beta \in \mathbb{R}^k$, and $\mathbf{u} \in \mathbb{R}^n$. Conditionally on $\mathbf{X}$, we assume:

1. $E[\mathbf{u}|\mathbf{X}] = \mathbf{0}$
2. $E[\mathbf{u}\mathbf{u}'|\mathbf{X}] = \sigma^2\mathbf{\Omega}$, with $\mathbf{\Omega}$ symmetric and positive definite.

Theorem 3.6.1 (Generalized Least Squares Estimator). *The Generalized Least Squares estimator is defined by:*

$$\boxed{\hat{\beta}_{GLS} = (\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{y}} \quad (3.30)$$

Objective: To demonstrate, via linear transformation of the model, that $\hat{\beta}_{GLS}$ is unbiased and to derive its exact finite-sample variance-covariance matrix.

3.6.3.2 Formal Derivation

Step 1: Factorization of the Covariance Matrix

Since $\mathbf{\Omega}$ is symmetric and positive definite, its inverse $\mathbf{\Omega}^{-1}$ shares the same properties. By the Cholesky decomposition, there exists a non-singular square matrix $\mathbf{P}$ of dimension $n \times n$ such that:

$$\Omega^{-1} = \mathbf{P}'\mathbf{P} \quad (3.31)$$

From this factorization, the sphericity property follows:

$$\mathbf{P}\Omega\mathbf{P}' = \mathbf{P}(\mathbf{P}'\mathbf{P})^{-1}\mathbf{P}' = \mathbf{P}\mathbf{P}^{-1}(\mathbf{P}')^{-1}\mathbf{P}' = \mathbf{I}_n \quad (3.32)$$

Step 2: Linear Transformation of the Model

Premultiplying the structural equation (3.29) by $\mathbf{P}$:

$$\mathbf{P}\mathbf{y} = \mathbf{P}\mathbf{X}\beta + \mathbf{P}\mathbf{u} \quad (3.33)$$

Defining the transformed objects:

$$\mathbf{y}^* = \mathbf{P}\mathbf{y}, \quad \mathbf{X}^* = \mathbf{P}\mathbf{X}, \quad \mathbf{u}^* = \mathbf{P}\mathbf{u} \quad (3.34)$$

we obtain the transformed model:

$$\mathbf{y}^* = \mathbf{X}^*\beta + \mathbf{u}^* \quad (3.35)$$

Step 3: Verification of Gauss-Markov Conditions in the Transformed Model

Conditional mean:

$$E[\mathbf{u}^*|\mathbf{X}] = \mathbf{P} E[\mathbf{u}|\mathbf{X}] = \mathbf{0} \quad (3.36)$$

Conditional variance:

$$\text{Var}[\mathbf{u}^*|\mathbf{X}] = \mathbf{P} E[\mathbf{u}\mathbf{u}'|\mathbf{X}] \mathbf{P}' = \mathbf{P}(\sigma^2\Omega)\mathbf{P}' = \sigma^2(\mathbf{P}\Omega\mathbf{P}')$$

Applying identity (3.32):

$$\text{Var}[\mathbf{u}^*|\mathbf{X}] = \sigma^2\mathbf{I}_n \quad (3.37)$$

Partial conclusion: The transformed model (3.35) satisfies the Gauss-Markov conditions of homoskedasticity and no serial correlation.

Step 4: Application of OLS to the Transformed Model

Once the classical assumptions hold for (3.35), OLS applied to this model is the Best Linear Unbiased Estimator (BLUE):

$$\hat{\beta} = (\mathbf{X}^{*'}\mathbf{X}^*)^{-1}\mathbf{X}^{*'}\mathbf{y}^* \quad (3.38)$$

Step 5: Return to the Original Variables

Substituting the definitions (3.34) into (3.38):

$$\hat{\beta}_{GLS} = [(\mathbf{P}\mathbf{X})'(\mathbf{P}\mathbf{X})]^{-1}(\mathbf{P}\mathbf{X})'(\mathbf{P}\mathbf{y}) = [\mathbf{X}'\mathbf{P}'\mathbf{P}\mathbf{X}]^{-1}\mathbf{X}'\mathbf{P}'\mathbf{P}\mathbf{y}$$

Applying the factorization $\mathbf{P}'\mathbf{P} = \mathbf{\Omega}^{-1}$ from (3.31):

$$\hat{\beta}_{GLS} = (\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{y}$$

which coincides with the statement (3.30). $\square$

Step 6: Finite-Sample Variance

Since $\hat{\beta}_{GLS}$ operates as OLS on the transformed model, its conditional variance is given directly by the classical formula applied to spherical variables:

$$\text{Var}[\hat{\beta}_{GLS}|\mathbf{X}] = \sigma^2(\mathbf{X}^{*'}\mathbf{X}^*)^{-1}$$

Substituting $\mathbf{X}^* = \mathbf{P}\mathbf{X}$ and applying factorization (3.31):

$$\boxed{\text{Var}[\hat{\beta}_{GLS}|\mathbf{X}] = \sigma^2[\mathbf{X}'\mathbf{P}'\mathbf{P}\mathbf{X}]^{-1} = \sigma^2(\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})^{-1}} \quad (3.39)$$

3.6.4 Relative Efficiency: The Generalized Gauss-Markov Theorem

3.6.4.1 Statement of the Theorem

We will formally prove that $\hat{\beta}_{GLS}$ is strictly more efficient than OLS ($\hat{\beta}_{OLS}$) under non-spherical errors. This is equivalent to proving that the difference between their variance-covariance matrices is positive semidefinite (PSD):

$$\Delta = \text{Var}(\hat{\beta}_{OLS}|\mathbf{X}) - \text{Var}(\hat{\beta}_{GLS}|\mathbf{X}) \succeq \mathbf{0} \quad (3.40)$$

3.6.4.2 Formal Derivation

Notation and Preliminary Lemmas:

1. **Lemma 1 (Spectral Decomposition):** Since $\mathbf{\Omega}$ is SPD, there exists a unique SPD matrix $\mathbf{\Omega}^{1/2}$ such that $\mathbf{\Omega}^{1/2}\mathbf{\Omega}^{1/2} = \mathbf{\Omega}$ and $\mathbf{\Omega}^{-1/2}\mathbf{\Omega}^{-1/2} = \mathbf{\Omega}^{-1}$.
2. **Lemma 2 (Inversion of Inequalities):** For two SPD matrices A and B , $A \succeq B \iff B^{-1} \succeq A^{-1}$.
3. **Lemma 3 (Projection Matrix):** The matrix $P_Z = Z(Z'Z)^{-1}Z'$ is symmetric and idempotent ($P_Z^2 = P_Z$), and its complement $M_Z = I - P_Z$ is also symmetric and idempotent, hence PSD.

Step 1: Expression of the Variances

The variance of the OLS estimator under non-spherical errors is:

$$\text{Var}(\hat{\beta}_{OLS}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{\Omega}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \quad (3.41)$$

The variance of the GLS estimator is:

$$\text{Var}(\hat{\beta}_{GLS}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})^{-1} \quad (3.42)$$

Step 2: Comparison of Inverses

By **Lemma 2**, proving $\text{Var}(\hat{\beta}_{OLS}) \succeq \text{Var}(\hat{\beta}_{GLS})$ is equivalent to proving:

$$[\text{Var}(\hat{\beta}_{GLS})]^{-1} \succeq [\text{Var}(\hat{\beta}_{OLS})]^{-1} \quad (3.43)$$

Substituting expressions (3.41) and (3.42) and simplifying σ^2 :

$$\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X} \succeq (\mathbf{X}'\mathbf{X})(\mathbf{X}'\mathbf{\Omega}\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X}) \quad (3.44)$$

Step 3: Identification of the Projection Matrix

Define the auxiliary matrices $A = \mathbf{\Omega}^{-1/2}\mathbf{X}$ and $B = \mathbf{\Omega}^{1/2}\mathbf{X}$.

We rewrite the terms of inequality (3.44):

$$\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X} = \mathbf{X}'\mathbf{\Omega}^{-1/2}\mathbf{\Omega}^{-1/2}\mathbf{X} = A'A \quad (3.45)$$

$$\mathbf{X}'\mathbf{X} = \mathbf{X}'\mathbf{\Omega}^{-1/2}\mathbf{\Omega}^{1/2}\mathbf{X} = A'B \quad (3.46)$$

$$\mathbf{X}'\mathbf{\Omega}\mathbf{X} = \mathbf{X}'\mathbf{\Omega}^{1/2}\mathbf{\Omega}^{1/2}\mathbf{X} = B'B \quad (3.47)$$

Substituting (3.45), (3.46), and (3.47) into inequality (3.44), the difference between the terms becomes:

$$\mathbf{\Delta} = A'A - (A'B)(B'B)^{-1}(B'A) \quad (3.48)$$

Factoring A' on the left and A on the right:

$$\mathbf{\Delta} = A'[I - B(B'B)^{-1}B']A \quad (3.49)$$

Step 4: Proof of Positive Semidefiniteness

The term in brackets in (3.49) is exactly the annihilator matrix $M_B = I - P_B$, where $P_B = B(B'B)^{-1}B'$ is the projection matrix onto the column space of B :

$$\Delta = A'M_B A \quad (3.50)$$

Verification of properties of M_B :

1. **Symmetry:** $M'_B = (I - P_B)' = I' - P'_B = I - P_B = M_B$.
2. **Idempotence:** $M_B M_B = M_B$.
3. **Positive Semidefiniteness:** For any vector $v \in \mathbb{R}^k$, define $z = Av$. Then:

$$v' \Delta v = v' A' M_B A v = z' M_B z \quad (3.51)$$

Since M_B is idempotent and symmetric:

$$z' M_B z = z' M'_B M_B z = \|M_B z\|^2 \geq 0 \quad (3.52)$$

Conclusion:

Since $v' \Delta v \geq 0$ for any vector v , the matrix Δ is PSD. This proves that the inverse of the GLS variance dominates the inverse of the OLS variance, confirming the superior efficiency of GLS. The **SPD assumption of Ω** was indispensable for guaranteeing the existence of $\Omega^{1/2}$ and the validity of the orthogonal projection in the transformed space.

3.6.5 Asymptotic Properties of GLS

3.6.5.1 Asymptotic Statement

Theorem 3.6.2 (Asymptotic Properties of GLS). *In finite samples, GLS requires complete knowledge of the parametric structure of Ω . Under usual regularity conditions, as $n \rightarrow \infty$ the GLS estimator satisfies:*

1. **Consistency:** $\text{plim}(\hat{\beta}_{GLS}) = \beta$
2. **Asymptotic Normality:** $\sqrt{n}(\hat{\beta}_{GLS} - \beta) \xrightarrow{d} N(\mathbf{0}, \sigma^2 \mathbf{Q}_{GLS}^{-1})$

3.6.5.2 Formal Derivation

Step 1: Formulation of the Estimation Deviation

Substituting $\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$ into (3.30) and canceling the term $(\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}(\mathbf{X}'\Omega^{-1}\mathbf{X}) = \mathbf{I}_k$:

$$\hat{\beta}_{GLS} - \beta = (\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\mathbf{X}'\Omega^{-1}\mathbf{u} \quad (3.53)$$

Step 2: Consistency

Normalizing (3.53) by n :

$$\hat{\beta}_{GLS} - \beta = \left(\frac{\mathbf{X}'\Omega^{-1}\mathbf{X}}{n} \right)^{-1} \left(\frac{\mathbf{X}'\Omega^{-1}\mathbf{u}}{n} \right) \quad (3.54)$$

By the Weak Law of Large Numbers, under regularity conditions:

$$\text{plim} \left(\frac{\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X}}{n} \right) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n E[\mathbf{x}_i^* \mathbf{x}_i^{*'}] = \mathbf{Q}_{GLS} \quad (3.55)$$

where $\mathbf{Q}_{GLS}$ is finite and positive definite. For the average score, by strict exogeneity:

$$\text{plim} \left(\frac{\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{u}}{n} \right) = E[\mathbf{x}_i^* u_i^*] = \mathbf{0} \quad (3.56)$$

By the Continuous Mapping Theorem applied to (3.54):

$$\text{plim}(\hat{\beta}_{GLS} - \beta) = \mathbf{Q}_{GLS}^{-1} \cdot \mathbf{0} = \mathbf{0} \implies \hat{\beta}_{GLS} \xrightarrow{p} \beta \quad (3.57)$$

Step 3: Asymptotic Distribution

Premultiplying (3.53) by $\sqrt{n}$:

$$\sqrt{n}(\hat{\beta}_{GLS} - \beta) = \left(\frac{\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X}}{n} \right)^{-1} \left(\frac{\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{u}}{\sqrt{n}} \right) \quad (3.58)$$

For the rescaled score vector $\mathbf{w}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{x}_i^* u_i^*$, its exact conditional variance is:

$$\text{Var}[\mathbf{w}_n | \mathbf{X}] = \frac{1}{n} \mathbf{X}'\boldsymbol{\Omega}^{-1} E[\mathbf{u}\mathbf{u}' | \mathbf{X}] \boldsymbol{\Omega}^{-1} \mathbf{X} = \sigma^2 \left(\frac{\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X}}{n} \right)$$

which converges to $\sigma^2 \mathbf{Q}_{GLS}$. Under finite fourth-order moment conditions, the Multivariate Central Limit Theorem provides:

$$\frac{\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{u}}{\sqrt{n}} \xrightarrow{d} N(\mathbf{0}, \sigma^2 \mathbf{Q}_{GLS}) \quad (3.59)$$

Step 4: Application of Slutsky's Lemma

Combining (3.55) and (3.59) into (3.58) via Slutsky's Lemma:

$$\sqrt{n}(\hat{\beta}_{GLS} - \beta) \xrightarrow{d} \mathbf{Q}_{GLS}^{-1} \cdot N(\mathbf{0}, \sigma^2 \mathbf{Q}_{GLS})$$

By the property of linear transformations of Gaussian vectors:

$$\boxed{\sqrt{n}(\hat{\beta}_{GLS} - \beta) \xrightarrow{d} N(\mathbf{0}, \sigma^2 \mathbf{Q}_{GLS}^{-1})} \quad (3.60)$$

3.6.6 Role of Assumptions in GLS Properties

The **symmetry and positive definiteness of Ω** (Assumption 3.16) are the formal core of the entire derivation. Without them:

- the factorization $\mathbf{P}'\mathbf{P} = \Omega^{-1}$ is undefined or generates a singular transformation matrix;
- mapping to the spherical space becomes impossible;
- the inversion of the moment matrix in (3.39) loses validity.

The **asymptotic full rank of $\mathbf{Q}_{GLS}$** is indispensable for the validity of matrix inversion in the limit. If $\mathbf{Q}_{GLS}$ were singular, the asymptotic variance would be undefined, invalidating inference in large samples.

3.7 Feasible Generalized Least Squares (FGLS) and the Relationship with EHW

3.7.1 Motivation and Geometric Interpretation

The Generalized Least Squares (**GLS**) estimator is an idealized theoretical construction: it requires full knowledge of the population variance-covariance matrix Ω , a condition rarely met in empirical data. The Feasible Generalized Least Squares (**FGLS**) estimator addresses this practical limitation by substituting the unknown matrix Ω with a consistent estimator $\hat{\Omega} = \Omega(\hat{\theta})$, constructed from the residuals of an initial parametric specification estimated by OLS.

Comparative Geometric Interpretation:

Imagine the uncertainty cloud of the original data in $\mathbb{R}^n$ as an elongated ellipsoid due to heteroskedasticity.

- **GLS/FGLS** applies a linear transformation that “spherifies” this deformed cloud back to a perfectly spherical geometry before projecting the vector $\mathbf{y}$ onto the column space of $\mathbf{X}$.
- **EHW** accepts the standard Euclidean projection of OLS onto the original elliptical cloud and merely inflates or rotates the axes of the confidence ellipsoid of $\hat{\beta}_{OLS}$ so that the reported confidence intervals are not misleadingly narrow.

Fragility of FGLS in Finite Samples:

While theoretical GLS is guaranteed to be the Best Linear Unbiased Estimator (**BLUE**) in finite samples, FGLS **loses the unbiasedness property** in finite samples due to the randomness introduced by the estimation of $\hat{\theta}$. Under small samples or if the heteroskedasticity structure is misspecified, FGLS may exhibit sample variance significantly **greater than that of ordinary OLS**, becoming inefficient.

Crucial Distinction between FGLS and EHW:

EHW does not constitute a variant or subclass of GLS:

- The GLS/FGLS framework acts directly on the **core of the estimator**, altering the coefficient vector $\hat{\beta}$ to extract efficiency gains through orthogonal reweighting of the data.
 - EHW **preserves the OLS estimate vector intact** — accepting its inefficiency under non-spherical errors — and merely corrects the variance-covariance matrix *ex-post*, ensuring only that inference remains asymptotically valid.
-

3.7.2 Assumptions of FGLS

Assumption 3.17: Structural Consistency of the Variance Estimator ($\text{plim } \hat{\Omega} = \Omega$)

- **Role:** Ensures that, as the sample size grows, the estimated matrix $\hat{\Omega}$ collapses onto the true population structure, allowing FGLS to recover the equivalence and asymptotic efficiency of idealized GLS.
- **Counterexample:** If the parametric function chosen to model the variance is incorrect (e.g., modeling heteroskedasticity as linear when it is exponential), $\hat{\Omega}$ will converge to an erroneous matrix, and asymptotic FGLS will be less efficient than simple OLS.

Assumption 3.18: Strict Exogeneity ($E[\mathbf{u}|\mathbf{X}] = \mathbf{0}$)

- **Role:** Indispensable condition for ensuring FGLS consistency. Unlike ordinary OLS — which requires only contemporaneous orthogonality $E[\mathbf{x}_i u_i] = \mathbf{0}$ for consistency — the linear transformation applied by FGLS mixes residuals and regressors from different observations.
 - **Counterexample:** In time series models with lagged dependent variables (e.g., autoregressive models), the errors are orthogonal to contemporaneous regressors but correlated with future regressors. In this scenario, OLS with EHW correction remains consistent, while FGLS becomes severely **inconsistent**.
-

3.7.3 Formal Statement and Derivation

3.7.3.1 Formal Statement

Consider the linear population model:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{u} \quad (3.61)$$

where $E[\mathbf{u}|\mathbf{X}] = \mathbf{0}$ and $\text{Var}[\mathbf{u}|\mathbf{X}] = \sigma^2 \Omega(\theta)$, with $\theta \in \mathbb{R}^p$ containing the unknown variance parameters.

Theorem 3.7.1 (Feasible Generalized Least Squares Estimator). *The FGLS estimator is defined by:*

$$\boxed{\hat{\beta}_{FGLS} = (\mathbf{X}'\hat{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\hat{\Omega}^{-1}\mathbf{y}} \quad (3.62)$$

3.7. FEASIBLE GENERALIZED LEAST SQUARES (FGLS) AND THE RELATIONSHIP WITH EHW

where $\hat{\Omega} = \Omega(\hat{\theta})$ and $\hat{\theta}$ is a consistent estimator such that $\hat{\theta} \xrightarrow{p} \theta$.

Objectives:

1. Demonstrate the asymptotic equivalence between $\hat{\beta}_{FGLS}$ and $\hat{\beta}_{GLS}$.
2. Prove by algebraic contradiction that EHW cannot be mapped as a special case of GLS reweighting.

3.7.3.2 Formal Derivation

Step 1: Decomposition and Scaling of the Estimation Error

Substituting $\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$ into (3.62):

$$\hat{\beta}_{FGLS} = \beta + (\mathbf{X}'\hat{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\hat{\Omega}^{-1}\mathbf{u}$$

Multiplying by $\sqrt{n}$:

$$\sqrt{n}(\hat{\beta}_{FGLS} - \beta) = \left(\frac{\mathbf{X}'\hat{\Omega}^{-1}\mathbf{X}}{n} \right)^{-1} \left(\frac{\mathbf{X}'\hat{\Omega}^{-1}\mathbf{u}}{\sqrt{n}} \right) \quad (3.63)$$

Step 2: Application of Slutsky's Lemma

Since $\hat{\theta} \xrightarrow{p} \theta$, the Continuous Mapping Theorem guarantees that $\hat{\Omega}^{-1} \xrightarrow{p} \Omega^{-1}$. We evaluate the limits:

1. By the Law of Large Numbers:

$$\text{plim} \left(\frac{\mathbf{X}'\hat{\Omega}^{-1}\mathbf{X}}{n} \right) = \text{plim} \left(\frac{\mathbf{X}'\Omega^{-1}\mathbf{X}}{n} \right) = \mathbf{Q}_{GLS} \quad (3.64)$$

2. By the consistency of $\hat{\Omega}$ and strict exogeneity:

$$\frac{\mathbf{X}'\hat{\Omega}^{-1}\mathbf{u}}{\sqrt{n}} - \frac{\mathbf{X}'\Omega^{-1}\mathbf{u}}{\sqrt{n}} \xrightarrow{p} \mathbf{0} \quad (3.65)$$

By **Slutsky's Lemma**, the asymptotic distribution of FGLS converges exactly to that of theoretical GLS:

$$\sqrt{n}(\hat{\beta}_{FGLS} - \beta) \xrightarrow{d} N(\mathbf{0}, \sigma^2 \mathbf{Q}_{GLS}^{-1}) \quad (3.66)$$

Partial Conclusion: The equivalence and efficiency of FGLS are strictly **asymptotic** properties. In finite samples, $\text{Var}[\hat{\beta}_{FGLS}|\mathbf{X}] \neq \sigma^2(\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}$ due to the remaining sample variance from $\hat{\theta}$.

Step 3: Algebraic Distinction of EHW

The EHW estimator (HC0) is defined by:

$$\widehat{\text{Var}}_{EHW} = (\mathbf{X}'\mathbf{X})^{-1} (\mathbf{X}'\hat{\Phi}\mathbf{X}) (\mathbf{X}'\mathbf{X})^{-1}, \quad \hat{\Phi} = \text{diag}(\hat{u}_1^2, \dots, \hat{u}_n^2) \quad (3.67)$$

Proof by Contradiction: Suppose EHW is identical to a GLS estimation with arbitrary weight matrix $\mathbf{W}$. Then the coefficient estimator should be:

$$\hat{\beta}_{EHW} = (\mathbf{X}'\mathbf{W}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}^{-1}\mathbf{y} \quad (3.68)$$

But by definition, the estimator that accompanies the EHW matrix is ordinary OLS:

$$\hat{\beta}_{EHW} = \hat{\beta}_{OLS} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \quad (3.69)$$

For (3.68) and (3.69) to be algebraically identical for any $\mathbf{y}$, the equality of operators requires:

$$(\mathbf{X}'\mathbf{W}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}^{-1} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

This identity is satisfied **if, and only if**, $\mathbf{W} = \sigma^2\mathbf{I}_n$ (pure homoskedasticity). Under heteroskedasticity, there is no $\mathbf{W} \neq \sigma^2\mathbf{I}_n$ that preserves the OLS estimator. This proves that EHW operates **exclusively as an *ex-post* inference correction**, and not as an efficient reweighting of the data.

3.8 Inference Foundation: χ^2 versus F in Each Framework

3.8.1 Unified Definition of the Wald Statistic

Under the null hypothesis of q linear restrictions $H_0 : \mathbf{R}\beta - \mathbf{r} = \mathbf{0}$, with $\text{rank}(\mathbf{R}) = q$, the Wald statistic is:

$$W = (\mathbf{R}\hat{\beta} - \mathbf{r})' [\mathbf{R}\widehat{\text{Var}}(\hat{\beta})\mathbf{R}']^{-1} (\mathbf{R}\hat{\beta} - \mathbf{r}) \quad (3.70)$$

3.8.2 The EHW Scenario: Purely Asymptotic Inference

In the EHW scenario, we maintain $\hat{\beta} = \hat{\beta}_{OLS}$ and $\widehat{\text{Var}}(\hat{\beta}) = \widehat{\text{Var}}_{EHW}$.

3.8.2.1 Convergence to χ_q^2

Under H_0 :

$$\mathbf{R}\hat{\beta}_{OLS} - \mathbf{r} = \mathbf{R}(\hat{\beta}_{OLS} - \beta)$$

Multiplying by $\sqrt{n}$ and invoking the CLT:

$$\sqrt{n}\mathbf{R}(\hat{\beta}_{OLS} - \beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{R}\mathbf{V}_{EHW}\mathbf{R}')$$

Since $\text{plim}(n\widehat{\text{Var}}_{EHW}) = \mathbf{V}_{EHW}$, Slutsky's Lemma provides:

$$\boxed{W \xrightarrow{d} \chi_q^2} \quad (3.71)$$

3.8.2.2 Limitation in Finite Samples

Under EHW with generic heteroskedasticity, the exact distribution of (3.70) in finite samples is **completely unknown**. There is no mathematical proof that the denominator of the sandwich form is distributed as a Chi-square independent of the numerator.

Practical Note: The practice of computing $F = W/q$ and testing against $F_{q,n-k}$ in the EHW context is a **heuristic and conservative approximation** for small samples, lacking support in finite-sample theory.

3.8.3 The GLS Scenario: Exact Finite-Sample Inference

In theoretical GLS, the data are filtered by $\mathbf{P}$, restoring sphericity.

3.8.3.1 Derivation of the F Statistic

1. In the transformed model $\mathbf{y}^* = \mathbf{X}^*\beta + \mathbf{u}^*$, we assume normality: $\mathbf{u}^* \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

2. Under normality:

$$\hat{\beta}_{GLS} \sim N\left(\beta, \sigma^2(\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\right)$$

3. The restriction error is:

$$\mathbf{m} = \mathbf{R}\hat{\beta}_{GLS} - \mathbf{r} \sim N\left(\mathbf{0}, \sigma^2 \mathbf{R}(\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\mathbf{R}'\right)$$

4. The normalized quadratic form:

$$\frac{W_{\sigma^2}}{q} = \frac{(\mathbf{R}\hat{\beta}_{GLS} - \mathbf{r})' [\mathbf{R}(\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\mathbf{R}']^{-1} (\mathbf{R}\hat{\beta}_{GLS} - \mathbf{r})}{q \cdot \sigma^2} \sim \frac{\chi_q^2}{q} \quad (3.72)$$

5. The estimator of σ^2 is:

$$s_{GLS}^2 = \frac{\hat{\mathbf{u}}_{GLS}'^* \hat{\mathbf{u}}_{GLS}^*}{n - k} \implies \frac{(n - k)s_{GLS}^2}{\sigma^2} \sim \chi_{n-k}^2 \quad (3.73)$$

6. By independence between numerator and denominator (property of orthogonal projections under normality):

$$F = \frac{W_{\sigma^2}/q}{s_{GLS}^2/\sigma^2} = \frac{(\mathbf{R}\hat{\beta}_{GLS} - \mathbf{r})' [\mathbf{R}(\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\mathbf{R}']^{-1} (\mathbf{R}\hat{\beta}_{GLS} - \mathbf{r})}{q \cdot s_{GLS}^2} \quad (3.74)$$

7. Conclusion:

$$\boxed{F \sim F_{q,n-k}} \quad (3.75)$$

3.8.4 Comparative Summary of Inference Environments

Dimension	EHW (Robust OLS)	GLS (Theoretical)
Coefficient Estimator	$\hat{\beta}_{OLS}$ (inefficient under non-spherical errors)	$\hat{\beta}_{GLS}$ (BLUE in finite samples)
Theoretical Foundation	Pure Asymptotic Theory ($n \rightarrow \infty$)	Exact Finite-Sample Distribution
Statistic Distribution	χ_q^2 (CLT collapse)	$F_{q,n-k}$ (exact ratio of two χ^2)
Assumption about Errors	Ignores and tolerates the fine structure	Requires normality and knowledge of Ω

Indispensable Conclusion:

- In **GLS**, the transition to the F distribution is algebraically legitimate: the linear transformation cleans the covariance structure, restoring sphericity and allowing the emergence of an independent Chi-square in the denominator (s_{GLS}^2).
- In **EHW**, we stop at χ^2 because the refusal to model the error structure prevents the factorization and isolation of independent Chi-square terms in small samples, forcing inference to depend exclusively on the Gaussian asymptotic limit.

3.9 Comparative Table: Treatment of Pure Heteroskedasticity

The scope is restricted to pure heteroskedasticity: $\text{Var}(\mathbf{u}|\mathbf{X}) = \sigma^2\Omega$, with Ω diagonal and entries $\omega_{ii} = h(\mathbf{x}_i) > 0$. EHW refers to OLS with White's robust adjustment.

Criterion	EHW (Robust OLS)	GLS (Theoretical)	FGLS (Feasible)
Estimator of β and Stages	$\hat{\beta}_{OLS} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. 1 stage: direct orthogonal projection.	$\hat{\beta}_{GLS} = (\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\mathbf{X}'\Omega^{-1}\mathbf{y}$. 1 stage: requires known Ω .	$\hat{\beta}_{FGLS} = (\mathbf{X}'\hat{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\hat{\Omega}^{-1}\mathbf{y}$. 2 stages: estimates $\hat{\Omega}$ via OLS residuals; then estimates β via WLS.

Criterion	EHW (Robust OLS)	GLS (Theoretical)	FGLS (Feasible)
Assumption about Ω	Arbitrary and unknown. Imposes no functional form on $h(\mathbf{x}_i)$. Consistency depends only on contemporaneous exogeneity.	Known <i>ex ante</i>. Ω known up to σ^2 . Impractical in real contexts.	Parametric structure. Assumes $\omega_{ii} = h(\mathbf{x}_i, \theta)$ with θ consistently estimated. Misspecification causes inefficiency; in dynamic models, may generate inconsistency.
Finite Samples	Unbiased but inefficient. $E[\hat{\beta}_{OLS} \mathbf{X}] = \beta$. Not BLUE.	BLUE — Aitken's Theorem. Best linear unbiased estimator under metric weighted by Ω^{-1} .	Generally biased. Dependence between $\hat{\Omega}$ and errors invalidates unbiasedness. No analogue to Aitken's Theorem.
Asymptotic Properties	Consistent and $\sqrt{n}$-normal. $\sqrt{n}(\hat{\beta}_{OLS} - \beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{V}_{EHW})$, with $\mathbf{V}_{EHW} = \mathbf{Q}^{-1}\mathbf{B}\mathbf{Q}^{-1}$.	Consistent and asymptotically efficient. $\mathbf{V}_{GLS} = (E[\mathbf{x}_i\mathbf{x}_i'/h(\mathbf{x}_i)])^{-1} \preceq \mathbf{V}_{EHW}$.	Asymptotically equivalent to GLS. If $\text{plim}(\hat{\Omega}) = \Omega$, achieves the same efficiency as theoretical GLS.
Rank Requirements	$\text{rank}(\mathbf{X}) = k$; $\mathbf{Q} = \text{plim}(\mathbf{X}'\mathbf{X}/n)$ SPD.	Same as EHW, plus Ω SPD for existence of $\mathbf{P}$.	GLS requirements + identifiability of θ in $h(\mathbf{x}_i, \theta)$.
Inference	Asymptotic and doubly robust. Based on χ^2 or Normal. Dispenses with specification of $h(\cdot)$.	Exact under Normality. t and F with exact distributions in the transformed space.	Asymptotic and sensitive. t and F valid only for large n . Sensitive to misspecification of $h(\cdot)$.
Main Limitation	Efficiency loss. $\mathbf{V}_{EHW} - \mathbf{V}_{GLS} \succeq \mathbf{0}$. Greater cost with pronounced heteroskedasticity.	Practical infeasibility. Ω rarely known in empirical contexts.	Risk of misspecification. Mild misspecification generates inefficiency; severe misspecification — especially in dynamic models — generates inconsistency.

3.9.1 Methodological Notes

1. **Sandwich Estimator (EHW):** The asymptotic variance $\mathbf{V}_{EHW} = \mathbf{Q}^{-1}\mathbf{BQ}^{-1}$ is estimated by $\hat{\mathbf{V}}_{EHW} = (\mathbf{X}'\mathbf{X})^{-1}(\sum_i \hat{u}_i^2 \mathbf{x}_i \mathbf{x}_i')(\mathbf{X}'\mathbf{X})^{-1}$, without imposing structure on Ω .
2. **Aitken's Transformation:** GLS premultiplies the model by $\mathbf{P}$ such that $\mathbf{P}'\mathbf{P} = \Omega^{-1}$, producing spherical errors $\text{Var}(\mathbf{Pu}|\mathbf{X}) = \sigma^2\mathbf{I}$ and restoring the Gauss-Markov conditions in the transformed space.
3. **FGLS Usage Frontier:** In static models with strict exogeneity, FGLS is asymptotically preferable to EHW if $h(\cdot)$ is reasonably specified. In dynamic models, the recommendation reverses: EHW with OLS is methodologically safer.

Chapter 4

Endogeneity

4.1 Sources of Endogeneity

The problem of **endogeneity** arises when the error term of a regression model is correlated with the regressors — violating the condition of strict or contemporaneous exogeneity. Three classical sources are responsible for this violation: (i) omission of a relevant variable, (ii) measurement error in the regressors, and (iii) simultaneity (or reverse causality). Each of these sources corrupts the consistency of the Ordinary Least Squares (OLS) estimator, albeit through distinct mechanisms.

4.1.1 Omitted Variable

4.1.1.1 Motivation and Geometric Interpretation

The omitted variable problem emerges when the researcher wishes to estimate the *ceteris paribus* effect of a set of regressors $\mathbf{X}_1$ on $\mathbf{y}$, but is unable to observe or include in the model a matrix of covariates $\mathbf{X}_2$ that simultaneously affects $\mathbf{y}$ and is correlated with $\mathbf{X}_1$.

Geometric Interpretation in $\mathbb{R}^n$:

The OLS estimator of the complete model — the *long regression* — orthogonally projects $\mathbf{y}$ onto the hyperplane defined by the combined column space $\text{Col}(\mathbf{X}_1, \mathbf{X}_2)$. By omitting $\mathbf{X}_2$, we force the projection of $\mathbf{y}$ solely onto the subspace $\text{Col}(\mathbf{X}_1)$. If $\mathbf{X}_1$ and $\mathbf{X}_2$ are not orthogonal, the portion of the variation in $\mathbf{y}$ that should be attributed to $\text{Col}(\mathbf{X}_2)$ is partially absorbed by $\text{Col}(\mathbf{X}_1)$. The coefficient associated with $\mathbf{X}_1$ improperly incorporates the effect of $\mathbf{X}_2$, producing an estimate that fails to isolate the true structural parameter.

4.1.1.2 Assumptions and Their Roles

Assumption 4.1: Linearity in Parameters and Structural Specification

- **Role:** Defines that the population Data Generating Process (DGP) obeys a linear and additive relationship in the observed and unobserved components.

- **Counterexample:** If the underlying relationship is strictly non-linear (e.g., $\mathbf{y} = \exp(\mathbf{X}\beta)$), OLS coefficients lose the interpretation of constant marginal effects, invalidating the additive decomposition of the error.

Assumption 4.2: No Perfect Multicollinearity

- **Role:** Ensures that the Gram matrix $\mathbf{X}'\mathbf{X}$ has full column rank ($\text{rank}(\mathbf{X}) = k$), guaranteeing the existence of its inverse.
- **Counterexample:** If a column of $\mathbf{X}_1$ is an exact linear combination of the others, the determinant of the Gram matrix collapses ($|\mathbf{X}'\mathbf{X}| = 0$) and the estimator becomes algebraically indeterminate.

Assumption 4.3: Strict Exogeneity of the Long Model

- **Role:** Establishes that, in the model conditioned on all structurally relevant variables, the disturbance vector ϵ has zero conditional mean: $E[\epsilon \mid \mathbf{X}_1, \mathbf{X}_2] = \mathbf{0}$.
- **Counterexample:** Without this assumption, even the inclusion of $\mathbf{X}_2$ does not restore the consistency of the estimator, as the error would retain factors correlated with the regressors.

4.1.1.3 Formal Statement and Derivation

Consider the true structural model — the *long regression*:

$$\mathbf{y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \epsilon \quad (4.1)$$

where $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X}_1 \in \mathbb{R}^{n \times k_1}$, $\mathbf{X}_2 \in \mathbb{R}^{n \times k_2}$, and $\epsilon \in \mathbb{R}^n$. We assume:

- (i) $E[\epsilon \mid \mathbf{X}_1, \mathbf{X}_2] = \mathbf{0}$ (strict exogeneity in the long model);
- (ii) $\text{rank}(\mathbf{X}_1'\mathbf{X}_1) = k_1$ and $\text{rank}(\mathbf{X}_2'\mathbf{X}_2) = k_2$ (full rank);
- (iii) $\beta_2 \neq \mathbf{0}$ (structural relevance of the omitted variable);
- (iv) $\mathbf{X}_1'\mathbf{X}_2 \neq \mathbf{0}$ (sample non-orthogonality between regressor blocks).

The researcher estimates the *short regression*:

$$\mathbf{y} = \mathbf{X}_1\gamma_1 + \mathbf{u} \quad (4.2)$$

where $\mathbf{u} = \mathbf{X}_2\beta_2 + \epsilon$ is the composite error. The OLS estimator is $\tilde{\gamma}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}$.

Theorem 4.1.1 (Omitted Variable Bias). *Under conditions (iii) and (iv), $E[\tilde{\gamma}_1 \mid \mathbf{X}_1, \mathbf{X}_2] \neq \beta_1$; the estimator $\tilde{\gamma}_1$ is biased for the vector of structural parameters β_1 .*

Proof. **Step 1.** Write the short regression estimator:

$$\tilde{\gamma}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y} \quad (4.3)$$

Step 2. Substitute the structural DGP (4.1) into (4.3):

$$\tilde{\gamma}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'(\mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \epsilon) \quad (4.4)$$

Step 3. Distribute the matrix product:

$$\tilde{\gamma}_1 = \beta_1 + (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{X}_2 \beta_2 + (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \epsilon \quad (4.5)$$

Step 4. Apply the conditional expectation on $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$:

$$E[\tilde{\gamma}_1 | \mathbf{X}] = \beta_1 + (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{X}_2 \beta_2 + (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 E[\epsilon | \mathbf{X}]$$

The third block vanishes by Assumption 4.3, resulting in:

$$\boxed{E[\tilde{\gamma}_1 | \mathbf{X}] = \beta_1 + (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{X}_2 \beta_2} \quad (4.6)$$

Analysis of the Bias Term:

The term $\mathbf{B} = (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{X}_2 \beta_2$ is the **omitted variable bias** (OVB). The expression $(\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{X}_2$ is algebraically equivalent to the OLS coefficient of an auxiliary regression of $\mathbf{X}_2$ on $\mathbf{X}_1$, denoted $\tilde{\delta}_{21}$. Hence:

$$E[\tilde{\gamma}_1 | \mathbf{X}] = \beta_1 + \tilde{\delta}_{21} \beta_2 \quad (4.7)$$

The bias exists if, and only if, $\tilde{\delta}_{21} \neq \mathbf{0}$ (non-orthogonality) **and** $\beta_2 \neq \mathbf{0}$ (structural relevance). If the included and omitted regressors are sample-orthogonal, $\tilde{\delta}_{21} = \mathbf{0}$ and the estimator remains unbiased — although the error variance is still affected. $\square$

4.1.1.4 Asymptotic Properties under Omission

Theorem 4.1.2 (Inconsistency and Limiting Distribution). *If $E[\mathbf{x}_{1i} \mathbf{x}'_{2i}] = \mathbf{Q}_{12} \neq \mathbf{0}$ and $\beta_2 \neq \mathbf{0}$:*

$$\boxed{\text{plim } \hat{\gamma}_1 = \beta_1 + \mathbf{Q}_{11}^{-1} \mathbf{Q}_{12} \beta_2 \neq \beta_1} \quad (4.8)$$

$$\boxed{\sqrt{n} (\hat{\gamma}_1 - \text{plim } \hat{\gamma}_1) \xrightarrow{d} N(\mathbf{0}, \mathbf{Q}_{11}^{-1} \mathbf{\Omega} \mathbf{Q}_{11}^{-1})} \quad (4.9)$$

where $\mathbf{Q}_{11} = E[\mathbf{x}_{1i} \mathbf{x}'_{1i}]$, $\mathbf{Q}_{12} = E[\mathbf{x}_{1i} \mathbf{x}'_{2i}]$, and $\mathbf{\Omega} = E[u_i^2 \mathbf{x}_{1i} \mathbf{x}'_{1i}]$.

Proof (sketch). Applying the Weak Law of Large Numbers and Slutsky's Theorem to the scaled form of the estimator yields (4.8). The Central Limit Theorem provides (4.9). The complete detail follows the pattern established in the heteroskedasticity chapter. $\square$

4.1.1.5 Inference Consequences

Under endogeneity by omission, the t -statistic for $H_0 : \beta_{1,j} = \beta_{1,j}^0$ satisfies:

$$\boxed{|T| \xrightarrow{p} \infty \quad \text{as } n \rightarrow \infty} \quad (4.10)$$

The test ceases to be an instrument of scientific inference and becomes a detector of omitted correlations. The same holds for the F -test, which diverges to $+\infty$ at a rate n , producing spurious rejections of any linear restrictions on β_1 .

4.1.2 Measurement Error

4.1.2.1 Motivation and Geometric Interpretation

The measurement error problem arises from the discrepancy between the theoretical variable prescribed by economic theory (such as “permanent income” or “human capital”) and the variable actually recorded in the data (such as “declared monthly income” or “years of schooling”). While measurement error in the dependent variable only expands the residual variance without compromising consistency, measurement error in the regressors makes the OLS estimator biased and inconsistent even in infinite samples.

Geometric Interpretation in $\mathbb{R}^2$:

Adding noise to the horizontal coordinate x_i^* artificially increases the horizontal dispersion of the observed sample. Since OLS minimizes the sum of squared vertical deviations, the estimated line is “pulled” toward the horizontal axis to accommodate this greater fictitious lateral dispersion. The result is a flatter line than the true structural relationship. This phenomenon of parameter compression toward zero is called **attenuation bias**.

4.1.2.2 Assumptions of the Measurement Error Model

Assumption 4.4: Classical Measurement Error (CME) Model

- **Role:** The measurement noise is orthogonal to the true value of the latent variable and to the structural error.
- **Counterexample:** If the error were correlated with the true value (e.g., high-income individuals underreporting earnings proportionally to wealth), the bias would no longer be attenuation and could inflate or invert the sign of the parameter unpredictably.

Assumption 4.5: Mutual Independence of Noises

- **Role:** The measurement noise $\mathbf{w}$ and the structural disturbance ϵ are mutually independent: $\text{Cov}(\mathbf{w}, \epsilon) = \mathbf{0}$.
- **Counterexample:** If the factor causing measurement error in $\mathbf{x}$ also directly affected $\mathbf{y}$ (e.g., a negligent interviewer who erroneously records both education and salary), an additional spurious correlation would emerge, preventing the analytical isolation of the attenuation effect.

4.1.2.3 Formal Statement and Derivation

Consider the latent structural model:

$$y_i = \beta x_i^* + \epsilon_i \quad (4.11)$$

where x_i^* is the unobserved true regressor and $\beta \in \mathbb{R}$ is the parameter of interest. The observed proxy variable is:

$$x_i = x_i^* + w_i \quad (4.12)$$

We assume:

- (i) $E[\epsilon_i] = E[w_i] = 0$;
- (ii) $\text{Cov}(x_i^*, \epsilon_i) = 0$;
- (iii) $\text{Cov}(x_i^*, w_i) = 0$ and $\text{Cov}(w_i, \epsilon_i) = 0$;
- (iv) $\text{Var}(x_i^*) = \sigma_{x^*}^2 > 0$ and $\text{Var}(w_i) = \sigma_w^2 > 0$.

Theorem 4.1.3 (Attenuation). *The estimator $\hat{\beta} = \sum x_i y_i / \sum x_i^2$ satisfies:*

$$\boxed{\text{plim } \hat{\beta} = \beta \left(\frac{\sigma_{x^*}^2}{\sigma_{x^*}^2 + \sigma_w^2} \right) = \beta \lambda} \quad (4.13)$$

with $\lambda \in (0, 1)$, proving strict inconsistency by attenuation.

Proof. **Step 1.** Substitute $x_i^* = x_i - w_i$ into (4.11):

$$y_i = \beta x_i + (\epsilon_i - \beta w_i) \quad (4.14)$$

Step 2. Define the composite error $v_i \equiv \epsilon_i - \beta w_i$, so that $y_i = \beta x_i + v_i$.

Step 3. Compute $\text{Cov}(x_i, v_i)$ using bilinearity:

$$\text{Cov}(x_i, v_i) = \text{Cov}(x_i^* + w_i, \epsilon_i - \beta w_i) = -\beta \sigma_w^2 \quad (4.15)$$

Step 4. Compute $\text{Var}(x_i)$:

$$\text{Var}(x_i) = \text{Var}(x_i^* + w_i) = \sigma_{x^*}^2 + \sigma_w^2 \quad (4.16)$$

Step 5. Apply the result $\text{plim } \hat{\beta} = \beta + \text{Cov}(x, v) / \text{Var}(x)$:

$$\text{plim } \hat{\beta} = \beta + \frac{-\beta \sigma_w^2}{\sigma_{x^*}^2 + \sigma_w^2} = \beta \left(\frac{\sigma_{x^*}^2}{\sigma_{x^*}^2 + \sigma_w^2} \right) \quad (4.17)$$

□

4.1.2.4 Inference Consequences

Measurement error simultaneously corrupts the centrality of the limiting distribution and the consistency of the residual variance:

1. **Power collapse** ($H_0 : \beta = 0$): The attenuation bias compresses the estimator toward zero, reducing test power and increasing the Type II error rate.
2. **Type I error explosion** ($H_0 : \beta = \beta_{\text{structural}}$): The numerator of the t -statistic ceases to be centered at zero; as n grows, the test diverges and systematically rejects true hypotheses. Formally:

$$\boxed{|t_n| \xrightarrow{p} \infty \quad \text{as } n \rightarrow \infty \text{ under } H_0 : \beta = \beta_0} \quad (4.18)$$

4.1.3 Simultaneity

4.1.3.1 Motivation and Geometric Interpretation

The simultaneous equations problem arises when explanatory variables are treated as exogenously fixed when they are actually determined by an integrated systemic equilibrium. The canonical example is the competitive market: price P affects quantity demanded Q , but Q feeds back into price through the market equilibrium mechanism.

Geometric Interpretation in $\mathbb{R}^2$:

The observed data do not belong individually to either the demand or supply curves; they constitute equilibrium points resulting from the stochastic intersection of both. An OLS line fitted to these points does not represent the price elasticity of demand nor that of supply, but an ambiguous linear combination of the two, without structural interpretation.

4.1.3.2 Assumptions of the Simultaneous System

Assumption 4.6: Structural Completeness of the System

- **Role:** Requires as many independent equations as there are endogenous variables, ensuring that the matrix of structural coefficients is invertible and that the system has a unique reduced-form solution.
- **Counterexample:** Without this condition, the structural parameters are not identifiable.

Assumption 4.7: Exogeneity of Predetermined Variables

- **Role:** The exogenous variables $\mathbf{z}$ are orthogonal to the contemporaneous structural shocks: $E[\mathbf{z}_t \mathbf{u}_t] = \mathbf{0}$.
 - **Counterexample:** Without this condition, the absence of a stable informational anchor prevents the identification of the system.
-

4.1.3.3 Formal Statement and Derivation

Consider the two-equation structural system:

$$y_{1t} = \alpha_1 y_{2t} + \beta_1 x_t + u_{1t} \quad (4.19)$$

$$y_{2t} = \alpha_2 y_{1t} + u_{2t} \quad (4.20)$$

where y_{1t} and y_{2t} are endogenous, x_t is strictly exogenous, and $1 - \alpha_1 \alpha_2 \neq 0$.

The OLS estimator of α_1 in (4.19) is $\hat{\alpha}_1 = \sum y_{2t} y_{1t} / \sum y_{2t}^2$.

Theorem 4.1.4 (Inconsistency by Simultaneity). *The estimator $\hat{\alpha}_1$ converges in probability to $\alpha^* \neq \alpha_1$:*

$$\text{plim } \hat{\alpha}_1 = \alpha_1 + \frac{\alpha_2 \sigma_1^2 + \sigma_{12}}{(1 - \alpha_1 \alpha_2) \text{Var}(y_{2t})} \quad (4.21)$$

Proof (sketch). Solving the system to obtain the reduced form of y_{2t} and computing $\text{Cov}(y_{2t}, u_{1t})$ yields (4.21). The complete detail follows the established pattern. $\square$

4.1.3.4 Inference Consequences

Under simultaneity, the convergence of the estimator to a displaced pseudo-parameter corrupts inference:

1. **Distortion of the nominal level:** The OLS estimator converges to $\beta^* \neq \beta$; the limiting distribution of T is shifted away from $N(0, 1)$.
2. **Spurious rejection of structural models:** The standard error converges to zero at rate $O(1/\sqrt{n})$, while the distance $|\beta^* - \beta|$ remains constant. The test statistic diverges:

$$|t_n| \xrightarrow{p} \infty \quad \text{as } n \rightarrow \infty \quad (4.22)$$

The solution is to migrate to Instrumental Variable estimators, which restore consistency by instrumenting the endogenous variables with regressors exogenous to the structural shock.

4.2 The Instrumental Variables (IV) and Two-Stage Least Squares (2SLS) Estimators

This section presents the mathematical and geometric formalization of the **Instrumental Variables (IV)** estimator and its generalization to overidentified models via **Two-Stage Least Squares (2SLS)**. The objective is to establish the statistical properties in finite

samples and the asymptotic limiting behavior from the perspective of classical estimation theory.

4.2.1 The Identification Problem and the Geometry of Instruments

4.2.1.1 Motivation and Geometric Interpretation

Identification is a logical prerequisite to estimation: it determines whether the information contained in the joint distribution of observable data, combined with the theoretical restrictions imposed by the economic model, is sufficient to uniquely deduce the structural parameters of interest.

The Ordinary Least Squares (OLS) estimator relies on the assumption of strict orthogonality between the regressors and the error term. When **endogeneity** occurs — whether through omitted variables, measurement error, or simultaneity — this condition is violated.

Consider the population structural model in matrix form:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{u} \quad (4.23)$$

where $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times k}$, $\beta \in \mathbb{R}^k$, and $\mathbf{u} \in \mathbb{R}^n$. The classical exogeneity assumption is violated in the population, that is, $E[\mathbf{X}'\mathbf{u}] \neq \mathbf{0}$.

Geometric Interpretation in $\mathbb{R}^n$:

If the space spanned by the regressors $\mathbf{X}$ is correlated with the error vector $\mathbf{u}$, the orthogonal projection of $\mathbf{y}$ directly onto $\text{Col}(\mathbf{X})$ captures not only the causal effect of β , but also the variation of the stochastic disturbance $\mathbf{u}$ transmitted through $\mathbf{X}$. OLS becomes inconsistent because it attributes to the parameter vector β covariances that actually belong to the population error.

The introduction of an instrument matrix $\mathbf{Z} \in \mathbb{R}^{n \times L}$ functions as an external reference axis in the Euclidean space. The instrument $\mathbf{Z}$ provides a source of variation for $\mathbf{X}$ that is perfectly orthogonal to $\mathbf{u}$. The central idea is to use only the portion of the variability of $\mathbf{X}$ that is provably exogenous to identify and isolate the true causal effect on $\mathbf{y}$.

4.2.1.2 Identification Assumptions

Let $\mathbf{X}$ be the $n \times k$ matrix of regressors and $\mathbf{Z}$ the $n \times L$ matrix of instruments. For the vector of structural parameters β to be identified, two classical conditions must be satisfied:

Assumption 4.13: Order Condition ($L \geq k$)

- **Role:** This is a necessary (but not sufficient) condition for the system of population moment equations to have a unique solution. It requires that the number of external exogenous pieces of information (instruments) be at least equal to the number of unknown structural parameters to be estimated.

- **Identification Failure:** If $L < k$, the model is *underidentified*. There will be infinitely many parameter combinations consistent with the data, making any unique point estimate impossible.

Assumption 4.14: Rank Condition (Relevance)

- **Role:** Ensures that the instruments have a real and relevant linear statistical relationship with the endogenous regressors. Mathematically, it ensures that the population cross-moment matrix $\mathbf{Q}_{ZX} = E[\mathbf{z}_i \mathbf{x}_i']$ has full column rank, that is, $\text{rank}(\mathbf{Q}_{ZX}) = k$.
- **Identification Failure:** If the instrument $\mathbf{Z}$ is linearly independent of or completely orthogonal to $\mathbf{X}$ in the population ($\mathbf{Q}_{ZX} = \mathbf{0}$), the model is *unidentified*. Algebraically, the linear system will not have a unique solution and the estimator will collapse.

Assumption 4.15: Exogeneity of Instruments ($E[\mathbf{z}_i u_i] = \mathbf{0}$)

- **Role:** Fundamental condition for the consistency of the IV estimator: the instruments are not correlated with the structural error.
- **Counterexample:** If $E[\mathbf{z}_i u_i] \neq \mathbf{0}$, the IV estimator inherits the endogeneity of the instruments and becomes inconsistent — the bias may be even more severe than that of OLS.

4.2.1.3 Analytical Identification Scenarios

Consider the linear structural model defined in (4.23). The identification of the vector β from the instrument matrix $\mathbf{Z}$ falls into three distinct analytical scenarios:

Scenario	Condition	Implication
Underidentification	$L < k$	Number of moment conditions is less than the number of unknowns. The structural parameter cannot be determined .
Exact Identification	$L = k$	There is exactly one instrument for each regressor. The classical IV estimator is obtained by direct algebraic solution.
Overidentification	$L > k$	There are more instruments than parameters. The system has more equations than unknowns, requiring optimal weighting (2SLS or GMM).

4.2.1.4 Algebraic Derivation under Exact Identification ($L = k$)

We postulate the population moment condition based on the strict exogeneity of the instrument:

$$E[\mathbf{Z}'\mathbf{u}] = \mathbf{0} \implies E[\mathbf{Z}'(\mathbf{y} - \mathbf{X}\beta)] = \mathbf{0} \quad (4.24)$$

By the principle of sample analogy, we replace the population expectation with its respective sample moment under exact identification ($L = k$):

$$\frac{1}{n}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\hat{\beta}_{IV}) = \mathbf{0} \quad (4.25)$$

Multiplying by n and applying the distributive property:

$$\mathbf{Z}'\mathbf{y} - \mathbf{Z}'\mathbf{X}\hat{\beta}_{IV} = \mathbf{0} \implies \mathbf{Z}'\mathbf{X}\hat{\beta}_{IV} = \mathbf{Z}'\mathbf{y} \quad (4.26)$$

Under **Assumption 4.14 (Rank Condition)** with $L = k$, the square matrix $\mathbf{Z}'\mathbf{X}$ of dimension $k \times k$ has full rank ($\text{rank}(\mathbf{Z}'\mathbf{X}) = k$), hence its inverse exists uniquely. Premultiplying by $(\mathbf{Z}'\mathbf{X})^{-1}$:

$$\boxed{\hat{\beta}_{IV} = (\mathbf{Z}'\mathbf{X})^{-1} \mathbf{Z}'\mathbf{y}} \quad (4.27)$$

4.2.2 Finite-Sample Properties: Expectation and Variance

4.2.2.1 Motivation

The proof of asymptotic consistency guarantees the appropriate behavior of the IV estimator as $n \rightarrow \infty$, but does not ensure the absence of bias in small samples. Unlike the OLS estimator under strict exogeneity, the IV estimator is **known to be biased in finite samples**.

4.2.2.2 Assumptions for Variance Derivation

Assumption 4.16: Conditional Homoskedasticity and No Autocorrelation

$$E[\mathbf{u}\mathbf{u}'|\mathbf{Z}, \mathbf{X}] = \sigma^2 \mathbf{I}_n \quad (4.28)$$

where $\sigma^2 > 0$ is the scalar conditional variance.

4.2.2.3 Formal Statement and Derivation

Theorem 4.2.1 (Finite-Sample Bias). *The IV estimator is biased in finite samples:*

$$E[\hat{\beta}_{IV}|\mathbf{Z}, \mathbf{X}] = \beta + (\mathbf{Z}'\mathbf{X})^{-1} \mathbf{Z}' E[\mathbf{u}|\mathbf{Z}, \mathbf{X}] \quad (4.29)$$

Proof. Substituting the structural equation (4.23) into the definition of the estimator (4.27):

$$\hat{\beta}_{IV} = (\mathbf{Z}'\mathbf{X})^{-1} \mathbf{Z}'(\mathbf{X}\beta + \mathbf{u}) = \beta + (\mathbf{Z}'\mathbf{X})^{-1} \mathbf{Z}'\mathbf{u} \quad (4.30)$$

Applying the conditional expectation $E[\cdot|\mathbf{Z}, \mathbf{X}]$:

$$E[\hat{\beta}_{IV}|\mathbf{Z}, \mathbf{X}] = \beta + (\mathbf{Z}'\mathbf{X})^{-1} \mathbf{Z}' E[\mathbf{u}|\mathbf{Z}, \mathbf{X}] \quad (4.31)$$

Since $\mathbf{X}$ contains endogenous regressors correlated with the error, $E[\mathbf{u}|\mathbf{Z}, \mathbf{X}] \neq \mathbf{0}$. Therefore, the conditional bias is:

$$\boxed{\text{Bias}(\hat{\beta}_{IV}|\mathbf{Z}, \mathbf{X}) = (\mathbf{Z}'\mathbf{X})^{-1} \mathbf{Z}' E[\mathbf{u}|\mathbf{Z}, \mathbf{X}]} \quad (4.32)$$

□

Note: This finite-sample bias severely approaches the OLS bias as the relevance of the instruments diminishes, characterizing the *weak instruments* problem.

Theorem 4.2.2 (Conditional Variance of the IV Estimator). *Under Assumption 4.16 (conditional homoskedasticity):*

$$\boxed{\text{Var}(\hat{\beta}_{IV}|\mathbf{Z}, \mathbf{X}) = \sigma^2 (\mathbf{Z}'\mathbf{X})^{-1} (\mathbf{Z}'\mathbf{Z}) (\mathbf{X}'\mathbf{Z})^{-1}} \quad (4.33)$$

Proof. By definition:

$$\text{Var}(\hat{\beta}_{IV}|\mathbf{Z}, \mathbf{X}) = E \left[\left(\hat{\beta}_{IV} - E[\hat{\beta}_{IV}|\mathbf{Z}, \mathbf{X}] \right) \left(\hat{\beta}_{IV} - E[\hat{\beta}_{IV}|\mathbf{Z}, \mathbf{X}] \right)' \middle| \mathbf{Z}, \mathbf{X} \right] \quad (4.34)$$

Substituting (4.30) and (4.31) into the definition and centering the analysis on the conditional distribution of the error:

$$\text{Var}(\hat{\beta}_{IV}|\mathbf{Z}, \mathbf{X}) = (\mathbf{Z}'\mathbf{X})^{-1} \mathbf{Z}' E[\mathbf{u}\mathbf{u}'|\mathbf{Z}, \mathbf{X}] \mathbf{Z} (\mathbf{X}'\mathbf{Z})^{-1} \quad (4.35)$$

Invoking Assumption 4.16, the core of the expectation collapses to $\sigma^2 \mathbf{I}_n$:

$$\text{Var}(\hat{\beta}_{IV}|\mathbf{Z}, \mathbf{X}) = \sigma^2 (\mathbf{Z}'\mathbf{X})^{-1} \mathbf{Z}'\mathbf{Z} (\mathbf{X}'\mathbf{Z})^{-1} \quad (4.36)$$

□

4.2.3 Generalization to the Overidentified Case: The 2SLS Estimator

4.2.3.1 Motivation and Geometric Interpretation

When the number of instruments exceeds the number of regressors ($L > k$), the matrix $\mathbf{Z}'\mathbf{X}$ becomes rectangular of dimension $L \times k$ and cannot be directly inverted. The Two-Stage Least Squares (**2SLS**) estimator solves this problem through a double geometric projection procedure that efficiently combines all available information from the instruments.

Geometric Interpretation in $\mathbb{R}^n$:

The sample space $\mathbb{R}^n$ contains three fundamental subspaces:

1. $\text{Col}(\mathbf{Z})$: the space spanned by the instruments (exogenous);
2. $\text{Col}(\mathbf{X})$: the space spanned by the original regressors (contaminated by endogeneity);
3. $\text{Col}(\hat{\mathbf{X}})$: the space spanned by the projected values of $\mathbf{X}$ onto $\text{Col}(\mathbf{Z})$ — the “clean part” of the regressors.

2SLS operates in two geometric stages:

- **First Stage:** Orthogonally project each column of $\mathbf{X}$ onto $\text{Col}(\mathbf{Z})$, eliminating the component correlated with the structural error. The result is $\hat{\mathbf{X}} = \mathbf{P}_Z \mathbf{X}$, the best linear approximation of $\mathbf{X}$ in the exogenous subspace.
- **Second Stage:** Project $\mathbf{y}$ onto $\text{Col}(\hat{\mathbf{X}})$, obtaining the estimator that efficiently combines all available instruments.

4.2.3.2 Additional Assumptions for 2SLS

Assumption 4.17: Rank Condition in the Overidentified Case

- **Role:** Ensures that $\text{rank}(\mathbf{Z}'\mathbf{X}) = k$, guaranteeing that the instruments are sufficiently correlated with the endogenous regressors. In the overidentified case ($L > k$), this condition implies that $\mathbf{X}'\mathbf{P}_Z\mathbf{X}$ is non-singular.
- **Identification Failure:** If $\text{rank}(\mathbf{Z}'\mathbf{X}) < k$, the instruments are weak or irrelevant for some regressors, and the matrix $\mathbf{X}'\mathbf{P}_Z\mathbf{X}$ becomes singular, preventing computation of the estimator.

Assumption 4.18: Finite Fourth-Order Moments

- **Role:** Guarantees the existence of asymptotic variances and the applicability of the Central Limit Theorem.
- **Counterexample:** With heavy-tailed distributions, the sample moment matrices may not converge properly, invalidating asymptotic inference.

4.2.3.3 Formal Statement and Derivation

We define the fundamental orthogonal projectors in the Euclidean space $\mathbb{R}^n$:

4.2. THE INSTRUMENTAL VARIABLES (IV) AND TWO-STAGE LEAST SQUARES (2SLS) ESTIMATOR

- Projection matrix onto $\text{Col}(\mathbf{Z})$:

$$\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}' \quad (4.37)$$

- Complementary annihilator matrix:

$$\mathbf{M}_Z = \mathbf{I}_n - \mathbf{P}_Z \quad (4.38)$$

First Stage: Project the matrix of endogenous regressors $\mathbf{X}$ onto the exogenous space generated by $\mathbf{Z}$:

$$\hat{\mathbf{X}} = \mathbf{P}_Z \mathbf{X} = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X} \quad (4.39)$$

The orthogonal decomposition ensures that $\mathbf{X} = \hat{\mathbf{X}} + \mathbf{M}_Z \mathbf{X}$, where $\hat{\mathbf{X}}$ represents the clean exogenous portion of the regressors, and $\mathbf{M}_Z \mathbf{X}$ captures the discarded endogenous variation.

Second Stage: Perform the regression of $\mathbf{y}$ on $\hat{\mathbf{X}}$ via OLS:

$$\hat{\beta}_{2SLS} = (\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}\hat{\mathbf{X}}'\mathbf{y} \quad (4.40)$$

Substituting (4.39) into (4.40) and using the idempotence and symmetry of $\mathbf{P}_Z$ ($\mathbf{P}_Z'\mathbf{P}_Z = \mathbf{P}_Z$):

$$\hat{\beta}_{2SLS} = (\mathbf{X}'\mathbf{P}_Z'\mathbf{P}_Z\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_Z'\mathbf{y} = (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_Z\mathbf{y} \quad (4.41)$$

Expanding the projection matrix expression yields the traditional closed form of the 2SLS estimator:

$$\boxed{\hat{\beta}_{2SLS} = [\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}]^{-1} \mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}} \quad (4.42)$$

4.2.3.4 Asymptotic Properties of 2SLS

Under the instrument validity conditions (Assumptions 4.13, 4.14, and 4.15) and asymptotic regularity, the 2SLS estimator satisfies:

1. **Consistency:** $\hat{\beta}_{2SLS} \xrightarrow{p} \beta$

2. **Asymptotic Normality:**

$$\sqrt{n}(\hat{\beta}_{2SLS} - \beta) \xrightarrow{d} N\left(\mathbf{0}, \sigma^2 \left(\mathbf{Q}_{XZ}\mathbf{Q}_{ZZ}^{-1}\mathbf{Q}_{ZX}\right)^{-1}\right) \quad (4.43)$$

where $\mathbf{Q}_{XZ} = E[\mathbf{x}_i\mathbf{z}_i']$, $\mathbf{Q}_{ZZ} = E[\mathbf{z}_i\mathbf{z}_i']$, and $\mathbf{Q}_{ZX} = \mathbf{Q}_{XZ}'$.

3. **Variance Estimator (under homoskedasticity):**

$$\widehat{\text{Var}}(\hat{\beta}_{2SLS}) = \hat{\sigma}^2 (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}, \quad \hat{\sigma}^2 = \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{n - k} \quad (4.44)$$

4. Robust Variance to Heteroskedasticity:

$$\widehat{\text{Var}}_{\text{robust}}(\hat{\beta}_{2SLS}) = (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1} \left(\sum_{i=1}^n \hat{u}_i^2 \hat{\mathbf{x}}_i \hat{\mathbf{x}}_i' \right) (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1} \quad (4.45)$$

4.2.4 Relative Efficiency and the Cost of Instrumentation

4.2.4.1 Motivation

Under the validity of the classical Gauss-Markov assumptions — where all original regressors are strictly exogenous — the OLS estimator is the Best Linear Unbiased Estimator (BLUE). Therefore, the substitution of $\mathbf{X}$ by its instrumented projection $\hat{\mathbf{X}}$ generates an inevitable loss of information, inflating the sample variance.

Geometric Interpretation:

In $\mathbb{R}^n$, $\text{Col}(\hat{\mathbf{X}})$ is a subspace of $\text{Col}(\mathbf{X})$ when $\mathbf{X}$ is exogenous (since $\hat{\mathbf{X}}$ is a linear combination of $\mathbf{X}$ through projection). Projecting $\mathbf{y}$ onto a smaller subspace reduces the explained variation and increases the uncertainty about the parameters. The smaller the correlation between $\mathbf{X}$ and $\mathbf{Z}$ (weak instruments), the smaller $\text{Col}(\hat{\mathbf{X}})$ and the greater the efficiency loss.

4.2.4.2 Assumptions for Efficiency Comparison

Assumption 4.19: Exogeneity of $\mathbf{X}$ (Benchmark)

- **Role:** Establishes the counterfactual scenario where OLS is consistent, allowing variance comparison.
- **Counterexample:** If $\mathbf{X}$ is endogenous, OLS is inconsistent and variance comparison loses practical meaning.

Assumption 4.20: Conditional Homoskedasticity

- **Role:** Ensures that $\text{Var}(\mathbf{u}|\mathbf{X}, \mathbf{Z}) = \sigma^2 \mathbf{I}_n$, simplifying variance expressions.
- **Counterexample:** Under heteroskedasticity, variance formulas require robust adjustments, and the ordering relationship may be more complex.

4.2.4.3 Formal Statement and Derivation

Theorem 4.2.3 (Cost of Instrumentation). *Under the assumption of exogeneity of $\mathbf{X}$ and conditional homoskedasticity, the variance-covariance matrices satisfy the ordering:*

$$\boxed{\text{Var}(\hat{\beta}_{2SLS}|\mathbf{X}, \mathbf{Z}) \succeq \text{Var}(\hat{\beta}_{OLS}|\mathbf{X})} \quad (4.46)$$

Proof. Auxiliary Lemma: If $\mathbf{A}$ and $\mathbf{B}$ are symmetric and strictly positive definite matrices such that $\mathbf{A} \succeq \mathbf{B}$, then $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$.

Step 1: Formulation of conditional variances

Under homoskedasticity ($\text{Var}(\mathbf{u}|\mathbf{X}, \mathbf{Z}) = \sigma^2 \mathbf{I}_n$):

$$\text{Var}(\hat{\beta}_{OLS}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} \quad (4.47)$$

$$\text{Var}(\hat{\beta}_{2SLS}|\mathbf{X}, \mathbf{Z}) = \sigma^2(\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1} \quad (4.48)$$

Step 2: Analysis of the difference of information matrices

We evaluate the difference matrix Δ between the inverses of the variances (scaled by $1/\sigma^2$):

$$\Delta = \mathbf{X}'\mathbf{X} - \mathbf{X}'\mathbf{P}_Z\mathbf{X} = \mathbf{X}'(\mathbf{I}_n - \mathbf{P}_Z)\mathbf{X} = \mathbf{X}'\mathbf{M}_Z\mathbf{X} \quad (4.49)$$

Step 3: Demonstration of the positive semidefinite character of $\mathbf{M}_Z$

The annihilator matrix $\mathbf{M}_Z$ is symmetric and idempotent ($\mathbf{M}_Z = \mathbf{M}_Z'\mathbf{M}_Z$). Taking an arbitrary non-zero vector $\mathbf{v} \in \mathbb{R}^n$:

$$\mathbf{v}'\mathbf{M}_Z\mathbf{v} = \mathbf{v}'\mathbf{M}_Z'\mathbf{M}_Z\mathbf{v} = (\mathbf{M}_Z\mathbf{v})'(\mathbf{M}_Z\mathbf{v}) = \|\mathbf{M}_Z\mathbf{v}\|^2 \geq 0 \quad (4.50)$$

Since the quadratic form is non-negative for any vector $\mathbf{v}$, we conclude that $\mathbf{M}_Z \succeq \mathbf{0}$. Consequently:

$$\mathbf{X}'\mathbf{M}_Z\mathbf{X} \succeq \mathbf{0} \implies \mathbf{X}'\mathbf{X} \succeq \mathbf{X}'\mathbf{P}_Z\mathbf{X} \quad (4.51)$$

Step 4: Inversion and conclusion

Invoking the Auxiliary Lemma on (4.51):

$$(\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1} \succeq (\mathbf{X}'\mathbf{X})^{-1} \quad (4.52)$$

Multiplying by the scalar $\sigma^2 > 0$:

$$\sigma^2(\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1} \succeq \sigma^2(\mathbf{X}'\mathbf{X})^{-1} \implies \text{Var}(\hat{\beta}_{2SLS}) \succeq \text{Var}(\hat{\beta}_{OLS}) \quad (4.53)$$

□

4.2.4.4 Practical Implications

The efficiency cost of 2SLS relative to OLS is greater when:

1. **The correlation between $\mathbf{X}$ and $\mathbf{Z}$ is smaller** (weak instruments): $\text{Col}(\hat{\mathbf{X}})$ shrinks, reducing the information available for estimation.
2. **The number of redundant instruments is larger**: irrelevant instruments add noise without increasing the exogenous variation of $\mathbf{X}$.
3. **The sample size n is smaller**: in small samples, the loss of degrees of freedom in the first stage becomes more severe.

This *trade-off* between **consistency** (2SLS) and **efficiency** (OLS) is the core of the choice between estimators: the researcher accepts greater sample variance in exchange for eliminating endogeneity bias.

4.2.5 Asymptotic Properties of the IV Estimator

4.2.5.1 Motivation

To overcome finite-sample pathologies — such as the absence of defined moments under weak identification — statistical inference relies strictly on the asymptotic limiting behavior as $n \rightarrow \infty$.

4.2.5.2 Asymptotic Assumptions

Assumption 4.21: Ergodically Random Sampling

- **Role:** The observations of the data vector $(\mathbf{y}_i, \mathbf{x}_i, \mathbf{z}_i)$ are independent and identically distributed (i.i.d.) with finite fourth-order moments, enabling the application of the Weak Law of Large Numbers and the Central Limit Theorem.
- **Counterexample:** In time series with dependence or non-stationarity, sample means may not converge to population expectations.

Assumption 4.22: Asymptotic Exogeneity of the Instrument

$$\text{plim} \left(\frac{1}{n} \mathbf{Z}' \mathbf{u} \right) = E[\mathbf{z}_i u_i] = \mathbf{0} \quad (4.54)$$

- **Role:** Fundamental condition for the consistency of the IV estimator: the population correlation between instruments and error is zero.
- **Counterexample:** If $E[\mathbf{z}_i u_i] \neq \mathbf{0}$, the IV estimator is inconsistent.

Assumption 4.23: Asymptotic Relevance of the Instrument

$$\text{plim} \left(\frac{1}{n} \mathbf{Z}' \mathbf{X} \right) = E[\mathbf{z}_i \mathbf{x}_i'] = \boldsymbol{\Sigma}_{ZX} \quad (4.55)$$

where $\boldsymbol{\Sigma}_{ZX}$ is a finite population coefficient matrix with full rank equal to k .

- **Role:** Ensures that the instruments are sufficiently correlated with the endogenous regressors to identify the parameters.
 - **Counterexample:** If $\text{rank}(\boldsymbol{\Sigma}_{ZX}) < k$, the model is underidentified and the estimator does not converge.
-

4.2.5.3 Formal Statement and Derivation of Consistency

Theorem 4.2.4 (Consistency of IV). *Under Assumptions 4.21 to 4.23, the classical instrumental variables estimator is consistent in probability:*

$$\boxed{\hat{\beta}_{IV} \xrightarrow{p} \beta} \quad (4.56)$$

Proof. Starting from the stochastic form derived in (4.30), we rescale the sample operators by dividing by the sample size n :

$$\hat{\beta}_{IV} = \beta + \left(\frac{\mathbf{Z}'\mathbf{X}}{n} \right)^{-1} \left(\frac{\mathbf{Z}'\mathbf{u}}{n} \right) \quad (4.57)$$

We apply the plim operator to both sides. By Slutsky's Theorem, since the inverse function is continuous over non-singular matrices:

$$\text{plim}(\hat{\beta}_{IV}) = \beta + \left[\text{plim} \left(\frac{\mathbf{Z}'\mathbf{X}}{n} \right) \right]^{-1} \cdot \text{plim} \left(\frac{\mathbf{Z}'\mathbf{u}}{n} \right) \quad (4.58)$$

Invoking the Weak Law of Large Numbers:

1. By asymptotic relevance (Assumption 4.23): $\text{plim} \left(\frac{\mathbf{Z}'\mathbf{X}}{n} \right) = \Sigma_{ZX}$ (non-singular).
2. By asymptotic exogeneity (Assumption 4.22): $\text{plim} \left(\frac{\mathbf{Z}'\mathbf{u}}{n} \right) = \mathbf{0}$.

Substituting:

$$\text{plim}(\hat{\beta}_{IV}) = \beta + \Sigma_{ZX}^{-1} \cdot \mathbf{0} = \beta \quad (4.59)$$

□

4.2.5.4 Asymptotic Normality

Theorem 4.2.5 (Asymptotic Normality of IV). *Under Assumptions 4.21 to 4.23 and assuming population homoskedasticity, the limiting distribution of the estimation error scaled by $\sqrt{n}$ satisfies:*

$$\boxed{\sqrt{n}(\hat{\beta}_{IV} - \beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{V})} \quad (4.60)$$

where the asymptotic variance matrix is defined by:

$$\mathbf{V} = \sigma^2 \Sigma_{ZX}^{-1} \Sigma_{ZZ} \Sigma_{XZ}^{-1} \quad (4.61)$$

with $\Sigma_{ZZ} = E[\mathbf{z}_i \mathbf{z}_i']$ and $\Sigma_{XZ} = \Sigma'_{ZX}$.

Proof. Subtracting β from both sides of (4.57) and scaling by $\sqrt{n}$:

$$\sqrt{n}(\hat{\beta}_{IV} - \beta) = \left(\frac{\mathbf{Z}'\mathbf{X}}{n} \right)^{-1} \left(\frac{\mathbf{Z}'\mathbf{u}}{\sqrt{n}} \right) \quad (4.62)$$

We analyze the limiting behavior of each block:

1. By the Continuous Mapping Theorem: $\left(\frac{\mathbf{Z}'\mathbf{X}}{n}\right)^{-1} \xrightarrow{p} \Sigma_{ZX}^{-1}$.
2. Define $\mathbf{w}_i = \mathbf{z}_i u_i$. By exogeneity, $E[\mathbf{w}_i] = \mathbf{0}$. Under conditional homoskedasticity:

$$\Omega = E[\mathbf{w}_i \mathbf{w}_i'] = E[u_i^2 \mathbf{z}_i \mathbf{z}_i'] = \sigma^2 E[\mathbf{z}_i \mathbf{z}_i'] = \sigma^2 \Sigma_{ZZ} \quad (4.63)$$

By the Lindeberg-Lévy Central Limit Theorem:

$$\frac{\mathbf{Z}'\mathbf{u}}{\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{z}_i u_i \xrightarrow{d} N(\mathbf{0}, \sigma^2 \Sigma_{ZZ}) \quad (4.64)$$

Combining the limits via Slutsky's Lemma:

$$\sqrt{n}(\hat{\beta}_{IV} - \beta) \xrightarrow{d} \Sigma_{ZX}^{-1} \cdot N(\mathbf{0}, \sigma^2 \Sigma_{ZZ}) \quad (4.65)$$

By the properties of linear transformations of Gaussian vectors:

$$\sqrt{n}(\hat{\beta}_{IV} - \beta) \xrightarrow{d} N\left(\mathbf{0}, \sigma^2 \Sigma_{ZX}^{-1} \Sigma_{ZZ} \Sigma_{XZ}^{-1}\right) \quad (4.66)$$

□

4.2.6 Hypothesis Testing and Asymptotic Inference

4.2.6.1 Motivation

The formulation of asymptotic test statistics aims to formally quantify the statistical uncertainty around the point estimator. It assesses whether the distance between the sample estimate and the restriction vector arises solely from sampling fluctuations or reflects a population discrepancy.

4.2.6.2 Assumptions of the Wald Test

Assumption 4.24: Independent Linear Restrictions

- **Role:** Defines q simultaneous linear conditions with $\mathbf{R} \in \mathbb{R}^{q \times k}$ of full row rank ($\text{rank}(\mathbf{R}) = q$).
- **Counterexample:** If $\text{rank}(\mathbf{R}) < q$, the restrictions are redundant or contradictory, making the covariance matrix singular and the statistic undefined.

4.2.6.3 Formal Statement and Derivation

Consider the multivariate linear null hypothesis:

$$H_0 : \mathbf{R}\beta = \mathbf{r} \quad (4.67)$$

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where $\mathbf{R} \in \mathbb{R}^{q \times k}$ with $\text{rank}(\mathbf{R}) = q$, and $\mathbf{r} \in \mathbb{R}^q$.

Theorem 4.2.6 (Wald Statistic for IV). *Under the asymptotic regularity conditions of the IV estimator, the quadratic Wald statistic:*

$$W = (\mathbf{R}\hat{\beta}_{IV} - \mathbf{r})' [\mathbf{R} (\widehat{\text{Avar}}(\hat{\beta}_{IV})) \mathbf{R}']^{-1} (\mathbf{R}\hat{\beta}_{IV} - \mathbf{r}) \quad (4.68)$$

converges asymptotically in distribution to a Chi-square variable with q degrees of freedom under the null hypothesis:

$$\boxed{W \xrightarrow{d} \chi_q^2} \quad (4.69)$$

Proof. Theoretical Lemma: If $\mathbf{v} \sim N(\mathbf{0}, \Sigma)$ with Σ non-singular of order q , then $\mathbf{v}'\Sigma^{-1}\mathbf{v} \sim \chi_q^2$.

Step 1: Limiting behavior under the null hypothesis

By the asymptotic normality theorem (4.60):

$$\sqrt{n}(\hat{\beta}_{IV} - \beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{V}) \quad (4.70)$$

Premultiplying by $\mathbf{R}$:

$$\sqrt{n}(\mathbf{R}\hat{\beta}_{IV} - \mathbf{R}\beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{RVR}') \quad (4.71)$$

Under H_0 , substitute $\mathbf{R}\beta = \mathbf{r}$:

$$\sqrt{n}(\mathbf{R}\hat{\beta}_{IV} - \mathbf{r}) \xrightarrow{d} N(\mathbf{0}, \mathbf{RVR}') \quad (4.72)$$

Step 2: Consistent estimator of the asymptotic variance

We define the consistent estimator:

$$\hat{\mathbf{V}} = \hat{\sigma}^2 \left(\frac{\mathbf{Z}'\mathbf{X}}{n} \right)^{-1} \left(\frac{\mathbf{Z}'\mathbf{Z}}{n} \right) \left(\frac{\mathbf{X}'\mathbf{Z}}{n} \right)^{-1} \quad (4.73)$$

where $\hat{\sigma}^2 = \frac{1}{n}(\mathbf{y} - \mathbf{X}\hat{\beta}_{IV})'(\mathbf{y} - \mathbf{X}\hat{\beta}_{IV}) \xrightarrow{p} \sigma^2$.

Since $\hat{\mathbf{V}} \xrightarrow{p} \mathbf{V}$, the Continuous Mapping Theorem ensures:

$$[\mathbf{R}\hat{\mathbf{V}}\mathbf{R}']^{-1} \xrightarrow{p} [\mathbf{RVR}']^{-1} \quad (4.74)$$

Step 3: Construction of the quadratic form

Rewriting (4.68) to highlight the $\sqrt{n}$ factors:

$$W = [\sqrt{n}(\mathbf{R}\hat{\beta}_{IV} - \mathbf{r})]' [\mathbf{R}\hat{\mathbf{V}}\mathbf{R}']^{-1} [\sqrt{n}(\mathbf{R}\hat{\beta}_{IV} - \mathbf{r})] \quad (4.75)$$

Substituting (4.72) and (4.74), Slutsky's Lemma guarantees:

$$W \xrightarrow{d} \mathbf{v}' [\mathbf{RVR}']^{-1} \mathbf{v}, \quad \mathbf{v} \sim N(\mathbf{0}, \mathbf{RVR}') \quad (4.76)$$

Invoking the Theoretical Lemma:

$$W \xrightarrow{d} \chi_q^2 \quad (4.77)$$

□

4.2.7 Instrument Validation: The Sargan Overidentification Test

4.2.7.1 Motivation

The Sargan test (or overidentifying restrictions test) is applicable exclusively to the scenario where the model has more instruments than endogenous regressors ($L > k$). When the model is exactly identified ($L = k$), the sample residuals are orthogonal to the instruments by algebraic construction, making any test of hypothesis validation impossible.

In the overidentified scenario, the 2SLS estimator selects a linear combination of the instruments. Therefore, it becomes possible to test whether the null hypothesis of full exogeneity of the instrument block is statistically sustainable. The test principle is based on projecting the residuals of the estimated model onto the full matrix of available instruments $\mathbf{Z}$. If the instruments are indeed exogenous, the explanatory power of this auxiliary regression should be statistically close to zero.

4.2.7.2 Assumptions of the Sargan Test

Assumption 4.25: Overidentification ($L > k$)

- **Role:** Necessary condition for the applicability of the test. Requires that there be more instruments than endogenous regressors.
- **Counterexample:** If $L = k$, the test is degenerate and cannot be computed.

Assumption 4.26: Joint Exogeneity of Instruments (H_0)

- **Role:** Establishes that all instruments are orthogonal to the structural error: $E[\mathbf{z}_i u_i] = \mathbf{0}$ for all i .
- **Counterexample:** If at least one instrument is endogenous, the Sargan statistic detects the violation and rejects H_0 .

4.2.7.3 Computation Algorithm and Test Statistic

The Sargan test is operationalized by the following algorithmic protocol:

1. Estimate the structural model via 2SLS and extract the sample residual vector:

$$\hat{\mathbf{u}} = \mathbf{y} - \mathbf{X}\hat{\beta}_{2SLS} \quad (4.78)$$

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2. Run an auxiliary regression of the residuals on the full instrument matrix $\mathbf{Z}$ via OLS:

$$\hat{\mathbf{u}} = \mathbf{Z}\gamma + \epsilon \quad (4.79)$$

3. Compute the coefficient of determination (R_u^2) of this auxiliary projection:

$$R_u^2 = \frac{\hat{\mathbf{u}}' \mathbf{P}_Z \hat{\mathbf{u}}}{\hat{\mathbf{u}}' \hat{\mathbf{u}}} \quad (4.80)$$

4. The Sargan test statistic (S) is obtained by multiplying the sample size by the coefficient of determination:

$$S = nR_u^2 = \frac{\hat{\mathbf{u}}' \mathbf{P}_Z \hat{\mathbf{u}}}{\hat{\sigma}^2} \quad (4.81)$$

where $\hat{\sigma}^2 = \hat{\mathbf{u}}' \hat{\mathbf{u}}/n$ is the estimator of the variance of the auxiliary regression error.

4.2.7.4 Asymptotic Distribution of the Sargan Test

Theorem 4.2.7 (Distribution of the Sargan Test). *Under the null hypothesis of total exogeneity of the instruments ($H_0 : E[\mathbf{z}_i u_i] = \mathbf{0}$), the Sargan statistic converges asymptotically in distribution to a Chi-square variable whose degrees of freedom correspond exactly to the number of overidentifying restrictions (excess instruments):*

$$\boxed{S \xrightarrow{d} \chi_{L-k}^2} \quad (4.82)$$

Interpretation: Rejection of the null hypothesis ($S > \chi_{\text{critical}}^2$) indicates that at least one of the instruments fails the asymptotic exogeneity condition (being correlated with the structural error term) or that the structural model has been incorrectly specified due to omitted variables.

4.2.7.5 Limitations and Practical Considerations

1. **Power of the test:** The Sargan test has low power to detect weakly endogenous instruments in small samples.
2. **Correct model specification:** Rejection may arise from either invalid instruments or misspecification of the structural equation (e.g., omission of relevant variables).
3. **Robust version to heteroskedasticity:** The Hansen test (J-statistic) is a generalization of the Sargan test that does not require homoskedasticity, being preferable in contexts with heteroskedastic errors.
4. **Interpretation of non-rejection:** Non-rejection of the null hypothesis **does not** prove that the instruments are exogenous; it only indicates that the test did not find sufficient statistical evidence to reject the joint validity of the instruments.

Chapter 5

Generalized Method of Moments (GMM) Estimator

5.1 Construction and Properties of the Estimator

5.1.1 Motivation and Geometric Interpretation

5.1.1.1 The Overidentification Problem

The **Generalized Method of Moments (GMM)**, proposed by Lars Hansen (1982), represents a fundamental milestone in modern econometrics by solving the **overidentification** problem in parametric models. In the classical method of moments (Pearson, 1894), the researcher seeks an estimator that satisfies a system of equations where the number of moment conditions (L) is exactly equal to the number of unknown parameters (K). In such **just-identified** systems, there is typically a unique solution that zeros out the sample moment conditions.

However, economic theory or statistical properties often provide more restrictions (moments) than strictly necessary to identify the parameters ($L > K$). In these cases, the system of equations becomes **overdetermined** and, due to inherent sample variability, it is impossible to find a single parameter vector that satisfies all equations simultaneously. GMM resolves this difficulty by transforming a root-finding problem into an **optimization problem**, minimizing a metric of “distance” between the sample moments and the null vector.

5.1.1.2 Geometric Interpretation

To develop intuition, consider the following geometric scenario:

- In $\mathbb{R}^2$, we seek a point θ that should lie at the intersection of three lines (representing three moment conditions).
- In a finite sample, these lines will rarely intersect at a single point due to stochastic noise, forming a small triangle instead of a common vertex.
- Geometrically, the GMM estimator locates the point $\hat{\theta}$ that minimizes the weighted sum of squared distances to these lines.

The **weighting matrix $\mathbf{W}$** acts as a metric that deforms the space, assigning greater

weight to moment conditions that have smaller variance (are more “reliable”). In $\mathbb{R}^3$, GMM can be visualized as the search for the point in a subspace that minimizes the orthogonal distance to a series of hyperplanes defined by the moment conditions.

5.1.2 Fundamental Assumptions of GMM

For the formal construction of the estimator, we define:

- $\theta \in \Theta \subseteq \mathbb{R}^K$: vector of parameters
- $\mathbf{w}_i$: vector of observed data
- $\mathbf{g}(\mathbf{w}_i, \theta)$: moment function of dimension $L \times 1$

Assumption 5.1: Population Orthogonality (Moment Condition)

- **Role:** Constitutes the theoretical foundation of the model. It states that, at the true parameter value θ_0 , the mathematical expectation of the moment function is zero:

$$E[\mathbf{g}(\mathbf{w}_i, \theta_0)] = \mathbf{0} \quad (5.1)$$

- **Counterexample:** In a linear regression model $\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$ where we omit a variable $\mathbf{Z}$ correlated with $\mathbf{X}$, the error $\mathbf{u}$ will have $E[\mathbf{u}|\mathbf{X}] \neq \mathbf{0}$. In this case, the moment condition $E[\mathbf{X}'\mathbf{u}] = \mathbf{0}$ fails and any estimator based on it will be inconsistent.

Assumption 5.2: Identification (Rank Condition)

- **Role:** Ensures that the moment condition provides sufficient information to distinguish θ_0 from any other value in the parameter space. Mathematically, it requires that the matrix of expected derivatives has full rank K :

$$\mathbf{G} = E \left[\frac{\partial \mathbf{g}(\mathbf{w}_i, \theta_0)}{\partial \theta'} \right], \quad \text{rank}(\mathbf{G}) = K \quad (5.2)$$

- **Counterexample:** If we try to estimate the return to education using an instrument that has no correlation with schooling, the rank of the covariance matrix between the instrument and the regressor will be zero. The parameter cannot be located because the objective function will be “flat” with respect to it.

Assumption 5.3: Positive Definite Weighting Matrix

- **Role:** Ensures that the quadratic objective function is strictly convex (in the linear case) and has a well-defined global minimum.
- **Counterexample:** If we use a zero or singular matrix $\mathbf{W}$, the objective function would not penalize deviations in certain directions of the moment conditions, allowing infinitely many solutions or preventing numerical convergence of the optimizer.

5.1.3 Formal Formulation of the Estimator

5.1.3.1 General Definition

Let $\{\mathbf{w}_i\}_{i=1}^n$ be a sequence of independent and identically distributed (i.i.d.) random vectors belonging to a probability space $(\Omega, \mathcal{F}, P)$. Let $\theta_0 \in \Theta$ be a vector of parameters in a compact parameter space $\Theta \subset \mathbb{R}^K$. We define a measurable moment function $\mathbf{g} : \mathbb{R}^M \times \Theta \rightarrow \mathbb{R}^L$ such that $L \geq K$.

The GMM estimator $\hat{\theta}_n$ is defined as the argument that minimizes the quadratic objective function:

$$Q_n(\theta) = \bar{\mathbf{g}}_n(\theta)' \mathbf{W}_n \bar{\mathbf{g}}_n(\theta) \quad (5.3)$$

where:

$$\bar{\mathbf{g}}_n(\theta) = \frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{w}_i, \theta) \quad (5.4)$$

and $\mathbf{W}_n$ is a stochastic weighting matrix such that $\mathbf{W}_n \xrightarrow{p} \mathbf{W}$, with $\mathbf{W}$ being symmetric and positive definite.

5.1.3.2 Derivation of the Estimator for the Linear Case

Consider the linear case where:

$$\mathbf{g}(\mathbf{w}_i, \theta) = \mathbf{z}_i(y_i - \mathbf{x}_i'\theta) \quad (5.5)$$

with $\mathbf{z}_i$ an $L \times 1$ vector of instruments and $\mathbf{x}_i$ a $K \times 1$ vector of regressors.

Step 1: Definition of the Sample Moment

In matrix notation, defining $\mathbf{y}$ as the $n \times 1$ vector, $\mathbf{X}$ as the $n \times K$ matrix, and $\mathbf{Z}$ as the $n \times L$ instrument matrix:

$$\bar{\mathbf{g}}_n(\theta) = \frac{1}{n} \mathbf{Z}'(\mathbf{y} - \mathbf{X}\theta) \quad (5.6)$$

Step 2: Substitution into the Objective Function

Substituting (5.6) into (5.3):

$$Q_n(\theta) = \left[\frac{1}{n} \mathbf{Z}'(\mathbf{y} - \mathbf{X}\theta) \right]' \mathbf{W}_n \left[\frac{1}{n} \mathbf{Z}'(\mathbf{y} - \mathbf{X}\theta) \right] \quad (5.7)$$

Step 3: First Order Condition (FOC)

Differentiating $Q_n(\theta)$ with respect to θ and setting it to the null vector:

$$\frac{\partial Q_n(\theta)}{\partial \theta} = -\frac{2}{n^2} \mathbf{X}' \mathbf{Z} \mathbf{W}_n \mathbf{Z}' \mathbf{y} + \frac{2}{n^2} \mathbf{X}' \mathbf{Z} \mathbf{W}_n \mathbf{Z}' \mathbf{X} \theta = \mathbf{0} \quad (5.8)$$

Step 4: Isolating the Estimator

$$\mathbf{X}' \mathbf{Z} \mathbf{W}_n \mathbf{Z}' \mathbf{X} \hat{\theta}_n = \mathbf{X}' \mathbf{Z} \mathbf{W}_n \mathbf{Z}' \mathbf{y} \quad (5.9)$$

Assuming that $(\mathbf{X}' \mathbf{Z} \mathbf{W}_n \mathbf{Z}' \mathbf{X})$ is non-singular (Assumption 5.2):

$$\boxed{\hat{\theta}_n = (\mathbf{X}' \mathbf{Z} \mathbf{W}_n \mathbf{Z}' \mathbf{X})^{-1} \mathbf{X}' \mathbf{Z} \mathbf{W}_n \mathbf{Z}' \mathbf{y}} \quad (5.10)$$

5.1.3.3 Role of Assumptions in the Derivation

The above derivation highlights the critical role of each assumption:

1. **Identification (Assumption 5.2):** Was indispensable in Step 4. If $(\mathbf{X}' \mathbf{Z} \mathbf{W}_n \mathbf{Z}' \mathbf{X})$ is singular ($\text{rank} < K$), the inverse does not exist and the estimator cannot be uniquely determined.
2. **Orthogonality (Assumption 5.1):** Ensures that, when substituting $\mathbf{y} = \mathbf{X}\theta_0 + \mathbf{u}$ into (5.10), the term associated with the error will collapse asymptotically, validating the consistency of the estimator:

$$\text{plim } \hat{\theta}_n = \theta_0 \quad (5.11)$$

3. **Positive Definite Matrix (Assumption 5.3):** Ensures that the FOC (5.8) characterizes a global minimum (not just a saddle point or maximum), guaranteeing the convexity of the objective function.
-

5.1.4 Moments of the GMM Estimator

5.1.4.1 Motivation

While the asymptotic properties of the GMM estimator (consistency and asymptotic normality) describe its limiting behavior as $n \rightarrow \infty$, the **conditional expectation and variance** allow us to analyze the statistical properties of the estimator in **finite samples**.

The central problem is **finite-sample bias**. In Instrumental Variables (IV) and overidentified GMM models, although the estimator is consistent, it often exhibits bias in small samples if the instruments are “weak” or if there is strong correlation between the endogenous regressors and the structural error.

Geometric Interpretation in $\mathbb{R}^K$:

Geometrically, the conditional expectation locates the “center of mass” of the distribution of the estimator $\hat{\beta}$ in the parameter space $\mathbb{R}^K$, given a fixed set of instruments $\mathbf{Z}$. If the estimator is unbiased, this center coincides with β_0 . The conditional variance defines the

concentration ellipsoid around this expectation; the volume and orientation of this ellipsoid are determined by the curvature of the objective function and the dispersion of the errors, weighted by the metric structured in $\mathbf{W}$.

5.1.4.2 Additional Assumptions

Assumption 5.4: Conditional Strict Exogeneity

- **Role:** Ensures that the error vector $\mathbf{u}$ has zero conditional mean for any realization of the instruments and regressors, allowing the expectation operator to pass through the bilinear form of the estimator. Mathematically: $E[\mathbf{u}|\mathbf{Z}, \mathbf{X}] = \mathbf{0}$.
- **Counterexample:** In models with predetermined regressors (such as lags of the dependent variable y_{t-1}), $E[u_t|y_{t-1}] = 0$ may hold contemporaneously, but $E[u_{t-1}|y_{t-1}] \neq 0$. This implies that $E[\hat{\beta}|\mathbf{X}] \neq \beta_0$, breaking the unbiasedness property.

Assumption 5.5: Conditional Homoskedasticity (Sphericity)

- **Role:** Simplifies the structure of the conditional variance-covariance matrix of the estimator, assuming that $\text{Var}(\mathbf{u}|\mathbf{Z}, \mathbf{X}) = \sigma^2 \mathbf{I}_n$. Without this, the conditional variance assumes the expanded “sandwich” form.
- **Counterexample:** If $\text{Var}(u_i|\mathbf{z}_i) = \sigma^2 z_i^2$, the precision of the estimator will vary with the magnitude of $\mathbf{Z}$. Ignoring this heteroskedasticity leads to incorrect standard errors and invalid inference in finite samples.

Assumption 5.6: Non-Stochasticity of $\mathbf{W}$ (or Conditioning)

- **Role:** For the exact derivation of the expectation in finite samples, the weighting matrix $\mathbf{W}$ must be treated as fixed or dependent only on $\mathbf{Z}$.
 - **Counterexample:** If $\mathbf{W}$ is estimated iteratively in two steps ($\mathbf{W} = \hat{\Omega}^{-1}$), it becomes a function of $\hat{\mathbf{u}}$, which depends on $\mathbf{y}$. This creates a complex stochastic dependence between $\mathbf{W}$ and $\mathbf{u}$, making linear factorization of the expectation impossible.
-

5.1.4.3 Formal Derivation

Statement: Consider the linear model $\mathbf{y} = \mathbf{X}\beta_0 + \mathbf{u}$. Let $\hat{\beta}_{GMM}$ be the estimator defined in (5.10). Under the assumptions of strict exogeneity (Assumption 5.4), rank identification ($\text{rank}(\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{X}) = K$), and non-stochasticity of $\mathbf{W}$ conditioned on $\mathbf{Z}$ (Assumption 5.6), the estimator takes the form:

$$\hat{\beta} = (\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{y} \quad (5.12)$$

Theorem 5.1.1 (Conditional Expectation of GMM). *Under Assumption 5.4 (strict exogeneity):*

$$\boxed{E[\hat{\beta}|\mathbf{Z}, \mathbf{X}] = \beta_0} \quad (5.13)$$

Proof. Substituting $\mathbf{y} = \mathbf{X}\beta_0 + \mathbf{u}$ into (5.12):

$$\hat{\beta} = \beta_0 + (\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{u} \quad (5.14)$$

Applying $E[\cdot|\mathbf{Z}, \mathbf{X}]$ and using linearity:

$$E[\hat{\beta}|\mathbf{Z}, \mathbf{X}] = \beta_0 + (\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'E[\mathbf{u}|\mathbf{Z}, \mathbf{X}] \quad (5.15)$$

By Assumption 5.4, $E[\mathbf{u}|\mathbf{Z}, \mathbf{X}] = \mathbf{0}$:

$$E[\hat{\beta}|\mathbf{Z}, \mathbf{X}] = \beta_0 \quad (5.16)$$

□

Theorem 5.1.2 (Conditional Variance of GMM). *Under Assumption 5.5 (homoskedasticity), the conditional variance of the GMM estimator is:*

$$\boxed{\text{Var}(\hat{\beta}|\mathbf{Z}, \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}(\mathbf{Z}'\mathbf{Z})\mathbf{W}\mathbf{Z}'\mathbf{X}(\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{X})^{-1}} \quad (5.17)$$

Proof. By definition:

$$\text{Var}(\hat{\beta}|\mathbf{Z}, \mathbf{X}) = E[(\hat{\beta} - \beta_0)(\hat{\beta} - \beta_0)'|\mathbf{Z}, \mathbf{X}] \quad (5.18)$$

Defining $\mathbf{M} = (\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'$:

$$\text{Var}(\hat{\beta}|\mathbf{Z}, \mathbf{X}) = \mathbf{M}E[\mathbf{u}\mathbf{u}'|\mathbf{Z}, \mathbf{X}]\mathbf{M}' \quad (5.19)$$

Under homoskedasticity (Assumption 5.5): $E[\mathbf{u}\mathbf{u}'|\mathbf{Z}, \mathbf{X}] = \sigma^2\mathbf{I}_n$.

Substituting $\mathbf{M}$ and simplifying:

$$\text{Var}(\hat{\beta}|\mathbf{Z}, \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}(\mathbf{Z}'\mathbf{Z})\mathbf{W}\mathbf{Z}'\mathbf{X}(\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{X})^{-1} \quad (5.20)$$

□

Special Case: 2SLS Estimator

If we choose $\mathbf{W} = (\mathbf{Z}'\mathbf{Z})^{-1}$, the conditional variance simplifies to:

$$\boxed{\text{Var}(\hat{\beta}_{2SLS}|\mathbf{Z}, \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}} \quad (5.21)$$

where $\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'$ is the orthogonal projection matrix onto $\text{Col}(\mathbf{Z})$.

5.1.4.4 Role of Assumptions in the Derivation

1. **Strict Exogeneity (Assumption 5.4):** Was indispensable in Step 2 of the expectation proof. If the error is conditionally correlated with the instruments or regressors, $E[\mathbf{u}|\mathbf{Z}, \mathbf{X}] \neq \mathbf{0}$, generating bias in finite samples.
2. **Non-Stochasticity of $\mathbf{W}$ (Assumption 5.6):** Allowed $\mathbf{W}$ to be treated deterministically in the expectation operator. In the classical two-step GMM, this premise is violated, introducing additional bias in small samples.
3. **Homoskedasticity (Assumption 5.5):** Was used to simplify the conditional variance. Under heteroskedasticity, the general expression (5.17) requires the substitution of $\sigma^2 \mathbf{I}_n$ by $\mathbf{\Omega}$:

$$\text{Var}(\hat{\beta}|\mathbf{Z}, \mathbf{X}) = (\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{\Omega}\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{X}(\mathbf{X}'\mathbf{Z}\mathbf{W}\mathbf{Z}'\mathbf{X})^{-1} \quad (5.22)$$

5.1.5 Relative Efficiency of the GMM Estimator

5.1.5.1 Motivation

The question of **relative efficiency** in the Generalized Method of Moments (GMM) lies in the choice of the ideal “metric” to optimally combine the information contained in different moment conditions. When the model is overidentified ($L > K$), there are infinitely many ways to weight the sample moments to obtain a consistent estimator. However, not all choices of weights are equivalent in terms of asymptotic variance.

The central problem is to determine whether an estimator structured with a weight matrix $\mathbf{W}_1$ is statistically superior to another structured with $\mathbf{W}_2$. In multivariate statistics, this is formalized by the **extended Gauss-Markov efficiency**: an estimator is globally more efficient than another if the difference between their respective variance-covariance matrices is a **Positive Semidefinite (PSD)** matrix.

Geometric Interpretation: Projection in Hilbert Space

Geometrically, the search for the optimal GMM estimator can be visualized as an **orthogonal projection** in a Hilbert space deformed by the intrinsic variance of the moments. If we think of the moments as vectors in $\mathbb{R}^L$, the GMM estimator projects the data onto the subspace spanned by the derivatives of the moments (the Jacobian matrix $\mathbf{G}$). The choice of the weight matrix $\mathbf{W}$ defines the angle and metric of this projection. The “Optimal” estimator is the one that adopts the inverse of the moment covariance matrix ($\mathbf{\Lambda}^{-1}$) as the metric, performing an orthogonal projection in the statistical sense, minimizing the volume of the error ellipsoid. Any other choice of $\mathbf{W}$ results in an “oblique” projection, presenting necessarily larger or equal variance.

5.1.5.2 Assumptions for Efficiency

Assumption 5.7: Comparability of Variances (PSD Ordering)

- **Role:** Allows establishing an unambiguous partial order between estimators. We define that $\text{Var}(\hat{\theta}_2) \succeq \text{Var}(\hat{\theta}_1)$ if, for any non-zero constant vector $\mathbf{c} \in \mathbb{R}^K$, the variance of the linear combination $\mathbf{c}'\hat{\theta}_2$ is greater than or equal to that of $\mathbf{c}'\hat{\theta}_1$.
- **Counterexample:** Without ordering based on the PSD property, we could have an estimator with smaller variance for the first parameter but larger for the second, preventing a global declaration of superiority between estimation methods.

Assumption 5.8: Invertibility of the Moment Covariance Matrix (Λ)

- **Role:** Strictly necessary to define the optimal weight matrix $\mathbf{W}^* = \Lambda^{-1}$. Ensures the absence of perfectly redundant moment conditions that would generate singularity in the matrix.
- **Counterexample:** If two moment conditions were perfectly correlated in the population, Λ would be singular and the weight assigned to that specific direction would be mathematically undefined, collapsing the determination of the efficient estimator.

5.1.5.3 Formal Statement and Derivation

Let $\hat{\theta}_A$ be the consistent GMM estimator that uses a symmetric positive definite weight matrix $\mathbf{A}$, and $\hat{\theta}_{opt}$ be the estimator that adopts the optimal weight matrix $\mathbf{W}^* = \Lambda^{-1}$. Their respective asymptotic variances are given by:

1. **Variance for $\mathbf{W} = \mathbf{A}$ (“Sandwich” Form):**

$$\mathbf{V}(\mathbf{A}) = (\mathbf{G}'\mathbf{A}\mathbf{G})^{-1}\mathbf{G}'\mathbf{A}\Lambda\mathbf{A}\mathbf{G}(\mathbf{G}'\mathbf{A}\mathbf{G})^{-1} \quad (5.23)$$

2. **Variance for $\mathbf{W} = \Lambda^{-1}$ (Optimal Form):**

$$\mathbf{V}(\Lambda^{-1}) = (\mathbf{G}'\Lambda^{-1}\mathbf{G})^{-1} \quad (5.24)$$

Theorem 5.1.3 (Optimality of GMM - Hansen, 1982). *Under regularity conditions, the difference matrix $\mathbf{D} = \mathbf{V}(\mathbf{A}) - \mathbf{V}(\Lambda^{-1})$ is Positive Semidefinite ($\mathbf{D} \succeq \mathbf{0}$) for any choice of $\mathbf{A}$.*

$$\boxed{\mathbf{V}(\mathbf{A}) \succeq \mathbf{V}(\Lambda^{-1})} \quad (5.25)$$

Proof. We will prove that $\mathbf{V}(\Lambda^{-1})^{-1} \succeq \mathbf{V}(\mathbf{A})^{-1}$, which is equivalent for positive definite matrices.

Step 1: Cholesky Decomposition and Change of Basis

Let $\Lambda = \Lambda^{1/2}\Lambda^{1/2}$ be the Cholesky decomposition. We define:

$$\mathbf{H} = \Lambda^{-1/2}\mathbf{G}, \quad \mathbf{K} = \Lambda^{1/2}\mathbf{A}\mathbf{G} \quad (5.26)$$

Step 2: Rewriting the Inverse Variances

The inverse of the optimal variance:

$$\mathbf{V}(\mathbf{\Lambda}^{-1})^{-1} = \mathbf{G}'\mathbf{\Lambda}^{-1}\mathbf{G} = \mathbf{H}'\mathbf{H} \quad (5.27)$$

The inverse of the “sandwich” variance:

$$\mathbf{V}(\mathbf{A})^{-1} = (\mathbf{G}'\mathbf{A}\mathbf{G})(\mathbf{G}'\mathbf{A}\mathbf{\Lambda}\mathbf{A}\mathbf{G})^{-1}(\mathbf{G}'\mathbf{A}\mathbf{G}) \quad (5.28)$$

Expanding with the auxiliary definitions:

$$\mathbf{G}'\mathbf{A}\mathbf{G} = \mathbf{H}'\mathbf{K}, \quad \mathbf{G}'\mathbf{A}\mathbf{\Lambda}\mathbf{A}\mathbf{G} = \mathbf{K}'\mathbf{K} \quad (5.29)$$

Substituting into (5.28):

$$\mathbf{V}(\mathbf{A})^{-1} = \mathbf{H}'\mathbf{K}(\mathbf{K}'\mathbf{K})^{-1}\mathbf{K}'\mathbf{H} = \mathbf{H}'\mathbf{P}_K\mathbf{H} \quad (5.30)$$

where $\mathbf{P}_K = \mathbf{K}(\mathbf{K}'\mathbf{K})^{-1}\mathbf{K}'$ is the orthogonal projection matrix onto $\text{Col}(\mathbf{K})$.

Step 3: Analysis of the Matrix Difference

$$\mathbf{\Delta} = \mathbf{V}(\mathbf{\Lambda}^{-1})^{-1} - \mathbf{V}(\mathbf{A})^{-1} = \mathbf{H}'\mathbf{H} - \mathbf{H}'\mathbf{P}_K\mathbf{H} = \mathbf{H}'(\mathbf{I} - \mathbf{P}_K)\mathbf{H} \quad (5.31)$$

Step 4: Property of Positive Semidefiniteness

Let $\mathbf{M}_K = \mathbf{I} - \mathbf{P}_K$ be the annihilator matrix. By the property of orthogonal projections, $\mathbf{M}_K$ is symmetric and idempotent ($\mathbf{M}_K^2 = \mathbf{M}_K$), hence $\mathbf{M}_K \succeq \mathbf{0}$.

For any vector $\mathbf{z} \in \mathbb{R}^K$:

$$\mathbf{z}'\mathbf{\Delta}\mathbf{z} = \mathbf{z}'\mathbf{H}'\mathbf{M}_K\mathbf{H}\mathbf{z} = (\mathbf{H}\mathbf{z})'\mathbf{M}_K(\mathbf{H}\mathbf{z}) \geq 0 \quad (5.32)$$

Hence, $\mathbf{\Delta} \succeq \mathbf{0}$, which implies $\mathbf{V}(\mathbf{A}) \succeq \mathbf{V}(\mathbf{\Lambda}^{-1})$. □

5.1.5.4 Role of Assumptions in the Derivation

The assumption of **Invertibility of $\mathbf{\Lambda}$ (Assumption 5.8)** was indispensable for performing the structural decomposition described in Step 1, enabling the construction of the complementary projection matrix $\mathbf{M}_K$. Without the validity of this premise, the geometric equivalence between statistical optimality and projection orthogonality would be broken, and the optimal estimator could not be uniquely defined.

5.1.6 Asymptotic Properties of the GMM Estimator

5.1.6.1 Motivation

The **asymptotic properties** of the GMM estimator describe its limiting behavior as $n \rightarrow \infty$. While finite-sample analysis deals with the exact distribution — often unknown or analytically intractable for non-linear models — asymptotic theory provides standardized and tractable approximations for conducting valid statistical inference.

The central problem solved here is guaranteeing **consistency** and deriving a **stable limiting distribution**. Consistency ensures that, as the sample size grows, the estimator will converge in probability to θ_0 . Asymptotic normality guarantees that the distribution of the estimation error scaled by $\sqrt{n}$ approaches a multivariate normal distribution, underpinning conventional hypothesis tests.

5.1.6.2 Assumptions for Asymptotic Properties

Assumption 5.9: Global Identification (Uniqueness)

- **Role:** Ensures that θ_0 is the only point in the parameter space capable of zeroing the population moment conditions: $E[\mathbf{f}(\mathbf{w}_i, \theta)] = \mathbf{0} \iff \theta = \theta_0$.
- **Counterexample:** If there are multiple candidate values that simultaneously satisfy the population moment conditions, the objective function will exhibit multiple global minima and the estimator may oscillate indefinitely between them, breaking consistency.

Assumption 5.10: Stationarity and Ergodicity

- **Role:** Allows the application of the Law of Large Numbers, ensuring that the sample means of the moment conditions and their derivatives converge in probability to their population expectations.
- **Counterexample:** If the data follow non-stationary processes (such as random walks with unit roots), the sample means will not converge to finite constants, invalidating the premise that sample moments mimic the population structure.

Assumption 5.11: Compactness of the Parameter Space Θ

- **Role:** Indispensable topological condition to ensure **uniform convergence** of the sample objective function to the population one. Prevents the numerical optimizer from diverging to infinity.
- **Counterexample:** In an objective function that is openly decreasing toward the infinite boundaries of the parameter space, the optimization algorithm would fail to establish a stopping point, invalidating the formal proof of consistency.

Assumption 5.12: Existence of Second-Order Moments

- **Role:** Requires that $\mathbf{S} = E[\mathbf{f}(\mathbf{w}_i, \theta_0)\mathbf{f}(\mathbf{w}_i, \theta_0)']$ be finite and positive definite, enabling the application of the Central Limit Theorem.
- **Counterexample:** If the errors follow heavy-tailed distributions with infinite variance (such as the Cauchy distribution), the $\sqrt{n}$ normalization would fail and the estimator would exhibit a non-Gaussian limiting distribution.

5.1.6.3 Formal Statement and Derivation

Let $\hat{\theta}_n = \operatorname{argmin}_{\theta \in \Theta} Q_n(\theta)$, where $Q_n(\theta) = \mathbf{g}_n(\theta)' \mathbf{W}_n \mathbf{g}_n(\theta)$. We define $\mathbf{G}_n(\theta) = \partial \mathbf{g}_n(\theta) / \partial \theta'$ ($L \times K$) and assume that its population limit $\mathbf{G}_0 = E[\partial \mathbf{f}(\mathbf{w}_i, \theta_0) / \partial \theta']$ has full rank K . Under regularity conditions, $\mathbf{W}_n \xrightarrow{p} \mathbf{W}$ (positive definite) and $\sqrt{n} \mathbf{g}_n(\theta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{S})$.

Theorem 5.1.4 (Asymptotic Normality of GMM). *The scaled estimation error of the GMM estimator satisfies:*

$$\boxed{\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N\left(\mathbf{0}, (\mathbf{G}_0' \mathbf{W} \mathbf{G}_0)^{-1} \mathbf{G}_0' \mathbf{W} \mathbf{S} \mathbf{W} \mathbf{G}_0 (\mathbf{G}_0' \mathbf{W} \mathbf{G}_0)^{-1}\right)} \quad (5.33)$$

Proof. **Step 1: First Order Condition (FOC)**

Assuming that $\hat{\theta}_n$ lies in the interior of the parameter space:

$$\frac{\partial Q_n(\hat{\theta}_n)}{\partial \theta} = 2 \mathbf{G}_n(\hat{\theta}_n)' \mathbf{W}_n \mathbf{g}_n(\hat{\theta}_n) = \mathbf{0} \quad (5.34)$$

Step 2: Expansion via the Mean Value Theorem

We expand $\mathbf{g}_n(\hat{\theta}_n)$ around θ_0 :

$$\mathbf{g}_n(\hat{\theta}_n) = \mathbf{g}_n(\theta_0) + \mathbf{G}_n(\bar{\theta})(\hat{\theta}_n - \theta_0) \quad (5.35)$$

where $\bar{\theta}$ lies on the segment between $\hat{\theta}_n$ and θ_0 .

Step 3: Substitution and Scaling

Substituting (5.35) into (5.34) and multiplying by $\sqrt{n}$:

$$\mathbf{G}_n(\hat{\theta}_n)' \mathbf{W}_n \sqrt{n} \mathbf{g}_n(\theta_0) + \mathbf{G}_n(\hat{\theta}_n)' \mathbf{W}_n \mathbf{G}_n(\bar{\theta}) \sqrt{n}(\hat{\theta}_n - \theta_0) = \mathbf{0} \quad (5.36)$$

Step 4: Isolating the Scaled Error

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = - \left[\mathbf{G}_n(\hat{\theta}_n)' \mathbf{W}_n \mathbf{G}_n(\bar{\theta}) \right]^{-1} \mathbf{G}_n(\hat{\theta}_n)' \mathbf{W}_n \sqrt{n} \mathbf{g}_n(\theta_0) \quad (5.37)$$

Step 5: Application of Stochastic Limits

- $\hat{\theta}_n \xrightarrow{p} \theta_0$ and $\bar{\theta} \xrightarrow{p} \theta_0$, hence $\mathbf{G}_n(\hat{\theta}_n) \xrightarrow{p} \mathbf{G}_0$ and $\mathbf{G}_n(\bar{\theta}) \xrightarrow{p} \mathbf{G}_0$.
- $\mathbf{W}_n \xrightarrow{p} \mathbf{W}$.
- $\sqrt{n} \mathbf{g}_n(\theta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{S})$.

By Slutsky's Theorem:

$$\left[\mathbf{G}_n(\hat{\theta}_n)' \mathbf{W}_n \mathbf{G}_n(\bar{\theta}) \right]^{-1} \mathbf{G}_n(\hat{\theta}_n)' \mathbf{W}_n \xrightarrow{p} (\mathbf{G}_0' \mathbf{W} \mathbf{G}_0)^{-1} \mathbf{G}_0' \mathbf{W} \quad (5.38)$$

Step 6: Consolidation of the Limiting Variance

Combining (5.38) with the CLT via continuous mapping:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N\left(\mathbf{0}, (\mathbf{G}'_0 \mathbf{W} \mathbf{G}_0)^{-1} \mathbf{G}'_0 \mathbf{W} \mathbf{S} \mathbf{W} \mathbf{G}_0 (\mathbf{G}'_0 \mathbf{W} \mathbf{G}_0)^{-1}\right) \quad (5.39)$$

□

5.1.6.4 Special Case: Optimal Weighting ($\mathbf{W} = \mathbf{S}^{-1}$)

Under the optimal choice of the weight matrix, the asymptotic variance simplifies dramatically:

$$\mathbf{V}_{opt} = (\mathbf{G}'_0 \mathbf{S}^{-1} \mathbf{G}_0)^{-1} \quad (5.40)$$

and the limiting distribution becomes:

$$\boxed{\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N\left(\mathbf{0}, (\mathbf{G}'_0 \mathbf{S}^{-1} \mathbf{G}_0)^{-1}\right)} \quad (5.41)$$

5.1.6.5 Role of Assumptions in the Derivation

1. **Stationarity and Ergodicity (Assumption 5.10):** Was indispensable in Step 5 to ensure that the sample moment matrices collapsed to stable and finite limits.
2. **Full Rank of $\mathbf{G}_0$:** Ensured the existence of the inversion operation performed in Step 4, allowing the isolation of the scaled error.
3. **Global Identification (Assumption 5.9):** Guaranteed that the population objective function has a unique minimum at θ_0 , ensuring the consistency of the estimator.

5.1.7 Hansen Test (Overidentification Test)

5.1.7.1 Motivation

The fundamental specification test in the GMM universe is the **Overidentifying Restrictions Test**, routinely called the **Hansen J Test**. The problem this test solves is the statistical verification of the orthogonality assumptions of the model when $L > K$.

In just-identified systems ($L = K$), the classical estimator has sufficient degrees of freedom to strictly zero out the sample equations. However, under overidentification, GMM minimizes a weighted distance metric, making the sample moment vector as close as possible to the null vector, but rarely zero in its entirety. The J Test formally evaluates whether this observed “residual distance” is statistically negligible (purely due to sample noise) or sufficiently large to indicate that the population moment conditions are violated.

Geometric Interpretation in $\mathbb{R}^L$:

Visualizing the sample moment conditions as a vector positioned in a space of dimension L , the GMM estimator projects the origin onto the subspace parameterized by the model. The

J statistic quantifies the squared Euclidean distance (corrected by the optimal covariance matrix metric) between the optimized point and the absolute origin. Excessive distances indicate that the generated subspace does not encompass the population target, leading to the rejection of the specification.

5.1.7.2 Assumptions for the J Test

Assumption 5.13: Orthogonality under the Null Hypothesis (H_0)

- **Role:** Sets the scenario of validity of the structural model. Assumes that there exists $\theta_0 \in \Theta$ such that $E[\mathbf{g}(\mathbf{w}_i, \theta_0)] = \mathbf{0}$.
- **Counterexample:** If any instrument contains direct correlation with the structural error, the respective moment condition will fail to converge to zero. The J statistic will diverge with n , forcing the rejection of the model.

Assumption 5.14: Optimal Weighting

- **Role:** Mathematical requirement for the test statistic to collapse into a standard chi-square distribution. Requires that $\mathbf{W}_n \xrightarrow{p} \mathbf{S}^{-1}$.
- **Counterexample:** If an arbitrary weight matrix (such as $\mathbf{I}_L$) is adopted under heteroskedasticity, the moments will receive distorted weights, causing the statistic to follow a linear combination of chi-square variables with unknown weights.

Assumption 5.15: Overidentification ($L > K$)

- **Role:** Ensures the existence of positive degrees of freedom for the test, defined by $df = L - K$.
- **Counterexample:** If $L = K$, the minimized objective function will reach zero by algebraic construction ($J = 0$), nullifying the power of the test.

5.1.7.3 Formal Statement and Derivation

Let $\hat{\theta}_n$ be the efficient GMM estimator with $\mathbf{W}_n \xrightarrow{p} \mathbf{S}^{-1}$, where $\mathbf{S} = E[\mathbf{g}(\mathbf{w}_i, \theta_0)\mathbf{g}(\mathbf{w}_i, \theta_0)']$. The Hansen J Test statistic is:

$$J_n = n \cdot \mathbf{g}_n(\hat{\theta}_n)' \hat{\mathbf{S}}_n^{-1} \mathbf{g}_n(\hat{\theta}_n) \quad (5.42)$$

Theorem 5.1.5 (Hansen J Test - Hansen, 1982). *Under the null hypothesis of correct specification of the moment conditions:*

$$\boxed{J_n \xrightarrow{d} \chi_{L-K}^2} \quad (5.43)$$

Proof. **Step 1: Linear Expansion of the Optimized Moment Vector**

$$\mathbf{g}_n(\hat{\theta}_n) \approx \mathbf{g}_n(\theta_0) + \mathbf{G}_n(\theta_0)(\hat{\theta}_n - \theta_0) \quad (5.44)$$

Step 2: Substitution of the Asymptotic Estimation Error

Under optimal weighting, the estimation error is:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -(\mathbf{G}_0' \mathbf{S}^{-1} \mathbf{G}_0)^{-1} \mathbf{G}_0' \mathbf{S}^{-1} \sqrt{n} \mathbf{g}_n(\theta_0) + o_p(1) \quad (5.45)$$

Step 3: Consolidation in Projection Form

Multiplying (5.44) by $\sqrt{n}$ and substituting (5.45):

$$\sqrt{n} \mathbf{g}_n(\hat{\theta}_n) \approx [\mathbf{I}_L - \mathbf{G}_0(\mathbf{G}_0' \mathbf{S}^{-1} \mathbf{G}_0)^{-1} \mathbf{G}_0' \mathbf{S}^{-1}] \sqrt{n} \mathbf{g}_n(\theta_0) \quad (5.46)$$

Step 4: Standardization and Change of Basis

Premultiplying by $\mathbf{S}^{-1/2}$ via Cholesky decomposition:

$$\mathbf{S}^{-1/2} \sqrt{n} \mathbf{g}_n(\hat{\theta}_n) \approx [\mathbf{I}_L - \mathbf{H}(\mathbf{H}' \mathbf{H})^{-1} \mathbf{H}'] \mathbf{S}^{-1/2} \sqrt{n} \mathbf{g}_n(\theta_0) = \mathbf{P}_M \mathbf{Z}_n \quad (5.47)$$

where $\mathbf{H} = \mathbf{S}^{-1/2} \mathbf{G}_0$, $\mathbf{P}_M = \mathbf{I}_L - \mathbf{H}(\mathbf{H}' \mathbf{H})^{-1} \mathbf{H}'$ is the projection matrix onto the null subspace of $\mathbf{H}'$, and $\mathbf{Z}_n = \mathbf{S}^{-1/2} \sqrt{n} \mathbf{g}_n(\theta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{I}_L)$.

Step 5: Identification of the Quadratic Form

Rewriting J_n :

$$J_n = (\mathbf{S}^{-1/2} \sqrt{n} \mathbf{g}_n(\hat{\theta}_n))' (\mathbf{S}^{-1/2} \sqrt{n} \mathbf{g}_n(\hat{\theta}_n)) \approx \mathbf{Z}_n' \mathbf{P}_M' \mathbf{P}_M \mathbf{Z}_n = \mathbf{Z}_n' \mathbf{P}_M \mathbf{Z}_n \quad (5.48)$$

since $\mathbf{P}_M$ is symmetric and idempotent ($\mathbf{P}_M^2 = \mathbf{P}_M$).

Step 6: Derivation of Degrees of Freedom

The trace of $\mathbf{P}_M$ is:

$$\text{tr}(\mathbf{P}_M) = \text{tr}(\mathbf{I}_L) - \text{tr}(\mathbf{H}(\mathbf{H}' \mathbf{H})^{-1} \mathbf{H}') = L - \text{tr}((\mathbf{H}' \mathbf{H})^{-1} \mathbf{H}' \mathbf{H}) = L - K \quad (5.49)$$

Therefore, by the theorem of quadratic forms of spherical normal vectors:

$$J_n \xrightarrow{d} \chi_{L-K}^2 \quad (5.50)$$

□

5.1.7.4 Role of Assumptions in the Derivation

1. **Optimal Weighting (Assumption 5.14):** Was indispensable in Steps 4 and 5. If the weight matrix differed from $\mathbf{S}^{-1}$, the central matrix in the final quadratic form would lose the idempotence property, preventing simplification into a clean chi-square distribution.
2. **Overidentification (Assumption 5.15):** Ensures that the dimension of the null subspace is positive ($L - K > 0$), providing degrees of freedom for the test.

5.1.8 Parametric Hypothesis Tests

5.1.8.1 Motivation

Parametric hypothesis testing solves the fundamental problem of validating economic theories under statistical uncertainty. Suppose a theory stipulates a specific restriction on the parameters (for example, constant returns to scale imposing that the sum of certain coefficients equals one). The unrestricted sample estimator $\hat{\theta}_n$ will rarely fall exactly on the restriction due to sampling fluctuations. The statistical test quantifies whether the observed deviation is probabilistically tolerable or indicates rejection of the theoretical hypothesis.

Geometric Interpretation in $\mathbb{R}^K$:

Consider the topography of the GMM objective function, $Q_n(\theta)$, as a hyperbolic surface (“bowl”) in the parameter space. The unrestricted estimator $\hat{\theta}_n$ lies exactly at the lower vertex (bottom of the bowl). When we impose a set of theoretical restrictions defining a subspace or hyperplane, the restricted estimator $\tilde{\theta}_n$ will correspond to the lowest point on the surface *that intersects the restriction hyperplane*. The tests evaluate, from different perspectives, the loss of optimality (the gain in height in the “bowl”) resulting from the displacement from the unrestricted minimum to the restricted minimum.

5.1.8.2 Assumptions for Parametric Tests

Assumption 5.16: Asymptotic Normality of the Estimator

- **Role:** Ensures that, in large samples, the scaled sampling distribution of the estimation error follows a Gaussian law, allowing quadratic forms to be mapped directly to stable chi-square distributions.
- **Counterexample:** Under violation of fourth-order moments, the limiting distribution of the estimator would exhibit non-standardized variability, invalidating p-values derived from the χ^2 table.

Assumption 5.17: Smoothness and Full Rank of the Restriction

- **Role:** The set of restrictions structured as $\mathbf{r}(\theta) = \mathbf{0}$ must be continuously differentiable to allow stable linear approximations via Taylor Expansion (Delta Method).
- **Counterexample:** If the restriction contains points of discontinuity or geometric corners, the derivative matrix will fail to be defined, collapsing the calculation of the test covariance matrix.

5.1.8.3 Wald Test

Theorem 5.1.6 (Wald Test for GMM). *Under the null hypothesis $H_0 : \mathbf{r}(\theta_0) = \mathbf{0}$, the Wald statistic:*

$$W_n = n \cdot \mathbf{r}(\hat{\theta}_n)' \left[\mathbf{R}(\hat{\theta}_n) \hat{\mathbf{V}}_{\theta} \mathbf{R}(\hat{\theta}_n)' \right]^{-1} \mathbf{r}(\hat{\theta}_n) \quad (5.51)$$

where $\mathbf{R}(\theta) = \partial \mathbf{r}(\theta) / \partial \theta'$ ($q \times K$), satisfies:

$$\boxed{W_n \xrightarrow{d} \chi_q^2} \quad (5.52)$$

Proof (sketch). From the asymptotic normality result, $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{V}_{\theta})$. By the Delta Method, $\sqrt{n}\mathbf{r}(\hat{\theta}_n) \xrightarrow{d} N(\mathbf{0}, \mathbf{R}\mathbf{V}_{\theta}\mathbf{R}')$. The quadratic form with the inverse variance converges to χ_q^2 . $\square$

5.1.8.4 GMM Distance Test

An alternative robust methodology consists of the direct evaluation of the increment suffered by the minimized objective function when moving from the unrestricted to the restricted model:

$$D_n = n \left[Q_n(\tilde{\theta}_n) - Q_n(\hat{\theta}_n) \right] \quad (5.53)$$

where $\tilde{\theta}_n$ denotes the minimizer of the objective function under the imposition of the restriction $\mathbf{r}(\theta) = \mathbf{0}$.

Theorem 5.1.7 (GMM Distance Test). *Under the null hypothesis and using the **same optimal weighting matrix** for both models:*

$$\boxed{D_n \xrightarrow{d} \chi_q^2} \quad (5.54)$$

Indispensable Assumption: For the distance statistic to converge rigorously to χ_q^2 , it is **strictly mandatory** that the **same optimal sample weighting matrix** ($\hat{\mathbf{S}}_n^{-1}$) be employed in the estimation of both models (restricted and unrestricted). If different weight matrices were used, the difference between the objective functions would incorporate spurious metric variations, destroying the idempotent structure necessary for convergence to a pure χ^2 .

5.2 Special Cases of GMM

The Generalized Method of Moments (GMM) constitutes a unifying framework that encompasses various classical estimators as special cases, depending on the choice of the weighting matrix $\mathbf{W}_n$ and the definition of the moment conditions. This section explores nine special cases, organized in increasing order of generality, demonstrating how GMM subsumes them as particular cases.

5.2.1 Perfectly Identified GMM (Just-Identified)

5.2.1.1 Motivation and Geometric Interpretation

Perfectly identified GMM (or just-identified) solves the problem of estimating parameters in systems where the number of moment conditions is strictly equal to the number of parameters ($L = K$). While general GMM deals with overidentification by transforming it into a quadratic optimization problem, the perfectly identified case reduces to a classical root-finding problem for a system of simultaneous equations.

Geometric Interpretation in $\mathbb{R}^K$:

Each sample moment condition can be visualized as a surface (or hyperplane, in the linear case) of dimension $K - 1$ in $\mathbb{R}^K$. In a perfectly identified model ($L = K$), the GMM estimator locates the exact point where these K surfaces intersect. Unlike the overidentified case ($L > K$), where surfaces rarely meet at a single point due to sample noise, in the $L = K$ case there exists a $\hat{\theta}$ that zeros all equations simultaneously. Geometrically, the estimator is the **common vertex** of these sample surfaces, making the choice of the distance metric (the matrix $\mathbf{W}$) irrelevant for the final solution.

5.2.1.2 Assumptions

Assumption 5.18: Exact Identification (Just-Identification)

- **Role:** Ensures that the number of moment conditions L is strictly equal to the number of parameters K ($L = K$). This allows the system of equations to be solved without the need to weight the relative statistical importance of different moments.
- **Counterexample:** If $L > K$, the system becomes overdetermined and does not have a common root for all sample moments, requiring a weight matrix $\mathbf{W}$ to minimize a quadratic distance.

Assumption 5.19: Non-Singular Jacobian

- **Role:** Ensures that the K moment conditions are not linearly dependent. Formally, $\text{rank}(\mathbf{G}_0) = K$, where $\mathbf{G}_0 = E[\partial \mathbf{g}(\mathbf{w}_i, \theta_0) / \partial \theta']$.
 - **Counterexample:** If one moment condition is an exact linear function of another, the system will have infinitely many solutions, preventing the location of a unique value for $\hat{\theta}$.
-

5.2.1.3 Formal Statement and Derivation

Let $\{\mathbf{w}_i\}_{i=1}^n$ be a sequence of stationary and ergodic random vectors. Let $\theta_0 \in \Theta \subset \mathbb{R}^K$ be the true vector. We define $\mathbf{g} : \mathbb{R}^M \times \Theta \rightarrow \mathbb{R}^L$ with $L = K$.

The perfectly identified GMM estimator $\hat{\theta}_n$ is defined as the argument that minimizes:

$$Q_n(\theta) = \bar{\mathbf{g}}_n(\theta)' \mathbf{W}_n \bar{\mathbf{g}}_n(\theta) \quad (5.159)$$

where $\bar{\mathbf{g}}_n(\theta) = \frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{w}_i, \theta)$ and $\mathbf{W}_n$ is positive definite. Under just-identification, the estimator satisfies $\bar{\mathbf{g}}_n(\hat{\theta}_n) = \mathbf{0}$.

Theorem 5.2.1 (Independence of the Weight Matrix). *Under $L = K$ and Assumption 5.19, the estimator $\hat{\theta}_n$ is independent of $\mathbf{W}_n$ and satisfies:*

$$\boxed{\bar{\mathbf{g}}_n(\hat{\theta}_n) = \mathbf{0}} \quad (5.160)$$

Proof. By the definition of a positive definite matrix, $\mathbf{x}'\mathbf{W}\mathbf{x} = 0 \iff \mathbf{x} = \mathbf{0}$. If $L = K$ and Assumption 5.19 is satisfied, there exists $\hat{\theta}_n$ such that $\bar{\mathbf{g}}_n(\hat{\theta}_n) = \mathbf{0}$ (by the Inverse Function Theorem). Substituting into (5.159), we obtain $Q_n(\hat{\theta}_n) = 0$, which is the global minimum, regardless of $\mathbf{W}_n$. $\square$

Theorem 5.2.2 (Asymptotic Normality of Just-Identified GMM). *Under regularity conditions:*

$$\boxed{\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{G}_0^{-1} \mathbf{S} (\mathbf{G}_0')^{-1})} \quad (5.161)$$

where $\mathbf{S} = E[\mathbf{g}(\mathbf{w}_i, \theta_0)\mathbf{g}(\mathbf{w}_i, \theta_0)']$.

Proof (sketch). Expanding $\bar{\mathbf{g}}_n(\hat{\theta}_n)$ around θ_0 via the Mean Value Theorem, isolating $\sqrt{n}(\hat{\theta}_n - \theta_0)$, and applying the CLT with $\sqrt{n}\bar{\mathbf{g}}_n(\theta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{S})$ yields (5.161). $\square$

5.2.1.4 Role of Assumptions

The assumption of **Exact Identification** ($L = K$) was indispensable for ensuring that the Jacobian matrix $\mathbf{G}_0$ was square, allowing direct inversion in (5.161). If $L > K$, the matrix would not be square, preventing inversion and requiring the use of the weight matrix $\mathbf{W}$ to define the estimator through the first-order conditions of the quadratic form.

5.2.2 Optimal GMM (Efficient)

5.2.2.1 Motivation and Geometric Interpretation

Optimal GMM solves the problem of statistical inefficiency in **overidentified** systems ($L > K$). In basic GMM, any positive definite weighting matrix $\mathbf{W}_n$ produces a consistent estimator, but the precision (variance) of the estimates depends critically on how different moments are weighted.

Geometric Interpretation in $\mathbb{R}^L$:

Visualize the sample moment vector $\bar{\mathbf{g}}_n(\theta)$ as a point in a space of dimension L . The weighting matrix $\mathbf{W}$ defines the **distance metric** (the geometric deformation of the error ellipsoid). If the moments have distinct variances or are correlated, simple Euclidean distance ($\mathbf{W} = \mathbf{I}$) ignores that some information is more “noisy” than others. Optimal

GMM uses $\mathbf{S}^{-1}$ to rotate and scale the space, so that the projection occurs in the direction of **minimum variance**.

5.2.2.2 Assumptions

Assumption 5.20: Overidentification

- **Role:** Ensures that $L \geq K$, allowing the choice of $\mathbf{W}$ to be statistically relevant for asymptotic efficiency.
- **Counterexample:** If $L = K$, the system has a unique solution with $\bar{\mathbf{g}}_n(\hat{\theta}) = \mathbf{0}$, making $\mathbf{W}$ irrelevant.

Assumption 5.21: Consistency of the Weighting Matrix

- **Role:** Requires that $\mathbf{W}_n \xrightarrow{p} \mathbf{S}^{-1}$, where $\mathbf{S} = E[\mathbf{m}(\mathbf{w}_i, \theta_0)\mathbf{m}(\mathbf{w}_i, \theta_0)']$.
- **Counterexample:** If $\mathbf{W}_n$ converges to $\mathbf{A} \neq \mathbf{S}^{-1}$, the estimator will still be consistent, but the asymptotic variance will be larger (in the PSD sense) than that of the optimal GMM.

5.2.2.3 Formal Statement and Derivation

The optimal GMM estimator $\hat{\theta}_{opt}$ minimizes:

$$Q_n(\theta) = \bar{\mathbf{g}}_n(\theta)' \hat{\mathbf{S}}_n^{-1} \bar{\mathbf{g}}_n(\theta) \quad (5.162)$$

where $\hat{\mathbf{S}}_n \xrightarrow{p} \mathbf{S} = E[\mathbf{m}(\mathbf{w}_i, \theta_0)\mathbf{m}(\mathbf{w}_i, \theta_0)']$.

Theorem 5.2.3 (Efficiency of Optimal GMM). *Under regularity conditions:*

$$\boxed{\sqrt{n}(\hat{\theta}_{opt} - \theta_0) \xrightarrow{d} N(\mathbf{0}, (\mathbf{G}_0' \mathbf{S}^{-1} \mathbf{G}_0)^{-1})} \quad (5.163)$$

and for any other positive definite matrix $\mathbf{W}$:

$$\boxed{\mathbf{V}(\mathbf{W}) \succeq \mathbf{V}(\mathbf{S}^{-1})} \quad (5.164)$$

Proof (sketch). The FOC of GMM is $\mathbf{G}_n(\hat{\theta}_n)' \mathbf{W}_n \bar{\mathbf{g}}_n(\hat{\theta}_n) = \mathbf{0}$. Expanding $\bar{\mathbf{g}}_n(\hat{\theta}_n)$ around θ_0 and applying Slutsky yields the variance $\mathbf{V}(\mathbf{W}) = (\mathbf{G}_0' \mathbf{W} \mathbf{G}_0)^{-1} \mathbf{G}_0' \mathbf{W} \mathbf{S} \mathbf{W} \mathbf{G}_0 (\mathbf{G}_0' \mathbf{W} \mathbf{G}_0)^{-1}$. For $\mathbf{W} = \mathbf{S}^{-1}$, $\mathbf{V}(\mathbf{S}^{-1}) = (\mathbf{G}_0' \mathbf{S}^{-1} \mathbf{G}_0)^{-1}$. The proof of $\mathbf{V}(\mathbf{W}) \succeq \mathbf{V}(\mathbf{S}^{-1})$ follows by the projection matrix argument presented in Section 5.1.5. $\square$

5.2.2.4 Role of Assumptions

The assumption of **Consistency of $\mathbf{W}_n$ to $\mathbf{S}^{-1}$** was indispensable. If $\mathbf{W}$ does not collapse exactly to the inverse of the population variance of the moments, the algebraic simplification fails, preventing the estimator from reaching the lower bound of asymptotic variance.

5.2.3 Method of Moments (MM)

5.2.3.1 Motivation and Geometric Interpretation

The **Method of Moments (MM)**, systematized by Karl Pearson (1895), solves the problem of estimating population parameters when one does not wish (or cannot) assume a strict functional form for the distribution of the data, but has grounded knowledge about the mathematical expectations of certain functions of the random variables.

Geometric Interpretation in $\mathbb{R}^K$:

Each population moment condition $E[\mathbf{g}(\mathbf{w}_i, \theta)] = \mathbf{0}$ establishes a surface (or hyperplane) of dimension $K - 1$ in the parameter space. In MM, with exactly K equations for K parameters, the estimator $\hat{\theta}_{MM}$ is located at the **unique intersection** of these K sample-estimated surfaces.

5.2.3.2 Assumptions

Assumption 5.22: Exact Identification

- **Role:** Ensures that $L = K$, allowing a uniquely determined solution.
- **Counterexample:** If $L > K$, the system becomes overdetermined and there will be no θ that zeros all sample moments simultaneously.

Assumption 5.23: Rank Condition

- **Role:** Ensures that $\mathbf{G}_0 = E[\partial \mathbf{g}(\mathbf{w}_i, \theta_0) / \partial \theta']$ is non-singular.
 - **Counterexample:** If we try to estimate the mean and variance using only the first moment and a linear function of it, the Jacobian matrix will have rank 1, making the parameters indistinguishable.
-

5.2.3.3 Formal Statement and Derivation

The method of moments estimator $\hat{\theta}_n$ is defined as the exact solution of the system of K sample equations:

$$\bar{\mathbf{g}}_n(\hat{\theta}_n) = \frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{w}_i, \hat{\theta}_n) = \mathbf{0} \quad (5.165)$$

Theorem 5.2.4 (Asymptotic Normality of MM). *Under regularity conditions:*

$$\boxed{\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{G}_0^{-1} \mathbf{S} (\mathbf{G}_0')^{-1})} \quad (5.166)$$

Proof. Expanding $\bar{\mathbf{g}}_n(\hat{\theta}_n)$ around θ_0 via the Mean Value Theorem:

$$\mathbf{0} = \bar{\mathbf{g}}_n(\theta_0) + \mathbf{G}_n(\bar{\theta})(\hat{\theta}_n - \theta_0)$$

Multiplying by $\sqrt{n}$, isolating, and applying CLT + Slutsky yields (5.166). $\square$

5.2.3.4 Role of Assumptions

The assumption of **Just-Identification** ($L = K$) was indispensable for ensuring that the Jacobian matrix $\mathbf{G}_0$ was square, allowing its direct inversion without resorting to projections based on external weighting matrices.

5.2.4 Ordinary Least Squares (OLS)

5.2.4.1 Motivation and Geometric Interpretation

OLS solves the problem of estimating the linear relationship between a dependent variable and regressors under the assumption of exogeneity. In the GMM context, it is the special case of **exact identification** ($L = K$) where the moment function is defined by the orthogonality condition between the regressors and the error.

Geometric Interpretation in $\mathbb{R}^n$:

The OLS estimator performs an **orthogonal projection** of $\mathbf{y}$ onto the linear subspace spanned by the columns of $\mathbf{X}$. The residual vector $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ is perpendicular to each column of $\mathbf{X}$, validating the sample moment condition $\mathbf{X}'\mathbf{e} = \mathbf{0}$.

5.2.4.2 Assumptions

Assumption 5.24: Population Orthogonality

- **Role:** States that $E[\mathbf{x}_i \varepsilon_i] = \mathbf{0}$.
- **Counterexample:** In a supply and demand model, if price is determined simultaneously with quantity, OLS will be inconsistent.

Assumption 5.25: No Perfect Multicollinearity

- **Role:** Ensures that $\mathbf{Q}_{xx} = E[\mathbf{x}_i \mathbf{x}_i']$ is invertible.
- **Counterexample:** Including “weight in kg” and “weight in pounds” as distinct regressors generates singularity.

5.2.4.3 Formal Statement and Derivation

The OLS estimator $\hat{\beta}_n$ is the GMM that uses $\mathbf{g}(\mathbf{w}_i, \beta) = \mathbf{x}_i(y_i - \mathbf{x}_i'\beta)$ and $\mathbf{W}_n = \mathbf{I}_K$.

Defining $\bar{\mathbf{g}}_n(\beta) = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i(y_i - \mathbf{x}_i'\beta)$, the FOC is $\bar{\mathbf{g}}_n(\hat{\beta}_n) = \mathbf{0}$:

$$\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i(y_i - \mathbf{x}_i'\hat{\beta}_n) = \mathbf{0} \quad (5.167)$$

Isolating:

$$\hat{\beta}_n = \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \left(\sum_{i=1}^n \mathbf{x}_i y_i \right) \quad (5.168)$$

Theorem 5.2.5 (Consistency and Normality of OLS as GMM). *Under exogeneity:*

$$\text{plim } \hat{\beta}_n = \beta_0 \quad (5.169)$$

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{Q}_{xx}^{-1} \mathbf{\Omega} \mathbf{Q}_{xx}^{-1}) \quad (5.170)$$

where $\mathbf{\Omega} = E[\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i']$.

5.2.4.4 Role of Assumptions

The assumption of **Orthogonality** ($E[\mathbf{x}\varepsilon] = \mathbf{0}$) was indispensable. Without it, the term $\frac{1}{n} \mathbf{X}'\varepsilon$ would converge to a non-zero value, generating inconsistency.

5.2.5 Robust Variance (Eicker-Huber-White)

5.2.5.1 Motivation and Geometric Interpretation

The **EHW** estimator solves the problem of **invalid inference under unknown heteroskedasticity**. OLS remains consistent under heteroskedasticity, but the classical formula $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ becomes biased.

Geometric Interpretation in $\mathbb{R}^n$:

Under homoskedasticity, the error vector $\mathbf{u}$ resides in a **probability sphere**. Under heteroskedasticity, this sphere deforms into an **ellipsoid** with non-uniform axes. EHW projects the variability of these individual deviations back to $\mathbb{R}^K$, adjusting the “width” of the confidence ellipses for each direction.

5.2.5.2 Assumptions

Assumption 5.26: Independence of Observations

- **Role:** Ensures that $\mathbf{\Omega} = E[\mathbf{u}\mathbf{u}']$ is diagonal.
- **Counterexample:** In time series with autocorrelation, the standard EHW would underestimate the variance.

Assumption 5.27: Existence of Fourth-Order Moments

- **Role:** Ensures the convergence of $\frac{1}{n} \sum \mathbf{x}_i \mathbf{x}_i' \hat{e}_i^2$.
 - **Counterexample:** Heavy-tailed distributions prevent stable convergence.
-

5.2.5.3 Formal Statement and Derivation

Let $\hat{\beta}_n$ be the OLS estimator and $\hat{e}_i = y_i - \mathbf{x}_i' \hat{\beta}_n$. The EHW estimator is:

$$\hat{\mathbf{V}}_{EHW} = n(\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \hat{e}_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1} \quad (5.171)$$

Theorem 5.2.6 (Consistency of EHW). *Under Assumptions 5.26 and 5.27:*

$$\hat{\mathbf{V}}_{EHW} \xrightarrow{p} \mathbf{Q}^{-1} \mathbf{S} \mathbf{Q}^{-1} \quad (5.172)$$

where $\mathbf{Q} = E[\mathbf{x}_i \mathbf{x}_i']$ and $\mathbf{S} = E[u_i^2 \mathbf{x}_i \mathbf{x}_i']$.

Proof (sketch). $\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{Q}^{-1} \mathbf{S} \mathbf{Q}^{-1})$. The consistency of $\hat{\mathbf{S}} = \frac{1}{n} \sum \mathbf{x}_i \mathbf{x}_i' \hat{e}_i^2$ for $\mathbf{S}$ follows from the expansion $\hat{e}_i = u_i - \mathbf{x}_i'(\hat{\beta}_n - \beta_0)$ and the consistency of $\hat{\beta}_n$. $\square$

5.2.5.4 Role of Assumptions

Independence (Assumption 5.26) was indispensable. If there were serial correlation, the “filling” of the sandwich would accumulate cross-covariance terms, invalidating the purely additive form of (5.171).

5.2.6 Generalized Least Squares (GLS)

5.2.6.1 Motivation and Geometric Interpretation

GLS solves the problem of **efficient estimation** in the presence of a known covariance structure $\mathbf{\Omega}$ for the errors, allowing observations with higher variance to receive lower relative weight.

Geometric Interpretation in $\mathbb{R}^n$:

In GLS, the space is modified by a transformation $\mathbf{P}$ such that $\mathbf{P}'\mathbf{P} = \mathbf{\Omega}^{-1}$. This is equivalent to performing an **orthogonal projection in a non-Euclidean metric space**, where distances are weighted by the inverse of the covariance matrix.

5.2.6.2 Assumptions

Assumption 5.28: Positive Definite Covariance Structure

- **Role:** Assumes $\text{Var}(\mathbf{u}|\mathbf{X}) = \sigma^2\mathbf{\Omega}$, with $\mathbf{\Omega}$ SPD.
 - **Counterexample:** If $\mathbf{\Omega}$ is singular, correct weighting of observations becomes impossible.
-

5.2.6.3 Formal Statement and Derivation

GLS is the GMM with $\mathbf{g}_n(\beta) = \frac{1}{n}\mathbf{X}'\mathbf{\Omega}^{-1}(\mathbf{y} - \mathbf{X}\beta)$ and $\mathbf{W}_n = \mathbf{I}_K$ (just-identified).

The FOC $\mathbf{g}_n(\hat{\beta}_{GLS}) = \mathbf{0}$ yields:

$$\boxed{\hat{\beta}_{GLS} = (\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{y}} \quad (5.173)$$

Theorem 5.2.7 (Asymptotic Normality of GLS). *Under exogeneity:*

$$\sqrt{n}(\hat{\beta}_{GLS} - \beta_0) \xrightarrow{d} N(\mathbf{0}, \sigma^2\mathbf{Q}_{glS}^{-1}) \quad (5.174)$$

where $\mathbf{Q}_{glS} = E[\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X}/n]$.

5.2.6.4 Role of Assumptions

The assumption of **Exogeneity** ($E[\mathbf{x}_i u_i] = \mathbf{0}$) was indispensable. If the regressors are endogenous, GLS will converge to a biased value, and the efficiency correction will only alter the weight of the bias, without removing it.

5.2.7 Instrumental Variables (IV)

5.2.7.1 Motivation and Geometric Interpretation

IV solves the problem of **endogeneity** when $E[\mathbf{x}_i \varepsilon_i] \neq \mathbf{0}$. In GMM, it is the case of **exact identification** ($L = K$) with instruments $\mathbf{Z}$.

Geometric Interpretation in $\mathbb{R}^n$:

OLS projects $\mathbf{y}$ onto $\text{Col}(\mathbf{X})$. If $\mathbf{X}$ is correlated with the error, the subspace is “tilted.” IV uses $\mathbf{Z}$ as an external reference, projecting $\mathbf{y}$ onto $\text{Col}(\mathbf{X})$ only in the direction that is orthogonal to $\mathbf{Z}$.

5.2.7.2 Assumptions

Assumption 5.29: Exogeneity of the Instrument

- **Role:** $E[\mathbf{z}_i \varepsilon_i] = \mathbf{0}$.
- **Counterexample:** If the instrument is endogenous, IV will be inconsistent.

Assumption 5.30: Relevance of the Instrument

- **Role:** $\text{rank}(E[\mathbf{z}_i \mathbf{x}_i']) = K$.
 - **Counterexample:** Weak instruments generate infinite variance.
-

5.2.7.3 Formal Statement and Derivation

IV is the GMM with $\mathbf{g}_n(\beta) = \frac{1}{n} \mathbf{Z}'(\mathbf{y} - \mathbf{X}\beta)$ and $L = K$.

The FOC $\mathbf{g}_n(\hat{\beta}_{IV}) = \mathbf{0}$ yields:

$$\boxed{\hat{\beta}_{IV} = (\mathbf{Z}'\mathbf{X})^{-1} \mathbf{Z}'\mathbf{y}} \quad (5.175)$$

Theorem 5.2.8 (Consistency and Normality of IV as GMM). *Under Assumptions 5.29 and 5.30:*

$$\text{plim } \hat{\beta}_{IV} = \beta_0 \quad (5.176)$$

$$\sqrt{n}(\hat{\beta}_{IV} - \beta_0) \xrightarrow{d} N(\mathbf{0}, \sigma^2 \mathbf{Q}_{zx}^{-1} \mathbf{Q}_{zz} \mathbf{Q}_{xz}^{-1}) \quad (5.177)$$

5.2.7.4 Role of Assumptions

Relevance (Assumption 5.30) was indispensable. If $\mathbf{Z}'\mathbf{X}$ were singular, the inverse would not exist and the IV estimator would be undefined.

5.2.8 Two-Stage Least Squares (2SLS)

5.2.8.1 Motivation and Geometric Interpretation

2SLS simultaneously solves **endogeneity** and **overidentification**. When $L > K$, the system is overidentified, and 2SLS emerges as the special case of GMM with $\mathbf{W}_n = (\mathbf{Z}'\mathbf{Z})^{-1}$.

Geometric Interpretation in $\mathbb{R}^n$:

1. **First Stage:** Project $\mathbf{X}$ onto $\text{Col}(\mathbf{Z})$: $\hat{\mathbf{X}} = \mathbf{P}_Z \mathbf{X}$.
 2. **Second Stage:** Regress $\mathbf{y}$ on $\hat{\mathbf{X}}$.
-

5.2.8.2 Assumptions

Assumption 5.31: Exogeneity of the Instrument

- **Role:** $E[\mathbf{Z}'\mathbf{u}] = \mathbf{0}$.

Assumption 5.32: Relevance of the Instrument

- **Role:** $\text{rank}(E[\mathbf{Z}'\mathbf{X}]) = K$.

Assumption 5.33: Homoskedasticity

- **Role:** Ensures that $\mathbf{W}_n = (\mathbf{Z}'\mathbf{Z})^{-1}$ is optimal.

5.2.8.3 Formal Statement and Derivation

2SLS is the GMM with $\bar{\mathbf{g}}_n(\beta) = \frac{1}{n}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\beta)$ and $\mathbf{W}_n = (\mathbf{Z}'\mathbf{Z})^{-1}$.

Minimizing $Q_n(\beta) = (\mathbf{y} - \mathbf{X}\beta)' \mathbf{P}_Z (\mathbf{y} - \mathbf{X}\beta)$:

$$\boxed{\hat{\beta}_{2SLS} = (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_Z\mathbf{y}} \quad (5.178)$$

Theorem 5.2.9 (Asymptotic Normality of 2SLS as GMM). *Under Assumptions 5.31 to 5.33:*

$$\sqrt{n}(\hat{\beta}_{2SLS} - \beta_0) \xrightarrow{d} N\left(\mathbf{0}, \sigma^2(\boldsymbol{\Sigma}_{xz}\boldsymbol{\Sigma}_{zz}^{-1}\boldsymbol{\Sigma}_{zx})^{-1}\right) \quad (5.179)$$

5.2.8.4 Role of Assumptions

Relevance (Assumption 5.32) was indispensable. If $\mathbf{Z}$ is not correlated with $\mathbf{X}$, the matrix $(\mathbf{X}'\mathbf{P}_Z\mathbf{X})$ will be singular, making the estimator impossible to define.

5.2.9 Three-Stage Least Squares (3SLS)

5.2.9.1 Motivation and Geometric Interpretation

3SLS solves the problem of **statistical inefficiency** in overidentified systems of simultaneous equations. While 2SLS estimates each equation in isolation, 3SLS treats the system as a single block, using the covariance matrix of the residuals to weight the global system.

Geometric Interpretation in $\mathbb{R}^{nL}$:

3SLS operates in a subspace of dimension nL where the equations are “coupled.” The geometry involves an **oblique projection** adjusted by the cross-covariance matrix $\boldsymbol{\Sigma}$, resulting in a confidence ellipsoid that is “contracted” relative to 2SLS.

5.2.9.2 Assumptions

Assumption 5.34: Systemic Exogeneity

- **Role:** $E[\mathbf{z}_t u_{jt}] = \mathbf{0}$ for all j .
- **Counterexample:** If an instrument is correlated with the shock of another equation, 3SLS will spread the bias throughout the entire system.

Assumption 5.35: Non-Singularity of the Cross-Covariance

- **Role:** $\Sigma = E[\mathbf{u}_t \mathbf{u}_t']$ must be SPD.
- **Counterexample:** If the errors of two equations are perfectly correlated, Σ will be singular.

5.2.9.3 Formal Statement and Derivation

Consider the stacked system $\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$, with $\mathbf{X} = \text{diag}(\mathbf{X}_1, \dots, \mathbf{X}_L)$ and $\bar{\mathbf{Z}} = \mathbf{I}_L \otimes \mathbf{Z}$. 3SLS is the GMM with $\mathbf{g}_n(\beta) = \frac{1}{n} \bar{\mathbf{Z}}'(\mathbf{y} - \mathbf{X}\beta)$ and $\mathbf{W}_n = [\frac{1}{n} \bar{\mathbf{Z}}'(\hat{\Sigma} \otimes \mathbf{I}_n) \bar{\mathbf{Z}}]^{-1} = \hat{\Sigma}^{-1} \otimes (\mathbf{Z}'\mathbf{Z})^{-1}$. The FOC yields:

$$\boxed{\hat{\beta}_{3SLS} = [\mathbf{X}'(\hat{\Sigma}^{-1} \otimes \mathbf{P}_Z)\mathbf{X}]^{-1} \mathbf{X}'(\hat{\Sigma}^{-1} \otimes \mathbf{P}_Z)\mathbf{y}} \quad (5.180)$$

Theorem 5.2.10 (Asymptotic Normality of 3SLS). *Under Assumptions 5.34 and 5.35:*

$$\sqrt{n}(\hat{\beta}_{3SLS} - \beta_0) \xrightarrow{d} N\left(\mathbf{0}, \{E[\mathbf{X}'(\Sigma^{-1} \otimes \mathbf{P}_Z)\mathbf{X}]\}^{-1}\right) \quad (5.181)$$

5.2.9.4 Role of Assumptions

The assumption of **Non-Singularity of Σ** (Assumption 5.35) was indispensable. Without the invertibility of Σ , we could not define the optimal weight metric $\hat{\Sigma}^{-1}$ that couples the equations. Without this coupling, the estimator would collapse to equation-by-equation 2SLS, losing the efficiency gains from the correlation between errors.

5.2.10 Generalized Nonlinear Least Squares (GNLLS)

5.2.10.1 Motivation and Geometric Interpretation

Generalized Nonlinear Least Squares (GNLLS) solves the problem of estimating parameters in models where the functional relationship between variables is inherently **non-linear** and the stochastic errors exhibit a **non-spherical** structure (heteroskedasticity or autocorrelation). While classical OLS and GLS are limited to linear response surfaces (hyperplanes), GNLLS allows the researcher to model complex phenomena, such as logistic growth curves or exponential decay, without sacrificing statistical efficiency in the presence of correlated errors.

In the **GMM** context, GNLLS emerges as a special case where the moment conditions are derived from the gradient of the non-linear function, weighted by the inverse of the error covariance matrix.

Geometric Interpretation in $\mathbb{R}^3$:

Imagine a cloud of points in $\mathbb{R}^3$ that appears to be distributed along a curved surface (such as a saddle or a rotated parabola). The GNLLS estimator seeks the parametric surface that best fits these points. Due to the covariance structure $\mathbf{\Omega}$, the “best” proximity is not measured by simple vertical Euclidean distance, but by a metric that “stretches” or “compresses” the sample space to compensate for the unequal variability of the errors. Visually, GNLLS performs an **oblique projection** onto the non-linear manifold, where the projection angle is dictated by the correlation structure among the errors.

5.2.10.2 Assumptions

Assumption 5.36: Differentiability and Continuity of the Model

- **Role:** Ensures that the response surface $h(\mathbf{x}_i, \theta)$ is smooth enough to allow the use of gradient-based optimization methods and the application of Taylor expansion in asymptotic theory.
- **Counterexample:** If $h(\cdot)$ were a step function (discrete), the gradient would be zero or undefined over almost the entire domain, making it impossible to locate the minimum via first-order conditions.

Assumption 5.37: Global Identification (Uniqueness)

- **Role:** Ensures that there exists only one vector θ_0 in the parameter space Θ that minimizes the population cost function.
- **Counterexample:** If the model is overparameterized (e.g., $y = \theta_1\theta_2x + \varepsilon$), there will be a continuum of combinations of θ_1 and θ_2 that produce the same fit, making the Jacobian matrix singular.

Assumption 5.38: Non-Singularity of the Covariance Matrix $\mathbf{\Omega}$

- **Role:** Ensures that the optimal weight matrix of GNLLS, $\mathbf{W} = \mathbf{\Omega}^{-1}$, exists and is positive definite, allowing the “spherification” of the transformed errors.
- **Counterexample:** If two errors were perfectly correlated, $\mathbf{\Omega}$ would be singular, preventing the numerical computation of the objective function.

5.2.10.3 Formal Statement and Derivation

Consider the non-linear model $y_i = h(\mathbf{x}_i, \theta_0) + \varepsilon_i$, where $E[\varepsilon_i|\mathbf{x}_i] = 0$. Let $\mathbf{y} \in \mathbb{R}^n$ be the dependent vector and $\mathbf{h}(\mathbf{X}, \theta) \in \mathbb{R}^n$ the vector of functions. Assume that $E[\varepsilon\varepsilon'|\mathbf{X}] = \sigma^2\mathbf{\Omega}$, where $\mathbf{\Omega}$ is a known SPD matrix.

The GNLLS estimator $\hat{\theta}_n$ minimizes:

$$Q_n(\theta) = [\mathbf{y} - \mathbf{h}(\mathbf{X}, \theta)]'\mathbf{\Omega}^{-1}[\mathbf{y} - \mathbf{h}(\mathbf{X}, \theta)] \quad (5.186)$$

Step 1: First Order Condition (FOC)

Differentiating and setting to zero:

$$\nabla_{\theta} Q_n(\hat{\theta}_n) = -2\mathbf{J}(\hat{\theta}_n)' \mathbf{\Omega}^{-1} [\mathbf{y} - \mathbf{h}(\mathbf{X}, \hat{\theta}_n)] = \mathbf{0} \quad (5.187)$$

where $\mathbf{J}(\theta) = \partial \mathbf{h}(\mathbf{X}, \theta) / \partial \theta'$ is the $n \times K$ Jacobian matrix.

Step 2: Taylor Expansion (Linearization)

Expanding $\mathbf{h}(\mathbf{X}, \hat{\theta}_n)$ around θ_0 :

$$\mathbf{h}(\mathbf{X}, \hat{\theta}_n) \approx \mathbf{h}(\mathbf{X}, \theta_0) + \mathbf{J}(\theta_0)(\hat{\theta}_n - \theta_0) \quad (5.188)$$

Substituting into (5.187) and using $\mathbf{y} = \mathbf{h}(\mathbf{X}, \theta_0) + \varepsilon$:

$$\mathbf{J}(\hat{\theta}_n)' \mathbf{\Omega}^{-1} [\varepsilon - \mathbf{J}(\theta_0)(\hat{\theta}_n - \theta_0)] \approx \mathbf{0} \quad (5.189)$$

Step 3: Isolating the Estimation Error

$$\mathbf{J}(\hat{\theta}_n)' \mathbf{\Omega}^{-1} \mathbf{J}(\theta_0)(\hat{\theta}_n - \theta_0) \approx \mathbf{J}(\hat{\theta}_n)' \mathbf{\Omega}^{-1} \varepsilon \quad (5.190)$$

Multiplying by $\sqrt{n}$ and premultiplying by the inverse:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \approx \left[\frac{1}{n} \mathbf{J}(\hat{\theta}_n)' \mathbf{\Omega}^{-1} \mathbf{J}(\theta_0) \right]^{-1} \frac{1}{\sqrt{n}} \mathbf{J}(\hat{\theta}_n)' \mathbf{\Omega}^{-1} \varepsilon \quad (5.191)$$

Step 4: Stochastic Limits and Asymptotic Distribution

- $\hat{\theta}_n \xrightarrow{p} \theta_0 \Rightarrow \mathbf{J}(\hat{\theta}_n) \xrightarrow{p} \mathbf{J}_0$
- $\frac{1}{n} \mathbf{J}_0' \mathbf{\Omega}^{-1} \mathbf{J}_0 \xrightarrow{p} \mathbf{H}_0$
- $\frac{1}{\sqrt{n}} \mathbf{J}_0' \mathbf{\Omega}^{-1} \varepsilon \xrightarrow{d} N(\mathbf{0}, \sigma^2 \mathbf{H}_0)$

By Slutsky's Theorem:

$$\boxed{\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(\mathbf{0}, \sigma^2 \mathbf{H}_0^{-1})} \quad (5.192)$$

5.2.10.4 Role of Assumptions

The **Non-Singularity of $\mathbf{\Omega}$ (Assumption 5.38)** was indispensable for defining the metric of the objective function and allowing the scaling of the error. **Differentiability (Assumption 5.36)** was essential for the linear expansion; without it, the estimator could not be approximated by a normal form.

5.2.11 Maximum Likelihood Estimator (MLE)

5.2.11.1 Motivation and Geometric Interpretation

The **Maximum Likelihood Method (MLE)** solves the problem of estimating population parameters when the researcher assumes a complete functional form for the conditional or joint probability distribution of the data. While pure GMM is often used in semiparametric contexts (knowing only moment conditions about unknown distributions), MLE is the parametric reference estimator that, under correct specification, attains the **Cramér-Rao lower bound**, being the asymptotically most efficient estimator possible.

In the **GMM** context, MLE can be derived as a special case of **exact identification** ($L = K$), where the population moment conditions are not chosen arbitrarily but extracted directly from the adopted probabilistic structure: the moments are the components of the **score function** (the gradient vector of the log-likelihood).

Geometric Interpretation in $\mathbb{R}^K$:

Consider the log-likelihood function as a hyperbolic “mountain surface” projected over the compact parameter space of dimension K . The MLE estimator locates the **peak (global maximum)** of this surface. Geometrically, at this peak, the tangent plane is perfectly horizontal, meaning that the **gradient vector (score)** $\mathbf{s}(\mathbf{w}, \theta)$ is zero. The **Fisher Information Matrix** $\mathcal{I}(\theta)$ acts as a measure of curvature (the negative Hessian matrix) of this surface at the top: the sharper the mountain (greater negative curvature), the more information the data provide about the precise location of the parameter.

5.2.11.2 Assumptions

Let $f(\mathbf{x}, \theta)$ be the probability density and $\theta \in \Theta$ the parameter vector.

Assumption 5.39: Correct Model Specification

- **Role:** Ensures that the assumed parametric density $f(\mathbf{x}, \theta)$ coincides with the true population density generating the data at $\theta = \theta_0$. It is necessary to ensure consistency and the attainment of maximum Fisher efficiency.
- **Counterexample:** If the actual data follow a Student’s t distribution (heavy tails), but we model via MLE assuming a Normal distribution, the estimator may be inconsistent. In the case of Quasi-MLE (QMLE), it may retain consistency under certain conditions but will lose efficiency, requiring robust variance-covariance matrices.

Assumption 5.40: Identifiability (Uniqueness)

- **Role:** Ensures an injective correspondence between the parameter space and the family of distributions, such that $f(\mathbf{x}, \theta_1) \neq f(\mathbf{x}, \theta_2)$ if and only if $\theta_1 \neq \theta_2$.
- **Counterexample:** In a linear model where $y_i = (\theta_1 + \theta_2)x_i + \epsilon_i$, it is impossible to isolate the individual effects. The likelihood surface will exhibit a perfectly flat “ridge” (infinite maxima), resulting in a singular information matrix.

Assumption 5.41: Fisher Regularity Conditions

- **Role:** Allows the interchange between integration over the support and differentiation in the parameter space. It is indispensable for proving that the mathematical expectation of the score is identically zero and for establishing the equality of the information matrix.
- **Counterexample:** If the support of the distribution intrinsically depends on the parameter (for example, a Uniform distribution on the interval $[0, \theta]$), the density will not be continuously differentiable at the boundaries of the support, breaking the standard asymptotic normality with convergence rates different from $\sqrt{n}$.

5.2.11.3 Formal Statement and Derivation

Let $\{\mathbf{x}_i\}_{i=1}^n$ be a simple random sample (i.i.d.) with population density given by $f(\mathbf{x}_i, \theta_0)$. We define:

1. **Likelihood Function:** $L(\theta) = \prod_{i=1}^n f(\mathbf{x}_i, \theta)$.
2. **Log-Likelihood:** $\ell(\theta) = \ln L(\theta) = \sum_{i=1}^n \ln f(\mathbf{x}_i, \theta)$.
3. **Score Function:** $\mathbf{s}(\mathbf{x}_i, \theta) = \nabla_{\theta} \ln f(\mathbf{x}_i, \theta)$ ($K \times 1$).
4. **Fisher Information Matrix (Unit):** $\mathcal{I}(\theta) = E[\mathbf{s}(\mathbf{x}_i, \theta)\mathbf{s}(\mathbf{x}_i, \theta)'] = -E[\nabla_{\theta\theta'}^2 \ln f(\mathbf{x}_i, \theta)]$.

Theorem 5.2.11 (Asymptotic Normality of MLE). *Under Fisher's regularity conditions, the maximum likelihood estimator $\hat{\theta}_n$ satisfies:*

$$\boxed{\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(\mathbf{0}, \mathcal{I}(\theta_0)^{-1})} \quad (5.186)$$

Proof. **Step 1: Sample Moment Condition**

MLE can be defined as the estimator that solves the system of sample moment equations $\mathbf{g}_n(\hat{\theta}_n) = \frac{1}{n} \sum_{i=1}^n \mathbf{s}(\mathbf{x}_i, \hat{\theta}_n) = \mathbf{0}$. By the necessary condition for the existence of a maximum at an interior point:

$$\sum_{i=1}^n \mathbf{s}(\mathbf{x}_i, \hat{\theta}_n) = \mathbf{0} \quad (5.187)$$

Step 2: Score Property (Zero Expectation)

We prove that $E[\mathbf{s}(\mathbf{x}_i, \theta_0)] = \mathbf{0}$.

By the regularity assumption (Assumption 5.41), we interchange integral and derivative:

$$E[\mathbf{s}(\mathbf{x}_i, \theta_0)] = \int \frac{\nabla_{\theta} f(\mathbf{x}, \theta_0)}{f(\mathbf{x}, \theta_0)} f(\mathbf{x}, \theta_0) d\mathbf{x} = \nabla_{\theta} \int f(\mathbf{x}, \theta_0) d\mathbf{x}$$

Since $\int f(\mathbf{x}, \theta_0) d\mathbf{x} = 1$ over the entire support by definition of density:

$$E[\mathbf{s}(\mathbf{x}_i, \theta_0)] = \nabla_{\theta}(1) = \mathbf{0} \quad (5.188)$$

Step 3: Linear Expansion via the Mean Value Theorem

We expand the system of sample score equations (5.187) around the true parameter θ_0 :

$$\mathbf{0} = \sum_{i=1}^n \mathbf{s}(\mathbf{x}_i, \theta_0) + \left[\sum_{i=1}^n \nabla_{\theta'} \mathbf{s}(\mathbf{x}_i, \bar{\theta}) \right] (\hat{\theta}_n - \theta_0) \quad (5.189)$$

where $\bar{\theta}$ is a stochastic mean vector lying on the segment between $\hat{\theta}_n$ and θ_0 .

Step 4: Scaling and Matrix Isolation

Multiplying expression (5.189) by $\frac{1}{\sqrt{n}}$ and isolating the scaled sample error:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{s}(\mathbf{x}_i, \theta_0) = - \left[\frac{1}{n} \sum_{i=1}^n \nabla_{\theta'} \mathbf{s}(\mathbf{x}_i, \bar{\theta}) \right] \sqrt{n}(\hat{\theta}_n - \theta_0) \quad (5.190)$$

Premultiplying by the inverse of the average sample Hessian matrix (guaranteed asymptotically by full rank):

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = - \left[\frac{1}{n} \sum_{i=1}^n \nabla_{\theta\theta'}^2 \ln f(\mathbf{x}_i, \bar{\theta}) \right]^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{s}(\mathbf{x}_i, \theta_0) \right) \quad (5.191)$$

Step 5: Application of Limit Theorems

- **CLT:** Since the individual scores $\mathbf{s}(\mathbf{x}_i, \theta_0)$ are i.i.d. with zero expectation and covariance matrix $\mathcal{I}(\theta_0)$:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{s}(\mathbf{x}_i, \theta_0) \xrightarrow{d} N(\mathbf{0}, \mathcal{I}(\theta_0))$$

- **WLLN:** By the consistency of $\hat{\theta}_n \xrightarrow{p} \theta_0$ and continuity of the second derivatives:

$$\frac{1}{n} \sum_{i=1}^n \nabla_{\theta\theta'}^2 \ln f(\mathbf{x}_i, \bar{\theta}) \xrightarrow{p} E \left[\nabla_{\theta\theta'}^2 \ln f(\mathbf{x}_i, \theta_0) \right] = -\mathcal{I}(\theta_0) \quad (5.192)$$

Step 6: Final Result via Slutsky

Applying Slutsky's Theorem:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} -[-\mathcal{I}(\theta_0)]^{-1} N(\mathbf{0}, \mathcal{I}(\theta_0)) \quad (5.193)$$

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(\mathbf{0}, \mathcal{I}(\theta_0)^{-1}) \quad (5.194)$$

□

5.2.11.4 Role of Assumptions

The **Regularity assumption (Assumption 5.41)** was indispensable for geometrically sustaining Fisher's identity ($\mathbf{S} = -\mathbf{G}$). Without this interchange of operators, the limiting variance of the score would not coincide with the negative population Hessian, forcing the appearance of a sandwich-type structure and breaking the direct asymptotic efficiency.

5.2.12 Logit Model

5.2.12.1 Motivation and Geometric Interpretation

The **Logit** model solves the problem of modeling the conditional probability of occurrence of a strictly binary dependent variable ($y_i \in \{0, 1\}$) from a set of exogenous regressor vectors. The traditional Linear Probability Model (LPM), estimated via OLS, exhibits severe structural flaws: it generates predictions outside the unit interval $[0, 1]$ and is intrinsically heteroskedastic. Logit overcomes these constraints by mapping the linear index of regressors to the interval $[0, 1]$ using the logistic Cumulative Distribution Function (CDF).

In the unified **GMM** view, Logit represents a case of **exact identification** ($L = K$), where the population moment condition is defined by the score of a conditional Bernoulli distribution.

Geometric Interpretation in $\mathbb{R}^2$:

The observed data lie exclusively at the discrete limits $y = 0$ or $y = 1$. Logit fits an S-shaped sigmoid curve to approximate the conditional expectation $E[y_i|\mathbf{x}_i]$. From the GMM perspective, estimation seeks to rotate and translate this sigmoidal profile so that $\frac{1}{n} \sum_{i=1}^n (y_i - \hat{p}_i)\mathbf{x}_i = \mathbf{0}$.

5.2.12.2 Assumptions

Let $\mathbf{x}_i \in \mathbb{R}^K$ be the vector of regressors and $y_i \in \{0, 1\}$ the observed binary response.

Assumption 5.42: Stochastic Independence (Random Sampling)

- **Role:** Allows the additive factorization of the log-likelihood as the sum of individual log-densities, enabling the application of the Law of Large Numbers over sums of sample moments.
- **Counterexample:** Under active temporal or spatial dependence, the simple Fisher covariance matrix would fail by ignoring cross-interactions, requiring robust variance corrections of the HAC type.

Assumption 5.43: Rank Identifiability (No Collinearity)

- **Role:** Requires that the population moment matrix of the regressors $\mathbf{Q}_{xx} = E[\mathbf{x}_i\mathbf{x}_i']$ have full rank K . Ensures that the Hessian matrix is strictly negative definite over the entire support, guaranteeing the existence of a unique stable global maximum.

- **Counterexample:** In the presence of perfect multicollinearity, the likelihood surface will exhibit an infinite flat valley. The Jacobian moment matrix will become singular, preventing analytical inversion and identification of the vector β .

Assumption 5.44: Logistic CDF Specification

- **Role:** Postulates that the true data generating process obeys the structural relationship $P(y_i = 1|\mathbf{x}_i) = \Lambda(\mathbf{x}_i'\beta_0)$, where $\Lambda(z) = e^z/(1 + e^z)$. Fixes the exact functional form of the orthogonality conditions.
- **Counterexample:** If the actual generating process follows a Probit (Normal) distribution, Logit estimators will maintain directional consistency of signs, but the calculated marginal effects will exhibit systematic biases caused by the relatively heavier tails of the logistic distribution.

5.2.12.3 Formal Statement and Derivation

Let $\{y_i, \mathbf{x}_i\}_{i=1}^n$ be an i.i.d. sample where $y_i|\mathbf{x}_i \sim \text{Bernoulli}(p_i)$ with probability parameterized by $p_i = \Lambda(\mathbf{x}_i'\beta_0)$. We define:

1. **Logistic Function:** $\Lambda(z) = \frac{e^z}{1+e^z}$.
2. **Derivative Identity:** $\frac{d\Lambda(z)}{dz} = \Lambda(z)(1 - \Lambda(z))$.
3. **GMM Moment Vector:** $\mathbf{g}(\mathbf{w}_i, \beta) = \mathbf{x}_i(y_i - \Lambda(\mathbf{x}_i'\beta))$ ($K \times 1$).

Theorem 5.2.12 (Asymptotic Normality of Logit). *The GMM/MLE estimator $\hat{\beta}_n$ obtained by solving $\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i(y_i - \Lambda(\mathbf{x}_i'\hat{\beta}_n)) = \mathbf{0}$ converges in the limiting distribution to:*

$$\boxed{\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(\mathbf{0}, \mathcal{I}(\beta_0)^{-1})} \quad (5.195)$$

where the Fisher Information Matrix is expressed as $\mathcal{I}(\beta_0) = E[\Lambda(\mathbf{x}_i'\beta_0)(1 - \Lambda(\mathbf{x}_i'\beta_0))\mathbf{x}_i\mathbf{x}_i']$.

Proof. Step 1: Extraction of the Moment via Bernoulli Log-Likelihood

The probability density function for a conditional observation is given by $f(y_i|\mathbf{x}_i; \beta) = p_i^{y_i}(1 - p_i)^{1-y_i}$. Its respective individual log-likelihood is:

$$\ell_i(\beta) = y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i) \quad (5.196)$$

Substituting the sigmoidal parameterization $p_i = \Lambda(\mathbf{x}_i'\beta)$ and differentiating:

$$\nabla_{\beta} \ell_i(\beta) = y_i \frac{1}{p_i} \nabla_{\beta} p_i - (1 - y_i) \frac{1}{1 - p_i} \nabla_{\beta} p_i \quad (5.197)$$

Employing $\nabla_{\beta} p_i = p_i(1 - p_i)\mathbf{x}_i$:

$$\nabla_{\beta} \ell_i(\beta) = y_i \frac{p_i(1 - p_i)\mathbf{x}_i}{p_i} - (1 - y_i) \frac{p_i(1 - p_i)\mathbf{x}_i}{1 - p_i} \quad (5.198)$$

Performing the algebraic cancellations:

$$\mathbf{s}_i(\beta) = y_i(1 - p_i)\mathbf{x}_i - (1 - y_i)p_i\mathbf{x}_i = \mathbf{x}_i(y_i - p_i) = \mathbf{x}_i(y_i - \Lambda(\mathbf{x}_i'\beta)) \quad (5.199)$$

This constitutes the exact-identification GMM moment condition vector: $\mathbf{g}_i(\beta) = \mathbf{x}_i(y_i - \Lambda(\mathbf{x}_i'\beta))$.

Step 2: Verification of the Population Orthogonality Condition

We evaluate the conditional expectation of the constructed moment under the true parameter β_0 :

$$E[\mathbf{g}_i(\beta_0)|\mathbf{x}_i] = \mathbf{x}_i (E[y_i|\mathbf{x}_i] - \Lambda(\mathbf{x}_i'\beta_0))$$

Under Assumption 5.44, $E[y_i|\mathbf{x}_i] = \Lambda(\mathbf{x}_i'\beta_0)$. Therefore:

$$E[\mathbf{g}_i(\beta_0)|\mathbf{x}_i] = \mathbf{x}_i(\mathbf{0}) = \mathbf{0} \quad (5.200)$$

By direct application of the Law of Iterated Expectations:

$$E[\mathbf{g}_i(\beta_0)] = E[E[\mathbf{g}_i(\beta_0)|\mathbf{x}_i]] = \mathbf{0} \quad (5.201)$$

Step 3: Linearization and Expansion via MVT

We expand the system of sample normal equations via the Mean Value Theorem:

$$\mathbf{0} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{g}_i(\beta_0) + \left[\frac{1}{n} \sum_{i=1}^n \nabla_{\beta'} \mathbf{g}_i(\bar{\beta}) \right] \sqrt{n}(\hat{\beta}_n - \beta_0) \quad (5.202)$$

We compute the sample Jacobian of the moment vector:

$$\nabla_{\beta'} \mathbf{g}_i(\beta) = -\Lambda(\mathbf{x}_i'\beta)(1 - \Lambda(\mathbf{x}_i'\beta))\mathbf{x}_i\mathbf{x}_i' \quad (5.203)$$

Step 4: Convergence and Variance Structure

- **WLLN:** The average sample Hessian matrix converges to $\mathbf{G}_0 = -\mathcal{I}(\beta_0)$.
- **CLT:** $\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{g}_i(\beta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{S})$, where $\mathbf{S} = \mathcal{I}(\beta_0)$.

Step 5: Isolation and Limiting Mapping

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} \mathcal{I}(\beta_0)^{-1} N(\mathbf{0}, \mathcal{I}(\beta_0)) \quad (5.204)$$

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(\mathbf{0}, \mathcal{I}(\beta_0)^{-1}) \quad (5.205)$$

□

5.2.12.4 Role of Assumptions

The premise of **Correct CDF Specification (Assumption 5.44)** was indispensable in Step 4. If the distribution were incorrect, the conditional equality $\text{Var}(y_i|\mathbf{x}_i) = p_i(1 - p_i)$ would be violated, generating a mismatch between the moment covariance matrix $\mathbf{S}$ and the population Jacobian $\mathbf{G}_0$. The model would require the robust sandwich formulation (QMLE), under penalty of underestimating or overestimating standard errors.

5.2.13 Probit Model

5.2.13.1 Motivation and Geometric Interpretation

The **Probit** model addresses the same problem of modeling binary discrete choices ($y_i \in \{0, 1\}$). Probit differs from the Logit model by mapping the linear index of exogenous regressors $\mathbf{x}_i'\beta$ to the probabilistic interval $[0, 1]$ through the Cumulative Distribution Function (CDF) of the Standard Normal distribution, denoted by $\Phi(\cdot)$.

Geometric Interpretation in $\mathbb{R}^K$:

In the vector space of explanatory variables, the Probit model establishes a **separation hyperplane** geometrically defined by the equation $\mathbf{x}_i'\beta = 0$. The conditional probability of success is represented by a **sigmoidal surface** that asymptotically extends between the ceiling and floor probabilistic boundaries. Estimation via GMM seeks to rotate and displace this separation hyperplane in the K -dimensional space of regressors so that the weighted score vector is zero in the sample.

5.2.13.2 Assumptions

Let $\mathbf{x}_i \in \mathbb{R}^K$ be the vector of regressors and $y_i \in \{0, 1\}$ the discrete response.

Assumption 5.45: Identifiability (Full Rank of the Regressor Matrix)

- **Role:** Requires that the population moment matrix $E[\mathbf{x}_i\mathbf{x}_i']$ be strictly positive definite, ensuring linearly independent statistical information to discriminate the effects of each parametric component.
- **Counterexample:** If there is perfect collinearity among the components of the regressor vector, the Jacobian moment matrix will become singular, generating a linear trough of infinite solutions and making inversion for asymptotic variance calculation impossible.

Assumption 5.46: Gaussian Specification of the Latent Error

- **Role:** Assumes that the model is generated by a latent variable structure $y_i^* = \mathbf{x}_i'\beta_0 + \epsilon_i$, where the unobserved errors strictly follow the distribution $\epsilon_i \sim N(0, 1)$. Ensures that the moment conditions extracted from the score are centered at zero in the population.
- **Counterexample:** If the actual underlying disturbances follow asymmetric or heavy-tailed distributions (such as the Cauchy distribution), the use of the normal

CDF $\Phi(\cdot)$ will cause a break in the probabilistic bridge, generating inconsistent and severely biased estimators.

Assumption 5.47: Strict Exogeneity of Regressors

- **Role:** Requires that the distribution of the latent error be independent of the regressors, implying the conditional condition $E[\epsilon_i|\mathbf{x}_i] = 0$.
- **Counterexample:** If one of the regressors is endogenous (such as omitting ability variables in educational choice models), the population orthogonality condition will fail because the latent error term will carry systematic information about $\mathbf{x}_i$, making the GMM/MLE estimator inconsistent.

5.2.13.3 Formal Statement and Derivation

Let $\{y_i, \mathbf{x}_i\}_{i=1}^n$ be a simple random sample (i.i.d.) with $y_i \in \{0, 1\}$ and $\mathbf{x}_i \in \mathbb{R}^K$. We define $\Phi(z) = \int_{-\infty}^z \phi(t)dt$, where $\phi(t)$ represents the probability density function (PDF) of the standard normal.

Theorem 5.2.13 (Asymptotic Normality of Probit). *Under regularity conditions and correct Gaussian specification, the estimator $\hat{\beta}_n$ satisfies the convergence:*

$$\boxed{\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(\mathbf{0}, \mathcal{I}(\beta_0)^{-1})} \quad (5.206)$$

where:

$$\mathcal{I}(\beta_0) = E \left[\frac{\phi(\mathbf{x}_i' \beta_0)^2}{\Phi(\mathbf{x}_i' \beta_0)(1 - \Phi(\mathbf{x}_i' \beta_0))} \mathbf{x}_i \mathbf{x}_i' \right] \quad (5.207)$$

Proof. Step 1: Formulation of the Structural Score

The conditional density of Probit for an observation is expressed as:

$$f(y_i|\mathbf{x}_i, \beta) = [\Phi(\mathbf{x}_i' \beta)]^{y_i} [1 - \Phi(\mathbf{x}_i' \beta)]^{1-y_i} \quad (5.208)$$

Its individual log-likelihood is:

$$\ln L_i(\beta) = y_i \ln \Phi(\mathbf{x}_i' \beta) + (1 - y_i) \ln [1 - \Phi(\mathbf{x}_i' \beta)] \quad (5.209)$$

The score vector $\mathbf{s}_i(\beta)$ is:

$$\mathbf{s}_i(\beta) = \left[\frac{y_i \phi(\mathbf{x}_i' \beta)}{\Phi(\mathbf{x}_i' \beta)} - \frac{(1 - y_i) \phi(\mathbf{x}_i' \beta)}{1 - \Phi(\mathbf{x}_i' \beta)} \right] \mathbf{x}_i \quad (5.210)$$

Reducing to the common denominator:

$$\mathbf{s}_i(\beta) = \frac{[y_i - \Phi(\mathbf{x}_i' \beta)] \phi(\mathbf{x}_i' \beta)}{\Phi(\mathbf{x}_i' \beta)[1 - \Phi(\mathbf{x}_i' \beta)]} \mathbf{x}_i \quad (5.211)$$

Step 2: Numerical Equivalence via Just-Identified GMM

Since the number of moments generated by the score exactly matches the dimension of the parameter vector ($L = K$), the GMM estimator simply zeros the sample moment vector:

$$\mathbf{g}_n(\hat{\beta}_n) = \frac{1}{n} \sum_{i=1}^n \mathbf{s}_i(\hat{\beta}_n) = \mathbf{0} \quad (5.212)$$

Since equation (5.212) represents exactly the First Order Conditions (FOC) of log-likelihood optimization, it follows that $\hat{\beta}_{GMM} = \hat{\beta}_{MLE}$ for any choice of positive definite weighting matrix $\mathbf{W}_n$.

Step 3: Linearization via the Mean Value Theorem

We perform the first-order linear expansion of $\mathbf{g}_n(\hat{\beta}_n)$ around the true point β_0 :

$$\mathbf{0} = \mathbf{g}_n(\hat{\beta}_n) = \mathbf{g}_n(\beta_0) + \mathbf{G}_n(\bar{\beta})(\hat{\beta}_n - \beta_0) \quad (5.213)$$

where $\bar{\beta}$ lies between the estimator and the population target. Multiplying by $\sqrt{n}$ and isolating the sample error:

$$\sqrt{n}(\hat{\beta}_n - \beta_0) = -[\mathbf{G}_n(\bar{\beta})]^{-1} \sqrt{n}\mathbf{g}_n(\beta_0) \quad (5.214)$$

Step 4: Mapping of Stochastic Limits

We define the population limit Jacobian as $\mathbf{G} = E[\partial \mathbf{s}_i(\beta_0)/\partial \beta']$ and the limit variance of the moments as $\mathbf{S} = E[\mathbf{s}_i(\beta_0)\mathbf{s}_i(\beta_0)']$.

- **WLLN:** $\mathbf{G}_n(\bar{\beta}) \xrightarrow{p} \mathbf{G}$.
- **CLT:** $\sqrt{n}\mathbf{g}_n(\beta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{S})$.

Applying Slutsky's Theorem:

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{G}^{-1}\mathbf{S}(\mathbf{G}')^{-1}) \quad (5.215)$$

Step 5: Fisher Information Matrix Identity and Simplification

Since the moment conditions were derived from the analytical score of a perfectly specified density (Assumption 5.46), the Fisher Information Matrix Identity holds, establishing that $\mathbf{S} = -\mathbf{G} = \mathcal{I}(\beta_0)$. Substituting:

$$\text{Asy.Var}(\hat{\beta}_n) = \mathcal{I}(\beta_0)^{-1}\mathcal{I}(\beta_0)\mathcal{I}(\beta_0)^{-1} = \mathcal{I}(\beta_0)^{-1} \quad (5.216)$$

□

5.2.13.4 Role of Assumptions

The assumption of **Rank Identifiability** (**Assumption 5.45**) is strictly indispensable for the validity of the estimator. If the regressors exhibited linear dependence, the population Jacobian matrix $\mathbf{G}$ would be singular. This would prevent the matrix inversion operation performed in Step 3, collapsing the derivation of the Gaussian limiting distribution.

5.3 Advanced Reference Guide: Unification of Estimators under GMM and M-Estimation

This reference guide analyzes the main econometric estimators through the lenses of the Generalized Method of Moments (GMM) and M-Estimation, unifying geometric, algebraic, and asymptotic intuition under a single framework.

5.3.1 Notation and Preliminary Definitions

To ensure rigorous consistency and avoid classical literature conflicts, the following strict symbol convention is adopted:

Symbol	Definition
$\mathbf{y}$	Vector of observations of the dependent variable ($n \times 1$)
$\mathbf{X}$	Matrix of regressors ($n \times k$)
$\mathbf{Z}$	Matrix of instruments ($n \times l$)
$\mathbf{u} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}$	Vector of structural disturbances/errors
$\mathbf{g}_n(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^n \mathbf{g}_i(\boldsymbol{\theta})$	Sample moment function ($l \times 1$). In the linear case: $\mathbf{g}_n(\boldsymbol{\beta}) = \frac{1}{n} \mathbf{Z}'\mathbf{u} = \frac{1}{n} \mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$
$\mathbf{S} = E[\mathbf{g}_i(\boldsymbol{\theta})\mathbf{g}_i(\boldsymbol{\theta})']$	Population covariance matrix of the moment conditions ($l \times l$). Consistent estimator: $\hat{\mathbf{S}}$
$\boldsymbol{\Omega} = E[\mathbf{u}\mathbf{u}' \mid \mathbf{X}]$	Population covariance matrix of the model errors/disturbances ($n \times n$)
$\mathbf{W}$	Positive definite stochastic weight matrix ($l \times l$) used to weight moments in overidentified GMM
$\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'$	Orthogonal projection matrix onto $\text{Col}(\mathbf{Z})$ (instrument <i>hat matrix</i>)

5.3.2 Comparative Table of Estimators

The following table organizes the main econometric estimators according to their formulation as a special case of GMM or M-Estimation, detailing moment conditions, identification structure, weight matrix, closed-form expression (when available), and asymptotic variance.

Table 5.2: Comparative Table of Estimators under GMM and M-Estimation

Estimator	Moment Condition	Identification	Matrices	Weight Matrix	Formula	Derivation	Variance
MM	$E[\mathbf{h}(\mathbf{y}_i, \boldsymbol{\theta})] = 0$	Exactly ($l = k$)	$\mathbf{X}$: Model; $\mathbf{Z}$: N/A	$\mathbf{W} = \mathbf{I}_k$	Implicit: $\mathbf{g}_n(\hat{\boldsymbol{\theta}}) = 0$	F.O.C. yields $\mathbf{g}_n = 0$; $\mathbf{W}$ irrelevant	$\frac{1}{n} \mathbf{G}_0^{-1} \mathbf{S}(\mathbf{G}_0')^{-1}$
OLS	$E[\mathbf{x}_i(y_i - \mathbf{x}_i' \boldsymbol{\beta})] = 0$	Exactly ($l = k$)	$\mathbf{Z} = \mathbf{X}$	$\mathbf{W} = \mathbf{I}_k$	$\hat{\boldsymbol{\beta}}_{OLS} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{y}$	$\frac{1}{n} \mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = 0$	$\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$
IV	$E[\mathbf{z}_i(y_i - \mathbf{x}_i' \boldsymbol{\beta})] = 0$	Exactly ($l = k$)	$\mathbf{Z}$: Instru- ments; $\mathbf{X}$: Regressors	$\mathbf{W} = \mathbf{I}_k$	$\hat{\boldsymbol{\beta}}_{IV} = (\mathbf{Z}'\mathbf{X})^{-1} \mathbf{Z}'\mathbf{y}$	$\frac{1}{n} \mathbf{Z}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = 0$	$\sigma^2(\mathbf{Z}'\mathbf{X})^{-1}(\mathbf{Z}'\mathbf{Z})(\mathbf{X}'\mathbf{Z})^{-1}$
2SLS	$E[\mathbf{z}_i(y_i - \mathbf{x}_i' \boldsymbol{\beta})] = 0$	Overid. ($l > k$)	$\mathbf{Z}$: All in- str.; $\mathbf{X}$: Re- gressors	$\mathbf{W} = (\frac{1}{n} \mathbf{Z}'\mathbf{Z})^{-1}$	$\hat{\boldsymbol{\beta}}_{2SLS} = (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1} \mathbf{X}'\mathbf{P}_Z\mathbf{y}$	Minimizes $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})' \mathbf{P}_Z(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$	$\sigma^2(\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}$
GMM (2- Step)	$E[\mathbf{z}_i(y_i - \mathbf{x}_i' \boldsymbol{\beta})] = 0$	Overid. ($l > k$)	$\mathbf{Z}$: Instru- ments; $\mathbf{X}$: Regressors	$\mathbf{W} = \hat{\mathbf{S}}^{-1}$	$\hat{\boldsymbol{\beta}}_{GMM} = (\mathbf{X}'\hat{\mathbf{Z}}\hat{\mathbf{S}}^{-1}\mathbf{Z}'\mathbf{X})^{-1} \mathbf{X}'\hat{\mathbf{Z}}\hat{\mathbf{S}}^{-1}\mathbf{Z}'\mathbf{y}$	Minimizes $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})' \hat{\mathbf{Z}}\hat{\mathbf{S}}^{-1}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$	$(\mathbf{X}'\hat{\mathbf{Z}}\hat{\mathbf{S}}^{-1}\mathbf{Z}'\mathbf{X})^{-1}$
EHW	$E[\mathbf{x}_i u_i] = 0$	Exactly ($l = k$)	$\mathbf{Z} = \mathbf{X}$	$\mathbf{W} = \mathbf{I}_k$	$\hat{\boldsymbol{\beta}}_{EHW} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{y}$	Same as OLS	$(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\hat{\boldsymbol{\Omega}}_{diag}\mathbf{X})^{-1}$
GLS	$E[\mathbf{x}_i^*(y_i^* - \mathbf{x}_i^{*'} \boldsymbol{\beta})] = 0$	Exactly (trans- formed)	$\mathbf{X}^* = \boldsymbol{\Omega}^{-1/2} \mathbf{X}$; $\mathbf{y}^* = \boldsymbol{\Omega}^{-1/2} \mathbf{y}$	$\mathbf{W} = \mathbf{I}_k$	$\hat{\boldsymbol{\beta}}_{GLS} = (\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1} \mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{y}$	Minimizes $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})' \boldsymbol{\Omega}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$	$(\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}$
FGLS	$E[\mathbf{x}_i u_i / \sigma_i^2] = 0$	Exactly Identified	Same as GLS with $\hat{\boldsymbol{\Omega}}$	$\mathbf{W} = \mathbf{I}_k$	$\hat{\boldsymbol{\beta}}_{FGLS} = (\mathbf{X}'\hat{\boldsymbol{\Omega}}^{-1}\mathbf{X})^{-1} \mathbf{X}'\hat{\boldsymbol{\Omega}}^{-1}\mathbf{y}$	Substitute $\boldsymbol{\Omega}$ with $\hat{\boldsymbol{\Omega}}$ only consistent estimator	$(\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}$ (asymptotically)
3SLS	$E[\mathbf{Z} \otimes \varepsilon] = 0$	Overid. (System)	$\mathbf{Z}$: Block instr.; $\mathbf{X}$: Block-diagonal	$\mathbf{W} = (\hat{\boldsymbol{\Sigma}}^{-1} \otimes (\mathbf{Z}'\mathbf{Z})^{-1})$	$\hat{\boldsymbol{\beta}}_{3SLS} = [\mathbf{X}'(\hat{\boldsymbol{\Sigma}}^{-1} \otimes \mathbf{P}_Z)\mathbf{X}]^{-1} \mathbf{X}'(\hat{\boldsymbol{\Sigma}}^{-1} \otimes \mathbf{P}_Z)\mathbf{y}$	GMM on simultaneous system	$[\mathbf{X}'(\hat{\boldsymbol{\Sigma}}^{-1} \otimes \mathbf{P}_Z)\mathbf{X}]^{-1}$
GNLLS	$E[\nabla_{\boldsymbol{\beta}} h(\mathbf{x}_i, \boldsymbol{\beta}) \cdot \sigma_i^{-2} u_i] = 0$	Exactly Identified	$\mathbf{X}$: Non-linear; $\mathbf{Z} = \nabla_{\boldsymbol{\beta}} h$	$\mathbf{W} = \boldsymbol{\Omega}^{-1}$	Implicit: $\sum_i \nabla h(\mathbf{x}_i, \hat{\boldsymbol{\beta}}) \sigma_i^{-2} (\mathbf{y}_i - \mathbf{h}(\mathbf{x}_i, \hat{\boldsymbol{\beta}})) = 0$	Minimizes $(\mathbf{y} - \mathbf{h})' \boldsymbol{\Omega}^{-1}(\mathbf{y} - \mathbf{h})$	$(\mathbf{G}_0' \boldsymbol{\Omega}^{-1} \mathbf{G}_0)^{-1}$

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Estimator	Moment Condition	Identification	Matrices	Weight Matrix	Formula	Derivation	Variance	
MLE	$E[\mathbf{s}(w_i, \boldsymbol{\theta})] = 0$	N/A (Score-Exact)	N/A (Based on density $f(y \mid \mathbf{x})$)	N/A	$\sum_i \mathbf{s}_i(\hat{\boldsymbol{\theta}}) = \sum_i \frac{\partial \ln f(y_i \mid \mathbf{x}_i, \hat{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta}}$	Maximizes $\sum \ln f_i(\boldsymbol{\theta})$	$\mathcal{I}(\boldsymbol{\theta}_0)^{-1}$ (or $\mathbf{H}^{-1} \mathbf{J} \mathbf{H}^{-1}$ for QMLE)	
Logit	$E[\mathbf{x}_i(y_i - \Lambda(\mathbf{x}_i' \boldsymbol{\beta}))] = 0$	N/A (M-Estimator)	$\mathbf{X}$: Logistic regressors	N/A	$\sum_i \mathbf{x}_i(y_i - \frac{\exp(\mathbf{x}_i' \hat{\boldsymbol{\beta}})}{1 + \exp(\mathbf{x}_i' \hat{\boldsymbol{\beta}})}) = 0$	Bernoulli MLE with logistic CDF	$(\mathbf{X}' \mathbf{V} \mathbf{X})^{-1}$, where $V_{ii} = \Lambda_i(1 - \Lambda_i)$	
Probit	$E[\mathbf{x}_i \frac{\phi_i}{\Phi_i(1 - \Phi_i)}(y_i - \Phi_i)] = 0$	N/A (M-Estimator)	$\mathbf{X}$: Normal CDF regressors	N/A	$\sum_i \mathbf{x}_i \frac{\phi(\mathbf{x}_i' \hat{\boldsymbol{\beta}})[y_i - \Phi(\mathbf{x}_i' \hat{\boldsymbol{\beta}})]}{\Phi(\mathbf{x}_i' \hat{\boldsymbol{\beta}})[1 - \Phi(\mathbf{x}_i' \hat{\boldsymbol{\beta}})]} = 0$	Bernoulli MLE with normal CDF	$(\mathbf{X}' \mathbf{V} \mathbf{X})^{-1}$, where $V_{ii} = \frac{\phi_i^2}{\Phi_i(1 - \Phi_i)}$	

5.3.3 Methodological Observations

1. **GMM as a Unifying Framework:** The Generalized Method of Moments (GMM) subsumes all listed estimators as special cases, depending on the choice of the weighting matrix $\mathbf{W}$ and the definition of the moment conditions. The table above explicitly shows this hierarchy, from traditional MM (exact identification) to optimal GMM (overidentification with efficient weighting).
2. **M-Estimation and the Score:** Estimators based on objective functions (MLE, Logit, Probit) can be reinterpreted as cases of M-Estimation where the moment condition is the gradient of the objective function (score). Under correct specification, the asymptotic variance simplifies to the inverse of the Fisher information matrix; under misspecification, the robust sandwich form (QMLE) is necessary.
3. **Identification and Weight Matrix:** The crucial distinction between exact identification ($l = k$) and overidentification ($l > k$) determines the relevance of the weight matrix $\mathbf{W}$. In exactly identified systems, $\mathbf{W}$ is irrelevant for point estimation; in overidentified systems, the choice of $\mathbf{W}$ affects both efficiency and (in finite samples) point estimation.
4. **Special Case of 2SLS:** 2SLS emerges as the one-step GMM with $\mathbf{W} = (\mathbf{Z}'\mathbf{Z})^{-1}$, which is the optimal weight matrix under homoskedasticity. Under heteroskedasticity, the two-step GMM with $\mathbf{W} = \hat{\mathbf{S}}^{-1}$ is asymptotically more efficient.
5. **Robustness and QMLE:** When parametric specification is doubtful (misspecification of density in MLE, or heteroskedasticity in GMM), the asymptotic variance assumes the sandwich form $\mathbf{H}^{-1}\mathbf{J}\mathbf{H}^{-1}$, ensuring valid inference even under more flexible conditions.
6. **GNLLS and Non-linearity:** In non-linear models, the Jacobian matrix $\mathbf{G}_0 = E[\nabla_{\boldsymbol{\beta}} h(\mathbf{x}_i, \boldsymbol{\beta}_0)]$ acts as the analog of instruments, and the asymptotic variance follows the structure $(\mathbf{G}_0'\boldsymbol{\Omega}^{-1}\mathbf{G}_0)^{-1}$, generalizing GLS to curved surfaces.

Part II

Econometrics II

Chapter 6

Endogeneity: Solutions and Properties

6.1 Introduction: The Endogeneity Problem

6.1.1 The Core Issue: Correlation Between Regressors and Error

When the exogeneity assumption fails, OLS becomes inadequate for causal inference. In the linear model

$$y_i = \mathbf{x}_i' \boldsymbol{\beta} + u_i,$$

exogeneity requires $\mathbb{E}[u_i | \mathbf{x}_i] = 0$ or, in weaker form, $\text{Cov}(\mathbf{x}_i, u_i) = \mathbf{0}$.

When a regressor is correlated with the error (an **endogenous regressor**), OLS attributes to it all variation in y that correlates with it, including variation from unobserved shocks in u .

6.1.2 Inconsistency of OLS

The OLS estimator can be written as

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = \boldsymbol{\beta} + \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1} \left(\frac{\mathbf{X}'\mathbf{u}}{n} \right).$$

If $\text{plim} \frac{\mathbf{X}'\mathbf{u}}{n} = \boldsymbol{\gamma} \neq \mathbf{0}$, then

$$\text{plim} \hat{\boldsymbol{\beta}}_{\text{OLS}} = \boldsymbol{\beta} + \mathbf{Q}^{-1} \boldsymbol{\gamma} \neq \boldsymbol{\beta}.$$

Key insight: Endogeneity causes **inconsistency**. Unlike efficiency problems (heteroskedasticity), the asymptotic bias does not vanish with larger samples.

6.1.3 Classical Sources of Endogeneity

1. **Omitted variables:** Unobserved factors in u affect y and correlate with x (e.g., ability in wage equations).

2. **Measurement error:** Error in measuring x enters the residual and correlates with observed x , typically causing **attenuation bias**.
3. **Simultaneity:** x affects y but y also affects x (e.g., supply and demand equilibrium).

6.1.4 The Modern Framework: Moment Conditions

The modern response to endogeneity is based on **moment conditions**:

$$\mathbb{E}[\mathbf{m}(\mathbf{w}_i; \boldsymbol{\theta}_0)] = \mathbf{0}.$$

Different estimators correspond to different moment conditions:

- **OLS:** $\mathbb{E}[\mathbf{x}_i u_i] = \mathbf{0}$
- **IV:** $\mathbb{E}[\mathbf{z}_i u_i] = \mathbf{0}$
- **GMM:** Minimization of quadratic form of sample moments

6.2 Instrumental Variables and 2SLS

6.2.1 Instrument Validity Conditions

For an instrument $\mathbf{z}_i$ to be valid, two conditions are necessary:

Definition 6.2.1 (Relevance). The instrument must be correlated with the endogenous regressor:

$$\text{Cov}(\mathbf{z}_i, \mathbf{x}_i) \neq \mathbf{0}.$$

Empirical check: First-stage F -statistic. Rule of thumb: $F > 10$, though Stock-Yogo critical values are more appropriate.

Definition 6.2.2 (Exogeneity and Exclusion Restriction). The instrument must be orthogonal to the structural error:

$$\mathbb{E}[\mathbf{z}_i u_i] = \mathbf{0}.$$

Nature: Not directly testable in exactly identified models. Depends on economic theory and institutional design.

6.2.2 Derivation of IV and 2SLS

Let $\mathbf{Z}$ be the instrument matrix. The first stage projects $\mathbf{X}$ onto $\mathbf{Z}$:

$$\widehat{\mathbf{X}} = \mathbf{P}_Z \mathbf{X}, \quad \text{where} \quad \mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'.$$

The second stage regresses $\mathbf{y}$ on $\widehat{\mathbf{X}}$:

$$\hat{\boldsymbol{\beta}}_{2SLS} = (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_Z\mathbf{y}.$$

For exactly identified models ($L = K$), IV coincides with 2SLS:

$$\hat{\beta}_{IV} = (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{y}.$$

Important: Do not manually run two stages and use second-stage standard errors. Use dedicated IV/2SLS routines.

6.2.3 Asymptotic Properties

Theorem 6.2.1 (Consistency of IV). *Under relevance and exogeneity:*

$$\hat{\beta}_{IV} \xrightarrow{p} \beta.$$

Theorem 6.2.2 (Asymptotic Normality of 2SLS). *Under regularity conditions:*

$$\sqrt{n}(\hat{\beta}_{2SLS} - \beta) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{V}).$$

Trade-off: IV variance is typically larger than OLS variance (when OLS would be valid), but IV provides consistency.

6.2.4 Diagnostics and Tests

6.2.4.1 Weak Instruments

Weak instruments cause:

- Finite-sample bias (potentially approaching OLS bias)
- Poor normal approximation
- Unreliable t -tests and confidence intervals

Alternatives: LIML, Fuller estimator, Anderson-Rubin inference.

6.2.4.2 Hausman Test for Endogeneity

Compare OLS and IV estimators:

$$H = (\hat{\beta}_{IV} - \hat{\beta}_{OLS})' [\widehat{\text{Var}}(\hat{\beta}_{IV}) - \widehat{\text{Var}}(\hat{\beta}_{OLS})]^{-1} (\hat{\beta}_{IV} - \hat{\beta}_{OLS}) \xrightarrow{d} \chi_k^2.$$

6.2.4.3 Overidentification Test: Sargan-Hansen J

For $L > K$:

$$J = n \mathbf{g}_n(\hat{\beta})' \widehat{\mathbf{W}} \mathbf{g}_n(\hat{\beta}) \xrightarrow{d} \chi_{L-K}^2.$$

Limitation: Non-rejection does not prove all instruments are valid.

6.3 System Estimation: SIV and 3SLS

6.3.1 Motivation: Interdependence Between Equations

In simultaneous equations systems, endogenous variables from one equation may be jointly determined with variables from other equations. **System IV (SIV)** and **3SLS** combine IV's endogeneity correction with SUR's systemic efficiency.

6.3.2 Why Estimate the System?

1. **Efficiency:** If errors are contemporaneously correlated, the matrix Σ contains useful information. 3SLS exploits this to reduce variances.
2. **Cross-equation restrictions:** Economic theory may impose restrictions across equations. System estimation allows testing these restrictions.

Cost: Greater fragility. If one equation is misspecified, the problem can contaminate others.

6.3.3 The 3SLS Estimator

The 3SLS estimator combines three steps:

1. Project endogenous regressors onto instruments
2. Estimate contemporaneous covariance matrix Σ
3. Apply GLS to the instrumented system

The estimator is:

$$\hat{\beta}_{3SLS} = \left[\mathbf{X}'(\hat{\Sigma}^{-1} \otimes \mathbf{P}_Z)\mathbf{X} \right]^{-1} \mathbf{X}'(\hat{\Sigma}^{-1} \otimes \mathbf{P}_Z)\mathbf{y}.$$

6.3.4 When Does 3SLS Gain Efficiency?

3SLS is more efficient than equation-by-equation 2SLS when:

- Errors are contemporaneously correlated
- The system is overidentified
- Systemic exogeneity assumptions hold

No gain when: Σ is diagonal, all equations are exactly identified, or specification fails.

6.4 Generalized Method of Moments (GMM)

6.4.1 Why GMM?

GMM generalizes moment-based estimators and is especially useful when:

1. There is overidentification ($L > K$)
2. There is unknown heteroskedasticity or autocorrelation
3. One wants to choose an optimal weighting matrix
4. One wants to unify different estimators under the same logic

Key insight: 2SLS is a special case of GMM with a specific weight matrix.

6.4.2 General Formulation

Suppose the model implies population moment conditions:

$$\mathbb{E}[\mathbf{f}(\mathbf{w}_i, \boldsymbol{\theta}_0)] = \mathbf{0}.$$

The GMM estimator minimizes:

$$\hat{\boldsymbol{\theta}}_{GMM} = \arg \min_{\boldsymbol{\theta}} \mathbf{g}_n(\boldsymbol{\theta})' \mathbf{W}_n \mathbf{g}_n(\boldsymbol{\theta}),$$

where $\mathbf{g}_n(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^n \mathbf{f}(\mathbf{w}_i, \boldsymbol{\theta})$ and $\mathbf{W}_n$ is a positive definite weight matrix.

6.4.3 Optimal Weight Matrix and Efficient GMM

The optimal weight matrix is $\mathbf{W}_{\text{opt}} = \mathbf{S}^{-1}$, where $\mathbf{S}$ is the asymptotic variance of the moments.

Two-step GMM procedure:

1. Estimate with simple weight matrix (e.g., $\mathbf{W} = \mathbf{I}$)
2. Use residuals to construct $\hat{\mathbf{S}}$
3. Re-estimate with $\mathbf{W}_n = \hat{\mathbf{S}}^{-1}$

6.4.4 Asymptotic Properties

Theorem 6.4.1 (Consistency of GMM). *Under identification and moment validity:*

$$\hat{\boldsymbol{\theta}}_{GMM} \xrightarrow{p} \boldsymbol{\theta}_0.$$

Theorem 6.4.2 (Asymptotic Normality of GMM). *Under regularity conditions:*

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_{GMM} - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{V}),$$

where

$$\mathbf{V} = (\mathbf{G}'\mathbf{W}\mathbf{G})^{-1}\mathbf{G}'\mathbf{W}\mathbf{S}\mathbf{W}\mathbf{G}(\mathbf{G}'\mathbf{W}\mathbf{G})^{-1}.$$

With optimal weight $\mathbf{W} = \mathbf{S}^{-1}$:

$$\mathbf{V}_{\text{opt}} = (\mathbf{G}'\mathbf{S}^{-1}\mathbf{G})^{-1}.$$

6.4.5 GMM as a Unifying Framework

6.5 Synthesis: Choosing an Estimator

6.5.1 Comparison Table

6.5.2 Decision Guide

- Single equation with endogeneity and valid instruments: Use IV/2SLS.

Estimator	Typical Moments	Identification	Weight Matrix
OLS	$\mathbf{x}_i u_i$	Exact, if $\dim(x) = \dim(\beta)$	Irrelevant
IV	$\mathbf{z}_i u_i$	Exact, if $L = K$	Irrelevant
2SLS	$\mathbf{z}_i u_i$	Over, if $L > K$	$(\mathbf{Z}'\mathbf{Z})^{-1}$
GLS/SUR	Moments weighted by $\mathbf{\Omega}^{-1}$	System	Error covariance structure
MLE	Likelihood score	Model-dependent	Fisher information

Table 6.1: Classical estimators as special cases of moment logic.

Aspect	IV / 2SLS	SIV / 3SLS	GMM
Scope	Single equation	System of equations	One or multiple equations
Handles endogeneity?	Yes, via instruments	Yes, via systemic instruments	Yes, via moment conditions
Uses cross-equation error correlation?	No	Yes	Can use
Weight matrix	Implicit: $(\mathbf{Z}'\mathbf{Z})^{-1}$	Based on $\mathbf{\Sigma}^{-1}$	Optimal: $\mathbf{S}^{-1}$
Efficiency	Under homoskedasticity	Under systemic homoskedasticity	Under optimal weighting
Robustness	Greater than 3SLS	Lower, full information	Depends on moments and weights
Main risk	Weak instruments	Systemic contamination	Small samples and many moments

Table 6.2: Comparison between IV/2SLS, SIV/3SLS, and GMM.

- **Suspected heteroskedasticity and overidentification:** Consider efficient GMM.
- **Simultaneous equations system with reliable instruments:** 3SLS may gain efficiency.
- **Doubt about specification of some equation:** Prefer equation-by-equation 2SLS to avoid systemic contamination.
- **Weak instruments:** Use LIML, Fuller, or Anderson-Rubin robust inference.
- **Many instruments:** Reduce moment set and report robustness.
- **Dynamic panel model:** Consider Arellano-Bond or Blundell-Bond with appropriate tests.

6.5.3 Conclusion

Endogeneity breaks the link between correlation and causality in OLS. The econometric response is to replace the regressor exogeneity assumption with moment conditions based on instruments or other sources of exogenous variation.

- **IV/2SLS** solves endogeneity in single equations.
- **SIV/3SLS** extends this logic to systems, exploiting error correlation for efficiency gains.
- **GMM** unifies all these methods as solutions to weighted moment minimization problems.

The practical choice depends on trade-offs between:

- Consistency
- Efficiency
- Robustness to misspecification
- Instrument strength
- Sample size
- Number of moments

Key takeaway: In applied research, it is not enough to estimate; one must defend identification. Causal credibility depends less on estimator sophistication and more on the plausibility of the moment conditions that support it.

Chapter 7

Efficiency: Sphericity and Its Violations

7.1 Introduction: Identification First, Efficiency Second

7.1.1 The Common Problem of This Block

In previous blocks, the focus was **identification**: ensuring that

$$\mathbb{E}[\mathbf{x}_i u_i] = \mathbf{0},$$

so that the parameter β is recoverable from observable data. Once consistency is guaranteed, the question changes nature:

The parameter is identified. Is it being estimated with the smallest possible variance?

This block deals exclusively with **efficiency**. It does not solve endogeneity problems. If $\mathbb{E}[\mathbf{u} | \mathbf{X}] \neq \mathbf{0}$, no procedure in this block restores consistency. What this block does is correct the **error variance-covariance matrix** when it ceases to be spherical, seeking estimators with lower variance and valid inference.

7.1.2 The Target Analogy

In the identification block, the concern was ensuring shots were centered on the target (absence of bias). In this block, assuming the target is already centered, the concern is reducing shot dispersion (variance). A consistent but inefficient estimator produces wide confidence intervals, reducing statistical power and potentially leading to non-rejection of false hypotheses.

7.1.3 Motivating Example

Consider a study on the effect of infrastructure spending on municipal GDP. The data exhibit two simultaneous violations of sphericity:

- Larger municipalities have higher error variance (heteroskedasticity).

- Management shocks persist over time within the same municipality (autocorrelation).

The OLS estimator remains consistent under $\mathbb{E}[\mathbf{u} \mid \mathbf{X}] = \mathbf{0}$, but:

1. It is not the minimum variance estimator (not BLUE).
2. The classical formula $\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}$ is an inconsistent estimator of the true variance.
3. Conventional t and F tests have incorrect size, typically rejecting H_0 too frequently (Type I Error).

7.1.4 The Assumption Being Relaxed

This entire block revolves around relaxing the **spherical errors** assumption:

$$\text{Var}(\mathbf{u} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n. \quad (7.1)$$

The generalization is:

$$\text{Var}(\mathbf{u} \mid \mathbf{X}) = \sigma^2 \mathbf{\Omega}, \quad \mathbf{\Omega} \neq \mathbf{I}_n, \quad (7.2)$$

where $\mathbf{\Omega}$ is symmetric, positive definite, and captures the structure of heteroskedasticity, autocorrelation, cross-equation correlation, or spatial dependence.

7.2 Error Sphericity: Formal Definition

7.2.1 Definition

Definition 7.2.1 (Error Sphericity). Let $\mathbf{u} = (u_1, u_2, \dots, u_n)'$ be the $n \times 1$ vector of errors from the linear model, conditional on the regressor matrix $\mathbf{X}$. The sphericity hypothesis states:

$$\text{Var}(\mathbf{u} \mid \mathbf{X}) = \mathbb{E}[\mathbf{u}\mathbf{u}' \mid \mathbf{X}] = \sigma^2 \mathbf{I}_n, \quad (7.3)$$

where $\sigma^2 > 0$ is a finite scalar and $\mathbf{I}_n$ is the identity matrix of order n .

This condition is equivalent to two simultaneous restrictions:

1. **Homoskedasticity** (constant diagonal):

$$\mathbb{E}[u_i^2 \mid \mathbf{X}] = \sigma^2, \quad \forall i = 1, \dots, n. \quad (7.4)$$

2. **Absence of correlation** (zeros off-diagonal):

$$\mathbb{E}[u_i u_j \mid \mathbf{X}] = 0, \quad \forall i \neq j. \quad (7.5)$$

7.2.2 Role in the Gauss-Markov Theorem

Under the assumptions of linearity, strict exogeneity, full rank, and sphericity, the Gauss-Markov Theorem guarantees that the OLS estimator is the **Best Linear Unbiased Estimator** (BLUE):

$$\text{Var}(\hat{\boldsymbol{\beta}}_{\text{OLS}} \mid \mathbf{X}) \preceq \text{Var}(\tilde{\boldsymbol{\beta}} \mid \mathbf{X}) \quad (7.6)$$

for any other linear unbiased estimator $\tilde{\boldsymbol{\beta}}$ (in the Loewner sense: the difference is positive semidefinite).

When sphericity fails, OLS remains **consistent** under $\mathbb{E}[\mathbf{u} \mid \mathbf{X}] = \mathbf{0}$, but ceases to be BLUE and classical inference becomes invalid.

7.2.3 What Changes in the Variance-Covariance Matrix

The table below summarizes how each violation affects the structure of Ω :

Violation	Effect on Diagonal	Effect Off-Diagonal
Heteroskedasticity	Distinct diagonal elements	Zeros maintained
Temporal autocorrelation	Constant diagonal (stationary)	Non-zero elements between periods
Compound error in panel	Diagonal: $\sigma_c^2 + \sigma_u^2$	Constant covariance σ_c^2 within i
Cross-equation correlation	Diagonal: variance by equation	Covariance between equations at same t
Spatial dependence	Constant or varying diagonal	Covariance between neighboring units

Table 7.1: Structure of Ω under different sphericity violations.

7.3 GLS with Known Covariance Matrix

7.3.1 Model and Assumptions

Consider the linear model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad (7.7)$$

with $\mathbf{y}$ of dimension $n \times 1$, $\mathbf{X}$ of dimension $n \times K$, $\boldsymbol{\beta}$ of dimension $K \times 1$, and $\mathbf{u}$ of dimension $n \times 1$.

The assumptions are:

Definition 7.3.1 (GLS.1: Strict Exogeneity).

$$\mathbb{E}[\mathbf{u} \mid \mathbf{X}] = \mathbf{0}. \quad (7.8)$$

Role: guarantee consistency and unbiasedness.

Definition 7.3.2 (GLS.2: Non-Spherical Covariance Structure).

$$\text{Var}(\mathbf{u} \mid \mathbf{X}) = \sigma^2 \Omega, \quad (7.9)$$

where Ω is $n \times n$, symmetric, positive definite, and **known**. **Role:** generalize Gauss-Markov to non-spherical errors.

Definition 7.3.3 (GLS.3: Full Rank).

$$\text{rank}(\mathbf{X}) = K. \quad (7.10)$$

Role: guarantee identification and invertibility.

7.3.2 Aitken's Theorem

Theorem 7.3.1 (Aitken, 1935). *Under GLS.1, GLS.2, and GLS.3, the minimum variance estimator in the class of linear unbiased estimators is:*

$$\hat{\boldsymbol{\beta}}_{GLS} = (\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1} \mathbf{X}'\Omega^{-1}\mathbf{y}. \quad (7.11)$$

7.3.3 Proof: Spherification Transformation

Step 1: Existence of the spherification matrix.

Since Ω is symmetric and positive definite (by GLS.2), there exists a symmetric square root $\Omega^{1/2}$ such that $\Omega^{1/2}\Omega^{1/2} = \Omega$. Define:

$$\mathbf{C} = \Omega^{-1/2}. \quad (7.12)$$

Then $\mathbf{C}$ is symmetric and satisfies:

$$\mathbf{C}\Omega\mathbf{C}' = \Omega^{-1/2}\Omega\Omega^{-1/2} = \mathbf{I}_n, \quad (7.13)$$

(by the property $\Omega^{-1/2}\Omega^{1/2} = \mathbf{I}$ and $\Omega^{1/2}\Omega^{-1/2} = \mathbf{I}$).

Moreover:

$$\mathbf{C}'\mathbf{C} = \Omega^{-1/2}\Omega^{-1/2} = \Omega^{-1}, \quad (7.14)$$

(since $\mathbf{C}$ is symmetric, so $\mathbf{C}' = \mathbf{C}$).

Step 2: Model transformation.

Pre-multiplying the model by $\mathbf{C}$:

$$\mathbf{C}\mathbf{y} = \mathbf{C}\mathbf{X}\beta + \mathbf{C}\mathbf{u}. \quad (7.15)$$

Defining $\mathbf{y}^* = \mathbf{C}\mathbf{y}$, $\mathbf{X}^* = \mathbf{C}\mathbf{X}$, and $\mathbf{u}^* = \mathbf{C}\mathbf{u}$:

$$\mathbf{y}^* = \mathbf{X}^*\beta + \mathbf{u}^*. \quad (7.16)$$

Step 3: Verification of transformed error sphericity.

$$\text{Var}(\mathbf{u}^* | \mathbf{X}) = \text{Var}(\mathbf{C}\mathbf{u} | \mathbf{X}) \quad (7.17)$$

$$= \mathbf{C} \text{Var}(\mathbf{u} | \mathbf{X}) \mathbf{C}' \quad (\text{by linearity of variance operator}) \quad (7.18)$$

$$= \mathbf{C}(\sigma^2\Omega)\mathbf{C}' \quad (\text{by GLS.2}) \quad (7.19)$$

$$= \sigma^2\mathbf{C}\Omega\mathbf{C}' \quad (\text{extracting scalar } \sigma^2) \quad (7.20)$$

$$= \sigma^2\mathbf{I}_n \quad (\text{by definition of } \mathbf{C} \text{ in Step 1}). \quad (7.21)$$

Therefore, the transformed model satisfies all Gauss-Markov assumptions (linearity, exogeneity, full rank, and spherical errors).

Step 4: Application of OLS to the transformed model.

By the Gauss-Markov Theorem, OLS applied to $(\mathbf{y}^*, \mathbf{X}^*)$ is BLUE:

$$\hat{\beta} = (\mathbf{X}^{*'}\mathbf{X}^*)^{-1}\mathbf{X}^{*'}\mathbf{y}^*. \quad (7.22)$$

Substituting the definitions:

$$\hat{\beta} = [(\mathbf{C}\mathbf{X})'(\mathbf{C}\mathbf{X})]^{-1}(\mathbf{C}\mathbf{X})'(\mathbf{C}\mathbf{y}) \quad (\text{by substitution}) \quad (7.23)$$

$$= (\mathbf{X}'\mathbf{C}'\mathbf{C}\mathbf{X})^{-1}\mathbf{X}'\mathbf{C}'\mathbf{C}\mathbf{y} \quad (\text{transposing the product}) \quad (7.24)$$

$$= (\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\mathbf{X}'\Omega^{-1}\mathbf{y} \quad (\text{since } \mathbf{C}'\mathbf{C} = \Omega^{-1} \text{ by Step 1}). \quad (7.25)$$

Indispensable assumption: Ω known. If Ω is estimated, finite-sample properties (exact unbiasedness) are lost, and we enter the domain of FGLS (Section 7.5).

7.3.4 Efficiency Comparison: GLS versus OLS

Under GLS.2, the conditional variances are:

$$\text{Var}(\hat{\beta}_{\text{OLS}} | \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{\Omega}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}, \quad (7.26)$$

$$\text{Var}(\hat{\beta}_{\text{GLS}} | \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})^{-1}. \quad (7.27)$$

The difference satisfies:

$$\text{Var}(\hat{\beta}_{\text{OLS}} | \mathbf{X}) - \text{Var}(\hat{\beta}_{\text{GLS}} | \mathbf{X}) \succeq \mathbf{0}, \quad (7.28)$$

where $\succeq \mathbf{0}$ denotes positive semidefiniteness in the Loewner sense (by Aitken's Theorem).

Degenerate case: if $\mathbf{\Omega} = \mathbf{I}_n$, then GLS coincides exactly with OLS. More generally, GLS and OLS coincide if the columns of $\mathbf{X}$ are eigenvectors of $\mathbf{\Omega}$ (since in this case the weighting does not alter the projection).

7.3.5 Asymptotic Properties of GLS

- **Consistency:** $\hat{\beta}_{\text{GLS}} \xrightarrow{p} \beta$, guaranteed by GLS.1 and the Law of Large Numbers applied to transformed moments.
- **Asymptotic normality:**

$$\sqrt{n}(\hat{\beta}_{\text{GLS}} - \beta) \xrightarrow{d} \mathcal{N}\left(\mathbf{0}, \sigma^2 \left(\text{plim} \frac{1}{n} \mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X}\right)^{-1}\right). \quad (7.29)$$

- **Efficiency:** GLS attains the lowest asymptotic variance among all linear estimators that weight by some positive definite matrix.

7.4 Sources of Non-Sphericity

7.4.1 Heteroskedasticity

7.4.1.1 Definition

Heteroskedasticity keeps $\mathbf{\Omega}$ diagonal, but with distinct elements:

$$\text{Var}(\mathbf{u} | \mathbf{X}) = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2) = \sigma^2 \mathbf{\Omega}, \quad (7.30)$$

where $\omega_{ii} = h(\mathbf{x}_i)$ for some function $h > 0$.

7.4.1.2 Where It Appears

- **Cross-section with heterogeneous scale:** household spending versus income; multinational versus small firm profits.
- **Aggregate data:** municipal averages calculated over populations of different sizes. The variance of the mean is inversely proportional to group size.
- **Group heteroskedasticity:** each industrial sector has its own variance, but errors are homoskedastic within each sector.

7.4.1.3 Consequences for OLS

The OLS estimator remains consistent under $\mathbb{E}[\mathbf{u} \mid \mathbf{X}] = \mathbf{0}$, but:

$$\text{Var}(\hat{\beta}_{\text{OLS}} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{\Sigma}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}, \quad (7.31)$$

where $\mathbf{\Sigma} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$.

The classical formula $\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}$ is only valid if $\mathbf{\Sigma} = \sigma^2\mathbf{I}$. Otherwise, t and F tests have incorrect size.

7.4.1.4 Correction via WLS

If $h(\mathbf{x}_i)$ is known, the WLS estimator weights each observation by $1/h(\mathbf{x}_i)$:

$$\hat{\beta}_{\text{WLS}} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{y}, \quad (7.32)$$

where $\mathbf{W} = \text{diag}(1/h(\mathbf{x}_1), \dots, 1/h(\mathbf{x}_n))$.

Degenerate case: if $h(\mathbf{x}_i) = 1$ for all i , WLS coincides with OLS.

7.4.2 Temporal Autocorrelation

7.4.2.1 Definition

Autocorrelation fills the off-diagonal elements of $\mathbf{\Omega}$. The most common case is the AR(1) process:

$$u_t = \rho u_{t-1} + \varepsilon_t, \quad |\rho| < 1, \quad (7.33)$$

where ε_t is white noise with variance σ_ε^2 .

7.4.2.2 Covariance Matrix Structure

For the stationary AR(1) process:

$$\text{Var}(u_t) = \frac{\sigma_\varepsilon^2}{1 - \rho^2}, \quad (7.34)$$

$$\text{Cov}(u_t, u_{t-k}) = \rho^k \text{Var}(u_t). \quad (7.35)$$

The covariance matrix $\mathbf{V} = \mathbb{E}[\mathbf{u}\mathbf{u}']$ is:

$$\mathbf{V} = \frac{\sigma_\varepsilon^2}{1 - \rho^2} \begin{bmatrix} 1 & \rho & \rho^2 & \dots & \rho^{n-1} \\ \rho & 1 & \rho & \dots & \rho^{n-2} \\ \rho^2 & \rho & 1 & \dots & \rho^{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{n-1} & \rho^{n-2} & \rho^{n-3} & \dots & 1 \end{bmatrix}. \quad (7.36)$$

Indispensable assumption: $|\rho| < 1$ (stationarity). If $\rho = 1$ (unit root), the variance explodes and modeling in levels is invalid; one must work in first differences.

7.4.2.3 Correction: Prais-Winsten and Cochrane-Orcutt

Both are FGLS procedures that estimate ρ from OLS residuals and apply the transformation:

$$y_t^* = y_t - \hat{\rho}y_{t-1}, \quad \mathbf{x}_t^* = \mathbf{x}_t - \hat{\rho}\mathbf{x}_{t-1}. \quad (7.37)$$

The difference is that Prais-Winsten preserves the first observation (special transformation), while Cochrane-Orcutt discards it.

Degenerate case: if $\rho = 0$, the transformation is null and GLS coincides with OLS.

7.4.3 Compound Error in Panel: Equicorrelation

7.4.3.1 Definition

In the panel model with compound error:

$$v_{it} = c_i + u_{it}, \quad (7.38)$$

the covariance matrix for each unit i is:

$$\mathbf{\Omega}_i = \sigma_u^2 \mathbf{I}_T + \sigma_c^2 \mathbf{j}_T \mathbf{j}_T', \quad (7.39)$$

where $\mathbf{j}_T$ is the $T \times 1$ vector of ones.

This implies:

$$\text{Cov}(v_{it}, v_{is}) = \sigma_c^2, \quad t \neq s, \quad (7.40)$$

(due to the constancy of c_i over time).

7.4.3.2 Correction: RE as FGLS

The Random Effects (RE) estimator is FGLS applied to this structure. The transformation is *quasi-demeaning*:

$$y_{it}^* = y_{it} - \theta \bar{y}_i, \quad (7.41)$$

where

$$\theta = 1 - \left(\frac{\sigma_u^2}{\sigma_u^2 + T\sigma_c^2} \right)^{1/2}. \quad (7.42)$$

Limiting cases:

- $\sigma_c^2 = 0$: $\theta = 0$, RE collapses to POLS.
- $\sigma_c^2 \rightarrow \infty$: $\theta \rightarrow 1$, RE converges to FE.

7.4.4 Cross-Equation Correlation (SUR)

In equation systems, errors from different equations may be correlated in the same period:

$$\mathbf{\Omega} = \mathbf{\Sigma} \otimes \mathbf{I}_T, \quad (7.43)$$

where $\mathbf{\Sigma}$ is $M \times M$ and captures contemporaneous correlation between the M equations.

The SUR estimator is GLS applied to the stacked system. If $\mathbf{\Sigma}$ is diagonal, SUR collapses to equation-by-equation OLS. If regressors are identical across all equations, SUR also coincides with equation-by-equation OLS.

7.4.5 Spatial Dependence and Cross-Sectional Correlation

When units are neighbors or suffer common shocks:

$$\text{Cov}(u_i, u_j \mid \mathbf{X}) \neq 0, \quad i \neq j. \quad (7.44)$$

The structure can be parameterized by a spatial weights matrix $\mathbf{W}$:

$$\mathbf{\Omega} = (\mathbf{I} - \lambda \mathbf{W})^{-1} (\mathbf{I} - \lambda \mathbf{W})^{-\prime} \sigma^2, \quad (7.45)$$

in spatial error models (SEM).

Alternatively, one can use cluster-robust or double-clustering to allow arbitrary correlation within groups.

7.4.6 Combined Case: Panel with Heteroskedasticity Between Units and Autocorrelation Within Each Unit

This is the most common structure in applied microeconometrics. The stacked covariance matrix is block-diagonal:

$$\mathbf{\Phi} = \text{diag}(\mathbf{\Sigma}_1, \mathbf{\Sigma}_2, \dots, \mathbf{\Sigma}_N), \quad (7.46)$$

where each $\mathbf{\Sigma}_i$ is $T \times T$, symmetric, and positive definite.

Definition 7.4.1 (Independence Between Units).

$$\mathbb{E}[\mathbf{u}_i \mathbf{u}_j' \mid \mathbf{X}] = \mathbf{0}, \quad i \neq j. \quad (7.47)$$

Role: guarantee the block-diagonal structure. **Counterexample:** spillovers between neighboring municipalities violate this condition.

The GLS estimator with this structure is:

$$\hat{\beta} = \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{\Sigma}_i^{-1} \mathbf{X}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{\Sigma}_i^{-1} \mathbf{y}_i \right). \quad (7.48)$$

Practical problem: estimating unrestricted $\mathbf{\Sigma}_i$ requires $T(T+1)/2$ parameters per unit. With fixed T , this is impossible. In practice, one assumes $\mathbf{\Sigma}_i = \mathbf{\Sigma}$ for all i , or uses cluster-robust.

7.5 FGLS: When the Covariance Matrix is Unknown

7.5.1 Definition

In applied research, $\mathbf{\Omega}$ is almost always unknown. The FGLS estimator replaces $\mathbf{\Omega}$ with an estimate $\hat{\mathbf{\Omega}}$:

$$\hat{\beta}_{\text{FGLS}} = \left(\mathbf{X}' \hat{\mathbf{\Omega}}^{-1} \mathbf{X} \right)^{-1} \mathbf{X}' \hat{\mathbf{\Omega}}^{-1} \mathbf{y}. \quad (7.49)$$

7.5.2 Asymptotic Equivalence with GLS

Theorem 7.5.1 (Asymptotic Equivalence FGLS-GLS). *If $\hat{\Omega} \xrightarrow{p} \Omega$ and regularity conditions are satisfied, then:*

$$\sqrt{n}(\hat{\beta}_{FGLS} - \beta) \xrightarrow{d} \mathcal{N}\left(\mathbf{0}, \sigma^2 \left(\text{plim} \frac{1}{n} \mathbf{X}' \Omega^{-1} \mathbf{X}\right)^{-1}\right), \quad (7.50)$$

the same limiting distribution as GLS with known Ω .

Proof sketch:

The scaled estimation error is:

$$\sqrt{n}(\hat{\beta}_{FGLS} - \beta) = \left(\frac{1}{n} \mathbf{X}' \hat{\Omega}^{-1} \mathbf{X}\right)^{-1} \left(\frac{1}{\sqrt{n}} \mathbf{X}' \hat{\Omega}^{-1} \mathbf{u}\right). \quad (7.51)$$

Since $\hat{\Omega}^{-1} \xrightarrow{p} \Omega^{-1}$ (by the Continuous Mapping Theorem), the first factor converges to $(\text{plim} n^{-1} \mathbf{X}' \Omega^{-1} \mathbf{X})^{-1}$ (by Slutsky).

For the second factor, decompose:

$$\frac{1}{\sqrt{n}} \mathbf{X}' \hat{\Omega}^{-1} \mathbf{u} = \frac{1}{\sqrt{n}} \mathbf{X}' \Omega^{-1} \mathbf{u} + \frac{1}{\sqrt{n}} \mathbf{X}' (\hat{\Omega}^{-1} - \Omega^{-1}) \mathbf{u}. \quad (7.52)$$

The second term converges in probability to zero if $\hat{\Omega}^{-1} - \Omega^{-1} = o_p(1)$ and the convergence rate is adequate (under regularity conditions). By Slutsky's Theorem, the limiting distribution is identical to that of GLS.

Indispensable assumption: $\hat{\Omega}$ consistent. If $\text{plim} \hat{\Omega} \neq \Omega$, FGLS remains consistent (under $\mathbb{E}[\mathbf{u} | \mathbf{X}] = \mathbf{0}$), but loses the maximum efficiency property.

7.5.3 Finite-Sample Risk

In small samples, the estimate $\hat{\Omega}$ is noisy. FGLS may have **higher** variance than robust OLS. The mean squared error is:

$$\text{MSE}(\hat{\beta}) = \text{Var}(\hat{\beta}) + [\text{Bias}(\hat{\beta})]^2. \quad (7.53)$$

If the modeling of Ω is incorrect, the variance gain may be dominated by the bias introduced by incorrect weighting.

7.6 Applications and Special Cases: Mapping

7.6.1 Observation on 3SLS

In 3SLS, endogeneity is resolved by instruments (as in 2SLS). Weighting by $\hat{\Sigma}^{-1}$ only addresses **efficiency**, exploiting correlation between equations. If instruments are invalid, 3SLS is inconsistent regardless of the structure of Ω .

Estimator	Structure of Ω	Feasible Step	Main Source
WLS	Diagonal: $\omega_{ii} = h(\mathbf{x}_i)$	Estimate $h(\cdot)$ via residuals	Heteroskedasticity
Prais-Winsten	AR(1): $\omega_{ts} = \rho^{ t-s }/(1-\rho^2)$	Estimate ρ via OLS	Temporal autocorrelation
RE (panel)	$\sigma_u^2 \mathbf{I}_T + \sigma_c^2 \mathbf{j}_T \mathbf{j}_T'$	Estimate σ_u^2, σ_c^2 via FE	Compound error / equicorrelation
SUR	$\Sigma \otimes \mathbf{I}_T$	Estimate Σ via OLS residuals	Cross-equation correlation
3SLS	$\Sigma \otimes \mathbf{I}_T + \text{instruments}$	2SLS residuals estimate $\hat{\Sigma}$	Endogeneity + cross-equation correlation
Spatial	$f(\mathbf{W}, \lambda)$	Estimate λ via ML or GMM	Spatial dependence

Table 7.2: Sources of non-sphericity and use of FGLS in different estimators.

7.6.2 Observation on RE

The RE estimator requires $\mathbb{E}[c_i | \mathbf{X}_i] = 0$. If this condition fails, RE is inconsistent and FE should be preferred. The structure of Ω in RE is an assumption about the **form** of correlation, not about **exogeneity**.

7.7 Diagnosing Sphericity Breakdown

7.7.1 Heteroskedasticity

7.7.1.1 Breusch-Pagan Test

Procedure:

1. Estimate the model by OLS and obtain residuals $\hat{u}_i$.
2. Regress $\hat{u}_i^2$ on a set of variables $\mathbf{z}_i$ (typically the regressors themselves or functions thereof).
3. Compute $LM = n \cdot R^2$ from the auxiliary regression.

Under H_0 : homoskedasticity,

$$LM \xrightarrow{d} \chi_q^2, \quad (7.54)$$

where q is the number of regressors in the auxiliary regression (by LM test theory).

7.7.1.2 White Test

Regress $\hat{u}_i^2$ on $\mathbf{x}_i$, $\mathbf{x}_i^2$, and interactions. It is consistent against any form of heteroskedasticity that correlates u^2 with functions of $\mathbf{x}$, but loses degrees of freedom rapidly.

7.7.1.3 Caution

Rejection of BP or White may indicate **functional misspecification** (omitted variable, incorrect functional form), not just heteroskedasticity. Ramsey's RESET test should be applied before concluding the problem is exclusively about variance.

7.7.2 Temporal Autocorrelation

7.7.2.1 Durbin-Watson

Tests AR(1) in models **without** lagged dependent variable. The statistic is:

$$DW = \frac{\sum_{t=2}^n (\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^n \hat{u}_t^2}. \quad (7.55)$$

$DW \approx 2$: absence of autocorrelation. $DW < 2$: positive autocorrelation. $DW > 2$: negative autocorrelation.

Limitation: invalid when y_{t-1} is a regressor. In this case, use Durbin h or Breusch-Godfrey.

7.7.2.2 Breusch-Godfrey

Regress $\hat{u}_t$ on the original regressors and $\hat{u}_{t-1}, \dots, \hat{u}_{t-p}$. Tests higher orders and allows stochastic regressors.

7.7.3 Autocorrelation in Panel

Wooldridge's test for panel examines serial correlation of residuals in first differences. Under H_0 of no serial correlation in u_{it} , the differenced residuals $\Delta \hat{u}_{it}$ exhibit predictable first-order autocorrelation. Rejection indicates that the structure of $\mathbf{\Omega}$ contains correlation within each unit i .

Consequence: rejection does not invalidate FE consistency (which depends on strict exogeneity), but invalidates inference based on classical standard errors. The correction is cluster-robust.

7.7.4 Spatial or Cross-Sectional Dependence

- Moran's test for spatial dependence.
- Cross-sectional dependence tests in panel (Pesaran CD).
- Cluster-robust by region, sector, or double-clustering.

7.7.5 Caveats on Diagnostics

- The mean of OLS residuals is zero by construction (normal equations). This is not evidence of homoskedasticity.
- In very large samples ($n > 10^5$), tests like BP will reject any trivial deviation. Assess economic relevance by comparing robust versus classical standard errors.
- In very small samples, diagnostic tests have low power. Non-rejection does not prove sphericity.

7.8 Two Correction Routes: FGLS versus Robust Standard Errors

7.8.1 Route A: FGLS (Model the Covariance Matrix and Re-estimate)

- **Advantage:** efficiency gain; smaller standard errors; higher statistical power.
- **Cost:** requires getting the functional form of Ω right. If wrong, it may be worse than OLS.
- **Key assumption:** $\hat{\Omega} \xrightarrow{p} \Omega$.

7.8.2 Route B: Robust Standard Errors (Keep the Estimator, Correct Inference)

- **Advantage:** does not require modeling Ω . Robust to any form of heteroskedasticity (White), autocorrelation (Newey-West), or intra-group correlation (cluster).
- **Cost:** does not gain efficiency. Standard errors may be larger.
- **Key assumption:** finite fourth-order moments: $\mathbb{E}[x_{ij}^2 u_i^2] < \infty$.

7.8.3 Formalizing Route B: Sandwich Estimator

The OLS estimator is maintained:

$$\hat{\beta}_{\text{OLS}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}. \quad (7.56)$$

The variance is estimated by:

$$\widehat{\text{Var}}(\hat{\beta}_{\text{OLS}}) = (\mathbf{X}'\mathbf{X})^{-1}\hat{\mathbf{S}}(\mathbf{X}'\mathbf{X})^{-1}, \quad (7.57)$$

where $\hat{\mathbf{S}}$ depends on the type of correction:

- **White (HC):** $\hat{\mathbf{S}} = \sum_{i=1}^n \hat{u}_i^2 \mathbf{x}_i \mathbf{x}_i'$.
- **Newey-West (HAC):** adds autocovariance terms up to lag L .
- **Cluster:** $\hat{\mathbf{S}} = \sum_{g=1}^G \mathbf{X}_g' \hat{\mathbf{u}}_g \hat{\mathbf{u}}_g' \mathbf{X}_g$, where g indexes clusters.

7.8.4 Formal Comparison

7.9 Decision Guide for Applied Research

7.9.1 When to Prefer FGLS

- The form of Ω has clear theoretical foundation (e.g., variance proportional to population in aggregate data).
- Heteroskedasticity is severe (variance varying by factor > 10) and the statistical power of robust OLS is insufficient.

Criterion	Route A (FGLS)	Route B (Robust SE)
Estimator	Altered (reweighted)	Maintained (OLS/FE/etc.)
Efficiency	Gains if Ω correct	Does not gain
Robustness	Fragile to misspecification	Robust to any form
Inference	Valid under correct model	Asymptotically valid
Small sample	May be unstable	HC2/HC3 recommended
Modern practice	Use when clear theory exists	Standard in microeconomics

Table 7.3: Comparison between FGLS and robust standard errors.

- The sample is large enough to estimate Ω precisely.
- The objective is prediction (minimize coefficient MSE).

7.9.2 When to Prefer Robust Standard Errors

- There is no clear theory about the form of Ω .
- The sample is large and the robust standard error is already small enough.
- The focus is causal identification and honest inference.
- There is risk of variance misspecification.
- Panel with small T : cluster-robust is safer than unrestricted FGLS.

7.9.3 Decision Table

Situation	Recommended Route	Justification
Heteroskedasticity without clear form	Robust SE	Avoids incorrectly specifying Ω
Variance linked to size/scale	WLS/FGLS	Theory for the form of Ω
Panel with correlation within i	Cluster-robust or RE	Cluster is robust; RE is efficient under orthogonality
Time series with AR(1)	Prais-Winsten or HAC	Models temporal dependence
Small sample ($n < 50$)	Robust SE (HC2/HC3)	FGLS may be noisy
Large sample and known structure	FGLS	Efficiency gain
Doubt about form of Ω	Robust SE	Lower risk of invalid inference

Table 7.4: Decision guide: FGLS versus robust standard errors.

7.10 Caveats and Common Errors

7.10.1 Misspecification of the Weight Matrix

Critique: “The author assumes $\text{Var}(u_i) = \sigma^2 x_i$ without theoretical foundation. If the form is exponential instead of linear, the weights are wrong.”

Symptom: FGLS coefficients drastically different from robust OLS, with implausibly small standard errors.

7.10.2 R^2 Not Comparable After GLS Transformation

Critique: “The author celebrates $R^2 = 0.99$ after the GLS transformation. Since the dependent variable was transformed, the R^2 is not comparable to the original model.”

7.10.3 FGLS with Small T in Panel

Critique: “The author estimates unrestricted Σ_i with $T = 2$. The estimate is purely noisy.”

Symptom: coefficients that change erratically with the inclusion of controls.

7.10.4 Ignoring Robustness in Favor of Efficiency

Critique: “The Hausman test rejects $\mathbb{E}[c_i | \mathbf{X}_i] = 0$. RE is inconsistent. Being efficient at the wrong parameter is useless.”

7.10.5 Robust Errors in Very Small Samples

Critique: “The author uses HC0 with $n = 30$. HC0 underestimates variance in finite samples. Use HC2 or HC3.”

7.10.6 Durbin-Watson with Lagged Dependent Variable

Critique: “The model contains y_{t-1} as a regressor. DW is biased toward 2. Use Breusch-Godfrey.”

7.11 Referee Questions

Question: “Given that OLS with robust standard errors (Eicker-Huber-White) already provides valid inference under heteroskedasticity, why estimate FGLS?”

Answer: “Robust standard errors correct only the *inference* of OLS, which remains inefficient. FGLS seeks *efficiency*: by reweighting observations by the variance structure, it reduces the estimator’s variance. In the present dataset, heterogeneity between firms is severe (profit variance grows with the cube of assets). By weighting via FGLS, the estimated variance fell 40% relative to robust OLS, allowing identification of a marginal effect that was previously not statistically significant. The assumption $\mathbb{E}[\mathbf{u} | \mathbf{X}] = \mathbf{0}$ was maintained in both cases; what changes is efficiency, not consistency.”

Question: “Your research finds a null effect. Is this because the effect is truly zero or because estimation is inefficient?”

Answer: “Identification was guaranteed by research design ($\mathbb{E}[\mathbf{u} | \mathbf{X}] = \mathbf{0}$). To address concerns about statistical power, I implemented FGLS modeling sectoral heteroskedasticity. Even after reducing variance, the coefficient remained close to zero and insignificant. This strengthens the evidence of a truly null effect, not an artifact of excessive noise.”

Question: “Why use cluster-robust instead of RE (FGLS), which would be theoretically more efficient?”

Answer: “RE efficiency depends on $\mathbb{E}[c_i | \mathbf{X}_i] = 0$ and the equicorrelation structure. Cluster-robust allows arbitrary intra-group correlation and heteroskedasticity of unknown form, without modeling Ω . Given that the main interest is valid inference under a potentially complex error structure, robustness dominates efficiency. If the Hausman test and Mundlak approach confirm orthogonality, RE can be reported as an efficiency specification.”

7.12 Block Synthesis

7.12.1 Unifying Principle

All estimators in this block answer the same question:

Given that the parameter is identified, how to estimate and make inference correctly when the error covariance matrix is not spherical?

The answer unfolds into two routes:

1. **Model Ω** (GLS/FGLS): gains efficiency, but requires getting the form right.
2. **Correct only inference** (robust SE): does not gain efficiency, but is robust.

7.12.2 Connection with Other Blocks

- **Panel:** RE is FGLS with $\Omega_i = \sigma_u^2 \mathbf{I}_T + \sigma_c^2 \mathbf{j}_T \mathbf{j}_T'$.
- **SUR:** Systemic GLS with $\Omega = \Sigma \otimes \mathbf{I}_T$.
- **3SLS:** FGLS with instruments for systems with endogeneity.
- **Time series:** Prais-Winsten and Cochrane-Orcutt are FGLS for AR(1).
- **GMM:** the optimal weight matrix in GMM ($\mathbf{S}^{-1}$) is analogous to Ω^{-1} in GLS.

7.12.3 Final Message

Efficiency is a conditional luxury. It is only sought **after** identification is guaranteed. And even when sought, it must be balanced against the risk of misspecification. In modern applied microeconomics, the standard route is:

Identification by design $\longrightarrow$ Consistent estimator $\longrightarrow$ Robust inference
(cluster) $\longrightarrow$ FGLS only as additional efficiency analysis.

Chapter 8

Linear Equation Systems

8.1 Fundamentals of the Kronecker Product in Systems and Panel

8.1.1 Motivation: Why Use the Kronecker Product?

In equation systems and panel data models, the **Kronecker product**, denoted by $\otimes$, is the algebraic operator that allows expanding local structures — such as a covariance matrix between equations or a temporal transformation — to the global dimension of the stacked system.

The main reason for its use is **notational economy** and **analytical tractability**. For example, in a system with M equations and T observations, the error covariance matrix has dimension $MT \times MT$. Writing it explicitly is impractical for large M or T . The Kronecker product allows constructing this matrix from smaller blocks, such as:

- an $M \times M$ matrix, capturing contemporaneous correlation between equations;
- a $T \times T$ matrix, capturing temporal correlation.

Thus, instead of manipulating an $MT \times MT$ matrix directly, one works with smaller and economically interpretable objects.

8.1.2 Formal Definition and Example

Definition 8.1.1 (Kronecker Product). Let $\mathbf{A}$ be an $m \times n$ matrix and $\mathbf{B}$ be a $p \times q$ matrix. The Kronecker product $\mathbf{A} \otimes \mathbf{B}$ is the $mp \times nq$ block matrix defined by

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \dots & a_{1n}\mathbf{B} \\ a_{21}\mathbf{B} & a_{22}\mathbf{B} & \dots & a_{2n}\mathbf{B} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}\mathbf{B} & a_{m2}\mathbf{B} & \dots & a_{mn}\mathbf{B} \end{bmatrix}. \quad (8.1)$$

Example. If

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix},$$

then

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{bmatrix}.$$

8.1.3 Fundamental Identities for Econometrics

The four properties below are the most useful in equation systems, SUR, GLS, and panel.

8.1.3.1 Inversion

$$(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}, \quad (8.2)$$

provided $\mathbf{A}$ and $\mathbf{B}$ are invertible.

Typical application: in SUR, if

$$\mathbf{\Omega} = \mathbf{\Sigma} \otimes \mathbf{I}_T,$$

then

$$\mathbf{\Omega}^{-1} = \mathbf{\Sigma}^{-1} \otimes \mathbf{I}_T.$$

Indeed,

$$(\mathbf{\Sigma} \otimes \mathbf{I}_T)(\mathbf{\Sigma}^{-1} \otimes \mathbf{I}_T) = (\mathbf{\Sigma}\mathbf{\Sigma}^{-1}) \otimes (\mathbf{I}_T\mathbf{I}_T) \quad (8.3)$$

$$= \mathbf{I}_M \otimes \mathbf{I}_T \quad (8.4)$$

$$= \mathbf{I}_{MT}. \quad (8.5)$$

8.1.3.2 Transposition

$$(\mathbf{A} \otimes \mathbf{B})' = \mathbf{A}' \otimes \mathbf{B}'. \quad (8.6)$$

Typical application: if the GLS transformation is

$$\mathbf{P} = \mathbf{\Sigma}^{-1/2} \otimes \mathbf{I}_T,$$

then

$$\mathbf{P}' = (\mathbf{\Sigma}^{-1/2})' \otimes \mathbf{I}_T.$$

8.1.3.3 Matrix Product

If dimensions are compatible,

$$(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC}) \otimes (\mathbf{BD}). \quad (8.7)$$

Typical application: verify that the GLS transformation makes the error covariance spherical. For example,

$$\mathbf{P}\mathbf{\Omega}\mathbf{P}' = (\mathbf{\Sigma}^{-1/2} \otimes \mathbf{I}_T)(\mathbf{\Sigma} \otimes \mathbf{I}_T)(\mathbf{\Sigma}^{-1/2})' \otimes \mathbf{I}_T \quad (8.8)$$

$$= (\mathbf{\Sigma}^{-1/2}\mathbf{\Sigma}(\mathbf{\Sigma}^{-1/2})') \otimes \mathbf{I}_T \quad (8.9)$$

$$= \mathbf{I}_M \otimes \mathbf{I}_T \quad (8.10)$$

$$= \mathbf{I}_{MT}. \quad (8.11)$$

8.1.3.4 Vec Operator

For compatible matrices,

$$\text{vec}(\mathbf{ABC}) = (\mathbf{C}' \otimes \mathbf{A}) \text{vec}(\mathbf{B}). \quad (8.12)$$

Typical application: linearize systems with matrix parameters. For example, if

$$\mathbf{Y} = \mathbf{XB} + \mathbf{U},$$

with $\mathbf{Y}$ of dimension $T \times M$, then

$$\text{vec}(\mathbf{Y}) = (\mathbf{I}_M \otimes \mathbf{X}) \text{vec}(\mathbf{B}) + \text{vec}(\mathbf{U}). \quad (8.13)$$

This representation is especially useful when there are cross-restrictions between coefficients.

8.1.4 Typical Applications in SUR and Panel

8.1.4.1 SUR: Contemporaneous Correlation Between Equations

In a SUR system with M equations and T periods, it is usually assumed that errors are correlated between equations in the same period, but independent over time. Ordering errors by equation, the covariance matrix can be written as

$$\mathbf{\Omega} = \mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \mathbf{X}] = \mathbf{\Sigma} \otimes \mathbf{I}_T, \quad (8.14)$$

where $\mathbf{\Sigma}$ is $M \times M$ and captures contemporaneous correlation between equations.

8.1.4.2 Panel: Random Effects

In a random effects model with N individuals and T periods, the covariance matrix of the compound error can be written as

$$\mathbf{\Omega} = \sigma_\mu^2(\mathbf{I}_N \otimes \mathbf{J}_T) + \sigma_\nu^2(\mathbf{I}_N \otimes \mathbf{I}_T), \quad (8.15)$$

where $\mathbf{J}_T$ is the $T \times T$ matrix of ones. The term $\mathbf{I}_N \otimes \mathbf{J}_T$ represents the individual effect constant over time.

8.1.4.3 Panel: Within Transformation

To estimate fixed effects by within transformation, the temporal demeaning matrix is applied to each individual:

$$\mathbf{M}_{FE} = \mathbf{I}_N \otimes \mathbf{Q}, \quad (8.16)$$

where

$$\mathbf{Q} = \mathbf{I}_T - \frac{1}{T}\mathbf{J}_T. \quad (8.17)$$

The identity $\mathbf{I}_N$ preserves separation between individuals, while $\mathbf{Q}$ removes individual means over time.

8.1.5 Simple Numerical Example

Consider $M = 2$ equations and $T = 2$ periods. Let

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}. \quad (8.18)$$

Then

$$\Sigma \otimes \mathbf{I}_2 = \begin{bmatrix} \sigma_{11} & 0 & \sigma_{12} & 0 \\ 0 & \sigma_{11} & 0 & \sigma_{12} \\ \sigma_{21} & 0 & \sigma_{22} & 0 \\ 0 & \sigma_{21} & 0 & \sigma_{22} \end{bmatrix}. \quad (8.19)$$

The matrix preserves correlation between equations in the same period, but imposes independence between different periods.

8.1.6 When the Kronecker Structure is Inadequate

The form $\Omega = \Sigma \otimes \mathbf{I}$ depends on correlation existing only in one dimension. If this fails, the structure must be generalized.

- **In SUR:** if there is temporal autocorrelation, the structure $\Sigma \otimes \mathbf{I}_T$ may be inadequate. A more general separable form would be

$$\Omega = \Sigma \otimes \Gamma,$$

where Γ captures temporal dependence.

- **In panel:** if there is spatial correlation between individuals, the structure based on $\mathbf{I}_N$ fails, requiring spatial dependence models or more general cross-sectional covariance.

Practical symptom: underestimated standard errors, inflated t -statistics, and residuals with systematic patterns over time or between individuals.

8.1.7 Synthesis of Most Common Structures

Context	Covariance Matrix	Interpretation
Classical SUR	$\Omega = \Sigma \otimes \mathbf{I}_T$	Correlation between equations, temporal independence
Random effects	$\Omega = \sigma_\mu^2(\mathbf{I}_N \otimes \mathbf{J}_T) + \sigma_\nu^2(\mathbf{I}_N \otimes \mathbf{I}_T)$	Constant intra-individual correlation over time
SUR with autocorrelation	$\Omega = \Sigma \otimes \Gamma$	Correlation between equations and over time
Panel with spatial dependence	$\Omega = \Sigma_{\text{spatial}} \otimes \mathbf{I}_T$	Correlation between individuals

Table 8.1: Covariance structures with Kronecker product.

Conclusion

The Kronecker product is not just a notational convenience. It is the mechanism that allows transporting the algebra of single-equation OLS to multi-equational systems and panels, maintaining block organization and making estimation and inference expressions tractable.

Remark 8.1.1. Before assuming a structure like $\Sigma \otimes \mathbf{I}$, it is advisable to perform diagnostic tests, such as tests for contemporaneous correlation, autocorrelation, or spatial dependence, depending on the application.

8.2 System Ordinary Least Squares (SOLS)

8.2.1 Motivation and Research Problem

The **System Ordinary Least Squares** (SOLS) estimator is used when the researcher wishes to estimate a system of linear equations as a unit, but without explicitly using the contemporaneous covariance matrix of errors, Σ , to weight the estimation.

A typical example is a study on the effect of an educational reform on grades in three subjects:

$$y_{i1} = \mathbf{x}'_{i1}\beta_1 + u_{i1}, \quad (8.20)$$

$$y_{i2} = \mathbf{x}'_{i2}\beta_2 + u_{i2}, \quad (8.21)$$

$$y_{i3} = \mathbf{x}'_{i3}\beta_3 + u_{i3}, \quad (8.22)$$

where y_{i1} , y_{i2} , and y_{i3} may represent, for example, grades in Mathematics, Portuguese, and Science.

Estimating each equation separately by OLS may be valid under exogeneity, but two important limitations arise:

1. **Cross-restriction tests:** to test hypotheses like

$$H_0 : \beta_{1,\text{reform}} = \beta_{3,\text{reform}},$$

one needs to know the covariance between estimators from different equations.

2. **Contemporaneous correlation of errors:** unobserved shocks may simultaneously affect several equations. If this correlation exists, separate OLS ignores potentially useful information for efficiency.

SOLS organizes the stacked system and allows joint estimation and inference, but, without cross-restrictions and without using Σ , it generally does not generate efficiency gains over equation-by-equation OLS.

Geometrically, SOLS projects the stacked vector $\mathbf{Y}$ onto the column space of $\mathbf{X}$, minimizing the Euclidean distance

$$\|\mathbf{Y} - \mathbf{X}\beta\|^2.$$

8.2.2 Model and Notation

Consider a system with G equations for N observational units. For each equation g ,

$$\mathbf{y}_g = \mathbf{X}_g \boldsymbol{\beta}_g + \mathbf{u}_g, \quad (8.23)$$

where $\mathbf{y}_g$ is $N \times 1$, $\mathbf{X}_g$ is $N \times K_g$, and $\boldsymbol{\beta}_g$ is $K_g \times 1$.

Define

$$K = \sum_{g=1}^G K_g.$$

Stacking the equations,

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{U}, \quad (8.24)$$

with

$$\mathbf{Y} = \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_G \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} \mathbf{X}_1 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{X}_2 & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{X}_G \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \\ \vdots \\ \boldsymbol{\beta}_G \end{bmatrix}. \quad (8.25)$$

Equivalently, in notation by observational unit,

$$\mathbf{y}_i = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{u}_i, \quad i = 1, \dots, N, \quad (8.26)$$

where $\mathbf{y}_i$ is $G \times 1$, $\mathbf{u}_i$ is $G \times 1$, and $\mathbf{X}_i$ is $G \times K$, usually block-diagonal.

8.2.3 SOLS Assumptions

Definition 8.2.1 (SOLS.1: Systemic Orthogonality). The fundamental consistency condition is

$$\mathbb{E}[\mathbf{X}_i' \mathbf{u}_i] = \mathbf{0}. \quad (8.27)$$

In equation-specific terms, this corresponds to

$$\mathbb{E}[\mathbf{x}_{ig} u_{ig}] = \mathbf{0}, \quad g = 1, \dots, G. \quad (8.28)$$

Role: guarantees estimator consistency.

Practice: requires exogeneity within each equation.

Important point: for SOLS consistency, it is not necessary that

$$\mathbb{E}[\mathbf{x}_{ig} u_{ih}] = 0 \quad \text{for } g \neq h.$$

This stronger requirement appears in systemic estimators that use cross-weighting, such as GLS/SUR.

Counterexample: in dynamic panel with y_{it} depending on $y_{i,t-1}$ and unobserved fixed effect, the lagged regressor may be correlated with the compound error, making pooled OLS inconsistent.

Definition 8.2.2 (SOLS.2: Rank Condition).

$$\text{rank}(\mathbb{E}[\mathbf{X}_i' \mathbf{X}_i]) = K. \quad (8.29)$$

Role: guarantees identification and invertibility of the moment matrix.

Practice: avoids perfect multicollinearity in the stacked system.

Counterexample: in budget share systems, estimating all equations whose dependent variables sum to 1 may generate linear dependence and singular matrix. A common solution is to exclude one equation.

Definition 8.2.3 (SOLS.3: Absence of Contemporaneous Correlation). The contemporaneous covariance matrix of errors is

$$\mathbf{\Sigma} = \mathbb{E}[\mathbf{u}_i \mathbf{u}_i' | \mathbf{X}_i]. \quad (8.30)$$

SOLS.3 assumption imposes

$$\sigma_{gh} = 0 \quad \text{for } g \neq h, \quad (8.31)$$

that is, $\mathbf{\Sigma}$ is diagonal.

Role: this assumption is not necessary for consistency, but is relevant for efficiency.

Practice: means there are no common unobserved shocks simultaneously affecting different equations.

Counterexample: in a demand system for several goods, an unobserved income shock may affect all goods, generating contemporaneous correlation between errors.

8.2.4 Derivation of the SOLS Estimator

The SOLS estimator minimizes the sum of squared residuals of the stacked system:

$$SSR(\mathbf{b}) = (\mathbf{Y} - \mathbf{Xb})'(\mathbf{Y} - \mathbf{Xb}). \quad (8.32)$$

Expanding,

$$SSR(\mathbf{b}) = \mathbf{Y}'\mathbf{Y} - 2\mathbf{Y}'\mathbf{Xb} + \mathbf{b}'\mathbf{X}'\mathbf{Xb}. \quad (8.33)$$

The first-order condition is

$$\frac{\partial SSR(\mathbf{b})}{\partial \mathbf{b}} = -2\mathbf{X}'\mathbf{Y} + 2\mathbf{X}'\mathbf{X}\hat{\mathbf{\beta}} = \mathbf{0}. \quad (8.34)$$

Thus, the normal equations are

$$\mathbf{X}'\mathbf{X}\hat{\mathbf{\beta}} = \mathbf{X}'\mathbf{Y}. \quad (8.35)$$

Under SOLS.2,

$$\hat{\mathbf{\beta}}_{\text{SOLS}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}. \quad (8.36)$$

In notation by observational unit,

$$\hat{\mathbf{\beta}}_{\text{SOLS}} = \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{X}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{y}_i \right). \quad (8.37)$$

8.2.5 SOLS Without Cross-Restrictions is Numerically Equal to Equation-by-Equation OLS

Theorem 8.2.1 (SOLS Equals OLS Without Cross-Restrictions). *If each equation has its own regressors and there are no cross-restrictions between parameters, then the SOLS estimator is numerically identical to the OLS estimator applied separately to each equation.*

Proof. Since $\mathbf{X}$ is block-diagonal,

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} \mathbf{X}'_1\mathbf{X}_1 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{X}'_2\mathbf{X}_2 & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{X}'_G\mathbf{X}_G \end{bmatrix}. \quad (8.38)$$

The inverse is

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{bmatrix} (\mathbf{X}'_1\mathbf{X}_1)^{-1} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & (\mathbf{X}'_2\mathbf{X}_2)^{-1} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & (\mathbf{X}'_G\mathbf{X}_G)^{-1} \end{bmatrix}. \quad (8.39)$$

Moreover,

$$\mathbf{X}'\mathbf{Y} = \begin{bmatrix} \mathbf{X}'_1\mathbf{y}_1 \\ \mathbf{X}'_2\mathbf{y}_2 \\ \vdots \\ \mathbf{X}'_G\mathbf{y}_G \end{bmatrix}. \quad (8.40)$$

Therefore,

$$\hat{\beta}_{\text{SOLS}} = \begin{bmatrix} (\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{y}_1 \\ (\mathbf{X}'_2\mathbf{X}_2)^{-1}\mathbf{X}'_2\mathbf{y}_2 \\ \vdots \\ (\mathbf{X}'_G\mathbf{X}_G)^{-1}\mathbf{X}'_G\mathbf{y}_G \end{bmatrix} = \begin{bmatrix} \hat{\beta}_{1,\text{OLS}} \\ \hat{\beta}_{2,\text{OLS}} \\ \vdots \\ \hat{\beta}_{G,\text{OLS}} \end{bmatrix}. \quad (8.41)$$

Thus, without cross-restrictions and without weighting by Σ , stacking only organizes the problem, it does not alter the estimates. $\square$

8.2.6 Intuition: Why Just Stacking Does Not Generate Efficiency Gains?

The central point is that SOLS does not exploit correlation between equation errors. Thus, equations remain operationally separate.

- If $\sigma_{gh} \neq 0$, the residual from equation g contains information about the likely error of equation h .
- GLS/SUR exploits this information by weighting the system by Σ^{-1} .
- SOLS implicitly uses an identity weight matrix, treating all stacked observations as if they were independent and equally informative.

Therefore, stacking equations without using Σ does not create efficiency gains. The informational gain depends on three conditions:

1. regressors not being identical between equations;
2. there being relevant contemporaneous correlation between errors;
3. using a GLS-type estimator, such as SUR.

8.2.7 Efficiency and Relation to GLS/SUR

Under independence between units i , the covariance matrix of stacked errors can be written as

$$\Omega = \mathbf{I}_N \otimes \Sigma. \quad (8.42)$$

The variance of SOLS is

$$\text{Var}(\hat{\beta}_{\text{SOLS}} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{I}_N \otimes \Sigma)\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}. \quad (8.43)$$

The systemic GLS estimator is

$$\hat{\beta}_{\text{GLS}} = [\mathbf{X}'(\mathbf{I}_N \otimes \Sigma^{-1})\mathbf{X}]^{-1} \mathbf{X}'(\mathbf{I}_N \otimes \Sigma^{-1})\mathbf{Y}. \quad (8.44)$$

If Σ is diagonal, then Σ^{-1} is also diagonal. In this case, GLS weighting only multiplies each equation by a scalar σ_{gg}^{-1} , which cancels in the estimate. Thus, GLS collapses to equation-by-equation OLS.

More generally, by the generalized Gauss-Markov Theorem,

$$\text{Var}(\hat{\beta}_{\text{SOLS}}) - \text{Var}(\hat{\beta}_{\text{GLS}}) \succeq \mathbf{0}, \quad (8.45)$$

when GLS assumptions are satisfied.

SOLS is efficient only in special cases, such as:

- Σ diagonal;
- all equations have exactly the same regressors.

Outside these cases, SOLS is consistent, but generally inefficient.

8.2.8 Asymptotic Properties

Define

$$\mathbf{A} = \mathbb{E}[\mathbf{X}_i'\mathbf{X}_i], \quad \mathbf{B} = \mathbb{E}[\mathbf{X}_i'\mathbf{u}_i\mathbf{u}_i'\mathbf{X}_i]. \quad (8.46)$$

Theorem 8.2.2 (Consistency of SOLS). *Under SOLS.1 and SOLS.2, and regular conditions for the law of large numbers,*

$$\hat{\beta}_{\text{SOLS}} \xrightarrow{p} \beta. \quad (8.47)$$

Proof. Starting from

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y},$$

and substituting $\mathbf{Y} = \mathbf{X}\beta + \mathbf{U}$,

$$\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}.$$

In terms per unit,

$$\hat{\beta} = \beta + \left(\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \mathbf{X}_i \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \mathbf{u}_i \right).$$

By the law of large numbers,

$$\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \mathbf{X}_i \xrightarrow{p} \mathbf{A},$$

and by SOLS.1,

$$\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \mathbf{u}_i \xrightarrow{p} \mathbf{0}.$$

By Slutsky,

$$\text{plim } \hat{\beta} = \beta + \mathbf{A}^{-1}\mathbf{0} = \beta.$$

Consistency does not depend on Σ being diagonal. □

Under additional conditions for the central limit theorem,

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{X}_i' \mathbf{u}_i \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{B}). \quad (8.48)$$

Thus,

$$\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}). \quad (8.49)$$

8.2.9 Robust Inference

The most common practical error is using the classical OLS variance formula as if stacked errors were independent and homoskedastic:

$$\widehat{\text{Var}}_{\text{naive}} = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}. \quad (8.50)$$

This formula is inadequate when there is contemporaneous correlation between equations or heteroskedasticity between equations.

The sandwich-type robust variance is

$$\widehat{\text{Var}}(\hat{\beta}_{\text{SOLS}}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^N \mathbf{X}_i' \hat{\mathbf{u}}_i \hat{\mathbf{u}}_i' \mathbf{X}_i \right) (\mathbf{X}'\mathbf{X})^{-1}. \quad (8.51)$$

This matrix is consistent under heteroskedasticity and contemporaneous correlation between equations, provided dependence is adequately treated in dimension i .

To test cross-restrictions,

$$H_0 : \mathbf{R}\boldsymbol{\beta} = \mathbf{r},$$

one uses the Wald statistic

$$W = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})' [\mathbf{R}\widehat{\text{Var}}(\hat{\boldsymbol{\beta}})\mathbf{R}']^{-1} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}). \quad (8.52)$$

Under H_0 ,

$$W \xrightarrow{d} \chi_q^2,$$

where q is the number of restrictions.

8.2.10 Caveats and Common Errors

8.2.10.1 Variance Underestimation

If errors are correlated between equations and the researcher uses the classical OLS variance, standard errors tend to be incorrect. The typical symptom is:

- very large t -statistics;
- artificially small p -values;
- inference sensitive to small sample changes.

8.2.10.2 Contamination by Misspecification

Without cross-restrictions, SOLS tends to preserve consistency of well-specified equations, even if another equation is misspecified.

However, if cross-restrictions are imposed, such as $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_2$, a misspecified equation can contaminate others. The bias of one equation can be transferred to shared coefficients.

8.2.10.3 False Sense of Information Gain

Stacking G equations for the same N units does not mean obtaining NG independent observations. If equations are highly correlated, the effective information gain may be small.

In extreme cases, such as accounting identities in demand systems, the moment matrix may become singular.

8.2.10.4 Diagnostic Table

8.2.11 Typical Committee or Referee Question

Question: If SOLS is inefficient relative to SUR when $\boldsymbol{\Sigma}$ is not diagonal, why use SOLS as the main specification?

Answer: The choice is based on the trade-off between efficiency and robustness.

SOLS requires, for consistency, basically orthogonality within each equation:

$$\mathbb{E}[\mathbf{x}_{ig}u_{ig}] = 0, \quad g = 1, \dots, G. \quad (8.53)$$

Problem	Symptom	Diagnosis
Ignored contemporaneous correlation	High t -stats, fragile inference	Breusch-Pagan test
Misspecification in one equation	Coefficients change between separate and joint estimation	Compare separate OLS and SOLS
Identical regressors	SUR does not generate gain over SOLS	Check if $\mathbf{X}_g$ are equal
Many equations for few observations	$\widehat{\Sigma}$ singular	Use SOLS or reduce system

Table 8.2: Common problems in system estimation.

SUR/GLS tends to require broader exogeneity, such as

$$\mathbb{E}[\mathbf{x}_{ig}u_{ih}] = 0, \quad \forall g, h, \quad (8.54)$$

when systemic weighting mixes errors from different equations.

If this stronger condition fails, SUR may become inconsistent, while SOLS may remain consistent.

Moreover, SOLS allows more easily isolating misspecified equations when there are no cross-restrictions. SUR, using $\widehat{\Sigma}^{-1}$, can propagate specification error between equations.

Therefore, in many applications, it is preferable to report SOLS with robust variance than to use SUR with efficiency gain based on possibly fragile cross-exogeneity assumptions.

8.2.12 Summary of SOLS Properties

Property	Assumption	Interpretation	If Fails
Consistency	SOLS.1: $\mathbb{E}[\mathbf{X}_i' \mathbf{u}_i] = \mathbf{0}$	Exogeneity within equations	Endogeneity bias
Identification	SOLS.2: full rank	No perfect multicollinearity	Estimator not computable
Efficiency	SOLS.3: Σ diagonal	Errors uncorrelated between equations	SOLS inefficient
Valid inference	Robust variance	Allows Σ non-diagonal	Incorrect standard errors

Table 8.3: SOLS properties and associated assumptions.

8.2.13 Complementary Topics

1. Breusch-Pagan test for residual independence:

$$LM = N \sum_{g=2}^G \sum_{h=1}^{g-1} \hat{\rho}_{gh}^2 \xrightarrow{d} \chi_{G(G-1)/2}^2. \quad (8.55)$$

The test evaluates whether Σ is diagonal. If the hypothesis is rejected, SOLS remains consistent, but SUR may be more efficient.

2. Degenerate cases:

- If Σ is diagonal, SOLS and SUR coincide.
 - If all equations have the same regressors, SUR collapses to equation-by-equation OLS.
 - If both conditions occur, the system does not generate gain over separate OLS.
3. **Cross-restrictions:** when there are restrictions like $\mathbf{R}\beta = \mathbf{r}$, the system estimator may differ from equation-by-equation OLS, as it imposes additional information on the problem.
 4. **Hypothesis tests between equations:** stacked SOLS, combined with robust variance matrix, allows Wald tests on equality of coefficients between equations.

8.3 Seemingly Unrelated Regressions (SUR)

8.3.1 Motivation and Research Problem

The **Seemingly Unrelated Regressions** (SUR) estimator, proposed by Zellner (1962), is a generalized least squares estimator applied to systems of linear equations. It is used when equations are related through contemporaneous correlation between their errors.

While SOLS estimates the system without exploiting the contemporaneous covariance matrix Σ , SUR uses this matrix to reweight the system and, under appropriate assumptions, obtain efficiency gains.

8.3.1.1 Example 1: Demand System

Consider a family's demand for three categories of goods: Housing (y_{1t}), Food (y_{2t}), and Clothing (y_{3t}):

$$y_{1t} = \mathbf{x}'_{1t}\beta_1 + \varepsilon_{1t}, \quad (8.56)$$

$$y_{2t} = \mathbf{x}'_{2t}\beta_2 + \varepsilon_{2t}, \quad (8.57)$$

$$y_{3t} = \mathbf{x}'_{3t}\beta_3 + \varepsilon_{3t}. \quad (8.58)$$

Since decisions are made by the same economic agent, unobserved shocks — such as preference changes or unexpected wealth gains — may simultaneously affect several equations. Thus,

$$\text{Cov}(\varepsilon_{1t}, \varepsilon_{2t}) \neq 0$$

is plausible.

In this case, equation-by-equation OLS remains consistent under exogeneity, but ignores information contained in error correlation. SUR exploits this correlation to reduce standard errors, when conditions for efficiency gain are present.

8.3.1.2 Example 2: Translog Cost Function

In firm microeconomics, a translog cost function generates cost share equations for inputs like Capital (s_K), Labor (s_L), and Materials (s_M). Since shares must sum to 1, the errors of the equations are linked by the budget constraint:

$$s_K + s_L + s_M = 1.$$

This generates two important points:

- errors are contemporaneously correlated;
- the matrix Σ is singular if all equations are estimated.

The usual solution is to discard one equation and recover its parameters through theoretical restrictions.

8.3.1.3 What Does SUR Solve?

OLS/SOLS is consistent, but may be inefficient when errors are correlated between equations. The intuition is that, if $\sigma_{gh} \neq 0$, the observed residual in one equation contains information about the likely error in another equation.

SUR uses this information through weighting based on Σ^{-1} . In simple terms, it stops treating equations as isolated compartments and starts recognizing that errors form a vector of contemporaneous shocks.

8.3.2 Model and Notation

Consider a system with M equations and T observations:

$$\mathbf{y}_m = \mathbf{X}_m \boldsymbol{\beta}_m + \boldsymbol{\varepsilon}_m, \quad m = 1, \dots, M, \quad (8.59)$$

where $\mathbf{y}_m$ is $T \times 1$, $\mathbf{X}_m$ is $T \times K_m$, and $\boldsymbol{\beta}_m$ is $K_m \times 1$.

Define

$$K = \sum_{m=1}^M K_m.$$

Stacking the equations,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (8.60)$$

with

$$\mathbf{y} = \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_M \end{bmatrix}, \quad \mathbf{X} = \text{diag}(\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_M), \quad \boldsymbol{\beta} = \begin{bmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \\ \vdots \\ \boldsymbol{\beta}_M \end{bmatrix}. \quad (8.61)$$

The contemporaneous covariance matrix of errors is

$$\Sigma = \mathbb{E}[\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t' | \mathbf{X}], \quad (8.62)$$

where $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \dots, \varepsilon_{Mt})'$.

Under the standard SUR structure,

$$\Omega = \mathbb{E}[\boldsymbol{\varepsilon} \boldsymbol{\varepsilon}' | \mathbf{X}] = \Sigma \otimes \mathbf{I}_T. \quad (8.63)$$

8.3.3 SUR Assumptions

Definition 8.3.1 (SUR.1: Broad Contemporaneous Exogeneity). For all $m, n = 1, \dots, M$,

$$\mathbb{E}[\mathbf{x}_{mt}\varepsilon_{nt}] = \mathbf{0}. \quad (8.64)$$

Equivalently, regressors from any equation must be orthogonal to errors from all equations in the same period.

Role: guarantee SUR estimator consistency.

Practice: requires that explanatory variables from one equation are not jointly determined with shocks from other equations.

Counterexample: in simultaneous supply and demand models, price may be correlated with the supply equation error and with the demand equation error. In this case, SUR may be inconsistent.

Definition 8.3.2 (SUR.2: Contemporaneous Covariance Structure). The covariance matrix of stacked errors is

$$\mathbf{\Omega} = \mathbf{\Sigma} \otimes \mathbf{I}_T. \quad (8.65)$$

This means:

- there is correlation between errors from different equations in the same period;
- there is no temporal autocorrelation;
- there is no cross-correlation between different periods.

Role: allow factorization of systemic covariance by Kronecker product and justify the GLS transformation.

Practice: in long time series, there is commonly autocorrelation. In this case, the form $\mathbf{\Sigma} \otimes \mathbf{I}_T$ may be inadequate.

Extension: if there is temporal autocorrelation, a more general separable form would be

$$\mathbf{\Omega} = \mathbf{\Sigma} \otimes \mathbf{\Gamma},$$

where $\mathbf{\Gamma}$ captures temporal dependence.

Definition 8.3.3 (SUR.3: Rank Condition). The system's regressor matrix must have full rank:

$$\text{rank}(\mathbf{X}) = K. \quad (8.66)$$

In asymptotic terms, it is usual to assume that

$$\text{plim} \frac{1}{T} \mathbf{X}'(\mathbf{\Sigma}^{-1} \otimes \mathbf{I}_T) \mathbf{X} \quad (8.67)$$

exists and is non-singular.

Role: guarantee identification and invertibility of the system's moment matrix.

Practice: avoids perfect multicollinearity within or between equations.

Counterexample: including redundant intercepts or linearly dependent variables between equations may make the moment matrix singular.

Definition 8.3.4 (SUR.4: Σ Positive Definite). The contemporaneous covariance matrix must be symmetric and positive definite:

$$\Sigma \succ \mathbf{0}. \quad (8.68)$$

Role: guarantee that Σ^{-1} exists.

Practice: in share systems that sum to 1, the matrix Σ is singular if all equations are included.

Solution: discard one equation or impose restrictions that eliminate singularity.

Condition for efficiency gain. For SUR to effectively generate gain over SOLS, two additional conditions are normally necessary:

1. non-zero contemporaneous correlation between some errors, that is, $\sigma_{mn} \neq 0$ for some $m \neq n$;
2. non-identical regressors between equations.

If Σ is diagonal, or if all equations have exactly the same regressors, SUR collapses to OLS/SOLS.

8.3.4 Efficiency Gain Intuition

SUR's efficiency gain comes from two related mechanisms.

8.3.4.1 Information Contained in Cross-Residuals

If $\sigma_{gh} \neq 0$, observing one equation's residual helps predict another equation's error in the same period.

In SOLS, each equation is treated separately. In SUR, the error vector is treated as a joint shock. Thus, estimation can use cross-correlation to reduce estimator variance.

8.3.4.2 Reweighting via GLS

SUR applies a transformation that makes the systemic error spherical. Instead of minimizing

$$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}),$$

SUR minimizes

$$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'\boldsymbol{\Omega}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

Since

$$\boldsymbol{\Omega}^{-1} = \Sigma^{-1} \otimes \mathbf{I}_T,$$

SUR assigns different weights to equations and exploits cross-correlations.

8.3.4.3 Gain Determinants

The magnitude of gain mainly depends on:

- intensity of contemporaneous correlation between errors;
- degree of difference between equation regressor matrices.

If regressors are identical across all equations, there is no efficiency gain. If Σ is diagonal, there is also no gain.

8.3.5 Comparison Between SOLS and SUR

Characteristic	SOLS	SUR
Use of Σ	Ignores contemporaneous correlation	Exploits Σ in weighting
Weights	Implicitly equal	Based on Σ^{-1}
Consistency	Requires exogeneity proper to each equation	Requires broad exogeneity between equations
Efficiency	Inefficient if $\sigma_{gh} \neq 0$	More efficient under correct assumptions
Risk	Robust inference simpler	Can propagate misspecification

Table 8.4: Comparison between SOLS and SUR.

8.3.6 Derivation of the SUR Estimator

Consider the stacked system

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\Omega} = \Sigma \otimes \mathbf{I}_T.$$

Since Σ is positive definite, there exists a matrix $\mathbf{C}$ such that

$$\mathbf{C}\Sigma\mathbf{C}' = \mathbf{I}_M, \quad \mathbf{C}'\mathbf{C} = \Sigma^{-1}. \quad (8.69)$$

Define the transformation matrix

$$\mathbf{P} = \mathbf{C} \otimes \mathbf{I}_T. \quad (8.70)$$

Pre-multiplying the model by $\mathbf{P}$,

$$\mathbf{P}\mathbf{y} = \mathbf{P}\mathbf{X}\boldsymbol{\beta} + \mathbf{P}\boldsymbol{\varepsilon}. \quad (8.71)$$

The transformed error is

$$\mathbf{u}^* = \mathbf{P}\boldsymbol{\varepsilon}.$$

Its conditional covariance matrix is

$$\mathbb{E}[\mathbf{u}^*\mathbf{u}^{*'} | \mathbf{X}] = \mathbf{P}\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' | \mathbf{X}]\mathbf{P}' \quad (8.72)$$

$$= (\mathbf{C} \otimes \mathbf{I}_T)(\Sigma \otimes \mathbf{I}_T)(\mathbf{C}' \otimes \mathbf{I}_T) \quad (8.73)$$

$$= (\mathbf{C}\Sigma\mathbf{C}') \otimes \mathbf{I}_T \quad (8.74)$$

$$= \mathbf{I}_M \otimes \mathbf{I}_T \quad (8.75)$$

$$= \mathbf{I}_{MT}. \quad (8.76)$$

Therefore, the transformed model has spherical error. Applying OLS to the transformed model,

$$\hat{\beta}_{\text{SUR}} = [(\mathbf{P}\mathbf{X})'(\mathbf{P}\mathbf{X})]^{-1}(\mathbf{P}\mathbf{X})'(\mathbf{P}\mathbf{y}). \quad (8.77)$$

Since

$$(\mathbf{P}\mathbf{X})'(\mathbf{P}\mathbf{X}) = \mathbf{X}'\mathbf{P}'\mathbf{P}\mathbf{X}$$

and

$$\mathbf{P}'\mathbf{P} = (\mathbf{C}'\mathbf{C}) \otimes \mathbf{I}_T = \boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_T,$$

one obtains the classical SUR form:

$$\hat{\beta}_{\text{SUR}} = [\mathbf{X}'(\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_T)\mathbf{X}]^{-1} \mathbf{X}'(\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_T)\mathbf{y}. \quad (8.78)$$

8.3.7 Feasible SUR

In practice, $\boldsymbol{\Sigma}$ is unknown. The usual procedure is feasible SUR, also called systemic FGLS.

The standard algorithm is:

1. Estimate each equation by OLS.
2. Obtain residuals $\hat{\epsilon}_m$.
3. Estimate the contemporaneous covariance matrix:

$$\hat{\sigma}_{mn} = \frac{\hat{\epsilon}_m' \hat{\epsilon}_n}{T}, \quad (8.79)$$

or with degrees of freedom correction:

$$\hat{\sigma}_{mn} = \frac{\hat{\epsilon}_m' \hat{\epsilon}_n}{T - \max(K_m, K_n)}. \quad (8.80)$$

4. Substitute $\boldsymbol{\Sigma}$ with $\hat{\boldsymbol{\Sigma}}$ in the SUR estimator:

$$\hat{\beta}_{\text{FSUR}} = [\mathbf{X}'(\hat{\boldsymbol{\Sigma}}^{-1} \otimes \mathbf{I}_T)\mathbf{X}]^{-1} \mathbf{X}'(\hat{\boldsymbol{\Sigma}}^{-1} \otimes \mathbf{I}_T)\mathbf{y}. \quad (8.81)$$

Under regularity conditions and if $\hat{\boldsymbol{\Sigma}} \xrightarrow{p} \boldsymbol{\Sigma}$, feasible SUR is asymptotically equivalent to SUR with known $\boldsymbol{\Sigma}$.

In small samples, however, estimation of $\boldsymbol{\Sigma}$ may be noisy, especially when M is large relative to T .

8.3.8 Asymptotic Properties

Theorem 8.3.1 (Consistency of SUR). *Under hypotheses SUR.1 (broad exogeneity), SUR.2 (contemporaneous covariance structure), SUR.3 (full rank), and SUR.4 ($\boldsymbol{\Sigma}$ positive definite), the SUR estimator is consistent:*

$$\hat{\beta}_{\text{SUR}} \xrightarrow{p} \beta. \quad (8.82)$$

Proof sketch. Starting from the closed form,

$$\hat{\beta}_{\text{SUR}} = \beta + [\mathbf{X}'(\Sigma^{-1} \otimes \mathbf{I}_T)\mathbf{X}]^{-1} \mathbf{X}'(\Sigma^{-1} \otimes \mathbf{I}_T)\epsilon.$$

Multiplying and dividing by T ,

$$\hat{\beta}_{\text{SUR}} - \beta = \left[\frac{1}{T} \mathbf{X}'(\Sigma^{-1} \otimes \mathbf{I}_T)\mathbf{X} \right]^{-1} \left[\frac{1}{T} \mathbf{X}'(\Sigma^{-1} \otimes \mathbf{I}_T)\epsilon \right].$$

By the broad exogeneity hypothesis,

$$\mathbb{E}[\mathbf{x}_{mt}\epsilon_{nt}] = 0 \quad \forall m, n.$$

Thus, by the law of large numbers,

$$\frac{1}{T} \mathbf{X}'(\Sigma^{-1} \otimes \mathbf{I}_T)\epsilon \xrightarrow{p} \mathbf{0}.$$

Moreover,

$$\frac{1}{T} \mathbf{X}'(\Sigma^{-1} \otimes \mathbf{I}_T)\mathbf{X} \xrightarrow{p} \mathbf{Q},$$

where $\mathbf{Q}$ is finite and non-singular by SUR.3. Therefore, by Slutsky,

$$\hat{\beta}_{\text{SUR}} \xrightarrow{p} \beta.$$

□

Under additional conditions for the central limit theorem,

$$\sqrt{T}(\hat{\beta}_{\text{SUR}} - \beta) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{V}_{\text{SUR}}), \quad (8.83)$$

with

$$\mathbf{V}_{\text{SUR}} = \mathbf{Q}^{-1}, \quad \mathbf{Q} = \text{plim} \frac{1}{T} \mathbf{X}'(\Sigma^{-1} \otimes \mathbf{I}_T)\mathbf{X}. \quad (8.84)$$

If Σ is consistently estimated, feasible SUR has the same asymptotic distribution.

8.3.9 Efficiency

Under classical SUR assumptions and with known Σ , the SUR estimator is the minimum variance linear unbiased estimator.

Formally,

$$\text{Var}(\hat{\beta}_{\text{SOLS}} | \mathbf{X}) - \text{Var}(\hat{\beta}_{\text{SUR}} | \mathbf{X}) \succeq \mathbf{0}. \quad (8.85)$$

The variance of SOLS, when errors have covariance $\Omega = \Sigma \otimes \mathbf{I}_T$, is

$$\text{Var}(\hat{\beta}_{\text{SOLS}} | \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'(\Sigma \otimes \mathbf{I}_T)\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}. \quad (8.86)$$

The variance of SUR is

$$\text{Var}(\hat{\beta}_{\text{SUR}} | \mathbf{X}) = [\mathbf{X}'(\Sigma^{-1} \otimes \mathbf{I}_T)\mathbf{X}]^{-1}. \quad (8.87)$$

The efficiency gain is greater when:

- contemporaneous correlation between errors is relevant;
- regressor matrices are not identical between equations.

8.3.10 Degenerate Cases: When SUR Coincides with SOLS

In some cases, SUR does not alter estimates relative to OLS/SOLS. The two most important cases are:

1. Σ diagonal;
2. identical regressors in all equations.

8.3.10.1 Case 1: Σ Diagonal

If

$$\Sigma = \text{diag}(\sigma_{11}, \dots, \sigma_{MM}),$$

then

$$\Sigma^{-1} = \text{diag}(\sigma_{11}^{-1}, \dots, \sigma_{MM}^{-1}).$$

Thus,

$$\Sigma^{-1} \otimes \mathbf{I}_T = \text{diag}(\sigma_{11}^{-1} \mathbf{I}_T, \dots, \sigma_{MM}^{-1} \mathbf{I}_T).$$

Since $\mathbf{X}$ is block-diagonal, the block of equation m in the SUR moment matrix is

$$\sigma_{mm}^{-1} \mathbf{X}_m' \mathbf{X}_m.$$

The inverse of this block is

$$\sigma_{mm} (\mathbf{X}_m' \mathbf{X}_m)^{-1}.$$

The corresponding term of the moment vector is

$$\sigma_{mm}^{-1} \mathbf{X}_m' \mathbf{y}_m.$$

Therefore,

$$\hat{\beta}_{m,\text{SUR}} = \sigma_{mm} (\mathbf{X}_m' \mathbf{X}_m)^{-1} \sigma_{mm}^{-1} \mathbf{X}_m' \mathbf{y}_m \quad (8.88)$$

$$= (\mathbf{X}_m' \mathbf{X}_m)^{-1} \mathbf{X}_m' \mathbf{y}_m \quad (8.89)$$

$$= \hat{\beta}_{m,\text{OLS}}. \quad (8.90)$$

Thus, if Σ is diagonal, SUR and equation-by-equation OLS coincide.

8.3.10.2 Case 2: Identical Regressors

Suppose all equations have the same regressor matrix:

$$\mathbf{X}_1 = \mathbf{X}_2 = \dots = \mathbf{X}_M = \mathbf{X}_0.$$

Then the system's regressor matrix can be written as

$$\mathbf{X} = \mathbf{I}_M \otimes \mathbf{X}_0.$$

The SUR moment term is

$$\mathbf{X}'(\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_T)\mathbf{X} = (\mathbf{I}_M \otimes \mathbf{X}'_0)(\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_T)(\mathbf{I}_M \otimes \mathbf{X}_0) \quad (8.91)$$

$$= \boldsymbol{\Sigma}^{-1} \otimes (\mathbf{X}'_0\mathbf{X}_0). \quad (8.92)$$

Its inverse is

$$\boldsymbol{\Sigma} \otimes (\mathbf{X}'_0\mathbf{X}_0)^{-1}.$$

Moreover,

$$\mathbf{X}'(\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_T)\mathbf{y} = (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{X}'_0)\mathbf{y}.$$

Thus,

$$\hat{\boldsymbol{\beta}}_{\text{SUR}} = [\boldsymbol{\Sigma} \otimes (\mathbf{X}'_0\mathbf{X}_0)^{-1}] (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{X}'_0)\mathbf{y} \quad (8.93)$$

$$= (\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-1}) \otimes [(\mathbf{X}'_0\mathbf{X}_0)^{-1}\mathbf{X}'_0] \mathbf{y} \quad (8.94)$$

$$= \mathbf{I}_M \otimes [(\mathbf{X}'_0\mathbf{X}_0)^{-1}\mathbf{X}'_0] \mathbf{y}. \quad (8.95)$$

This result is numerically equal to equation-by-equation OLS.

Intuition: if all equations use exactly the same set of regressors, correlation between errors does not create additional information for estimating coefficients, as all equations project dependent variables onto the same regressor space.

Example: in an unrestricted VAR, all equations usually have the same lags as regressors. In this case, SUR does not generate efficiency gain over equation-by-equation OLS.

8.3.11 Caveats and Common Errors

8.3.11.1 Using SUR with Identical Regressors as if There Were Gain

If all equations have the same regressors, SUR and equation-by-equation OLS produce identical estimates. Thus, one should not justify SUR only by efficiency gain in this case.

Still, SUR may be useful with identical regressors if the objective is to obtain covariance between estimators from different equations for cross-restriction tests.

8.3.11.2 Contamination by Misspecification

SUR weights the system by $\hat{\boldsymbol{\Sigma}}^{-1}$. If one equation is misspecified or contains endogenous regressors, the problem can propagate to other equations.

The broad exogeneity condition is stronger than exogeneity proper to each equation. For SUR, it is necessary that

$$\mathbb{E}[\mathbf{x}_{mt}\varepsilon_{nt}] = 0 \quad \forall m, n.$$

If this condition fails, SUR may become inconsistent for the entire system.

Practical symptom: coefficients from a previously stable equation change substantially when another equation is added to the system.

8.3.11.3 Singularity in Share Systems

In demand or cost systems, share equations sum to 1:

$$\sum_{m=1}^M y_{mt} = 1.$$

Then,

$$\sum_{m=1}^M \varepsilon_{mt} = 0,$$

which makes Σ singular.

Solution: estimate $M - 1$ equations and recover the remaining equation through sum restrictions.

8.3.11.4 FGLS in Small Samples

When T is small and M is large, estimation of $\hat{\Sigma}$ may be very noisy. Feasible SUR may perform worse than OLS in finite samples.

Symptom: standard errors larger than individual OLS or numerical instability.

Alternatives: use OLS, reduce the number of equations, or apply regularization methods for $\hat{\Sigma}$.

8.3.11.5 Diagnostic Table

Problem	Symptom	Diagnosis
Identical regressors	SUR equal to separate OLS	Check if $\mathbf{X}_m$ are equal
Endogeneity in one equation	Coefficients from other equations change	Test exogeneity; compare OLS and SUR
Σ singular	Inversion error or unstable SEs	Check sum restriction; discard equation
Small T and large M	Noisy FGLS	Compare with OLS; consider shrinkage
Autocorrelation	Structure $\Sigma \otimes \mathbf{I}_T$ inadequate	Test autocorrelation; use Γ

Table 8.5: Common problems in SUR estimation.

8.3.12 Typical Committee or Referee Question

Question: If one equation in the system contains an endogenous regressor, can SUR contaminate other equations?

Answer: Yes, this risk exists because SUR uses cross-weighting based on Σ^{-1} .

To see the point, consider a system with two equations. The matrix Σ^{-1} has elements

$$\Sigma^{-1} = \begin{bmatrix} \sigma^{11} & \sigma^{12} \\ \sigma^{21} & \sigma^{22} \end{bmatrix}.$$

The transformed moment associated with the second equation involves combinations of errors from both equations:

$$\mathbf{x}_{2t} (\sigma^{21} \varepsilon_{1t} + \sigma^{22} \varepsilon_{2t}).$$

Even if the second equation is isolatedly well-specified, with

$$\mathbb{E}[\mathbf{x}_{2t} \varepsilon_{2t}] = 0,$$

SUR also requires

$$\mathbb{E}[\mathbf{x}_{2t} \varepsilon_{1t}] = 0.$$

If this condition fails, the second equation's moment does not converge to zero, and the estimate of β_2 may be contaminated.

Therefore, when there is suspicion of endogeneity in one equation, it is safer to:

- estimate that equation by IV/2SLS;
- use systemic methods appropriate for endogeneity, such as 3SLS or GMM;
- or report SOLS/OLS with robust standard errors, prioritizing consistency over efficiency.

8.3.13 Tests and Extensions

8.3.13.1 Breusch-Pagan Test for Contemporaneous Correlation

To test whether Σ is diagonal, one uses the LM statistic

$$LM = T \sum_{m=2}^M \sum_{n=1}^{m-1} \hat{\rho}_{mn}^2, \quad (8.96)$$

where $\hat{\rho}_{mn}$ is the residual correlation between equations m and n .

Under the null hypothesis of contemporaneous independence,

$$LM \xrightarrow{d} \chi_{M(M-1)/2}^2. \quad (8.97)$$

If the null hypothesis is rejected, there is evidence that SUR may be more efficient than SOLS.

8.3.13.2 Cross-Restriction Tests

To test

$$H_0 : \mathbf{R}\beta = \mathbf{r},$$

one uses the Wald statistic

$$W = (\mathbf{R}\hat{\beta} - \mathbf{r})' [\mathbf{R}\widehat{\text{Var}}(\hat{\beta})\mathbf{R}']^{-1} (\mathbf{R}\hat{\beta} - \mathbf{r}). \quad (8.98)$$

Under H_0 ,

$$W \xrightarrow{d} \chi_q^2,$$

where q is the number of restrictions.

SUR is especially useful when one wishes to test equality of coefficients between equations, as it directly provides covariance between estimators.

8.3.13.3 Autocorrelation and Heteroskedasticity

If there is temporal autocorrelation, the standard structure

$$\mathbf{\Omega} = \mathbf{\Sigma} \otimes \mathbf{I}_T$$

may be replaced by

$$\mathbf{\Omega} = \mathbf{\Sigma} \otimes \mathbf{\Gamma},$$

where $\mathbf{\Gamma}$ models temporal dependence.

If there is heteroskedasticity or more general dependence, one can use sandwich-type robust standard errors for systems, for example:

$$\widehat{\text{Var}}(\hat{\beta}) = \widehat{\mathbf{Q}}^{-1} \widehat{\mathbf{S}} \widehat{\mathbf{Q}}^{-1}, \quad (8.99)$$

where $\widehat{\mathbf{Q}}$ estimates the moment matrix and $\widehat{\mathbf{S}}$ estimates the variance of moments.

8.3.13.4 Relation to Other Estimators

SUR can be seen as:

- a particular case of GLS;
- an extension of equation-by-equation OLS when there is contemporaneous correlation;
- a precursor of more robust systemic estimators to endogeneity, such as 3SLS and GMM.

In summary, SUR is efficient when broad exogeneity and covariance structure assumptions are correct, but may be risky when there is misspecification or endogeneity in some equation of the system.

Chapter 9

Panel Data

9.1 Introduction: Unobserved Heterogeneity in Panels

Consider a balanced panel with N cross-sectional units and T periods. For unit $i = 1, \dots, N$ in period $t = 1, \dots, T$, we write the linear model as

$$y_{it} = \mathbf{x}_{it}'\boldsymbol{\beta} + v_{it}, \quad (9.1)$$

where y_{it} is the dependent variable, $\mathbf{x}_{it}$ is a $K \times 1$ vector of regressors, $\boldsymbol{\beta}$ is the $K \times 1$ parameter vector, and v_{it} is the compound error.

The central panel data structure is the decomposition of the compound error:

$$v_{it} = c_i + u_{it}. \quad (9.2)$$

The term c_i represents unobserved individual heterogeneity, constant over time. The term u_{it} is the idiosyncratic shock, which varies across individuals and over time.

The main econometric difficulty is that c_i may be correlated with the regressors $\mathbf{x}_{it}$. When this occurs, estimators that treat c_i as part of the error without proper correction tend to be inconsistent.

In matrix notation by individual, we define:

$$\mathbf{y}_i = \begin{bmatrix} y_{i1} \\ y_{i2} \\ \vdots \\ y_{iT} \end{bmatrix}, \quad \mathbf{X}_i = \begin{bmatrix} \mathbf{x}_{i1}' \\ \mathbf{x}_{i2}' \\ \vdots \\ \mathbf{x}_{iT}' \end{bmatrix}, \quad \mathbf{v}_i = \begin{bmatrix} v_{i1} \\ v_{i2} \\ \vdots \\ v_{iT} \end{bmatrix}. \quad (9.3)$$

Thus,

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{v}_i, \quad i = 1, \dots, N. \quad (9.4)$$

Throughout this chapter, we will repeatedly use:

- $\mathbf{j}_T$: $T \times 1$ vector of ones;
- $\mathbf{P}_j = \frac{1}{T}\mathbf{j}_T\mathbf{j}_T'$: projection matrix onto the space of temporal constants;
- $\mathbf{Q} = \mathbf{I}_T - \mathbf{P}_j$: temporal mean annihilation matrix.

9.2 Pooled OLS (POLS)

9.2.1 Research Problem and Intuition

The Pooled OLS (POLS) estimator consists of stacking all (i, t) observations and applying OLS as if the panel were a large cross-sectional sample of size NT .

A typical example is estimating the return to education on wages using a panel of workers:

$$\log(\text{wage}_{it}) = \beta_0 + \beta_1 \text{educ}_{it} + c_i + u_{it}. \quad (9.5)$$

POLS is the natural starting point because it is simple and uses all available variation, both within individuals and between individuals. However, it ignores the compound error structure.

If c_i represents innate ability, motivation, or permanent unit characteristics, and if c_i is correlated with educ_{it} , then

$$\text{Cov}(\text{educ}_{it}, c_i) \neq 0. \quad (9.6)$$

Since c_i is contained in v_{it} , the regressor becomes correlated with the compound error:

$$\mathbb{E}[\mathbf{x}_{it} v_{it}] = \mathbb{E}[\mathbf{x}_{it} c_i] + \mathbb{E}[\mathbf{x}_{it} u_{it}]. \quad (9.7)$$

If the first term is non-zero, POLS is inconsistent.

Geometrically, POLS fits a single line to all stacked observations. It mixes within variation (changes within the same individual over time) and between variation (differences between individuals). If each individual has an unobserved intercept c_i , this mixing may generate an estimate that does not adequately represent either source of variation.

9.2.2 Assumptions and Their Roles

Definition 9.2.1 (POLS.1: Contemporaneous Orthogonality of Compound Error — Consistency). For all $t = 1, \dots, T$,

$$\mathbb{E}[\mathbf{x}_{it} v_{it}] = \mathbf{0}. \quad (9.8)$$

Since $v_{it} = c_i + u_{it}$, this condition simultaneously requires:

$$\mathbb{E}[\mathbf{x}_{it} c_i] = \mathbf{0} \quad \text{and} \quad \mathbb{E}[\mathbf{x}_{it} u_{it}] = \mathbf{0}. \quad (9.9)$$

Role: guarantee **consistency** of the POLS estimator when $N \rightarrow \infty$ with fixed T .

Translation to practice: time-invariant unobserved factors cannot be correlated with contemporaneous regressors. In wage applications, this would require that innate ability, for example, is not correlated with education in the same period.

What this assumption does *not* prohibit: idiosyncratic shocks u_{it} may affect regressors in *future* periods ($\mathbf{x}_{i,t+1}$). That is, dynamic feedbacks are allowed.

Counterexample: in a productivity equation, more efficient firms may invest more and also have higher unobserved productivity. If efficiency is in c_i and investment is in $\mathbf{x}_{it}$, then $\mathbb{E}[\mathbf{x}_{it} c_i] \neq 0$, and POLS is inconsistent.

Definition 9.2.2 (POLS.1': Strict Exogeneity — Unbiasedness in Finite Samples). For all $t = 1, \dots, T$,

$$\mathbb{E}[v_{it} \mid \mathbf{X}_i] = 0, \quad (9.10)$$

where $\mathbf{X}_i = (\mathbf{x}_{i1}, \dots, \mathbf{x}_{iT})'$ is individual i 's regressor matrix in all periods.

Equivalently, for all t and all s :

$$\mathbb{E}[\mathbf{x}_{is}v_{it}] = \mathbf{0}. \quad (9.11)$$

Role: guarantee that the estimator is **conditionally unbiased** given regressors in finite samples.

Relation to POLS.1: by the Law of Iterated Expectations, $\text{POLS.1}' \implies \text{POLS.1}$, but the converse is false.

Translation to practice: prohibits feedback mechanisms. The current shock cannot affect future regressors.

Counterexample: if a positive productivity shock today leads the firm to invest more tomorrow, then $\mathbb{E}[\mathbf{x}_{i,t+1}v_{it}] \neq 0$, and strict exogeneity fails. However, POLS may still be consistent under POLS.1.

Definition 9.2.3 (POLS.2: Rank Condition).

$$\text{rank} \left(\mathbb{E} \left[\sum_{t=1}^T \mathbf{x}_{it} \mathbf{x}_{it}' \right] \right) = K. \quad (9.12)$$

Role: guarantee identification and invertibility of the moment matrix.

Translation to practice: there must be sufficient variation in regressors and absence of perfect collinearity.

Counterexample: simultaneously including two variables that are perfect linear combinations, such as years and months of education, makes the moment matrix singular.

Definition 9.2.4 (POLS.3: Spherical Errors for Efficiency and Classical Inference). For POLS to be BLUE and for conventional standard errors to be valid, it would be necessary:

$$\mathbb{E}[\mathbf{v}_i \mathbf{v}_i' \mid \mathbf{X}_i] = \sigma^2 \mathbf{I}_T. \quad (9.13)$$

Role: justify classical inference and Gauss-Markov efficiency.

Translation to practice: requires absence of serial correlation and homoskedasticity.

Important: violation of POLS.3 does **not** generate bias or inconsistency. It only affects estimator efficiency and validity of conventional standard errors, which must be corrected via cluster-robust standard errors.

9.2.3 Global Variance-Covariance Matrix Structure

To internalize the content of sphericity assumptions, we abandon the usual simplifications of introductory textbooks and deduce the global variance-covariance matrix from its fundamental blocks. This construction maps three sources of correlation that often generate confusion: **intra-individual** correlation (within the same unit over time), **inter-individual** correlation (between different units), and **pure temporal** correlation (induced by common macroeconomic shocks).

9.2.3.1 Definition of the Stacked Error Vector

Consider a balanced panel with N cross-sectional units observed over T periods. The compound error v_{it} associated with individual i in period t forms, for each i , the $T \times 1$ vector:

$$\mathbf{v}_i = \begin{bmatrix} v_{i1} \\ v_{i2} \\ \vdots \\ v_{iT} \end{bmatrix}. \quad (9.14)$$

Stacking the N individuals, we construct the global vector $\mathbf{v}$ of dimension $NT \times 1$:

$$\mathbf{v} = \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \vdots \\ \mathbf{v}_N \end{bmatrix} = \begin{bmatrix} v_{11} \\ \vdots \\ v_{1T} \\ \vdots \\ v_{N1} \\ \vdots \\ v_{NT} \end{bmatrix}. \quad (9.15)$$

Under the zero-mean premise $\mathbb{E}[\mathbf{v}] = \mathbf{0}$, the global variance-covariance matrix is

$$\mathbf{\Omega} = \mathbb{E}[\mathbf{v}\mathbf{v}'] \in \mathbb{R}^{NT \times NT}. \quad (9.16)$$

9.2.3.2 Block Expansion

Applying the definition of outer product of partitioned vectors and linearity of expectation over submatrices,

$$\mathbf{\Omega} = \mathbb{E} \left[\begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \vdots \\ \mathbf{v}_N \end{bmatrix} \begin{bmatrix} \mathbf{v}'_1 & \mathbf{v}'_2 & \dots & \mathbf{v}'_N \end{bmatrix} \right] = \begin{bmatrix} \mathbf{\Sigma}_{11} & \mathbf{\Sigma}_{12} & \dots & \mathbf{\Sigma}_{1N} \\ \mathbf{\Sigma}_{21} & \mathbf{\Sigma}_{22} & \dots & \mathbf{\Sigma}_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{\Sigma}_{N1} & \mathbf{\Sigma}_{N2} & \dots & \mathbf{\Sigma}_{NN} \end{bmatrix}, \quad (9.17)$$

where each block $\mathbf{\Sigma}_{ij} \in \mathbb{R}^{T \times T}$ is

$$\mathbf{\Sigma}_{ij} = \mathbb{E}[\mathbf{v}_i \mathbf{v}'_j] = \begin{bmatrix} \mathbb{E}[v_{i1}v_{j1}] & \mathbb{E}[v_{i1}v_{j2}] & \dots & \mathbb{E}[v_{i1}v_{jT}] \\ \mathbb{E}[v_{i2}v_{j1}] & \mathbb{E}[v_{i2}v_{j2}] & \dots & \mathbb{E}[v_{i2}v_{jT}] \\ \vdots & \vdots & \ddots & \vdots \\ \mathbb{E}[v_{iT}v_{j1}] & \mathbb{E}[v_{iT}v_{j2}] & \dots & \mathbb{E}[v_{iT}v_{jT}] \end{bmatrix}. \quad (9.18)$$

9.2.3.3 Intra-Individual Correlation: Diagonal Blocks $\mathbf{\Sigma}_{ii}$

Intra-individual correlation (serial autocorrelation) lives in the diagonal blocks $\mathbf{\Sigma}_{ii}$ and measures how shocks from the same individual are related over time.

Scenario A — Classical Error Components Model. Assume the canonical random effects model:

$$v_{it} = c_i + u_{it}, \quad (9.19)$$

with $\mathbb{E}[c_i^2] = \sigma_c^2$, $\mathbb{E}[u_{it}^2] = \sigma_u^2$, $\mathbb{E}[u_{it}u_{is}] = 0$ for $t \neq s$, and $\mathbb{E}[c_i u_{it}] = 0$. Calculating the covariance terms:

$$t = s : \quad \mathbb{E}[v_{it}^2] = \mathbb{E}[(c_i + u_{it})^2] = \sigma_c^2 + \sigma_u^2, \quad (9.20)$$

$$t \neq s : \quad \mathbb{E}[v_{it}v_{is}] = \mathbb{E}[(c_i + u_{it})(c_i + u_{is})] = \sigma_c^2. \quad (9.21)$$

Therefore,

$$\Sigma_{ii} = \begin{bmatrix} \sigma_c^2 + \sigma_u^2 & \sigma_c^2 & \dots & \sigma_c^2 \\ \sigma_c^2 & \sigma_c^2 + \sigma_u^2 & \dots & \sigma_c^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_c^2 & \sigma_c^2 & \dots & \sigma_c^2 + \sigma_u^2 \end{bmatrix} = \sigma_u^2 \mathbf{I}_T + \sigma_c^2 \mathbf{J}_T, \quad (9.22)$$

where $\mathbf{J}_T = \mathbf{j}_T \mathbf{j}_T'$ is the $T \times T$ matrix of ones. Under unobserved individual heterogeneity, errors from the same individual exhibit constant positive covariance σ_c^2 , regardless of temporal distance between periods.

Scenario B — AR(1) Idiosyncratic Shock. If $u_{it} = \rho u_{i,t-1} + e_{it}$ with $|\rho| < 1$ and $e_{it} \sim \text{i.i.d.}(0, \sigma_e^2)$, covariance decays geometrically:

$$\mathbb{E}[v_{it}v_{is}] = \sigma_c^2 + \sigma_u^2 \rho^{|t-s|}, \quad (9.23)$$

and Σ_{ii} assumes the form

$$\Sigma_{ii} = \begin{bmatrix} \sigma_c^2 + \sigma_u^2 & \sigma_c^2 + \sigma_u^2 \rho & \dots & \sigma_c^2 + \sigma_u^2 \rho^{T-1} \\ \sigma_c^2 + \sigma_u^2 \rho & \sigma_c^2 + \sigma_u^2 & \dots & \sigma_c^2 + \sigma_u^2 \rho^{T-2} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_c^2 + \sigma_u^2 \rho^{T-1} & \sigma_c^2 + \sigma_u^2 \rho^{T-2} & \dots & \sigma_c^2 + \sigma_u^2 \end{bmatrix}. \quad (9.24)$$

9.2.3.4 Inter-Individual Correlation: Off-Diagonal Blocks Σ_{ij} , $i \neq j$

Inter-individual correlation measures cross-sectional dependence: whether shocks in individual i affect individual j .

Scenario A — Classical Random Sampling. Under independence between individuals,

$$\Sigma_{ij} = \mathbf{0}_{T \times T} \quad \forall i \neq j, \quad (9.25)$$

and the global matrix is perfectly block-diagonal:

$$\Omega = \mathbf{I}_N \otimes \Sigma_{ii} = \begin{bmatrix} \Sigma_{11} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \Sigma_{22} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \Sigma_{NN} \end{bmatrix}. \quad (9.26)$$

Scenario B — Spatial or Network Correlation. If there is spatial dependence or network contiguity, contemporaneous shocks correlate cross-sectionally. Defining σ_{ij} as the contemporaneous covariance between i and j , and assuming non-contemporaneous shocks are independent, off-diagonal blocks become

$$\Sigma_{ij} = \sigma_{ij} \mathbf{I}_T = \begin{bmatrix} \sigma_{ij} & 0 & \dots & 0 \\ 0 & \sigma_{ij} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{ij} \end{bmatrix} \quad \text{for } i \neq j. \quad (9.27)$$

9.2.3.5 Pure Temporal Correlation: Two-Way Error Components Model

To simultaneously visualize the three sources of correlation, consider the **two-way error components model**:

$$v_{it} = c_i + d_t + u_{it}, \quad (9.28)$$

where c_i is the time-invariant individual effect ($\mathbb{E}[c_i^2] = \sigma_c^2$), d_t is the temporal effect common to all individuals in period t ($\mathbb{E}[d_t^2] = \sigma_d^2$, $\mathbb{E}[d_t d_s] = 0$ for $t \neq s$), and u_{it} is the spherical idiosyncratic shock ($\mathbb{E}[u_{it}^2] = \sigma_u^2$). Under mutual orthogonality between all components:

$$\text{Total variance (same } i, \text{ same } t): \quad \mathbb{E}[v_{it}^2] = \sigma_c^2 + \sigma_d^2 + \sigma_u^2, \quad (9.29)$$

$$\text{Intra-individual (same } i, t \neq s): \quad \mathbb{E}[v_{it} v_{is}] = \sigma_c^2, \quad (9.30)$$

$$\text{Inter-individual/temporal (} i \neq j, \text{ same } t): \quad \mathbb{E}[v_{it} v_{jt}] = \sigma_d^2, \quad (9.31)$$

$$\text{Cross-orthogonality (} i \neq j, t \neq s): \quad \mathbb{E}[v_{it} v_{js}] = 0. \quad (9.32)$$

9.2.3.6 Global Matrix Visualization for $N = 2$ and $T = 3$

To fix the position of each correlation, we explicitly expand $\mathbf{\Omega}$ in the two-way model with $N = 2$ individuals and $T = 3$ periods. The stacked vector is $\mathbf{v} = (v_{11}, v_{12}, v_{13}, v_{21}, v_{22}, v_{23})'$, and the 6×6 matrix assumes the form:

$$\mathbf{\Omega} = \left[\begin{array}{ccc|ccc} \sigma_c^2 + \sigma_d^2 + \sigma_u^2 & \sigma_c^2 & \sigma_c^2 & \sigma_d^2 & 0 & 0 \\ \sigma_c^2 & \sigma_c^2 + \sigma_d^2 + \sigma_u^2 & \sigma_c^2 & 0 & \sigma_d^2 & 0 \\ \sigma_c^2 & \sigma_c^2 & \sigma_c^2 + \sigma_d^2 + \sigma_u^2 & 0 & 0 & \sigma_d^2 \\ \hline \sigma_d^2 & 0 & 0 & \sigma_c^2 + \sigma_d^2 + \sigma_u^2 & \sigma_c^2 & \sigma_c^2 \\ 0 & \sigma_d^2 & 0 & \sigma_c^2 & \sigma_c^2 + \sigma_d^2 + \sigma_u^2 & \sigma_c^2 \\ 0 & 0 & \sigma_d^2 & \sigma_c^2 & \sigma_c^2 & \sigma_c^2 + \sigma_d^2 + \sigma_u^2 \end{array} \right]. \quad (9.33)$$

Reading guide:

- **Main diagonal:** total variability of each observation ($\sigma_c^2 + \sigma_d^2 + \sigma_u^2$).
- **Off-diagonal in diagonal blocks (σ_c^2):** intra-individual correlation, induced by c_i over time.
- **Diagonal of off-diagonal blocks (σ_d^2):** contemporaneous inter-individual correlation, induced by common macroeconomic shock d_t .
- **Remaining zeros:** absence of cross-temporal correlation between different individuals in different periods.

Conclusion for practice. Even in the simplest case ($d_t = 0$, random sampling), the matrix $\mathbf{\Omega}$ is not proportional to the identity whenever $\sigma_c^2 > 0$. This invalidates classical inference and justifies using individual-level cluster-robust standard errors — a strategy that accommodates arbitrary correlation within each block $\mathbf{\Sigma}_{ii}$ without imposing specific structure.

9.2.4 Derivation of the POLS Estimator

The POLS estimator minimizes the sum of squared residuals of the stacked system:

$$Q(\boldsymbol{\beta}) = \sum_{i=1}^N (\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta})' (\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta}). \quad (9.34)$$

The first-order condition is

$$\frac{\partial Q(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -2 \sum_{i=1}^N \mathbf{X}_i' (\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta}) = \mathbf{0}. \quad (9.35)$$

Thus,

$$\left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{X}_i \right) \hat{\boldsymbol{\beta}}_{\text{POLS}} = \sum_{i=1}^N \mathbf{X}_i' \mathbf{y}_i. \quad (9.36)$$

Under POLS.2, the moment matrix is invertible. Therefore,

$$\hat{\boldsymbol{\beta}}_{\text{POLS}} = \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{X}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{y}_i \right). \quad (9.37)$$

Substituting $\mathbf{y}_i = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{v}_i$,

$$\hat{\boldsymbol{\beta}}_{\text{POLS}} = \boldsymbol{\beta} + \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{X}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{v}_i \right). \quad (9.38)$$

9.2.5 First-Order Moment Properties

9.2.5.1 Consistency Under Contemporaneous Exogeneity

Theorem 9.2.1 (POLS Consistency Under POLS.1). *Under POLS.1 (contemporaneous exogeneity) and POLS.2 (full rank), the POLS estimator is consistent when $N \rightarrow \infty$ with fixed T :*

$$\text{plim}_{N \rightarrow \infty} \hat{\boldsymbol{\beta}}_{\text{POLS}} = \boldsymbol{\beta}. \quad (9.39)$$

Proof. Normalizing by N :

$$\hat{\boldsymbol{\beta}}_{\text{POLS}} - \boldsymbol{\beta} = \left(\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \mathbf{X}_i \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \mathbf{v}_i \right). \quad (9.40)$$

Expanding the matrix products within the summations:

$$\hat{\boldsymbol{\beta}}_{\text{POLS}} - \boldsymbol{\beta} = \left(\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \mathbf{x}_{it}' \mathbf{x}_{it} \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \mathbf{x}_{it}' v_{it} \right). \quad (9.41)$$

By the Weak Law of Large Numbers applied to independent units:

$$\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \mathbf{x}_{it}' \mathbf{x}_{it} \xrightarrow{p} \sum_{t=1}^T \mathbb{E}[\mathbf{x}_{it}' \mathbf{x}_{it}] \equiv \mathbf{Q}_{XX}, \quad (9.42)$$

where $\mathbf{Q}_{XX}$ is a positive definite matrix under POLS.2.

Moreover:

$$\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \mathbf{x}'_{it} v_{it} \xrightarrow{p} \sum_{t=1}^T \mathbb{E}[\mathbf{x}'_{it} v_{it}]. \quad (9.43)$$

Under POLS.1 (contemporaneous exogeneity):

$$\mathbb{E}[\mathbf{x}_{it} v_{it}] = \mathbf{0}, \quad \forall t. \quad (9.44)$$

Therefore:

$$\sum_{t=1}^T \mathbb{E}[\mathbf{x}'_{it} v_{it}] = \mathbf{0}, \quad (9.45)$$

and by Slutsky's Theorem:

$$\text{plim } \hat{\boldsymbol{\beta}}_{\text{POLS}} = \boldsymbol{\beta} + \mathbf{Q}_{XX}^{-1} \cdot \mathbf{0} = \boldsymbol{\beta}. \quad (9.46)$$

□

Remark 9.2.1 (Consistency Under Dynamic Feedback). POLS consistency requires **only** contemporaneous exogeneity (POLS.1). Even if there is dynamic feedback ($\mathbb{E}[\mathbf{x}_{i,t+1} v_{it}] \neq 0$), the estimator remains consistent.

9.2.5.2 Unbiasedness Under Strict Exogeneity

Theorem 9.2.2 (Conditional Unbiasedness Under POLS.1'). *Under POLS.1' (strict exogeneity), the POLS estimator is conditionally unbiased in finite samples:*

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{POLS}} \mid \mathbf{X}] = \boldsymbol{\beta}. \quad (9.47)$$

Proof. To prove finite-sample unbiasedness, we condition on the complete regressor matrix $\mathbf{X}$:

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{POLS}} \mid \mathbf{X}] = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \mathbb{E}[\mathbf{v} \mid \mathbf{X}]. \quad (9.48)$$

Under POLS.1' (strict exogeneity), $\mathbb{E}[\mathbf{v} \mid \mathbf{X}] = \mathbf{0}$, and therefore:

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{POLS}} \mid \mathbf{X}] = \boldsymbol{\beta}. \quad (9.49)$$

□

Remark 9.2.2 (Hierarchy Between POLS.1 and POLS.1'). Unbiasedness requires the strong assumption (POLS.1'). If there is dynamic feedback, POLS is biased in finite samples, but remains asymptotically consistent under POLS.1.

9.2.5.3 Asymptotic Normality

Theorem 9.2.3 (Limiting Distribution of POLS). *Under regularity conditions, finite moments, and i.i.d. sampling in dimension i :*

$$\sqrt{N}(\hat{\beta}_{POLS} - \beta) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}), \quad (9.50)$$

where

$$\mathbf{A} = \mathbb{E}[\mathbf{X}_i' \mathbf{X}_i], \quad (9.51)$$

$$\mathbf{B} = \mathbb{E}[\mathbf{X}_i' \mathbf{v}_i \mathbf{v}_i' \mathbf{X}_i]. \quad (9.52)$$

Remark 9.2.3 (Sandwich Form). The asymptotic variance $\mathbf{V} = \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$ assumes a sandwich structure, where $\mathbf{A}$ is the "bread" (regressor moments) and $\mathbf{B}$ is the "meat" (cross-moments with the error). This form is the basis for robust standard errors.

9.2.6 Second-Order Anatomy and Non-Sphericity

9.2.6.1 Collapse of Classical Variance

Theorem 9.2.4 (Inconsistency of Classical Variance Under Non-Sphericity). *Under the presence of unobserved individual heterogeneity ($\sigma_c^2 > 0$), the classical POLS variance estimator is inconsistent:*

$$\text{plim}_{N \rightarrow \infty} N\widehat{\text{Var}}_{\text{class}}(\hat{\beta}_{POLS}) = \sigma_v^2 \mathbf{A}^{-1} \neq \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1} \equiv \mathbf{V}_{\text{asymptotic}}, \quad (9.53)$$

where $\sigma_v^2 = \sigma_c^2 + \sigma_u^2$ is the total variance of the compound error.

Proof. The classical variance estimator is:

$$\widehat{\text{Var}}_{\text{class}}(\hat{\beta}_{POLS}) = s^2 (\mathbf{X}'\mathbf{X})^{-1}, \quad (9.54)$$

where $s^2 = \frac{\hat{\mathbf{v}}'\hat{\mathbf{v}}}{NT-K}$ is the residual variance estimator.

By consistency of $\hat{\beta}_{POLS}$ and the Weak Law of Large Numbers:

$$\text{plim}_{N \rightarrow \infty} s^2 = \mathbb{E}[v_{it}^2] = \sigma_v^2. \quad (9.55)$$

Therefore:

$$\text{plim}_{N \rightarrow \infty} N\widehat{\text{Var}}_{\text{class}}(\hat{\beta}_{POLS}) = \sigma_v^2 \mathbf{A}^{-1}. \quad (9.56)$$

On the other hand, the true asymptotic variance is:

$$\mathbf{V}_{\text{asymptotic}} = \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}, \quad (9.57)$$

where $\mathbf{B} = \mathbb{E}[\mathbf{X}_i' \mathbf{v}_i \mathbf{v}_i' \mathbf{X}_i]$.

By the Law of Iterated Expectations:

$$\begin{aligned} \mathbf{B} &= \mathbb{E}[\mathbf{X}_i' \mathbb{E}[\mathbf{v}_i \mathbf{v}_i' \mid \mathbf{X}_i] \mathbf{X}_i] \\ &= \mathbb{E}[\mathbf{X}_i' (\sigma_u^2 \mathbf{I}_T + \sigma_c^2 \mathbf{1}_T \mathbf{1}_T') \mathbf{X}_i] \\ &= \sigma_u^2 \mathbb{E}[\mathbf{X}_i' \mathbf{X}_i] + \sigma_c^2 \mathbb{E}[\mathbf{X}_i' \mathbf{1}_T \mathbf{1}_T' \mathbf{X}_i]. \end{aligned} \quad (9.58)$$

On the other hand:

$$\sigma_v^2 \mathbf{A} = (\sigma_u^2 + \sigma_c^2) \mathbb{E}[\mathbf{X}_i' \mathbf{X}_i] = \sigma_u^2 \mathbb{E}[\mathbf{X}_i' \mathbf{X}_i] + \sigma_c^2 \mathbb{E}[\mathbf{X}_i' \mathbf{X}_i]. \quad (9.59)$$

The difference is:

$$\mathbf{B} - \sigma_v^2 \mathbf{A} = \sigma_c^2 (\mathbb{E}[\mathbf{X}_i' \mathbf{1}_T \mathbf{1}_T' \mathbf{X}_i] - \mathbb{E}[\mathbf{X}_i' \mathbf{X}_i]) = \sigma_c^2 \sum_{t \neq s} \mathbb{E}[\mathbf{x}_{it}' \mathbf{x}_{is}]. \quad (9.60)$$

Since regressors typically exhibit positive temporal correlation ($\mathbb{E}[\mathbf{x}_{it}' \mathbf{x}_{is}] > 0$ for $t \neq s$) and $\sigma_c^2 > 0$, we have $\mathbf{B} > \sigma_v^2 \mathbf{A}$, implying:

$$\sigma_v^2 \mathbf{A}^{-1} \neq \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}. \quad (9.61)$$

□

Remark 9.2.4 (Direction of Bias). Under positive intraclass correlation ($\sigma_c^2 > 0$) and temporal persistence of regressors ($\mathbb{E}[\mathbf{x}_{it}' \mathbf{x}_{is}] > 0$), the classical variance **underestimates** the true asymptotic variability:

$$\widehat{\text{Var}}_{\text{class}}(\hat{\boldsymbol{\beta}}_{\text{POLs}}) \ll \text{Var}(\hat{\boldsymbol{\beta}}_{\text{POLs}}). \quad (9.62)$$

This generates artificially low standard errors, inflated t -statistics, and rejection rates of H_0 much higher than the nominal level (excessive Type I Error).

9.2.7 Cluster-Robust Inference

9.2.7.1 Cluster-Robust Variance Estimator (CRVE)

Since v_{it} is serially correlated within the individual when $\text{Var}(c_i) > 0$, the classical OLS variance is inadequate. Arellano's (1987) cluster-robust estimator is:

$$\widehat{\text{Var}}_{\text{cluster}}(\hat{\boldsymbol{\beta}}_{\text{POLs}}) = \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{X}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{X}_i' \hat{\mathbf{v}}_i \hat{\mathbf{v}}_i' \mathbf{X}_i \right) \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{X}_i \right)^{-1}, \quad (9.63)$$

where $\hat{\mathbf{v}}_i = \mathbf{y}_i - \mathbf{X}_i \hat{\boldsymbol{\beta}}_{\text{POLs}}$ are the estimated residuals.

Theorem 9.2.5 (CRVE Consistency). *Under i.i.d. sampling in dimension i , finite moments, and $N \rightarrow \infty$ with fixed T :*

$$\text{plim}_{N \rightarrow \infty} N \cdot \widehat{\text{Var}}_{\text{cluster}}(\hat{\boldsymbol{\beta}}_{\text{POLs}}) = \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1} = \mathbf{V}_{\text{asymptotic}}. \quad (9.64)$$

Proof. The CRVE estimator consistently replaces the "meat" $\mathbf{B}$ with its sample counterpart:

$$\hat{\mathbf{B}}_N = \frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \hat{\mathbf{v}}_i \hat{\mathbf{v}}_i' \mathbf{X}_i. \quad (9.65)$$

By consistency of $\hat{\boldsymbol{\beta}}_{\text{POLs}}$, residuals converge: $\hat{\mathbf{v}}_i \xrightarrow{p} \mathbf{v}_i$. By the Weak Law of Large Numbers:

$$\hat{\mathbf{B}}_N \xrightarrow{p} \mathbb{E}[\mathbf{X}_i' \mathbf{v}_i \mathbf{v}_i' \mathbf{X}_i] = \mathbf{B}. \quad (9.66)$$

By the Continuous Mapping Theorem and Slutsky:

$$N \cdot \widehat{\text{Var}}_{\text{cluster}} \xrightarrow{p} \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}. \quad (9.67)$$

□

Remark 9.2.5 (CRVE Advantages). 1. **Robustness:** allows arbitrary correlation between errors from the same unit i , accommodating both heteroskedasticity and serial autocorrelation within the individual, without imposing specific structure on Ω .

2. **Non-parametric:** does not require parameterization of the covariance matrix Ω_i .

3. **Consistency:** is asymptotically consistent under $N \rightarrow \infty$ with fixed T .

Remark 9.2.6 (Finite-Sample Adjustments). To mitigate underestimation bias in small samples, correction factors are applied:

$$\widehat{\text{Var}}_{\text{adj}} = \frac{N}{N-1} \cdot \frac{NT-1}{NT-K} \cdot \widehat{\text{Var}}_{\text{cluster}}. \quad (9.68)$$

Moreover, under a small number of clusters ($G < 50$), it is recommended to use a t -distribution with $G - 1$ degrees of freedom instead of the standard normal for hypothesis testing.

9.2.8 Caveats, Common Errors, and Committee Questions

9.2.8.1 Caveats and Common Errors

Inference without clustering Even if POLS.1 is valid and the estimator is consistent, the presence of c_i induces serial correlation:

$$\text{Cov}(v_{it}, v_{is}) = \text{Var}(c_i), \quad t \neq s. \quad (9.69)$$

Ignoring this fact generates underestimated standard errors and inflated t -statistics.

Symptom: excessive statistical significance, which disappears after clustering.

Heterogeneity bias If c_i is correlated with $\mathbf{x}_{it}$, POLS captures part of the effect of c_i as if it were the effect of $\mathbf{x}_{it}$.

Symptom: coefficients change substantially when comparing POLS with FE or FD.

Dynamic models If the model contains $y_{i,t-1}$ as a regressor and c_i is present, then $y_{i,t-1}$ is correlated with v_{it} by construction. POLS tends to attribute the persistence of c_i to the true dynamics of the dependent variable.

Symptom: autoregressive coefficient very close to one, which falls substantially when using appropriate estimators, such as dynamic GMM.

Structural stability POLS assumes that β is constant between individuals and periods. If the effect of $\mathbf{x}_{it}$ varies systematically over time or between groups, POLS estimates a potentially uninterpretable structural average.

Usual solution: stability tests, such as Chow test or interactions.

9.2.8.2 Typical Committee Question

Question: Your model uses Pooled OLS, but c_i is plausibly correlated with regressors. How do you justify consistency and inference?

Answer: POLS consistency requires

$$\mathbb{E}[\mathbf{x}_{it}v_{it}] = \mathbf{0}, \quad (9.70)$$

which includes

$$\mathbb{E}[\mathbf{x}_{it}c_i] = \mathbf{0}. \quad (9.71)$$

If this condition is implausible, POLS should be treated only as a benchmark. For inference, even under exogeneity, it is necessary to use cluster-robust standard errors, since

$$\text{Cov}(v_{it}, v_{is}) = \text{Var}(c_i) > 0, \quad t \neq s. \quad (9.72)$$

As robustness, one should compare POLS with FE or FD. If coefficients change drastically, this indicates heterogeneity bias.

9.3 Fixed Effects (FE)

9.3.1 Research Problem and Intuition

The Fixed Effects (FE) estimator solves the endogeneity problem generated by time-invariant unobserved heterogeneity. It allows c_i to be arbitrarily correlated with $\mathbf{x}_{it}$ and eliminates this term through a data transformation.

Consider again:

$$y_{it} = \mathbf{x}_{it}'\boldsymbol{\beta} + c_i + u_{it}. \quad (9.73)$$

If c_i is constant over time, subtracting the time mean for each individual eliminates c_i :

$$y_{it} - \bar{y}_i = (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)'\boldsymbol{\beta} + (u_{it} - \bar{u}_i). \quad (9.74)$$

Thus, FE exploits only within variation: what happens to y_{it} when $\mathbf{x}_{it}$ changes within the same individual.

The price of this solution is that time-invariant variables are not identified, as they are eliminated along with c_i .

9.3.2 Linear Algebra and Geometry of the Within Operator

9.3.2.1 Specification and Matrix Representation

Let there be a random sample of N cross-sectional units (individuals, firms, countries) indexed by $i \in \{1, \dots, N\}$, each observed over T discrete and consecutive time periods indexed by $t \in \{1, \dots, T\}$. The population model with additive individual effects is:

$$y_{it} = \mathbf{x}_{it}'\boldsymbol{\beta} + c_i + u_{it}, \quad (9.75)$$

where $y_{it} \in \mathbb{R}$ is the dependent variable, $\mathbf{x}_{it} \in \mathbb{R}^{K \times 1}$ is the vector of K time-varying regressors, $\boldsymbol{\beta} \in \mathbb{R}^{K \times 1}$ is the vector of structural parameters, $c_i \in \mathbb{R}$ is the unobserved individual heterogeneity (time-invariant individual effect), and $u_{it} \in \mathbb{R}$ is the time-varying idiosyncratic shock.

Remark 9.3.1 (Contextualization). In applied microeconometrics, c_i captures intrinsic fixed characteristics (innate ability, organizational culture) and u_{it} captures purely transitory perturbations (sectoral shocks, weather variations). The central problem is that, under selection or economic optimization, agents often determine $\mathbf{x}_{it}$ based on c_i , implying $\mathbb{E}[\mathbf{x}'_{it}c_i] \neq 0$. This violation of orthogonality invalidates Pooled OLS, generating omitted variable bias.

To operate geometrically on the temporal trajectory of each unit, we stack the T observations of individual i :

$$\mathbf{y}_i = \begin{bmatrix} y_{i1} \\ y_{i2} \\ \vdots \\ y_{iT} \end{bmatrix} \in \mathbb{R}^{T \times 1}, \quad \mathbf{X}_i = \begin{bmatrix} \mathbf{x}'_{i1} \\ \mathbf{x}'_{i2} \\ \vdots \\ \mathbf{x}'_{iT} \end{bmatrix} \in \mathbb{R}^{T \times K}, \quad \mathbf{u}_i = \begin{bmatrix} u_{i1} \\ u_{i2} \\ \vdots \\ u_{iT} \end{bmatrix} \in \mathbb{R}^{T \times 1}, \quad \mathbf{j}_T = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^{T \times 1}. \quad (9.76)$$

The system of T simultaneous equations for individual i is rewritten as:

$$\mathbf{y}_i = \mathbf{X}_i \boldsymbol{\beta} + c_i \mathbf{j}_T + \mathbf{u}_i. \quad (9.77)$$

Proposition 9.3.1 (Equivalence between scalar and matrix representations). *The t -th row of the matrix representation (9.77) exactly reproduces the scalar specification (9.75).*

Proof. Let $t \in \{1, \dots, T\}$. Evaluating the t -th row of each term:

- $[\mathbf{y}_i]_t = y_{it}$ (definition of stacked vector);
- $[\mathbf{X}_i \boldsymbol{\beta}]_t = \mathbf{x}'_{it} \boldsymbol{\beta}$ (row-column matrix product);
- $[c_i \mathbf{j}_T]_t = c_i \cdot [\mathbf{j}_T]_t = c_i \cdot 1 = c_i$;
- $[\mathbf{u}_i]_t = u_{it}$ (definition of stacked vector).

Summing the terms row by row:

$$[\mathbf{y}_i]_t = [\mathbf{X}_i \boldsymbol{\beta}]_t + [c_i \mathbf{j}_T]_t + [\mathbf{u}_i]_t \implies y_{it} = \mathbf{x}'_{it} \boldsymbol{\beta} + c_i + u_{it},$$

which is exactly Equation (9.75). □

9.3.2.2 The Within Demeaning Operator ($\mathbf{Q}_T$)

To purify the model variables from their time-stable components, we introduce the Within operator, which performs the orthogonal projection of level observations onto the space of deviations from the time mean. We define:

- $\mathbf{I}_T \in \mathbb{R}^{T \times T}$: identity matrix of order T ;
- $\mathbf{J}_T = \mathbf{j}_T \mathbf{j}'_T \in \mathbb{R}^{T \times T}$: matrix of ones;
- $\mathbf{P}_T = \frac{1}{T} \mathbf{j}_T \mathbf{j}'_T = \frac{1}{T} \mathbf{J}_T \in \mathbb{R}^{T \times T}$: projection matrix that computes the time mean;

- $\mathbf{Q}_T = \mathbf{I}_T - \mathbf{P}_T = \mathbf{I}_T - \frac{1}{T} \mathbf{j}_T \mathbf{j}_T' \in \mathbb{R}^{T \times T}$: Within demeaning operator.

Lemma 9.3.1 (Fundamental geometric properties of $\mathbf{Q}_T$). *The operator $\mathbf{Q}_T$ is symmetric ($\mathbf{Q}_T' = \mathbf{Q}_T$), idempotent ($\mathbf{Q}_T^2 = \mathbf{Q}_T$), and orthogonal to the vector of ones ($\mathbf{Q}_T \mathbf{j}_T = \mathbf{0}$).*

Proof. Symmetry:

$$\begin{aligned} \mathbf{Q}_T' &= \left(\mathbf{I}_T - \frac{1}{T} \mathbf{j}_T \mathbf{j}_T' \right)' = \mathbf{I}_T' - \frac{1}{T} (\mathbf{j}_T \mathbf{j}_T')' \\ &= \mathbf{I}_T - \frac{1}{T} (\mathbf{j}_T')' \mathbf{j}_T = \mathbf{I}_T - \frac{1}{T} \mathbf{j}_T \mathbf{j}_T' = \mathbf{Q}_T. \end{aligned}$$

Idempotence: Note first that $\mathbf{j}_T' \mathbf{j}_T = \sum_{t=1}^T 1^2 = T$. Then:

$$\begin{aligned} \mathbf{Q}_T^2 &= \left(\mathbf{I}_T - \frac{1}{T} \mathbf{j}_T \mathbf{j}_T' \right) \left(\mathbf{I}_T - \frac{1}{T} \mathbf{j}_T \mathbf{j}_T' \right) \\ &= \mathbf{I}_T - \frac{1}{T} \mathbf{j}_T \mathbf{j}_T' - \frac{1}{T} \mathbf{j}_T \mathbf{j}_T' + \frac{1}{T^2} \mathbf{j}_T \underbrace{(\mathbf{j}_T' \mathbf{j}_T)}_{=T} \mathbf{j}_T' \\ &= \mathbf{I}_T - \frac{2}{T} \mathbf{j}_T \mathbf{j}_T' + \frac{T}{T^2} \mathbf{j}_T \mathbf{j}_T' = \mathbf{I}_T - \frac{1}{T} \mathbf{j}_T \mathbf{j}_T' = \mathbf{Q}_T. \end{aligned}$$

Orthogonality:

$$\mathbf{Q}_T \mathbf{j}_T = \mathbf{I}_T \mathbf{j}_T - \frac{1}{T} \mathbf{j}_T \underbrace{(\mathbf{j}_T' \mathbf{j}_T)}_{=T} = \mathbf{j}_T - \frac{T}{T} \mathbf{j}_T = \mathbf{j}_T - \mathbf{j}_T = \mathbf{0}.$$

□

9.3.2.3 Spectral Decomposition and Rank

The asymptotic analysis of the FE estimator requires understanding the eigenvalue structure of $\mathbf{Q}_T$. Since $\mathbf{Q}_T$ is real and symmetric, the Spectral Theorem guarantees its diagonalization by a basis of orthonormal eigenvectors.

Lemma 9.3.2 (Rank and trace of orthogonal projection matrices). *Let $\mathbf{A} \in \mathbb{R}^{M \times M}$ be a real, symmetric, and idempotent matrix. Then $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$.*

Proof. By the Spectral Theorem, $\mathbf{A} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}'$ with $\mathbf{U}' \mathbf{U} = \mathbf{I}_M$ and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_M)$. Idempotence implies:

$$\mathbf{A}^2 = \mathbf{A} \implies \mathbf{U} \mathbf{\Lambda}^2 \mathbf{U}' = \mathbf{U} \mathbf{\Lambda} \mathbf{U}' \implies \mathbf{\Lambda}^2 = \mathbf{\Lambda}.$$

Thus, $\lambda_i^2 = \lambda_i \implies \lambda_i \in \{0, 1\}$ for all i . Since the rank of $\mathbf{A}$ is the number of non-zero eigenvalues and the trace is the sum of eigenvalues:

$$\text{rank}(\mathbf{A}) = \sum_{i=1}^M \mathbf{1}(\lambda_i = 1) = \sum_{i=1}^M \lambda_i = \text{tr}(\mathbf{A}).$$

□

Theorem 9.3.1 (Spectral decomposition and rank of $\mathbf{Q}_T$). *The spectrum of $\mathbf{Q}_T$ is $\sigma(\mathbf{Q}_T) = \{\underbrace{1, \dots, 1}_{T-1}, 0\}$. Consequently, $\text{rank}(\mathbf{Q}_T) = T - 1$.*

Proof. Let $\mathbf{P}_T = \frac{1}{T}\mathbf{j}_T\mathbf{j}_T'$. If $\mathbf{v}$ is an eigenvector of $\mathbf{P}_T$ with eigenvalue μ , then $\mathbf{Q}_T\mathbf{v} = (\mathbf{I}_T - \mathbf{P}_T)\mathbf{v} = (1 - \mu)\mathbf{v}$. We identify the eigenvectors of $\mathbf{P}_T$:

- **Case A:** $\mathbf{v} = \mathbf{j}_T \implies \mathbf{P}_T\mathbf{j}_T = \frac{1}{T}\mathbf{j}_T(\mathbf{j}_T'\mathbf{j}_T) = \frac{T}{T}\mathbf{j}_T = 1 \cdot \mathbf{j}_T$. Thus $\mu_1 = 1$.
- **Case B:** Let $\{\mathbf{h}_1, \dots, \mathbf{h}_{T-1}\}$ be an orthonormal basis of the orthogonal complement of $\text{span}(\mathbf{j}_T)$, such that $\mathbf{j}_T'\mathbf{h}_k = 0$. Then $\mathbf{P}_T\mathbf{h}_k = \frac{1}{T}\mathbf{j}_T(\mathbf{j}_T'\mathbf{h}_k) = \mathbf{0} = 0 \cdot \mathbf{h}_k$. Thus $\mu_k = 0$ for $k = 1, \dots, T - 1$.

The eigenvalues of $\mathbf{Q}_T$ are $\lambda_1 = 1 - 1 = 0$ and $\lambda_k = 1 - 0 = 1$ for $k = 1, \dots, T - 1$. The spectral decomposition is:

$$\mathbf{Q}_T = \mathbf{H}\mathbf{A}\mathbf{H}', \quad \mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_{T-1}, \frac{1}{\sqrt{T}}\mathbf{j}_T], \quad \mathbf{A} = \text{diag}(1, \dots, 1, 0).$$

By the previous Lemma:

$$\text{rank}(\mathbf{Q}_T) = \text{tr}(\mathbf{Q}_T) = \text{tr}(\mathbf{I}_T) - \frac{1}{T} \text{tr}(\mathbf{j}_T\mathbf{j}_T') = T - \frac{1}{T} \text{tr}(\mathbf{j}_T'\mathbf{j}_T) = T - \frac{T}{T} = T - 1.$$

□

Corollary 9.3.1 (Dimension of the null space). *By the Rank-Nullity Theorem, $\dim(\ker(\mathbf{Q}_T)) = T - \text{rank}(\mathbf{Q}_T) = 1$, generated by $\mathbf{j}_T$:*

$$\ker(\mathbf{Q}_T) = \{\mathbf{z} \in \mathbb{R}^T : \mathbf{z} = c\mathbf{j}_T, c \in \mathbb{R}\}.$$

The loss of one degree of freedom reflects the perfect linear sum restriction $\sum_{t=1}^T \ddot{u}_{it} = \sum_{t=1}^T (u_{it} - \bar{u}_i) = 0$, reducing the temporally independent information from T to $T - 1$ per individual.

9.3.3 Assumptions and Their Roles

Definition 9.3.1 (FE.1: Strict Exogeneity of the Idiosyncratic Error).

$$\mathbb{E}[u_{it} \mid \mathbf{X}_i, c_i] = 0, \quad t = 1, \dots, T. \quad (9.78)$$

Equivalently, for all t and all s :

$$\mathbb{E}[\mathbf{x}_{is}u_{it}] = \mathbf{0}. \quad (9.79)$$

Role: guarantee consistency of FE after the within transformation.

Why FE requires strict exogeneity (unlike POLS): the within transformation computes $\ddot{\mathbf{x}}_{it} = \mathbf{x}_{it} - \bar{\mathbf{x}}_i$, where $\bar{\mathbf{x}}_i$ depends on **all** periods. Therefore, even if u_{it} is orthogonal to $\mathbf{x}_{it}$ contemporaneously, if u_{it} is correlated with $\mathbf{x}_{i,t+1}$ (dynamic feedback), then u_{it} will be correlated with $\bar{\mathbf{x}}_i$, and therefore with $\ddot{\mathbf{x}}_{it}$.

Translation to practice: regressors cannot react to past, present, or future idiosyncratic shocks.

Counterexample: if a productivity shock today increases investment tomorrow, then $\mathbb{E}[u_{it}\mathbf{x}_{i,t+1}] \neq 0$, and strict exogeneity fails. In this case, FE is inconsistent, while POLS remains consistent.

Definition 9.3.2 (FE.2: Full Rank of Within Variation).

$$\text{rank}(\mathbb{E}[\mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i]) = K. \quad (9.80)$$

Role: guarantee identification of coefficients based solely on temporal variation within units.

Translation to practice: regressors must vary over time for a sufficient portion of the sample.

Counterexample: time-invariant variables, such as gender or race, become null columns after the within transformation, violating the rank condition.

Definition 9.3.3 (FE.3: Spherical Idiosyncratic Errors for Efficiency). For FE to be BLUE conditional on the within transformation, we assume:

$$\mathbb{E}[\mathbf{u}_i \mathbf{u}_i' \mid \mathbf{X}_i, c_i] = \sigma_u^2 \mathbf{I}_T. \quad (9.81)$$

Role: justify conventional standard errors and efficiency.

Translation to practice: idiosyncratic shocks should not exhibit autocorrelation or heteroskedasticity.

Important: violation of FE.3 does **not** generate bias or inconsistency. It only affects efficiency and the validity of conventional standard errors.

Definition 9.3.4 (FE.4: Finite Moments). To guarantee the existence of expectations and the applicability of the Law of Large Numbers, we assume:

$$\mathbb{E}[\|\mathbf{X}_i\|^2] < \infty \quad \text{and} \quad \mathbb{E}[\|\mathbf{u}_i\|^2] < \infty, \quad (9.82)$$

where $\|\cdot\|$ denotes the Euclidean norm. For asymptotic normality, finite fourth-order moments are required: $\mathbb{E}[\|\mathbf{X}_i\|^4] < \infty$ and $\mathbb{E}[\|\mathbf{u}_i\|^4] < \infty$.

9.3.4 Derivation of the FE Estimator

The model by individual is

$$\mathbf{y}_i = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{j}_T c_i + \mathbf{u}_i. \quad (9.83)$$

Pre-multiplying by $\mathbf{Q}_T$,

$$\mathbf{Q}_T \mathbf{y}_i = \mathbf{Q}_T \mathbf{X}_i \boldsymbol{\beta} + \mathbf{Q}_T \mathbf{j}_T c_i + \mathbf{Q}_T \mathbf{u}_i. \quad (9.84)$$

Since $\mathbf{Q}_T \mathbf{j}_T = \mathbf{0}$,

$$\mathbf{Q}_T \mathbf{y}_i = \mathbf{Q}_T \mathbf{X}_i \boldsymbol{\beta} + \mathbf{Q}_T \mathbf{u}_i. \quad (9.85)$$

Defining

$$\ddot{\mathbf{y}}_i = \mathbf{Q}_T \mathbf{y}_i, \quad \ddot{\mathbf{X}}_i = \mathbf{Q}_T \mathbf{X}_i, \quad \ddot{\mathbf{u}}_i = \mathbf{Q}_T \mathbf{u}_i, \quad (9.86)$$

we have

$$\ddot{\mathbf{y}}_i = \ddot{\mathbf{X}}_i \boldsymbol{\beta} + \ddot{\mathbf{u}}_i. \quad (9.87)$$

The FE estimator minimizes

$$S(\beta) = \sum_{i=1}^N (\ddot{\mathbf{y}}_i - \ddot{\mathbf{X}}_i \beta)' (\ddot{\mathbf{y}}_i - \ddot{\mathbf{X}}_i \beta). \quad (9.88)$$

The first-order condition is

$$\sum_{i=1}^N \ddot{\mathbf{X}}_i' (\ddot{\mathbf{y}}_i - \ddot{\mathbf{X}}_i \beta) = \mathbf{0}. \quad (9.89)$$

Thus,

$$\hat{\beta}_{FE} = \left(\sum_{i=1}^N \ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i \right)^{-1} \left(\sum_{i=1}^N \ddot{\mathbf{X}}_i' \ddot{\mathbf{y}}_i \right). \quad (9.90)$$

Proposition 9.3.2 (Sandwich form of the FE estimator). *The FE estimator can be explicitly rewritten in terms of the operator $\mathbf{Q}_T$:*

$$\hat{\beta}_{FE} = \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{y}_i \right). \quad (9.91)$$

Proof. By symmetry ($\mathbf{Q}_T' = \mathbf{Q}_T$) and idempotence ($\mathbf{Q}_T^2 = \mathbf{Q}_T$):

$$\begin{aligned} \ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i &= (\mathbf{Q}_T \mathbf{X}_i)' (\mathbf{Q}_T \mathbf{X}_i) = \mathbf{X}_i' \mathbf{Q}_T' \mathbf{Q}_T \mathbf{X}_i = \mathbf{X}_i' \mathbf{Q}_T^2 \mathbf{X}_i = \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i, \\ \ddot{\mathbf{X}}_i' \ddot{\mathbf{y}}_i &= (\mathbf{Q}_T \mathbf{X}_i)' (\mathbf{Q}_T \mathbf{y}_i) = \mathbf{X}_i' \mathbf{Q}_T' \mathbf{Q}_T \mathbf{y}_i = \mathbf{X}_i' \mathbf{Q}_T \mathbf{y}_i. \end{aligned}$$

Substituting into the estimator expression yields the sandwich form. □

By the Frisch-Waugh-Lovell theorem, this estimator is algebraically identical to OLS with individual dummy variables for each unit i .

9.3.5 First-Order Moment Properties

9.3.5.1 Estimation Error Equation

Substituting the transformed model,

$$\hat{\beta}_{FE} = \beta + \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{u}_i \right). \quad (9.92)$$

9.3.5.2 Unbiasedness Under Strict Exogeneity

Theorem 9.3.2 (Finite-sample unbiasedness of the FE estimator). *Under hypotheses FE.1 (Strict Exogeneity), FE.2 (Within Rank Condition), FE.4 (Finite Moments), and independent sampling in the cross-sectional dimension i , the Within-Group estimator is strictly unbiased:*

$$\mathbb{E}[\hat{\beta}_{FE}] = \beta. \quad (9.93)$$

Proof. Step 1: Derivation of the finite-sample error equation.

Substituting the true model $\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + c_i\mathbf{j}_T + \mathbf{u}_i$ into the sandwich form:

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{\text{FE}} &= \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T (\mathbf{X}_i\boldsymbol{\beta} + c_i\mathbf{j}_T + \mathbf{u}_i) \\ &= \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \left[\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \boldsymbol{\beta} + \sum_{i=1}^N c_i \mathbf{X}_i' \mathbf{Q}_T \mathbf{j}_T + \sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{u}_i \right].\end{aligned}\quad (9.94)$$

Step 2: Annihilation of the individual effect.

Since $\boldsymbol{\beta}$ is common to all individuals, we extract it from the summation:

$$\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \boldsymbol{\beta} = \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \right) \boldsymbol{\beta}. \quad (9.95)$$

By the orthogonality of $\mathbf{Q}_T$ to the vector of ones, $\mathbf{Q}_T \mathbf{j}_T = \mathbf{0}$, so the second term collapses:

$$\sum_{i=1}^N c_i \mathbf{X}_i' \underbrace{\mathbf{Q}_T \mathbf{j}_T}_{=\mathbf{0}} = \mathbf{0}. \quad (9.96)$$

Step 3: Isolation of the sample error.

Multiplying by the inverse (which exists by FE.2):

$$\hat{\boldsymbol{\beta}}_{\text{FE}} = \boldsymbol{\beta} + \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{u}_i. \quad (9.97)$$

This is the **finite-sample error equation**, which decomposes the estimator into the true parameter plus a stochastic deviation term.

Step 4: Conditional expectation.

We apply $\mathbb{E}[\cdot \mid \mathbf{X}, \mathbf{c}]$ to both sides, where $\mathbf{X} = [\mathbf{X}_1', \dots, \mathbf{X}_N']'$ and $\mathbf{c} = [c_1, \dots, c_N]'$:

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{FE}} \mid \mathbf{X}, \mathbf{c}] = \boldsymbol{\beta} + \mathbb{E} \left[\left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{u}_i \mid \mathbf{X}, \mathbf{c} \right]. \quad (9.98)$$

Step 5: Extraction of deterministic terms under conditioning.

Under the conditioning set $(\mathbf{X}, \mathbf{c})$, the matrix $\mathbf{X}$ is treated as fixed. Consequently, $\mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i$, $\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i$, its inverse, and $\mathbf{X}_i' \mathbf{Q}_T$ are all deterministic. By the *taking out what is known* property of conditional expectation:

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{FE}} \mid \mathbf{X}, \mathbf{c}] = \boldsymbol{\beta} + \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbb{E}[\mathbf{u}_i \mid \mathbf{X}, \mathbf{c}]. \quad (9.99)$$

Step 6: Reduction by cross-sectional independence.

Since units i are sampled independently:

$$\mathbb{E}[\mathbf{u}_i \mid \mathbf{X}, \mathbf{c}] = \mathbb{E}[\mathbf{u}_i \mid \mathbf{X}_i, c_i]. \quad (9.100)$$

Under Hypothesis FE.1, $\mathbb{E}[\mathbf{u}_i \mid \mathbf{X}_i, c_i] = \mathbf{0}$, hence:

$$\mathbb{E}[\hat{\beta}_{FE} \mid \mathbf{X}, \mathbf{c}] = \beta + \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \cdot \mathbf{0} = \beta. \quad (9.101)$$

Step 7: Unconditional expectation via LIE.

By the Law of Iterated Expectations:

$$\mathbb{E}[\hat{\beta}_{FE}] = \mathbb{E}[\mathbb{E}[\hat{\beta}_{FE} \mid \mathbf{X}, \mathbf{c}]] = \mathbb{E}[\beta] = \beta. \quad (9.102)$$

□

9.3.5.3 Consistency Under Arbitrary Correlation with the Fixed Effect

Theorem 9.3.3 (Consistency of the FE estimator). *Under hypotheses FE.1, FE.2, FE.4, and i.i.d. sampling in dimension i , the FE estimator is consistent even when $\mathbb{E}[c_i \mid \mathbf{X}_i] = g(\mathbf{X}_i) \neq 0$ for an arbitrary function g :*

$$\text{plim}_{N \rightarrow \infty} \hat{\beta}_{FE} = \beta. \quad (9.103)$$

Proof. From the finite-sample error equation scaled by $1/N$:

$$\hat{\beta}_{FE} = \beta + \left(\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{u}_i \right). \quad (9.104)$$

Term 1 (Moment matrix): By Kolmogorov's WLLN (i.i.d. sampling in i , finite second moments by FE.4):

$$\text{plim}_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i = \mathbb{E}[\mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i]. \quad (9.105)$$

By the Within Rank Condition (FE.2), the limit matrix is of full rank K , hence invertible. By Slutsky's Lemma:

$$\text{plim}_{N \rightarrow \infty} \left(\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \right)^{-1} = (\mathbb{E}[\mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i])^{-1} < \infty. \quad (9.106)$$

Term 2 (Cross-moments with error): Similarly, by WLLN:

$$\text{plim}_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{u}_i = \mathbb{E}[\mathbf{X}_i' \mathbf{Q}_T \mathbf{u}_i]. \quad (9.107)$$

Via the Law of Iterated Expectations and Hypothesis FE.1:

$$\mathbb{E}[\mathbf{X}_i' \mathbf{Q}_T \mathbf{u}_i] = \mathbb{E}[\mathbf{X}_i' \mathbf{Q}_T \mathbb{E}[\mathbf{u}_i \mid \mathbf{X}_i, c_i]] = \mathbb{E}[\mathbf{X}_i' \mathbf{Q}_T \cdot \mathbf{0}] = \mathbf{0}. \quad (9.108)$$

Reassembling the limits: By Slutsky's Theorem:

$$\text{plim}_{N \rightarrow \infty} \hat{\beta}_{FE} = \beta + (\mathbb{E}[\mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i])^{-1} \cdot \mathbf{0} = \beta. \quad (9.109)$$

□

Remark 9.3.2 (Economic intuition: why FE works under arbitrary correlation?). 1.

Failure of Pooled OLS and Random Effects: Under POLS and RE, the individual effect c_i remains in the compound error. If $\mathbb{E}[c_i | \mathbf{X}_i] = g(\mathbf{X}_i) \neq 0$, regressors correlate with the error, generating omitted variable bias and inconsistency.

2. **Mechanical annihilation by Within:** The transformation $\mathbf{Q}_T$ projects observations onto the space orthogonal to $\mathbf{j}_T$. The component $c_i \mathbf{j}_T$ is mechanically annihilated ($\mathbf{Q}_T c_i \mathbf{j}_T = c_i \cdot \mathbf{0} = \mathbf{0}$), **regardless of the functional form of correlation between c_i and $\mathbf{X}_i$** . Consistency then depends only on the orthogonality of the transitory shock $\mathbf{u}_i$.

Remark 9.3.3 (Relationship between unbiasedness and consistency). 1. **Unbiasedness (finite sample):** Under FE.1, FE.2, and FE.4, we have $\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{FE}}] = \boldsymbol{\beta}$ for any finite N . This is an *exact* result, not asymptotic.

2. **Consistency (asymptotic):** Under the same hypotheses, $\text{plim}_{N \rightarrow \infty} \hat{\boldsymbol{\beta}}_{\text{FE}} = \boldsymbol{\beta}$. This is a *limit* result, describing behavior as $N \rightarrow \infty$.
3. **Logical hierarchy:** Unbiasedness $\nRightarrow$ consistency in general. However, in the case of FE, the combination of unbiasedness with the convergence of variance to zero (guaranteed by WLLN applied to the error equation) implies consistency.

9.3.6 Second-Order Anatomy and Mechanical Non-Sphericity

9.3.6.1 Classical Sphericity and Hypothesis FE.3

To perform valid statistical inference and discuss the Gauss-Markov efficiency of the FE estimator, we analyze the second-order moment structure of the idiosyncratic errors.

Remark 9.3.4 (Scalar decomposition of sphericity). Hypothesis FE.3 simultaneously imposes two restrictions:

1. **Temporal homoskedasticity:** $\text{Var}(u_{it} | \mathbf{X}_i, c_i) = \sigma_u^2 \quad \forall t \in \{1, \dots, T\}$
2. **Absence of serial autocorrelation:** $\text{Cov}(u_{it}, u_{is} | \mathbf{X}_i, c_i) = 0 \quad \forall t \neq s$

Economically, this ensures that transitory shocks are temporally independent once c_i is controlled for.

9.3.6.2 Theorem of Induced Negative Serial Autocorrelation

Theorem 9.3.4 (Induced negative serial autocorrelation by the within transformation). *Under Hypothesis FE.3 ($\mathbb{E}[\mathbf{u}_i \mathbf{u}_i' | \mathbf{X}_i, c_i] = \sigma_u^2 \mathbf{I}_T$), the Within transformation $\ddot{\mathbf{u}}_i = \mathbf{Q}_T \mathbf{u}_i$ mechanically induces:*

$$\text{Var}(\ddot{u}_{it} | \mathbf{X}_i, c_i) = \sigma_u^2 \left(\frac{T-1}{T} \right) \quad \forall t, \quad (9.110)$$

$$\text{Cov}(\ddot{u}_{it}, \ddot{u}_{is} | \mathbf{X}_i, c_i) = -\frac{\sigma_u^2}{T} \quad \forall t \neq s, \quad (9.111)$$

$$\text{Corr}(\ddot{u}_{it}, \ddot{u}_{is} | \mathbf{X}_i, c_i) = -\frac{1}{T-1} \quad \forall t \neq s. \quad (9.112)$$

Proof. Step 1: Compact variance-covariance matrix.

By strict exogeneity, $\mathbb{E}[\ddot{\mathbf{u}}_i \mid \mathbf{X}_i, c_i] = \mathbf{0}$, so:

$$\begin{aligned}\text{Var}(\ddot{\mathbf{u}}_i \mid \mathbf{X}_i, c_i) &= \mathbb{E}[\ddot{\mathbf{u}}_i \ddot{\mathbf{u}}_i' \mid \mathbf{X}_i, c_i] \\ &= \mathbb{E}[\mathbf{Q}_T \mathbf{u}_i \mathbf{u}_i' \mathbf{Q}_T' \mid \mathbf{X}_i, c_i] \\ &= \mathbf{Q}_T \mathbb{E}[\mathbf{u}_i \mathbf{u}_i' \mid \mathbf{X}_i, c_i] \mathbf{Q}_T.\end{aligned}$$

Substituting FE.3 ($\mathbb{E}[\mathbf{u}_i \mathbf{u}_i' \mid \mathbf{X}_i, c_i] = \sigma_u^2 \mathbf{I}_T$):

$$\text{Var}(\ddot{\mathbf{u}}_i \mid \mathbf{X}_i, c_i) = \mathbf{Q}_T (\sigma_u^2 \mathbf{I}_T) \mathbf{Q}_T = \sigma_u^2 \mathbf{Q}_T \mathbf{Q}_T.$$

By idempotence ($\mathbf{Q}_T^2 = \mathbf{Q}_T$):

$$\text{Var}(\ddot{\mathbf{u}}_i \mid \mathbf{X}_i, c_i) = \sigma_u^2 \mathbf{Q}_T. \quad (9.113)$$

Step 2: Element-by-element decomposition.

The generic element of $\mathbf{Q}_T = \mathbf{I}_T - \frac{1}{T} \mathbf{j}_T \mathbf{j}_T'$ is:

$$[\mathbf{Q}_T]_{ts} = \delta_{ts} - \frac{1}{T}, \quad (9.114)$$

where δ_{ts} is the Kronecker delta ($\delta_{ts} = 1$ if $t = s$, 0 otherwise).

Therefore, the (t, s) element of the covariance matrix is:

$$[\text{Var}(\ddot{\mathbf{u}}_i \mid \mathbf{X}_i, c_i)]_{ts} = \sigma_u^2 \left(\delta_{ts} - \frac{1}{T} \right). \quad (9.115)$$

Case A: Variance ($t = s \implies \delta_{tt} = 1$):

$$\text{Var}(\ddot{u}_{it} \mid \mathbf{X}_i, c_i) = \sigma_u^2 \left(1 - \frac{1}{T} \right) = \sigma_u^2 \left(\frac{T-1}{T} \right). \quad (9.116)$$

Case B: Covariance ($t \neq s \implies \delta_{ts} = 0$):

$$\text{Cov}(\ddot{u}_{it}, \ddot{u}_{is} \mid \mathbf{X}_i, c_i) = \sigma_u^2 \left(0 - \frac{1}{T} \right) = -\frac{\sigma_u^2}{T}. \quad (9.117)$$

Step 3: Correlation coefficient.

By definition:

$$\text{Corr}(\ddot{u}_{it}, \ddot{u}_{is} \mid \mathbf{X}_i, c_i) = \frac{\text{Cov}(\ddot{u}_{it}, \ddot{u}_{is} \mid \mathbf{X}_i, c_i)}{\sqrt{\text{Var}(\ddot{u}_{it} \mid \mathbf{X}_i, c_i)} \sqrt{\text{Var}(\ddot{u}_{is} \mid \mathbf{X}_i, c_i)}} = \frac{-\sigma_u^2/T}{\sigma_u^2(T-1)/T} = -\frac{1}{T-1}.$$

□

Remark 9.3.5 (Geometric intuition of negative autocorrelation). The negative serial autocorrelation is **not an economic phenomenon**, but rather a mathematical byproduct of the Within transformation. By subtracting the time mean, we impose the restriction:

$$\sum_{t=1}^T \ddot{u}_{it} = \sum_{t=1}^T (u_{it} - \bar{u}_i) = 0 \quad \forall i. \quad (9.118)$$

If a shock at $t = 1$ is extremely positive, the shocks in the remaining $T - 1$ periods are forced to be, on average, negative to satisfy the zero-sum restriction. This mathematical compensation generates a correlation of $-1/(T-1)$. Note that as $T \rightarrow \infty$, $-1/(T-1) \rightarrow 0$, evidencing that this artificial non-sphericity disappears in long panels, but is severe in short panels.

9.3.6.3 Degrees of Freedom and Variance Estimation

Definition 9.3.5 (Unbiased estimator of within residual variance). The correct estimator of σ_u^2 under Fixed Effects is:

$$\hat{\sigma}_u^2 = \frac{\sum_{i=1}^N \hat{\mathbf{u}}_i' \hat{\mathbf{u}}_i}{N(T-1) - K}, \quad (9.119)$$

where $\hat{\mathbf{u}}_i = \mathbf{y}_i - \ddot{\mathbf{X}}_i \hat{\boldsymbol{\beta}}_{\text{FE}}$ are the transformed Within residuals.

Remark 9.3.6 (Why the divisor $NT - K$ is biased). Within demeaning is equivalent to implicitly estimating N individual intercepts (c_i), imposing N linear restrictions:

$$\sum_{t=1}^T \hat{u}_{it} = 0 \quad \forall i \in \{1, \dots, N\}. \quad (9.120)$$

These N restrictions remove N degrees of freedom, reducing the available degrees from $NT - K$ to $N(T-1) - K$.

Theorem 9.3.5 (Finite-sample unbiasedness of $\hat{\sigma}_u^2$). *Under hypotheses FE.1, FE.2, and FE.3, the estimator $\hat{\sigma}_u^2$ is strictly unbiased:*

$$\mathbb{E}[\hat{\sigma}_u^2] = \sigma_u^2. \quad (9.121)$$

Proof. Step 1: Global structure via Kronecker product.

We stack the N units and define the global Within operator:

$$\mathbf{Q} = \mathbf{I}_N \otimes \mathbf{Q}_T \in \mathbb{R}^{NT \times NT}, \quad (9.122)$$

where $\otimes$ is the Kronecker product. The matrix $\mathbf{Q}$ is symmetric and idempotent, with:

$$\text{rank}(\mathbf{Q}) = \text{tr}(\mathbf{Q}) = \text{tr}(\mathbf{I}_N \otimes \mathbf{Q}_T) = N \cdot \text{tr}(\mathbf{Q}_T) = N(T-1). \quad (9.123)$$

Step 2: Global residual vector.

The global FE estimator is $\hat{\boldsymbol{\beta}}_{\text{FE}} = (\ddot{\mathbf{X}}' \ddot{\mathbf{X}})^{-1} \ddot{\mathbf{X}}' \ddot{\mathbf{y}}$. The residual vector is:

$$\begin{aligned} \hat{\mathbf{u}} &= \ddot{\mathbf{y}} - \ddot{\mathbf{X}} \hat{\boldsymbol{\beta}}_{\text{FE}} \\ &= \ddot{\mathbf{y}} - \ddot{\mathbf{X}} (\ddot{\mathbf{X}}' \ddot{\mathbf{X}})^{-1} \ddot{\mathbf{X}}' \ddot{\mathbf{y}} \\ &= (\mathbf{I}_{NT} - \ddot{\mathbf{X}} (\ddot{\mathbf{X}}' \ddot{\mathbf{X}})^{-1} \ddot{\mathbf{X}}') \ddot{\mathbf{y}} \\ &= \mathbf{M}_{\ddot{\mathbf{X}}} \ddot{\mathbf{y}}, \end{aligned}$$

where $\mathbf{M}_{\ddot{\mathbf{X}}} = \mathbf{I}_{NT} - \ddot{\mathbf{X}} (\ddot{\mathbf{X}}' \ddot{\mathbf{X}})^{-1} \ddot{\mathbf{X}}'$ is the orthogonal projection matrix onto the column space orthogonal to $\ddot{\mathbf{X}}$.

Substituting $\ddot{\mathbf{y}} = \ddot{\mathbf{X}} \boldsymbol{\beta} + \ddot{\mathbf{u}}$ and using $\mathbf{M}_{\ddot{\mathbf{X}}} \ddot{\mathbf{X}} = \mathbf{0}$:

$$\hat{\mathbf{u}} = \mathbf{M}_{\ddot{\mathbf{X}}} \ddot{\mathbf{u}} = \mathbf{M}_{\ddot{\mathbf{X}}} \mathbf{Q} \mathbf{u}. \quad (9.124)$$

Step 3: Sum of squared residuals (SSR).

$$\begin{aligned} SSR &= \hat{\mathbf{u}}' \hat{\mathbf{u}} = (\mathbf{M}_{\ddot{\mathbf{X}}} \mathbf{Q} \mathbf{u})' (\mathbf{M}_{\ddot{\mathbf{X}}} \mathbf{Q} \mathbf{u}) \\ &= \mathbf{u}' \mathbf{Q}' \mathbf{M}_{\ddot{\mathbf{X}}} \mathbf{M}_{\ddot{\mathbf{X}}} \mathbf{Q} \mathbf{u}. \end{aligned}$$

By symmetry ($\mathbf{M}_{\ddot{\mathbf{X}}} = \mathbf{M}_{\ddot{\mathbf{X}}}'$) and idempotence ($\mathbf{M}_{\ddot{\mathbf{X}}}^2 = \mathbf{M}_{\ddot{\mathbf{X}}}$):

$$SSR = \mathbf{u}' \mathbf{Q} \mathbf{M}_{\ddot{\mathbf{X}}} \mathbf{Q} \mathbf{u}.$$

We define $\mathbf{A} = \mathbf{Q} \mathbf{M}_{\ddot{\mathbf{X}}} \mathbf{Q}$. Expanding:

$$\begin{aligned} \mathbf{A} &= \mathbf{Q} (\mathbf{I}_{NT} - \ddot{\mathbf{X}} (\ddot{\mathbf{X}}' \ddot{\mathbf{X}})^{-1} \ddot{\mathbf{X}}') \mathbf{Q} \\ &= \mathbf{Q}^2 - \mathbf{Q} \ddot{\mathbf{X}} (\ddot{\mathbf{X}}' \ddot{\mathbf{X}})^{-1} \ddot{\mathbf{X}}' \mathbf{Q}. \end{aligned}$$

Since $\mathbf{Q}$ is idempotent ($\mathbf{Q}^2 = \mathbf{Q}$) and $\mathbf{Q} \ddot{\mathbf{X}} = \mathbf{Q} \mathbf{Q} \mathbf{X} = \mathbf{Q} \mathbf{X} = \ddot{\mathbf{X}}$:

$$\mathbf{A} = \mathbf{Q} - \ddot{\mathbf{X}} (\ddot{\mathbf{X}}' \ddot{\mathbf{X}})^{-1} \ddot{\mathbf{X}}'. \quad (9.125)$$

Step 4: Conditional expectation of SSR.

We apply $\mathbb{E}[\cdot \mid \mathbf{X}, \mathbf{c}]$:

$$\mathbb{E}[SSR \mid \mathbf{X}, \mathbf{c}] = \mathbb{E}[\mathbf{u}' \mathbf{A} \mathbf{u} \mid \mathbf{X}, \mathbf{c}].$$

Since $\mathbf{u}' \mathbf{A} \mathbf{u}$ is scalar, it equals its trace. By the cyclic property $\text{tr}(\mathbf{BC}) = \text{tr}(\mathbf{CB})$:

$$\begin{aligned} \mathbb{E}[SSR \mid \mathbf{X}, \mathbf{c}] &= \mathbb{E}[\text{tr}(\mathbf{u}' \mathbf{A} \mathbf{u}) \mid \mathbf{X}, \mathbf{c}] \\ &= \mathbb{E}[\text{tr}(\mathbf{A} \mathbf{u} \mathbf{u}') \mid \mathbf{X}, \mathbf{c}] \\ &= \text{tr}(\mathbf{A} \mathbb{E}[\mathbf{u} \mathbf{u}' \mid \mathbf{X}, \mathbf{c}]). \end{aligned}$$

Under FE.3, $\mathbb{E}[\mathbf{u} \mathbf{u}' \mid \mathbf{X}, \mathbf{c}] = \sigma_u^2 \mathbf{I}_{NT}$, so:

$$\mathbb{E}[SSR \mid \mathbf{X}, \mathbf{c}] = \text{tr}(\mathbf{A} \sigma_u^2 \mathbf{I}_{NT}) = \sigma_u^2 \text{tr}(\mathbf{A}).$$

Step 5: Calculation of the trace of $\mathbf{A}$.

$$\begin{aligned} \text{tr}(\mathbf{A}) &= \text{tr}(\mathbf{Q} - \ddot{\mathbf{X}} (\ddot{\mathbf{X}}' \ddot{\mathbf{X}})^{-1} \ddot{\mathbf{X}}') \\ &= \text{tr}(\mathbf{Q}) - \text{tr}(\ddot{\mathbf{X}} (\ddot{\mathbf{X}}' \ddot{\mathbf{X}})^{-1} \ddot{\mathbf{X}}'). \end{aligned}$$

We already know $\text{tr}(\mathbf{Q}) = N(T - 1)$. For the second term, by the cyclic property:

$$\text{tr}(\ddot{\mathbf{X}} (\ddot{\mathbf{X}}' \ddot{\mathbf{X}})^{-1} \ddot{\mathbf{X}}') = \text{tr}((\ddot{\mathbf{X}}' \ddot{\mathbf{X}})^{-1} \ddot{\mathbf{X}}' \ddot{\mathbf{X}}) = \text{tr}(\mathbf{I}_K) = K.$$

Therefore:

$$\text{tr}(\mathbf{A}) = N(T - 1) - K. \quad (9.126)$$

Substituting:

$$\mathbb{E}[SSR \mid \mathbf{X}, \mathbf{c}] = \sigma_u^2 [N(T - 1) - K]. \quad (9.127)$$

Step 6: Unconditional expectation via LIE.

$$\mathbb{E}[SSR] = \mathbb{E}[\mathbb{E}[SSR \mid \mathbf{X}, \mathbf{c}]] = \sigma_u^2[N(T-1) - K].$$

Step 7: Unbiasedness of $\hat{\sigma}_u^2$.

$$\mathbb{E}[\hat{\sigma}_u^2] = \mathbb{E}\left[\frac{SSR}{N(T-1) - K}\right] = \frac{\mathbb{E}[SSR]}{N(T-1) - K} = \frac{\sigma_u^2[N(T-1) - K]}{N(T-1) - K} = \sigma_u^2.$$

□

Remark 9.3.7 (Consequence of bias from omitting the correction). If we use the incorrect divisor $NT - K$, the estimator $\tilde{\sigma}_u^2 = \frac{SSR}{NT-K}$ has expectation:

$$\mathbb{E}[\tilde{\sigma}_u^2] = \sigma_u^2 \left(\frac{N(T-1) - K}{NT - K} \right) = \sigma_u^2 \left(1 - \frac{N}{NT - K} \right) < \sigma_u^2. \quad (9.128)$$

In short panels, the bias is severe. For example, if $T = 2$ and $K = 1$:

$$\frac{N(1) - 1}{2N - 1} \approx \frac{1}{2}.$$

The variance would be underestimated by half, artificially doubling the t -statistics and generating over-significance.

9.3.7 Relaxation of Sphericity and Cluster-Robust Inference

9.3.7.1 Breakdown of Sphericity and Sandwich Asymptotic Variance

Under the realistic microeconomic scenario, we relax hypothesis FE.3, allowing idiosyncratic shocks to exhibit temporal heteroskedasticity and genuine serial autocorrelation.

Definition 9.3.6 (Genuine non-sphericity). Under relaxation of the Gauss-Markov hypothesis, the conditional covariance matrix of the original errors assumes an arbitrary structure:

$$\boldsymbol{\Sigma}_i \equiv \text{Var}(\mathbf{u}_i \mid \mathbf{X}_i, c_i) = \mathbb{E}[\mathbf{u}_i \mathbf{u}_i' \mid \mathbf{X}_i, c_i] \in \mathbb{R}^{T \times T}, \quad (9.129)$$

where $[\boldsymbol{\Sigma}_i]_{ts} = \sigma_{its}$ can be non-zero for all t, s , simultaneously modeling:

1. **Temporal heteroskedasticity:** $\sigma_{itt} = \sigma_{it}^2 \neq \sigma_{is}^2$ for $t \neq s$
2. **Genuine serial autocorrelation:** $\sigma_{its} \neq 0$ for $t \neq s$, without imposing a parametric structure (AR(1), MA(1), etc.)

Remark 9.3.8 (Crucial distinction). **Genuine** serial autocorrelation ($\sigma_{its} \neq 0$ in the original errors $\mathbf{u}_i$) differs fundamentally from the **mechanical** autocorrelation of $-1/(T-1)$ induced by the Within transformation. The former is a population stochastic property; the latter is a mathematical artifact of demeaning.

Theorem 9.3.6 (Asymptotic distribution under general errors). *Under hypotheses FE.1 (Strict Exogeneity), FE.2 (Within Rank Condition), i.i.d. sampling in dimension i with $N \rightarrow \infty$ and fixed T , and finite fourth-order moments, the limiting distribution of the FE estimator is:*

$$\sqrt{N}(\hat{\boldsymbol{\beta}}_{FE} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{V}_{\text{sandwich}}), \quad (9.130)$$

where the asymptotic variance assumes a sandwich structure:

$$\mathbf{V}_{sandwich} = \left(\mathbb{E}[\ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i] \right)^{-1} \mathbb{E}[\ddot{\mathbf{X}}_i' \mathbf{u}_i \mathbf{u}_i' \ddot{\mathbf{X}}_i] \left(\mathbb{E}[\ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i] \right)^{-1}. \quad (9.131)$$

Proof. Step 1: Scaled deviation equation.

From the finite-sample error equation, multiplying by $\sqrt{N}$:

$$\sqrt{N}(\hat{\boldsymbol{\beta}}_{FE} - \boldsymbol{\beta}) = \left(\frac{1}{N} \sum_{i=1}^N \ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i \right)^{-1} \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \ddot{\mathbf{X}}_i' \mathbf{u}_i \right). \quad (9.132)$$

Step 2: Limit of the inverse term (WLLN + Slutsky).

Since $\{\ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i\}_{i=1}^N$ is i.i.d. with finite moments, by Kolmogorov's WLLN:

$$\frac{1}{N} \sum_{i=1}^N \ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i \xrightarrow{p} \mathbb{E}[\ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i] \equiv \mathbf{A}. \quad (9.133)$$

By the Within Rank Condition (FE.2), $\mathbf{A}$ is positive definite and invertible. By the Continuous Mapping Theorem (Slutsky):

$$\left(\frac{1}{N} \sum_{i=1}^N \ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i \right)^{-1} \xrightarrow{p} \mathbf{A}^{-1}. \quad (9.134)$$

Step 3: Limit of the numerator term (Multivariate CLT).

Define $\mathbf{w}_i \equiv \ddot{\mathbf{X}}_i' \mathbf{u}_i$. By strict exogeneity (FE.1), $\mathbb{E}[\mathbf{w}_i] = \mathbf{0}$. Since $\{\mathbf{w}_i\}$ is i.i.d. with finite moments, by the Lindeberg-Lévy CLT:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{w}_i \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{B}_0), \quad (9.135)$$

where $\mathbf{B}_0 \equiv \text{Var}(\mathbf{w}_i) = \mathbb{E}[\mathbf{w}_i \mathbf{w}_i'] = \mathbb{E}[\ddot{\mathbf{X}}_i' \mathbf{u}_i \mathbf{u}_i' \ddot{\mathbf{X}}_i]$.

By the Law of Iterated Expectations and $\text{Var}(\mathbf{u}_i | \mathbf{X}_i, c_i) = \boldsymbol{\Sigma}_i$:

$$\mathbf{B}_0 = \mathbb{E} \left[\ddot{\mathbf{X}}_i' \mathbb{E}[\mathbf{u}_i \mathbf{u}_i' | \mathbf{X}_i, c_i] \ddot{\mathbf{X}}_i \right] = \mathbb{E}[\ddot{\mathbf{X}}_i' \boldsymbol{\Sigma}_i \ddot{\mathbf{X}}_i]. \quad (9.136)$$

Step 4: Combination via Slutsky.

Applying Slutsky to the product of limits:

$$\sqrt{N}(\hat{\boldsymbol{\beta}}_{FE} - \boldsymbol{\beta}) \xrightarrow{d} \mathbf{A}^{-1} \cdot \mathcal{N}(\mathbf{0}, \mathbf{B}_0) = \mathcal{N}(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B}_0 \mathbf{A}^{-1}). \quad (9.137)$$

□

Remark 9.3.9 (Consequences of using classical variance). If the researcher ignores non-sphericity and uses $\widehat{\text{Var}}_{\text{class}}(\hat{\boldsymbol{\beta}}_{FE} | \mathbf{X}) = \hat{\sigma}_u^2 \left(\sum_{i=1}^N \ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i \right)^{-1}$, then:

$$\text{plim}_{N \rightarrow \infty} \left[N \cdot \widehat{\text{Var}}_{\text{class}} \right] = \sigma_u^2 \left(\mathbb{E}[\ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i] \right)^{-1}. \quad (9.138)$$

This expression equals $\mathbf{V}_{\text{sandwich}}$ only if $\mathbb{E}[\ddot{\mathbf{X}}_i' \mathbf{u}_i \mathbf{u}_i' \ddot{\mathbf{X}}_i] = \sigma_u^2 \mathbb{E}[\ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i]$, a condition violated under heteroskedasticity or genuine autocorrelation.

In applications with positive serial autocorrelation (the predominant pattern in longitudinal data), it can be shown that $\mathbf{V}_{\text{sandwich}} > \sigma_u^2 \mathbf{A}^{-1}$ in the matrix sense, implying that the classical formula **severely underestimates** the true standard errors. Consequences:

1. **Inflation of t -statistics:** $|t_{\text{class}}| \gg |t_{\text{correct}}|$
2. **Distortion of significance level:** Bertrand, Duflo & Mullainathan (2004) show rejection rates of true H_0 reaching 45% under a nominal $\alpha = 5\%$
3. **Invalid confidence intervals:** excessively narrow, with actual coverage below nominal

9.3.7.2 Temporal Heteroskedasticity

Definition 9.3.7 (Temporal heteroskedasticity). Isolating the effect of the breakdown of homoskedasticity (without serial autocorrelation), the conditional covariance matrix is diagonal with time-varying elements:

$$\Sigma_i = \text{Var}(\mathbf{u}_i \mid \mathbf{X}_i, c_i) = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_T^2), \quad (9.139)$$

where σ_t^2 is the variance of the shock in period t , and $\text{Cov}(u_{it}, u_{is} \mid \mathbf{X}_i, c_i) = 0$ for $t \neq s$.

Theorem 9.3.7 (Propagation of temporal heteroskedasticity on within errors). *Under temporal heteroskedasticity, the variance and covariance of the transformed errors $\ddot{u}_{it} = u_{it} - \bar{u}_i$ are:*

$$\text{Var}(\ddot{u}_{it} \mid \mathbf{X}_i, c_i) = \sigma_t^2 \left(1 - \frac{2}{T}\right) + \frac{\bar{\sigma}^2}{T}, \quad (9.140)$$

$$\text{Cov}(\ddot{u}_{it}, \ddot{u}_{is} \mid \mathbf{X}_i, c_i) = -\frac{\sigma_t^2 + \sigma_s^2 - \bar{\sigma}^2}{T} \quad \forall t \neq s, \quad (9.141)$$

where $\bar{\sigma}^2 = \frac{1}{T} \sum_{r=1}^T \sigma_r^2$ is the time average of the original variances.

Proof. **Variance** ($t = s$):

By the variance property of differences:

$$\begin{aligned} \text{Var}(\ddot{u}_{it} \mid \mathbf{X}_i, c_i) &= \text{Var}(u_{it} - \bar{u}_i \mid \mathbf{X}_i, c_i) \\ &= \text{Var}(u_{it}) + \text{Var}(\bar{u}_i) - 2 \text{Cov}(u_{it}, \bar{u}_i). \end{aligned}$$

We calculate each component:

- $\text{Var}(u_{it}) = \sigma_t^2$ (by definition)
- Since u_{ir} are temporally independent:

$$\text{Var}(\bar{u}_i) = \text{Var}\left(\frac{1}{T} \sum_{r=1}^T u_{ir}\right) = \frac{1}{T^2} \sum_{r=1}^T \sigma_r^2 = \frac{T \bar{\sigma}^2}{T^2} = \frac{\bar{\sigma}^2}{T}.$$

- By temporal independence ($\text{Cov}(u_{it}, u_{ir}) = 0$ for $r \neq t$):

$$\text{Cov}(u_{it}, \bar{u}_i) = \text{Cov}\left(u_{it}, \frac{1}{T} \sum_{r=1}^T u_{ir}\right) = \frac{1}{T} \text{Var}(u_{it}) = \frac{\sigma_t^2}{T}.$$

Substituting:

$$\text{Var}(\ddot{u}_{it}) = \sigma_t^2 + \frac{\bar{\sigma}^2}{T} - 2\frac{\sigma_t^2}{T} = \sigma_t^2 \left(1 - \frac{2}{T}\right) + \frac{\bar{\sigma}^2}{T}.$$

Covariance ($t \neq s$):

By bilinearity of covariance:

$$\begin{aligned} \text{Cov}(\ddot{u}_{it}, \ddot{u}_{is}) &= \text{Cov}(u_{it} - \bar{u}_i, u_{is} - \bar{u}_i) \\ &= \underbrace{\text{Cov}(u_{it}, u_{is})}_{=0} - \text{Cov}(u_{it}, \bar{u}_i) - \text{Cov}(u_{is}, \bar{u}_i) + \text{Var}(\bar{u}_i) \\ &= 0 - \frac{\sigma_t^2}{T} - \frac{\sigma_s^2}{T} + \frac{\bar{\sigma}^2}{T} \\ &= -\frac{\sigma_t^2 + \sigma_s^2 - \bar{\sigma}^2}{T}. \end{aligned}$$

□

Remark 9.3.10 (Homoskedastic case as a particular case). If $\sigma_t^2 = \sigma_s^2 = \bar{\sigma}^2 = \sigma_u^2$ for all t, s , the formulas collapse to the classical results:

$$\begin{aligned} \text{Var}(\ddot{u}_{it}) &= \sigma_u^2 \left(1 - \frac{2}{T}\right) + \frac{\sigma_u^2}{T} = \sigma_u^2 \left(\frac{T-1}{T}\right), \\ \text{Cov}(\ddot{u}_{it}, \ddot{u}_{is}) &= -\frac{2\sigma_u^2 - \sigma_u^2}{T} = -\frac{\sigma_u^2}{T}. \end{aligned}$$

9.3.7.3 Arellano's (1987) Cluster-Robust Estimator

Under arbitrary non-sphericity, Arellano (1987) extended White's (1980) approach to panel data, proposing a cluster-robust variance estimator (CRVE) that requires no parameterization of intra-individual dynamics.

Definition 9.3.8 (Arellano's cluster-robust estimator). The cluster-robust estimated covariance matrix is:

$$\hat{\mathbf{V}}_{\text{cluster}} = \left(\sum_{i=1}^N \ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i \right)^{-1} \left(\sum_{i=1}^N \ddot{\mathbf{X}}_i' \hat{\mathbf{u}}_i \hat{\mathbf{u}}_i' \ddot{\mathbf{X}}_i \right) \left(\sum_{i=1}^N \ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i \right)^{-1}, \quad (9.142)$$

where:

- $\mathbf{A}_N = \left(\sum_{i=1}^N \ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i \right)^{-1}$ is the "bread" of the sandwich
- $\mathbf{B}_N = \sum_{i=1}^N \ddot{\mathbf{X}}_i' \hat{\mathbf{u}}_i \hat{\mathbf{u}}_i' \ddot{\mathbf{X}}_i$ is the "meat" of the sandwich

Theorem 9.3.8 (Asymptotic consistency of $\hat{\mathbf{V}}_{\text{cluster}}$ under arbitrary non-sphericity). Under hypotheses FE.1, FE.2, i.i.d. sampling in dimension i with $N \rightarrow \infty$ and fixed T , finite fourth-order moments, and arbitrary conditional covariance matrix $\Sigma_i = \text{Var}(\mathbf{u}_i | \mathbf{X}_i, c_i)$, the scaled estimator converges in probability to the true asymptotic sandwich variance:

$$\text{plim}_{N \rightarrow \infty} (N \hat{\mathbf{V}}_{\text{cluster}}) = \mathbf{V}_{\text{sandwich}} \equiv \mathbf{A}^{-1} \mathbf{B}_0 \mathbf{A}^{-1}, \quad (9.143)$$

where $\mathbf{A} = \mathbb{E}[\ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i]$ and $\mathbf{B}_0 = \mathbb{E}[\ddot{\mathbf{X}}_i' \Sigma_i \ddot{\mathbf{X}}_i]$.

Proof. Step 1: Convergence of the bread (WLLN + Slutsky).

By Kolmogorov's WLLN applied to i.i.d. $\{\ddot{\mathbf{X}}'_i \ddot{\mathbf{X}}_i\}$:

$$\mathbf{A}_N = \frac{1}{N} \sum_{i=1}^N \ddot{\mathbf{X}}'_i \ddot{\mathbf{X}}_i \xrightarrow{p} \mathbb{E}[\ddot{\mathbf{X}}'_i \ddot{\mathbf{X}}_i] \equiv \mathbf{A}. \quad (9.144)$$

By the Within Rank Condition, $\mathbf{A}$ is invertible. By the Continuous Mapping Theorem:

$$\mathbf{A}_N^{-1} \xrightarrow{p} \mathbf{A}^{-1}. \quad (9.145)$$

Step 2: Convergence of the meat.

Define $\hat{\mathbf{B}}_N = \frac{1}{N} \sum_{i=1}^N \ddot{\mathbf{X}}'_i \hat{\mathbf{u}}_i \hat{\mathbf{u}}'_i \ddot{\mathbf{X}}_i$. Substituting $\hat{\mathbf{u}}_i = \ddot{\mathbf{u}}_i - \ddot{\mathbf{X}}_i(\hat{\beta}_{\text{FE}} - \beta)$:

$$\hat{\mathbf{B}}_N = \frac{1}{N} \sum_{i=1}^N \ddot{\mathbf{X}}'_i [\ddot{\mathbf{u}}_i - \ddot{\mathbf{X}}_i(\hat{\beta}_{\text{FE}} - \beta)] [\ddot{\mathbf{u}}_i - \ddot{\mathbf{X}}_i(\hat{\beta}_{\text{FE}} - \beta)]' \ddot{\mathbf{X}}_i.$$

Expanding the inner product and distributing $\ddot{\mathbf{X}}'_i$ and $\ddot{\mathbf{X}}_i$:

$$\hat{\mathbf{B}}_N = \mathbf{T}_{1,N} - \mathbf{T}_{2,N} - \mathbf{T}_{3,N} + \mathbf{T}_{4,N}, \quad (9.146)$$

where:

$$\begin{aligned} \mathbf{T}_{1,N} &= \frac{1}{N} \sum_{i=1}^N \ddot{\mathbf{X}}'_i \ddot{\mathbf{u}}_i \ddot{\mathbf{u}}'_i \ddot{\mathbf{X}}_i, \\ \mathbf{T}_{2,N} &= \left[\frac{1}{N} \sum_{i=1}^N \ddot{\mathbf{X}}'_i \ddot{\mathbf{u}}_i \ddot{\mathbf{X}}'_i \ddot{\mathbf{X}}_i \right] (\hat{\beta}_{\text{FE}} - \beta), \\ \mathbf{T}_{3,N} &= \mathbf{T}'_{2,N}, \\ \mathbf{T}_{4,N} &= \frac{1}{N} \sum_{i=1}^N \ddot{\mathbf{X}}'_i \ddot{\mathbf{X}}_i (\hat{\beta}_{\text{FE}} - \beta) (\hat{\beta}_{\text{FE}} - \beta)' \ddot{\mathbf{X}}'_i \ddot{\mathbf{X}}_i. \end{aligned}$$

Limit of $\mathbf{T}_{1,N}$: By symmetry and idempotence of $\mathbf{Q}_T$, $\ddot{\mathbf{X}}'_i \ddot{\mathbf{u}}_i = \mathbf{X}'_i \mathbf{Q}_T \mathbf{u}_i = \ddot{\mathbf{X}}'_i \mathbf{u}_i$. By WLLN:

$$\mathbf{T}_{1,N} = \frac{1}{N} \sum_{i=1}^N \ddot{\mathbf{X}}'_i \mathbf{u}_i \mathbf{u}'_i \ddot{\mathbf{X}}_i \xrightarrow{p} \mathbb{E}[\ddot{\mathbf{X}}'_i \mathbf{u}_i \mathbf{u}'_i \ddot{\mathbf{X}}_i] \equiv \mathbf{B}_0. \quad (9.147)$$

Limit of $\mathbf{T}_{2,N}$ and $\mathbf{T}_{3,N}$: The term in brackets converges to $\mathbf{C}_0 = \mathbb{E}[\ddot{\mathbf{X}}'_i \ddot{\mathbf{u}}_i \ddot{\mathbf{X}}'_i \ddot{\mathbf{X}}_i] < \infty$. By consistency of $\hat{\beta}_{\text{FE}}$, $(\hat{\beta}_{\text{FE}} - \beta) \xrightarrow{p} \mathbf{0}$. By Slutsky:

$$\mathbf{T}_{2,N} \xrightarrow{p} \mathbf{C}_0 \cdot \mathbf{0} = \mathbf{0}. \quad (9.148)$$

Similarly, $\mathbf{T}_{3,N} \xrightarrow{p} \mathbf{0}$.

Limit of $\mathbf{T}_{4,N}$: By $\sqrt{N}$ -consistency, $(\hat{\beta}_{\text{FE}} - \beta) = \mathcal{O}_p(N^{-1/2})$, so $(\hat{\beta}_{\text{FE}} - \beta)(\hat{\beta}_{\text{FE}} - \beta)' = \mathcal{O}_p(N^{-1})$. The sum of fourth-order regressors converges to $\mathbf{D}_0 < \infty$. By Slutsky:

$$\mathbf{T}_{4,N} \xrightarrow{p} \mathbf{0}. \quad (9.149)$$

Consolidation: Reassembling the limits:

$$\hat{\mathbf{B}}_N \xrightarrow{p} \mathbf{B}_0 - \mathbf{0} - \mathbf{0} + \mathbf{0} = \mathbf{B}_0 = \mathbb{E}[\ddot{\mathbf{X}}_i' \boldsymbol{\Sigma}_i \ddot{\mathbf{X}}_i]. \quad (9.150)$$

Step 3: Final combination (Slutsky).

Applying Slutsky to the triple product $N\hat{\mathbf{V}}_{\text{cluster}} = \mathbf{A}_N^{-1} \hat{\mathbf{B}}_N \mathbf{A}_N^{-1}$:

$$\text{plim}_{N \rightarrow \infty} (N\hat{\mathbf{V}}_{\text{cluster}}) = \mathbf{A}^{-1} \mathbf{B}_0 \mathbf{A}^{-1} = \mathbf{V}_{\text{sandwich}}. \quad (9.151)$$

□

Remark 9.3.11 (Analytical immunity and elegance of the proof). **1. Immunity to $\boldsymbol{\Sigma}_i$:** Consistency does not depend on any parametric format for $\boldsymbol{\Sigma}_i$. The matrix $\hat{\mathbf{B}}_N$ absorbs and consistently estimates the true variability of the cross-moments, whether under AR(p), MA(q), complex conditional heteroskedasticity, or seasonal variations.

2. Why estimated residuals do not contaminate the variance? The terms $\mathbf{T}_{2,N}$, $\mathbf{T}_{3,N}$, and $\mathbf{T}_{4,N}$ involving the difference $(\hat{\boldsymbol{\beta}}_{\text{FE}} - \boldsymbol{\beta})$ vanish asymptotically. Since the FE estimator is $\sqrt{N}$ -consistent, the estimation error converges to zero at rate $\mathcal{O}_p(N^{-1/2})$, neutralizing any substitution bias in large samples.

Remark 9.3.12 (Finite-sample adjustments). Although consistent under $N \rightarrow \infty$, the classical CRVE underestimates variances in small samples. Software applies scale factors a_N :

1. **C. Hansen (2007) correction:** $a_N^{\text{Hansen}} = \frac{N}{N-1}$
2. **areg (Stata) correction:** $a_N^{\text{areg}} = \left(\frac{NT-1}{N(T-1)-K} \right) \left(\frac{N}{N-1} \right)$

B. Hansen (2022) criticizes the **areg** adjustment as excessively conservative and lacking solid theoretical justification. Since the estimation of N individual fixed effects does not incur degrees of freedom bias on the cross-sectional sample variance (consistency relies strictly on $N \rightarrow \infty$), the ad hoc multiplicative factor is unnecessarily inflationary.

9.3.7.4 Inference with a Small Number of Clusters ($G < 50$)

Remark 9.3.13 (Asymptotic limitations). CRVE consistency requires $G \rightarrow \infty$ (number of clusters). The **effective sample size for inference is determined by G , not by the total number of observations n** . Under small G (typically $G < 50$):

1. **Variance underestimation:** Classical CRVE severely underestimates autocorrelation and intraclass correlation (Moulton)
2. **Size distortion (over-rejection):** Downward-biased standard errors inflate t -statistics, generating rejection rates of true H_0 much higher than nominal α
3. **Rank deficiency:** If $K > G$, the robust covariance matrix is singular, preventing joint Wald tests

Remark 9.3.14 (Impact of cluster size heterogeneity). In real data, cluster sizes n_g are typically heterogeneous. Inequality in n_g induces severe effective heteroskedasticity: the sums of residuals at the cluster level have variances proportional to n_g . High variation in

n_g reduces CRVE precision and accentuates heavy tails and asymmetry of test statistics, making the normal approximation even more misleading.

Definition 9.3.9 (Finite-sample adjustments for small G). To mitigate biases under a small number of clusters, correction factors a_n are applied:

1. **Hansen (2007) correction:** $a_n^{\text{Hansen}} = \frac{G}{G-1}$, with comparison to Student's t with $G - 1$ degrees of freedom
2. **Stata correction (CR1):** $a_n^{\text{Stata}} = \left(\frac{n-1}{n-K}\right) \left(\frac{G}{G-1}\right)$

Stata's $\frac{n-1}{n-K}$ factor is ad hoc and lacks solid asymptotic justification in the presence of intra-cluster dependence, frequently appearing as excessively conservative.

Definition 9.3.10 (Wild Cluster Bootstrap of Cameron, Gelbach & Miller (2008)). When $G < 30$, degrees of freedom multipliers fail to restore nominal size. The Wild Cluster Bootstrap provides exact inference via stochastic perturbation of residuals, keeping regressors and cluster structure fixed.

Restricted Wild Cluster Bootstrap

To test $H_0 : r(\beta) = 0$:

1. **Restricted estimation:** Estimate the model under H_0 , obtaining $\tilde{\beta}$ and restricted residuals $\tilde{\mathbf{e}}_g = \mathbf{Y}_g - \mathbf{X}_g \tilde{\beta}$ for each cluster g
2. **Bootstrap generation:** For each replication $b = 1, \dots, B$:
 - Draw ξ_g^* for each cluster g from the Rademacher distribution:

$$\xi_g^* = \begin{cases} +1 & \text{with prob. } 0.5 \\ -1 & \text{with prob. } 0.5 \end{cases} \quad (9.152)$$

- Generate $\mathbf{Y}_g^* = \mathbf{X}_g \tilde{\beta} + \tilde{\mathbf{e}}_g \xi_g^*$
3. **Estimation and test:** For each pseudo-sample, estimate the unrestricted model and compute $t^*(b) = \frac{\hat{\beta}^*(b) - \tilde{\beta}}{SE(\hat{\beta}^*(b))}$, where SE is obtained via recalculated CRVE
 4. **Bootstrap p -value:** After $B \geq 10,000$ replications:

$$p^* = \frac{1}{B} \sum_{b=1}^B \mathbf{1}(|t^*(b)| > |t_{\text{sample}}|) \quad (9.153)$$

Remark 9.3.15 (Properties of the Wild Cluster Bootstrap). 1. **Rademacher vs.**

Mammen distribution: Davidson & Flachaire (2008) recommend Rademacher in small samples because it is perfectly symmetric, nullifying third-order skewness bias terms in two-tailed tests

2. **Asymptotic refinements:** Under Edgeworth expansions, the Wild Bootstrap reduces size errors from $\mathcal{O}(G^{-1})$ to $o(G^{-1})$, ensuring exact control of empirical size
3. **Advantage over pairs bootstrap:** Cluster pairs bootstrap (resampling with replacement) fails under small G due to the probability of degenerate samples with identical group replication. The Wild Bootstrap keeps the original cluster structure intact

9.3.8 Non-Identification of Time-Invariant Variables

The impossibility of estimating coefficients for time-invariant variables is not a practical software limitation, but an inevitable algebraic consequence of orthogonal projection.

9.3.8.1 Annihilation of Invariant Regressors

Theorem 9.3.9 (Non-identification of invariant attributes). *Let $\mathbf{w}_i \in \mathbb{R}^{T \times 1}$ be the vector containing the temporal trajectory of a strictly time-invariant regressor for individual i , such that $w_{it} = \bar{w}_i$ for all t . Then,*

$$\mathbf{Q}_T \mathbf{w}_i = \mathbf{0}. \quad (9.154)$$

Proof. Time invariance implies $\mathbf{w}_i = \bar{w}_i \mathbf{j}_T$. By linearity of the operator and the orthogonality demonstrated in the Lemma:

$$\mathbf{Q}_T \mathbf{w}_i = \mathbf{Q}_T(\bar{w}_i \mathbf{j}_T) = \bar{w}_i(\mathbf{Q}_T \mathbf{j}_T) = \bar{w}_i \cdot \mathbf{0} = \mathbf{0}. \quad (9.155)$$

□

9.3.8.2 Collapse of Rank of the Moment Matrix

The annihilation of invariant variables has a direct consequence for identifiability. Consider the partitioned model:

$$\mathbf{y}_i = \mathbf{Z}_i \boldsymbol{\beta}_1 + \mathbf{W}_i \boldsymbol{\beta}_2 + c_i \mathbf{j}_T + \mathbf{u}_i, \quad (9.156)$$

where $\mathbf{Z}_i \in \mathbb{R}^{T \times K_1}$ contains time-varying regressors and $\mathbf{W}_i \in \mathbb{R}^{T \times K_2}$ contains K_2 strictly invariant regressors. Each column of $\mathbf{W}_i$ satisfies $\mathbf{w}_i^{(k)} = \bar{w}_i^{(k)} \mathbf{j}_T$, so $\mathbf{W}_i = \mathbf{j}_T \bar{\mathbf{w}}_i'$.

After the within transformation:

$$\ddot{\mathbf{y}}_i = \ddot{\mathbf{Z}}_i \boldsymbol{\beta}_1 + \underbrace{\mathbf{Q}_T \mathbf{W}_i}_{=\mathbf{0}} \boldsymbol{\beta}_2 + \underbrace{c_i \mathbf{Q}_T \mathbf{j}_T}_{=\mathbf{0}} + \ddot{\mathbf{u}}_i = \ddot{\mathbf{Z}}_i \boldsymbol{\beta}_1 + \ddot{\mathbf{u}}_i. \quad (9.157)$$

Defining the global matrix of transformed regressors as $\ddot{\mathbf{X}}_i = [\ddot{\mathbf{Z}}_i, \mathbf{Q}_T \mathbf{W}_i] = [\ddot{\mathbf{Z}}_i, \mathbf{0}_{T \times K_2}]$, the cross-moment matrix is

$$\sum_{i=1}^N \ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i = \begin{bmatrix} \sum_{i=1}^N \ddot{\mathbf{Z}}_i' \ddot{\mathbf{Z}}_i & \mathbf{0}_{K_1 \times K_2} \\ \mathbf{0}_{K_2 \times K_1} & \mathbf{0}_{K_2 \times K_2} \end{bmatrix}. \quad (9.158)$$

Since the last K_2 columns are entirely null, the matrix is singular:

$$\text{rank} \left(\sum_{i=1}^N \ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i \right) = \text{rank} \left(\sum_{i=1}^N \ddot{\mathbf{Z}}_i' \ddot{\mathbf{Z}}_i \right) \leq K_1 < K_1 + K_2. \quad (9.159)$$

The identification condition (FE.2) requires rank $K_1 + K_2$, which is violated. The inverse matrix does not exist and $\boldsymbol{\beta}_2$ is **algebraically non-identifiable** by the FE estimator.

9.3.8.3 Practical Implications

I. Mechanical disappearance of the intercept. The intercept β_0 is associated with a constant regressor ($x_{it,1} = 1$ for all i, t). Since this variable is perfectly time-invariant, $\mathbf{Q}_T$ transforms it into a vector of zeros. The FE estimator eliminates the common intercept by construction. Software reports as "intercept" the estimated mean of individual fixed effects ($\bar{c} = \frac{1}{N} \sum_i \hat{c}_i$), but the original parameter β_0 is not identified in isolation.

II. Treatment in econometric software. When including invariant variables (such as gender or race) in an FE regression (e.g., `xtreg`, `fe` in Stata or `plm` in R), the algorithm detects the column of zeros in the transformed matrix and automatically drops the variable, reporting `omitted due to collinearity`. This is not a bug, but the correct implementation of projection algebra.

9.3.9 Caveats, Common Errors, and Committee Questions

9.3.9.1 Caveats and Common Errors

Time-invariant variables If z_i does not vary over time, then $z_i - \bar{z}_i = 0$. Therefore, FE does not identify coefficients for invariant variables.

Symptom: variable is automatically excluded from the fixed effects regression. If the researcher needs to estimate the effect of invariant variables, they should consider RE (under $\mathbb{E}[c_i | \mathbf{X}_i] = 0$) or the Mundlak estimator, which includes $\bar{\mathbf{x}}_i$ as a proxy for c_i .

Nickell bias in dynamic panels Consider

$$y_{it} = \rho y_{i,t-1} + c_i + u_{it}. \quad (9.160)$$

After the within transformation,

$$\ddot{y}_{it} = \rho \ddot{y}_{i,t-1} + \ddot{u}_{it}. \quad (9.161)$$

The regressor $\ddot{y}_{i,t-1}$ depends on $u_{i,t-1}$, and the error $\ddot{u}_{it}$ contains $\bar{u}_i$, which also includes $u_{i,t-1}$. Thus,

$$\mathbb{E}[\ddot{y}_{i,t-1} \ddot{u}_{it}] \neq 0. \quad (9.162)$$

The FE estimator is inconsistent for fixed T . The bias is of order $O(1/T)$.

Symptom: autoregressive coefficient substantially biased in short panels.

Measurement error FE uses within variation. If the variable $\mathbf{x}_{it}$ is persistent and measured with error, the within transformation may reduce the signal-to-noise ratio.

Symptom: coefficient shrinks toward zero when moving from POLS to FE.

Non-linear models In non-linear models, such as Logit or Probit with fixed effects, estimating c_i can generate the incidental parameters problem. The linear within transformation is not available in the same way.

9.3.9.2 Typical Committee Question

Question: You use FE in a short panel with possible feedback from the error to future regressors. How do you defend consistency?

Answer: FE requires strict exogeneity:

$$\mathbb{E}[u_{it} \mid \mathbf{X}_i, c_i] = 0. \quad (9.163)$$

If there is feedback, for example,

$$\text{Cov}(\mathbf{x}_{i,t+1}, u_{it}) \neq 0, \quad (9.164)$$

then

$$\mathbb{E}[\ddot{\mathbf{x}}'_{it} \ddot{u}_{it}] \neq 0, \quad (9.165)$$

and FE is inconsistent.

As an empirical defense, one can:

- include lead regressors as a falsification test;
- compare FE with instrumental variable estimators;
- use dynamic GMM if there is a lagged dependent variable.

Table 9.1: Final Synthesis: Fixed Effects Estimator

Concept	Mathematical Expression	Eq.
Population Model	$y_{it} = \mathbf{x}'_{it}\boldsymbol{\beta} + c_i + u_{it}$	(1)
Within Operator	$\mathbf{Q}_T = \mathbf{I}_T - \frac{1}{T}\mathbf{j}_T\mathbf{j}'_T$	(5)
Properties of $\mathbf{Q}_T$	$\mathbf{Q}'_T = \mathbf{Q}_T, \mathbf{Q}_T^2 = \mathbf{Q}_T, \mathbf{Q}_T\mathbf{j}_T = \mathbf{0}$	(6)
Rank of $\mathbf{Q}_T$	$\text{rank}(\mathbf{Q}_T) = T - 1$	(10)
FE Estimator	$\hat{\boldsymbol{\beta}}_{\text{FE}} = (\sum \mathbf{X}'_i \mathbf{Q}_T \mathbf{X}_i)^{-1} \sum \mathbf{X}'_i \mathbf{Q}_T \mathbf{y}_i$	(15)
Unbiasedness	$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{FE}}] = \boldsymbol{\beta}$	(17)
Consistency	$\text{plim } \hat{\boldsymbol{\beta}}_{\text{FE}} = \boldsymbol{\beta}$	(18)
Induced Autocorrelation	$\text{Corr}(\ddot{u}_{it}, \ddot{u}_{is}) = -\frac{1}{T-1}$	(23)
Residual Variance	$\hat{\sigma}_u^2 = \frac{SSR}{N(T-1)-K}$	(25)
Sandwich Variance	$\mathbf{V}_{\text{sandwich}} = \mathbf{A}^{-1}\mathbf{B}_0\mathbf{A}^{-1}$	(31)
Arellano's CRVE	$\hat{\mathbf{V}}_{\text{cluster}} = \mathbf{A}_N^{-1}\mathbf{B}_N\mathbf{A}_N^{-1}$	(36)

Remark 9.3.16 (Final Summary for the Qualification Committee). 1. The **FE estimator** eliminates c_i via the within transformation $\mathbf{Q}_T$, allowing arbitrary correlation between c_i and $\mathbf{x}_{it}$.

2. **Critical hypotheses:** FE.1 (strict exogeneity) guarantees consistency; FE.2 (full rank) guarantees identification; FE.3 (sphericity) justifies classical standard errors; FE.4 (finite moments) guarantees applicability of WLLN.
3. **Unbiasedness and consistency:** Under FE.1-FE.4, $\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{FE}}] = \boldsymbol{\beta}$ (finite sample) and $\text{plim } \hat{\boldsymbol{\beta}}_{\text{FE}} = \boldsymbol{\beta}$ (asymptotic), even with $\text{Cov}(\mathbf{x}_{it}, c_i) \neq 0$.

4. **Mechanical autocorrelation:** The within transformation induces a correlation of $-1/(T - 1)$ between residuals, a mathematical (not economic) artifact that disappears as $T \rightarrow \infty$.
5. **Degrees of freedom:** The correct denominator is $N(T - 1) - K$, not $NT - K$, reflecting the loss of N degrees of freedom due to demeaning.
6. **Robust inference:** Under non-sphericity, Arellano's CRVE (sandwich) is used, consistent under arbitrary heteroskedasticity/autocorrelation. For $G < 50$ clusters, Wild Cluster Bootstrap is applied.
7. **FE limitations:** (i) does not identify time-invariant variables; (ii) inconsistent under dynamic feedback (Nickell bias); (iii) amplifies measurement error in persistent variables.

9.4 Random Effects (RE)

9.4.1 Population Specification and Identification Assumptions

9.4.1.1 Linear Model with Additive Compound Error

Definition 9.4.1 (Population Specification of the Panel Model). Let there be a balanced panel with N cross-sectional units $i \in \{1, \dots, N\}$ observed over T periods $t \in \{1, \dots, T\}$. The population model for observation (i, t) is:

$$y_{it} = \mathbf{x}_{it}\boldsymbol{\beta} + v_{it} \quad (9.166)$$

where $y_{it} \in \mathbb{R}$ is the dependent variable, $\mathbf{x}_{it} \in \mathbb{R}^{1 \times K}$ is the vector of K regressors, $\boldsymbol{\beta} \in \mathbb{R}^{K \times 1}$ is the vector of homogeneous and time-invariant structural parameters, and $v_{it} \in \mathbb{R}$ is the contemporaneous compound error.

Definition 9.4.2 (Additive Decomposition of the Compound Error). The compound error v_{it} is additively decomposed into two components:

$$v_{it} = c_i + u_{it} \quad (9.167)$$

where:

- $c_i \in \mathbb{R}$ is the **unobserved individual effect** (time-invariant individual heterogeneity), capturing constant latent factors such as innate ability or organizational culture
- $u_{it} \in \mathbb{R}$ is the **contemporaneous idiosyncratic error**, representing transitory random shocks that vary over time and across individuals

Substituting into the level equation:

$$y_{it} = \mathbf{x}_{it}\boldsymbol{\beta} + c_i + u_{it} \quad (9.168)$$

Remark 9.4.1 (Matrix Representation by Individual). Stacking the T observations of individual i :

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{v}_i \quad (9.169)$$

$$\mathbf{v}_i = c_i\mathbf{1}_T + \mathbf{u}_i \quad (9.170)$$

where $\mathbf{y}_i, \mathbf{v}_i, \mathbf{u}_i \in \mathbb{R}^{T \times 1}$, $\mathbf{X}_i \in \mathbb{R}^{T \times K}$, and $\mathbf{1}_T \in \mathbb{R}^{T \times 1}$ is the vector of ones. The final population vector representation is:

$$\mathbf{y}_i = \mathbf{X}_i \boldsymbol{\beta} + c_i \mathbf{1}_T + \mathbf{u}_i \quad (9.171)$$

Remark 9.4.2 (Distinction Between Fixed Effects and Random Effects). The separation of the compound error into c_i and u_{it} is crucial for distinguishing the approaches:

- **Fixed Effects (FE):** allows arbitrary correlation between c_i and $\mathbf{x}_{it}$ ($\text{Cov}(\mathbf{x}_{it}, c_i) \neq 0$)
- **Random Effects (RE):** requires strict orthogonality between c_i and $\mathbf{x}_{it}$ ($E[c_i | \mathbf{X}_i] = 0$), allowing c_i to remain in the compound error without generating inconsistency

9.4.1.2 Stochastic Nature of Individual Heterogeneity (c_i)

Remark 9.4.3 (Epistemological Justification for Stochastic Treatment of c_i). Under simple random sampling from the population, all individual characteristics $\mathbf{X}_i$ and $\mathbf{y}_i$ are random variables. By logical consistency, the unobserved heterogeneity c_i must be treated under the same stochastic ontology, being a latent random variable generated by the same sampling process.

This approach differs from the classical view that treated c_i as N fixed deterministic parameters. Under the standard microeconomic asymptotics ($N \rightarrow \infty$, T fixed), estimating each c_i individually generates the **incidental parameters problem**: the number of parameters grows linearly with N , making it impossible to accumulate historical information about each unit and infecting $\hat{\boldsymbol{\beta}}$ with severe inconsistency.

Remark 9.4.4 (Conceptual Distinction: “Fixed” vs “Random”). The distinction between “fixed effect” and “random effect” **does not lie in the mathematical nature of c_i** (stochastic under both approaches), but rather in the **conditional correlation structure** allowed between c_i and $\mathbf{X}_i$:

- **Fixed Effect:** allows arbitrary statistical dependence ($\text{Cov}(\mathbf{x}_{it}, c_i) \neq 0$)
- **Random Effect:** imposes strict exogeneity and conditional orthogonality ($E[c_i | \mathbf{X}_i] = 0$)

Remark 9.4.5 (Conditional vs Unconditional Inference). The choice between conditional (FE) and unconditional (RE) modeling has profound implications for the scope of scientific conclusions:

- **Conditional Inference (FE/LSDV):** the scope of inference is restricted to the N units present in the sample. The Within projection removes inter-individual variability and prevents extrapolation beyond sampled units.
- **Unconditional Inference (RE):** by conceiving c_i as a random variable with modeled moments, one performs marginal inference directed at the superpopulation from which the sample was drawn. The estimated parameters represent structural population moments, allowing extrapolation to unsampled units and counterfactual scenarios.

9.4.1.3 System of Orthogonality Assumptions

Definition 9.4.3 (Assumption RE.1: Orthogonality of Individual Heterogeneity). The defining assumption of the Random Effects model postulates that the conditional expectation of the individual effect c_i given the regressor matrix $\mathbf{X}_i$ is independent of it and equal to its unconditional expectation (assumed zero without loss of generality):

$$E[c_i | \mathbf{X}_i] = E[c_i] = 0 \quad (9.172)$$

Role: This is the assumption that distinguishes RE from FE. It requires that time-invariant omitted characteristics (innate ability) do not influence systematic choices of regressors in any period. If $\text{Cov}(\mathbf{x}_{itk}, c_i) \neq 0$, the compound error correlates with regressors, generating omitted variable bias and severe inconsistency of the RE estimator.

Counterexample: If innate ability affects wages and education, then c_i is correlated with educ_{it} , and RE is inconsistent.

Definition 9.4.4 (Assumption RE.2: Conditional Strict Exogeneity of the Idiosyncratic Error). The contemporaneous idiosyncratic shock u_{it} is orthogonal to regressors from all periods, conditional on individual heterogeneity c_i :

$$E[u_{it} | \mathbf{X}_i, c_i] = 0 \quad \forall t \in \{1, \dots, T\} \quad (9.173)$$

Role: Ensures that contemporaneous shocks are orthogonal to any measurable function of explanatory variables (present, past, or future), conditional on c_i . Prevents contemporaneous reverse causality and temporal feedback effects. It is violated by definition in dynamic models with the lagged dependent variable as regressor.

Definition 9.4.5 (Assumption RE.3: Homoskedasticity and Absence of Serial Correlation). The components of the compound error satisfy:

$$\text{Var}(u_{it} | \mathbf{X}_i, c_i) = \sigma_u^2 > 0 \quad \forall t \quad (\text{idiosyncratic homoskedasticity}) \quad (9.174)$$

$$\text{Cov}(u_{it}, u_{is} | \mathbf{X}_i, c_i) = 0 \quad \forall t \neq s \quad (\text{absence of serial autocorrelation}) \quad (9.175)$$

$$\text{Var}(c_i | \mathbf{X}_i) = \sigma_c^2 \geq 0 \quad (\text{individual effect homoskedasticity}) \quad (9.176)$$

Role: Allow construction of the covariance matrix $\mathbf{\Omega}$ and the efficient GLS transformation.

Definition 9.4.6 (Assumption RE.4: Rank Condition).

$$\text{rank} \left(E[\mathbf{X}_i' \mathbf{\Omega}^{-1} \mathbf{X}_i] \right) = K \quad (9.177)$$

Role: Guarantee identification and invertibility of the weighted moment matrix.

Theorem 9.4.1 (Conditional Moment of the Compound Error). *Under assumptions RE.1 and RE.2, the compound error vector satisfies:*

$$E[\mathbf{v}_i | \mathbf{X}_i] = \mathbf{0}_T \quad (9.178)$$

Proof. By the decomposition $\mathbf{v}_i = c_i \mathbf{1}_T + \mathbf{u}_i$ and linearity of conditional expectation:

$$\begin{aligned} E[\mathbf{v}_i | \mathbf{X}_i] &= E[c_i \mathbf{1}_T + \mathbf{u}_i | \mathbf{X}_i] \\ &= E[c_i | \mathbf{X}_i] \mathbf{1}_T + E[\mathbf{u}_i | \mathbf{X}_i] \end{aligned} \quad (9.179)$$

By assumption RE.1, $E[c_i | \mathbf{X}_i] = 0$, so the first term vanishes:

$$E[\mathbf{v}_i | \mathbf{X}_i] = \mathbf{0}_T + E[\mathbf{u}_i | \mathbf{X}_i] \quad (9.180)$$

To evaluate $E[\mathbf{u}_i | \mathbf{X}_i]$, we apply the Law of Iterated Expectations (LIE) conditioning on c_i :

$$\begin{aligned} E[\mathbf{u}_i | \mathbf{X}_i] &= E[E[\mathbf{u}_i | \mathbf{X}_i, c_i] | \mathbf{X}_i] \\ &= E[\mathbf{0}_T | \mathbf{X}_i] \quad (\text{by assumption RE.2}) \\ &= \mathbf{0}_T \end{aligned} \quad (9.181)$$

Therefore:

$$E[\mathbf{v}_i | \mathbf{X}_i] = \mathbf{0}_T + \mathbf{0}_T = \mathbf{0}_T \quad (9.182)$$

□

Remark 9.4.6 (Implication for Pooled OLS Consistency). Strict exogeneity $E[\mathbf{v}_i | \mathbf{X}_i] = \mathbf{0}_T$ implies $E[\mathbf{X}_i' \mathbf{v}_i] = \mathbf{0}_K$ by LIE, guaranteeing Pooled OLS consistency. However, the temporal persistence of c_i in $\mathbf{v}_i$ induces non-zero intraclass covariance, violating Gauss-Markov and requiring FGLS for efficiency.

Theorem 9.4.2 (Second-Order Mutual Orthogonality). *Under assumptions RE.1 and RE.2, the conditional covariance between c_i and u_{it} is zero:*

$$\text{Cov}(c_i, u_{it} | \mathbf{X}_i) = 0 \quad \forall t \quad (9.183)$$

Proof. By definition of conditional covariance:

$$\text{Cov}(c_i, u_{it} | \mathbf{X}_i) = E[c_i u_{it} | \mathbf{X}_i] - E[c_i | \mathbf{X}_i] E[u_{it} | \mathbf{X}_i] \quad (9.184)$$

First term: By LIE conditioning on c_i :

$$\begin{aligned} E[c_i u_{it} | \mathbf{X}_i] &= E[E[c_i u_{it} | \mathbf{X}_i, c_i] | \mathbf{X}_i] \\ &= E[c_i E[u_{it} | \mathbf{X}_i, c_i] | \mathbf{X}_i] \\ &= E[c_i \cdot 0 | \mathbf{X}_i] \quad (\text{by assumption RE.2}) \\ &= 0 \end{aligned} \quad (9.185)$$

Second term: By assumption RE.1, $E[c_i | \mathbf{X}_i] = 0$, and by LIE + RE.2, $E[u_{it} | \mathbf{X}_i] = 0$:

$$E[c_i | \mathbf{X}_i] E[u_{it} | \mathbf{X}_i] = 0 \cdot 0 = 0 \quad (9.186)$$

Therefore:

$$\text{Cov}(c_i, u_{it} | \mathbf{X}_i) = 0 - 0 = 0 \quad (9.187)$$

□

Theorem 9.4.3 (Structure of the Compound Error Variance-Covariance Matrix). *Under assumptions RE.1, RE.2, and RE.3, the conditional variance-covariance matrix of the compound error is:*

$$\mathbf{\Omega} \equiv \text{Var}(\mathbf{v}_i | \mathbf{X}_i) = \sigma_c^2 \mathbf{J}_T + \sigma_u^2 \mathbf{I}_T \quad (9.188)$$

where $\mathbf{I}_T$ is the identity matrix and $\mathbf{J}_T = \mathbf{1}_T \mathbf{1}_T'$ is the matrix of ones.

Proof. By the decomposition $\mathbf{v}_i = c_i \mathbf{1}_T + \mathbf{u}_i$ and linearity of variance:

$$\begin{aligned} \text{Var}(\mathbf{v}_i \mid \mathbf{X}_i) &= \text{Var}(c_i \mathbf{1}_T + \mathbf{u}_i \mid \mathbf{X}_i) \\ &= \text{Var}(c_i \mathbf{1}_T \mid \mathbf{X}_i) + \text{Var}(\mathbf{u}_i \mid \mathbf{X}_i) + 2 \text{Cov}(c_i \mathbf{1}_T, \mathbf{u}_i \mid \mathbf{X}_i) \end{aligned} \quad (9.189)$$

By the previous theorem (mutual orthogonality), $\text{Cov}(c_i \mathbf{1}_T, \mathbf{u}_i \mid \mathbf{X}_i) = \mathbf{0}_{T \times T}$, so:

$$\text{Var}(\mathbf{v}_i \mid \mathbf{X}_i) = \mathbf{1}_T \text{Var}(c_i \mid \mathbf{X}_i) \mathbf{1}_T' + \text{Var}(\mathbf{u}_i \mid \mathbf{X}_i) \quad (9.190)$$

By assumption RE.3:

$$\text{Var}(c_i \mid \mathbf{X}_i) = \sigma_c^2 \quad (9.191)$$

$$\text{Var}(\mathbf{u}_i \mid \mathbf{X}_i) = \sigma_u^2 \mathbf{I}_T \quad (\text{by homoskedasticity and absence of autocorrelation}) \quad (9.192)$$

Substituting:

$$\text{Var}(\mathbf{v}_i \mid \mathbf{X}_i) = \sigma_c^2 \mathbf{1}_T \mathbf{1}_T' + \sigma_u^2 \mathbf{I}_T = \sigma_c^2 \mathbf{J}_T + \sigma_u^2 \mathbf{I}_T \equiv \mathbf{\Omega} \quad (9.193)$$

□

Remark 9.4.7 (Explicit Structure of $\mathbf{\Omega}$). The matrix $\mathbf{\Omega}$ assumes the form:

$$\mathbf{\Omega} = \begin{bmatrix} \sigma_c^2 + \sigma_u^2 & \sigma_c^2 & \dots & \sigma_c^2 \\ \sigma_c^2 & \sigma_c^2 + \sigma_u^2 & \dots & \sigma_c^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_c^2 & \sigma_c^2 & \dots & \sigma_c^2 + \sigma_u^2 \end{bmatrix} \quad (9.194)$$

The conditional intraclass correlation between errors from distinct periods ($t \neq s$) is constant:

$$\text{Corr}(v_{it}, v_{is} \mid \mathbf{X}_i) = \frac{\sigma_c^2}{\sigma_c^2 + \sigma_u^2} \equiv \rho \quad (9.195)$$

This constant serial correlation induced by c_i violates Gauss-Markov, making Pooled OLS inefficient (though consistent under RE.1 and RE.2), justifying the application of FGLS to recover efficiency.

Table 9.2: Synthesis: RE Assumption System

Property	Moment Condition	Eq.
Heterogeneity Orthogonality	$E[c_i \mid \mathbf{X}_i] = 0$	(8)
Conditional Strict Exogeneity	$E[u_{it} \mid \mathbf{X}_i, c_i] = 0$	(9)
Compound Error Exogeneity	$E[\mathbf{v}_i \mid \mathbf{X}_i] = \mathbf{0}_T$	(14)
Idiosyncratic Homoskedasticity	$\text{Var}(u_{it} \mid \mathbf{X}_i, c_i) = \sigma_u^2$	(19)
Absence of Serial Autocorrelation	$\text{Cov}(u_{it}, u_{is} \mid \mathbf{X}_i, c_i) = 0, \forall t \neq s$	(20)
Individual Effect Homoskedasticity	$\text{Var}(c_i \mid \mathbf{X}_i) = \sigma_c^2$	(21)
Cross Mutual Orthogonality	$\text{Cov}(c_i, u_{it} \mid \mathbf{X}_i) = 0$	(26)
Intraclass Covariance Structure	$\mathbf{\Omega} = \sigma_c^2 \mathbf{J}_T + \sigma_u^2 \mathbf{I}_T$	(30)

9.4.2 Second-Order Structure and Spectral Decomposition

9.4.2.1 Conditional Moments of the Compound Error

Theorem 9.4.4 (Second-Order Moments of v_{it}). *Under assumptions RE.1-RE.3, the conditional moments of the compound error $v_{it} = c_i + u_{it}$ satisfy:*

$$\text{Var}(v_{it} \mid \mathbf{X}_i) = \sigma_c^2 + \sigma_u^2 \quad (9.196)$$

$$\text{Cov}(v_{it}, v_{is} \mid \mathbf{X}_i) = \sigma_c^2 \quad \forall t \neq s \quad (9.197)$$

$$\text{Corr}(v_{it}, v_{is} \mid \mathbf{X}_i) = \frac{\sigma_c^2}{\sigma_c^2 + \sigma_u^2} \equiv \rho \quad (9.198)$$

Proof. **Contemporaneous variance:** By the decomposition $v_{it} = c_i + u_{it}$ and quadratic linearity of conditional variance:

$$\text{Var}(v_{it} \mid \mathbf{X}_i) = \text{Var}(c_i \mid \mathbf{X}_i) + \text{Var}(u_{it} \mid \mathbf{X}_i) + 2 \text{Cov}(c_i, u_{it} \mid \mathbf{X}_i) \quad (9.199)$$

By assumption RE.3, $\text{Var}(c_i \mid \mathbf{X}_i) = \sigma_c^2$. For $\text{Var}(u_{it} \mid \mathbf{X}_i)$, we apply the Law of Total Conditional Variance:

$$\text{Var}(u_{it} \mid \mathbf{X}_i) = E[\text{Var}(u_{it} \mid \mathbf{X}_i, c_i) \mid \mathbf{X}_i] + \text{Var}(E[u_{it} \mid \mathbf{X}_i, c_i] \mid \mathbf{X}_i) \quad (9.200)$$

By assumption RE.2, $E[u_{it} \mid \mathbf{X}_i, c_i] = 0$, so the second term vanishes. By assumption RE.3, $\text{Var}(u_{it} \mid \mathbf{X}_i, c_i) = \sigma_u^2$, so:

$$\text{Var}(u_{it} \mid \mathbf{X}_i) = E[\sigma_u^2 \mid \mathbf{X}_i] + 0 = \sigma_u^2 \quad (9.201)$$

By second-order mutual orthogonality (proved in Section 9.4.1), $\text{Cov}(c_i, u_{it} \mid \mathbf{X}_i) = 0$. Substituting:

$$\text{Var}(v_{it} \mid \mathbf{X}_i) = \sigma_c^2 + \sigma_u^2 + 2(0) = \sigma_c^2 + \sigma_u^2 \quad (9.202)$$

Temporal covariance ($t \neq s$): By bilinearity of conditional covariance:

$$\text{Cov}(v_{it}, v_{is} \mid \mathbf{X}_i) = \text{Cov}(c_i + u_{it}, c_i + u_{is} \mid \mathbf{X}_i) \quad (9.203)$$

$$= \text{Cov}(c_i, c_i \mid \mathbf{X}_i) + \text{Cov}(c_i, u_{is} \mid \mathbf{X}_i) + \text{Cov}(u_{it}, c_i \mid \mathbf{X}_i) + \text{Cov}(u_{it}, u_{is} \mid \mathbf{X}_i) \quad (9.204)$$

We evaluate each term:

- $\text{Cov}(c_i, c_i \mid \mathbf{X}_i) = \text{Var}(c_i \mid \mathbf{X}_i) = \sigma_c^2$
- $\text{Cov}(c_i, u_{is} \mid \mathbf{X}_i) = 0$ (mutual orthogonality)
- $\text{Cov}(u_{it}, c_i \mid \mathbf{X}_i) = 0$ (symmetry of covariance)
- For $\text{Cov}(u_{it}, u_{is} \mid \mathbf{X}_i)$, we apply the Law of Total Conditional Covariance:

$$\text{Cov}(u_{it}, u_{is} \mid \mathbf{X}_i) = E[\text{Cov}(u_{it}, u_{is} \mid \mathbf{X}_i, c_i) \mid \mathbf{X}_i] + \text{Cov}(E[u_{it} \mid \mathbf{X}_i, c_i], E[u_{is} \mid \mathbf{X}_i, c_i] \mid \mathbf{X}_i) \quad (9.205)$$

By assumption RE.3, $\text{Cov}(u_{it}, u_{is} \mid \mathbf{X}_i, c_i) = 0$ for $t \neq s$. By assumption RE.2, $E[u_{it} \mid \mathbf{X}_i, c_i] = 0$. Thus:

$$\text{Cov}(u_{it}, u_{is} \mid \mathbf{X}_i) = E[0 \mid \mathbf{X}_i] + \text{Cov}(0, 0 \mid \mathbf{X}_i) = 0 \quad (9.206)$$

Substituting:

$$\text{Cov}(v_{it}, v_{is} \mid \mathbf{X}_i) = \sigma_c^2 + 0 + 0 + 0 = \sigma_c^2 \quad (9.207)$$

Intraclass correlation: By definition:

$$\rho \equiv \text{Corr}(v_{it}, v_{is} \mid \mathbf{X}_i) = \frac{\text{Cov}(v_{it}, v_{is} \mid \mathbf{X}_i)}{\sqrt{\text{Var}(v_{it} \mid \mathbf{X}_i)}\sqrt{\text{Var}(v_{is} \mid \mathbf{X}_i)}} = \frac{\sigma_c^2}{\sqrt{\sigma_c^2 + \sigma_u^2}\sqrt{\sigma_c^2 + \sigma_u^2}} = \frac{\sigma_c^2}{\sigma_c^2 + \sigma_u^2} \quad (9.208)$$

□

Remark 9.4.8 (Interpretation of ρ and Stochastic Inertia). The coefficient ρ quantifies the fraction of total compound error variance attributable to latent individual heterogeneity c_i . Unlike AR(1) processes where autocorrelation decays exponentially with $|t - s|$, the intraclass correlation remains **strictly constant** at ρ for any $t \neq s$, regardless of the temporal interval. This occurs because c_i enters additively and persistently in all periods, functioning as permanent non-decaying memory. This constant autocorrelation violates Gauss-Markov, making Pooled OLS inefficient (underestimated standard errors, inflated t -statistics) and justifying FGLS.

9.4.2.2 Matrix Formulation of Ω_i

Theorem 9.4.5 (Individual Conditional Covariance Matrix). *Under assumptions RE.1-RE.3, the conditional variance-covariance matrix of the compound error vector $\mathbf{v}_i = c_i \mathbf{1}_T + \mathbf{u}_i$ is:*

$$\Omega_i \equiv \text{Var}(\mathbf{v}_i \mid \mathbf{X}_i) = \sigma_u^2 \mathbf{I}_T + \sigma_c^2 \mathbf{1}_T \mathbf{1}_T' \quad (9.209)$$

Proof. By the decomposition $\mathbf{v}_i = c_i \mathbf{1}_T + \mathbf{u}_i$ and linearity of matrix covariance:

$$\Omega_i = \text{Var}(c_i \mathbf{1}_T \mid \mathbf{X}_i) + \text{Var}(\mathbf{u}_i \mid \mathbf{X}_i) + \text{Cov}(c_i \mathbf{1}_T, \mathbf{u}_i \mid \mathbf{X}_i) + \text{Cov}(\mathbf{u}_i, c_i \mathbf{1}_T \mid \mathbf{X}_i) \quad (9.210)$$

We evaluate each term:

- By quadratic homogeneity: $\text{Var}(c_i \mathbf{1}_T \mid \mathbf{X}_i) = \mathbf{1}_T \text{Var}(c_i \mid \mathbf{X}_i) \mathbf{1}_T' = \sigma_c^2 \mathbf{1}_T \mathbf{1}_T'$
- By assumption RE.3 (homoskedasticity + absence of autocorrelation): $\text{Var}(\mathbf{u}_i \mid \mathbf{X}_i) = \sigma_u^2 \mathbf{I}_T$
- By mutual orthogonality: $\text{Cov}(c_i \mathbf{1}_T, \mathbf{u}_i \mid \mathbf{X}_i) = \mathbf{1}_T \text{Cov}(c_i, \mathbf{u}_i \mid \mathbf{X}_i) = \mathbf{1}_T \mathbf{0}_T' = \mathbf{0}_{T \times T}$
- By symmetry: $\text{Cov}(\mathbf{u}_i, c_i \mathbf{1}_T \mid \mathbf{X}_i) = \mathbf{0}_{T \times T}$

Substituting:

$$\Omega_i = \sigma_c^2 \mathbf{1}_T \mathbf{1}_T' + \sigma_u^2 \mathbf{I}_T + \mathbf{0}_{T \times T} + \mathbf{0}_{T \times T} = \sigma_u^2 \mathbf{I}_T + \sigma_c^2 \mathbf{1}_T \mathbf{1}_T' \quad (9.211)$$

□

Remark 9.4.9 (Equicorrelated Structure). The matrix Ω_i assumes the form:

$$\Omega_i = \begin{bmatrix} \sigma_u^2 + \sigma_c^2 & \sigma_c^2 & \dots & \sigma_c^2 \\ \sigma_c^2 & \sigma_u^2 + \sigma_c^2 & \dots & \sigma_c^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_c^2 & \sigma_c^2 & \dots & \sigma_u^2 + \sigma_c^2 \end{bmatrix} \quad (9.212)$$

This configuration is called **equicorrelated structure** (or compound symmetry). The presence of σ_c^2 in all off-diagonal elements indicates that the covariance between any two distinct periods is constant and equal to the variance of individual heterogeneity, violating Gauss-Markov sphericity in the longitudinal dimension.

9.4.2.3 Projection Operators and Spectral Decomposition

Definition 9.4.7 (Within and Between Projection Operators). We define the orthogonal projection operators in the space $\mathbb{R}^T$:

$$\mathbf{Q}_T = \mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T' \quad (\text{Within-group operator}) \quad (9.213)$$

$$\mathbf{P}_T = \frac{1}{T} \mathbf{1}_T \mathbf{1}_T' \quad (\text{Between-group operator}) \quad (9.214)$$

Remark 9.4.10 (Geometric Interpretation). • $\mathbf{Q}_T$ projects any vector $\mathbf{z}_i$ onto the subspace orthogonal to $\mathbf{1}_T$, transforming raw values into deviations from the time mean: $\mathbf{Q}_T \mathbf{z}_i = \mathbf{z}_i - \bar{z}_i \mathbf{1}_T \equiv \check{\mathbf{z}}_i$

- $\mathbf{P}_T$ projects $\mathbf{z}_i$ onto the one-dimensional subspace spanned by $\mathbf{1}_T$, compressing the temporal dimension into individual means: $\mathbf{P}_T \mathbf{z}_i = \bar{z}_i \mathbf{1}_T$

Theorem 9.4.6 (Algebraic Properties of $\mathbf{P}_T$ and $\mathbf{Q}_T$). *The operators $\mathbf{P}_T$ and $\mathbf{Q}_T$ are symmetric, idempotent, and mutually orthogonal orthogonal projectors:*

$$\mathbf{P}_T' = \mathbf{P}_T \quad \text{and} \quad \mathbf{Q}_T' = \mathbf{Q}_T \quad (\text{symmetry}) \quad (9.215)$$

$$\mathbf{P}_T^2 = \mathbf{P}_T \quad \text{and} \quad \mathbf{Q}_T^2 = \mathbf{Q}_T \quad (\text{idempotence}) \quad (9.216)$$

$$\mathbf{Q}_T \mathbf{P}_T = \mathbf{P}_T \mathbf{Q}_T = \mathbf{0}_{T \times T} \quad (\text{mutual orthogonality}) \quad (9.217)$$

$$\mathbf{I}_T = \mathbf{P}_T + \mathbf{Q}_T \quad (\text{partition of identity}) \quad (9.218)$$

Proof. Symmetry: $\mathbf{P}_T' = (\frac{1}{T} \mathbf{1}_T \mathbf{1}_T')' = \frac{1}{T} (\mathbf{1}_T')' (\mathbf{1}_T)' = \frac{1}{T} \mathbf{1}_T \mathbf{1}_T' = \mathbf{P}_T$. Since $\mathbf{Q}_T = \mathbf{I}_T - \mathbf{P}_T$, we have $\mathbf{Q}_T' = \mathbf{I}_T' - \mathbf{P}_T' = \mathbf{I}_T - \mathbf{P}_T = \mathbf{Q}_T$.

Idempotence of $\mathbf{P}_T$: Since $\mathbf{1}_T' \mathbf{1}_T = T$:

$$\mathbf{P}_T^2 = \left(\frac{1}{T} \mathbf{1}_T \mathbf{1}_T' \right) \left(\frac{1}{T} \mathbf{1}_T \mathbf{1}_T' \right) = \frac{1}{T^2} \mathbf{1}_T (\mathbf{1}_T' \mathbf{1}_T) \mathbf{1}_T' = \frac{1}{T^2} \mathbf{1}_T (T) \mathbf{1}_T' = \frac{1}{T} \mathbf{1}_T \mathbf{1}_T' = \mathbf{P}_T \quad (9.219)$$

Idempotence of $\mathbf{Q}_T$:

$$\mathbf{Q}_T^2 = (\mathbf{I}_T - \mathbf{P}_T)(\mathbf{I}_T - \mathbf{P}_T) = \mathbf{I}_T - 2\mathbf{P}_T + \mathbf{P}_T^2 = \mathbf{I}_T - 2\mathbf{P}_T + \mathbf{P}_T = \mathbf{I}_T - \mathbf{P}_T = \mathbf{Q}_T \quad (9.220)$$

Mutual orthogonality:

$$\mathbf{Q}_T \mathbf{P}_T = (\mathbf{I}_T - \mathbf{P}_T) \mathbf{P}_T = \mathbf{P}_T - \mathbf{P}_T^2 = \mathbf{P}_T - \mathbf{P}_T = \mathbf{0}_{T \times T} \quad (9.221)$$

Similarly, $\mathbf{P}_T \mathbf{Q}_T = \mathbf{0}_{T \times T}$.

Partition of identity: $\mathbf{P}_T + \mathbf{Q}_T = \frac{1}{T} \mathbf{1}_T \mathbf{1}_T' + \mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T' = \mathbf{I}_T$. □

Theorem 9.4.7 (Spectral Decomposition of Ω_i). *The conditional covariance matrix Ω_i admits the spectral decomposition:*

$$\Omega_i = \lambda_1 \mathbf{P}_T + \lambda_2 \mathbf{Q}_T \quad (9.222)$$

where the eigenvalues are:

$$\lambda_1 = \sigma_u^2 + T\sigma_c^2 \quad (\text{Between eigenvalue, multiplicity } 1) \quad (9.223)$$

$$\lambda_2 = \sigma_u^2 \quad (\text{Within eigenvalue, multiplicity } T - 1) \quad (9.224)$$

Proof. **Step 1: Rewriting Ω_i .**

Using $\mathbf{1}_T \mathbf{1}_T' = T\mathbf{P}_T$ and $\mathbf{I}_T = \mathbf{Q}_T + \mathbf{P}_T$:

$$\begin{aligned} \Omega_i &= \sigma_u^2 \mathbf{I}_T + \sigma_c^2 \mathbf{1}_T \mathbf{1}_T' \\ &= \sigma_u^2 (\mathbf{Q}_T + \mathbf{P}_T) + \sigma_c^2 (T\mathbf{P}_T) \\ &= \sigma_u^2 \mathbf{Q}_T + (\sigma_u^2 + T\sigma_c^2) \mathbf{P}_T \end{aligned} \quad (9.225)$$

Step 2: Identification of eigenvalues.

Between eigenvalue: We evaluate Ω_i on $\mathbf{1}_T$:

$$\Omega_i \mathbf{1}_T = [\sigma_u^2 \mathbf{Q}_T + (\sigma_u^2 + T\sigma_c^2) \mathbf{P}_T] \mathbf{1}_T \quad (9.226)$$

Since $\mathbf{Q}_T \mathbf{1}_T = \mathbf{0}_T$ (orthogonality) and $\mathbf{P}_T \mathbf{1}_T = \frac{1}{T} \mathbf{1}_T (\mathbf{1}_T' \mathbf{1}_T) = \frac{1}{T} \mathbf{1}_T (T) = \mathbf{1}_T$:

$$\Omega_i \mathbf{1}_T = \sigma_u^2 (\mathbf{0}_T) + (\sigma_u^2 + T\sigma_c^2) \mathbf{1}_T = (\sigma_u^2 + T\sigma_c^2) \mathbf{1}_T \quad (9.227)$$

Thus, $\mathbf{1}_T$ is an eigenvector with eigenvalue $\lambda_1 = \sigma_u^2 + T\sigma_c^2$.

Within eigenvalue: Let $\mathbf{w}_j$ be orthogonal to $\mathbf{1}_T$ ($\mathbf{1}_T' \mathbf{w}_j = 0$). Then:

$$\mathbf{P}_T \mathbf{w}_j = \frac{1}{T} \mathbf{1}_T (\mathbf{1}_T' \mathbf{w}_j) = \frac{1}{T} \mathbf{1}_T (0) = \mathbf{0}_T \quad (9.228)$$

$$\mathbf{Q}_T \mathbf{w}_j = (\mathbf{I}_T - \mathbf{P}_T) \mathbf{w}_j = \mathbf{w}_j - \mathbf{0}_T = \mathbf{w}_j \quad (9.229)$$

Therefore:

$$\Omega_i \mathbf{w}_j = \sigma_u^2 \mathbf{w}_j + (\sigma_u^2 + T\sigma_c^2) \mathbf{0}_T = \sigma_u^2 \mathbf{w}_j \quad (9.230)$$

Thus, any $\mathbf{w}_j \perp \mathbf{1}_T$ is an eigenvector with eigenvalue $\lambda_2 = \sigma_u^2$. Since the orthogonal complement of $\mathbf{1}_T$ in $\mathbb{R}^T$ has dimension $T - 1$, the eigenvalue λ_2 has multiplicity $T - 1$. $\square$

Theorem 9.4.8 (Arbitrary Powers of Ω_i). *For any $r \in \mathbb{R}$, the matrix power is:*

$$\Omega_i^r = \lambda_1^r \mathbf{P}_T + \lambda_2^r \mathbf{Q}_T \quad (9.231)$$

Proof. We prove by induction for $m \in \mathbb{N}$.

Base case ($m = 1$): $\Omega_i^1 = \lambda_1 \mathbf{P}_T + \lambda_2 \mathbf{Q}_T$ (true by spectral decomposition).

Inductive step: Assume $\mathbf{\Omega}_i^k = \lambda_1^k \mathbf{P}_T + \lambda_2^k \mathbf{Q}_T$. Then:

$$\begin{aligned}\mathbf{\Omega}_i^{k+1} &= \mathbf{\Omega}_i^k \mathbf{\Omega}_i^k = (\lambda_1^k \mathbf{P}_T + \lambda_2^k \mathbf{Q}_T)(\lambda_1^k \mathbf{P}_T + \lambda_2^k \mathbf{Q}_T) \\ &= \lambda_1^{k+1} \mathbf{P}_T^2 + \lambda_1^k \lambda_2 \mathbf{P}_T \mathbf{Q}_T + \lambda_2^k \lambda_1 \mathbf{Q}_T \mathbf{P}_T + \lambda_2^{k+1} \mathbf{Q}_T^2\end{aligned}$$

By idempotence ($\mathbf{P}_T^2 = \mathbf{P}_T$, $\mathbf{Q}_T^2 = \mathbf{Q}_T$) and mutual orthogonality ($\mathbf{P}_T \mathbf{Q}_T = \mathbf{0}$):

$$\mathbf{\Omega}_i^{k+1} = \lambda_1^{k+1} \mathbf{P}_T + \lambda_2^{k+1} \mathbf{Q}_T \quad (9.232)$$

By induction, valid for all $m \in \mathbb{N}$. By analytic extension, valid for all $r \in \mathbb{R}$. $\square$

Remark 9.4.11 (Applications for GLS). The fundamental powers for estimation are:

$$\mathbf{\Omega}_i^{-1} = \frac{1}{\sigma_u^2 + T\sigma_c^2} \mathbf{P}_T + \frac{1}{\sigma_u^2} \mathbf{Q}_T \quad (9.233)$$

$$\mathbf{\Omega}_i^{-1/2} = \frac{1}{\sqrt{\sigma_u^2 + T\sigma_c^2}} \mathbf{P}_T + \frac{1}{\sqrt{\sigma_u^2}} \mathbf{Q}_T \quad (9.234)$$

The spectral decomposition eliminates the need for numerical inversion of $T \times T$ matrices for each individual, allowing efficient computation of the quasi-demeaning transformation of the RE-FGLS estimator.

9.4.3 Quasi-Demeaning Transformation, GLS, and FGLS

9.4.3.1 Variance Filter and Parameter θ

Theorem 9.4.9 (Variance Filter $\mathbf{\Omega}_i^{-1/2}$). *Applying the Powers Theorem via Spectral Decomposition ($r = -1/2$) to the representation $\mathbf{\Omega}_i = \lambda_1 \mathbf{P}_T + \lambda_2 \mathbf{Q}_T$:*

$$\mathbf{\Omega}_i^{-1/2} = \frac{1}{\sqrt{\sigma_u^2 + T\sigma_c^2}} \mathbf{P}_T + \frac{1}{\sigma_u} \mathbf{Q}_T \quad (9.235)$$

Proof. From the spectral decomposition, $\mathbf{\Omega}_i = \lambda_1 \mathbf{P}_T + \lambda_2 \mathbf{Q}_T$ with $\lambda_1 = \sigma_u^2 + T\sigma_c^2$ and $\lambda_2 = \sigma_u^2$. By the Matrix Powers Theorem:

$$\mathbf{\Omega}_i^{-1/2} = \lambda_1^{-1/2} \mathbf{P}_T + \lambda_2^{-1/2} \mathbf{Q}_T = \frac{1}{\sqrt{\sigma_u^2 + T\sigma_c^2}} \mathbf{P}_T + \frac{1}{\sigma_u} \mathbf{Q}_T \quad (9.236)$$

$\square$

Remark 9.4.12 (Geometric Interpretation). The operator $\mathbf{\Omega}_i^{-1/2}$ acts with weighted scaling on the two orthogonal dimensions of $\mathbb{R}^T$: it divides the Between projection by the standard deviation of the Between error ($\sqrt{\sigma_u^2 + T\sigma_c^2}$) and the Within projection by the standard deviation of the idiosyncratic shock (σ_u), eliminating all serial correlation induced by c_i .

Definition 9.4.8 (Quasi-Demeaning Parameter). We define the scale parameter $\theta \in [0, 1]$:

$$\theta \equiv 1 - \sqrt{\frac{\sigma_u^2}{\sigma_u^2 + T\sigma_c^2}} \quad (9.237)$$

Remark 9.4.13 (Asymptotic Limits of θ). 1. **Absence of heterogeneity** ($\sigma_c^2 = 0$): $\lim_{\sigma_c^2 \rightarrow 0} \theta = 1 - 1 = 0$. The transformation reduces to the identity, and GLS $\equiv$ Pooled OLS.

2. **Dominance of heterogeneity or $T \rightarrow \infty$** : $\lim_{T \rightarrow \infty} \theta = 1 - 0 = 1$. The transformation converges to Within, and GLS $\equiv$ FE.

The parameter θ functions as an optimal calibrator, continuously weighting Within and Between variations.

Theorem 9.4.10 (Compact Matrix Representation). *The variance filter can be factored as:*

$$\mathbf{\Omega}_i^{-1/2} = \frac{1}{\sigma_u} [\mathbf{I}_T - \theta \mathbf{P}_T] \quad (9.238)$$

Proof. Starting from the spectral expression and using $\mathbf{Q}_T = \mathbf{I}_T - \mathbf{P}_T$:

$$\begin{aligned} \mathbf{\Omega}_i^{-1/2} &= \frac{1}{\sqrt{\sigma_u^2 + T\sigma_c^2}} \mathbf{P}_T + \frac{1}{\sigma_u} (\mathbf{I}_T - \mathbf{P}_T) \\ &= \frac{1}{\sigma_u} \mathbf{I}_T + \left(\frac{1}{\sqrt{\sigma_u^2 + T\sigma_c^2}} - \frac{1}{\sigma_u} \right) \mathbf{P}_T \\ &= \frac{1}{\sigma_u} \left[\mathbf{I}_T + \left(\frac{\sigma_u}{\sqrt{\sigma_u^2 + T\sigma_c^2}} - 1 \right) \mathbf{P}_T \right] \\ &= \frac{1}{\sigma_u} [\mathbf{I}_T - \theta \mathbf{P}_T] \end{aligned}$$

where we recognize $\frac{\sigma_u}{\sqrt{\sigma_u^2 + T\sigma_c^2}} = \sqrt{\frac{\sigma_u^2}{\sigma_u^2 + T\sigma_c^2}} = 1 - \theta$. □

9.4.3.2 Scalar Transformation and Sphericity

Definition 9.4.9 (Scalar Quasi-Demeaning). Applying the operator $\sigma_u \mathbf{\Omega}_i^{-1/2} = \mathbf{I}_T - \theta \mathbf{P}_T$ to any variable z_{it} :

$$z_{it}^* = z_{it} - \theta \bar{z}_i \quad (9.239)$$

where $\bar{z}_i = \frac{1}{T} \sum_{t=1}^T z_{it}$ is the individual time mean.

Proof. By definition of the Between operator:

$$\mathbf{z}_i^* = (\mathbf{I}_T - \theta \mathbf{P}_T) \mathbf{z}_i = \mathbf{z}_i - \theta \left(\frac{1}{T} \mathbf{1}_T \mathbf{1}_T' \mathbf{z}_i \right) = \mathbf{z}_i - \theta \bar{z}_i \mathbf{1}_T \quad (9.240)$$

In scalar notation: $z_{it}^* = z_{it} - \theta \bar{z}_i$. □

Theorem 9.4.11 (Sphericity of the Transformed Error). *Under the quasi-demeaning transformation, the transformed compound error $\mathbf{v}_i^* = \mathbf{v}_i - \theta \bar{v}_i \mathbf{1}_T$ satisfies:*

$$\text{Var}(\mathbf{v}_i^* \mid \mathbf{X}_i) = \sigma_u^2 \mathbf{I}_T \quad (9.241)$$

Proof.

$$\begin{aligned}
 \text{Var}(\mathbf{v}_i^* \mid \mathbf{X}_i) &= \text{Var}(\sigma_u \boldsymbol{\Omega}_i^{-1/2} \mathbf{v}_i \mid \mathbf{X}_i) \\
 &= \sigma_u^2 \boldsymbol{\Omega}_i^{-1/2} \text{Var}(\mathbf{v}_i \mid \mathbf{X}_i) \boldsymbol{\Omega}_i^{-1/2} \\
 &= \sigma_u^2 \boldsymbol{\Omega}_i^{-1/2} \boldsymbol{\Omega}_i \boldsymbol{\Omega}_i^{-1/2} \\
 &= \sigma_u^2 \mathbf{I}_T
 \end{aligned}$$

□

Remark 9.4.14 (Transformed Model). Applying the transformation to the structural equation $y_{it} = \mathbf{x}_{it}\boldsymbol{\beta} + c_i + u_{it}$:

$$y_{it} - \theta \bar{y}_i = (\mathbf{x}_{it} - \theta \bar{\mathbf{x}}_i)\boldsymbol{\beta} + \beta_0(1 - \theta) + (v_{it} - \theta \bar{v}_i) \quad (9.242)$$

The intercept transforms into $\beta_0(1 - \theta)$ since $\bar{x}_{i0} = 1$ for the constant.

9.4.3.3 GLS Estimator and Aitken's Theorem (BLUE)

Definition 9.4.10 (Random Effects GLS Estimator). The GLS estimator is:

$$\hat{\boldsymbol{\beta}}_{RE, GLS} = \left(\sum_{i=1}^N \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{y}_i \quad (9.243)$$

Theorem 9.4.12 (Conditional Unbiasedness). *Under RE.1 and RE.2, the GLS estimator is conditionally unbiased:*

$$E[\hat{\boldsymbol{\beta}}_{RE, GLS} \mid \mathbf{X}] = \boldsymbol{\beta} \quad (9.244)$$

with conditional variance:

$$\text{Var}(\hat{\boldsymbol{\beta}}_{RE, GLS} \mid \mathbf{X}) = \left(\sum_{i=1}^N \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i \right)^{-1} \quad (9.245)$$

Proof. Substituting $\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{v}_i$:

$$\hat{\boldsymbol{\beta}}_{RE, GLS} = \boldsymbol{\beta} + \left(\sum_{i=1}^N \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{v}_i \quad (9.246)$$

Applying conditional expectation and treating $\mathbf{X}_i$ and $\boldsymbol{\Omega}_i$ as deterministic under conditioning:

$$E[\hat{\boldsymbol{\beta}}_{RE, GLS} \mid \mathbf{X}] = \boldsymbol{\beta} + \left(\sum_{i=1}^N \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \underbrace{E[\mathbf{v}_i \mid \mathbf{X}_i]}_{=\mathbf{0}_T} = \boldsymbol{\beta} \quad (9.247)$$

The variance follows from the standard weighted OLS formula:

$$\text{Var}(\hat{\boldsymbol{\beta}}_{RE, GLS} \mid \mathbf{X}) = \left(\sum_{i=1}^N \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \underbrace{\text{Var}(\mathbf{v}_i \mid \mathbf{X}_i)}_{=\boldsymbol{\Omega}_i} \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i \right) \left(\sum_{i=1}^N \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i \right)^{-1} \quad (9.248)$$

$$= \left(\sum_{i=1}^N \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i \right)^{-1} \quad (9.249)$$

□

Theorem 9.4.13 (BLUE under RE.1-RE.3: Aitken's Theorem). *The GLS estimator is the Best Linear Unbiased Estimator (BLUE) in the class of conditional linear estimators. For any alternative linear unbiased estimator $\tilde{\beta} = \sum_{i=1}^N \mathbf{A}_i' \mathbf{y}_i$:*

$$\text{Var}(\tilde{\beta} \mid \mathbf{X}) \succeq \text{Var}(\hat{\beta}_{RE, GLS} \mid \mathbf{X}) \quad (9.250)$$

in the positive semidefinite matrix sense.

Proof. For $\tilde{\beta}$ to be unbiased, it is required that $\sum_{i=1}^N \mathbf{A}_i' \mathbf{X}_i = \mathbf{I}_K$. We define the difference matrix:

$$\mathbf{D}_i \equiv \mathbf{A}_i - \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i \left(\sum_{j=1}^N \mathbf{X}_j' \boldsymbol{\Omega}_j^{-1} \mathbf{X}_j \right)^{-1} \quad (9.251)$$

We compute the weighted quadratic form:

$$\begin{aligned} \sum_{i=1}^N \mathbf{D}_i' \boldsymbol{\Omega}_i \mathbf{D}_i &= \sum_{i=1}^N \mathbf{A}_i' \boldsymbol{\Omega}_i \mathbf{A}_i - \left(\sum_{j=1}^N \mathbf{X}_j' \boldsymbol{\Omega}_j^{-1} \mathbf{X}_j \right)^{-1} \underbrace{\left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{A}_i \right)}_{=\mathbf{I}_K} \\ &\quad - \underbrace{\left(\sum_{i=1}^N \mathbf{A}_i' \mathbf{X}_i \right)}_{=\mathbf{I}_K} \left(\sum_{k=1}^N \mathbf{X}_k' \boldsymbol{\Omega}_k^{-1} \mathbf{X}_k \right)^{-1} + \left(\sum_{j=1}^N \mathbf{X}_j' \boldsymbol{\Omega}_j^{-1} \mathbf{X}_j \right)^{-1} \\ &= \text{Var}(\tilde{\beta} \mid \mathbf{X}) - \left(\sum_{i=1}^N \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i \right)^{-1} \\ &= \text{Var}(\tilde{\beta} \mid \mathbf{X}) - \text{Var}(\hat{\beta}_{RE, GLS} \mid \mathbf{X}) \end{aligned}$$

Since $\boldsymbol{\Omega}_i$ is positive definite, $\sum_{i=1}^N \mathbf{D}_i' \boldsymbol{\Omega}_i \mathbf{D}_i \succeq \mathbf{0}$, hence:

$$\text{Var}(\tilde{\beta} \mid \mathbf{X}) \succeq \text{Var}(\hat{\beta}_{RE, GLS} \mid \mathbf{X}) \quad (9.252)$$

□

9.4.3.4 Relative Efficiency: RE vs FE

Theorem 9.4.14 (Efficiency Dominance over FE). *Under RE.1-RE.3, the GLS estimator is asymptotically more efficient than the Within Fixed Effects estimator:*

$$\text{Var}(\hat{\beta}_{FE} \mid \mathbf{X}) \succeq \text{Var}(\hat{\beta}_{RE, GLS} \mid \mathbf{X}) \quad (9.253)$$

Proof. The FE estimator is linear and unbiased (Section 9.3), hence belongs to the class of Aitken's Theorem, implying the inequality immediately.

To quantify the efficiency gain, we compare precisions (inverses of variances). The variance of FE is:

$$\text{Var}(\hat{\beta}_{FE} \mid \mathbf{X}) = \sigma_u^2 \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \quad (9.254)$$

since $\mathbf{Q}_T \boldsymbol{\Omega}_i \mathbf{Q}_T = \sigma_u^2 \mathbf{Q}_T$.

The precision of RE is:

$$\text{Var}(\hat{\boldsymbol{\beta}}_{RE, GLS} | \mathbf{X})^{-1} = \sum_{i=1}^N \mathbf{X}_i' \left[\frac{1}{\sigma_u^2} \mathbf{Q}_T + \frac{1}{\lambda_1} \mathbf{P}_T \right] \mathbf{X}_i \quad (9.255)$$

Subtracting precisions:

$$\text{Var}(\hat{\boldsymbol{\beta}}_{RE, GLS} | \mathbf{X})^{-1} - \text{Var}(\hat{\boldsymbol{\beta}}_{FE} | \mathbf{X})^{-1} = \frac{1}{\lambda_1} \sum_{i=1}^N \mathbf{X}_i' \mathbf{P}_T \mathbf{X}_i$$

Since $\mathbf{P}_T$ is positive semidefinite, the precision difference is non-negative, implying:

$$\text{Var}(\hat{\boldsymbol{\beta}}_{FE} | \mathbf{X}) - \text{Var}(\hat{\boldsymbol{\beta}}_{RE, GLS} | \mathbf{X}) \succeq \mathbf{0} \quad (9.256)$$

□

Remark 9.4.15 (Interpretation of the Efficiency Gain). The superiority of RE stems from incorporating Between variation ($\mathbf{P}_T \mathbf{X}_i$) into estimation. While FE discards all inter-individual variation by pre-multiplying by $\mathbf{Q}_T$, RE optimally weights Within and Between variations based on spectral eigenvalues, recovering additional information.

9.4.3.5 Classical Variance of the GLS Estimator

Theorem 9.4.15 (Conditional Variance of the GLS Estimator). *The conditional variance of the GLS estimator in terms of the parameter θ is:*

$$\text{Var}(\hat{\boldsymbol{\beta}}_{RE, GLS} | \mathbf{X}) = \sigma_u^2 \left[\sum_{i=1}^N \mathbf{X}_i' (\mathbf{I}_T - \theta \mathbf{P}_T) \mathbf{X}_i \right]^{-1} \quad (9.257)$$

Proof. Under the quasi-demeaning transformation, the transformed model is $\mathbf{y}_i^* = \mathbf{X}_i^* \boldsymbol{\beta} + \mathbf{v}_i^*$ with $\text{Var}(\mathbf{v}_i^* | \mathbf{X}_i) = \sigma_u^2 \mathbf{I}_T$. The GLS estimator is OLS on the transformed data:

$$\hat{\boldsymbol{\beta}}_{RE, GLS} = \left(\sum_{i=1}^N \mathbf{X}_i^{*'} \mathbf{X}_i^* \right)^{-1} \sum_{i=1}^N \mathbf{X}_i^{*'} \mathbf{y}_i^* \quad (9.258)$$

The variance follows from the standard OLS formula:

$$\text{Var}(\hat{\boldsymbol{\beta}}_{RE, GLS} | \mathbf{X}) = \sigma_u^2 \left(\sum_{i=1}^N \mathbf{X}_i^{*'} \mathbf{X}_i^* \right)^{-1} \quad (9.259)$$

To maintain correspondence with the quasi-demeaning structure and computational simplification, the compact expression is:

$$\text{Var}(\hat{\boldsymbol{\beta}}_{RE, GLS} | \mathbf{X}) = \sigma_u^2 \left[\sum_{i=1}^N \mathbf{X}_i' (\mathbf{I}_T - \theta \mathbf{P}_T) \mathbf{X}_i \right]^{-1} \quad (9.260)$$

□

9.4.3.6 FGLS Estimation: Variance Components

The GLS estimator requires knowledge of the population parameters σ_u^2 and σ_c^2 , which are unknown in practice. The FGLS (Feasible GLS) estimator replaces these parameters with consistent first-stage estimates.

Definition 9.4.11 (Estimator of σ_u^2 via Within Residuals). The classical Within Gauss-Markov estimator for σ_u^2 is:

$$\hat{\sigma}_u^2 = \frac{SSR_{FE}}{N(T-1) - K} \quad (9.261)$$

where $SSR_{FE} = \sum_{i=1}^N \sum_{t=1}^T \hat{u}_{it}^2$ is the sum of squared residuals from the Within (FE) estimator.

Remark 9.4.16 (Degrees of Freedom Justification). The denominator $N(T-1) - K$ reflects the loss of N degrees of freedom due to estimation of the N individual effects (via demeaning) and an additional K degrees of freedom for the regressors. Under RE.1-RE.3, $\hat{\sigma}_u^2$ is unbiased in finite samples and consistent under $N \rightarrow \infty$ with fixed T .

Definition 9.4.12 (Estimator of σ_w^2 via Between Regression). The variance of the mean compound error $\bar{v}_i = c_i + \bar{u}_i$ is:

$$\sigma_w^2 \equiv \text{Var}(\bar{v}_i | \mathbf{X}_i) = \sigma_c^2 + \frac{\sigma_u^2}{T} \quad (9.262)$$

We estimate σ_w^2 via the Between regression $\bar{y}_i = \bar{\mathbf{x}}_i \boldsymbol{\beta} + \bar{v}_i$:

$$\hat{\sigma}_w^2 = \frac{SSR_{Between}}{N - (K+1)} \quad (9.263)$$

where $SSR_{Between} = \sum_{i=1}^N (\bar{y}_i - \bar{\mathbf{x}}_i \hat{\boldsymbol{\beta}}_{Between})^2$.

Definition 9.4.13 (Estimator of σ_c^2). Isolating σ_c^2 from the relation $\sigma_w^2 = \sigma_c^2 + \sigma_u^2/T$:

$$\hat{\sigma}_c^2 = \hat{\sigma}_w^2 - \frac{\hat{\sigma}_u^2}{T} \quad (9.264)$$

Remark 9.4.17 (Non-Positive Variance Problem). In finite samples, it may occur that $\hat{\sigma}_c^2 < 0$ when $\hat{\sigma}_w^2 < \hat{\sigma}_u^2/T$. This violates the theoretical restriction $\sigma_c^2 \geq 0$. The standard solution is truncation:

$$\hat{\sigma}_{c,\text{trunc}}^2 = \max \left(0, \hat{\sigma}_w^2 - \frac{\hat{\sigma}_u^2}{T} \right) \quad (9.265)$$

Remark 9.4.18 (Consequences of Truncation). 1. If $\sigma_c^2 > 0$, truncation is asymptotically irrelevant since $\lim_{N \rightarrow \infty} P(\hat{\sigma}_c^2 < 0) = 0$.

2. In finite samples, it introduces positive bias: $E[\hat{\sigma}_{c,\text{trunc}}^2] > \sigma_c^2$.

3. If $\hat{\sigma}_{c,\text{trunc}}^2 = 0$, then $\hat{\theta} = 0$ and FGLS $\equiv$ Pooled OLS.

9.4.3.7 Asymptotic Properties of FGLS

Definition 9.4.14 (Random Effects FGLS Estimator). Replacing Ω_i with $\hat{\Omega}_i$ constructed with $\hat{\sigma}_u^2$ and $\hat{\sigma}_c^2$:

$$\hat{\beta}_{RE,FGLS} = \left(\sum_{i=1}^N \mathbf{X}_i' \hat{\Omega}_i^{-1} \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \mathbf{X}_i' \hat{\Omega}_i^{-1} \mathbf{y}_i \quad (9.266)$$

Equivalently, applying OLS to quasi-demeaned data with $\hat{\theta} = 1 - \sqrt{\hat{\sigma}_u^2 / (\hat{\sigma}_u^2 + T\hat{\sigma}_c^2)}$:

$$\hat{\beta}_{RE,FGLS} = \left(\sum_{i=1}^N \mathbf{X}_i^{*'} \mathbf{X}_i^* \right)^{-1} \sum_{i=1}^N \mathbf{X}_i^{*'} \mathbf{y}_i^* \quad (9.267)$$

where $\mathbf{z}_i^* = \mathbf{z}_i - \hat{\theta} \bar{\mathbf{z}}_i \mathbf{1}_T$.

Theorem 9.4.16 (Consistency and Asymptotic Normality of FGLS). *Under RE.1-RE.3, i.i.d. sampling in dimension i , and finite moments, when $N \rightarrow \infty$ with fixed T :*

$$\sqrt{N}(\hat{\beta}_{RE,FGLS} - \beta) \xrightarrow{d} N\left(\mathbf{0}_K, \left(E[\mathbf{X}_i' \Omega_i^{-1} \mathbf{X}_i]\right)^{-1}\right) \quad (9.268)$$

Proof. Step 1: Asymptotic linear representation.

Substituting $\mathbf{y}_i = \mathbf{X}_i \beta + \mathbf{v}_i$ and scaling by $\sqrt{N}$:

$$\sqrt{N}(\hat{\beta}_{RE,FGLS} - \beta) = \left(\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \hat{\Omega}_i^{-1} \mathbf{X}_i \right)^{-1} \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{X}_i' \hat{\Omega}_i^{-1} \mathbf{v}_i \right) \quad (9.269)$$

Step 2: Consistency of variance estimators.

By consistency of $\hat{\sigma}_u^2$ and $\hat{\sigma}_c^2$ (proved via WLLN applied to first-stage residuals):

$$\text{plim}_{N \rightarrow \infty} \hat{\sigma}_u^2 = \sigma_u^2 \quad \text{and} \quad \text{plim}_{N \rightarrow \infty} \hat{\sigma}_c^2 = \sigma_c^2 \quad (9.270)$$

By the Continuous Mapping Theorem:

$$\hat{\Omega}_i^{-1} \xrightarrow{p} \Omega_i^{-1} \quad (9.271)$$

Step 3: Limit of the moment matrix.

By WLLN and Slutsky:

$$\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \hat{\Omega}_i^{-1} \mathbf{X}_i \xrightarrow{p} E[\mathbf{X}_i' \Omega_i^{-1} \mathbf{X}_i] \equiv \mathbf{A} \quad (9.272)$$

Step 4: Asymptotic irrelevance of Ω_i estimation error.

By Slutsky's Theorem, the estimation error of $\hat{\Omega}_i^{-1}$ is asymptotically irrelevant for the first-order distribution:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{X}_i' \hat{\Omega}_i^{-1} \mathbf{v}_i = \frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{X}_i' \Omega_i^{-1} \mathbf{v}_i + o_p(1) \quad (9.273)$$

Step 5: Application of CLT.

Defining $\mathbf{w}_i \equiv \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{v}_i$, we have $E[\mathbf{w}_i] = \mathbf{0}_K$ by strict exogeneity. The variance is:

$$\begin{aligned} \text{Var}(\mathbf{w}_i) &= E[\mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{v}_i \mathbf{v}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i] \\ &= E[\mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \underbrace{E[\mathbf{v}_i \mathbf{v}_i' | \mathbf{X}_i]}_{=\boldsymbol{\Omega}_i} \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i] \\ &= E[\mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i] = \mathbf{A} \end{aligned}$$

By the Lindeberg-Lévy CLT:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{v}_i \xrightarrow{d} N(\mathbf{0}_K, \mathbf{A}) \quad (9.274)$$

Step 6: Combination via Slutsky.

Combining the limits:

$$\sqrt{N}(\hat{\boldsymbol{\beta}}_{RE,FGLS} - \boldsymbol{\beta}) \xrightarrow{d} \mathbf{A}^{-1} \cdot N(\mathbf{0}_K, \mathbf{A}) = N(\mathbf{0}_K, \mathbf{A}^{-1} \mathbf{A} \mathbf{A}^{-1}) = N(\mathbf{0}_K, \mathbf{A}^{-1}) \quad (9.275)$$

□

Theorem 9.4.17 (Estimated Asymptotic Variance Matrix). *The feasible asymptotic variance matrix is:*

$$\widehat{\text{Avar}}(\sqrt{N}(\hat{\boldsymbol{\beta}}_{RE,FGLS} - \boldsymbol{\beta})) = \left(\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \hat{\boldsymbol{\Omega}}_i^{-1} \mathbf{X}_i \right)^{-1} \quad (9.276)$$

In terms of the parameter $\hat{\theta}$:

$$\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}_{RE,FGLS}) = \hat{\sigma}_u^2 \left[\sum_{i=1}^N \mathbf{X}_i' (\mathbf{I}_T - \hat{\theta} \mathbf{P}_T) \mathbf{X}_i \right]^{-1} \quad (9.277)$$

Proof. By consistency of $\hat{\boldsymbol{\Omega}}_i^{-1}$ and WLLN:

$$\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \hat{\boldsymbol{\Omega}}_i^{-1} \mathbf{X}_i \xrightarrow{p} E[\mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i] \quad (9.278)$$

By the Continuous Mapping Theorem (inversion is continuous in the space of non-singular matrices):

$$\left(\frac{1}{N} \sum_{i=1}^N \mathbf{X}_i' \hat{\boldsymbol{\Omega}}_i^{-1} \mathbf{X}_i \right)^{-1} \xrightarrow{p} \left(E[\mathbf{X}_i' \boldsymbol{\Omega}_i^{-1} \mathbf{X}_i] \right)^{-1} \quad (9.279)$$

The expression in terms of $\hat{\theta}$ follows from the compact representation $\hat{\boldsymbol{\Omega}}_i^{-1} = \frac{1}{\hat{\sigma}_u^2} (\mathbf{I}_T - \hat{\theta} \mathbf{P}_T)$. □

Remark 9.4.19 (Practical Inference). The square roots of the diagonal elements of $\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}_{RE,FGLS})$ are the asymptotic standard errors. By asymptotic normality, the statistics $t = (\hat{\beta}_j - \beta_j^0) / \text{se}(\hat{\beta}_j)$ converge to $N(0, 1)$ under H_0 , allowing asymptotically valid hypothesis tests and confidence interval construction.

9.4.4 Intuition and Practical Application

9.4.4.1 When to Use RE

The Random Effects model is appropriate when one assumes that individual heterogeneity c_i is not correlated with the regressors. In this case, c_i can be treated as part of the compound error, but with a specific covariance structure.

RE is useful when:

- one wishes to estimate coefficients of time-invariant variables;
- one believes in orthogonality between c_i and $\mathbf{x}_{it}$;
- one seeks efficiency superior to POLS, exploiting the serial correlation structure induced by c_i .

If $\text{Cov}(\mathbf{x}_{it}, c_i) \neq 0$, RE is inconsistent.

9.4.4.2 Quasi-Demeaning Intuition

Since $v_{it} = c_i + u_{it}$, observations from the same individual are correlated:

$$\text{Cov}(v_{it}, v_{is}) = \text{Var}(c_i), \quad t \neq s. \quad (9.280)$$

POLS ignores this correlation. FE completely eliminates c_i . RE performs an intermediate weighting.

The RE transformation is

$$y_{it}^* = y_{it} - \theta \bar{y}_i, \quad (9.281)$$

and analogously for $\mathbf{x}_{it}$.

The parameter θ is between zero and one:

- if $\theta = 0$, the transformation coincides with POLS;
- if $\theta = 1$, the transformation coincides with FE;
- if $0 < \theta < 1$, RE combines within and between variation.

9.4.5 Specification Tests: Hausman and Mundlak

9.4.5.1 Hausman Test (1978)

Contrast Principle The Hausman test is based on comparing two estimators: the Within Fixed Effects estimator ($\hat{\beta}_{FE}$) and the Random Effects estimator ($\hat{\beta}_{RE}$). We partition the regressor matrix as $\mathbf{X}_i = [\mathbf{X}_{i,1} \ \mathbf{X}_{i,2}]$, where $\mathbf{X}_{i,1}$ contains the K_1 time-varying variables and $\mathbf{X}_{i,2}$ contains the K_2 time-invariant variables. Since $\mathbf{Q}_T \mathbf{X}_{i,2} = \mathbf{0}$, the FE estimator identifies only β_1 .

Definition 9.4.15 (Contrast Vector). We define the contrast vector of time-varying variables:

$$\hat{\mathbf{q}} \equiv \hat{\beta}_{FE} - \hat{\beta}_{RE,1} \quad (9.282)$$

Under the null hypothesis of systemic orthogonality, both estimators are consistent for β_1 , hence:

$$\text{plim}_{N \rightarrow \infty} \hat{\mathbf{q}} = \beta_1 - \beta_1 = \mathbf{0}_{K_1} \quad (9.283)$$

Definition 9.4.16 (Test Hypotheses).

$$H_0 : E[c_i \mid \mathbf{X}_i] = 0 \quad (\text{RE consistent and efficient}) \quad (9.284)$$

$$H_1 : E[c_i \mid \mathbf{X}_i] \neq 0 \quad (\text{RE inconsistent, FE consistent}) \quad (9.285)$$

Under H_0 , the RE estimator achieves the lower variance bound (BLUE). Under H_1 , FE remains consistent since the Within transformation annihilates $c_i \mathbf{1}_T$ regardless of correlation with $\mathbf{X}_i$.

Hausman's Lemma and Covariance Simplification

Lemma 9.4.1 (Hausman's Lemma). *Under H_0 , the covariance between the efficient estimator (RE) and the contrast vector is zero:*

$$\text{Cov}(\hat{\beta}_{RE}, \hat{\beta}_{FE} - \hat{\beta}_{RE}) = \mathbf{0} \quad (9.286)$$

Proof. Under H_0 , we construct a generic alternative linear estimator:

$$\hat{\beta}^* = \hat{\beta}_{RE} + \mathbf{A}\hat{\mathbf{q}} \quad (9.287)$$

where $\mathbf{A} \in \mathbb{R}^{K_1 \times K_1}$ is an arbitrary constant matrix. Since $E[\hat{\mathbf{q}}] = \mathbf{0}$ under H_0 , $\hat{\beta}^*$ is unbiased:

$$E[\hat{\beta}^*] = E[\hat{\beta}_{RE}] + \mathbf{A}E[\hat{\mathbf{q}}] = \beta_1 \quad (9.288)$$

The variance matrix of $\hat{\beta}^*$ is:

$$\text{Var}(\hat{\beta}^*) = \text{Var}(\hat{\beta}_{RE}) + \mathbf{A} \text{Var}(\hat{\mathbf{q}}) \mathbf{A}' + \mathbf{A} \text{Cov}(\hat{\mathbf{q}}, \hat{\beta}_{RE}) + \text{Cov}(\hat{\beta}_{RE}, \hat{\mathbf{q}}) \mathbf{A}' \quad (9.289)$$

Since $\hat{\beta}_{RE}$ is efficient under H_0 , the difference $\mathbf{D} \equiv \text{Var}(\hat{\beta}^*) - \text{Var}(\hat{\beta}_{RE})$ must be positive semidefinite for any $\mathbf{A}$:

$$\mathbf{A} \text{Var}(\hat{\mathbf{q}}) \mathbf{A}' + \mathbf{A} \text{Cov}(\hat{\mathbf{q}}, \hat{\beta}_{RE}) + \text{Cov}(\hat{\beta}_{RE}, \hat{\mathbf{q}}) \mathbf{A}' \succeq \mathbf{0} \quad (9.290)$$

Defining $\mathbf{A} = \gamma \mathbf{C}$ with $\gamma \in \mathbb{R}$ and $\mathbf{C}$ arbitrary:

$$\gamma^2 \mathbf{C} \text{Var}(\hat{\mathbf{q}}) \mathbf{C}' + \gamma [\mathbf{C} \text{Cov}(\hat{\mathbf{q}}, \hat{\beta}_{RE}) + \text{Cov}(\hat{\beta}_{RE}, \hat{\mathbf{q}}) \mathbf{C}'] \succeq \mathbf{0} \quad (9.291)$$

For this quadratic form in γ to be non-negative for all $\gamma \in \mathbb{R}$, the linear coefficient must be zero:

$$\mathbf{C} \text{Cov}(\hat{\mathbf{q}}, \hat{\beta}_{RE}) + \text{Cov}(\hat{\beta}_{RE}, \hat{\mathbf{q}}) \mathbf{C}' = \mathbf{0} \quad (9.292)$$

Since $\mathbf{C}$ is arbitrary, we choose $\mathbf{C} = \text{Cov}(\hat{\beta}_{RE}, \hat{\mathbf{q}})'$:

$$\text{Cov}(\hat{\beta}_{RE}, \hat{\mathbf{q}})' \text{Cov}(\hat{\beta}_{RE}, \hat{\mathbf{q}}) + \text{Cov}(\hat{\beta}_{RE}, \hat{\mathbf{q}}) \text{Cov}(\hat{\beta}_{RE}, \hat{\mathbf{q}})' = \mathbf{0} \quad (9.293)$$

Since both terms are positive semidefinite (outer products), each must be individually zero:

$$\text{Cov}(\hat{\beta}_{RE}, \hat{\mathbf{q}}) = \mathbf{0} \quad (9.294)$$

Substituting $\hat{\mathbf{q}} = \hat{\beta}_{FE} - \hat{\beta}_{RE}$:

$$\text{Cov}(\hat{\beta}_{RE}, \hat{\beta}_{FE} - \hat{\beta}_{RE}) = \mathbf{0} \quad (9.295)$$

□

Theorem 9.4.18 (Hausman's Equality). *Under H_0 , the variance of the contrast vector simplifies to:*

$$\text{Var}(\hat{\mathbf{q}}) = \text{Var}(\hat{\boldsymbol{\beta}}_{FE}) - \text{Var}(\hat{\boldsymbol{\beta}}_{RE,1}) \quad (9.296)$$

Proof. By quadratic linearity of variance:

$$\text{Var}(\hat{\mathbf{q}}) = \text{Var}(\hat{\boldsymbol{\beta}}_{FE}) + \text{Var}(\hat{\boldsymbol{\beta}}_{RE}) - \text{Cov}(\hat{\boldsymbol{\beta}}_{FE}, \hat{\boldsymbol{\beta}}_{RE}) - \text{Cov}(\hat{\boldsymbol{\beta}}_{RE}, \hat{\boldsymbol{\beta}}_{FE}) \quad (9.297)$$

From Hausman's Lemma, $\text{Cov}(\hat{\boldsymbol{\beta}}_{RE}, \hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{RE}) = \mathbf{0}$, hence:

$$\text{Cov}(\hat{\boldsymbol{\beta}}_{RE}, \hat{\boldsymbol{\beta}}_{FE}) - \text{Var}(\hat{\boldsymbol{\beta}}_{RE}) = \mathbf{0} \implies \text{Cov}(\hat{\boldsymbol{\beta}}_{RE}, \hat{\boldsymbol{\beta}}_{FE}) = \text{Var}(\hat{\boldsymbol{\beta}}_{RE}) \quad (9.298)$$

By symmetry of covariance:

$$\text{Cov}(\hat{\boldsymbol{\beta}}_{FE}, \hat{\boldsymbol{\beta}}_{RE}) = [\text{Cov}(\hat{\boldsymbol{\beta}}_{RE}, \hat{\boldsymbol{\beta}}_{FE})]' = \text{Var}(\hat{\boldsymbol{\beta}}_{RE}) \quad (9.299)$$

Substituting:

$$\text{Var}(\hat{\mathbf{q}}) = \text{Var}(\hat{\boldsymbol{\beta}}_{FE}) + \text{Var}(\hat{\boldsymbol{\beta}}_{RE}) - \text{Var}(\hat{\boldsymbol{\beta}}_{RE}) - \text{Var}(\hat{\boldsymbol{\beta}}_{RE}) = \text{Var}(\hat{\boldsymbol{\beta}}_{FE}) - \text{Var}(\hat{\boldsymbol{\beta}}_{RE}) \quad (9.300)$$

□

Wald Test Statistic

Definition 9.4.17 (Hausman Statistic). The test statistic is a Wald quadratic form:

$$\hat{H} = \hat{\mathbf{q}}' [\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}_{FE}) - \widehat{\text{Var}}(\hat{\boldsymbol{\beta}}_{RE,1})]^{-1} \hat{\mathbf{q}} \quad (9.301)$$

Theorem 9.4.19 (Asymptotic Distribution). *Under H_0 and $N \rightarrow \infty$ with fixed T :*

$$\hat{H} \xrightarrow{d} \chi_{K_1}^2 \quad (9.302)$$

where K_1 is the number of time-varying variables tested.

Proof. Under H_0 , by the Central Limit Theorem:

$$\sqrt{N}\hat{\mathbf{q}} \xrightarrow{d} N(\mathbf{0}_{K_1}, \mathbf{V}_q) \quad (9.303)$$

where $\mathbf{V}_q = \text{Avar}(\hat{\boldsymbol{\beta}}_{FE}) - \text{Avar}(\hat{\boldsymbol{\beta}}_{RE,1})$ is positive definite.

By Slutsky's Theorem, if $\widehat{\text{Var}}(\hat{\mathbf{q}})$ is consistent for $\mathbf{V}_q/N$, the quadratic form converges to $\chi_{K_1}^2$. □

Remark 9.4.20 (Hausman Test Pathologies). 1. **Negative statistic in finite sam-**

ples: Since $\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}_{FE})$ and $\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}_{RE,1})$ are estimated separately, the difference matrix may not be positive definite in small samples, generating negative eigenvalues and $\hat{H} < 0$ (mathematical absurdity for a quadratic distance).

2. **Sensitivity to non-spherical errors:** The simplification $\text{Var}(\hat{\mathbf{q}}) = \text{Var}(\hat{\boldsymbol{\beta}}_{FE}) - \text{Var}(\hat{\boldsymbol{\beta}}_{RE})$ requires RE to be efficient under H_0 . Under heteroskedasticity or serial autocorrelation, RE loses efficiency and the equality fails.

3. **Inability to test time-invariant variables:** Since FE annihilates time-invariant variables, the test is restricted to the K_1 time-varying variables.

9.4.5.2 Mundlak's Projection Filter (1978)

Modeling Correlation via Linear Projection

Definition 9.4.18 (Mundlak Projection). Mundlak (1978) proposes modeling the conditional expectation of the individual effect as a linear projection onto the time means of the regressors:

$$E[c_i | \mathbf{X}_i] = \bar{\mathbf{x}}_i \boldsymbol{\gamma} \quad (9.304)$$

where $\bar{\mathbf{x}}_i = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_{it} \in \mathbb{R}^{1 \times K}$ and $\boldsymbol{\gamma} \in \mathbb{R}^{K \times 1}$.

Definition 9.4.19 (Decomposition of the Individual Effect). We decompose c_i into its conditional projection and an orthogonal residual:

$$c_i = \bar{\mathbf{x}}_i \boldsymbol{\gamma} + a_i \quad (9.305)$$

where, by properties of orthogonal projection:

$$E[a_i | \mathbf{X}_i] = 0 \quad (9.306)$$

Definition 9.4.20 (Mundlak's Augmented Model). Substituting into the level equation:

$$y_{it} = \mathbf{x}_{it} \boldsymbol{\beta} + \bar{\mathbf{x}}_i \boldsymbol{\gamma} + a_i + u_{it} \quad (9.307)$$

The term a_i acts as a new individual random effect orthogonal to $\mathbf{X}_i$, and $\bar{\mathbf{x}}_i$ functions as a control function that purifies the compound error of systematic correlation with the regressors.

Algebraic Equivalence with the Within Estimator

Theorem 9.4.20 (RE-Mundlak $\equiv$ FE Equivalence). *FGLS (RE) estimation of Mundlak's augmented model produces $\boldsymbol{\beta}$ estimates algebraically identical to the Within Fixed Effects estimator:*

$$\hat{\boldsymbol{\beta}}_{RE, Mundlak} \equiv \hat{\boldsymbol{\beta}}_{FE} \quad (9.308)$$

Proof. The augmented model in vector form is:

$$\mathbf{y}_i = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{P}_T \mathbf{X}_i \boldsymbol{\gamma} + \mathbf{v}_i \quad (9.309)$$

where $\mathbf{v}_i = a_i \mathbf{1}_T + \mathbf{u}_i$ with $\text{Var}(\mathbf{v}_i | \mathbf{X}_i) = \boldsymbol{\Omega}_i = \sigma_u^2 \mathbf{I}_T + \sigma_a^2 \mathbf{1}_T \mathbf{1}_T'$.

We apply the spectral weighting operator $\boldsymbol{\Omega}_i^{-1/2} = \frac{1}{\sigma_u} \mathbf{Q}_T + \frac{1}{\eta} \mathbf{P}_T$, where $\eta = \sqrt{\sigma_u^2 + T\sigma_a^2}$:

$$\boldsymbol{\Omega}_i^{-1/2} \mathbf{y}_i = \boldsymbol{\Omega}_i^{-1/2} \mathbf{X}_i \boldsymbol{\beta} + \boldsymbol{\Omega}_i^{-1/2} \mathbf{P}_T \mathbf{X}_i \boldsymbol{\gamma} + \boldsymbol{\Omega}_i^{-1/2} \mathbf{v}_i \quad (9.310)$$

We evaluate the transformed terms using $\mathbf{Q}_T \mathbf{P}_T = \mathbf{0}$ and $\mathbf{P}_T^2 = \mathbf{P}_T$:

$$\boldsymbol{\Omega}_i^{-1/2} \mathbf{P}_T \mathbf{X}_i = \left(\frac{1}{\sigma_u} \mathbf{Q}_T + \frac{1}{\eta} \mathbf{P}_T \right) \mathbf{P}_T \mathbf{X}_i = \frac{1}{\eta} \mathbf{P}_T \mathbf{X}_i \quad (9.311)$$

$$\boldsymbol{\Omega}_i^{-1/2} \mathbf{X}_i = \frac{1}{\sigma_u} \mathbf{Q}_T \mathbf{X}_i + \frac{1}{\eta} \mathbf{P}_T \mathbf{X}_i \quad (9.312)$$

Substituting and defining $\gamma^* \equiv \frac{1}{\eta}(\beta + \gamma)$:

$$\Omega_i^{-1/2} \mathbf{y}_i = \underbrace{\frac{1}{\sigma_u} \mathbf{Q}_T \mathbf{X}_i}_{\mathbf{Z}_{i,1}} \beta + \underbrace{\mathbf{P}_T \mathbf{X}_i}_{\mathbf{Z}_{i,2}} \gamma^* + \Omega_i^{-1/2} \mathbf{v}_i \quad (9.313)$$

We verify sample orthogonality between regressor blocks:

$$\sum_{i=1}^N \mathbf{Z}'_{i,1} \mathbf{Z}_{i,2} = \frac{1}{\sigma_u} \sum_{i=1}^N \mathbf{X}'_i \mathbf{Q}_T \mathbf{P}_T \mathbf{X}_i = \mathbf{0}_{K \times K} \quad (9.314)$$

By the Frisch-Waugh-Lovell Theorem, exact orthogonality allows estimating β in isolation:

$$\begin{aligned} \hat{\beta}_{RE, Mundlak} &= \left(\sum_{i=1}^N \mathbf{Z}'_{i,1} \mathbf{Z}_{i,1} \right)^{-1} \sum_{i=1}^N \mathbf{Z}'_{i,1} (\Omega_i^{-1/2} \mathbf{y}_i) \\ &= \left(\frac{1}{\sigma_u^2} \sum_{i=1}^N \mathbf{X}'_i \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \frac{1}{\sigma_u} \mathbf{X}'_i \mathbf{Q}_T \left(\frac{1}{\sigma_u} \mathbf{Q}_T + \frac{1}{\eta} \mathbf{P}_T \right) \mathbf{y}_i \\ &= \left(\frac{1}{\sigma_u^2} \sum_{i=1}^N \mathbf{X}'_i \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \left(\frac{1}{\sigma_u^2} \sum_{i=1}^N \mathbf{X}'_i \mathbf{Q}_T \mathbf{y}_i \right) \\ &= \left(\sum_{i=1}^N \mathbf{X}'_i \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \mathbf{X}'_i \mathbf{Q}_T \mathbf{y}_i \equiv \hat{\beta}_{FE} \end{aligned} \quad (9.315)$$

□

Robust Exogeneity Test via VAT

Theorem 9.4.21 (Mundlak Test as Hausman Substitute). *The exogeneity hypothesis $H_0 : E[c_i | \mathbf{X}_i] = 0$ is equivalent to $H_0 : \gamma = \mathbf{0}$ in the augmented model. The test is a Variable Addition Test (VAT) via the Wald statistic:*

$$W = \hat{\gamma}' [\widehat{\text{Var}}(\hat{\gamma})]^{-1} \hat{\gamma} \xrightarrow{d} \chi^2_{K_1} \quad (9.316)$$

Remark 9.4.21 (Advantages over the Hausman Test). 1. **Native robustness to non-spherical errors:** The matrix $\widehat{\text{Var}}(\hat{\gamma})$ can be computed via the cluster-robust estimator (CRVE) clustered at the i level:

$$\widehat{\text{Var}}_{\text{robust}}(\hat{\gamma}) = \left(\sum_{i=1}^N \mathbf{W}'_i \mathbf{W}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{W}'_i \hat{\mathbf{v}}_i \hat{\mathbf{v}}'_i \mathbf{W}_i \right) \left(\sum_{i=1}^N \mathbf{W}'_i \mathbf{W}_i \right)^{-1} \quad (9.317)$$

where $\mathbf{W}_i = [\mathbf{X}_i \quad \mathbf{1}_T \bar{\mathbf{x}}_i]$.

2. **Immunity to negative eigenvalue pathology:** Since the variance matrix is estimated directly (not by difference), it is guaranteed positive definite in finite samples.
3. **Algebraic equivalence:** Since $\hat{\beta}_{RE, Mundlak} \equiv \hat{\beta}_{FE}$, the Mundlak test is asymptotically equivalent to the Hausman test under homoskedasticity, but superior under heteroskedasticity/autocorrelation.

9.4.5.3 Chamberlain's Extension (1982)

General Projection of the Individual Effect

Definition 9.4.21 (Chamberlain Projection). Chamberlain (1982) generalizes Mundlak by projecting c_i onto **all leads and lags** of the regressors, relaxing the temporal symmetry restriction:

$$c_i = \mathbf{x}_{i1}\gamma_1 + \mathbf{x}_{i2}\gamma_2 + \cdots + \mathbf{x}_{iT}\gamma_T + a_i \quad (9.318)$$

where $\gamma_t \in \mathbb{R}^{K \times 1}$ for each $t \in \{1, \dots, T\}$, and $E[\mathbf{x}'_{it}a_i] = \mathbf{0}_K \ \forall t$.

Remark 9.4.22 (Mundlak vs Chamberlain Comparison). • **Mundlak:** imposes $\gamma_1 = \gamma_2 = \cdots = \gamma_T = \gamma/T$, restricting correlation to the time mean $\bar{\mathbf{x}}_i$.

- **Chamberlain:** allows arbitrary temporal dependence patterns, capturing whether past, present, or future shocks exert differentiated impacts on c_i .

Definition 9.4.22 (Chamberlain Regression System). Substituting into the level equation, we obtain a system of T equations:

$$y_{it} = \mathbf{x}_{it}\beta + \sum_{s=1}^T \mathbf{x}_{is}\gamma_s + e_{it} \quad \forall t \in \{1, \dots, T\} \quad (9.319)$$

where $e_{it} \equiv a_i + u_{it}$ satisfies $E[\mathbf{x}'_{is}e_{it}] = \mathbf{0}_K \ \forall s, t$ under strict exogeneity.

Minimum Distance Estimator (CMD)

Definition 9.4.23 (Unrestricted Reduced Form). We estimate each period equation separately via OLS, obtaining the unrestricted parameter vector $\hat{\pi} \in \mathbb{R}^{T^2K \times 1}$ from the reduced form:

$$\mathbf{y}_i = \mathbf{X}_i^* \pi + \mathbf{e}_i \quad (9.320)$$

where $\mathbf{X}_i^* = \mathbf{I}_T \otimes [\mathbf{x}_{i1} \ \dots \ \mathbf{x}_{iT}] \in \mathbb{R}^{T \times T^2K}$.

Definition 9.4.24 (Structural Restrictions). The structural model imposes $\pi = \mathbf{g}(\theta)$, where $\theta = (\beta', \gamma'_1, \dots, \gamma'_T)' \in \mathbb{R}^{(K+TK) \times 1}$ and $\mathbf{g} : \mathbb{R}^{K+TK} \rightarrow \mathbb{R}^{T^2K}$. For $T \geq 2$, the system is overidentified since $T^2K > K + TK$.

Definition 9.4.25 (Optimal CMD Estimator). Chamberlain proposes minimizing the weighted quadratic distance:

$$\hat{\theta}_{CMD} = \arg \min_{\theta} [\hat{\pi} - \mathbf{g}(\theta)]' \hat{\mathbf{W}}_N [\hat{\pi} - \mathbf{g}(\theta)] \quad (9.321)$$

For optimal asymptotic efficiency, we choose $\hat{\mathbf{W}}_N^* = \hat{\Omega}_N^{-1}$, where $\hat{\Omega}_N \equiv \widehat{\text{Var}}(\hat{\pi})$.

Theorem 9.4.22 (Asymptotic Distribution of CMD). *Under optimal weighting matrix choice:*

$$\sqrt{N}(\hat{\theta}_{CMD} - \theta) \xrightarrow{d} N(\mathbf{0}, (\mathbf{G}'\Omega^{-1}\mathbf{G})^{-1}) \quad (9.322)$$

where $\mathbf{G} \equiv \frac{\partial \mathbf{g}(\theta)}{\partial \theta} \in \mathbb{R}^{T^2K \times (K+TK)}$ is the Jacobian matrix.

Theorem 9.4.23 (Overidentification Test). *The minimum distance statistic evaluated at the optimum:*

$$J = N[\hat{\pi} - \mathbf{g}(\hat{\theta}_{CMD})]' \hat{\Omega}_N^{-1} [\hat{\pi} - \mathbf{g}(\hat{\theta}_{CMD})] \xrightarrow{d} \chi_{df}^2 \quad (9.323)$$

where the degrees of freedom are:

$$df = T^2K - (K + TK) = (T - 1)(1 + TK) - K \quad (9.324)$$

Remark 9.4.23 (Interpretation of the J Test). • Under H_0 of strict exogeneity and correct specification, J follows χ^2_{df} .

- If H_0 is violated (e.g., temporal feedback where shocks to y_{it} correlate with future regressors), the structural restrictions are rejected and J is significantly large.
- Chamberlain's J test is a Sargan-type test that evaluates the global validity of strict exogeneity assumptions, being more general than Hausman/Mundlak since it allows arbitrary temporal dependence patterns.

Table 9.3: Synthesis: Specification Tests in Panel Data

Test	Null Hypothesis	Statistic	Advantages
Hausman	$E[c_i \mathbf{X}_i] = 0$	$\hat{H} = \hat{\mathbf{q}}'[\widehat{\text{Var}}(\hat{\mathbf{q}})]^{-1}\hat{\mathbf{q}} \sim \chi^2_{K_1}$	Classical, intuitive
Mundlak	$\gamma = \mathbf{0}$	$W = \hat{\gamma}'[\widehat{\text{Var}}(\hat{\gamma})]^{-1}\hat{\gamma} \sim \chi^2_{K_1}$	Robust to cluster/heteroskedas
Chamberlain	Global strict exogeneity	$J \sim \chi^2_{df}$	Allows arbitrary temporal depe

9.4.6 Caveats, Common Errors, and Committee Questions

9.4.6.1 Caveats and Common Errors

Using RE only to keep invariant variables If RE.1 fails, the entire vector β may be inconsistent. Estimating an invariant variable at the cost of contaminating the remaining coefficients is a serious error.

Misspecification of the Ω matrix RE is GLS based on a specific covariance structure. If there is additional heteroskedasticity or autocorrelation in u_{it} , RE may lose efficiency.

Symptom: RE standard errors larger or unstable compared to POLS or FE.

Ignoring robust inference Even under RE, it is recommended to use cluster-robust standard errors to protect inference against residual violations.

Negative variance of c_i In small samples, $\hat{\sigma}_c^2$ may be negative. If the software imposes zero, RE collapses to POLS.

9.4.6.2 Typical Committee Question

Question: You use RE because the Hausman test did not reject, but N is small. How do you ensure this is not lack of power?

Answer: Non-rejection of Hausman does not prove $E[c_i | \mathbf{X}_i] = 0$. In small samples, the test may have low power.

The defense should include:

- Mundlak test with time means;
- Coefficient comparison between RE and FE;
- Institutional or experimental justification for exogeneity;

- Cluster-robust standard errors;
- Sensitivity analysis with alternative specifications.

If the orthogonality hypothesis is implausible, FE should be preferred, even if it sacrifices estimation of invariant variables.

Table 9.4: Final Synthesis: Quasi-Demeaning Transformation, GLS, and FGLS

Concept	Mathematical Expression	Eq.
Spectral Filter	$\Omega_i^{-1/2} = \frac{1}{\sqrt{\sigma_u^2 + T\sigma_c^2}} \mathbf{P}_T + \frac{1}{\sigma_u} \mathbf{Q}_T$	(42)
Parameter θ	$\theta = 1 - \sqrt{\frac{\sigma_u^2}{\sigma_u^2 + T\sigma_c^2}}$	(44)
Compact Representation	$\Omega_i^{-1/2} = \frac{1}{\sigma_u} [\mathbf{I}_T - \theta \mathbf{P}_T]$	(48)
Scalar Transformation	$z_{it}^* = z_{it} - \theta \bar{z}_i$	(50)
Transformed Sphericity	$\text{Var}(\mathbf{v}_i^* \mathbf{X}_i) = \sigma_u^2 \mathbf{I}_T$	(51)
GLS Variance	$\text{Var}(\hat{\beta}_{RE, GLS} \mathbf{X}) = \left(\sum_{i=1}^N \mathbf{X}_i' \Omega_i^{-1} \mathbf{X}_i \right)^{-1}$	(55)
BLUE (Aitken)	$\text{Var}(\hat{\beta} \mathbf{X}) \succeq \text{Var}(\hat{\beta}_{RE, GLS} \mathbf{X})$	(57)
Dominance over FE	$\text{Var}(\hat{\beta}_{FE} \mathbf{X}) \succeq \text{Var}(\hat{\beta}_{RE, GLS} \mathbf{X})$	(59)
Precision Gain	$\text{Var}(\hat{\beta}_{RE, GLS})^{-1} - \text{Var}(\hat{\beta}_{FE})^{-1} = \frac{1}{\lambda_1} \sum_{i=1}^N \mathbf{X}_i' \mathbf{P}_T \mathbf{X}_i$	(61)
Classical Variance	$\text{Var}(\hat{\beta}_{RE, GLS} \mathbf{X}) = \sigma_u^2 \left[\sum_{i=1}^N \mathbf{X}_i' (\mathbf{I}_T - \theta \mathbf{P}_T) \mathbf{X}_i \right]^{-1}$	(63)
$\hat{\sigma}_u^2$ Estimator	$\hat{\sigma}_u^2 = \frac{SSR_{FE}}{N(T-1)-K}$	(64)
$\hat{\sigma}_w^2$ Estimator	$\hat{\sigma}_w^2 = \frac{SSR_{Between}}{N-(K+1)}$	(66)
$\hat{\sigma}_c^2$ Estimator	$\hat{\sigma}_c^2 = \hat{\sigma}_w^2 - \frac{\hat{\sigma}_u^2}{T}$	(67)
Truncation	$\hat{\sigma}_{c, \text{trunc}}^2 = \max\left(0, \hat{\sigma}_w^2 - \frac{\hat{\sigma}_u^2}{T}\right)$	(68)
FGLS Asymptotic Normality	$\sqrt{N}(\hat{\beta}_{RE, FGLS} - \beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{A}^{-1})$	(71)
Estimated FGLS Variance	$\widehat{\text{Var}}(\hat{\beta}_{RE, FGLS}) = \hat{\sigma}_u^2 \left[\sum_{i=1}^N \mathbf{X}_i' (\mathbf{I}_T - \hat{\theta} \mathbf{P}_T) \mathbf{X}_i \right]^{-1}$	(73)

- Remark 9.4.24* (Final Summary for the Qualification Committee). 1. The **quasi-demeaning transformation** $\mathbf{z}_i^* = \mathbf{z}_i - \theta \bar{z}_i \mathbf{1}_T$ with $\theta = 1 - \sqrt{\sigma_u^2 / (\sigma_u^2 + T\sigma_c^2)}$ restores sphericity: $\text{Var}(\mathbf{v}_i^* | \mathbf{X}_i) = \sigma_u^2 \mathbf{I}_T$.
2. The **parameter** θ optimally calibrates Within/Between weighting: $\theta \rightarrow 0$ when $\sigma_c^2 = 0$ (GLS $\equiv$ Pooled OLS); $\theta \rightarrow 1$ when $\sigma_c^2 \rightarrow \infty$ or $T \rightarrow \infty$ (GLS $\equiv$ FE).
3. The **GLS estimator** $\hat{\beta}_{RE, GLS} = (\sum \mathbf{X}_i' \Omega_i^{-1} \mathbf{X}_i)^{-1} \sum \mathbf{X}_i' \Omega_i^{-1} \mathbf{y}_i$ is BLUE by Aitken's Theorem, dominating FE with precision gain $\frac{1}{\lambda_1} \sum \mathbf{X}_i' \mathbf{P}_T \mathbf{X}_i$.
4. The **FGLS estimator** replaces Ω_i with $\hat{\Omega}_i$ constructed with $\hat{\sigma}_u^2$ (via Within residuals) and $\hat{\sigma}_c^2$ (via difference $\hat{\sigma}_w^2 - \hat{\sigma}_u^2/T$).
5. Under $N \rightarrow \infty$ with fixed T , FGLS is **asymptotically equivalent to GLS**: $\sqrt{N}(\hat{\beta}_{RE, FGLS} - \beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{A}^{-1})$, with $\mathbf{A} = E[\mathbf{X}_i' \Omega_i^{-1} \mathbf{X}_i]$.
6. The **estimated variance matrix** $\hat{\sigma}_u^2 [\sum \mathbf{X}_i' (\mathbf{I}_T - \hat{\theta} \mathbf{P}_T) \mathbf{X}_i]^{-1}$ is consistent, allowing asymptotically valid inference (t -tests, F -tests, confidence intervals).
7. The **negative variance problem** ($\hat{\sigma}_c^2 < 0$) is circumvented via truncation $\max(0, \cdot)$, which is asymptotically irrelevant if $\sigma_c^2 > 0$.

8. The **Hausman test** compares $\hat{\beta}_{FE}$ and $\hat{\beta}_{RE,1}$ via the contrast vector $\hat{\mathbf{q}}$. Under H_0 , Hausman's Lemma proves that $\text{Cov}(\hat{\beta}_{RE}, \hat{\mathbf{q}}) = \mathbf{0}$, simplifying $\text{Var}(\hat{\mathbf{q}}) = \text{Var}(\hat{\beta}_{FE}) - \text{Var}(\hat{\beta}_{RE})$.
9. The **Mundlak filter** models $E[c_i | \mathbf{X}_i] = \bar{\mathbf{x}}_i \boldsymbol{\gamma}$, decomposing $c_i = \bar{\mathbf{x}}_i \boldsymbol{\gamma} + a_i$. The augmented model allows testing $H_0 : \boldsymbol{\gamma} = \mathbf{0}$ via cluster-robust VAT, with algebraic equivalence $\hat{\beta}_{RE, \text{Mundlak}} \equiv \hat{\beta}_{FE}$.
10. The **Chamberlain extension** projects c_i onto all leads/lags, relaxing Mundlak's temporal symmetry. The CMD estimator imposes structural restrictions efficiently, and the overidentification J test evaluates global strict exogeneity validity.

9.5 First Differences (FD)

9.5.1 Research Problem and Intuition

The First Differences estimator eliminates c_i by subtracting the observation from the previous period. Identification then depends on changes between consecutive periods.

The original model is

$$y_{it} = \mathbf{x}'_{it} \boldsymbol{\beta} + c_i + u_{it}. \quad (9.325)$$

For t and $t - 1$,

$$y_{i,t-1} = \mathbf{x}'_{i,t-1} \boldsymbol{\beta} + c_i + u_{i,t-1}. \quad (9.326)$$

Subtracting the second equation from the first,

$$y_{it} - y_{i,t-1} = (\mathbf{x}_{it} - \mathbf{x}_{i,t-1})' \boldsymbol{\beta} + (u_{it} - u_{i,t-1}), \quad (9.327)$$

since $c_i - c_i = 0$.

Defining the operator $\Delta z_{it} = z_{it} - z_{i,t-1}$,

$$\Delta y_{it} = \Delta \mathbf{x}'_{it} \boldsymbol{\beta} + \Delta u_{it}, \quad t = 2, \dots, T. \quad (9.328)$$

FD answers the question: what happens to the variation in y when the variation in $\mathbf{x}$ occurs?

9.5.2 Operator Algebra and FD Geometry

9.5.2.1 The Temporal Differentiation Operator (Δ)

Definition 9.5.1 (Scalar and Polynomial Representation). The first difference operator is defined scalarly by:

$$\Delta z_{it} \equiv z_{it} - z_{i,t-1} \quad (9.329)$$

Defining the Lag operator L such that $L z_{it} = z_{i,t-1}$, we have the polynomial representation:

$$\Delta \equiv 1 - L \quad (9.330)$$

Higher-order differentiations d are represented by the binomial expansion:

$$\Delta^d = (1 - L)^d = \sum_{j=0}^d \binom{d}{j} (-1)^j L^j \quad (9.331)$$

9.5.2.2 Matrix Construction of the Operator $\mathbf{D}_T$

Definition 9.5.2 (Linear Transformation Matrix). To stack the $T - 1$ differences of individual i , we define the matrix $\mathbf{D}_T \in \mathbb{R}^{(T-1) \times T}$:

$$\mathbf{D}_T \equiv \begin{bmatrix} -1 & 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & -1 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & -1 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & -1 & 1 \end{bmatrix} \quad (9.332)$$

Application to the vector system $\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + c_i\mathbf{1}_T + \mathbf{u}_i$ produces:

$$\mathbf{D}_T\mathbf{y}_i = \mathbf{D}_T\mathbf{X}_i\boldsymbol{\beta} + c_i(\mathbf{D}_T\mathbf{1}_T) + \mathbf{D}_T\mathbf{u}_i \quad (9.333)$$

Theorem 9.5.1 (Annihilation of the Individual Effect). *The vector of ones belongs to the null space of $\mathbf{D}_T$:*

$$\mathbf{D}_T\mathbf{1}_T = \mathbf{0}_{T-1} \quad (9.334)$$

Proof. Computing the matrix product:

$$\mathbf{D}_T\mathbf{1}_T = \begin{bmatrix} -1(1) + 1(1) \\ -1(1) + 1(1) \\ \vdots \\ -1(1) + 1(1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \mathbf{0}_{T-1} \quad (9.335)$$

Therefore:

$$\mathbf{D}_T(c_i\mathbf{1}_T) = c_i(\mathbf{D}_T\mathbf{1}_T) = c_i\mathbf{0}_{T-1} = \mathbf{0}_{T-1} \quad (9.336)$$

The purified system is:

$$\Delta\mathbf{y}_i = \Delta\mathbf{X}_i\boldsymbol{\beta} + \Delta\mathbf{u}_i \quad (9.337)$$

where $\Delta\mathbf{y}_i \equiv \mathbf{D}_T\mathbf{y}_i$, $\Delta\mathbf{X}_i \equiv \mathbf{D}_T\mathbf{X}_i$, $\Delta\mathbf{u}_i \equiv \mathbf{D}_T\mathbf{u}_i$. $\square$

9.5.2.3 Algebraic Properties of $\mathbf{D}_T$

Theorem 9.5.2 (Rank and Null Space). *The matrix $\mathbf{D}_T$ satisfies:*

$$\text{rank}(\mathbf{D}_T) = T - 1 \quad (9.338)$$

$$\dim(\text{Ker}(\mathbf{D}_T)) = 1 \quad (9.339)$$

$$\text{Ker}(\mathbf{D}_T) = \text{span}(\mathbf{1}_T) \quad (9.340)$$

Proof. The $T - 1$ rows of $\mathbf{D}_T$ are linearly independent by construction, hence $\text{rank}(\mathbf{D}_T) = T - 1$.

By the Rank-Nullity Theorem:

$$\dim(\text{Ker}(\mathbf{D}_T)) + \text{rank}(\mathbf{D}_T) = T \implies \dim(\text{Ker}(\mathbf{D}_T)) = T - (T - 1) = 1 \quad (9.341)$$

Since $\mathbf{D}_T\mathbf{1}_T = \mathbf{0}_{T-1}$ and the null space is one-dimensional, $\{\mathbf{1}_T\}$ forms a basis of $\text{Ker}(\mathbf{D}_T)$. $\square$

Remark 9.5.1 (Oblique vs Orthogonal Projection). Unlike the Within operator $\mathbf{Q}_T = \mathbf{I}_T - \frac{1}{T}\mathbf{1}_T\mathbf{1}_T'$, which is a symmetric and idempotent orthogonal projector ($\mathbf{Q}_T' = \mathbf{Q}_T$, $\mathbf{Q}_T^2 = \mathbf{Q}_T$), the matrix $\mathbf{D}_T$ performs a **non-orthogonal oblique projection** that reduces the dimension from T to $T - 1$. This projection eliminates the need to estimate the N individual intercepts c_i , circumventing the incidental parameters problem.

The matrix $\mathbf{D}_T$ is neither symmetric ($\mathbf{D}_T' \neq \mathbf{D}_T$) nor idempotent ($\mathbf{D}_T^2 \neq \mathbf{D}_T$), since it maps $\mathbb{R}^T \rightarrow \mathbb{R}^{T-1}$. However, the matrix $\mathbf{M}_T \equiv \mathbf{D}_T'(\mathbf{D}_T\mathbf{D}_T')^{-1}\mathbf{D}_T \in \mathbb{R}^{T \times T}$ is symmetric, idempotent, and coincides exactly with the Within operator:

$$\mathbf{M}_T = \mathbf{Q}_T \quad (9.342)$$

This identity reveals that the Within estimator is algebraically equivalent to GLS applied to the first-differenced model under spherical errors.

9.5.2.4 The Problem of Identifying Invariant Parameters

Theorem 9.5.3 (Null Space Lemma of $\mathbf{D}_T$). *Let $\mathbf{z}_i = [z_{i1}, \dots, z_{iT}]'$ be a vector of a strictly time-invariant variable ($z_{it} = \bar{z}_i$ for all t). Then:*

$$\mathbf{D}_T\mathbf{z}_i = \mathbf{0}_{T-1} \quad (9.343)$$

Proof. Since $z_{it} = \bar{z}_i$ for all t , we have $\mathbf{z}_i = \bar{z}_i\mathbf{1}_T$. By linearity:

$$\mathbf{D}_T\mathbf{z}_i = \mathbf{D}_T(\bar{z}_i\mathbf{1}_T) = \bar{z}_i(\mathbf{D}_T\mathbf{1}_T) = \bar{z}_i\mathbf{0}_{T-1} = \mathbf{0}_{T-1} \quad (9.344)$$

□

Remark 9.5.2 (Implications for Stable Individual Characteristics). The mechanical annihilation proved above implies that stable individual characteristics — such as gender, race, geographic heritage, or schooling completed prior to the panel — **cannot be identified** by the FD estimator.

If we partition the regressor matrix as $\mathbf{X}_i = [\mathbf{X}_{i,1} \quad \mathbf{1}_T\mathbf{w}_i']$, where $\mathbf{X}_{i,1}$ contains time-varying variables and $\mathbf{w}_i$ contains K_2 fixed attributes, the differenced moment matrix assumes the form:

$$\sum_{i=1}^N \Delta\mathbf{X}_i' \Delta\mathbf{X}_i = \begin{bmatrix} \sum_{i=1}^N \Delta\mathbf{X}_{i,1}' \Delta\mathbf{X}_{i,1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0}_{K_2 \times K_2} \end{bmatrix} \quad (9.345)$$

The block of zeros implies that $\text{rank}(\sum_{i=1}^N \Delta\mathbf{X}_i' \Delta\mathbf{X}_i) \leq K_1 < K$, characterizing perfect collinearity induced by the transformation. The matrix is singular and $\boldsymbol{\gamma}$ (coefficients of invariant variables) is algebraically non-identifiable.

Remark 9.5.3 (Alternative Strategies: Hausman-Taylor). To circumvent the identification collapse while still controlling for correlation between c_i and $\mathbf{x}_{it}$, Hausman and Taylor (1981) proposed a multi-stage instrumental variables estimator. The model is partitioned into four groups:

$$y_{it} = \mathbf{x}_{1,it}\boldsymbol{\beta}_1 + \mathbf{x}_{2,it}\boldsymbol{\beta}_2 + \mathbf{w}_{1,i}\boldsymbol{\gamma}_1 + \mathbf{w}_{2,i}\boldsymbol{\gamma}_2 + c_i + u_{it} \quad (9.346)$$

where $\mathbf{x}_{1,it}$ and $\mathbf{w}_{1,i}$ are exogenous (orthogonal to c_i), while $\mathbf{x}_{2,it}$ and $\mathbf{w}_{2,i}$ are endogenous (correlated with c_i).

Identification of γ_2 uses time means $\bar{\mathbf{x}}_{1,i}$ as instruments for $\mathbf{w}_{2,i}$. The order condition requires:

$$K_1 \geq G_2 \quad (9.347)$$

If $K_1 = G_2$, the system is exactly identified; if $K_1 > G_2$, it is overidentified, allowing overidentification tests (Hansen's J statistic).

9.5.3 Assumptions and Their Roles

Definition 9.5.3 (FD.1: Additive and Constant Individual Effect).

$$v_{it} = c_i + u_{it}, \quad c_i \text{ constant over time} \quad (9.348)$$

Role: guarantee that first differencing eliminates c_i .

Counterexample: if the unobserved characteristic changes abruptly between $t - 1$ and t , then $\Delta c_i \neq 0$, and differencing does not remove heterogeneity.

Definition 9.5.4 (FD.2: Independent Sampling in the Cross-Sectional Dimension). Units $i = 1, \dots, N$ are independent of each other, or sufficiently weakly dependent to allow the law of large numbers.

Role: allow consistency when $N \rightarrow \infty$ and T is fixed.

Counterexample: strong spatial dependence between units may invalidate the estimated variance and, in extreme cases, consistency itself.

Definition 9.5.5 (FD.3: Full Rank of Differenced Variation).

$$\text{rank} \left(\mathbb{E} \left[\sum_{t=2}^T \Delta \mathbf{x}_{it} \Delta \mathbf{x}_{it}' \right] \right) = K \quad (9.349)$$

Role: guarantee identification.

Translation to practice: regressors must change over time.

Counterexample: time-invariant variables have $\Delta z_{it} = 0$ and are not identified.

Definition 9.5.6 (FD.4: Strict Exogeneity).

$$\mathbb{E}[u_{it} \mid \mathbf{X}_i, c_i] = 0, \quad t = 1, \dots, T \quad (9.350)$$

Role: guarantee

$$\mathbb{E}[\Delta \mathbf{x}_{it} \Delta u_{it}] = \mathbf{0} \quad (9.351)$$

Counterexample: in dynamic models, $\Delta y_{i,t-1}$ is correlated with Δu_{it} by construction.

Theorem 9.5.4 (Orthogonality of Differenced Regressors). *Under FD.4, for all $t \geq 2$:*

$$\mathbb{E}[\Delta \mathbf{x}_{it}' \Delta u_{it}] = \mathbf{0} \quad (9.352)$$

Proof. Expanding the product:

$$\begin{aligned}\mathbb{E}[\Delta \mathbf{x}'_{it} \Delta u_{it}] &= \mathbb{E}[(\mathbf{x}_{it} - \mathbf{x}_{i,t-1})'(u_{it} - u_{i,t-1})] \\ &= \mathbb{E}[\mathbf{x}'_{it} u_{it}] - \mathbb{E}[\mathbf{x}'_{it} u_{i,t-1}] - \mathbb{E}[\mathbf{x}'_{i,t-1} u_{it}] + \mathbb{E}[\mathbf{x}'_{i,t-1} u_{i,t-1}]\end{aligned}\quad (9.353)$$

By strict exogeneity (FD.4), $\mathbb{E}[\mathbf{x}'_{is} u_{it}] = \mathbf{0}$ for all s, t . Hence all four terms are zero:

$$\mathbb{E}[\Delta \mathbf{x}'_{it} \Delta u_{it}] = \mathbf{0} - \mathbf{0} - \mathbf{0} + \mathbf{0} = \mathbf{0} \quad (9.354)$$

□

Remark 9.5.4 (Failure Under Dynamic Feedback). If there is contemporaneous feedback ($\mathbb{E}[\mathbf{x}'_{it} u_{i,t-1}] \neq \mathbf{0}$), then:

$$\mathbb{E}[\Delta \mathbf{x}'_{it} \Delta u_{it}] = -\mathbb{E}[\mathbf{x}'_{it} u_{i,t-1}] \neq \mathbf{0} \quad (9.355)$$

The FD estimator becomes inconsistent. The inconsistency is:

$$\text{plim}_{N \rightarrow \infty} \hat{\beta}_{FD} = \beta - (\mathbb{E}[\Delta \mathbf{X}'_i \Delta \mathbf{X}_i])^{-1} \sum_{t=2}^T \mathbb{E}[\mathbf{x}'_{it} u_{i,t-1}] \neq \beta \quad (9.356)$$

Definition 9.5.7 (FD.5: Error Structure for Efficiency). For FD to be efficient by Gauss-Markov, the differenced error Δu_{it} must be spherical.

Implication: this occurs if u_{it} follows a random walk:

$$u_{it} = u_{i,t-1} + e_{it} \quad (9.357)$$

since

$$\Delta u_{it} = e_{it} \quad (9.358)$$

If u_{it} is white noise, FD introduces negative autocorrelation in Δu_{it} .

9.5.4 Derivation of the FD Estimator

The differenced model stacked by individual is

$$\Delta \mathbf{y}_i = \Delta \mathbf{X}_i \beta + \Delta \mathbf{u}_i \quad (9.359)$$

where $\Delta \mathbf{y}_i$ and $\Delta \mathbf{X}_i$ have $T - 1$ rows.

The FD estimator minimizes

$$S_{FD}(\beta) = \sum_{i=1}^N (\Delta \mathbf{y}_i - \Delta \mathbf{X}_i \beta)' (\Delta \mathbf{y}_i - \Delta \mathbf{X}_i \beta) \quad (9.360)$$

The first-order condition is

$$\sum_{i=1}^N \Delta \mathbf{X}'_i (\Delta \mathbf{y}_i - \Delta \mathbf{X}_i \beta) = \mathbf{0} \quad (9.361)$$

Hence, under FD.3,

$$\hat{\beta}_{FD} = \left(\sum_{i=1}^N \Delta \mathbf{X}'_i \Delta \mathbf{X}_i \right)^{-1} \left(\sum_{i=1}^N \Delta \mathbf{X}'_i \Delta \mathbf{y}_i \right) \quad (9.362)$$

Substituting $\Delta \mathbf{y}_i = \Delta \mathbf{X}_i \beta + \Delta \mathbf{u}_i$,

$$\hat{\beta}_{FD} = \beta + \left(\sum_{i=1}^N \Delta \mathbf{X}'_i \Delta \mathbf{X}_i \right)^{-1} \left(\sum_{i=1}^N \Delta \mathbf{X}'_i \Delta \mathbf{u}_i \right) \quad (9.363)$$

9.5.5 First-Order Moment Properties

9.5.5.1 Unbiasedness

Theorem 9.5.5 (Conditional Unbiasedness). *Under FD.4,*

$$\mathbb{E}[\hat{\beta}_{FD} \mid \Delta \mathbf{X}] = \beta \quad (9.364)$$

Proof. From the sample error equation:

$$\hat{\beta}_{FD} = \beta + \left(\sum_{i=1}^N \Delta \mathbf{X}_i' \Delta \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \Delta \mathbf{X}_i' \Delta \mathbf{u}_i \quad (9.365)$$

Applying conditional expectation:

$$\mathbb{E}[\hat{\beta}_{FD} \mid \Delta \mathbf{X}] = \beta + \left(\sum_{i=1}^N \Delta \mathbf{X}_i' \Delta \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \Delta \mathbf{X}_i' \mathbb{E}[\Delta \mathbf{u}_i \mid \Delta \mathbf{X}] \quad (9.366)$$

Under FD.4, $\mathbb{E}[\mathbf{u}_i \mid \mathbf{X}_i, c_i] = \mathbf{0}_T$, hence:

$$\mathbb{E}[\Delta \mathbf{u}_i \mid \Delta \mathbf{X}] = \mathbb{E}[\mathbf{D}_T \mathbf{u}_i \mid \mathbf{X}_i, c_i] = \mathbf{D}_T \mathbb{E}[\mathbf{u}_i \mid \mathbf{X}_i, c_i] = \mathbf{D}_T \mathbf{0}_T = \mathbf{0}_{T-1} \quad (9.367)$$

Therefore:

$$\mathbb{E}[\hat{\beta}_{FD} \mid \Delta \mathbf{X}] = \beta + \mathbf{0} = \beta \quad (9.368)$$

□

9.5.5.2 Asymptotic Consistency

Theorem 9.5.6 (Consistency Under $N \rightarrow \infty$). *Under FD.2-FD.4,*

$$\text{plim}_{N \rightarrow \infty} \hat{\beta}_{FD} = \beta \quad (9.369)$$

Proof. Normalizing by N :

$$\hat{\beta}_{FD} - \beta = \left(\frac{1}{N} \sum_{i=1}^N \Delta \mathbf{X}_i' \Delta \mathbf{X}_i \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \Delta \mathbf{X}_i' \Delta \mathbf{u}_i \right) \quad (9.370)$$

By Kolmogorov's WLLN (i.i.d. sampling in i):

$$\frac{1}{N} \sum_{i=1}^N \Delta \mathbf{X}_i' \Delta \mathbf{X}_i \xrightarrow{p} \mathbb{E}[\Delta \mathbf{X}_i' \Delta \mathbf{X}_i] \quad (9.371)$$

$$\frac{1}{N} \sum_{i=1}^N \Delta \mathbf{X}_i' \Delta \mathbf{u}_i \xrightarrow{p} \mathbb{E}[\Delta \mathbf{X}_i' \Delta \mathbf{u}_i] \quad (9.372)$$

By Slutsky's Theorem:

$$\text{plim}_{N \rightarrow \infty} \hat{\beta}_{FD} = \beta + (\mathbb{E}[\Delta \mathbf{X}_i' \Delta \mathbf{X}_i])^{-1} \mathbb{E}[\Delta \mathbf{X}_i' \Delta \mathbf{u}_i] \quad (9.373)$$

By the orthogonality proved earlier, $\mathbb{E}[\Delta \mathbf{X}_i' \Delta \mathbf{u}_i] = \mathbf{0}$. By the rank condition (FD.3), the inverse matrix exists. Hence:

$$\text{plim}_{N \rightarrow \infty} \hat{\beta}_{FD} = \beta + \mathbf{0} = \beta \quad (9.374)$$

□

9.5.5.3 Asymptotic Normality

Theorem 9.5.7 (Limiting Distribution). *Under regularity conditions,*

$$\sqrt{N}(\hat{\beta}_{FD} - \beta) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{A}_{FD}^{-1} \mathbf{B}_{FD} \mathbf{A}_{FD}^{-1}) \quad (9.375)$$

where

$$\mathbf{A}_{FD} = \mathbb{E}[\Delta \mathbf{X}_i' \Delta \mathbf{X}_i] \quad (9.376)$$

$$\mathbf{B}_{FD} = \mathbb{E}[\Delta \mathbf{X}_i' \Delta \mathbf{u}_i \Delta \mathbf{u}_i' \Delta \mathbf{X}_i] \quad (9.377)$$

9.5.6 Second-Order Anatomy and Non-Sphericity

9.5.6.1 Mechanically Induced Serial Autocorrelation

Theorem 9.5.8 (MA(1) Structure of the Differenced Error). *If u_{it} is white noise with $\text{Var}(u_{it} | \mathbf{X}_i, c_i) = \sigma_u^2$ and $\text{Cov}(u_{it}, u_{is} | \mathbf{X}_i, c_i) = 0$ for $t \neq s$, then:*

$$\text{Var}(\Delta u_{it} | \mathbf{X}_i, c_i) = 2\sigma_u^2 \quad (9.378)$$

$$\text{Cov}(\Delta u_{it}, \Delta u_{i,t-1} | \mathbf{X}_i, c_i) = -\sigma_u^2 \quad (9.379)$$

$$\text{Corr}(\Delta u_{it}, \Delta u_{i,t-1} | \mathbf{X}_i, c_i) = -0.5 \quad (9.380)$$

$$\text{Cov}(\Delta u_{it}, \Delta u_{i,t-s} | \mathbf{X}_i, c_i) = 0 \quad \forall s \geq 2 \quad (9.381)$$

Proof. **Variance:**

$$\begin{aligned} \text{Var}(\Delta u_{it} | \mathbf{X}_i, c_i) &= \text{Var}(u_{it} - u_{i,t-1} | \mathbf{X}_i, c_i) \\ &= \text{Var}(u_{it}) + \text{Var}(u_{i,t-1}) - 2 \text{Cov}(u_{it}, u_{i,t-1}) \\ &= \sigma_u^2 + \sigma_u^2 - 0 = 2\sigma_u^2 \end{aligned} \quad (9.382)$$

First-order covariance:

$$\begin{aligned} \text{Cov}(\Delta u_{it}, \Delta u_{i,t-1} | \mathbf{X}_i, c_i) &= \mathbb{E}[(u_{it} - u_{i,t-1})(u_{i,t-1} - u_{i,t-2}) | \mathbf{X}_i, c_i] \\ &= \mathbb{E}[u_{it}u_{i,t-1}] - \mathbb{E}[u_{it}u_{i,t-2}] - \mathbb{E}[u_{i,t-1}^2] + \mathbb{E}[u_{i,t-1}u_{i,t-2}] \\ &= 0 - 0 - \sigma_u^2 + 0 = -\sigma_u^2 \end{aligned} \quad (9.383)$$

Correlation:

$$\text{Corr}(\Delta u_{it}, \Delta u_{i,t-1}) = \frac{-\sigma_u^2}{\sqrt{(2\sigma_u^2)(2\sigma_u^2)}} = \frac{-\sigma_u^2}{2\sigma_u^2} = -0.5 \quad (9.384)$$

Absence of correlation for $s \geq 2$:

$$\begin{aligned} \text{Cov}(\Delta u_{it}, \Delta u_{i,t-s}) &= \mathbb{E}[(u_{it} - u_{i,t-1})(u_{i,t-s} - u_{i,t-s-1})] \\ &= \mathbb{E}[u_{it}u_{i,t-s}] - \mathbb{E}[u_{it}u_{i,t-s-1}] - \mathbb{E}[u_{i,t-1}u_{i,t-s}] + \mathbb{E}[u_{i,t-1}u_{i,t-s-1}] \end{aligned} \quad (9.385)$$

For $s \geq 2$, all four terms involve distinct periods, hence all are zero by serial independence of u_{it} . $\square$

Remark 9.5.5 (Interpretation as MA(1) Process). The differenced error Δu_{it} follows a first-order moving average process MA(1) with parameter $\theta = 1$. The autocorrelation of -0.5 is **mechanical**, not economic: it results from the sharing of the shock $u_{i,t-1}$ between Δu_{it} (with negative sign) and $\Delta u_{i,t-1}$ (with positive sign).

This non-sphericity induced by the differencing operator implies that applying Pooled OLS to the first-differenced model results in efficiency loss relative to a procedure that properly treats the error correlation structure (such as GLS).

9.5.6.2 Matrix Formulation of Variance-Covariance

Theorem 9.5.9 (Tridiagonal Covariance Matrix). *The variance-covariance matrix of the differenced error is:*

$$\mathbf{\Omega}_{\Delta \mathbf{u}} \equiv \text{Var}(\Delta \mathbf{u}_i \mid \mathbf{X}_i) = \sigma_u^2 \mathbf{D}_T \mathbf{D}_T' \quad (9.386)$$

which assumes the Toeplitz tridiagonal form:

$$\mathbf{\Omega}_{\Delta \mathbf{u}} = \sigma_u^2 \begin{bmatrix} 2 & -1 & 0 & 0 & \dots & 0 \\ -1 & 2 & -1 & 0 & \dots & 0 \\ 0 & -1 & 2 & -1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 2 \end{bmatrix} \quad (9.387)$$

Proof. By the linear transformation $\Delta \mathbf{u}_i = \mathbf{D}_T \mathbf{u}_i$:

$$\text{Var}(\Delta \mathbf{u}_i \mid \mathbf{X}_i) = \mathbf{D}_T \text{Var}(\mathbf{u}_i \mid \mathbf{X}_i) \mathbf{D}_T' = \mathbf{D}_T (\sigma_u^2 \mathbf{I}_T) \mathbf{D}_T' = \sigma_u^2 \mathbf{D}_T \mathbf{D}_T' \quad (9.388)$$

The product $\mathbf{D}_T \mathbf{D}_T'$ has elements:

$$[\mathbf{D}_T \mathbf{D}_T']_{t,t} = (-1)^2 + 1^2 = 2 \quad (\text{main diagonal}) \quad (9.389)$$

$$[\mathbf{D}_T \mathbf{D}_T']_{t,t-1} = [\mathbf{D}_T \mathbf{D}_T']_{t-1,t} = (-1)(1) = -1 \quad (\text{adjacent diagonals}) \quad (9.390)$$

$$[\mathbf{D}_T \mathbf{D}_T']_{t,s} = 0 \quad \text{for } |t - s| \geq 2 \quad (9.391)$$

The matrix $\mathbf{\Omega}_{\Delta \mathbf{u}}$ is strictly symmetric and positive definite, since $\mathbf{D}_T$ has full row rank $T - 1$, guaranteeing that the quadratic form $\mathbf{a}' \mathbf{\Omega}_{\Delta \mathbf{u}} \mathbf{a} = \sigma_u^2 \|\mathbf{D}_T' \mathbf{a}\|^2 > 0$ for any $\mathbf{a} \neq \mathbf{0}_{T-1}$. $\square$

9.5.6.3 Corrected Estimation of Residual Variance

Definition 9.5.8 (Estimator of σ_u^2 with Degrees of Freedom Correction).

$$\hat{\sigma}_u^2 = \frac{\sum_{i=1}^N \hat{\Delta \mathbf{u}}_i' \hat{\Delta \mathbf{u}}_i}{2(N(T-1) - K)} \quad (9.392)$$

where $\hat{\Delta \mathbf{u}}_i = \Delta \mathbf{y}_i - \Delta \mathbf{X}_i \hat{\boldsymbol{\beta}}_{FD}$ are the FD residuals.

Theorem 9.5.10 (Unbiasedness of $\hat{\sigma}_u^2$). *Under FD.1-FD.5,*

$$\mathbb{E}[\hat{\sigma}_u^2 \mid \mathbf{X}] = \sigma_u^2 \quad (9.393)$$

Proof. Defining the projection matrix $\mathbf{M}_{\Delta\mathbf{X}} = \mathbf{I}_{N(T-1)} - \Delta\mathbf{X}(\Delta\mathbf{X}'\Delta\mathbf{X})^{-1}\Delta\mathbf{X}'$, the residuals are:

$$\hat{\Delta\mathbf{u}} = \mathbf{M}_{\Delta\mathbf{X}}\Delta\mathbf{u} \quad (9.394)$$

The sum of squared residuals is:

$$SSR_{FD} = \hat{\Delta\mathbf{u}}' \hat{\Delta\mathbf{u}} = \Delta\mathbf{u}' \mathbf{M}_{\Delta\mathbf{X}} \Delta\mathbf{u} \quad (9.395)$$

Applying conditional expectation and using the cyclic property of the trace:

$$\begin{aligned} \mathbb{E}[SSR_{FD} \mid \mathbf{X}] &= \mathbb{E}[\text{tr}(\mathbf{M}_{\Delta\mathbf{X}}\Delta\mathbf{u}\Delta\mathbf{u}') \mid \mathbf{X}] \\ &= \text{tr}(\mathbf{M}_{\Delta\mathbf{X}}\mathbb{E}[\Delta\mathbf{u}\Delta\mathbf{u}' \mid \mathbf{X}]) \end{aligned} \quad (9.396)$$

Under FD.5, $\mathbb{E}[\Delta\mathbf{u}\Delta\mathbf{u}' \mid \mathbf{X}] = 2\sigma_u^2\mathbf{I}_{N(T-1)}$:

$$\mathbb{E}[SSR_{FD} \mid \mathbf{X}] = 2\sigma_u^2 \text{tr}(\mathbf{M}_{\Delta\mathbf{X}}) \quad (9.397)$$

Since $\mathbf{M}_{\Delta\mathbf{X}}$ is an idempotent projector, $\text{tr}(\mathbf{M}_{\Delta\mathbf{X}}) = \text{rank}(\mathbf{M}_{\Delta\mathbf{X}}) = N(T-1) - K$:

$$\mathbb{E}[SSR_{FD} \mid \mathbf{X}] = 2\sigma_u^2[N(T-1) - K] \quad (9.398)$$

Therefore:

$$\mathbb{E}[\hat{\sigma}_u^2 \mid \mathbf{X}] = \frac{2\sigma_u^2[N(T-1) - K]}{2[N(T-1) - K]} = \sigma_u^2 \quad (9.399)$$

□

Remark 9.5.6 (Role of the Factor 2). The denominator contains the factor 2 because $\text{Var}(\Delta u_{it}) = 2\sigma_u^2$. Without this correction, the estimator would overestimate the variance of the original shock by a factor of 2. The denominator $N(T-1) - K$ reflects:

- Loss of the first period per individual: $T \rightarrow T-1$
- Loss of K degrees of freedom from estimating the regressors

9.5.7 Asymptotic Efficiency Comparison: FE vs FD

9.5.7.1 Scenario 1: Spherical Idiosyncratic Shocks

Theorem 9.5.11 (FE Dominance Under i.i.d. u_{it}). *If $\text{Var}(\mathbf{u}_i \mid \mathbf{X}_i, c_i) = \sigma_u^2\mathbf{I}_T$, then:*

$$\text{Avar}(\sqrt{N}(\hat{\beta}_{FD} - \beta)) - \text{Avar}(\sqrt{N}(\hat{\beta}_{FE} - \beta)) \succeq \mathbf{0} \quad (9.400)$$

in the positive semidefinite matrix sense.

Proof. The FE estimator is equivalent to GLS applied to the differenced model. Defining $\mathbf{M}_T \equiv \mathbf{D}_T'(\mathbf{D}_T\mathbf{D}_T')^{-1}\mathbf{D}_T$, we prove that:

$$\mathbf{M}_T = \mathbf{Q}_T \quad (9.401)$$

The FD-GLS estimator is:

$$\begin{aligned}
 \hat{\beta}_{FD, GLS} &= \left(\sum_{i=1}^N \Delta \mathbf{X}_i' (\sigma_u^2 \mathbf{D}_T \mathbf{D}_T')^{-1} \Delta \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \Delta \mathbf{X}_i' (\sigma_u^2 \mathbf{D}_T \mathbf{D}_T')^{-1} \Delta \mathbf{y}_i \\
 &= \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{D}_T' (\mathbf{D}_T \mathbf{D}_T')^{-1} \mathbf{D}_T \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \mathbf{X}_i' \mathbf{D}_T' (\mathbf{D}_T \mathbf{D}_T')^{-1} \mathbf{D}_T \mathbf{y}_i \\
 &= \left(\sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{X}_i \right)^{-1} \sum_{i=1}^N \mathbf{X}_i' \mathbf{Q}_T \mathbf{y}_i \equiv \hat{\beta}_{FE}
 \end{aligned} \tag{9.402}$$

By the Gauss-Markov Theorem, GLS is BLUE, hence dominates Pooled OLS (FD) in the class of linear unbiased estimators:

$$\text{Var}(\hat{\beta}_{FD}) - \text{Var}(\hat{\beta}_{FE}) \succeq \mathbf{0} \tag{9.403}$$

□

Remark 9.5.7 (Special Case $T = 2$). For $T = 2$, the Within and FD transformations are algebraically identical, hence $\hat{\beta}_{FE} = \hat{\beta}_{FD}$ and the variances coincide. For $T > 2$, the inequality is strict (positive definite), implying that FE standard errors are strictly smaller than those of FD.

9.5.7.2 Scenario 2: Shocks with Unit Root (Random Walk)

Theorem 9.5.12 (FD Dominance Under $u_{it} = u_{i,t-1} + e_{it}$). *If u_{it} follows a random walk with e_{it} i.i.d., then:*

$$\text{Var}(\hat{\beta}_{FE} \mid \mathbf{X}) - \text{Var}(\hat{\beta}_{FD} \mid \mathbf{X}) \succeq \mathbf{0} \tag{9.404}$$

Proof. Under random walk, $\Delta u_{it} = e_{it}$, hence:

$$\text{Var}(\Delta \mathbf{u}_i \mid \mathbf{X}_i) = \sigma_e^2 \mathbf{I}_{T-1} \tag{9.405}$$

The differenced error is spherical, so Pooled OLS (FD) is BLUE by Gauss-Markov:

$$\text{Var}(\hat{\beta}_{FD} \mid \mathbf{X}) = \sigma_e^2 \left(\sum_{i=1}^N \Delta \mathbf{X}_i' \Delta \mathbf{X}_i \right)^{-1} \tag{9.406}$$

The FE estimator becomes inefficient since the Within transformation does not remove the unit root persistence. Under random walk, $\text{Var}(u_{it} \mid \mathbf{X}_i, c_i) = t\sigma_e^2$, violating sphericity. The Within transformation produces demeaned errors with a complex covariance structure, while first differencing completely annihilates the unit root, reinstating Gauss-Markov for FD. □

9.5.7.3 General Scenario: AR(1) Process

For $u_{it} = \rho u_{i,t-1} + e_{it}$ with $0 < \rho < 1$:

- If $\rho \rightarrow 0$ (white noise): FE is more efficient
- If $\rho \rightarrow 1$ (random walk): FD is more efficient

- There exists a cutoff point ρ^* that dictates the dominance transition

For intermediate ρ , relative efficiency depends on T , ρ , and the regressor structure. In general:

- For small T and ρ close to 1, FD tends to be more efficient
- For large T and ρ close to 0, FE tends to be more efficient

9.5.7.4 Wooldridge Test for Error Structure

Definition 9.5.9 (Test Procedure). To test the autocorrelation structure of errors in levels:

1. Estimate the FD model by Pooled OLS and obtain residuals $\hat{e}_{it} = \Delta y_{it} - \Delta \mathbf{x}'_{it} \hat{\beta}_{FD}$
2. Regress $\hat{e}_{it}$ on $\hat{e}_{i,t-1}$ (without intercept):

$$\hat{e}_{it} = \rho \hat{e}_{i,t-1} + r_{it} \quad (9.407)$$

3. Test $H_0 : \rho = -0.5$ vs $H_1 : \rho \neq -0.5$

Remark 9.5.8 (Interpretation). • If $\hat{\rho} \approx -0.5$: errors in levels are i.i.d., FE is preferable

- If $\hat{\rho} > -0.5$: errors in levels have positive autocorrelation, FD or FE with correction may be preferable
- If $\hat{\rho} \approx 0$: errors in levels follow a random walk, FD is BLUE

Important: use cluster-robust standard errors at the i level for the auxiliary regression, since residuals $\hat{e}_{it}$ have mechanical autocorrelation under H_0 . The t -statistic is:

$$t_{\rho} = \frac{\hat{\rho} - (-0.5)}{se(\hat{\rho})} = \frac{\hat{\rho} + 0.5}{se(\hat{\rho})} \xrightarrow{d} N(0, 1) \quad (9.408)$$

9.5.8 Practical Comparison: FE vs FD

9.5.8.1 Case $T = 2$

For $T = 2$, FE and FD are numerically identical, provided they are specified with the same relevant temporal regressors.

In FE,

$$y_{it} - \bar{y}_i \quad (9.409)$$

produces variation around the individual mean. With two periods, this transformation is proportional to the first difference:

$$y_{i2} - \frac{y_{i1} + y_{i2}}{2} = \frac{y_{i2} - y_{i1}}{2} = \frac{\Delta y_{i2}}{2} \quad (9.410)$$

9.5.8.2 Efficiency for $T \geq 3$

The choice depends on the serial structure of u_{it} :

- if u_{it} is white noise, FE is more efficient;
- if u_{it} is a random walk, FD is more efficient.

9.5.8.3 Unbalanced Panels

FE uses all available observations from individuals with at least two periods. FD requires observations in consecutive periods. Therefore, in panels with irregular gaps, FE tends to preserve more observations.

9.5.8.4 Dynamic Models

In models with $y_{i,t-1}$, both FE and FD are inconsistent under fixed T . However, FE bias is frequently of order $O(1/T)$, while FD may have more severe bias under certain violations of strict exogeneity.

The usual solution is to use instruments, as in Anderson-Hsiao or Arellano-Bond.

9.5.8.5 Measurement Error

FD tends to exacerbate attenuation when regressors are persistent and measured with error. Differencing reduces the signal, but preserves or amplifies the noise.

If $\mathbf{x}_{it}$ is persistent, $\Delta\mathbf{x}_{it}$ has low signal variance, while the differenced measurement error may have high variance. This reduces the signal-to-noise ratio and pushes the coefficient toward zero (attenuation bias).

9.5.8.6 Non-Stationarity

In long panels with $I(1)$ variables, FD transforms integrated variables into stationary variables, reducing risks of spurious regression.

9.5.9 Caveats, Common Errors, and Committee Questions

9.5.9.1 Caveats and Common Errors

Loss of observations FD discards the first period of each unit. In short panels, this may substantially reduce the sample.

Invariant variables Like FE, FD does not identify time-invariant variables.

Mechanical autocorrelation If u_{it} is white noise, the differenced error follows an MA(1) process with negative autocorrelation of $-1/2$. Conventional standard errors are invalid.

Solution: use cluster-robust or appropriate GLS.

Feedback and dynamics If $\mathbf{x}_{it}$ reacts to $u_{i,t-1}$, then

$$\mathbb{E}[\Delta\mathbf{x}_{it}\Delta u_{it}] \neq 0 \quad (9.411)$$

FD becomes inconsistent.

9.5.9.2 Typical Committee Question

Question: Your regressor is persistent and possibly measured with error. Why isn't FD just producing attenuation?

Answer: The risk exists. If $\mathbf{x}_{it}$ is persistent, $\Delta \mathbf{x}_{it}$ has low signal variance, while the differenced measurement error may have high variance. This reduces the signal-to-noise ratio and pushes the coefficient toward zero.

To defend the specification, one should:

- compare FD with long differences, e.g., $y_{it} - y_{i,t-k}$;
- compare FE and FD;
- use instruments when there is dynamic endogeneity;
- report cluster-robust standard errors.

Table 9.5: Final Synthesis: First Differences Estimator

Concept	Mathematical Expression	Eq.
Difference Operator	$\Delta z_{it} = z_{it} - z_{i,t-1}$	(4)
Matrix $\mathbf{D}_T$	$\mathbf{D}_T \in \mathbb{R}^{(T-1) \times T}$	(6)
Annihilation of c_i	$\mathbf{D}_T \mathbf{1}_T = \mathbf{0}_{T-1}$	(8)
Rank of $\mathbf{D}_T$	$\text{rank}(\mathbf{D}_T) = T - 1$	(12)
Oblique Projection	$\mathbf{M}_T = \mathbf{D}_T' (\mathbf{D}_T \mathbf{D}_T')^{-1} \mathbf{D}_T = \mathbf{Q}_T$	—
FD Estimator	$\hat{\beta}_{FD} = (\sum \Delta \mathbf{X}_i' \Delta \mathbf{X}_i)^{-1} \sum \Delta \mathbf{X}_i' \Delta \mathbf{y}_i$	(20)
Unbiasedness	$\mathbb{E}[\hat{\beta}_{FD} \mid \Delta \mathbf{X}] = \beta$	(22)
Consistency	$\text{plim } \hat{\beta}_{FD} = \beta$	(24)
Differenced Error Variance	$\text{Var}(\Delta u_{it}) = 2\sigma_u^2$	(30)
Induced Autocorrelation	$\text{Corr}(\Delta u_{it}, \Delta u_{i,t-1}) = -0.5$	(32)
Covariance Matrix	$\boldsymbol{\Omega}_{\Delta \mathbf{u}} = \sigma_u^2 \mathbf{D}_T \mathbf{D}_T'$	(34)
Estimator of σ_u^2	$\hat{\sigma}_u^2 = \frac{SSR_{FD}}{2(N(T-1)-K)}$	(36)
FE Dominance (i.i.d.)	$\text{Var}(\hat{\beta}_{FD}) \succeq \text{Var}(\hat{\beta}_{FE})$	(40)
FD Dominance (RW)	$\text{Var}(\hat{\beta}_{FE}) \succeq \text{Var}(\hat{\beta}_{FD})$	(42)

Remark 9.5.9 (Final Summary for the Qualification Committee). 1. The **FD estimator** eliminates c_i via temporal differencing Δ , exploiting only within-individual variation between consecutive periods.

2. **Operator $\mathbf{D}_T$ algebra:** $(T-1) \times T$ matrix with $\text{rank}(\mathbf{D}_T) = T-1$ and $\text{Ker}(\mathbf{D}_T) = \text{span}(\mathbf{1}_T)$. Performs oblique (non-orthogonal) projection that reduces dimension from T to $T-1$. The matrix $\mathbf{M}_T = \mathbf{D}_T' (\mathbf{D}_T \mathbf{D}_T')^{-1} \mathbf{D}_T$ coincides with the Within operator $\mathbf{Q}_T$.
3. **Identification problem:** time-invariant variables belong to the null space of $\mathbf{D}_T$ and are not identified. To estimate effects of invariant variables under correlation with c_i , use Hausman-Taylor with internal instruments.
4. **Critical hypotheses:** FD.1 (constant individual effect) guarantees elimination of c_i ; FD.3 (full rank) guarantees identification; FD.4 (strict exogeneity) guarantees consistency; FD.5 (error structure) determines relative efficiency.

5. **Mechanical autocorrelation:** if u_{it} is i.i.d., then Δu_{it} follows MA(1) with $\text{Corr}(\Delta u_{it}, \Delta u_{i,t-1}) = -0.5$. This is non-sphericity induced by the operator, not an economic property. The covariance matrix is Toeplitz tridiagonal: $\mathbf{\Omega}_{\Delta \mathbf{u}} = \sigma_u^2 \mathbf{D}_T \mathbf{D}_T'$.
6. **Variance estimation:** the correct denominator is $2(N(T-1)-K)$, not $N(T-1)-K$, reflecting that $\text{Var}(\Delta u_{it}) = 2\sigma_u^2$.
7. **FE vs FD comparison:** under i.i.d. u_{it} , FE is more efficient (equivalent to FD-GLS); under random walk u_{it} , FD is more efficient (BLUE); for $T = 2$, they are identical. For $T > 2$, the choice depends on the serial structure of u_{it} .
8. **Wooldridge test:** auxiliary regression of $\hat{e}_{it}$ on $\hat{e}_{i,t-1}$ tests $H_0 : \rho = -0.5$. Rejection in favor of $\hat{\rho} > -0.5$ indicates positive autocorrelation in level errors. Use cluster-robust standard errors in the auxiliary regression.
9. **FD limitations:** (i) loses first observation; (ii) does not identify temporal invariants; (iii) exacerbates measurement error in persistent variables (attenuation bias); (iv) inconsistent under dynamic feedback.

9.6 Dynamic Fixed Effects: Arellano-Bond (Panel GMM)

9.6.1 Dynamic Panel Specification and Nickell Bias

9.6.1.1 AR(1) Autoregressive Model with Individual Effects

Definition 9.6.1 (Dynamic Model in Levels). The first-order autoregressive model with individual effects is:

$$y_{it} = \rho y_{i,t-1} + c_i + u_{it}, \quad |\rho| < 1 \quad (9.412)$$

where y_{it} is the dependent variable, $y_{i,t-1}$ is the autoregressive regressor, c_i is the individual fixed effect, u_{it} is the idiosyncratic shock, and ρ is the stationary autoregressive parameter.

Remark 9.6.1 (Breakdown of Strict Exogeneity). By recursive substitution, $y_{i,t-1}$ depends on c_i :

$$y_{i,t-1} = \rho^{t-1} y_{i0} + c_i \sum_{s=0}^{t-2} \rho^s + \sum_{s=0}^{t-2} \rho^s u_{i,t-1-s} \quad (9.413)$$

Hence $\text{Cov}(y_{i,t-1}, c_i) \neq 0$, invalidating Pooled OLS and Within in levels.

9.6.1.2 Nickell Bias in the Within Estimator

Theorem 9.6.1 (Within Inconsistency Under Dynamics). *Under i.i.d. shocks with variance σ_u^2 , the asymptotic bias of the Within estimator is:*

$$\text{plim}_{N \rightarrow \infty} (\hat{\rho}_{FE} - \rho) = -\frac{1+\rho}{T-1} \left[1 - \frac{2\rho}{(1-\rho)(T-1)} (1-\rho^{T-1}) \right] \left[1 - \frac{2\rho}{(1-\rho)(T-1)} + \frac{2\rho^2(1-\rho^T)}{(1-\rho^2)(1-\rho)(T-1)} \right] \quad (9.414)$$

For moderately large T , this simplifies to:

$$\text{plim}_{N \rightarrow \infty} (\hat{\rho}_{FE} - \rho) \approx -\frac{1+\rho}{T-1} \quad (9.415)$$

Proof. The Within estimator is $\hat{\rho}_{FE} = \rho + (\sum_{i=1}^N \ddot{\mathbf{y}}'_{i,-1} \ddot{\mathbf{y}}_{i,-1})^{-1} \sum_{i=1}^N \ddot{\mathbf{y}}'_{i,-1} \ddot{\mathbf{u}}_i$. Applying $\text{plim}_{N \rightarrow \infty}$:

$$\text{plim}_{N \rightarrow \infty} (\hat{\rho}_{FE} - \rho) = \frac{\sum_{t=2}^T E[\ddot{y}_{i,t-1} \ddot{u}_{it}]}{\sum_{t=2}^T E[\ddot{y}_{i,t-1}^2]} \quad (9.416)$$

The numerator contains the dominant term $-E[y_{i,t-1} \bar{u}_i]$:

$$-E[y_{i,t-1} \bar{u}_i] = -\frac{1}{T} \sum_{s=1}^{t-1} \text{Cov}(y_{i,t-1}, u_{is}) = -\frac{\sigma_u^2}{T} \sum_{s=1}^{t-1} \rho^{t-1-s} = -\frac{\sigma_u^2}{T} \left(\frac{1 - \rho^{t-1}}{1 - \rho} \right) \quad (9.417)$$

Integrating over t and normalizing by the variance of the demeaned regressor, Nickell (1981) obtains the exact expression above. The bias is strictly negative for $\rho > 0$ and vanishes only when $T \rightarrow \infty$. $\square$

Remark 9.6.2 (FD-OLS Inconsistency). Applying first differences: $\Delta y_{it} = \rho \Delta y_{i,t-1} + \Delta u_{it}$. The FD-OLS estimator is inconsistent because:

$$\text{Cov}(\Delta y_{i,t-1}, \Delta u_{it}) = \text{Cov}(y_{i,t-1} - y_{i,t-2}, u_{it} - u_{i,t-1}) = -\sigma_u^2 \neq 0 \quad (9.418)$$

Unlike Within, this inconsistency **persists** even when $T \rightarrow \infty$.

9.6.2 Orthogonality Conditions and Instruments

9.6.2.1 Sequential Exogeneity and Instrument Space

Definition 9.6.2 (Conditional Sequential Exogeneity). The identification hypothesis is:

$$E[u_{it} \mid y_{i,t-1}, y_{i,t-2}, \dots, y_{i1}, c_i] = 0 \quad \forall t \quad (9.419)$$

That is, contemporaneous shocks are orthogonal to the entire past history of the dependent variable, conditional on c_i .

Theorem 9.6.2 (Valid Moment Conditions). *After differencing* ($\Delta y_{it} = \rho \Delta y_{i,t-1} + \Delta u_{it}$), for $t \geq 3$ and $s \geq 2$:

$$E[y_{i,t-s} \Delta u_{it}] = E[y_{i,t-s} (u_{it} - u_{i,t-1})] = 0 \quad (9.420)$$

The valid instruments in levels are $y_{i,t-2}, y_{i,t-3}, \dots, y_{i1}$.

Proof. Under sequential exogeneity:

- $E[y_{i,t-s} u_{it}] = 0$ for $s \geq 2$ (past lags orthogonal to contemporaneous shock)
- $E[y_{i,t-s} u_{i,t-1}] = 0$ for $s \geq 2$ (lags up to $t-2$ are predetermined relative to $u_{i,t-1}$)

Therefore $E[y_{i,t-s} \Delta u_{it}] = 0 - 0 = 0$. $\square$

Remark 9.6.3 (Quadratic Growth of Instruments). The total number of moment conditions is:

$$g = \sum_{t=3}^T (t-2) = \frac{(T-2)(T-1)}{2} \quad (9.421)$$

For $T = 10$, we have $g = 36$ instruments (massive overidentification).

9.6.2.2 Arellano-Bond Instrument Matrix

Definition 9.6.3 (Block-Diagonal Instrument Matrix). For individual i :

$$\mathbf{Z}_i \equiv \text{diag}([y_{i1}], [y_{i1}, y_{i2}], \dots, [y_{i1}, y_{i2}, \dots, y_{i,T-2}]) \in \mathbb{R}^{(T-2) \times g} \quad (9.422)$$

Remark 9.6.4 (Population Moment Conditions). The orthogonal expectation is:

$$E[\mathbf{Z}_i' \Delta \mathbf{u}_i] = \mathbf{0}_g \quad (9.423)$$

where $\Delta \mathbf{u}_i = \mathbf{D}_T \mathbf{u}_i \in \mathbb{R}^{(T-2) \times 1}$ and $\mathbf{D}_T$ is the differencing operator.

9.6.3 Two-Step Generalized Method of Moments (GMM)

9.6.3.1 First-Step Estimator (One-Step GMM)

Definition 9.6.4 (Sample Moment Matrix).

$$\mathbf{Q}_{ZN}(\rho) \equiv \frac{1}{N} \sum_{i=1}^N \mathbf{Z}_i' \Delta \mathbf{u}_i(\rho) \quad (9.424)$$

where $\Delta \mathbf{u}_i(\rho) = \Delta \mathbf{y}_i - \rho \Delta \mathbf{y}_{i,-1}$.

Definition 9.6.5 (Initial Weighting Matrix). Under homoskedasticity of level shocks ($\text{Var}(\mathbf{u}_i) = \sigma_u^2 \mathbf{I}_T$):

$$\mathbf{W}_{1,N} = \left(\frac{1}{N} \sum_{i=1}^N \mathbf{Z}_i' \mathbf{H} \mathbf{Z}_i \right)^{-1} \quad (9.425)$$

where $\mathbf{H} \equiv \mathbf{D}_T \mathbf{D}_T'$ is the Toeplitz tridiagonal matrix:

$$\mathbf{H} = \begin{bmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & -1 & \dots & 0 \\ 0 & -1 & 2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 2 \end{bmatrix} \quad (9.426)$$

Theorem 9.6.3 (One-Step GMM Estimator). *The first-step estimator is:*

$$\hat{\rho}_{GMM1} = \left(\Delta \mathbf{y}_{-1}' \mathbf{Z} (\mathbf{Z}' (\mathbf{I}_N \otimes \mathbf{H}) \mathbf{Z})^{-1} \mathbf{Z}' \Delta \mathbf{y}_{-1} \right)^{-1} \Delta \mathbf{y}_{-1}' \mathbf{Z} (\mathbf{Z}' (\mathbf{I}_N \otimes \mathbf{H}) \mathbf{Z})^{-1} \mathbf{Z}' \Delta \mathbf{y} \quad (9.427)$$

Remark 9.6.5 (One-Step Properties). • Consistent under sequential exogeneity

- Efficient if level shocks are homoskedastic and non-autocorrelated
- Provides consistent residuals $\hat{\Delta} \mathbf{u}_i$ for second step

9.6.3.2 Second-Step Estimator (Efficient Two-Step GMM)

Definition 9.6.6 (Hansen Robust Weighting Matrix). Using first-step residuals $\hat{\Delta} \mathbf{u}_i = \Delta \mathbf{y}_i - \hat{\rho}_{GMM1} \Delta \mathbf{y}_{i,-1}$:

$$\mathbf{W}_{2,N} = \hat{\mathbf{\Lambda}}_N^{-1} = \left(\frac{1}{N} \sum_{i=1}^N \mathbf{Z}_i' \hat{\Delta} \mathbf{u}_i \hat{\Delta} \mathbf{u}_i' \mathbf{Z}_i \right)^{-1} \quad (9.428)$$

Theorem 9.6.4 (Efficient Two-Step GMM Estimator). *The second-step estimator is:*

$$\hat{\rho}_{GMM2} = \left(\Delta \mathbf{y}'_{-1} \mathbf{Z} \hat{\mathbf{\Lambda}}_N^{-1} \mathbf{Z}' \Delta \mathbf{y}_{-1} \right)^{-1} \Delta \mathbf{y}'_{-1} \mathbf{Z} \hat{\mathbf{\Lambda}}_N^{-1} \mathbf{Z}' \Delta \mathbf{y} \quad (9.429)$$

Proof. The estimator minimizes the quadratic objective function:

$$J_{2,N}(\rho) = \mathbf{Q}_{ZN}(\rho)' \hat{\mathbf{\Lambda}}_N^{-1} \mathbf{Q}_{ZN}(\rho) \quad (9.430)$$

By Hansen's (1982) GMM theory, $\hat{\rho}_{GMM2}$ attains the lower bound of asymptotic variance (Hansen-GMM efficiency) under any heteroskedasticity/autocorrelation structure. $\square$

Definition 9.6.7 (Uncorrected Asymptotic Variance).

$$\widehat{\text{Var}}_{\text{asympt}}(\hat{\rho}_{GMM2}) = \left(\Delta \mathbf{y}'_{-1} \mathbf{Z} \hat{\mathbf{\Lambda}}_N^{-1} \mathbf{Z}' \Delta \mathbf{y}_{-1} \right)^{-1} \quad (9.431)$$

Remark 9.6.6 (Finite-Sample Problem). The conventional asymptotic variance **severely underestimates** the actual variability in finite samples, since it treats $\mathbf{W}_{2,N}$ as deterministic, ignoring the first-step estimation error.

9.6.3.3 Windmeijer Correction (2005)

Theorem 9.6.5 (Windmeijer Corrected Variance). *The finite-sample correction is:*

$$\widehat{\text{Var}}_{\text{Wind}}(\hat{\rho}_{GMM2}) = \mathbf{V}_2 + \mathbf{D} \mathbf{V}_1 \mathbf{D}' + \mathbf{D} \mathbf{C} + \mathbf{C}' \mathbf{D}' \quad (9.432)$$

where:

- $\mathbf{V}_2 = \widehat{\text{Var}}_{\text{asympt}}(\hat{\rho}_{GMM2})$ (uncorrected variance)
- $\mathbf{V}_1 = \widehat{\text{Var}}(\hat{\rho}_{GMM1})$ (first-step variance)
- $\mathbf{C}$ = mutual covariance between steps
- $\mathbf{D}$ = matrix of partial derivatives evaluating second-step sensitivity to first step

Proof. Windmeijer (2005) applies first-order Taylor expansion over the first-step moment conditions, incorporating the estimation error of $\hat{\rho}_{GMM1}$ into the second-step variance. The matrix $\mathbf{D}$ captures how $\hat{\rho}_{GMM2}$ varies when $\hat{\rho}_{GMM1}$ fluctuates sample-wise. $\square$

Remark 9.6.7 (Correction Impact). The Windmeijer correction:

- Eliminates downward bias of two-step standard errors
- Restores empirical Type I Error rates aligned with nominal levels
- Is mandatory in practical applications with moderate N

9.6.4 Specification Tests and Instrument Validation

9.6.4.1 Residual Autocorrelation Test (m_2)

Theorem 9.6.6 (First-Order Mechanical Autocorrelation). *Under i.i.d. shocks in levels, the first-order correlation in differenced residuals is:*

$$\text{Corr}(\Delta u_{it}, \Delta u_{i,t-1}) = \frac{\text{Cov}(\Delta u_{it}, \Delta u_{i,t-1})}{\sqrt{\text{Var}(\Delta u_{it}) \text{Var}(\Delta u_{i,t-1})}} = \frac{-\sigma_u^2}{\sqrt{2\sigma_u^2 \cdot 2\sigma_u^2}} = -0.5 \quad (9.433)$$

Proof. We calculate:

$$\begin{aligned} \text{Cov}(\Delta u_{it}, \Delta u_{i,t-1}) &= \text{Cov}(u_{it} - u_{i,t-1}, u_{i,t-1} - u_{i,t-2}) \\ &= \text{Cov}(u_{it}, u_{i,t-1}) - \text{Cov}(u_{it}, u_{i,t-2}) - \text{Cov}(u_{i,t-1}, u_{i,t-1}) + \text{Cov}(u_{i,t-1}, u_{i,t-2}) \\ &= 0 - 0 - \sigma_u^2 + 0 = -\sigma_u^2 \end{aligned} \quad (9.434)$$

Since $\text{Var}(\Delta u_{it}) = \text{Var}(u_{it} - u_{i,t-1}) = 2\sigma_u^2$:

$$\text{Corr}(\Delta u_{it}, \Delta u_{i,t-1}) = \frac{-\sigma_u^2}{2\sigma_u^2} = -0.5 \quad (9.435)$$

□

Remark 9.6.8 (Implication for Tests). Detection of first-order autocorrelation (m_1) is **expected** and does not invalidate instruments. The relevant test is for **second-order** autocorrelation (m_2).

Definition 9.6.8 (Arellano-Bond m_2 Statistic).

$$m_2 = \frac{\hat{d}}{\sqrt{\hat{V}}} \xrightarrow{d} N(0, 1) \quad (9.436)$$

where $\hat{d} = \frac{1}{N} \sum_{i=1}^N \sum_{t=4}^T \hat{e}_{it} \hat{e}_{i,t-2}$ is the second-order sample covariance of differenced residuals $\hat{e}_{it} = \Delta \hat{u}_{it}$, and $\hat{V}$ is its estimated asymptotic variance (incorporating correction for first-step estimation).

Theorem 9.6.7 (Null Hypothesis of m_2 Test).

$$H_0 : \text{Cov}(u_{it}, u_{i,t-2}) = 0 \quad \forall t \quad (9.437)$$

Proof. Under H_0 (absence of autocorrelation in levels):

$$\begin{aligned} \text{Cov}(\Delta u_{it}, \Delta u_{i,t-2}) &= \text{Cov}(u_{it} - u_{i,t-1}, u_{i,t-2} - u_{i,t-3}) \\ &= \text{Cov}(u_{it}, u_{i,t-2}) - \text{Cov}(u_{it}, u_{i,t-3}) - \text{Cov}(u_{i,t-1}, u_{i,t-2}) + \text{Cov}(u_{i,t-1}, u_{i,t-3}) \\ &= 0 - 0 - 0 + 0 = 0 \end{aligned} \quad (9.438)$$

□

Remark 9.6.9 (Consequences of m_2 Rejection). If m_2 is significant ($|m_2| > 1.96$ for $\alpha = 5\%$):

- There is second-order autocorrelation in levels
- Instruments $y_{i,t-2}$ become **invalid** since $\text{Cov}(y_{i,t-2}, \Delta u_{it}) \approx -E[u_{i,t-2} u_{i,t-1}] \neq 0$
- One should use only deeper lags ($y_{i,t-3}, y_{i,t-4}, \dots$) as instruments

9.6.4.2 Hansen Overidentification Test

Definition 9.6.9 (Hansen J Statistic).

$$J_N = N \cdot \mathbf{Q}_{ZN}(\hat{\rho}_{\text{GMM2}})' \hat{\mathbf{\Lambda}}_N^{-1} \mathbf{Q}_{ZN}(\hat{\rho}_{\text{GMM2}}) \quad (9.439)$$

Equivalently:

$$J_N = \Delta \hat{\mathbf{u}}' \mathbf{Z} \left(\sum_{i=1}^N \mathbf{Z}_i' \hat{\Delta} \mathbf{u}_i \hat{\Delta} \mathbf{u}_i' \mathbf{Z}_i \right)^{-1} \mathbf{Z}' \Delta \hat{\mathbf{u}} \quad (9.440)$$

Theorem 9.6.8 (Asymptotic Distribution Under H_0). *Under the null hypothesis of joint instrument exogeneity ($H_0 : E[\mathbf{Z}_i' \Delta \mathbf{u}_i] = \mathbf{0}_g$):*

$$J_N \xrightarrow{d} \chi_{g-K}^2 \quad (9.441)$$

where $g = \frac{(T-2)(T-1)}{2}$ is the number of instruments and K is the number of structural parameters.

Proof. Under H_0 and regularity conditions (full rank, convergence of $\hat{\mathbf{\Lambda}}_N$), the minimized quadratic form follows chi-square with $g - K$ degrees of freedom. The loss of K degrees occurs because estimation of the K parameters consumes K identifying moment conditions, leaving $g - K$ overidentifying restrictions for testing. $\square$

Remark 9.6.10 (Test Interpretation). • If $J_N > \chi_{g-K, 1-\alpha}^2$: reject H_0 , indicating at least one instrument is endogenous or the model is misspecified

• If $J_N \leq \chi_{g-K, 1-\alpha}^2$: do not reject H_0 , instruments are jointly valid

9.6.4.3 Instrument Proliferation

Remark 9.6.11 (Finite-Sample Pathologies). When T is moderately long ($T > 10$), the number of instruments $g = O(T^2)$ grows quadratically, generating:

1. **Overfitting Bias:** GMM approaches biased estimators (Pooled OLS/Within) because excess instruments extract all sample variation from the endogenous regressor
2. **Collapse of J Test Power:** The matrix $\hat{\mathbf{\Lambda}}_N \in \mathbb{R}^{g \times g}$ becomes nearly singular, mechanically forcing $\hat{\Delta} \mathbf{u}_i \perp \mathbf{Z}_i$ by sample construction. The J_N statistic collapses to values near zero, generating p -values ≈ 1.000 and eliminating power to detect endogenous instruments

Definition 9.6.10 (Mitigation Strategies). To combat instrument proliferation:

- **Limit lags:** Use only $y_{i,t-2}$ and $y_{i,t-3}$ (not entire history until y_{i1})
- **Collapsed instruments:** Sum instruments by period, reducing dimension from g to $T - 2$
- **Hansen difference test:** Compare J with and without instrument subsets to validate specific groups

Table 9.6: Final Synthesis: Arellano-Bond (Dynamic Panel GMM)

Concept	Mathematical Expression	Eq.
Dynamic Model	$y_{it} = \rho y_{i,t-1} + c_i + u_{it}$	(1)
Nickell Bias	$\text{plim}(\hat{\rho}_{FE} - \rho) \approx -\frac{1+\rho}{T-1}$	(3)
Sequential Exogeneity	$E[u_{it} \mid y_{i,t-1}, \dots, y_{i1}, c_i] = 0$	(5)
Moment Conditions	$E[y_{i,t-s} \Delta u_{it}] = 0 \quad \forall s \geq 2$	(6)
Number of Instruments	$g = \frac{(T-2)(T-1)}{2}$	(8)
Instrument Matrix	$\mathbf{Z}_i = \text{diag}([y_{i1}], [y_{i1}, y_{i2}], \dots)$	(9)
One-Step GMM	$\hat{\rho}_{\text{GMM1}}$ (homoskedastic weight $\mathbf{H}$)	(13)
Two-Step GMM	$\hat{\rho}_{\text{GMM2}}$ (robust weight $\hat{\mathbf{\Lambda}}_N^{-1}$)	(16)
Windmeijer Correction	$\widehat{\text{Var}}_{\text{Wind}} = \mathbf{V}_2 + \mathbf{D}\mathbf{V}_1\mathbf{D}' + \mathbf{D}\mathbf{C} + \mathbf{C}'\mathbf{D}'$	(19)
Mechanical Autocorrelation	$\text{Corr}(\Delta u_{it}, \Delta u_{i,t-1}) = -0.5$	(21)
m_2 Test	$H_0 : \text{Cov}(u_{it}, u_{i,t-2}) = 0$	(24)
Hansen J Test	$J_N \xrightarrow{d} \chi_{g-K}^2$	(28)

Remark 9.6.12 (Final Summary for the Qualification Committee). 1. **Nickell Bias:**

In dynamic panels ($y_{i,t-1}$ as regressor), the Within estimator is inconsistent with bias $-\frac{1+\rho}{T-1}$ that vanishes only when $T \rightarrow \infty$. FD-OLS is also inconsistent (bias persists even with $T \rightarrow \infty$).

2. **Sequential Exogeneity:** The hypothesis $E[u_{it} \mid y_{i,t-1}, \dots, y_{i1}, c_i] = 0$ allows level lags ($y_{i,t-2}, y_{i,t-3}, \dots$) to be valid instruments for the first-differenced equation.
3. **Instrument Matrix:** $\mathbf{Z}_i$ has block-diagonal structure with quadratic growth $g = \frac{(T-2)(T-1)}{2}$. For $T = 10$, $g = 36$ (massive overidentification).
4. **Two-Step GMM:** One-Step uses homoskedastic weight $\mathbf{H} = \mathbf{D}_T \mathbf{D}_T'$ (efficient if level shocks are i.i.d.). Two-Step uses robust weight $\hat{\mathbf{\Lambda}}_N^{-1}$ based on first-step residuals (efficient under arbitrary heteroskedasticity/autocorrelation).
5. **Windmeijer Correction:** Mandatory in finite samples. Corrects underestimation of two-step standard errors by incorporating first-step estimation error via Taylor expansion.
6. **m_2 Test:** Tests $H_0 : \text{Cov}(u_{it}, u_{i,t-2}) = 0$. First-order autocorrelation (m_1) is expected (-0.5) and does not invalidate instruments. Rejection of m_2 indicates $y_{i,t-2}$ is invalid; one should use deeper lags.
7. **Hansen J Test:** Tests joint exogeneity of all instruments ($J_N \sim \chi_{g-K}^2$). Under instrument proliferation (large T), the test loses power (p -values ≈ 1.000) due to overfitting. Solutions: limit lags or collapse instruments.
8. **Recommended Practice:** (i) Report One-Step and Two-Step; (ii) Use Windmeijer standard errors; (iii) Test m_2 to validate instruments; (iv) Test J for overidentification; (v) Monitor number of instruments vs N (practical rule: $g < N$); (vi) If T is large, collapse instruments or limit lags.

9.7 Panel Data Block Synthesis

9.7.1 Reference Model and Notation

To consolidate the comparison between estimators in this block, we start from the linear compound error model:

$$y_{it} = \mathbf{x}_{it}'\boldsymbol{\beta} + c_i + u_{it}, \quad i = 1, \dots, N; \quad t = 1, \dots, T, \quad (9.442)$$

where $\mathbf{x}_{it}$ is the $K \times 1$ vector of regressors, c_i is the unobserved individual heterogeneity (time-invariant), and u_{it} is the idiosyncratic error. The compound error is

$$v_{it} = c_i + u_{it}. \quad (9.443)$$

The choice between POLS, RE, FE, and FD is not merely a technical matter, but concerns **which sources of variation in the data** the researcher is willing to assume as exogenous.

9.7.2 Unified Comparative Summary

Estimator	Transformation	Assumption about c_i	Assumption about u_{it}	Identifies in-variants?
POLS	None (stacks)	$\mathbb{E}[\mathbf{x}_{it}c_i] = 0$	Contemporaneous: $\mathbb{E}[\mathbf{x}_{it}u_{it}] = 0$	Yes
RE	Quasi-demeaning	$\mathbb{E}[c_i \mathbf{X}_i] = 0$	Strict: $\mathbb{E}[u_{it} \mathbf{X}_i, c_i] = 0$	Yes
FE	Within ($\mathbf{Q}$)	Arbitrary	Strict: $\mathbb{E}[u_{it} \mathbf{X}_i, c_i] = 0$	No
FD	First differences (Δ)	Arbitrary	Strict: $\mathbb{E}[u_{it} \mathbf{X}_i, c_i] = 0$	No

Table 9.7: Unified comparison of basic panel data estimators.

9.7.3 Exogeneity Hierarchy

The exogeneity assumptions form a clear hierarchy that determines each estimator's consistency.

9.7.3.1 Contemporaneous Exogeneity (Weak) — Sufficient for POLS

Necessary and sufficient for POLS consistency when $N \rightarrow \infty$:

$$\mathbb{E}[\mathbf{x}_{it}v_{it}] = \mathbf{0}, \quad \forall t. \quad (9.444)$$

Since $v_{it} = c_i + u_{it}$, this simultaneously requires:

$$\mathbb{E}[\mathbf{x}_{it}c_i] = \mathbf{0} \quad \text{and} \quad \mathbb{E}[\mathbf{x}_{it}u_{it}] = \mathbf{0}. \quad (9.445)$$

What it allows: dynamic feedbacks, where u_{it} affects $\mathbf{x}_{i,t+1}$. POLS is robust to this type of endogeneity.

Counterexample: if more productive firms (high c_i) invest more in R&D ($\mathbf{x}_{it}$), POLS will attribute to investment the effect that actually comes from base productivity.

9.7.3.2 Strict Exogeneity (Strong) — Necessary for FE, FD, and RE

For all t and all s :

$$\mathbb{E}[u_{it} \mid \mathbf{X}_i, c_i] = 0 \quad \Longleftrightarrow \quad \mathbb{E}[\mathbf{x}_{is}u_{it}] = \mathbf{0}. \quad (9.446)$$

What it prohibits: *feedback* mechanisms where u_{it} affects $\mathbf{x}_{i,t+1}$.

Why FE and FD require this hypothesis: the within transformation computes $\tilde{\mathbf{x}}_{it} = \mathbf{x}_{it} - \bar{\mathbf{x}}_i$, and the FD transformation computes $\Delta\mathbf{x}_{it} = \mathbf{x}_{it} - \mathbf{x}_{i,t-1}$. Both mix information from multiple periods. If u_{it} is correlated with $\mathbf{x}_{i,t+1}$, then it will be correlated with $\bar{\mathbf{x}}_i$ and with $\tilde{\mathbf{x}}_{it}$, making FE inconsistent.

Counterexample: in a crime model, if policing ($\mathbf{x}$) reacts to past crime shocks (u_{t-1}), strict exogeneity fails and FE/FD are inconsistent.

9.7.4 Variance-Covariance Matrix and Relative Efficiency

Estimator efficiency depends on the structure of $\mathbf{\Omega} = \mathbb{E}[\mathbf{v}\mathbf{v}']$. As discussed previously, under the classical error components model ($v_{it} = c_i + u_{it}$), the covariance matrix for individual i is

$$\mathbf{\Sigma}_{ii} = \sigma_u^2 \mathbf{I}_T + \sigma_c^2 \mathbf{J}_T, \quad (9.447)$$

where $\mathbf{J}_T = \mathbf{j}_T \mathbf{j}_T'$.

This structure has direct implications for efficiency:

- **POLS:** inefficient, since it ignores serial correlation induced by c_i .
- **RE:** efficient (BLUE) under $\mathbb{E}[c_i \mid \mathbf{X}_i] = 0$, since it applies GLS to matrix $\mathbf{\Sigma}_{ii}$ via quasi-demeaning.
- **FE:** efficient if u_{it} is white noise. If u_{it} is AR(1) or random walk, FE is consistent but inefficient.
- **FD:** efficient if u_{it} follows a random walk, since Δu_{it} becomes white noise.

For $T = 2$, FE and FD are numerically identical. For $T > 2$, the choice depends on the serial structure of u_{it} :

- If u_{it} is white noise: FE is more efficient than FD.
- If u_{it} is a random walk: FD is more efficient than FE.

9.7.5 Practical Choice Guide

The choice between estimators should be guided by three fundamental questions:

9.7.5.1 Question 1: Is c_i plausibly correlated with $\mathbf{x}_{it}$?

- **Yes:** use FE or FD. RE and POLS will be inconsistent.
- **No (or strong justification):** RE is preferable for being more efficient. POLS can be used as a benchmark, but requires clustering.
- **Test:** Hausman test compares FE and RE. If it rejects H_0 , there is correlation between c_i and $\mathbf{x}_{it}$.

9.7.5.2 Question 2: What is the serial structure of u_{it} ?

- **White noise:** FE is more efficient than FD.
- **Random walk or highly persistent:** FD is more efficient than FE.
- **Test:** examine autocorrelation of FE residuals. If there is strong serial correlation, consider FD.

9.7.5.3 Question 3: Is the model dynamic?

- **Yes (contains $y_{i,t-1}$):** strict exogeneity fails mechanically. FE and FD are inconsistent for fixed T (Nickell bias).
- **Solution:** use dynamic GMM (Arellano-Bond or Blundell-Bond) with lagged instruments.

9.7.5.4 Additional Considerations

- **Time-invariant variables:** if they are essential for the research (gender, race, geographic location), use RE or POLS, provided there is justification for $\mathbb{E}[c_i | \mathbf{X}_i] = 0$.
- **Measurement error:** FD amplifies measurement noise when differencing. If the variable is persistent and measured with error, FD may suffer severe attenuation bias.
- **Short panels (small T):** FE and FD lose degrees of freedom. Consider RE if there is strong theoretical justification.
- **Inference:** in all cases, use cluster-robust standard errors at the unit i level.

9.7.6 Caveats and Common Pitfalls

- **Omission of Hausman test:** “The author uses RE to estimate returns to education, but does not report the Hausman test. If unobserved ability is correlated with schooling, RE coefficients are inconsistent.”
- **Dynamic endogeneity:** “The model includes lagged GDP as a regressor. In this scenario, the strict exogeneity hypothesis is violated by construction, making FE and FD inconsistent (Nickell bias).”
- **Weak instruments in FD:** “When differencing variables with high temporal persistence, the author eliminated almost all signal variation, leaving only measurement noise, which exacerbates attenuation bias.”
- **Negative variance in RE:** “In small samples, the estimate of $\hat{\sigma}_c^2$ may be negative. The software zeros this component, and RE silently collapses to POLS.”
- **Absence of clustering:** “Even under FE, idiosyncratic errors may exhibit residual serial correlation. Cluster-robust standard errors at the unit i level are mandatory.”
- **Confusion between weak and strong exogeneity:** “The author justifies POLS by saying u_{it} is orthogonal to $\mathbf{x}_{it}$, but does not discuss whether c_i is correlated with

$\mathbf{x}_{it}$. Contemporaneous exogeneity is not sufficient if there is correlated unobserved heterogeneity.”

9.7.7 Typical Referee Question

Question: “Why did you choose FE instead of FD, given that your idiosyncratic error u_{it} appears to have a unit root in diagnostic tests?”

Answer: “The choice between FE and FD for $T > 2$ is based on efficiency under strict exogeneity. If u_{it} is a random walk, Δu_{it} becomes white noise, making FD the BLUE estimator (by the Gauss-Markov Theorem applied to differences). Using FE in this case would introduce negative serial correlation in the transformed errors, reducing statistical power. I acknowledge that, given the unit root evidence, FD is the more efficient choice. As robustness, I report both specifications with cluster-robust standard errors and confirm that results are qualitatively similar.”

9.7.8 Block Conclusion

The panel data block is organized around a central trade-off: **efficiency versus robustness to unobserved heterogeneity**.

- **POLS** is the starting point, but is fragile when c_i is correlated with regressors. It requires only contemporaneous exogeneity, but is inefficient and requires clustering.
- **RE** is efficient under c_i orthogonality, but becomes inconsistent if this hypothesis fails. It allows estimation of invariant variable effects.
- **FE** solves the endogeneity problem from unobserved heterogeneity, at the cost of losing invariant variables and requiring strict exogeneity. It is the gold standard for causal identification in panels.
- **FD** also eliminates c_i , but its efficiency depends on the serial structure of errors. It is preferable when u_{it} is highly persistent or when there is concern about $I(1)$ variables.

Rigorous practice requires:

- explicitly stating which source of variation identifies the parameter (within, between, or both);
- justifying strict exogeneity or presenting an instrumental strategy;
- using cluster-robust standard errors at the unit level;
- comparing POLS, FE, FD, and RE specifications;
- treating dynamic models with appropriate instruments (dynamic GMM).

Chapter 10

Limited Dependent Variable Models

10.1 Block Introduction

10.1.1 Common Problem: Discrete, Censored, Truncated, or Non-Randomly Selected Dependent Variables

In applied microeconomic research, the classical linear regression model estimated by Ordinary Least Squares (OLS) assumes that the dependent variable Y has continuous support over the entire real line $\mathbb{R}$. Under this premise, along with the conditional mean exogeneity assumption ($\mathbb{E}[U \mid \mathbf{X}] = 0$), the OLS estimator is consistent and asymptotically normal for the structural parameters β .

However, modeling individual choices, consumption decisions, and market participation frequently results in dependent variables whose support is restricted or exhibits discontinuities. These variables are called **Limited Dependent Variables** (LDV). When the support of Y is limited, the population conditional expectation relationship $\mathbb{E}[Y \mid \mathbf{X}]$ becomes intrinsically non-linear, violating the OLS linearity assumption and causing simple linear regression to produce severely biased, inconsistent estimates lacking structural economic validity.

To systematize the study of this block, we analytically classify the four most common support pathologies in microdata:

10.1.1.1 Discrete Dependent Variables

These occur when the response variable assumes only integer values in a countable set. The limiting case is the binary response, where $Y_i \in \{0, 1\}$, representing participation choices or dichotomous decisions. In count data cases, the support extends over non-negative integers ($Y_i \in \mathbb{N}_0 = \{0, 1, 2, \dots\}$), such as the number of firm patents or doctor visits.

The observation rule is:

$$Y_i = \mathbb{I}(Y_i^* > 0) \quad \text{where} \quad Y_i^* = \mathbf{X}_i^T \beta + e_i, \quad (10.1)$$

where Y_i^* is an unobserved continuous latent utility or propensity and $\mathbb{I}(\cdot)$ is the binary indicator function.

Why OLS fails (Linear Probability Model): Although OLS can be algebraically computed (LPM), the conditional expectation $\mathbb{E}[Y_i | \mathbf{X}_i] = P(Y_i = 1 | \mathbf{X}_i) = \mathbf{X}_i^T \boldsymbol{\beta}$ imposes an inadequate linear restriction. Since the probability support is strictly bounded to the interval $[0, 1]$, the linear specification generates absurd predictions of negative probabilities or probabilities greater than one for extreme values of $\mathbf{X}_i$. Additionally, errors are inherently heteroskedastic and non-normal, invalidating classical test statistics in finite samples.

10.1.1.2 Censored Data

Censoring occurs when the latent dependent variable of interest, $Y_i^* \in \mathbb{R}$, is defined and endowed with economic meaning for the entire population, but its physical measurement is restricted above or below a certain threshold τ due to sample design or institutional constraints. The observed support is hybrid: it exhibits a discrete probability mass at the censoring point and a continuous density outside it.

The observation rule (censoring at the origin, $\tau = 0$) is:

$$Y_i = \max(0, Y_i^*) = \begin{cases} Y_i^*, & \text{if } Y_i^* > 0, \\ 0, & \text{if } Y_i^* \leq 0. \end{cases} \quad (10.2)$$

Why OLS fails: If we estimate the model by OLS using the entire sample (including zeros), coefficient estimates will suffer from severe attenuation bias (underestimation in absolute value). The estimator tries to fit a continuous line over a mixed distribution, failing to capture the true slope of the latent tail. If we discard zeros and estimate OLS only on the positive subsample ($Y_i > 0$), estimators will remain inconsistent, since error truncation induces mechanical correlation between regressors and the remaining stochastic disturbance.

10.1.1.3 Truncated Data

Unlike censoring, in truncation we have no information about individuals who fall outside the support limit. If the latent dependent variable does not satisfy the cutoff criterion, the individual is entirely excluded from the database, making it impossible to observe even their exogenous covariates $\mathbf{X}_i$.

The observation rule (left truncation at the origin) is: the vector $(\mathbf{X}_i, Y_i)$ is observed if and only if

$$Y_i^* > 0 \quad \text{where} \quad Y_i \equiv Y_i^* = \mathbf{X}_i^T \boldsymbol{\beta} + e_i. \quad (10.3)$$

Why OLS fails: Since the sample is a non-random selection from the population of interest, the conditional mean of the error in the truncated sample is no longer zero, becoming non-linearly dependent on regressors through the truncated density:

$$\mathbb{E}[e_i | \mathbf{X}_i, Y_i > 0] = \sigma \lambda \left(\frac{\mathbf{X}_i^T \boldsymbol{\beta}}{\sigma} \right) \neq 0, \quad (10.4)$$

where $\lambda(\cdot)$ is the Inverse Mills Ratio. OLS ignores this correction term, generating estimates severely biased toward zero for all slope parameters.

10.1.1.4 Sample Selection (Incidental Truncation)

This occurs when the dependent variable of interest Y_i^* is observed only for a non-random subsample of the population, but the mechanism determining this observability (selection, S_i) is governed by a behavioral decision process of the agent themselves that differs from the outcome equation. This phenomenon is known as incidental truncation.

The observation rule is:

$$S_i = \mathbb{I}(\mathbf{Z}_i^T \boldsymbol{\gamma} + V_i > 0) \quad (\text{Selection Mechanism}), \quad (10.5)$$

$$Y_i = \begin{cases} Y_i^* = \mathbf{X}_i^T \boldsymbol{\beta} + U_i, & \text{if } S_i = 1, \\ \text{not observed (missing)}, & \text{if } S_i = 0. \end{cases} \quad (10.6)$$

Why OLS fails: If we estimate the outcome equation by OLS using only the selected sample ($S_i = 1$), mean exogeneity is violated whenever the unobserved shocks of selection and outcome are correlated ($\text{Cov}(U_i, V_i) = \sigma_{UV} \neq 0$). Conditioned on participation, the expectation of the outcome error is non-zero:

$$\mathbb{E}[U_i \mid \mathbf{X}_i, \mathbf{Z}_i, S_i = 1] = \sigma_{UV} \lambda(\mathbf{Z}_i^T \boldsymbol{\gamma}) \neq 0. \quad (10.7)$$

Omission of $\lambda(\mathbf{Z}_i^T \boldsymbol{\gamma})$ acts exactly as an omitted variable bias, correlating the remaining error term with the regressor vector $\mathbf{X}_i$ in the active subsample and rendering the OLS estimator inconsistent.

10.1.1.5 The Theoretical Block of Identification and Consistency

Resolving these four support pathologies constitutes the core of large-sample microeconomic theory for limited data. Since the non-linearity of conditional expectations prevents direct application of classical linear projections, consistent identification of structural parameters $\boldsymbol{\beta}$ requires alternative estimation methods, specifically Maximum Likelihood or semiparametric absolute loss methods.

Historically, this identification rests upon two distinct methodological approaches:

- **Structural Parametric Identification:** Based on imposing strong distributional assumptions (such as bivariate joint normality of errors) to exactly deduce and correct the non-linear curvature of the tail under truncation (as in classical Tobit and two-stage Heckman).
- **Semiparametric and Exclusion Identification:** Relaxes normality and homoskedasticity assumptions in favor of robust estimators (such as Powell's CLAD) or requires the presence of valid exclusion restrictions (instruments that exogenously determine participation/selection probability without directly affecting the structural outcome), ensuring that identification is statistically stable and immune to collinearity in real samples.

10.1.2 Why OLS Fails Here

The classical linear regression model estimated by OLS relies on the premise that the population conditional expectation relationship $\mathbb{E}[Y \mid \mathbf{X}]$ is linear and continuous over

the entire real support. When we encounter Limited Dependent Variables (LDV) or data under Non-Random Sample Selection, this structural linearity is mechanically violated.

Below, we detail the three fundamental channels of OLS estimator collapse under these circumstances.

10.1.2.1 Predictions Outside the Coherent Support

Being an unrestricted linear projection, OLS projects predicted values $\hat{Y}_i = \mathbf{X}_i^T \hat{\beta}_{OLS}$ that can assume any real value over the line $\mathbb{R}$, ignoring the physical and probabilistic restrictions inherent to the actual dependent variable support.

The Linear Probability Model (LPM) Case: When the dependent variable is binary ($Y_i \in \{0, 1\}$), the population conditional expectation exactly equals the conditional probability of success: $\mathbb{E}[Y_i | \mathbf{X}_i] = P(Y_i = 1 | \mathbf{X}_i)$. When fitting the model by OLS, the estimated probability varies linearly with covariates. Consequently, for extreme values of $\mathbf{X}_i$, the LPM will inevitably generate absurd predictions, providing negative probabilities or probabilities exceeding 1.

Fractional, Censored, and Count Variables: If the variable of interest is strictly non-negative (such as working hours in Tobit or patent counts in Poisson models), OLS projects negative values for substantial portions of the sample. Since these negative values lack physical or economic interpretation, the predictive power of the linear model collapses outside the sample mean.

10.1.2.2 Parametric Inconsistency Due to DGP Non-Linearity

In applied microeconometrics, the researcher's central interest lies in identifying the structural (causal) behavior parameters, β_0 . Although OLS always provides a consistent estimate of the Best Linear Projection (BLP) parameters of Y given $\mathbf{X}$, it is systematically inconsistent for recovering the true structural coefficients β_0 if the population DGP is strictly non-linear.

Distortion by Inadequate Linear Approximation: If the true DGP is governed by a Probit or Logit distribution, such that $\mathbb{E}[Y_i | \mathbf{X}_i] = G(\mathbf{X}_i^T \beta_0)$ with $G(\cdot)$ being a strictly non-linear CDF, the true marginal effect

$$\frac{\partial \mathbb{E}[Y_i | \mathbf{X}_i]}{\partial X_{ik}} = g(\mathbf{X}_i^T \beta_0) \beta_k \quad (10.8)$$

will vary continuously over the entire regressor support. By forcing a linear fit with constant slope coefficient ($\hat{\beta}_{OLS}$), OLS will report a weighted average coefficient suffering from so-called attenuation bias.

Fit Collapse in Corner Solution Models (Tobit): Let $Y_i = \max(0, Y_i^*)$ be a censored variable or one with zero accumulation at the origin. The true unconditional population expected value follows Tobin's non-linear mixed relationship:

$$\mathbb{E}[Y_i | \mathbf{X}_i] = \mathbf{X}_i^T \beta_0 \Phi \left(\frac{\mathbf{X}_i^T \beta_0}{\sigma} \right) + \sigma \phi \left(\frac{\mathbf{X}_i^T \beta_0}{\sigma} \right). \quad (10.9)$$

If we estimate this relationship by OLS using the entire sample, the linear estimator will try to fit a continuous line connecting the probability mass concentrated at zero to the

positive continuous data. This linear fit generates a slope severely attenuated toward zero, failing to recover β_0 .

10.1.2.3 Sample Selection Bias (Incidental Truncation)

In sample selection or incidental truncation problems, the available subsample for estimation is systematically different from the target population due to agents' self-selection choices. OLS collapse in this context stems from the fact that incidental truncation generates a mechanical violation of the mean error exogeneity hypothesis conditioned on selection.

Let the outcome and selection equations be represented by the bivariate latent system:

$$Y_i^* = \mathbf{X}_i^T \beta_0 + U_i \quad (\text{Outcome Equation}), \quad (10.10)$$

$$S_i^* = \mathbf{Z}_i^T \gamma_0 + V_i \quad (\text{Selection Equation}), \quad (10.11)$$

where the econometrician observes $Y_i = Y_i^*$ if and only if the participation indicator is active, $S_i = \mathbb{I}(S_i^* > 0) = 1$.

If the latent error terms are correlated in the population ($\text{Cov}(U_i, V_i) = \sigma_{UV} \neq 0$), the conditional expectation of the structural error term in the selected sample is non-zero:

$$\mathbb{E}[U_i \mid \mathbf{X}_i, \mathbf{Z}_i, S_i = 1] = \mathbb{E}[U_i \mid \mathbf{X}_i, \mathbf{Z}_i, V_i > -\mathbf{Z}_i^T \gamma_0] = \sigma_{UV} \lambda(\mathbf{Z}_i^T \gamma_0), \quad (10.12)$$

where $\lambda(z) \equiv \phi(z)/\Phi(z)$ represents the Inverse Mills Ratio.

The OLS estimator is therefore inconsistent and permanently biased, converging in probability to a pseudo-true vector distorted by selectivity:

$$\text{plim } \hat{\beta}_{\text{OLS}} = \beta_0 + \sigma_{UV} \left(\mathbb{E}[\mathbf{X}_i \mathbf{X}_i^T \mid S_i = 1] \right)^{-1} \mathbb{E}[\mathbf{X}_i \lambda_i \mid S_i = 1] \neq \beta_0. \quad (10.13)$$

10.1.3 Identification Lens: The Tension Between Distributional Assumption Identification vs. Exclusion Restriction Identification

In the historical development and contemporary practice of applied microeconometrics, the researcher faces an inevitable structural dilemma: how to ensure that estimated parameters truly represent the actual population Data Generating Process (DGP) and not an artificial byproduct of imposed mathematical assumptions?

This tension splits econometric methodology into two distinct approaches to achieve identification:

- **Identification by Distributional Assumption (Parametric):** Where identification relies on imposing a functional form and strict probability distribution for error terms (typically the Gaussian normal distribution).
- **Identification by Exclusion Restriction (Exogenous Variation / Semiparametric):** Where identification requires the existence of an exogenous variable that alters the selection or participation process without directly affecting the outcome equation.

10.1.3.1 The Identification Paradigm and Manski's Law of Decreasing Credibility

The concept of identification establishes whether it is possible to uniquely recover the parameters of interest from a model given the observable joint distribution of data. Let P be the probability distribution of observed data and $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ be the vector of structural parameters of the theoretical model. We say that $\boldsymbol{\theta}$ is identified if there exists a unique mapping from P to $\boldsymbol{\theta}$. If two distinct parameter vectors, $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$, generate exactly the same observed distribution P , the model is said to be unidentified (or observationally equivalent).

In the context of selection and LDV models, Charles Manski formulated the **Law of Decreasing Credibility**, which states:

“The credibility of structural estimates from an econometric model decreases as the strength of identification assumptions increases.”

Models that achieve identification by imposing rigid distributional assumptions (such as bivariate normality of errors in Tobit and Probit) lie at the extreme of high assumption strength and low credibility. If the normality assumption is violated — which frequently occurs in microdata due to skewness, kurtosis, and heavy tails — the identification point collapses and the maximum likelihood estimator becomes globally inconsistent.

To restore credibility, modern econometrics requires that identification be anchored in exogenous exclusion restrictions, transferring the weight of identification from mathematical curvature to actual physical variation generated by research design.

10.1.3.2 The Fragility of Identification by Distributional Assumption: The Type I Tobit Case

In the classical Type I Tobit model, the observed variable is $Y_i = \max(0, Y_i^*)$, where $Y_i^* = \mathbf{X}_i^T \boldsymbol{\beta}_0 + e_i$. The conditional expectation is given by Tobin's equation:

$$\mathbb{E}[Y_i | \mathbf{X}_i] = \mathbf{X}_i^T \boldsymbol{\beta}_0 \Phi\left(\frac{\mathbf{X}_i^T \boldsymbol{\beta}_0}{\sigma_0}\right) + \sigma_0 \phi\left(\frac{\mathbf{X}_i^T \boldsymbol{\beta}_0}{\sigma_0}\right). \quad (10.14)$$

Identification of the structural parameter vector $\boldsymbol{\beta}_0$ relies entirely on the validity of the hypothesis that $e_i | \mathbf{X}_i \sim N(0, \sigma_0^2)$.

The Failure Mechanism Under Non-Normality: If the true distribution of the latent error deviates from the Gaussian normal distribution, the true truncated conditional expectation function will no longer be proportional to the Inverse Mills Ratio (λ). Unlike the classical OLS linear model (where error non-normality does not affect parameter consistency), in Tobit, breaking the distributional assumption destroys asymptotic consistency of the entire coefficient vector:

$$\text{plim } \hat{\boldsymbol{\beta}}_{\text{Tobit_MLE}} \neq \boldsymbol{\beta}_0 \quad \text{if} \quad e_i | \mathbf{X}_i \not\sim N(0, \sigma_0^2). \quad (10.15)$$

10.1.3.3 The Illusion of Identification by Non-Linearity in the Heckman Model (Type II Tobit)

The Heckman selection model (Type II Tobit) can be estimated without exclusion variables if the selection equation regressor vector is identical to that of the outcome equation

($\mathbf{Z}_i \equiv \mathbf{X}_i$). The estimated equation in the second stage is:

$$Y_i = \mathbf{X}_i^T \boldsymbol{\beta}_0 + \sigma_{UV} \lambda(\mathbf{X}_i^T \boldsymbol{\gamma}_0) + e_i. \quad (10.16)$$

From a purely mathematical standpoint, the vector $(\boldsymbol{\beta}_0^T, \sigma_{UV})^T$ is identified in the population because the Inverse Mills Ratio is a strictly non-linear function. However, this identification constitutes a numerical illusion in finite samples:

- **The Near-Linearity of Mills:** For moderate participation probabilities (in the central range of the normal distribution), the curvature of $\lambda(z)$ is extremely smooth, behaving as $\lambda(z) \approx c_0 + c_1 z$.
- **Variance Collapse:** Without an exclusion restriction, the generated correction term $\hat{\lambda}_i \equiv \lambda(\mathbf{X}_i^T \hat{\boldsymbol{\gamma}})$ becomes almost perfectly correlated with linear regressors $\mathbf{X}_i$. This elevates the Variance Inflation Factor (VIF) to pathological levels, inflating standard errors to infinity.

Proof of Semiparametric Non-Identification Without Exclusion: If we relax the bivariate normality assumption of errors, the true selected conditional relationship assumes the general semiparametric form of Gallant and Nychka (1987):

$$\mathbb{E}[Y_i | \mathbf{X}_i, \mathbf{Z}_i, S_i = 1] = \mathbf{X}_i^T \boldsymbol{\beta}_0 + g(\mathbf{Z}_i^T \boldsymbol{\gamma}_0), \quad (10.17)$$

where $g(\cdot)$ is an unknown continuous selection function.

Suppose there exists an alternative parameter vector $\tilde{\boldsymbol{\beta}} \equiv \boldsymbol{\beta}_0 - \mathbf{M}\boldsymbol{\gamma}_0 \neq \boldsymbol{\beta}_0$ and an alternative selection function $\tilde{g}(z) \equiv g(z) + \mathbf{M}z$, where $\mathbf{M} \in \mathbb{R}^K$ is a non-zero column vector of real constants. Substituting:

$$\mathbf{X}_i^T \tilde{\boldsymbol{\beta}} + \tilde{g}(\mathbf{X}_i^T \boldsymbol{\gamma}_0) = \mathbf{X}_i^T (\boldsymbol{\beta}_0 - \mathbf{M}\boldsymbol{\gamma}_0) + g(\mathbf{X}_i^T \boldsymbol{\gamma}_0) + \mathbf{M}(\mathbf{X}_i^T \boldsymbol{\gamma}_0) = \mathbf{X}_i^T \boldsymbol{\beta}_0 + g(\mathbf{X}_i^T \boldsymbol{\gamma}_0). \quad (10.18)$$

Therefore, without exogenous exclusion restriction, identification of the Heckman selection model depends 100% on the arbitrary imposition of the bivariate Gaussian normal distribution.

10.1.3.4 The Exclusion Restriction Paradigm: Identification Anchored in Research Design

The existence of an exogenous variable W_{i1} contained in $\mathbf{Z}_i$ but strictly excluded from $\mathbf{X}_i$ dictates that:

$$\mathbf{Z}_i = [\mathbf{X}_i, W_{i1}] \quad \text{with} \quad \gamma_W \neq 0. \quad (10.19)$$

This exogenous variable acts as a physical participation lever:

- **Relevance (Effect on Selection):** Systematically alters the individual's participation probability ($\gamma_W \neq 0$).
- **Structural Exclusion (Outcome Orthogonality):** Has no direct causal impact on the outcome variable of interest.

The inclusion of W_{i1} ensures that the Mills ratio indexing term varies independently relative to second-stage regressors, guaranteeing non-singularity of the expected conditional moment matrix:

$$\text{rank} \left(\mathbb{E}[\mathbf{W}_i \mathbf{W}_i^T | S_i = 1] \right) = K + 1, \quad (10.20)$$

where $\mathbf{W}_i \equiv [\mathbf{X}_i^T, \lambda_i]^T$.

10.1.3.5 Structural and Epistemological Comparison

Dimension	Parametric Identification (Probit/Tobit)	Exclusion Identification (Heckman)
Identification Source	Non-linear curvature of Gaussian CDF (Φ)	Physical exogenous variation provided by instrument (W)
Mathematical Premise	Strictly Gaussian and homoskedastic errors	Conditional mean exogeneity of instrument ($W \perp U$)
Robustness to Misspecification	Very low	Very high
Finite Sample Precision	Low standard errors, but masking biases	Realistic standard errors
Credibility (Manski)	Low	High

Table 10.1: Methodological tension: parametric identification vs. exclusion restriction.

10.1.3.6 The Partial Identification Perspective (Manski-Molinari)

When the researcher does not have a strong exogenous exclusion restriction and refuses to impose strict distributional premises, modern econometrics offers the **Partial Identification** (*Set Identification*) framework, developed by Manski and Molinari (2020).

Instead of assuming that the population error is normal to force convergence of $\hat{\beta}$ to a single pseudo-true point, the researcher assumes only weak regularity conditions. The objective becomes identifying the *Sharp* Identification Region $HP[\theta]$:

$$HP[\theta] = \left\{ \theta \in \Theta : \mathbb{E}[Y_{Li} | \mathbf{X}_i] \leq \mathbf{X}_i^T \theta \leq \mathbb{E}[Y_{Ui} | \mathbf{X}_i] \right\}, \quad (10.21)$$

where Y_{Li} and Y_{Ui} represent the lower and upper bounds of observable physical realizations of the limited dependent variable.

10.2 Binary Choice Models and Panel Extensions

10.2.1 Practical Problem: Labor Force Participation Decision

In applied microeconomic analysis, interest frequently shifts from quantitative and continuous dependent variables to qualitative decisions, of a discrete or dichotomous nature. This conceptual scenario is designated as **binary choice models** or qualitative response models.

The observed dependent variable assumes only two possible values:

$$Y_i = \begin{cases} 1, & \text{if the event of interest (success) occurs,} \\ 0, & \text{if the event of interest (failure) does not occur,} \end{cases} \quad (10.22)$$

where the numerical values $\{0, 1\}$ are purely labels of mathematical convenience.

10.2.1.1 The Canonical Example: Labor Force Participation

Consider the decision of a married woman to enter the formal labor market (represented by the dummy variable $\text{inlf}_i \in \{0, 1\}$). According to microeconomic theory of labor supply and time allocation, labor market entry results from individual utility maximization over consumption and leisure.

This process is governed by the interaction of two fundamental economic variables:

- **The Offered Market Wage** (W_i^o): Represents the monetary compensation per hour that firms are willing to pay the individual based on their productivity attributes and human capital.
- **The Reservation Wage** (W_i^r): Represents the minimum subjective wage that would induce the agent to give up leisure and domestic activities to work in the formal market.

Under classical economic rationality:

$$Y_i = \mathbb{I}(W_i^o > W_i^r), \quad (10.23)$$

where $\mathbb{I}(\cdot)$ denotes the structural indicator function.

10.2.1.2 The Conceptual Failure of OLS: The Linear Probability Model (LPM)

The naive approach would be to directly estimate:

$$Y_i = \mathbf{X}_i^T \boldsymbol{\beta} + u_i \implies P(Y_i = 1 \mid \mathbf{X}_i) = \mathbf{X}_i^T \boldsymbol{\beta}. \quad (10.24)$$

Although the LPM provides simple estimates, it presents serious limitations:

- **Predictions Outside the Coherent Support:** The model will generate predicted values $\hat{Y}_i < 0$ or $\hat{Y}_i > 1$ for observations with regressors at extreme levels.
- **Intrinsic Heteroskedasticity:**

$$\text{Var}(u_i \mid \mathbf{X}_i) = \mathbf{X}_i^T \boldsymbol{\beta} (1 - \mathbf{X}_i^T \boldsymbol{\beta}). \quad (10.25)$$

- **Error Non-Normality:** The disturbance u_i assumes a strictly discrete two-point distribution.
- **Inconsistency:** OLS converges to the Best Linear Projection, which depends on the marginal distribution of regressors and suffers from systematic attenuation bias.

10.2.2 Latent Utility Model and Link Function

The central intuition consists of modeling an unobservable continuous indicator of behavioral propensity and mapping this index onto a symmetric-tailed cumulative distribution function (CDF), ensuring that estimated probabilities remain strictly contained within the unit interval.

10.2.2.1 Latent Propensity Model Formulation

Let $Y_i^* \in \mathbb{R}$ be an unobserved latent variable denoting the net benefit or expected net utility:

$$Y_i^* = \mathbf{X}_i^T \boldsymbol{\beta}_0 + e_i, \quad e_i | \mathbf{X}_i \sim G(e), \quad (10.26)$$

where $G(\cdot)$ represents the conditional CDF of the idiosyncratic shock, assumed to be continuous and strictly increasing.

The empirical bridge between the invisible latent utility Y_i^* and the observed physical decision Y_i is given by the sign partition rule:

$$Y_i = \mathbb{I}(Y_i^* > 0) = \begin{cases} 1, & \text{if } Y_i^* > 0 \iff e_i > -\mathbf{X}_i^T \boldsymbol{\beta}_0, \\ 0, & \text{if } Y_i^* \leq 0 \iff e_i \leq -\mathbf{X}_i^T \boldsymbol{\beta}_0. \end{cases} \quad (10.27)$$

10.2.2.2 Probabilistic Mapping and the Role of the Non-Linear Link

The econometrician's objective is to estimate the conditional probability:

$$P(Y_i = 1 | \mathbf{X}_i) = P(Y_i^* > 0 | \mathbf{X}_i) = P(e_i > -\mathbf{X}_i^T \boldsymbol{\beta}_0 | \mathbf{X}_i) = 1 - G(-\mathbf{X}_i^T \boldsymbol{\beta}_0). \quad (10.28)$$

Assuming symmetry ($G(-u) = 1 - G(u)$):

$$P(Y_i = 1 | \mathbf{X}_i) = G(\mathbf{X}_i^T \boldsymbol{\beta}_0). \quad (10.29)$$

Probit Model: $e_i | \mathbf{X}_i \sim \mathcal{N}(0, 1)$, where $G(\cdot) = \Phi(\cdot)$:

$$P(Y_i = 1 | \mathbf{X}_i) = \Phi(\mathbf{X}_i^T \boldsymbol{\beta}_0) \equiv \int_{-\infty}^{\mathbf{X}_i^T \boldsymbol{\beta}_0} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt. \quad (10.30)$$

Logit Model: $e_i | \mathbf{X}_i \sim \text{Logistic}(0, \pi^2/3)$, where $G(\cdot) = \Lambda(\cdot)$:

$$P(Y_i = 1 | \mathbf{X}_i) = \Lambda(\mathbf{X}_i^T \boldsymbol{\beta}_0) \equiv \frac{\exp(\mathbf{X}_i^T \boldsymbol{\beta}_0)}{1 + \exp(\mathbf{X}_i^T \boldsymbol{\beta}_0)}. \quad (10.31)$$

10.2.2.3 Scale Non-Identifiability and the Need for Normalization

Let us multiply the latent equation by an arbitrary scale constant $k > 0$:

$$\tilde{Y}_i^* = \mathbf{X}_i^T (k\boldsymbol{\beta}_0) + ke_i. \quad (10.32)$$

Evaluating the observability rule:

$$Y_i = \mathbb{I}(\tilde{Y}_i^* > 0) = \mathbb{I}(kY_i^* > 0) \equiv \mathbb{I}(Y_i^* > 0). \quad (10.33)$$

Therefore, we uniquely identify the slope vector standardized by the scale factor:

$$\boldsymbol{\theta}_0 = \frac{\boldsymbol{\beta}_0}{\sigma}. \quad (10.34)$$

In the Probit model, we fix $\sigma^2 = 1$. In the Logit model, we fix the variance at $\sigma^2 = \pi^2/3$. Under consistency, coefficients approximately satisfy Amemiya's rule:

$$\hat{\boldsymbol{\beta}}_{\text{Logit}} \approx 1.6 \times \hat{\boldsymbol{\beta}}_{\text{Probit}}. \quad (10.35)$$

10.2.3 Identification and Behavioral Assumptions

10.2.3.1 The Latent Error Distribution Hypothesis

The conditional CDF of ε_i is assumed to be continuous, increasing, and symmetric around zero:

$$G(-z) = 1 - G(z), \quad \forall z \in \mathbb{R}. \quad (10.36)$$

Probit Model: $\varepsilon_i \mid \mathbf{X}_i \sim \mathcal{N}(0, 1)$, with:

$$G(z) = \Phi(z) \equiv \int_{-\infty}^z \phi(t) dt = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt. \quad (10.37)$$

Logit Model: $\varepsilon_i \mid \mathbf{X}_i \sim \text{Logistic}(0, \pi^2/3)$, with:

$$G(z) = \Lambda(z) \equiv \frac{\exp(z)}{1 + \exp(z)} = \frac{1}{1 + \exp(-z)}. \quad (10.38)$$

The logistic density has slightly heavier tails than the normal.

10.2.3.2 The Strict Exogeneity Hypothesis

$$\varepsilon_i \perp \mathbf{X}_i, \quad (10.39)$$

which imposes:

$$\mathbb{E}[\varepsilon_i \mid \mathbf{X}_i] = 0, \quad \text{Var}(\varepsilon_i \mid \mathbf{X}_i) = \sigma^2 \quad (\text{constant}). \quad (10.40)$$

Consequences of Exogeneity Collapse: If $\text{Var}(\varepsilon_i \mid \mathbf{X}_i) = h(\mathbf{X}_i)^2 \neq 1$, the true response probability becomes:

$$P(Y_i = 1 \mid \mathbf{X}_i) = G\left(\frac{\mathbf{X}_i^T \boldsymbol{\beta}}{h(\mathbf{X}_i)}\right). \quad (10.41)$$

The Maximum Likelihood Estimator loses consistency globally.

10.2.3.3 Rank Condition and Conditional Support

The population moment matrix of regressors must be non-singular:

$$\boldsymbol{\Sigma}_{XX} \equiv \mathbb{E}[\mathbf{X}_i \mathbf{X}_i^T] \quad \text{must be positive definite (full rank)}. \quad (10.42)$$

Moreover, stable identification requires that at least one regressor be continuous and present sufficiently broad support:

$$P(Y_i = 1 \mid \mathbf{X}_i) = G(\mathbf{X}_i^T \boldsymbol{\beta}) \in (\epsilon, 1 - \epsilon) \quad \text{for some } \epsilon > 0. \quad (10.43)$$

10.2.4 Identification Failure Scenarios

10.2.4.1 Complete Separation (Perfect Prediction)

Complete separation occurs when a regressor perfectly divides sample observations:

$$Y_i = 1 \quad \forall i \text{ where } X_{ik} \geq \tau \quad \text{and} \quad Y_i = 0 \quad \forall i \text{ where } X_{ik} < \tau. \quad (10.44)$$

Since sigmoid functions only assume the limiting values 1 and 0 asymptotically:

$$\lim_{z \rightarrow \infty} G(z) = 1 \quad \text{and} \quad \lim_{z \rightarrow -\infty} G(z) = 0, \quad (10.45)$$

the point estimator of the coefficient diverges:

$$\hat{\beta}_k \rightarrow \infty \quad (\text{or } -\infty). \quad (10.46)$$

Symptoms: Algorithm non-convergence; monumental coefficients ($\hat{\beta}_k > 25$); exploding standard errors ($SE(\hat{\beta}_k) > 2000$).

10.2.4.2 Omission of Orthogonal Relevant Variables

In the Probit/Logit model, omitting a relevant variable, even if strictly orthogonal to included regressors, generates systematic attenuation bias.

Proof: Let the DGP be:

$$Y_i^* = \mathbf{X}_i^T \boldsymbol{\beta}_0 + \gamma_0 Z_i + e_i, \quad e_i \mid \mathbf{X}_i, Z_i \sim \mathcal{N}(0, 1), \quad (10.47)$$

with $\text{Cov}(\mathbf{X}_i, Z_i) = 0$. Omitting $Z_i \sim \mathcal{N}(0, \sigma_Z^2)$, the compound error is $v_i \equiv \gamma_0 Z_i + e_i$, with:

$$v_i \mid \mathbf{X}_i \sim \mathcal{N}(0, \gamma_0^2 \sigma_Z^2 + 1) \equiv \mathcal{N}(0, \sigma_v^2), \quad \sigma_v > 1. \quad (10.48)$$

The omitted conditional probability converges to:

$$\text{plim } \hat{\boldsymbol{\beta}} = \boldsymbol{\beta}^* = \frac{\boldsymbol{\beta}_0}{\sqrt{1 + \gamma_0^2 \sigma_Z^2}}. \quad (10.49)$$

Since $\sqrt{1 + \gamma_0^2 \sigma_Z^2} > 1$:

$$|\boldsymbol{\beta}^*| < |\boldsymbol{\beta}_0| \quad (\text{Attenuation Bias}). \quad (10.50)$$

10.2.5 Likelihood Derivation and Asymptotic Properties

10.2.5.1 Construction of the Conditional Likelihood Function

The observed response variable Y_i , conditional on the covariate vector $\mathbf{X}_i$, follows a Bernoulli distribution:

$$Y_i \mid \mathbf{X}_i \sim \text{Bernoulli}(p_i) \quad \text{where} \quad p_i \equiv P(Y_i = 1 \mid \mathbf{X}_i) = G(\mathbf{X}_i^T \boldsymbol{\beta}_0). \quad (10.51)$$

The conditional Probability Mass Function (PMF) is:

$$f(y_i \mid \mathbf{X}_i; \boldsymbol{\beta}) = [G(\mathbf{x}_i^T \boldsymbol{\beta})]^{y_i} [1 - G(\mathbf{x}_i^T \boldsymbol{\beta})]^{1-y_i}. \quad (10.52)$$

Under i.i.d. random sampling, the Likelihood Function is:

$$\mathcal{L}(\boldsymbol{\beta} \mid \mathbf{X}, \mathbf{Y}) = \prod_{i=1}^n [G(\mathbf{x}_i^T \boldsymbol{\beta})]^{y_i} [1 - G(\mathbf{x}_i^T \boldsymbol{\beta})]^{1-y_i}. \quad (10.53)$$

The Log-Likelihood Function is:

$$\ln \mathcal{L}(\boldsymbol{\beta}) = \sum_{i=1}^n \left\{ y_i \ln G(\mathbf{x}_i^T \boldsymbol{\beta}) + (1 - y_i) \ln [1 - G(\mathbf{x}_i^T \boldsymbol{\beta})] \right\}. \quad (10.54)$$

10.2.5.2 Score Vector Derivation and First-Order Conditions

The Score vector is:

$$\mathbf{s}_n(\boldsymbol{\beta}) \equiv \frac{\partial \ln \mathcal{L}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \sum_{i=1}^n \frac{y_i - G(\mathbf{x}_i^T \boldsymbol{\beta})}{G(\mathbf{x}_i^T \boldsymbol{\beta})[1 - G(\mathbf{x}_i^T \boldsymbol{\beta})]} g(\mathbf{x}_i^T \boldsymbol{\beta}) \mathbf{x}_i = \mathbf{0}. \quad (10.55)$$

Probit Score ($G = \Phi$, $g = \phi$):

$$\mathbf{s}_n(\boldsymbol{\beta})_{\text{probit}} = \sum_{i=1}^n \frac{y_i - \Phi(\mathbf{x}_i^T \boldsymbol{\beta})}{\Phi(\mathbf{x}_i^T \boldsymbol{\beta})[1 - \Phi(\mathbf{x}_i^T \boldsymbol{\beta})]} \phi(\mathbf{x}_i^T \boldsymbol{\beta}) \mathbf{x}_i = \mathbf{0}. \quad (10.56)$$

Logit Score ($G = \Lambda$, $g = \Lambda(1 - \Lambda)$):

$$\mathbf{s}_n(\boldsymbol{\beta})_{\text{logit}} = \sum_{i=1}^n (y_i - \Lambda(\mathbf{x}_i^T \boldsymbol{\beta})) \mathbf{x}_i = \mathbf{0}. \quad (10.57)$$

There is no closed-form analytical solution; iterative algorithms (Newton-Raphson or Fisher Scoring) are required.

10.2.5.3 Demonstration of OLS Inconsistency

The OLS estimator is:

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^T \right)^{-1} \sum_{i=1}^n \mathbf{x}_i y_i. \quad (10.58)$$

By the Weak Law of Large Numbers:

$$\text{plim } \hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbb{E}[\mathbf{X}_i \mathbf{X}_i^T])^{-1} \mathbb{E}[\mathbf{X}_i Y_i]. \quad (10.59)$$

By the Law of Iterated Expectations:

$$\mathbb{E}[\mathbf{X}_i Y_i] = \mathbb{E}[\mathbf{X}_i G(\mathbf{X}_i^T \boldsymbol{\beta}_0)]. \quad (10.60)$$

Therefore:

$$\text{plim } \hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbb{E}[\mathbf{X}_i \mathbf{X}_i^T])^{-1} \mathbb{E}[\mathbf{X}_i G(\mathbf{X}_i^T \boldsymbol{\beta}_0)] \neq \boldsymbol{\beta}_0. \quad (10.61)$$

OLS converges to the Best Linear Projection (BLP), which depends on the marginal distribution of regressors and suffers from systematic attenuation bias.

10.2.5.4 MLE Asymptotic Properties

Under regularity conditions, correct specification, and identification:

1. **Score moment:** $\mathbb{E}[\mathbf{s}_i(\boldsymbol{\beta}_0) \mid \mathbf{X}_i] = \mathbf{0}$.
2. **Information equality:** $\mathbb{E}[\mathbf{s}_i \mathbf{s}_i'] = -\mathbb{E}[\mathbf{H}_i]$, where $\mathbf{H}_i$ is the Hessian.
3. **Consistency:** $\hat{\boldsymbol{\beta}}_{\text{MLE}} \xrightarrow{p} \boldsymbol{\beta}_0$.
4. **Asymptotic normality:** $\sqrt{N}(\hat{\boldsymbol{\beta}}_{\text{MLE}} - \boldsymbol{\beta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathcal{I}(\boldsymbol{\beta}_0)^{-1})$.
5. **Efficiency:** Attains the Cramér-Rao lower bound.

10.2.6 Economic Interpretation: Marginal Effects

The golden rule of applied microeconometrics is: **only marginal effects are interpretable**. The raw coefficients $\boldsymbol{\beta}$ measure the effect on latent utility, not on probability.

10.2.6.1 Instantaneous Marginal Effect

The marginal effect of a continuous regressor X_j is:

$$\text{ME}_{ij} \equiv \frac{\partial P(Y_i = 1 \mid \mathbf{X}_i)}{\partial X_{ij}} = \beta_j \cdot g(\mathbf{X}_i^T \boldsymbol{\beta}), \quad (10.62)$$

where $g(z) = G'(z)$ is the probability density function (PDF).

Important note: The marginal effect depends on the evaluation point $\mathbf{X}_i^T \boldsymbol{\beta}$ through the density $g(\cdot)$. For individuals with probability already close to 0 or 1, the marginal effect will be small (the density is low in the tails).

10.2.6.2 Marginal Effect for Dummy Variables

For binary variables $D_i \in \{0, 1\}$, the derivative is not appropriate. The correct marginal effect is the discrete difference:

$$\text{ME}_{iD} \equiv P(Y_i = 1 \mid \mathbf{X}_i, D_i = 1) - P(Y_i = 1 \mid \mathbf{X}_i, D_i = 0). \quad (10.63)$$

10.2.6.3 Summary Measures: AME and MEM

Average Marginal Effect (AME/APE):

$$\text{AME}_j = \frac{1}{N} \sum_{i=1}^N \hat{\beta}_j \cdot g(\mathbf{X}_i^T \hat{\boldsymbol{\beta}}). \quad (10.64)$$

Marginal Effect at the Mean (MEM):

$$\text{MEM}_j = \hat{\beta}_j \cdot g(\bar{\mathbf{X}}^T \hat{\boldsymbol{\beta}}). \quad (10.65)$$

Contemporary microeconometrics recommends AME as the standard reporting measure, since it captures sample heterogeneity and is more robust to extreme values.

10.2.7 Extensions for Panel Data

In many microeconomic applications, we observe the same individual i making decisions in multiple periods $t = 1, \dots, T_i$. This panel data structure requires extensions of binary choice models to accommodate unobserved heterogeneity and intra-individual correlation.

10.2.7.1 The Incidental Parameters Problem in Binary Panel

Consider the latent variable model for binary choice in panel:

$$y_{it}^* = \alpha_i + \mathbf{x}_{it}'\boldsymbol{\beta} + \varepsilon_{it}, \quad y_{it} = \mathbb{I}(y_{it}^* > 0), \quad (10.66)$$

where α_i is the individual-specific unobserved heterogeneity.

In the linear model, the *within* transformation (demeaning) perfectly eliminates α_i . However, in non-linear models like Logit and Probit, there is no mathematical transformation that eliminates α_i without distorting the non-linear link function.

If we estimate the model by Unconditional Maximum Likelihood (Fixed Effects), we need to estimate N intercepts α_i . Since $N \rightarrow \infty$ and T is fixed (and generally small), the estimators $\hat{\alpha}_i$ are inconsistent. Since $\hat{\boldsymbol{\beta}}$ functionally depends on $\hat{\alpha}_i$, the estimator of $\boldsymbol{\beta}$ also becomes inconsistent. This is the **incidental parameters problem** (Neyman and Scott, 1948).

The bias of the Fixed Effects Maximum Likelihood estimator depends on T . For static models, the general rule is $\text{plim}(\hat{\boldsymbol{\beta}}_{FE}) = S(T)\boldsymbol{\beta}$, where $S(2) \approx 2$ (100% bias if $T = 2$), and $\lim_{T \rightarrow \infty} S(T) = 1$.

10.2.7.2 Chamberlain's Conditional Logit

For the Logit model, there is an elegant solution developed by Chamberlain (1980). The sum of choices over time, $S_i = \sum_{t=1}^T y_{it}$, is a **sufficient statistic** for α_i . Conditioning the likelihood function on S_i , the parameter α_i is eliminated.

Example for $T = 2$: If $S_i = 1$, the individual changed state (chose 1 in one period and 0 in the other). The conditional probability is:

$$P(y_{i1} = 1, y_{i2} = 0 \mid S_i = 1) = \frac{P(y_{i1} = 1)P(y_{i2} = 0)}{P(y_{i1} = 1)P(y_{i2} = 0) + P(y_{i1} = 0)P(y_{i2} = 1)}. \quad (10.67)$$

Substituting the Logit probabilities, the term $\exp(\alpha_i)$ appears in both numerator and denominator and cancels perfectly:

$$P(y_{i1} = 1, y_{i2} = 0 \mid S_i = 1) = \frac{\exp(\mathbf{x}_{i1}'\boldsymbol{\beta})}{\exp(\mathbf{x}_{i1}'\boldsymbol{\beta}) + \exp(\mathbf{x}_{i2}'\boldsymbol{\beta})}. \quad (10.68)$$

Limitations:

- Individuals who do not vary ($S_i = 0$ or $S_i = T$) are discarded from estimation, which may generate selection bias.
- Since α_i is eliminated, it is not possible to calculate marginal effects or predicted probabilities, only coefficient ratios.
- There is no exact analog for the Probit model.

10.2.7.3 Mundlak's Approach (CRE)

To overcome Conditional Logit limitations and allow calculation of marginal effects, Mundlak (1978) proposes projecting unobserved heterogeneity onto time means of regressors:

$$\alpha_i = \bar{\alpha} + \bar{\mathbf{x}}_i' \boldsymbol{\delta} + u_i, \quad (10.69)$$

where $\bar{\mathbf{x}}_i = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_{it}$.

Substituting into the original model:

$$y_{it}^* = \bar{\alpha} + \bar{\mathbf{x}}_i' \boldsymbol{\delta} + \mathbf{x}_{it}' \boldsymbol{\beta} + (u_i + \varepsilon_{it}). \quad (10.70)$$

The model is estimated as Random Effects, but including time means $\bar{\mathbf{x}}_i$ as additional regressors. The term $\bar{\mathbf{x}}_i' \boldsymbol{\delta}$ absorbs the correlation between α_i and $\mathbf{x}_{it}$.

Advantages:

- Allows calculation of marginal effects and probabilities.
- Uses all data (does not discard individuals without variation).
- The Hausman test for choosing between Random Effects and Fixed Effects reduces to a Wald test for $H_0 : \boldsymbol{\delta} = \mathbf{0}$.

10.2.8 Mixed Logit and Preference Heterogeneity

10.2.8.1 Formulation with Random Coefficients

In panel data with multiple choices per individual, or in stated choice experiments, the same individual i faces multiple choice situations $t = 1, \dots, T_i$. Unobserved preferences are constant over time, generating intra-individual correlation.

The **Mixed Logit** model allows coefficients to vary across individuals according to a probability distribution:

$$U_{ijt} = \mathbf{x}_{ijt}' \boldsymbol{\beta}_i + \varepsilon_{ijt}, \quad \boldsymbol{\beta}_i = \bar{\boldsymbol{\beta}} + \boldsymbol{\Gamma} \mathbf{u}_i, \quad (10.71)$$

where $\mathbf{u}_i \sim D(\mathbf{0}, \mathbf{I})$ and $\boldsymbol{\Gamma}$ is a scale matrix allowing correlation between random coefficients.

The unconditional choice probability is a multivariate integral:

$$P_{ijt} = \int \frac{\exp(\mathbf{x}_{ijt}' \boldsymbol{\beta})}{\sum_{m=1}^J \exp(\mathbf{x}_{imt}' \boldsymbol{\beta})} f(\boldsymbol{\beta} \mid \bar{\boldsymbol{\beta}}, \boldsymbol{\Omega}) d\boldsymbol{\beta}, \quad (10.72)$$

where $\boldsymbol{\Omega} = \boldsymbol{\Gamma} \boldsymbol{\Gamma}'$.

Estimation is done by **Simulated Maximum Likelihood (SML)**, using Halton sequences or the GHK simulator to approximate the integral. Mixed Logit relaxes IIA, allows arbitrary substitution patterns, and captures preference heterogeneity.

10.2.8.2 Connection with Nested Logit and Error Components

Nested Logit can be seen as a restricted case of Mixed Logit (error components model). If only alternative-specific constants are random, and correlation is imposed by nests:

$$U_{ij} = \alpha_j + u_{i,\text{nest}(j)} + \mathbf{x}'_{ij}\boldsymbol{\beta} + \varepsilon_{ij}. \quad (10.73)$$

The term $u_{i,\text{nest}(j)}$ is shared by all alternatives within the same nest, generating error correlation and relaxing IIA only between nests. Estimating Nested Logit as a Mixed Logit with grouped random constants produces virtually identical results.

10.2.9 Stated Choice Experiments and Willingness to Pay

10.2.9.1 Revealed vs Stated Preference

In many contexts, stated choice experiments are used, where respondents evaluate hypothetical scenarios. The advantage is allowing analysis of non-existent alternatives and controlled variation in attributes. The disadvantage is statement bias.

Frequently, revealed preference (RP) and stated preference (SP) data are combined. To accommodate possible differences in utility scales between the two data types, one can use a Nested Logit model grouping RP and SP into distinct nests, allowing different scale parameters (λ) for each nest.

10.2.9.2 Willingness to Pay (WTP) and Distribution Choice

In unlabeled experiments, the focus is on Willingness to Pay (WTP) for attributes. Given a price coefficient β_{price} , the WTP for attribute k is:

$$\text{WTP}_k = -\frac{\beta_k}{\beta_{\text{price}}}. \quad (10.74)$$

In Mixed Logit, since β_k is a random variable, WTP is also a distribution. The choice of distribution is crucial for behavioral plausibility:

- **Normal:** allows the coefficient to change sign in the population (suitable for ambiguous attributes, such as brand preference).
- **Log-Normal:** imposes that the coefficient has a constant sign (suitable for universally undesirable attributes, such as cost or travel time). However, it produces thick tails, generating implausibly high WTP values.
- **Triangular:** limits coefficient dispersion, avoiding thick tails and producing more behaviorally plausible WTP distributions.

10.2.9.3 Attribute Non-Attendance (ANA)

In stated surveys, some respondents ignore certain attributes when making decisions (heuristics). This can be modeled via latent class models, where a fraction of the population has coefficients for certain attributes forced to zero, while the other fraction considers all attributes.

10.2.10 Synthesis and Practical Guide

The choice of binary choice model and its extensions must respect three principles:

1. **Data nature:** simple cross-sectional data require standard Logit/Probit. Panel data require treatment of unobserved heterogeneity (Chamberlain, Mundlak, or Random Effects).
2. **Research interest:** if only coefficient ratios matter, Chamberlain's Conditional Logit is sufficient. If marginal effects and predicted probabilities are needed, use Mundlak or Random Effects.
3. **Interpretation by marginal effects:** raw coefficients are not effects on probabilities. Always report AME (Average Marginal Effects) as the standard measure.

Chapter 11

Limited Variable Models

11.1 Truncated Data

The central objective of this section is to infer the characteristics of a complete population from a sample drawn only from a restricted part (subpopulation) of that population.

11.1.1 Truncated Distributions

A truncated distribution is the part of an original (untruncated) distribution that lies strictly above or below a specified value.

Example: Analyzing only incomes above \$100,000 means observing a subset of the complete income distribution (which ranges from zero to infinity).

Theorem 19.1: Density of a Truncated Random Variable

Statement. If a continuous random variable x has probability density function (pdf) $f(x)$ and a is a constant, then the density of the truncated variable (conditioned on $x > a$) is:

$$f(x | x > a) = \begin{cases} \frac{f(x)}{P(x > a)} & \text{if } x > a, \\ 0 & \text{if } x \leq a. \end{cases} \quad (11.1)$$

Rigorous Proof via CDF (Recommended Classical Approach).

We first define the conditional CDF $F(x | X > a) = P(X \leq x | X > a)$ and then differentiate it with respect to x to obtain the pdf. By the definition of conditional probability:

$$F(x | X > a) = \frac{P(X \leq x \cap X > a)}{P(X > a)}. \quad (11.2)$$

We analyze the behavior in two regions of the support:

- **Case 1: For $x \leq a$.** The events $\{X \leq x\}$ and $\{X > a\}$ are mutually exclusive (empty intersection). Therefore:

$$F(x | X > a) = 0. \quad (11.3)$$

- **Case 2: For $x > a$.** The intersection between $\{X \leq x\}$ and $\{X > a\}$ is the interval $(a, x]$. Thus:

$$F(x | X > a) = \frac{P(a < X \leq x)}{P(X > a)} = \frac{F(x) - F(a)}{P(X > a)}. \quad (11.4)$$

The complete conditional CDF is:

$$F(x | X > a) = \begin{cases} 0 & \text{if } x \leq a, \\ \frac{F(x) - F(a)}{P(X > a)} & \text{if } x > a. \end{cases} \quad (11.5)$$

To obtain the pdf, we differentiate $F(x | X > a)$ with respect to x using the Fundamental Theorem of Calculus (noting that $F(a)$ and $P(X > a)$ are constants with respect to x):

- For $x \leq a$: $f(x | X > a) = \frac{d}{dx}(0) = 0$
- For $x > a$: $f(x | X > a) = \frac{d}{dx} \left(\frac{F(x) - F(a)}{P(X > a)} \right) = \frac{f(x)}{P(X > a)}$

This completes the proof. ■

Alternative Approach: Formalization via Infinitesimal Limit.

The conditional pdf is formally defined by the limit:

$$f(x | X > a) = \lim_{dx \rightarrow 0^+} \frac{P(x \leq X < x + dx | X > a)}{dx}. \quad (11.6)$$

Applying conditional probability:

$$f(x | X > a) = \lim_{dx \rightarrow 0^+} \frac{P(\{x \leq X < x + dx\} \cap \{X > a\})}{dx \cdot P(X > a)}. \quad (11.7)$$

- **If $x < a$:** For dx sufficiently small ($dx < a - x$), the interval $[x, x + dx)$ does not intersect the region $X > a$. The intersection is empty, so $f(x | X > a) = 0$.
- **If $x > a$:** For dx sufficiently small, $[x, x + dx) \subset (a, \infty)$. The numerator is $\int_x^{x+dx} f(u) du$. By the **Mean Value Theorem for Integrals**, there exists $c \in [x, x + dx]$ such that the integral equals $f(c) dx$. Substituting:

$$f(x | X > a) = \lim_{dx \rightarrow 0^+} \frac{f(c) dx}{dx \cdot P(X > a)} = \lim_{dx \rightarrow 0^+} \frac{f(c)}{P(X > a)} = \frac{f(x)}{P(X > a)}, \quad (11.8)$$

since $c \rightarrow x$ as $dx \rightarrow 0^+$ and f is continuous. ■

Interpretation Note. The term $P(X > a)$ in the denominator acts as a **normalization constant (scale factor)**. Since the area under $f(x)$ to the right of a is less than 1, dividing by $P(X > a)$ rescales the density so that the new pdf integrates exactly to 1 over the restricted support (a, ∞) .

Continuous Application: The Truncated Normal Distribution

Most recent applications with continuous variables use the **truncated normal distribution**. If $x \sim N(\mu, \sigma^2)$, we define the standardized truncation point as $\alpha = \frac{a-\mu}{\sigma}$. The probability that $x > a$ is $P(x > a) = 1 - \Phi(\alpha)$, where $\Phi(\cdot)$ is the CDF of the standard normal.

The density of the normal truncated from below at a is:

$$f(x | x > a) = \frac{1}{1 - \Phi(\alpha)} \cdot \frac{1}{\sigma} \phi\left(\frac{x - \mu}{\sigma}\right), \quad x > a, \quad (11.9)$$

where $\phi(\cdot)$ is the pdf of the standard normal.

Interpretation: As we truncate the left tail (increasing a), the observed distribution not only loses the tail but **shifts its mean to the right** and alters its variance, since the density is rescaled so that the total area equals 1.

Discrete Application: The Poisson Distribution Truncated at Zero

For discrete variables, the **Poisson truncated at zero** stands out. If $Y \sim \text{Poisson}(\lambda)$, we want $P(Y = y | Y > 0)$.

Complete and Rigorous Proof.

By the definition of conditional probability for discrete variables:

$$P(Y = y | Y > 0) = \frac{P(Y = y \cap Y > 0)}{P(Y > 0)}. \quad (11.10)$$

1. The Numerator: Intersection of Events

- **Case 1: For $y \leq 0$.** Since Y has support on the non-negative integers, if $y \leq 0$, the events $\{Y = y\}$ and $\{Y > 0\}$ are mutually exclusive (empty intersection):

$$P(Y = y \cap Y > 0) = 0. \quad (11.11)$$

- **Case 2: For $y \in \{1, 2, 3, \dots\}$.** Since $y > 0$, the event $\{Y = y\}$ is fully contained in $\{Y > 0\}$. The intersection is the event itself:

$$P(Y = y \cap Y > 0) = P(Y = y) = \frac{e^{-\lambda} \lambda^y}{y!}. \quad (11.12)$$

2. The Denominator: Scale Factor

The denominator is the probability of the complement of $Y = 0$:

$$P(Y > 0) = 1 - P(Y = 0) = 1 - \frac{e^{-\lambda} \lambda^0}{0!} = 1 - e^{-\lambda}. \quad (11.13)$$

3. Final Conditional Probability Function

Combining the cases of the numerator with the denominator:

$$P(Y = y \mid Y > 0) = \begin{cases} \frac{e^{-\lambda} \lambda^y}{y! (1 - e^{-\lambda})} & \text{if } y = 1, 2, 3, \dots \\ 0 & \text{if } y \leq 0. \end{cases} \quad (11.14)$$

Proof that the pmf sums to 1 (validity check).

$$\sum_{y=1}^{\infty} P(Y = y \mid Y > 0) = \sum_{y=1}^{\infty} \frac{e^{-\lambda} \lambda^y}{y! (1 - e^{-\lambda})} = \frac{e^{-\lambda}}{1 - e^{-\lambda}} \sum_{y=1}^{\infty} \frac{\lambda^y}{y!}. \quad (11.15)$$

By the Taylor series expansion of e^λ :

$$e^\lambda = \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} = 1 + \sum_{y=1}^{\infty} \frac{\lambda^y}{y!} \implies \sum_{y=1}^{\infty} \frac{\lambda^y}{y!} = e^\lambda - 1. \quad (11.16)$$

Substituting:

$$\frac{e^{-\lambda}}{1 - e^{-\lambda}} \cdot (e^\lambda - 1) = \frac{e^{-\lambda} \cdot e^\lambda - e^{-\lambda}}{1 - e^{-\lambda}} = \frac{1 - e^{-\lambda}}{1 - e^{-\lambda}} = 1. \quad \blacksquare \quad (11.17)$$

Practical Application: This distribution is used in count models of recreational or tourist facility usage, where observations of “zero uses” are discarded (the sample includes only individuals who used the facility at least once).

Terminological Convention: A random variable whose underlying distribution has been truncated will be called simply a **truncated random variable**.

11.1.2 Moments of Truncated Distributions

The main interest lies in the **mean** and **variance** of the truncated random variable. They are obtained by the general formulas:

$$E[x \mid x > a] = \int_a^{\infty} x f(x \mid x > a) dx, \quad (11.18)$$

and analogously for the variance.

Example 19.1: Truncated Uniform Distribution

If x has a standard uniform distribution, $x \sim U(0, 1)$, then:

$$f(x) = 1, \quad 0 \leq x \leq 1. \quad (11.19)$$

The distribution truncated at $x = 1/3$ (from below) is also uniform:

$$f(x \mid x > 1/3) = \frac{f(x)}{P(x > 1/3)} = \frac{1}{2/3} = \frac{3}{2}, \quad 1/3 \leq x \leq 1. \quad (11.20)$$

Proof of the Truncated Mean.

$$E[x \mid x > 1/3] = \int_{1/3}^1 x \cdot \frac{3}{2} dx = \frac{3}{2} \left[\frac{x^2}{2} \right]_{1/3}^1 = \frac{3}{2} \left(\frac{1}{2} - \frac{1}{18} \right) = \frac{3}{2} \cdot \frac{8}{18} = \frac{2}{3}. \quad (11.21)$$

For a variable uniformly distributed between L and U , the variance is $(U - L)^2/12$. Thus:

Proof of the Truncated Variance.

$$\text{Var}[x \mid x > 1/3] = \frac{(1 - 1/3)^2}{12} = \frac{(2/3)^2}{12} = \frac{4/9}{12} = \frac{1}{27}. \quad (11.22)$$

The mean and variance of the **untruncated** distribution are $1/2$ and $1/12$, respectively.

Two General Conclusions from Example 19.1

1. **Mean Shift:** If truncation is from below, the mean of the truncated variable is **greater** than the mean of the original. If truncation is from above, the mean of the truncated variable is **smaller** than the mean of the original.
2. **Variance Reduction:** Truncation **reduces the variance** compared to the variance of the untruncated distribution.

From this point onward, we will use the terms **truncated mean** and **truncated variance** to refer to the mean and variance of the random variable with a truncated distribution.

Theorem 19.2: Moments of the Truncated Normal Distribution

Theorem 19.2 is a fundamental analytical result in microdata econometrics and discrete choice decisions, serving as the theoretical basis for classic models such as the **Tobit** and the **Heckman Sample Selection Model (Heckit)**. The great theoretical elegance of the equations presented lies in their ability to **unify the two types of truncation** — lower/left ($x > a$) and upper/right ($x < a$) — into a single simple, consistent, and symmetric parametric structure.

Statement. If $x \sim N[\mu, \sigma^2]$ and a is a constant, then:

$$E[x \mid \text{truncation}] = \mu + \sigma\lambda(\alpha), \quad (19-1)$$

$$\text{Var}[x \mid \text{truncation}] = \sigma^2[1 - \delta(\alpha)], \quad (19-2)$$

where $\alpha = (a - \mu)/\sigma$, $\phi(\alpha)$ is the density of the standard normal, and:

$$\lambda(\alpha) = \frac{\phi(\alpha)}{1 - \Phi(\alpha)} \quad \text{if truncation is } x > a, \quad (19-3a)$$

$$\lambda(\alpha) = -\frac{\phi(\alpha)}{\Phi(\alpha)} \quad \text{if truncation is } x < a, \quad (19-3b)$$

and

$$\delta(\alpha) = \lambda(\alpha)[\lambda(\alpha) - \alpha]. \quad (19-4)$$

Rigorous Proof of the Moments of the Truncated Normal.

To establish the proof with maximum rigor, we begin by defining the standardized normal variable $Z = \frac{x-\mu}{\sigma} \sim N(0, 1)$. The cutoff point a on the original scale of x is equivalent to $\alpha = \frac{a-\mu}{\sigma}$ on the standardized scale of Z .

A. Conditional Expectation (Equation 19-1)

Consider first the case of lower (below) truncation, where the condition is $x > a$ (or $Z > \alpha$):

The conditional probability density function (pdf) of Z given $Z > \alpha$ is:

$$f(z | Z > \alpha) = \frac{\phi(z)}{1 - \Phi(\alpha)}, \quad z > \alpha. \quad (11.23)$$

Where $1 - \Phi(\alpha)$ is the normalization scale constant. The conditional expectation of the standardized variable is:

$$E[Z | Z > \alpha] = \int_{\alpha}^{\infty} z \frac{\phi(z)}{1 - \Phi(\alpha)} dz = \frac{1}{1 - \Phi(\alpha)} \int_{\alpha}^{\infty} z \phi(z) dz. \quad (11.24)$$

Knowing that the standard normal density is $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$, its derivative with respect to z is:

$$\frac{d\phi(z)}{dz} = -z\phi(z) \implies z\phi(z) = -\frac{d\phi(z)}{dz}. \quad (11.25)$$

Substituting this identity into the integral, we obtain directly by the Fundamental Theorem of Calculus:

$$\int_{\alpha}^{\infty} z\phi(z) dz = [-\phi(z)]_{\alpha}^{\infty} = 0 - (-\phi(\alpha)) = \phi(\alpha). \quad (11.26)$$

Therefore:

$$E[Z | Z > \alpha] = \frac{\phi(\alpha)}{1 - \Phi(\alpha)} \equiv \lambda(\alpha). \quad (11.27)$$

Since $x = \mu + \sigma Z$, by linearity of expectation:

$$E[x | x > a] = \mu + \sigma E[Z | Z > \alpha] = \mu + \sigma \lambda(\alpha). \quad (11.28)$$

This perfectly demonstrates equation (19-1) using the definition of lower truncation (19-3a). The derivation for upper truncation $x < a$ follows analogously, yielding $E[Z | Z < \alpha] = -\frac{\phi(\alpha)}{\Phi(\alpha)}$, which mathematically validates the opposite sign definition in (19-3b). ■

B. Conditional Variance (Equation 19-2)

To find the conditional variance, we first compute the second moment of Z conditional on $Z > \alpha$:

$$E[Z^2 | Z > \alpha] = \frac{1}{1 - \Phi(\alpha)} \int_{\alpha}^{\infty} z^2 \phi(z) dz. \quad (11.29)$$

Applying integration by parts with $u = z \implies du = dz$ and $dv = z\phi(z) dz \implies v = -\phi(z)$:

$$\int_{\alpha}^{\infty} z(z\phi(z)) dz = [-z\phi(z)]_{\alpha}^{\infty} - \int_{\alpha}^{\infty} (-\phi(z)) dz = \alpha\phi(\alpha) + (1 - \Phi(\alpha)). \quad (11.30)$$

Dividing by the scale term $1 - \Phi(\alpha)$, we obtain:

$$E[Z^2 \mid Z > \alpha] = \frac{\alpha\phi(\alpha)}{1 - \Phi(\alpha)} + 1 = \alpha\lambda(\alpha) + 1. \quad (11.31)$$

Now, the conditional variance of the standardized variable Z is given by:

$$\begin{aligned} \text{Var}[Z \mid Z > \alpha] &= E[Z^2 \mid Z > \alpha] - (E[Z \mid Z > \alpha])^2 \\ &= 1 + \alpha\lambda(\alpha) - \lambda(\alpha)^2 = 1 - \lambda(\alpha)[\lambda(\alpha) - \alpha]. \end{aligned} \quad (11.32)$$

Substituting the definition of $\delta(\alpha) = \lambda(\alpha)[\lambda(\alpha) - \alpha]$ established in **(19-4)**:

$$\text{Var}[Z \mid Z > \alpha] = 1 - \delta(\alpha). \quad (11.33)$$

Finally, applying the scale properties of variance, where $\text{Var}[\mu + \sigma Z] = \sigma^2 \text{Var}[Z]$:

$$\text{Var}[x \mid x > a] = \sigma^2[1 - \delta(\alpha)]. \quad (11.34)$$

This demonstrates in a unified and definitive manner equation **(19-2)** for both cases of truncation. ■

Theoretical and Conceptual Analysis of the Theorem Notes

1. The Fundamental Property of $\delta(\alpha)$

The statement that $0 < \delta(\alpha) < 1$ **for all values of α** has enormous practical relevance.

- **Geometric Intuition:** Truncation removes one or both tails of the distribution, concentrating the remaining probability density on a smaller support. Since extreme and dispersed observations are discarded, the dispersion of the truncated variable is strictly smaller than that of the original population.
- **Algebraic Verification:** This guarantees that the truncated variance $\sigma^2[1 - \delta(\alpha)]$ is always strictly smaller than the original unconditional variance σ^2 . The variance ratio $[1 - \delta(\alpha)]$ always lies in the open interval $(0, 1)$, decreasing as truncation becomes more severe (i.e., the smaller the portion of data retained, the smaller the remaining variance).

2. Remarkable Connection: $\delta(\alpha)$ as the Derivative of the Inverse Mills Ratio

A mathematical property of extreme elegance that enriches this theorem is that $\delta(\alpha)$ **is, in fact, the derivative of the Inverse Mills Ratio with respect to the cutoff point α !**

If we differentiate $\lambda(\alpha) = \frac{\phi(\alpha)}{1 - \Phi(\alpha)}$ with respect to α using the quotient rule:

$$\lambda'(\alpha) = \frac{\phi'(\alpha)(1 - \Phi(\alpha)) - \phi(\alpha)(-\phi(\alpha))}{(1 - \Phi(\alpha))^2}. \quad (11.35)$$

Substituting the derivative $\phi'(\alpha) = -\alpha\phi(\alpha)$:

$$\lambda'(\alpha) = \frac{-\alpha\phi(\alpha)}{1 - \Phi(\alpha)} + \left(\frac{\phi(\alpha)}{1 - \Phi(\alpha)} \right)^2 = -\alpha\lambda(\alpha) + \lambda(\alpha)^2, \quad (11.36)$$

$$\lambda'(\alpha) = \lambda(\alpha)[\lambda(\alpha) - \alpha] \equiv \delta(\alpha). \quad (11.37)$$

Since $\lambda(\alpha)$ is strictly decreasing and convex for upper truncation (and strictly increasing for structured lower truncation), the fact that $\lambda'(\alpha) = \delta(\alpha)$ belongs to the interval $(0, 1)$ confirms the mathematical stability of the index model over the entire real line.

3. Economic Application and Hazard Function

- **The Inverse Mills Ratio (IMR):** Represented by $\lambda(\alpha)$, the IMR is the heart of the **Heckman two-step estimator**. If we perform an ordinary least squares (OLS) regression using only the selected subsample (where $y > 0$), the term $\sigma\lambda(\alpha)$ acts as an **omitted variable**. Since the IMR is correlated with the other independent variables, its omission causes the classic selection bias in the coefficients.
- **Hazard Function:** In the context of survival analysis (duration models), the IMR $\frac{\phi(\alpha)}{1-\Phi(\alpha)}$ is called the **hazard rate** or instantaneous failure rate. It measures the conditional probability that a “failure” occurs at instant α , given that the individual survived until that period.

Example 19.2: A Truncated Lognormal Income Distribution

An article in the *New York Post* in 1987 stated: “The typical ‘upper-class wealthy American’ ... earns \$142,000 per year ... The people surveyed had family incomes of at least \$100,000.” Would this statistic tell us anything about the typical American? Probably not, since only 2% of American families fell into this category. With a truncation degree of 98%, the \$142,000 was probably very far from the mean of the complete population.

Practical Application: Recovering the Population Mean.

Suppose that incomes x in the population are **lognormally distributed**. Then the log of income, $y = \ln(x)$, has a normal distribution with mean μ and standard deviation σ .

Data from the article:

- $P[x \geq 100] = P[\exp(y) \geq 100] = 0.02$, that is, $P[y < 4.605] = 0.98$ (since $\ln 100 = 4.605$).
- $E[x \mid x > 100] = E[\exp(y) \mid \exp(y) > 100] = 142$ (since $\ln 142 = 4.956$).

Step 1: Find a relationship between μ and σ

From the first piece of information:

$$P\left[\frac{y - \mu}{\sigma} < \frac{4.605 - \mu}{\sigma}\right] = 0.98. \quad (11.38)$$

Since $\Phi\left(\frac{4.605 - \mu}{\sigma}\right) = 0.98$, we know that $\Phi^{-1}(0.98) = 2.054 = \frac{4.605 - \mu}{\sigma}$.

Therefore:

$$4.605 = \mu + 2.054\sigma. \quad (\text{Equation 1})$$

Step 2: Use the truncated lognormal result

For the lognormal distribution, we have the following result:

$$\text{If } y \sim N[\mu, \sigma^2], \text{ then } E[\exp(y) \mid y > a] = \exp(\mu + \sigma^2/2) \cdot \frac{\Phi(\sigma - (a - \mu)/\sigma)}{1 - \Phi((a - \mu)/\sigma)}. \quad (11.39)$$

For our application, we set this expression equal to 142, with $a = \ln 100 = 4.605$:

$$142 = \exp(\mu + \sigma^2/2) \cdot \frac{\Phi(\sigma - (4.605 - \mu)/\sigma)}{1 - \Phi((4.605 - \mu)/\sigma)}. \quad (\text{Equation 2})$$

Step 3: Method of Moments Estimation

To estimate the two parameters μ and σ , we solve the minimization problem:

$$\min_{\mu, \sigma} [4.605 - (\mu + 2.054\sigma)]^2 + \left[142\Phi\left(\frac{\mu - 4.605}{\sigma}\right) - \exp(\mu + \sigma^2/2)\Phi\left(\sigma - \frac{4.605 - \mu}{\sigma}\right) \right]^2. \quad (11.40)$$

The two solutions are $\mu = 2.89372$ and $\sigma = 0.83314$.

Step 4: Calculate the population mean

We use the result that if $y \sim N[\mu, \sigma^2]$ and $x = \exp(y)$, then:

$$E[x] = \exp(\mu + \sigma^2/2). \quad (11.41)$$

Inserting our values:

$$E[x] = \exp(2.89372 + 0.83314^2/2) = \exp(2.89372 + 0.34705) = \exp(3.24077) \approx \$25,554. \quad (11.42)$$

The *Statistical Abstract of the United States* of 1987 gave the mean family income across all groups as approximately \$25,000. Therefore, the estimate based on surprisingly little information was relatively good.

Conclusion of the Example: These scarce data, in fact, told us something about the American mean. We were able to deduce the overall mean from estimates of the truncated mean and variance and the theoretical relationships between truncated and untruncated means and variances.

11.1.3 The Truncated Regression Model

In the previous models, we assumed that the population mean μ was constant. Now, we incorporate the classic regression structure, where the deterministic part is indexed by a vector of observable characteristics.

1. Definition of the Model and Conditional pdf

The classic linear regression model in the population is defined by:

$$y_i = x_i' \beta + \varepsilon_i, \quad (11.43)$$

where $\varepsilon_i|x_i \sim N[0, \sigma^2]$, so that the unconditional (in terms of selection) distribution is:

$$y_i|x_i \sim N[x'_i\beta, \sigma^2]. \quad (19-5)$$

In the truncated regression model, the sample is selected so that observations are only recorded if the dependent variable satisfies a cutoff criterion. Assuming lower (left) truncation at a point a , the conditional truncated pdf of y_i for each individual is:

$$f(y_i|y_i > a, x_i) = \begin{cases} \frac{1}{\sigma} \frac{\phi\left(\frac{y_i - x'_i\beta}{\sigma}\right)}{1 - \Phi(\alpha_i)} & \text{if } y_i > a, \\ 0 & \text{if } y_i \leq a, \end{cases} \quad (11.44)$$

where $\alpha_i = \frac{a - x'_i\beta}{\sigma}$ is the standardized truncation limit for individual i .

2. Conditional Mean in the Subsample

Applying Theorem 19.2 to the conditional distribution of $y_i|x_i$, the conditional mean of the observations included in the selected sample is:

$$E[y_i|y_i > a, x_i] = x'_i\beta + \sigma\lambda(\alpha_i), \quad (19-7)$$

where:

- $\alpha_i = \frac{a - x'_i\beta}{\sigma}$ is the standardized truncation point (which varies across individuals)
- $\lambda(\alpha_i) = \frac{\phi(\alpha_i)}{1 - \Phi(\alpha_i)}$ is the **Inverse Mills Ratio (IMR)**

Rigor Note: The conditional mean is a **non-linear** function of the parameters, contrasting with the classic linear regression where $E[y|x] = x'\beta$ is linear.

3. Partial Effects in the Subpopulation: Greene's Attenuation

The partial effects (partial derivatives of the conditional mean with respect to the regressor vector x) are obtained by differentiating Equation (19-7).

Rigorous Proof of the Partial Effects.

Adopting the *denominator layout* convention (gradient as a $K \times 1$ column vector), we seek $\frac{\partial E[y|y > a, x]}{\partial x}$.

$$\frac{\partial E[y|y > a, x]}{\partial x} = \frac{\partial(x'\beta)}{\partial x} + \sigma \frac{\partial\lambda(\alpha)}{\partial x} = \beta + \sigma \frac{\partial\lambda(\alpha)}{\partial x}. \quad (11.45)$$

By the chain rule for vector differentiation:

$$\frac{\partial\lambda(\alpha)}{\partial x} = \lambda'(\alpha) \cdot \frac{\partial\alpha}{\partial x}. \quad (11.46)$$

Step 1: Determination of $\frac{\partial\alpha}{\partial x}$

$$\alpha = \frac{a - x'\beta}{\sigma} \implies \frac{\partial\alpha}{\partial x} = -\frac{\beta}{\sigma}. \quad (11.47)$$

Step 2: Determination of $\lambda'(\alpha)$

From Theorem 19.2, the derivative of the IMR is:

$$\lambda'(\alpha) = \lambda(\alpha)[\lambda(\alpha) - \alpha] \equiv \delta(\alpha). \quad (11.48)$$

Step 3: Substitution and Simplification

$$\frac{\partial E[y|y > a, x]}{\partial x} = \beta + \sigma \cdot \delta(\alpha) \cdot \left(-\frac{\beta}{\sigma}\right) = \beta - \beta\delta(\alpha) = \beta[1 - \delta(\alpha)]. \quad (19-8)$$

This formally demonstrates the result. ■

Crucial Economic Interpretation. Since $0 < \delta(\alpha) < 1$ for the entire real line, the scale factor $[1 - \delta(\alpha)] \in (0, 1)$. This implies that:

$$\left| \frac{\partial E[y|y > a, x]}{\partial x_k} \right| < |\beta_k| \quad \forall k. \quad (11.49)$$

The marginal effect of any explanatory variable on the conditional mean of the truncated subpopulation is strictly smaller (in absolute value) than the respective structural population coefficient β_k . There is a **systematic attenuation** of the partial effects.

4. Conditional Variance and Intrinsic Heteroskedasticity

The conditional variance of the truncated regression on the selected support $y > a$ is:

$$\text{Var}[y|y > a, x] = \sigma^2[1 - \delta(\alpha)]. \quad (19-9)$$

Rigor of Interpretation and Heteroskedasticity.

- **Dispersion Reduction:** The conditional variance is reduced by the same factor $[1 - \delta(\alpha)] \in (0, 1)$. Truncation discards observations in the lower tail, compressing the distribution.
- **Intrinsic Heteroskedasticity:** A fundamental result is that **the truncated variance is inherently heteroskedastic**. Since $\alpha = \frac{a-x'\beta}{\sigma}$ depends on x , the factor $[1 - \delta(\alpha)]$ varies systematically across observations. Even if the population error ε is homoskedastic, truncation induces heteroskedasticity in the subsample.
- **Implication for Estimation:** The presence of intrinsic heteroskedasticity and an omitted regressor (the IMR) invalidates classic OLS inference, justifying the need for estimation by **Maximum Likelihood (MLE)**.

5. Which Parameter Matters?

The choice between analyzing the **partial effects** $\beta(1 - \delta)$ or the **coefficients** β depends on the objective of the inference:

- **If the analysis is restricted to the subpopulation** (e.g., only individuals with income above the poverty line), the partial effects are of interest.
- **If the study aims to extend conclusions to the entire population**, the structural coefficients β are what matter.

6. The OLS Problem and the Olsen Reparametrization

If we estimate the model by OLS on the subsample, we rewrite (19-6) as:

$$y|y > a = x'\beta + \sigma\lambda(\alpha) + u, \quad (19-10)$$

where $u = y - E[y|y > a]$. By construction, $E[u] = 0$, but:

1. **Heteroskedasticity:** $\text{Var}[u] = \sigma^2(1 - \delta)$ is a function of x .
2. **Omitted Variable:** The non-linear term $\sigma\lambda(\alpha)$ is omitted, generating omitted variable bias (Heckman, 1979).

Log-Likelihood Function and Olsen's Reparametrization (1978).

Consistent estimation must be done via MLE. The conditional log-likelihood is:

$$\ln L(\beta, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - x'_i\beta)^2 - \sum_{i=1}^n \ln \left[1 - \Phi \left(\frac{a - x'_i\beta}{\sigma} \right) \right]. \quad (11.50)$$

The Challenge: The non-linear term in the denominator of the pdf makes the log-likelihood not globally concave, causing numerical instability.

Olsen's Solution: Reparametrize by defining $\theta = \frac{\beta}{\sigma}$ and $\gamma = \frac{1}{\sigma}$. The Hessian becomes **negative definite over the entire parameter space**, ensuring global and stable convergence.

7. The Chung and Goldberger (1984) Proportionality Theorem

Without knowledge of the distribution of x , it is not possible to determine the exact magnitude of the OLS bias. However, under general conditions, the Chung and Goldberger (1984) theorem provides a fundamental result.

Formal Statement.

If $E[x|y]$ in the complete population is a linear function of y (automatically satisfied if x and y have a Multivariate Normal joint distribution), then:

$$\text{plim } \hat{\beta}_{\text{OLS}} = \tau\beta, \quad (11.51)$$

where τ is a proportionality constant with $\tau \in (0, 1)$.

Analytical Proof and Derivation of τ .

Assuming $E[x] = 0$ and $\text{Var}(x) = \Sigma_{xx}$, the unconditional variance of y is $\sigma_y^2 = \beta'\Sigma_{xx}\beta + \sigma^2$. The population R^2 is $R^2 = \frac{\beta'\Sigma_{xx}\beta}{\sigma_y^2}$.

Step 1: Linear Projection of x on y

By joint normality:

$$E[x|y] = \left(\frac{\Sigma_{xx}\beta}{\sigma_y^2} \right) y. \quad (11.52)$$

Decomposing $x = \left(\frac{\Sigma_{xx}\beta}{\sigma_y^2} \right) y + u$, where u is independent of y .

Step 2: Properties in the Truncated Subsample

Since u is independent of y , its conditional distribution given $y > a$ is identical to the unconditional one. We calculate:

$$\begin{aligned}\text{Cov}(x, y \mid y > a) &= \Sigma_{xx}\beta[1 - \delta(\alpha)], \\ \text{Var}(x \mid y > a) &= \Sigma_{xx} - \delta(\alpha) \frac{\Sigma_{xx}\beta\beta'\Sigma_{xx}}{\sigma_y^2}.\end{aligned}\tag{11.53}$$

Step 3: Probability Limit of $\hat{\beta}_{\text{OLS}}$

$$\text{plim } \hat{\beta}_{\text{OLS}} = [\text{Var}(x|y > a)]^{-1} \text{Cov}(x, y|y > a).\tag{11.54}$$

Applying the **Sherman-Morrison-Woodbury Identity** to invert $\text{Var}(x|y > a)$ and simplifying (algebraic details omitted for brevity), we obtain:

$$\boxed{\text{plim } \hat{\beta}_{\text{OLS}} = \left[\frac{1 - \delta(\alpha)}{1 - \delta(\alpha)R^2} \right] \beta \equiv \tau\beta.}\tag{19-11}$$

Where $\tau = \frac{1 - \delta(\alpha)}{1 - \delta(\alpha)R^2}$. ■

Analysis and Implications of τ .

- **Attenuation Bias Confirmed** ($\tau < 1$): Since $\delta(\alpha) \in (0, 1)$ and $R^2 \in [0, 1]$, the denominator satisfies $1 - \delta(\alpha)R^2 > 1 - \delta(\alpha)$, guaranteeing $\tau < 1$. OLS is always attenuated toward zero.
- **Limiting Case $R^2 \rightarrow 1$** : If the fit is perfect ($\sigma^2 \rightarrow 0$), then $\lim_{R^2 \rightarrow 1} \tau = 1$. OLS becomes **consistent**!
- **Limiting Case $R^2 \rightarrow 0$** : If there is no explanatory power, $\lim_{R^2 \rightarrow 0} \tau = 1 - \delta(\alpha)$. Attenuation is maximal.
- **Preservation of Relative Effects**: Since τ is common to all regressors:

$$\frac{\text{plim } \hat{\beta}_{\text{OLS},j}}{\text{plim } \hat{\beta}_{\text{OLS},k}} = \frac{\tau\beta_j}{\tau\beta_k} = \frac{\beta_j}{\beta_k}.\tag{11.55}$$

OLS preserves the relative importance and sign of the variables, being useful for qualitative analyses.

Summary of Key Points of Subsection 19.2.3

1. **Index Model**: $y_i|x_i \sim N[x'_i\beta, \sigma^2]$, under lower truncation $y_i > a$.
2. **Non-Linear Conditional Mean**: $E[y_i|y_i > a, x_i] = x'_i\beta + \sigma\lambda(\alpha_i)$, where $\lambda(\alpha_i)$ is the IMR.
3. **Attenuated Partial Effects**: $\frac{\partial E[y_i|y_i > a, x_i]}{\partial x_i} = \beta[1 - \delta(\alpha_i)]$, with $\delta(\alpha_i) \in (0, 1)$, proving systematic attenuation.
4. **Intrinsic Heteroskedasticity**: $\text{Var}[y_i|y_i > a, x_i] = \sigma^2[1 - \delta(\alpha_i)]$ varies with x_i , invalidating OLS.

5. **OLS Inconsistency:** Estimating by OLS omits $\lambda(\alpha_i)$, generating omitted variable bias.
6. **Proportionality Theorem:** Under joint normality, $\text{plim } \hat{\beta}_{\text{OLS}} = \tau\beta$ with $\tau = \frac{1-\delta(\alpha)}{1-\delta(\alpha)R^2} \in (0, 1)$.
7. **Consistent Estimation:** Requires **Maximum Likelihood (MLE)**, preferably with Olsen's reparametrization for numerical stability.

11.2 Censored Data

11.2.1 The Censored Normal Distribution and Its Moments

This section presents a mathematically rigorous review of the censored normal distribution, enriched with contributions from classic econometrics works (Bruce Hansen, Jeffrey Wooldridge, Hayashi, and Angrist & Pischke).

Mathematical Structure of the Distribution

Let y^* be a latent (unobservable) random variable that follows a normal distribution conditioned on a vector of regressors $\mathbf{x}$, such that $y^* | \mathbf{x} \sim N(\mu, \sigma^2)$, where $\mu = \mathbf{x}'\boldsymbol{\beta}$. Defining a constant censoring point a (often normalized to $a = 0$), the observed variable y is generated by the following left-censoring mechanism:

$$y = \begin{cases} a, & \text{if } y^* \leq a \\ y^*, & \text{if } y^* > a \end{cases} \quad (11.56)$$

The Mixed Probability Density Function (Mass and Density) Statistically, the censored variable y presents a hybrid (mixed) distribution, possessing a discrete probability mass exactly at the censoring point a and a continuous tail for values strictly greater than a .

The accumulated probability at the censoring point is given by:

$$P(y = a) = P(y^* \leq a) = \Phi\left(\frac{a - \mu}{\sigma}\right) = \Phi(\alpha) \quad (11.57)$$

Where $\alpha = \frac{a - \mu}{\sigma}$ is the standardized censoring point, and $\Phi(\cdot)$ represents the cumulative distribution function (CDF) of the standard normal.

For values in the continuous region ($y > a$), the density of y is identical to the original normal density of y^* . Using the Dirac delta function ($\delta_a(y)$) to represent the point mass, we can write the complete conditional probability density function (pdf) of $y | \mathbf{x}$ with respect to a mixed measure as:

$$f_y(y | \mathbf{x}) = \Phi(\alpha)\delta_a(y) + \mathbf{1}(y > a)\frac{1}{\sigma}\phi\left(\frac{y - \mu}{\sigma}\right) \quad (11.58)$$

Rigorous Derivation of the Moments (Theorem 19.3)

Theorem 19.3 of Greene establishes the exact expressions for the Expectation and Variance of the censored variable y . To present these derivations with maximum rigor, we define the following auxiliary variables based on the standardized censoring point $\alpha = \frac{a - \mu}{\sigma}$:

- $\Phi = \Phi(\alpha) = P(y^* \leq a)$ (censoring probability).
- $\phi = \phi(\alpha)$ (standard normal density evaluated at the standardized point).
- $l = \lambda(-\alpha) = \frac{\phi(\alpha)}{1-\Phi(\alpha)}$ (Inverse Mills Ratio associated with the upper tail).
- $d = l(l - \alpha) = l^2 - l\alpha$ (variance scale factor, where $0 < d < 1$).

A. Proof of the Conditional Expectation $E[y]$ The mathematical expectation of y is derived by applying the Law of Total Expectation with respect to the regime indicator variable $I = \mathbf{1}(y^* > a)$:

$$E[y | \mathbf{x}] = P(I = 0) \cdot E[y | I = 0, \mathbf{x}] + P(I = 1) \cdot E[y | I = 1, \mathbf{x}] \quad (11.59)$$

- **Censored Regime ($I = 0$):** Occurs with probability $P(y^* \leq a) = \Phi$. In this regime, y deterministically assumes the value a , so that $E[y | I = 0, \mathbf{x}] = a$.
- **Uncensored Regime ($I = 1$):** Occurs with probability $P(y^* > a) = 1 - \Phi$. In this regime, $y = y^*$, which follows a normal distribution truncated from the left at point a . Using the classic result for the mean of the truncated normal, we have $E[y | I = 1, \mathbf{x}] = E[y^* | y^* > a, \mathbf{x}] = \mu + \sigma \lambda(-\alpha) = \mu + \sigma l$.

Substituting both terms into the total expectation equation:

$$E[y | \mathbf{x}] = \Phi a + (1 - \Phi)(\mu + \sigma l) \quad (11.60)$$

Simplified Form of the Expectation: Since $(1 - \Phi)l = (1 - \Phi)\frac{\phi}{1-\Phi} = \phi$, we can rewrite the expectation in linearized form:

$$E[y | \mathbf{x}] = \Phi a + (1 - \Phi)\mu + \sigma \phi \quad (11.61)$$

For the standard case of censoring at zero ($a = 0$), the equation simplifies directly to the famous expression of the Type 1 Tobit model:

$$E[y | \mathbf{x}] = \mu(1 - \Phi) + \sigma \phi = \mu \Phi \left(\frac{\mu}{\sigma} \right) + \sigma \phi \left(\frac{\mu}{\sigma} \right) \quad (11.62)$$

B. Proof of the Conditional Variance $Var[y]$ The variance of the censored variable y is obtained with absolute mathematical rigor through the Law of Total Variance:

$$Var[y | \mathbf{x}] = E[Var(y | I, \mathbf{x})] + Var(E[y | I, \mathbf{x}]) \quad (11.63)$$

Step 1: Calculation of the first term, $E[Var(y | I, \mathbf{x})]$

- In the censored regime ($I = 0$): $y = a$ (constant), so $Var[y | I = 0, \mathbf{x}] = 0$.
- In the uncensored regime ($I = 1$): y follows a truncated normal distribution. The variance of the truncated normal is $Var[y | I = 1, \mathbf{x}] = \sigma^2(1 - d) = \sigma^2[1 - l(l - \alpha)]$.

Taking the weighted expectation of these within-regime variances:

$$E[Var(y | I, \mathbf{x})] = \Phi \cdot 0 + (1 - \Phi)\sigma^2(1 - d) = \sigma^2(1 - \Phi)(1 - d) \quad (11.64)$$

Step 2: Calculation of the second term, $Var(E[y | I, \mathbf{x}])$

The conditional expectation $E[y | I, \mathbf{x}]$ is a discrete random variable that assumes $w_0 = a$ (with probability Φ) and $w_1 = \mu + \sigma l$ (with probability $1 - \Phi$). The variance of a binary variable is $Var[W] = p(1 - p)(w_1 - w_0)^2$. Applying this identity:

$$Var(E[y | I, \mathbf{x}]) = \Phi(1 - \Phi) [(\mu + \sigma l) - a]^2 \quad (11.65)$$

Rewriting the term in brackets in terms of the standardized scale: $(\mu + \sigma l) - a = \sigma \left(l - \frac{a - \mu}{\sigma} \right) = \sigma(l - \alpha)$. Squaring:

$$Var(E[y | I, \mathbf{x}]) = \Phi(1 - \Phi)\sigma^2(l - \alpha)^2 \quad (11.66)$$

Step 3: Combination of terms

Adding the results and factoring out $\sigma^2(1 - \Phi)$:

$$Var[y | \mathbf{x}] = \sigma^2(1 - \Phi) [(1 - d) + \Phi(l - \alpha)^2] \quad (11.67)$$

Given that $(l - \alpha)^2 = (\alpha - l)^2$, we arrive at Greene's exact formula for the variance:

$$Var[y | \mathbf{x}] = \sigma^2(1 - \Phi) [(1 - d) + (\alpha - l)^2\Phi] \quad (11.68)$$

Important Correction Note: For the formula to be dimensionally correct and mathematically valid for any normal $N(\mu, \sigma^2)$, all variables inside the brackets must be calculated using the standardized scale $\alpha = \frac{a - \mu}{\sigma}$, and not the original scale a .

Detailed Practical Example: The Arena Example

To consolidate the theory, this example illustrates how the correct specification of the model affects the identification of the latent parameters of demand for seats in an arena.

Parameters and Data of the Problem:

- Maximum physical capacity of the arena (Upper Censoring): $C = 20,000$ seats.
- Sellout rate (Censoring): The arena sells out 25% of the time, meaning the regime is censored with probability 0.25.
- Mean tickets sold (censored): $E[y] = 18,000$ tickets.

Scenario 1: Censored Model (Tobit) In this scenario, the observed variable is censored at the upper limit $C = 20,000$. The censoring probability is $P(y^* \geq C) = 1 - \Phi(C^*) = 0.25$, which gives us $\Phi(C^*) = 0.75$.

1. **Find the standardized cutoff point (C^*):** Using the standard normal table, $C^* = \Phi^{-1}(0.75) \approx 0.6745$. Therefore,

$$\frac{20,000 - \mu}{\sigma} = 0.6745 \implies \mu = 20,000 - 0.6745\sigma \quad (\text{Eq. I}) \quad (11.69)$$

2. **Use the censored expectation:** $E[y] = \mu\Phi(C^*) - \sigma\phi(C^*) + C(1 - \Phi(C^*))$. Substituting the values ($\phi(0.6745) \approx 0.3178$):

$$18,000 = 0.75\mu - 0.3178\sigma + 20,000(0.25) \implies 13,000 = 0.75\mu - 0.3178\sigma \quad (\text{Eq. II}) \quad (11.70)$$

3. **Solution of the System:** Substituting (I) into (II):

$$13,000 = 0.75(20,000 - 0.6745\sigma) - 0.3178\sigma \implies 0.8237\sigma = 2,000 \implies \sigma \approx 2,428 \quad (11.71)$$

Substituting σ into (I): $\mu = 20,000 - 0.6745(2,428) \approx 18,362$.

Interpretation: The true mean of the unconstrained latent demand is 18,362 seats, with a standard deviation of 2,428.

Scenario 2: Truncated Model (Conditional Mean) Suppose the researcher takes the mean of 18,000 considering only the days when the arena *did not* sell out. This configures an upper-truncated distribution at $C = 20,000$.

1. **Scale relationship:** $\mu = 20,000 - 0.6745\sigma$ (Eq. I)

2. **Conditional (truncated) expectation:** $E[y^* | y^* < C] = \mu - \sigma \frac{\phi(C^*)}{\Phi(C^*)}$. Substituting:

$$18,000 = \mu - \sigma \left(\frac{0.3178}{0.75} \right) \implies 18,000 = \mu - 0.4237\sigma \quad (\text{Eq. III}) \quad (11.72)$$

3. **Solution of the System:** Substituting (I) into (III):

$$18,000 = (20,000 - 0.6745\sigma) - 0.4237\sigma \implies 1.0982\sigma = 2,000 \implies \sigma \approx 1,821 \quad (11.73)$$

Substituting into (I): $\mu = 20,000 - 0.6745(1,821) \approx 18,772$.

Interpretation: By ignoring sellout days, we estimate an artificially higher mean latent demand (18,772) and lower volatility (1,821), due to the elimination of variability from the right tail.

Advanced Perspectives and Points from Other Sources

A. Greene's "Rule of Thumb" under Normal Regressors (Hansen) Bruce Hansen highlights an analytical discovery by Greene (1981) regarding the OLS bias in censored models. If the regressors follow a multivariate normal distribution, the slope coefficient estimated by OLS in the complete censored model is given by:

$$\beta_{\text{OLS}} = \beta(1 - \pi) \quad (11.74)$$

Where $\pi = P(y = 0)$ represents the exact proportion of censored observations. The OLS coefficients are attenuated in direct proportion to the censoring rate.

B. Data Censoring vs. Corner Solutions (Wooldridge)

- **Real Data Censoring:** Missing data problem (e.g., top-coding). The latent variable y^* is the variable of interest and the objective is to recover its structural parameters.
- **Corner Solution Responses:** Result of optimal economic choice (e.g., hours of work). The latent variable has no real existence. The interest lies entirely in the behavior of the observed mean $E[y | \mathbf{x}]$.

C. The Angrist-Pischke Critique and the CLAD Estimator Angrist and Pischke (2009) are skeptical of the Tobit applied to corner decisions due to its strong dependence on normality and homoskedasticity. As a robust alternative, they highlight Powell's CLAD (Censored Least Absolute Deviations) estimator. Due to the equivariance of quantiles under monotone transformations:

$$Q_\tau[y \mid \mathbf{x}] = \max(Q_\tau[y^* \mid \mathbf{x}], 0) = \max(\mathbf{x}'\boldsymbol{\beta}, 0) \quad (\text{for } \tau = 0.5) \quad (11.75)$$

This allows consistent and semiparametric identification of $\boldsymbol{\beta}$ without requiring normality or homoskedasticity.

11.2.2 The Censored Regression Model: Type 1 Tobit Model

This section deepens the analytical derivations of the marginal effects, presents the formal proof of Theorem 19.4 using Leibniz's Theorem, and details the McDonald-Moffitt (1980) Decomposition, consolidating theoretical bridges with Wooldridge, Hansen, Hayashi, and Angrist & Pischke.

Structure and Motivation of the Type 1 Tobit Model

The Type 1 Tobit model formalizes the censored normal distribution within a linear regression framework. We consider a latent variable y_i^* :

$$y_i^* = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \mid \mathbf{x}_i \sim N(0, \sigma^2) \quad (11.76)$$

The observed variable y_i is generated through the left-censoring mechanism at zero:

$$y_i = \max(y_i^*, 0) = \begin{cases} y_i^* & \text{if } y_i^* > 0 \\ 0 & \text{if } y_i^* \leq 0 \end{cases} \quad (11.77)$$

The econometric challenge is that the distribution of y_i is mixed. Estimating this model by OLS while ignoring censoring results in inconsistent and severely attenuated estimates.

The Three Conditional Mean Functions (CEFs)

Defining the normalized index $w_i = \frac{\mathbf{x}_i'\boldsymbol{\beta}}{\sigma}$, the three conditional means are:

A. Mean of the Latent Variable: $E[y_i^* \mid \mathbf{x}_i]$

$$E[y_i^* \mid \mathbf{x}_i] = \mathbf{x}_i'\boldsymbol{\beta} \quad (11.78)$$

Practical Use: Generally of less interest in corner solution studies, since y_i^* is not observable.

B. Conditional (Truncated) Mean: $E[y_i \mid y_i > 0, \mathbf{x}_i]$

$$E[y_i \mid y_i > 0, \mathbf{x}_i] = \mathbf{x}_i'\boldsymbol{\beta} + \sigma\lambda(w_i) \quad (11.79)$$

Where $\lambda(w_i) = \frac{\phi(w_i)}{\Phi(w_i)}$ is the Inverse Mills Ratio. *Practical Use:* Crucial for analyzing the intensity of choice of active economic agents.

C. Unconditional (Censored) Mean: $E[y_i | \mathbf{x}_i]$

$$E[y_i | \mathbf{x}_i] = \Phi(w_i) \left[\mathbf{x}'_i \boldsymbol{\beta} + \sigma \frac{\phi(w_i)}{\Phi(w_i)} \right] = \mathbf{x}'_i \boldsymbol{\beta} \Phi(w_i) + \sigma \phi(w_i) \quad (11.80)$$

Practical Use: It is the unconditional prediction function par excellence, essential for aggregate projections.

Theorem 19.4: Partial Effects and Derivation with Leibniz's Rule

Theorem 19.4 establishes that the derivative of the observed mean is equal to the structural coefficient weighted by the probability that the observation lies in the uncensored region.

A. Analytical Proof (General Case: Censoring between a and b) Let $y^* = \mathbf{x}'\boldsymbol{\beta} + \varepsilon$, with $\varepsilon \sim f(\varepsilon)$. The observed variable y is censored at a and b :

$$E[y | \mathbf{x}] = aF(a - \mathbf{x}'\boldsymbol{\beta}) + b[1 - F(b - \mathbf{x}'\boldsymbol{\beta})] + \int_{a - \mathbf{x}'\boldsymbol{\beta}}^{b - \mathbf{x}'\boldsymbol{\beta}} (\mathbf{x}'\boldsymbol{\beta} + \varepsilon) f(\varepsilon) d\varepsilon \quad (11.81)$$

Differentiating with respect to $\mathbf{x}$ (knowing that $\frac{\partial \mathbf{x}'\boldsymbol{\beta}}{\partial \mathbf{x}} = \boldsymbol{\beta}$):

$$\begin{aligned} \frac{\partial}{\partial \mathbf{x}} [aF(a - \mathbf{x}'\boldsymbol{\beta})] &= -af(a - \mathbf{x}'\boldsymbol{\beta})\boldsymbol{\beta} \\ \frac{\partial}{\partial \mathbf{x}} [b[1 - F(b - \mathbf{x}'\boldsymbol{\beta})]] &= bf(b - \mathbf{x}'\boldsymbol{\beta})\boldsymbol{\beta} \end{aligned} \quad (11.82)$$

For the integral term, we apply **Leibniz's Theorem**:

$$\frac{\partial I(\mathbf{x})}{\partial \mathbf{x}} = \int_{u(\mathbf{x})}^{v(\mathbf{x})} \frac{\partial h(\mathbf{x}, \varepsilon)}{\partial \mathbf{x}} d\varepsilon + h(\mathbf{x}, v(\mathbf{x})) \frac{\partial v(\mathbf{x})}{\partial \mathbf{x}} - h(\mathbf{x}, u(\mathbf{x})) \frac{\partial u(\mathbf{x})}{\partial \mathbf{x}} \quad (11.83)$$

Where $u(\mathbf{x}) = a - \mathbf{x}'\boldsymbol{\beta}$, $v(\mathbf{x}) = b - \mathbf{x}'\boldsymbol{\beta}$ and $h(\mathbf{x}, \varepsilon) = (\mathbf{x}'\boldsymbol{\beta} + \varepsilon)f(\varepsilon)$. Substituting:

$$\frac{\partial I(\mathbf{x})}{\partial \mathbf{x}} = \boldsymbol{\beta} [F(b - \mathbf{x}'\boldsymbol{\beta}) - F(a - \mathbf{x}'\boldsymbol{\beta})] - bf(b - \mathbf{x}'\boldsymbol{\beta})\boldsymbol{\beta} + af(a - \mathbf{x}'\boldsymbol{\beta})\boldsymbol{\beta} \quad (11.84)$$

Adding the three terms, the boundaries cancel perfectly, leaving:

$$\frac{\partial E[y | \mathbf{x}]}{\partial \mathbf{x}} = \boldsymbol{\beta} [F(b - \mathbf{x}'\boldsymbol{\beta}) - F(a - \mathbf{x}'\boldsymbol{\beta})] = \boldsymbol{\beta} \times P(a < y^* < b | \mathbf{x}) \quad \blacksquare \quad (11.85)$$

B. The Special Case of the Standard Tobit Model (Censoring at Zero) For $a = 0$ and $b = \infty$, under normal errors, Theorem 19.4 simplifies to:

$$\frac{\partial E[y_i | \mathbf{x}_i]}{\partial x_{ik}} = \beta_k \Phi \left(\frac{\mathbf{x}'_i \boldsymbol{\beta}}{\sigma} \right) = \beta_k \Phi(w_i) \quad (11.86)$$

C. The Great Intuitive Lesson: The Attenuation Phenomenon The marginal effect on the observed variable y is equal to the structural coefficient β_k multiplied by the probability of non-censoring ($\Phi(w_i) \in (0, 1)$). The effect on the observable mean is always attenuated. Under normal regressors, OLS suffers systematic attenuation $\beta_{OLS} \approx \beta \cdot (1 - \pi)$.

The McDonald-Moffitt (1980) Decomposition

McDonald and Moffitt (1980) decomposed the total marginal effect into two behavioral channels by applying the product rule to $E[y_i | \mathbf{x}_i] = P(y_i > 0 | \mathbf{x}_i) \cdot E[y_i | y_i > 0, \mathbf{x}_i]$:

$$\frac{\partial E[y_i | \mathbf{x}_i]}{\partial x_{ik}} = \underbrace{P(y_i > 0 | \mathbf{x}_i) \frac{\partial E[y_i | y_i > 0, \mathbf{x}_i]}{\partial x_{ik}}}_{\text{Intensive Margin (Volume)}} + \underbrace{E[y_i | y_i > 0, \mathbf{x}_i] \frac{\partial P(y_i > 0 | \mathbf{x}_i)}{\partial x_{ik}}}_{\text{Extensive Margin (Participation)}} \quad (11.87)$$

A. Proof of Algebraic Equivalence Knowing that the derivative of the truncated mean is $\beta_k[1 - d(w_i)]$ and the derivative of the probability is $\phi(w_i)\frac{\beta_k}{\sigma}$:

$$\text{Intensive Margin} = \Phi(w_i)\beta_k[1 - \lambda(w_i)(w_i + \lambda(w_i))] = \beta_k\Phi(w_i) - \beta_k\phi(w_i)(w_i + \lambda(w_i))$$

$$\text{Extensive Margin} = [\mathbf{x}'_i\boldsymbol{\beta} + \sigma\lambda(w_i)]\phi(w_i)\frac{\beta_k}{\sigma} = \beta_k\phi(w_i)(w_i + \lambda(w_i)) \quad (11.88)$$

The sum of the margins results exactly in $\beta_k\Phi(w_i)$, proving the consistency of the decomposition. ■

B. Structural Limitations and Extensions The decomposition reveals that the two margins are governed by the same $\boldsymbol{\beta}$. To circumvent this:

- **Heckman Model (Type 2 Tobit):** Separates the selection equation ($d_i = 1(\mathbf{z}'_i\boldsymbol{\gamma} + v_i > 0)$) from the outcome equation.
- **Powell's CLAD Estimator:** Semiparametric estimation based on the median, robust to heteroskedasticity and non-normality.

Advanced Perspectives and Critical Dialogue with the Literature

A. Parametric Fragility (Hayashi) The Maximum Likelihood (MLE) estimator of the Tobit is highly sensitive to violations of normality and homoskedasticity. Under heteroskedasticity ($\sigma_i^2 = h(\mathbf{x}_i)$), the MLE estimator becomes inconsistent, since the scores lose their zero conditional expectation:

$$E\left[\frac{\partial \ln L_i}{\partial \boldsymbol{\beta}} \middle| \mathbf{x}_i\right] \neq \mathbf{0} \quad \text{if } \sigma^2 \neq \text{constant} \quad (11.89)$$

Tests such as Pagan-Vella (based on generalized residuals) and Bera-Jarque-Lee are essential for detection.

B. Wooldridge's View: Corner Solutions vs. Censoring In **Data Censoring**, the focus is on the parameters of y^* . In **Corner Solutions**, y^* does not exist in the real world; the focus should be exclusively on the observable CEFs ($E[y | \mathbf{x}]$ and $E[y | y > 0, \mathbf{x}]$) and the McDonald-Moffitt decomposition for public policy analysis.

C. The Bridge to Maximum Likelihood Estimation The log-likelihood function is additively divided into a Probit component (for zeros) and a Classical Regression component (for positives):

$$\ln L(\boldsymbol{\beta}, \sigma^2) = \sum_{y_i=0} \ln \left[1 - \Phi\left(\frac{\mathbf{x}'_i\boldsymbol{\beta}}{\sigma}\right) \right] + \sum_{y_i>0} \left\{ -\frac{1}{2} \left[\ln(2\pi) + \ln \sigma^2 + \frac{(y_i - \mathbf{x}'_i\boldsymbol{\beta})^2}{\sigma^2} \right] \right\} \quad (11.90)$$

Due to non-linearity, optimization requires algorithms such as Newton-Raphson or BHHH. Olsen's (1978) reparametrization ($\boldsymbol{\theta} = \boldsymbol{\beta}/\sigma$ and $\gamma = 1/\sigma$) guarantees the strict global concavity of the log-likelihood, ensuring convergence to the global maximum.

Verdict and Synthesis of the Section

The Type 1 Tobit model constitutes the true watershed in the econometrics of limited dependent variables. It performs the fundamental transition between abstract mathematical statistics and structured applied econometrics. The understanding that the structural coefficients $\boldsymbol{\beta}$ do not directly measure observable effects, and the consequent need to master the McDonald-Moffitt Decomposition and the Three CEFs, represent the threshold of technical maturity of the researcher, enabling them to align the mathematical specification with the true nature of economic behavior.

11.2.3 Estimation and Inference in the Tobit Model

This section focuses on the formulation of the log-likelihood function under a mixed measure, the asymptotic properties established by Amemiya (1973), Olsen's (1978) linearizing reparametrization with the formal proof of global concavity, and the recovery of the original economic parameters via the Delta Method. Additionally, this version incorporates the mathematical derivations of the OLS Attenuation Bias (Greene, 1981), the Goldberger (1981) Rule of Thumb, the proof that discarding censored observations induces truncation bias, and the exact resolution of Example 19.5 (Quester and Greene, 1982). Theoretical connections are also established with the works of Fumio Hayashi, Jeffrey Wooldridge, and Lars Peter Hansen.

The Log-Likelihood Function and the Mixed Measure

Consistent estimation of the Type 1 Tobit model requires the Maximum Likelihood (ML) method. Due to the hybrid nature of the observed dependent variable y_i — which exhibits a point of discontinuity (probability mass) exactly at zero and a strictly positive continuous density — the density cannot be defined with respect to the usual Lebesgue measure.

Formulation under Mixed Measure Formally, we define the conditional density of $y_i \mid \mathbf{x}_i$ with respect to a mixed (discrete/continuous) measure $\nu(dy) = \delta_0(dy) + \lambda(dy)$, where δ_0 represents the Dirac measure concentrated at the origin and λ denotes the Lebesgue measure on the real line:

$$f(y_i \mid \mathbf{x}_i) = \left[1 - \Phi\left(\frac{\mathbf{x}_i' \boldsymbol{\beta}}{\sigma}\right) \right]^{\mathbf{1}(y_i=0)} \cdot \left[\frac{1}{\sigma} \phi\left(\frac{y_i - \mathbf{x}_i' \boldsymbol{\beta}}{\sigma}\right) \right]^{\mathbf{1}(y_i>0)} \quad (11.91)$$

Where $\mathbf{x}_i$ is the $K \times 1$ vector of exogenous regressors, $\boldsymbol{\beta}$ is the $K \times 1$ vector of structural population parameters, σ represents the standard deviation of the homoskedastic normal error, and $\Phi(\cdot)$ and $\phi(\cdot)$ are, respectively, the C.D.F. and the p.d.f. of the standard normal.

The Sample Log-Likelihood Function By applying the logarithm to the product of densities for a sample of independent observations and using the symmetry of the normal,

$1 - \Phi(z) = \Phi(-z)$, we obtain the classic log-likelihood function of the Tobit model:

$$\ln L(\boldsymbol{\beta}, \sigma^2) = \sum_{y_i=0} \ln \Phi\left(-\frac{\mathbf{x}_i' \boldsymbol{\beta}}{\sigma}\right) + \sum_{y_i>0} \left[-\frac{1}{2} \ln(2\pi) - \ln \sigma - \frac{(y_i - \mathbf{x}_i' \boldsymbol{\beta})^2}{2\sigma^2} \right] \quad (11.92)$$

This additive structure divides perfectly coherently into two distinct economic and statistical components:

- **The Discrete Part** ($y_i = 0$): Equivalent exactly to the participation probability of a binary Probit decision model, where the latent variable lies below the censoring threshold.
- **The Continuous Part** ($y_i > 0$): Equivalent to the density of a classic normal linear regression minimizing squared residuals, with an effective sample size restricted to the subsample of active participants.

Asymptotic Properties and Regularity (Amemiya, 1973)

Takeshi Amemiya (1973) formally established that, although the likelihood has a non-standard (mixed) nature, the Maximum Likelihood Estimator (MLE) enjoys all ideal asymptotic properties:

- **Consistency:** $\hat{\boldsymbol{\beta}}_{\text{MLE}} \xrightarrow{p} \boldsymbol{\beta}_0$ and $\hat{\sigma}_{\text{MLE}} \xrightarrow{p} \sigma_0$.
- **Asymptotic Normality:**

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_{\text{MLE}} - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{I}^{-1}(\boldsymbol{\theta}_0)) \quad (11.93)$$

Where $\mathbf{I}(\boldsymbol{\theta}_0) = -E[\mathbf{H}(\boldsymbol{\theta}_0)]$ represents the population Fisher Information Matrix.

- **Asymptotic Efficiency:** The estimator asymptotically attains the lower bound established by the Cramér-Rao Bound.

Amemiya's Strict Regularity Conditions The validity of these asymptotic properties rests strictly on the following regularity hypotheses:

1. **Exogeneity and Full Rank:** The moment matrix of the regressors, $\frac{1}{n} \mathbf{X}' \mathbf{X}$, converges in probability to a finite, non-singular positive definite limit matrix ($\mathbf{Q}$) as $n \rightarrow \infty$.
2. **Interior Parameters:** The vector of true parameters $\boldsymbol{\theta}_0 = (\boldsymbol{\beta}_0', \sigma_0)'$ lies in the interior of a compact parameter space Θ .
3. **Support Independence:** The censoring point zero is fixed and strictly exogenous, ensuring that the support of the distribution of y_i does not depend on the parameters to be estimated.
4. **Strict Homoskedasticity and Normality:** The conditional distribution of the idiosyncratic error $\varepsilon_i \mid \mathbf{x}_i$ must be strictly normal and homoskedastic. If the errors are non-normal or heteroskedastic, the Tobit maximum likelihood estimator becomes inconsistent and severely biased.

Estimators of the Asymptotic Covariance Matrix For practical inference, the asymptotic covariance matrix can be consistently estimated by three classic approximations at the optimum point:

$$\begin{aligned}\hat{\mathbf{V}}_H &= \left[-\sum_{i=1}^n \frac{\partial^2 \ln f_i(\hat{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \right]^{-1} \quad (\text{Observed Hessian}) \\ \hat{\mathbf{V}}_E &= \left[-E \left(\sum_{i=1}^n \frac{\partial^2 \ln f_i(\hat{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \right) \right]^{-1} \quad (\text{Expected Fisher Information}) \\ \hat{\mathbf{V}}_B &= \left[\sum_{i=1}^n \left(\frac{\partial \ln f_i(\hat{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta}} \right) \left(\frac{\partial \ln f_i(\hat{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta}'} \right) \right]^{-1} \quad (\text{BHHH})\end{aligned} \quad (11.94)$$

Practical Note: Although asymptotically equivalent under correct specification, the Observed Hessian ($\hat{\mathbf{V}}_H$) is preferred in applied microeconometrics in finite samples, due to its better numerical performance and because it does not require the analytical integration of the expected second derivatives.

Olsen's (1978) Reparametrization and the Proof of Global Concavity

The numerical optimization of the original log-likelihood in the parameter scale $(\boldsymbol{\beta}, \sigma)$ presents severe convergence obstacles. Since the original likelihood surface is not globally concave, algorithms such as the Newton-Raphson method are highly unstable. To solve this obstacle, Olsen (1978) proposed a linearizing reparametrization, defining the new structural parameters:

$$\boldsymbol{\theta} = \frac{\boldsymbol{\beta}}{\sigma} \quad \text{and} \quad \gamma = \frac{1}{\sigma} \quad (11.95)$$

Substituting this transformation into the previous equation, the reparametrized log-likelihood function assumes the globally concave form:

$$\ln L(\boldsymbol{\theta}, \gamma) = \sum_{y_i=0} \ln \Phi(-\mathbf{x}_i' \boldsymbol{\theta}) + \sum_{y_i>0} \left[-\frac{1}{2} \ln(2\pi) + \ln \gamma - \frac{1}{2} (\gamma y_i - \mathbf{x}_i' \boldsymbol{\theta})^2 \right] \quad (11.96)$$

The Score Vector (Analytical Gradient) Differentiating the reparametrized log-likelihood analytically with respect to $\boldsymbol{\theta}$ and γ , we obtain the individual score vector $s_i(\boldsymbol{\theta}, \gamma)$:

For uncensored observations ($y_i > 0$):

$$\begin{aligned}\frac{\partial \ln L_i}{\partial \boldsymbol{\theta}} &= (\gamma y_i - \mathbf{x}_i' \boldsymbol{\theta}) \mathbf{x}_i \\ \frac{\partial \ln L_i}{\partial \gamma} &= \frac{1}{\gamma} - (\gamma y_i - \mathbf{x}_i' \boldsymbol{\theta}) y_i\end{aligned} \quad (11.97)$$

For censored observations ($y_i = 0$):

$$\begin{aligned}\frac{\partial \ln L_i}{\partial \boldsymbol{\theta}} &= -\lambda(-\mathbf{x}_i' \boldsymbol{\theta}) \mathbf{x}_i \\ \frac{\partial \ln L_i}{\partial \gamma} &= 0\end{aligned} \quad (11.98)$$

Where $\lambda(-\mathbf{x}'_i\boldsymbol{\theta}) = \frac{\phi(-\mathbf{x}'_i\boldsymbol{\theta})}{\Phi(-\mathbf{x}'_i\boldsymbol{\theta})}$ represents the Inverse Mills Ratio calculated at the censored cutoff point. Note that observations censored at the origin do not provide direct marginal information for the estimation of the scale parameter γ .

Analytical and Rigorous Proof of Global Concavity To prove that the reparametrized log-likelihood is globally concave, it suffices to demonstrate that the complete sample Hessian matrix $\mathbf{H}(\boldsymbol{\theta}, \gamma) = \sum_{i=1}^n \mathbf{H}_i(\boldsymbol{\theta}, \gamma)$ is strictly negative definite for any point in the parameter space.

1. Uncensored Regime ($y_i > 0$):

$$\mathbf{H}_{i,\text{uncensored}}(\boldsymbol{\theta}, \gamma) = - \begin{bmatrix} \mathbf{x}_i \\ -y_i \end{bmatrix} \begin{bmatrix} \mathbf{x}'_i & -y_i \end{bmatrix} - \begin{bmatrix} \mathbf{0}_{K \times K} & \mathbf{0} \\ \mathbf{0}' & \frac{1}{\gamma^2} \end{bmatrix} \quad (11.99)$$

The first term is the negative of the outer product of a vector with itself, being a negative semidefinite matrix of rank 1. The second term has only one non-zero element in the position of the scale parameter. Since $\gamma > 0$, the term $\frac{1}{\gamma^2} > 0$ makes the subtracted matrix negative semidefinite of rank 1. The sum results in a negative semidefinite matrix.

2. Censored Regime ($y_i = 0$):

$$\mathbf{H}_{i,\text{censored}}(\boldsymbol{\theta}, \gamma) = \lambda'(-\mathbf{x}'_i\boldsymbol{\theta}) \begin{bmatrix} \mathbf{x}_i\mathbf{x}'_i & \mathbf{0} \\ \mathbf{0}' & 0 \end{bmatrix} \quad (11.100)$$

Where $\lambda'(z) = -\lambda(z)[z + \lambda(z)]$. Since the standard normal is strictly log-concave, its C.D.F. $\Phi(z)$ implies that the second derivative of the log-CDF is strictly less than zero: $\lambda'(z) < 0$ for all $z \in \mathbb{R}$. Since the data matrix is positive semidefinite, its multiplication by a strictly negative scalar $\lambda'(-\mathbf{x}'_i\boldsymbol{\theta})$ makes the Hessian contribution of censored observations negative semidefinite.

3. Complete Sample Aggregation: Adding the contributions of all sample observations and given that the complete data matrix $\mathbf{X}'\mathbf{X}$ has full rank K , the sample sum results in a strictly negative definite matrix over the entire parameter space $(\boldsymbol{\theta}, \gamma)$:

$$\mathbf{H}(\boldsymbol{\theta}, \gamma) \text{ is strictly negative definite } \forall (\boldsymbol{\theta}, \gamma) \quad (11.101)$$

This establishes the global and strict concavity of the reparametrized log-likelihood, guaranteeing that numerical optimization algorithms will converge quickly and stably to a single global maximum point.

Recovery of the Original Parameters and the Delta Method

After convergence of the algorithm in the reparametrized scale, obtaining $(\hat{\boldsymbol{\theta}}, \hat{\gamma})$, the original economic parameters $(\hat{\boldsymbol{\beta}}, \hat{\sigma})$ are recovered through the inverse bijection:

$$\hat{\boldsymbol{\beta}} = \frac{\hat{\boldsymbol{\theta}}}{\hat{\gamma}} \quad \text{and} \quad \hat{\sigma} = \frac{1}{\hat{\gamma}} \quad (11.102)$$

To calculate the consistent asymptotic covariance matrix in the original structural scale, we apply the Delta Method:

$$\hat{\mathbf{V}}_{(\hat{\boldsymbol{\beta}}, \hat{\sigma})} = \mathbf{J} \cdot \hat{\mathbf{V}}_{(\hat{\boldsymbol{\theta}}, \hat{\gamma})} \cdot \mathbf{J}' \quad (11.103)$$

Where $\mathbf{J}$ represents the Jacobian matrix of the inverse transformation:

$$\mathbf{J} = \begin{bmatrix} \frac{\partial \boldsymbol{\beta}}{\partial \boldsymbol{\theta}'} & \frac{\partial \boldsymbol{\beta}}{\partial \gamma} \\ \frac{\partial \sigma}{\partial \boldsymbol{\theta}'} & \frac{\partial \sigma}{\partial \gamma} \end{bmatrix} = \begin{bmatrix} \sigma \mathbf{I}_K & -\sigma \boldsymbol{\beta} \\ \mathbf{0}' & -\sigma^2 \end{bmatrix} \quad (11.104)$$

The OLS Bias and the Empirical “Golden Rule” (Greene-Goldberger)

The OLS Attenuation Bias If we estimate the model $y_i = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i$ by OLS on the complete sample (including observations equal to zero), the estimated coefficients will be inconsistent and severely attenuated. Conceptually, OLS attempts to fit a linear line over a conditional expectation function (CEF) that is inherently non-linear and flattened in the left tail.

Analytical Proof of the Rule of Thumb (Greene, 1981) Theorem (Greene, 1981): Let the latent variable model be $y_i^* = \alpha + \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i$ with homoskedastic normal error and censoring rule $y_i = \max(0, y_i^*)$. If the regressors $\mathbf{x}_i$ follow a multivariate normal distribution in the population, then the probability limit of the OLS slope coefficient (the Best Linear Predictor, BLP) is given by:

$$\boldsymbol{\beta}_{\text{BLP}} = \boldsymbol{\beta} \cdot (1 - \pi) = \boldsymbol{\beta} \cdot P(y_i > 0) \quad (11.105)$$

Where $\pi = P(y_i = 0)$ represents the population censoring probability.

Analytical Proof: By the definition of the Best Linear Predictor: $\boldsymbol{\beta}_{\text{BLP}} = E[\mathbf{xx}']^{-1} E[\mathbf{xy}]$. Since $y_i = \max(0, y_i^*)$, the expectation of $\mathbf{xy}$ depends only on the regime where $y_i = y_i^* > 0$:

$$E[\mathbf{xy}] = E[\mathbf{xy}^* \mid y^* > 0] \cdot P(y^* > 0) \quad (11.106)$$

Using the linear projection under multivariate normality of $\mathbf{x}$ with respect to y^* , we can express $\mathbf{x}$ as:

$$\mathbf{x} = E[\mathbf{xy}^*] \sigma_{y^*}^{-2} (y^* - E[y^*]) + \mathbf{u} \quad (11.107)$$

Where $\mathbf{u}$ is an orthogonal projection error term independent of y^* . Taking the expectation conditioned on $y^* > 0$ and multiplying by y^* :

$$E[\mathbf{xy}^* \mid y^* > 0] = E[\mathbf{xy}^*] \sigma_{y^*}^{-2} E[(y^* - E[y^*]) y^* \mid y^* > 0] = E[\mathbf{xy}^*] \quad (11.108)$$

Substituting back:

$$E[\mathbf{xy}] = E[\mathbf{xy}^*] \cdot P(y^* > 0) = E[\mathbf{xy}^*] \cdot (1 - \pi) \quad (11.109)$$

Since $\boldsymbol{\beta} = E[\mathbf{xx}']^{-1} E[\mathbf{xy}^*]$, we substitute into $\boldsymbol{\beta}_{\text{BLP}}$:

$$\boldsymbol{\beta}_{\text{BLP}} = E[\mathbf{xx}']^{-1} E[\mathbf{xy}^*] (1 - \pi) = \boldsymbol{\beta} \cdot (1 - \pi) \quad \blacksquare \quad (11.110)$$

This proof establishes the mathematical foundation for the Goldberger (1981) Rule of Thumb:

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} \xrightarrow{p} \boldsymbol{\beta} \cdot P(y_i > 0) \implies \hat{\boldsymbol{\beta}}_{\text{MLE}} \approx \frac{\hat{\boldsymbol{\beta}}_{\text{OLS}}}{P(y_i > 0)} \quad (11.111)$$

The Methodological Error of Discarding Zeros: Truncation Bias Discarding observations equal to zero and running OLS only on the truncated subsample ($y_i > 0$) does not eliminate the bias; it merely replaces censoring bias with truncation bias.

Analytical Proof of Inconsistency due to Truncation: By restricting the sample to $y_i > 0$, the conditional expectation of the dependent variable becomes:

$$E[y_i \mid \mathbf{x}_i, y_i > 0] = \mathbf{x}_i' \boldsymbol{\beta} + \sigma \lambda \left(\frac{\mathbf{x}_i' \boldsymbol{\beta}}{\sigma} \right) \quad (11.112)$$

If we estimate the regression $y_i = \mathbf{x}_i' \boldsymbol{\beta} + u_i$ by OLS using only the positive observations, the effective error term is $u_i = \varepsilon_i \mid \varepsilon_i > -\mathbf{x}_i' \boldsymbol{\beta}$. The conditional expectation of this new error is:

$$E[u_i \mid \mathbf{x}_i] = \sigma \lambda \left(\frac{\mathbf{x}_i' \boldsymbol{\beta}}{\sigma} \right) \neq 0 \quad (11.113)$$

Since the Inverse Mills Ratio $\lambda(z)$ is a non-linear decreasing function of the index $z_i = \mathbf{x}_i' \boldsymbol{\beta} / \sigma$, the error of the truncated regression is systematically correlated with the regressors $\mathbf{x}_i$. The omission of the non-linear term directly violates the strict exogeneity hypothesis, generating inconsistent and severely biased estimates.

Empirical Application: The Divorce “Hedge” (Example 19.5)

Example 19.5 (Quester and Greene, 1982) illustrates the application of the Tobit model in the analysis of the labor supply of married women.

Economic Logic of the Model The economic hypothesis investigates whether married women facing a higher statistical probability of divorce react by increasing their labor supply in the formal market. This increase in hours worked acts as a self-insurance strategy (*hedge*) against the future loss of household income.

Sample Analysis and Statistical Necessity of the Tobit The database is segmented into two distinct demographic groups:

- **White Wives:** $N = 7,459$, where the proportion of working women is only 29%. This implies that 71% of the sample has zero work hours.
- **Black Wives:** $N = 2,798$, where the proportion of working women is 46%. This implies that 54% of the sample has zero hours.

This gigantic proportion of observations censored at the origin completely invalidates the use of linear regression by OLS on the complete sample.

Estimation Results Greene’s Table 19.2 summarizes the structural coefficients estimated by ML of the Tobit (effect on y^*) and the *Slopes* (unconditional marginal effect on $E[y \mid \mathbf{x}]$):

For **White Wives** ($P(y > 0) = 0.29$):

- Mean Divorce Probability: Tobit coefficient = 417.39 ($t = 6.52$); Partial effect (slope) = 121.58.
- High Divorce Probability: Tobit coefficient = 670.22 ($t = 8.40$); Partial effect (slope) = 195.22.
- Scale parameter: $\sigma = 1,559$.

For **Black Wives** ($P(y > 0) = 0.46$):

- Mean Divorce Probability: Tobit coefficient = 481.02 ($t = 5.28$); Partial effect (slope) = 219.75.
- High Divorce Probability: Tobit coefficient = 578.66 ($t = 5.33$); Partial effect (slope) = 264.36.

- Scale parameter: $\sigma = 1,511$.

Crucial Lessons and Interpretation Validation of the Rule of Thumb: For White Wives ($P(y > 0) = 0.29$), the partial effect for high divorce probability is predicted by:

$$\text{Slope} \approx 670.22 \times 0.29 = 194.36 \quad (\text{Exact value: } 195.22) \quad (11.114)$$

For Black Wives ($P(y > 0) = 0.46$):

$$\text{Slope} \approx 578.66 \times 0.46 = 266.18 \quad (\text{Exact value: } 264.36) \quad (11.115)$$

The Meaning of σ : The estimated standard deviation (1,559 and 1,511) does NOT represent the standard deviation of the observed hours y . It is, strictly, the standard deviation of the error of the unobservable latent variable ε_i . The actual standard deviation of y is drastically smaller, since censoring compresses the left tail of the distribution.

Economic Results: The probability of divorce exhibits a positive and robust impact on hours worked. For white women, being in the high-probability group increases annual hours by 195.22 hours. For black women, the effect is 264.36 hours. This confirms the *hedge* hypothesis.

Advanced Perspectives and Dialogue with Modern Literature

The Tobit Formulation via GMM Lars Peter Hansen (1982) illustrates that the first-order conditions of the likelihood can be interpreted as moment orthogonality conditions for GMM estimation. Defining the vector of individual moments based on Olsen's scores:

$$\mathbf{g}_i(\boldsymbol{\theta}, \gamma) = \begin{bmatrix} (1 - D_i)(\gamma y_i - \mathbf{x}_i' \boldsymbol{\theta}) \mathbf{x}_i - D_i \lambda(-\mathbf{x}_i' \boldsymbol{\theta}) \mathbf{x}_i \\ (1 - D_i) \left[\frac{1}{\gamma} - (\gamma y_i - \mathbf{x}_i' \boldsymbol{\theta}) y_i \right] \end{bmatrix} \quad (11.116)$$

The maximum likelihood estimator is equivalent to the exactly identified GMM estimator that solves $E[\mathbf{g}_i(\boldsymbol{\theta}, \gamma)] = \mathbf{0}$. The GMM formulation opens the way for semiparametric estimation or estimation with heteroskedasticity correction.

The Critique of Parametric Fragility (Hayashi and Wooldridge) While in the classic linear model deviations from normality and homoskedasticity affect only efficiency, in the Tobit maximum likelihood model the consistency properties collapse completely under non-normality or heteroskedasticity. If $\sigma_i^2 = \sigma^2 \exp(\mathbf{z}_i' \boldsymbol{\alpha})$, the conditional expectation of the score ceases to be zero, violating Fisher's identity and making the estimators inconsistent. This justifies the preference for rigorous specification tests or Powell's CLAD estimator.

Verdict and Synthesis of the Section

This section closes the fundamental cycle of the Tobit censored regression model. It performs the crucial transition between the algebra of the log-likelihood and practical empirical implications. The joint mastery of Olsen's reparametrization, the Delta Method, the exact understanding of the OLS attenuation bias, and the correct interpretation of the parameter σ constitute the true watershed in applied microeconometrics.

11.2.4 Two-Part Models and Corner Solutions

This section expands the conceptual and mathematical framework of the two-part model, proving the fundamental log-likelihood identity that enables the Fin and Schmidt (1984) Likelihood Ratio (LR) Test, strictly establishes the Amemiya (1985) classification of Tobit models (Types I to V), details the Murphy-Topel (2002) framework for the Heckman (Heckit) two-step estimator, and introduces a deep discussion on visual specification diagnostics and the empirical application of extramarital *affairs* (Fair, 1977 vs. Fair, 1978). The conceptual dialogue is enriched with contributions from Jeffrey Wooldridge (2010), Bruce Hansen (2022), Fumio Hayashi (2000), and *Mostly Harmless Econometrics* (2009).

The Implicit Restriction of the Type 1 Tobit and the Economic Motivation

As established previously, the Type 1 Tobit model imposes an extremely rigid parametric structure on the behavior of economic agents. Under this specification, the observed dependent variable is $y_i = \max(0, y_i^*)$, where the latent variable is $y_i^* = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i$ with $\varepsilon_i \mid \mathbf{x}_i \sim N(0, \sigma^2)$.

The fundamental limitation of the Type 1 Tobit lies in the fact that a single parameter vector $\boldsymbol{\beta}$ governs simultaneously the participation decision (extensive margin) and the consumption intensity decision (intensive margin). Strictly, the two partial derivatives (marginal effects) with respect to a continuous regressor x_{ik} are expressed by:

Marginal Effect on the Extensive Margin (Participation):

$$\frac{\partial P(y_i > 0 \mid \mathbf{x}_i)}{\partial x_{ik}} = \frac{\phi(w_i)}{\sigma} \beta_k \quad (11.117)$$

Where $w_i = \frac{\mathbf{x}_i' \boldsymbol{\beta}}{\sigma}$ is the normalized index and $\phi(\cdot) > 0$ is the p.d.f. of the standard normal. Since $\sigma > 0$, the sign of this derivative is strictly identical to the sign of β_k .

Marginal Effect on the Conditional Mean (Intensity):

$$\frac{\partial E[y_i \mid y_i > 0, \mathbf{x}_i]}{\partial x_{ik}} = \beta_k [1 - d(w_i)] \quad (11.118)$$

Where $d(w_i) = \lambda(w_i) [\lambda(w_i) + w_i]$ is the truncated variance scale factor (with $0 < d(w_i) < 1$) and $\lambda(\cdot)$ is the Inverse Mills Ratio. Since $1 - d(w_i) > 0$, the sign of this derivative is also strictly determined by the sign of β_k .

Restrictive Implications and Economic Counterexamples This dependence on a single parameter β_k generates two severe and implausible behavioral restrictions in applied economics:

1. **Unified Sign Restriction:** If a regressor x_k increases the probability of an individual entering the market (participation), it must obligatorily increase the quantity consumed or hours worked by that individual (volume). It is impossible to model a factor that stimulates entry but discourages subsequent consumption.
2. **Strict Proportionality Restriction (Scale Sufficiency):** For any two continuous regressors x_j and x_h , the ratio between their marginal effects on participation is

exactly equal to the ratio of their effects on intensity, collapsing into the ratio of their structural coefficients:

$$\frac{\partial P(y > 0 \mid \mathbf{x}) / \partial x_j}{\partial P(y > 0 \mid \mathbf{x}) / \partial x_h} = \frac{\partial E[y \mid y > 0, \mathbf{x}] / \partial x_j}{\partial E[y \mid y > 0, \mathbf{x}] / \partial x_h} = \frac{\beta_j}{\beta_h} \quad (11.119)$$

Classic Counterexamples in the Literature:

- **Charitable Donations:** The Age of the individual may positively affect the decision to make a donation (extensive margin). However, conditional on donating, age may have a negative effect on the volume donated (intensive margin), due to income decline after retirement.
- **Monetary Fire Damage (Fin & Schmidt, 1984):** Building age has opposite effects on the two margins. Older buildings have a higher probability of suffering a fire (positive effect on the probability of loss). However, new buildings, having a higher incorporated market value, suffer much larger monetary damages when a fire occurs.
- **Female Labor Supply:** The presence of small school-age children in the household may not significantly impact the mother's decision to participate in the labor market, but drastically reduce the number of hours worked conditional on participation.

The Cragg (1971) Two-Part Model: Truncated Normal Hurdle Model

To break the parametric rigidity of the Tobit, John Cragg (1971) proposed the truncated tail Hurdle model, formulating two distinct behavioral equations under the assumption of conditional independence between the errors of the two regimes.

Mathematical Structure of the Two Parts Part 1: Participation Equation (Binary Probit Regime): Let $s_i \in \{0, 1\}$ be the indicator of non-zero observation ($s_i = \mathbf{1}(y_i > 0)$). The participation probability is governed by a parameter vector $\boldsymbol{\gamma}$:

$$\begin{aligned} P(s_i = 1 \mid \mathbf{x}_i) &= \Phi(\mathbf{x}_i' \boldsymbol{\gamma}) \\ P(s_i = 0 \mid \mathbf{x}_i) &= 1 - \Phi(\mathbf{x}_i' \boldsymbol{\gamma}) \end{aligned} \quad (11.120)$$

Part 2: Intensity Equation (Positive Consumption Regime): Conditional on participation ($y_i > 0$), the dependent variable follows a normal distribution truncated from the left at zero, governed by an independent parameter vector $\boldsymbol{\beta}$ and standard deviation σ :

$$f(y_i \mid y_i > 0, \mathbf{x}_i) = \frac{\frac{1}{\sigma} \phi\left(\frac{y_i - \mathbf{x}_i' \boldsymbol{\beta}}{\sigma}\right)}{\Phi\left(\frac{\mathbf{x}_i' \boldsymbol{\beta}}{\sigma}\right)} \quad (11.121)$$

The Complete Unconditional Probability Density Function Multiplying the conditional probabilities, the unconditional mixed p.d.f. of $y_i \mid \mathbf{x}_i$ under the Cragg model is expressed by:

$$f_y(y_i \mid \mathbf{x}_i) = [1 - \Phi(\mathbf{x}_i' \boldsymbol{\gamma})]^{\mathbf{1}(y_i=0)} \cdot \left[\Phi(\mathbf{x}_i' \boldsymbol{\gamma}) \frac{\frac{1}{\sigma} \phi\left(\frac{y_i - \mathbf{x}_i' \boldsymbol{\beta}}{\sigma}\right)}{\Phi\left(\frac{\mathbf{x}_i' \boldsymbol{\beta}}{\sigma}\right)} \right]^{\mathbf{1}(y_i>0)} \quad (11.122)$$

The Fin and Schmidt (1984) Specification Test and the Log-Likelihood Identity

Proof of the Algebraic Identity The algebraic beauty of the relationship between the Type 1 Tobit model and the Cragg model lies in the fact that the Tobit is a direct parametric restriction of Cragg under the null hypothesis:

$$H_0 : \gamma = \frac{\beta}{\sigma} \quad (11.123)$$

Let us prove analytically that, under this restriction, the Cragg density collapses exactly into the Type 1 Tobit density:

- **For censored observations** ($y_i = 0$): Substituting $\gamma = \frac{\beta}{\sigma}$ into the Cragg censoring probability:

$$1 - \Phi(\mathbf{x}'_i \gamma) \xrightarrow{H_0} 1 - \Phi\left(\frac{\mathbf{x}'_i \beta}{\sigma}\right) \quad (11.124)$$

Which is identically equal to the discrete term of the Tobit.

- **For uncensored observations** ($y_i > 0$): Substituting $\gamma = \frac{\beta}{\sigma}$ into the Cragg continuous term:

$$\Phi(\mathbf{x}'_i \gamma) \frac{\frac{1}{\sigma} \phi\left(\frac{y_i - \mathbf{x}'_i \beta}{\sigma}\right)}{\Phi\left(\frac{\mathbf{x}'_i \beta}{\sigma}\right)} \xrightarrow{H_0} \Phi\left(\frac{\mathbf{x}'_i \beta}{\sigma}\right) \frac{\frac{1}{\sigma} \phi\left(\frac{y_i - \mathbf{x}'_i \beta}{\sigma}\right)}{\Phi\left(\frac{\mathbf{x}'_i \beta}{\sigma}\right)} = \frac{1}{\sigma} \phi\left(\frac{y_i - \mathbf{x}'_i \beta}{\sigma}\right) \quad (11.125)$$

Which simplifies perfectly through the cancellation of the term $\Phi(\cdot)$, exactly reproducing the continuous p.d.f. of the classic normal linear regression.

Thus, the restricted log-likelihood function of the Tobit is the exact sum of the Probit and Truncated Regression components when evaluated under the same parametric scale:

$$\ln L_{\text{Tobit}}(\beta, \sigma^2) = \ln L_{\text{Probit}}\left(\gamma = \frac{\beta}{\sigma}\right) + \ln L_{\text{Truncated}}(\beta, \sigma^2) \quad (11.126)$$

The Likelihood Ratio (LR) Test Since the Cragg model assumes conditional independence between the two equations, the unrestricted likelihood is calculated by estimating the parts separately. The Likelihood Ratio (LR) Test to test the validity of the Tobit specification is given by:

$$LR = -2 [\ln L_{\text{Tobit}} - (\ln L_{\text{Probit}} + \ln L_{\text{Truncated}})] \sim \chi_K^2 \quad (11.127)$$

Where K is the number of restrictions imposed by H_0 , which is exactly equal to the dimension of the parameter vector β (number of regressors including the intercept, $K_{\mathbf{x}} + 1$). If the calculated value of LR exceeds the critical value from the χ_K^2 table, we reject H_0 , concluding that the parametric coupling of the Tobit is invalid.

Generalization: Different Regressors and the Five Tobit Models of Amemiya (1985)

The Cragg (1971) model, although flexible, still imposes that the same regressor vector $\mathbf{x}$ governs both decisions. The generalization of Amemiya (1985) formalizes a robust framework for censored regressions and sample selection, classifying limited models into five major categories:

The Structural Classification of Amemiya (1985)

1. **Type I Tobit (Classic Censoring):** $y_i^* = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i$, $y_i = \max(0, y_i^*)$. The dependent variable is observed if positive; otherwise, zero is observed.
2. **Type II Tobit (Heckman Selection Model):** Consists of two equations and a binary indicator variable structure for participation.

$$\begin{aligned} \text{Selection: } d_i^* &= \mathbf{z}_i' \boldsymbol{\gamma} + u_i, & d_i &= \mathbf{1}(d_i^* > 0) \\ \text{Intensity: } y_i^* &= \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i \end{aligned} \quad (11.128)$$

Observation Rule: $y_i = y_i^*$ if $d_i = 1$; y_i is *unobserved* (missing) if $d_i = 0$.

3. **Type III Tobit (Continuous Censoring as Selection):** The selection indicator is itself an observable censored variable of the Type 1 Tobit type.
4. **Type IV Tobit:** The selection indicator is a censored variable, and the final outcome is divided into different continuous observation regimes.
5. **Type V Tobit:** The selection indicator is a conventional binary Probit choice, but there are two distinct latent intensity equations.

Conceptual Differentiation: Type II Tobit vs. Heckman Selection Model

- **Type II Tobit (Sample Censoring / Corner Solution):** The variable y is observed for the entire population, accumulating a real point mass at zero. The regressors $\mathbf{x}$ and $\mathbf{z}$ are fully observed for all individuals. E.g., Travel expenditures (those who do not travel report zero expenditure, but their characteristics are known).
- **Heckman Selection Model (Incidental Truncation):** The outcome variable y^* is *unobserved/missing* when the individual chooses not to participate. Zero is not a real economic value. E.g., The wage of those who do not work simply does not physically exist in the market.

Type II Estimation: The Heckman Two-Step Method (Heckit)

Rigorous Derivation of the Conditional Expectation Let the bivariate normal selection structural model with error correlation be:

$$\begin{aligned} s_i^* &= \mathbf{z}_i' \boldsymbol{\gamma} + u_i, & s_i &= \mathbf{1}(s_i^* > 0), & u_i &\sim N(0, 1) \\ y_i^* &= \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i, & \varepsilon_i &\sim N(0, \sigma^2) \\ \text{Cov}(u_i, \varepsilon_i) &= \sigma_{u\varepsilon} = \rho\sigma \end{aligned} \quad (11.129)$$

By the classic properties of the bivariate normal distribution, we can linearly decompose ε_i with respect to the selection error u_i :

$$\varepsilon_i = \frac{\text{Cov}(\varepsilon_i, u_i)}{\text{Var}(u_i)} u_i + \eta_i = \rho\sigma u_i + \eta_i \quad (11.130)$$

Where η_i is normally distributed with mean zero and independent of u_i . Applying the conditional expectation:

$$E[\varepsilon_i \mid u_i > -\mathbf{z}_i' \boldsymbol{\gamma}] = \rho\sigma E[u_i \mid u_i > -\mathbf{z}_i' \boldsymbol{\gamma}] \quad (11.131)$$

By the classic left-tail normal truncation theorem, using the symmetry of the standard normal distribution:

$$E[u_i \mid u_i > -\mathbf{z}'_i \boldsymbol{\gamma}] = \frac{\phi(\mathbf{z}'_i \boldsymbol{\gamma})}{\Phi(\mathbf{z}'_i \boldsymbol{\gamma})} = \lambda(\mathbf{z}'_i \boldsymbol{\gamma}) \quad (11.132)$$

Where $\lambda(\cdot)$ is the Inverse Mills Ratio. Substituting:

$$E[\varepsilon_i \mid s_i = 1, \mathbf{x}_i, \mathbf{z}_i] = \rho\sigma\lambda(\mathbf{z}'_i \boldsymbol{\gamma}) \quad (11.133)$$

Therefore, the conditional expectation of the observed variable is expressed by:

$$E[y_i \mid s_i = 1, \mathbf{x}_i, \mathbf{z}_i] = \mathbf{x}'_i \boldsymbol{\beta} + \rho\sigma\lambda(\mathbf{z}'_i \boldsymbol{\gamma}) \quad (11.134)$$

Two-Step Algorithm (Heckit)

1. **Step 1 (Selection Probit):** Estimate the Probit model of s_i with respect to $\mathbf{z}_i$ for the entire sample (n), obtaining $\hat{\boldsymbol{\gamma}}$. Calculate the estimated Inverse Mills Ratio $\hat{\lambda}_i = \lambda(\mathbf{z}'_i \hat{\boldsymbol{\gamma}})$.
2. **Step 2 (Second-Stage OLS Regression):** Estimate by OLS the outcome equation using only the active observations (n_1 where $s_i = 1$):

$$y_i = \mathbf{x}'_i \boldsymbol{\beta} + \theta \hat{\lambda}_i + e_i \quad (11.135)$$

Where $\theta = \rho\sigma$ is the estimated covariance coefficient.

The Murphy-Topel (2002) Correction for Consistent Standard Errors In the second stage, the Inverse Mills Ratio $\hat{\lambda}_i$ is a regressor generated in the first stage. Ignoring the variability of $\hat{\boldsymbol{\gamma}}$ results in severe underestimation of the error. The Murphy-Topel correction rewrites the estimator as a GMM system to derive the robust asymptotic covariance matrix:

$$\mathbf{V}_2^* = \mathbf{V}_2 + \mathbf{V}_2 [\mathbf{C}\mathbf{V}_1\mathbf{C}' - \mathbf{R}\mathbf{V}_1\mathbf{C}' - \mathbf{C}\mathbf{V}_1\mathbf{R}'] \mathbf{V}_2 \quad (11.136)$$

Where the transition and cross-covariance matrices are estimated at the optimum by:

$$\begin{aligned} \mathbf{C} &= \sum_{i \in \{s_i=1\}} \frac{\partial \mathbf{s}_{2i}(\hat{\boldsymbol{\beta}}, \hat{\theta}, \hat{\boldsymbol{\gamma}})}{\partial \boldsymbol{\gamma}'} \\ \mathbf{R} &= \sum_{i \in \{s_i=1\}} \mathbf{s}_{2i}(\hat{\boldsymbol{\beta}}, \hat{\theta}, \hat{\boldsymbol{\gamma}}) \cdot \mathbf{s}_{1i}(\hat{\boldsymbol{\gamma}})' \end{aligned} \quad (11.137)$$

Visual Diagnostics: Real Censoring vs. Corner Solution

The theoretical distinction between data censoring and corner solution is accompanied by a very clear empirical visual signature when we plot the distribution of the observed dependent variable y .

- **Corner Solution:** Large cluster accumulated at the economic zero by utility choice. The positive distribution ($y > 0$) is continuous and smooth, with no apparent sign of truncation. There is no artificial accumulation of observations immediately above zero.

- **Censored Data:** Large artificial cluster at the cutoff point imposed by the sampling design. The positive distribution is clearly “cut” or abruptly sliced at the censoring point. There is dense and abnormal accumulation of observations adjacent to the censoring point.

Greene’s Warning: The Physics of Specification Error. When we force the Tobit on Corner Solution data, the ML algorithm is forced to “flatten” the slope of the regression line and artificially inflate the estimated standard deviation ($\hat{\sigma}$) to compensate for the lack of positive observations near zero, resulting in severe attenuation bias of the structural coefficients $\hat{\beta}$.

Empirical Application: Extramarital Affairs (Fair, 1977 vs. Fair, 1978)

The study on time devoted to extramarital *affairs* conducted by Ray Fair illustrates the failure of the Tobit and the victory of two-part models. The *Redbook Magazine* database (Fair, 1977) contains $N = 6,366$ observations of married women (4,313 zeros, i.e., 68% censoring rate).

Critical Analysis of the Comparison Models

- **OLS:** Suffers from severe attenuation bias and drastically underestimates the standard deviation of the errors.
- **Type 1 Tobit:** Collapses by imposing the proportionality and sign equality restriction.
- **Truncated Regression:** Discarding 68% of the sample results in extreme numerical instability. The coefficient of “years married” reverses sign.
- **Cragg Hurdle / Two-Part:** Estimates the decisions separately, freeing the parameters from scale and sign restrictions.

Mathematical Proof: Rejection of the Tobit Restriction via LR Test From the reported maximum log-likelihoods:

$$\begin{aligned}\ln L_{\text{Tobit}} &= -7,804.38 \\ \ln L_{\text{Probit}} &= -3,463.71 \\ \ln L_{\text{Truncated}} &= -3,469.58\end{aligned}\tag{11.138}$$

We construct the Likelihood Ratio (LR) Test statistic:

$$LR = 2[-3,463.71 + (-3,469.58) - (-7,804.38)] = 2[871.09] = 1,742.18 \tag{11.139}$$

The obtained value of $LR = 1,742.18$ is astronomically higher than the critical value of the chi-square distribution with 9 degrees of freedom ($\chi^2_{0.05,9} = 16.92$). The Tobit restriction is overwhelmingly rejected.

The Victory of the Hurdle: Heterogeneity and Sign Reversal The Hurdle model reveals a rich behavioral reality that the Tobit concealed. The coefficient of the variable “years married” exhibits opposite effects in each stage:

- **In the Participation Equation (Probit):** The coefficient is strictly negative (-0.168). The probability of initiating an *affair* decreases with marriage duration.
- **In the Intensity Equation (Truncated):** The coefficient is strongly positive ($+0.140$). For those who have already decided to participate, the time devoted increases as the marriage is older.

In the standard Tobit model, this richness was compressed into a single positive coefficient of $+0.128$, concealing the real dynamics.

Synthesis and Methodological Comparison

The methodological comparison between the main limited dependent variable models can be summarized as follows:

Type 1 Tobit: Participation is rigid (governed by β); Intensity ($y > 0$) follows a truncated normal; error correlation is unified; the exclusion restriction is not applicable; predictions are non-negative; covariance correction is not applicable.

Cragg Hurdle: Participation is free (governed by γ); Intensity follows a free truncated normal; errors are independent; the exclusion restriction is not applicable; predictions are non-negative; covariance correction is not applicable.

Hurdle Lognormal: Participation is free (governed by γ); Intensity follows a lognormal distribution; errors are independent; the exclusion restriction is not applicable; predictions are non-negative; covariance correction is not applicable.

Type II Tobit (ET2T): Participation is free (governed by γ); Intensity follows a corrected lognormal; errors are correlated ($\rho\sigma$); the exclusion restriction is strictly necessary; predictions are non-negative; covariance correction requires Murphy-Topel (2002).

Verdict and Conclusion

This section consolidates the definitive abandonment of the blind and purely mechanical application of the Type 1 Tobit model in modern applied microeconometrics. Mastery of the McDonald-Moffitt Decomposition and the Empirical Analysis of Extramarital *Affairs* proves that participation and intensity decisions are distinct behavioral phenomena governed by frequently opposing economic forces.

By mastering the Cragg (1971) Model, the Hurdle Lognormal (Duan et al., 1984), and the Amemiya (1985) generalizations, the researcher is empowered to model each margin with total theoretical independence and statistical precision. Finally, the critical understanding brought by Jeffrey Wooldridge regarding the need for solid exclusion restrictions and the limitations of the Type II Tobit at extreme levels protects the econometrician from the pitfalls of functional multicollinearity and incoherent negative predictions at the origin, raising empirical analysis to the highest standards of rigor and scientific relevance.

11.2.5 Applications in Panel Data

This section presents a structured, rigorous, and logically chained review of panel data applications, maintaining the standard of academic excellence and dialogue with modern literature established in the previous sections. The analysis addresses the extension of the

Tobit model to panel data, the fixed effects enigma, Greene's (2004a) surprising discovery, the analogy with linear regression, and semiparametric and dynamic approaches.

Motivation and Extension of the Tobit to Panel Data

The extension of the Tobit model to panel data follows the same logic as discrete choice models (Probit/Logit). The basic structure incorporates unobserved individual effects (α_i), which capture the unobserved heterogeneity specific to each cross-sectional unit.

Latent Model with Individual Effects The latent variable is specified as:

$$y_{it}^* = \alpha_i + \mathbf{x}_{it}'\boldsymbol{\beta} + \varepsilon_{it}, \quad \varepsilon_{it} \sim N(0, \sigma^2) \quad (11.140)$$

Observation Mechanism (Left Censoring at Zero) The observed variable is generated by the standard censoring mechanism:

$$y_{it} = \max(0, y_{it}^*) \quad (11.141)$$

The fundamental challenge lies in how to treat α_i : as fixed parameters to be estimated (fixed effects) or as random variables correlated or not with $\mathbf{x}_{it}$ (random effects).

Random Effects Models: Standard Approach

For random effects models, the strategy is analogous to the Probit with random effects.

Specification It is assumed that the individual effects are random variables independent of the regressors:

$$\alpha_i \sim N(0, \sigma_\alpha^2), \quad \alpha_i \perp \mathbf{x}_{it} \quad (11.142)$$

Likelihood Function The individual likelihood integrates over the distribution of α_i :

$$L_i = \int \prod_{t=1}^T f(y_{it} | \alpha_i, \mathbf{x}_{it}, \boldsymbol{\beta}, \sigma) \cdot g(\alpha_i) d\alpha_i \quad (11.143)$$

Computational Challenge The multidimensional integral has no closed form. The standard solutions are:

1. **Gaussian Quadrature:** Numerical approximation of the integral (efficient for small T).
2. **Simulation (Simulated Maximum Likelihood — SML):** Uses Monte Carlo simulation to approximate the integral (more flexible, but computationally intensive).

The Mundlak (1978) Correction To accommodate correlation between α_i and $\mathbf{x}_{it}$ without losing the random effects structure, the Mundlak correction is applied:

$$\alpha_i = \bar{\mathbf{x}}_i' \boldsymbol{\gamma} + \eta_i, \quad \eta_i \sim N(0, \sigma_\eta^2) \quad (11.144)$$

Where $\bar{\mathbf{x}}_i = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_{it}$ are the time means of the regressors. This allows α_i to be correlated with $\mathbf{x}_{it}$ in a systematic way, preserving the efficiency of the random effects estimator.

The Fixed Effects Enigma in the Tobit

The theoretical literature has focused intensely on the problem of **fixed effects** in the Tobit. The natural approach would be the Maximum Likelihood (ML) estimator for the model with fixed effects.

Model with Fixed Effects The specification is:

$$y_{it}^* = \alpha_i + \mathbf{x}_{it}'\boldsymbol{\beta} + \varepsilon_{it} \quad (11.145)$$

The ML estimator simultaneously estimates $(\hat{\alpha}_1, \dots, \hat{\alpha}_N, \hat{\boldsymbol{\beta}}, \hat{\sigma})$.

The Incidental Parameters Problem (Neyman-Scott, 1948) When $N \rightarrow \infty$ with T fixed, the number of parameters to estimate grows with the sample size (N individual effects α_i). This classically leads to inconsistent estimators, as demonstrated in the Probit with fixed effects (where $\hat{\boldsymbol{\beta}}$ is biased away from zero — reverse attenuation bias).

Greene's (2004a) Surprising Discovery

Contrary to the intuition based on the Probit, Greene (2004a) discovered in extensive Monte Carlo experiments that the ML estimator of the Tobit with fixed effects **does not present substantial bias in $\hat{\boldsymbol{\beta}}$** .

The Empirical Result

- $\hat{\boldsymbol{\beta}}$ is approximately consistent (small or zero bias).
- $\hat{\sigma}$ is **severely biased downward**, persistently and substantially.

Why Is This Surprising? In the Probit with fixed effects, the estimator of $\boldsymbol{\beta}$ is inconsistently biased away from zero (reverse attenuation bias). The same was expected in the Tobit, but this is not the case.

The Intuition Behind the Result Greene argues that the behavior of the Tobit with fixed effects resembles more that of the **linear regression with fixed effects** than the Probit. This analogy is fundamental to understanding the apparent contradiction.

The Analogy with Linear Regression and Its Implications

To understand why $\hat{\boldsymbol{\beta}}$ is consistent but $\hat{\sigma}$ is biased, Greene makes a revealing analogy with the classic linear model.

Linear Model with Fixed Effects (LSDV) The specification is:

$$y_{it} = \alpha_i + \mathbf{x}_{it}'\boldsymbol{\beta} + \varepsilon_{it} \quad (11.146)$$

The ML estimator is the *Least Squares Dummy Variables* (LSDV) or *within-groups* estimator:

$$\hat{\boldsymbol{\beta}}_{\text{LSDV}} = \left[\sum_{i=1}^N \sum_{t=1}^T (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)' \right]^{-1} \sum_{i=1}^N \sum_{t=1}^T (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)(y_{it} - \bar{y}_i) \quad (11.147)$$

Properties of LSDV

- $\hat{\beta}_{\text{LSDV}}$ is **unbiased and consistent** (even with T fixed and $N \rightarrow \infty$).
- The ML estimator of the error variance is:

$$\hat{\sigma}_{\text{LSDV}}^2 = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \hat{\varepsilon}_{it}^2 \quad (11.148)$$

This estimator is **biased downward** by a factor of $(T-1)/T$, since it does not correct for the degrees of freedom lost in estimating the N fixed effects α_i .

The Connection with the Tobit Greene argues that the Tobit with fixed effects behaves analogously:

- $\hat{\beta}$ is consistent (as in LSDV).
- $\hat{\sigma}$ is biased downward (as in LSDV), because the estimation of the N incidental parameters α_i “consumes” degrees of freedom.

Implications for Statistical Inference Although $\hat{\beta}$ is consistent, the bias in $\hat{\sigma}$ **contaminates the standard errors and hypothesis tests**. Since the partial effects in the Tobit depend on σ (via $\Phi(\mathbf{x}'\beta/\sigma)$ and $\lambda(\mathbf{x}'\beta/\sigma)$), inference on marginal effects is compromised.

The Impossibility of Conditional Transformations

In the linear model with fixed effects, the incidental parameters problem is solved by the **within transformation** (subtraction of individual means $\bar{y}_i$), which eliminates α_i :

$$y_{it} - \bar{y}_i = (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)' \beta + (\varepsilon_{it} - \bar{\varepsilon}_i) \quad (11.149)$$

Why Does This Not Work in the Tobit? The latent variable y_{it}^* is **censored before observation**. We do not observe y_{it}^* , but only $y_{it} = \max(0, y_{it}^*)$. Therefore:

- **First-differencing:** $y_{it} - y_{i,t-1}$ does not eliminate α_i in a useful way, since censoring makes the difference non-linear and non-additive.
- **Within-transformation:** $y_{it} - \bar{y}_i$ also does not preserve the model structure, since the mean $\bar{y}_i$ is a non-linear function of α_i , $\mathbf{x}_{it}$, and ε_{it} .

Conclusion: There is no simple conditional estimator for β in the Tobit with fixed effects, unlike the linear model.

Semiparametric Estimators and Dynamic Models

Given the impossibility of conditional transformations, the theoretical literature has developed alternative approaches.

Semiparametric Estimators

- **Honoré and Kyriazidou (2000):** Develop semiparametric estimators for the Tobit with fixed effects that do not require specifying the distribution of ε_{it} nor knowledge of σ . These estimators exploit temporal variation within individuals to identify β consistently, even with small T .
- **Advantage:** Consistency without requiring normality or homoskedasticity.
- **Disadvantage:** Computational complexity and loss of efficiency relative to the parametric ML (if the parametric assumptions are valid).

Dynamic Models When the lagged dependent variable is included as a regressor:

$$y_{it}^* = \alpha_i + \gamma y_{i,t-1} + \mathbf{x}_{it}'\beta + \varepsilon_{it} \quad (11.150)$$

Greene's (2004a) benign result **is lost**. The combination of:

1. Fixed effects (α_i),
2. Lagged dependent variable ($y_{i,t-1}$), and
3. Censoring ($y_{it} = \max(0, y_{it}^*)$)

generates a problem of **severe endogeneity**: $y_{i,t-1}$ is correlated with α_i and with ε_{it} (via censoring).

Theoretical Contributions

- **Arellano (2001):** General results on the estimation of dynamic models with fixed effects and limited variables.
- **Hahn and Kuersteiner (2004):** Characterize the bias of the ML estimator in dynamic binary choice and censored regression models, and propose **bias reduction** methods based on asymptotic expansions.

Verdict and Synthesis of the Section

This section consolidates the extension of the Tobit model to the panel data context, revealing profound theoretical nuances that distinguish panel analysis from cross-sectional analysis. Greene's (2004a) discovery — that the ML estimator of the Tobit with fixed effects produces $\hat{\beta}$ approximately consistent but $\hat{\sigma}$ severely biased — represents a rupture with the intuition based on the Probit and establishes a fundamental analogy with the LSDV estimator of the classic linear regression.

The impossibility of simple conditional transformations (such as the *within* transformation) in the Tobit, due to the non-linear nature of censoring, has driven the development of sophisticated semiparametric estimators (Honoré and Kyriazidou, 2000) and bias correction methods for dynamic models (Hahn and Kuersteiner, 2004). For the applied researcher, mastery of these subtleties is essential: the choice between fixed effects, random effects (with the Mundlak correction), and semiparametric approaches should be guided by the nature of the unobserved heterogeneity, the time horizon T , and the presence or absence of dynamic dependence. Inference on marginal effects, in particular, requires redoubled

caution, since the bias in $\hat{\sigma}$ directly contaminates the measures of economic impact derived from the model.

11.3 Sample Selection

11.3.1 Fundamental Concepts: Direct vs. Incidental Truncation

Before formalizing the model, it is indispensable to establish the methodological distinction between two types of truncation frequently confused in empirical practice:

- **Direct Truncation:** Occurs when the sample is selected directly based on the values of the variable of interest y itself. The researcher does not observe any information (not even the regressors X) for individuals outside the limit.
 - *Example:* A study on poverty that selects only families with income up to 1.5 times the poverty level. Direct truncation of y from above flattens the regression line estimated by OLS, biasing the coefficients toward zero (attenuation effect).
- **Incidental Truncation or Sample Selection:** Occurs when the observability of the variable of interest y_1 depends on the outcome of another selection variable, y_2 (or s). We observe y_1 if, and only if, the selection variable crosses a certain threshold. The truncation of y_1 is, therefore, a “byproduct” or incidental effect of the selection that occurred in y_2 .
 - *Classic example:* In labor economics, the offer wage (y_1) is only observed for individuals who participate in the labor force ($s = 1$). Since the decision to work (s) is correlated with skills and unobservable characteristics that also affect the wage (u_1), estimation by OLS on the sample of workers will generate severely biased and inconsistent estimates.

11.3.2 Mathematical Basis: Moments of the Bivariate Truncated Normal

To establish the mathematical basis of incidental truncation, we analyze two continuous random variables, y (of interest) and z (of selection), which follow a bivariate normal distribution in the population:

$$\begin{pmatrix} y \\ z \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_y \\ \mu_z \end{pmatrix}, \begin{pmatrix} \sigma_y^2 & \rho\sigma_y\sigma_z \\ \rho\sigma_y\sigma_z & \sigma_z^2 \end{pmatrix} \right) \quad (11.151)$$

The objective is to determine the mean and variance of y , given that the selection variable exceeds a specific threshold value: $z > a$.

The Standardization Process We define the standardized selection variable $z^* = \frac{z - \mu_z}{\sigma_z} \sim \mathcal{N}(0, 1)$. The truncation limit $z > a$ is equivalent to $z^* > a_z$, where:

$$a_z = \frac{a - \mu_z}{\sigma_z} \quad (\text{Standardized Truncation Point}) \quad (11.152)$$

Derivation of the Conditional Mean $\mathbb{E}[y \mid z > a]$ By the properties of the bivariate normal, we express y as a classic linear projection with respect to z^* plus an orthogonal error term e :

$$y = \mu_y + \rho\sigma_y z^* + e \quad (11.153)$$

Where e has the following fundamental properties:

- $\mathbb{E}[e] = 0$
- $\text{Var}[e] = \sigma_y^2(1 - \rho^2)$
- e is statistically independent of z^* (and therefore independent of any truncation applied to z^*).

Applying the conditional expectation with respect to the truncated regime $z^* > a_z$:

$$\mathbb{E}[y \mid z > a] = \mu_y + \rho\sigma_y \mathbb{E}[z^* \mid z^* > a_z] + \mathbb{E}[e \mid z^* > a_z] \quad (11.154)$$

Since e is independent of z^* , the term $\mathbb{E}[e \mid z^* > a_z]$ reduces to $\mathbb{E}[e] = 0$. The truncated expectation term of the standard normal is given exactly by the Inverse Mills Ratio $\lambda(a_z)$:

$$\mathbb{E}[z^* \mid z^* > a_z] = \lambda(a_z) = \frac{\phi(a_z)}{1 - \Phi(a_z)} \quad (11.155)$$

Where $\phi(\cdot)$ is the p.d.f. and $\Phi(\cdot)$ is the C.D.F. of the standard normal. Substituting:

$$\mathbb{E}[y \mid z > a] = \mu_y + \rho\sigma_y \lambda(a_z) \quad (11.156)$$

Economic Interpretation: The shift in the conditional mean of y depends crucially on the sign of the correlation ρ . If $\rho > 0$, truncation of z from below ($z > a$) shifts the conditional mean of y upward ($\mathbb{E}[y \mid z > a] > \mu_y$), indicating positive selection.

Derivation of the Conditional Variance $\text{Var}[y \mid z > a]$ Using the same orthogonal decomposition and the independence between e and z^* :

$$\text{Var}[y \mid z > a] = \rho^2 \sigma_y^2 \text{Var}[z^* \mid z^* > a_z] + \text{Var}[e] \quad (11.157)$$

We know that $\text{Var}[e] = \sigma_y^2(1 - \rho^2)$ and the variance of a standard normal truncated from the left at a_z is:

$$\text{Var}[z^* \mid z^* > a_z] = 1 - \delta(a_z) \quad (11.158)$$

Where $\delta(a_z) = \lambda(a_z)[\lambda(a_z) - a_z]$ is the variance decay factor. Substituting:

$$\text{Var}[y \mid z > a] = \rho^2 \sigma_y^2 [1 - \delta(a_z)] + \sigma_y^2(1 - \rho^2) = \sigma_y^2 [1 - \rho^2 \delta(a_z)] \quad (11.159)$$

Interpretation: Since $0 < \rho^2 \delta(a_z) < 1$, incidental truncation always reduces the variance of the variable of interest relative to its untruncated population variance.

11.3.3 The Heckman Selection Model (Type II Tobit)

Economic Motivation: The Labor Supply Model

The contemporary econometric formulation of sample selection dates back to the classic labor supply model of Gronau (1974) and Heckman (1974). The offer wage (w) is only

observed for individuals who participate in the labor force ($z = 1$). The participation decision occurs if the market wage exceeds the reservation wage (w^*):

$$z = 1 \iff w > w^* \quad (11.160)$$

The selection bias arises because the unobserved variables that determine the market wage (ability, motivation) are correlated with the unobserved factors that affect the decision to work (preference for leisure, household productivity).

General Structure of the Model

The system is structured as a model of triangular simultaneous equations with correlated errors:

- **Equation of Interest (Outcome):** $y = x'\beta + e$, $\mathbb{E}[e | x] = 0$
- **Selection Equation (Participation):** $z^* = w'\gamma + u$, $z = \mathbf{1}[z^* > 0]$

Strict exogeneity of the regressors with respect to the errors is assumed, but with fundamental stochastic dependence: $\text{Cov}(e, u) = \sigma_{eu} \neq 0$. Under joint bivariate normality:

$$\begin{pmatrix} e \\ u \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_e^2 & \rho\sigma_e\sigma_u \\ \rho\sigma_e\sigma_u & \sigma_u^2 \end{pmatrix} \right) \quad (11.161)$$

The Selection Bias Problem

If we estimate the equation of interest by OLS using only the selected observations ($z = 1$), the conditional expectation of the error in the selected subsample is not zero:

$$\mathbb{E}[y | x, z = 1] = x'\beta + \mathbb{E}[e | x, u > -w'\gamma] \quad (11.162)$$

By the orthogonal linear projection, $e = \rho\sigma_e u^* + \eta$, where $u^* = \frac{u}{\sigma_u} \sim \mathcal{N}(0, 1)$. Applying the conditional expectation in the selected sample:

$$\mathbb{E} \left[e \mid u^* > -\frac{w'\gamma}{\sigma_u} \right] = \rho\sigma_e \lambda \left(\frac{w'\gamma}{\sigma_u} \right) \quad (11.163)$$

The corrected regression equation becomes:

$$y \mid z^* > 0 = x'\beta + \rho\sigma_e \lambda \left(\frac{w'\gamma}{\sigma_u} \right) + v \quad (11.164)$$

The omission of the Inverse Mills Ratio (λ) in the regression generates a classic omitted variable bias, where OLS confounds the direct causal effect of the regressors x with changes in the selection probability.

Inherent Heteroskedasticity

The conditional variance of the error in the selected sample is:

$$\text{Var}[y \mid x, w, z^* > 0] = \sigma_e^2 [1 - \rho^2 \delta(c)], \quad \text{where } c = \frac{w'\gamma}{\sigma_u}, \quad \delta(c) = \lambda(c)[\lambda(c) + c] \quad (11.165)$$

This proves the structural heteroskedasticity of the model, requiring correction of the covariance matrix for valid inference.

Partial Effects: Decomposition into Two Components

If x_k also appears in the selection equation, the total partial effect is:

$$\frac{\partial \mathbb{E}[y \mid z^* > 0, x, w]}{\partial x_k} = \beta_k - \left(\frac{\rho \sigma_e \gamma_k}{\sigma_u} \right) \delta \left(\frac{w' \gamma}{\sigma_u} \right) \quad (11.166)$$

The total effect decomposes into the **Direct Causal Effect** (β_k) and the **Indirect Selection Effect**, which alters the composition of unobserved characteristics of the sample.

The Critical Role of the Exclusion Restriction

The formal identification of the model requires that the vector w contains at least one variable that affects the selection probability but has no causal effect on y (being excluded from x). Without this restriction, identification occurs only through the non-linearity of $\lambda(\cdot)$, which generates extreme multicollinearity, inflating standard errors and making estimates unstable.

11.3.4 Estimation Methods

The Heckman Two-Stage Procedure (Heckit)

The Heckman (1979) two-stage estimator was developed as a computationally simple and consistent alternative to FIML.

Step 1: Estimation of the Selection Equation The observability process is modeled using the complete sample:

$$z_i^* = w_i' G + u_i, \quad u_i \sim \mathcal{N}(0, 1) \quad (11.167)$$

G is estimated by Probit (Maximum Likelihood). From the estimates $\hat{G}$, we calculate for each selected observation ($z_i = 1$):

- **Estimated Inverse Mills Ratio:** $\hat{\lambda}_i = \frac{\phi(w_i' \hat{G})}{\Phi(w_i' \hat{G})}$
- **Variance Decay Factor:** $\hat{\delta}_i = \hat{\lambda}_i(\hat{\lambda}_i + w_i' \hat{G})$

Step 2: Corrected OLS Regression The equation of interest is estimated by OLS only on the selected sample, including $\hat{\lambda}_i$ as an additional regressor:

$$y_i = x_i' B + \beta_\lambda \hat{\lambda}_i + \eta_i \quad (11.168)$$

Where $\beta_\lambda = \rho \sigma_e$ and η_i satisfies $\mathbb{E}[\eta_i \mid x_i, w_i, z_i = 1] = 0$.

Recovery of the Structural Parameters The residual variance of the second stage ($\hat{\sigma}_\eta^2$) is not consistent for σ_e^2 . From Theorem 19.5:

$$\text{plim } \hat{\sigma}_\eta^2 = \sigma_e^2(1 - \rho^2 \bar{\delta}) \quad (11.169)$$

Given that $\text{plim } \hat{\beta}_\lambda^2 = \rho^2 \sigma_e^2$, we construct the consistent estimator:

$$\hat{\sigma}_e^2 = \frac{e'e}{n_1} + \bar{\delta}_{n_1} \hat{\beta}_\lambda^2 \quad (11.170)$$

The correlation coefficient is recovered by:

$$\hat{\rho} = \frac{\hat{\beta}_\lambda}{\hat{\sigma}_e} \quad (11.171)$$

Note: Since the Heckit estimator recovers ρ indirectly and non-linearly, $\hat{\rho}$ is not mathematically restricted to $[-1, 1]$. Values $|\hat{\rho}| > 1$ indicate model misspecification.

Full Information Maximum Likelihood (FIML) Estimation

The FIML estimator simultaneously estimates all structural parameters (B, G, σ_e, ρ) by maximizing the joint log-likelihood function.

Derivation of the Joint Log-Likelihood The sample is divided into two mutually exclusive observable regimes:

Regime 1: Non-Selected Observations ($z_i = 0$)

$$P(z_i = 0 \mid w_i) = \Phi(-w_i'G) \quad (11.172)$$

Regime 2: Selected Observations ($z_i = 1$) The joint density is $f(y_i, z_i = 1 \mid x_i, w_i) = f(y_i \mid x_i) \cdot P(z_i = 1 \mid y_i, x_i, w_i)$, where:

- $f(y_i \mid x_i) = \frac{1}{\sigma_e} \phi\left(\frac{y_i - x_i' B}{\sigma_e}\right)$
- By the bivariate normal, $u_i \mid e_i \sim \mathcal{N}\left(\frac{\rho}{\sigma_e} e_i, 1 - \rho^2\right)$, therefore:

$$P(z_i = 1 \mid y_i, x_i, w_i) = \Phi\left(\frac{w_i'G + \frac{\rho}{\sigma_e}(y_i - x_i' B)}{\sqrt{1 - \rho^2}}\right) \quad (11.173)$$

The Complete Log-Likelihood Function

$$\ln L(B, G, \sigma_e, \rho) = \sum_{i \in \{z_i=0\}} \ln \Phi(-w_i'G) + \sum_{i \in \{z_i=1\}} \left[\ln \Phi\left(\frac{w_i'G + \frac{\rho}{\sigma_e}(y_i - x_i' B)}{\sqrt{1 - \rho^2}}\right) - \ln \sigma_e + \ln \phi\left(\frac{y_i - x_i' B}{\sigma_e}\right) \right] \quad (11.174)$$

Numerical Implementation and Reparametrization To ensure numerical stability, two reparametrizations are applied:

1. **Correlation Restriction** ($\rho \in (-1, 1)$): Fisher transformation $\tau = \text{atanh}(\rho) = \frac{1}{2} \ln\left(\frac{1+\rho}{1-\rho}\right)$, optimizing $\tau \in (-\infty, \infty)$ and recovering $\rho = \tanh(\tau)$.
2. **Variance Positivity**: Reparametrize $\omega = \ln(\sigma_e)$, ensuring $\sigma_e = e^\omega > 0$.

Consolidated Comparison: Heckit vs. FIML

Asymptotic Consistency: Heckit is consistent under bivariate normality and linear projection; FIML is consistent under strict joint bivariate normality.

Asymptotic Efficiency: Heckit is inefficient (uses only 1st-order moments); FIML is fully efficient (attains Cramér-Rao).

Robustness to Specification: Heckit is highly robust (marginal normality of u and linear projection suffice); FIML is very low (sensitive to any deviation).

Estimation of ρ : Heckit is indirect, not restricted to $[-1, 1]$; FIML is direct, restricted to $(-1, 1)$.

Numerical Stability: Heckit is excellent (concave Probit + linear OLS); FIML is problematic (non-concave likelihood).

Standard Errors: Heckit requires complex corrections (Greene or bootstrap); FIML is direct (inversion of the Hessian).

Reparametrizations: Heckit requires none; FIML requires mandatory reparametrizations ($\tau = \text{atanh}(\rho)$, $\omega = \ln(\sigma_e)$).

Exclusion Restriction: Heckit requires it vitally for precision; FIML requires it critically for convergence.

Outlier Treatment: Heckit is relatively stable; FIML is highly sensitive.

Practical Use: Heckit is the dominant standard; FIML is reserved for large samples.

11.3.5 Inference and Variance Correction

Corrected Asymptotic Covariance Matrix (Greene, 1981)

Valid inference in the second stage of Heckit requires dealing with two simultaneous problems that invalidate conventional OLS standard errors:

1. **Inherent Heteroskedasticity:** The conditional error has non-constant variance, depending on the index $w_i'G$ through the factor δ_i .
2. **Error Propagation from Generated Regressors:** Since $\hat{\lambda}_i$ is estimated in the first stage, it introduces additional sampling variability ignored by conventional OLS.

Matrix Formulation Greene (1981) derived the correct asymptotic covariance matrix:

$$\widehat{\text{Var}}_{asy}[\hat{B}, \hat{\beta}_\lambda] = \hat{\sigma}_e^2 (X^{*'} X^*)^{-1} [X^{*'} (I - \hat{\rho}^2 \Delta) X^* + Q] (X^{*'} X^*)^{-1} \quad (11.175)$$

Where:

- $X^* = [X, \hat{\lambda}]$ is the augmented second-stage regressor matrix ($n_1 \times (K + 1)$).
- $\Delta = \text{diag}(\hat{\delta}_1, \hat{\delta}_2, \dots, \hat{\delta}_{n_1})$ contains the variance decay factors.
- Q is the generated regressor correction matrix:

$$Q = \hat{\rho}^2 (X^{*'} \Delta W) \widehat{\text{Var}}[\hat{G}] (W' \Delta X^*) \quad (11.176)$$

Where W is the selection equation regressor matrix and $\widehat{\text{Var}}[\hat{G}]$ is the covariance matrix of the Probit coefficients.

Theoretical Connection: The Murphy-Topel (1985) Theorem

The Heckit estimator is a classic application of the two-step procedures formalized by Murphy and Topel (1985). The corrected asymptotic variance of $\hat{\theta}_2$ is:

$$V_2^* = V_2 + V_2 [CV_1C' - RV_1C' - CV_1R'] V_2 \quad (11.177)$$

Where V_1 and V_2 are the covariance matrices of stages 1 and 2, C is the cross-sensitivity matrix, and R is the contemporaneous covariance between the score functions.

Proof of Score Orthogonality ($R = 0$): In the selection model, the second-stage score is $g_{2i} = x_i^* \eta_i$ and the first-stage score is $g_{1i} = w_i \lambda_i^{gen}$. By conditional orthogonality $\mathbb{E}[\eta_i \mid x_i, w_i, z_i = 1] = 0$, applying the Law of Iterated Expectations:

$$R = \mathbb{E}[g_{2i} \cdot g_{1i}'] = \mathbb{E}\{x_i^* w_i' \lambda_i^{gen} \cdot \mathbb{E}[\eta_i \mid x_i, w_i, z_i = 1]\} = 0 \quad (11.178)$$

The formula collapses to $V_2^* = V_2 + V_2 CV_1 C' V_2$, which results exactly in the Greene (1981) formulation.

The Exception under the Null Hypothesis

Under $H_0 : \beta_\lambda = 0 \iff \rho = 0$, the heteroskedasticity and generated regressor terms vanish ($\rho^2 \Delta = 0$ and $Q = 0$). Consequently, the Greene matrix collapses to the classic OLS formula:

$$\text{Var}[\hat{B}, \hat{\beta}_\lambda] = \sigma_e^2 (X^{*'} X^*)^{-1} \quad (11.179)$$

Practical Implication: The conventional t -test on $\hat{\beta}_\lambda$ is asymptotically valid under the null to test for the presence of selection bias. If H_0 is rejected, the Greene correction or bootstrap are mandatory.

Contemporary Alternatives

- **Control Function Estimator:** Heckit is classified as a control function estimator, where λ_i acts as a control variable to capture endogeneity (*residual inclusion* method).
- **Bootstrap Approach:** Must be executed completely in two stages. For each replication, re-estimate the Probit, recalculate $\hat{\lambda}_i$, and re-estimate the regression. Bootstrapping only the second stage is a serious methodological error.

11.3.6 Parametric Limitations and Alternatives

Limitations of Bivariate Normality

The assumption of joint bivariate normality is the pillar of the classic estimators, but also their main vulnerability:

- **Inconsistency under Deviations from Normality:** Unlike the classic linear model, in models with incidental truncation, the breakdown of normality destroys the consistency of both Heckit and FIML.

- **Extreme Sensitivity to Heteroskedasticity:** The derivation of the IMR assumes $\sigma_u^2 = 1$. Heteroskedasticity invalidates the linearized correction.
- **Dependence on the Correlation Parametrization:** The model is extremely fragile with respect to the estimate of ρ , sensitive to *outliers*.

Semiparametric and Non-Parametric Estimators

Heckman and Singer (1984) / Gallant and Nychka (1987) Sieve Estimator Relaxes the normality assumption by approximating the unknown joint density with a flexible polynomial expansion:

$$h_n(u) = \left[\sum_{j=0}^{J_n} \gamma_j u^j \right]^2 \phi(u) \quad (11.180)$$

The estimator is consistent and asymptotically normal for a wide range of smooth distributions.

Ahn and Powell (1993) Estimator Dispenses with parametric assumptions by replacing the Probit with local non-parametric methods:

- *First Stage:* Non-parametrically estimate the *propensity score* $p(Z_i) = P(z_i = 1 \mid Z_i)$ via Kernel.
- *Second Stage:* Estimate B by weighted least squares applied to paired differences of individuals with identical propensities ($p(Z_i) \approx p(Z_j)$), eliminating the selection term.

Copula Function Approach (Smith, 2003; Trivedi and Zimmer, 2007)

Sklar's (1973) Theorem Any joint distribution $F_{12}(y_1, y_2)$ can be expressed by a copula C evaluated at the marginals:

$$F_{12}(y_1, y_2) = C(F_1(y_1), F_2(y_2); \theta) \quad (11.181)$$

The joint density is $f_{12}(y_1, y_2) = c(F_1(y_1), F_2(y_2); \theta) \cdot f_1(y_1) \cdot f_2(y_2)$, where c is the copula density.

Application to Selection Models It allows specifying arbitrary marginal distributions (e.g., Student-t, logistic) and choosing a copula to capture dependence:

$$\ln L = \sum_{z_i=0} \ln[1 - F_u(-w_i'G)] + \sum_{z_i=1} \left[\ln f_e(e_i) + \ln \left(\frac{\partial C(u_1, u_2; \theta)}{\partial u_1} \bigg|_{u_1=F_e(e_i), u_2=F_u(w_i'G)} \right) \right] \quad (11.182)$$

Common Copula Families

- **Gaussian Copula:** Returns the classic bivariate normal structure.
- **Clayton Copula:** $C(u_1, u_2; \theta) = [u_1^{-\theta} + u_2^{-\theta} - 1]^{-1/\theta}$ for $\theta > 0$. Models asymmetric dependence in the left tail.

- **Frank Copula:** Symmetric dependence, suitable when association occurs equally in both tails.
- **Plackett Copula:** Based on the odds ratio of dependence, allowing robust linear correlations.

11.3.7 Classic Empirical Applications

Example 1: Female Labor Supply (Mroz, 1987)

Mroz's (1987) work is a milestone in empirical econometrics, investigating the sensitivity of the labor supply model of married women.

Dataset Structure

- Sample: 753 white married women (30–60 years) from the 1975 PSID.
- 428 with active participation ($z = 1$) and 325 with zero hours ($z = 0$).

Equation of Interest (Wage Rate)

$$\ln(\text{Wage}_i) = \beta_1 + \beta_2 \text{Exper}_i + \beta_3 \text{Exper}_i^2 + \beta_4 \text{Educ}_i + e_i \quad (11.183)$$

Selection Equation The vector w includes the variables of interest plus the exclusion restrictions: husband's income ($nwifeinc$), age (age), number of children under 6 years ($kidslt6$) and between 6 and 18 years ($kidsge6$).

Numerical Results

- *Estimate of ρ :* Heckit: $\hat{\rho} = 0.049$ (s.e. = 0.182); FIML: $\hat{\rho} = 0.027$ (s.e. = 0.134). Both fail to reject $H_0 : \rho = 0$.
- *Education Coefficients (β_4):* OLS: 0.1075; Heckit: 0.1067; FIML: 0.1082.

Conclusion: The estimates are practically identical. When the selection bias is zero ($\rho \approx 0$), conventional OLS is consistent and provides a robust approximation.

Example 2: Mover-Stayer Migration Model (Nakosteen and Zimmer, 1980)

Extends the Heckman model to choices with distinct outcomes in each decision branch (Roy Model).

Theoretical Structure

- *Selection Equation:* $M_i^* = w_i' \gamma + u_i$, with $z_i = \mathbf{1}[M_i^* > 0]$.
- *Equation of Interest 1 (Migrant):* $I_{1,i} = x_{1,i}' \beta_1 + e_{1,i}$, observed if $z_i = 1$.
- *Equation of Interest 2 (Non-Migrant):* $I_{0,i} = x_{0,i}' \beta_0 + e_{0,i}$, observed if $z_i = 0$.

Double Truncation Problem The income of migrants is truncated from the left ($u_i > -w_i'\gamma$) and that of non-migrants from the right ($u_i \leq -w_i'\gamma$). The corrections are:

$$\mathbb{E}[I_{1,i} | z_i = 1] = x'_{1,i}\beta_1 + \rho_{1u}\sigma_1\lambda_1(w_i'\gamma), \quad \lambda_1(c) = \frac{\phi(c)}{\Phi(c)} \quad (11.184)$$

$$\mathbb{E}[I_{0,i} | z_i = 0] = x'_{0,i}\beta_0 + \rho_{0u}\sigma_0\lambda_0(w_i'\gamma), \quad \lambda_0(c) = \frac{-\phi(c)}{1 - \Phi(c)} \quad (11.185)$$

Note that the IMR assumes a negative sign for the upper-truncated tail.

Selection Variables Self-employment, state employment growth (ΔEMP), per capita income growth (ΔPCI), industry change (ΔSIC), and demographic characteristics.

11.3.8 Applications in Panel Data and the Attrition Problem

This section presents a deep theoretical, conceptually expanded, and mathematically rigorous review of the extension of sample selection models to the panel data context and the treatment of the attrition problem. The content integrates the classic developments of William Greene, Jeffrey Wooldridge, Manuel Arellano and Bo Honoré, Badi Baltagi, and the seminal contributions of Kyriazidou (1997), Zabel (1992), and Hausman and Wise (1979).

Methodological Challenges in the Extension to Panel Data

The transition from cross-sectional sample selectivity models to panel data enriches the identification possibilities, allowing control for the presence of unobserved individual heterogeneity (fixed/random effects) that interacts simultaneously with the selection mechanism and the equation of interest. However, the temporal structure imposes the need to carefully distinguish the nature of the selection process:

- **Initial Sample Selection:** The selection mechanism operates once, typically in the baseline period ($t = 1$) or immediately before the start of follow-up. Once selected to participate, the individual is followed over all subsequent periods (panel balanced conditional on entry).
- **Sample Attrition:** The selection process occurs continuously and dynamically over time. Attrition is characterized as an **absorbing state**: if an individual decides to leave the survey, moves out of the geographic coverage area, or dies in period t , they are permanently lost for subsequent periods ($s_{it} = 0 \implies s_{i,t+k} = 0$ for all $k \geq 0$). When attrition is correlated with the contemporaneous idiosyncratic error terms of the equation of interest, the exclusion of individuals who leave the panel induces a non-random attrition bias that destroys the consistency of conventional estimators.

Parametric Models with Common Effects: Zabel (1992)

The first attempts to model selectivity in panel data faced serious specification asymmetry problems. Verbeek (1990) formulated a hybrid specification that included random effects in the selection equation and fixed effects in the outcome equation. Zabel (1992) pointed out that this asymmetry was conceptually inconsistent and proposed a symmetric model with individual effects in both equations, resolving the correlation between unobserved effects and regressors through the Mundlak (1978) linear projection.

Mathematical Structure of the Zabel Model The system of simultaneous equations in levels is specified for $i = 1, \dots, N$ and $t = 1, \dots, T$:

- **Outcome Equation:** $y_{it}^* = \eta_i + x'_{it}\beta + e_{it}, \quad y_{it} = y_{it}^* \cdot d_{it}$
- **Selection Equation:** $d_{it}^* = u_i + z'_{it}\alpha + v_{it}, \quad d_{it} = \mathbf{1}[d_{it}^* > 0]$

The idiosyncratic error terms follow a bivariate normal distribution:

$$\begin{pmatrix} e_{it} \\ v_{it} \end{pmatrix} \sim \mathcal{N}_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_e^2 & \rho\sigma_e \\ \rho\sigma_e & 1 \end{pmatrix} \right) \quad (11.186)$$

Mundlak Projection for Fixed Effects To circumvent the inconsistency problem arising from treating (η_i, u_i) as unstructured fixed effects (incidental parameters problem with short T), Zabel projects the individual effects onto the time means of all regressors:

$$\eta_i = \bar{x}'_i \pi + \tau w_i, \quad w_i \sim \mathcal{N}(0, \sigma_w^2) \quad (11.187)$$

$$u_i = \bar{z}'_i \delta + \nu v_i, \quad v_i \sim \mathcal{N}(0, \sigma_v^2) \quad (11.188)$$

Computational Challenge Maximum Likelihood estimation requires the evaluation of the joint density conditioned on selectivity. The log-likelihood function requires solving a bivariate integral for each individual, corresponding to the projected random effects (w_i, v_i) . Assuming independence between w_i and v_i , the double integral can be factored into the product of two univariate integrals, calculated efficiently by Simulated Maximum Likelihood (SML) with the GHK simulator or by Gauss-Hermite quadrature.

The Wooldridge (1995a) Selection Test and Correction

Wooldridge (1995a) developed an extremely influential parametric procedure to test and correct for panel selection bias without requiring the complex bivariate integration of the Zabel model. The strategy extends the Chamberlain (1980, 1984) projection to the Probit selection equation.

Chamberlain Projection in the Selection Equation Let the individual effect u_i of the selection equation be projected unrestrictedly onto the complete history of exogenous regressors $z_i \equiv (z'_{i1}, z'_{i2}, \dots, z'_{iT})'$:

$$u_i = \phi_0 + z'_{i1}\phi_1 + z'_{i2}\phi_2 + \dots + z'_{iT}\phi_T + a_i = \phi_0 + \bar{z}'_i\phi + a_i \quad (11.189)$$

Where $a_i \sim \mathcal{N}(0, \sigma_a^2)$ is independent of z_i and v_{it} . Dividing the selection equation by $\sigma_{wt} = \sqrt{\sigma_a^2 + 1}$, we obtain the identified Probit formulation:

$$P(d_{it} = 1 \mid z_i) = \Phi(z'_{it}\alpha_t^s + \bar{z}'_i\phi_t^s) \quad (11.190)$$

The Selectivity Test Under $H_0 : \rho = 0$, the idiosyncratic errors are independent with respect to the selection rule, and the conventional Fixed Effects (Within) estimator is consistent.

Test Procedure:

1. **First Step:** For each period $t = 1, \dots, T$, estimate a cross-sectional Probit of d_{it} against z_{it} and $\bar{z}_i$, obtaining the estimated Inverse Mills Ratios:

$$\hat{\lambda}_{it} = \frac{\phi(z'_{it}\hat{\alpha}_t^s + \bar{z}'_i\hat{\phi}_t^s)}{\Phi(z'_{it}\hat{\alpha}_t^s + \bar{z}'_i\hat{\phi}_t^s)} \quad (11.191)$$

2. **Second Step:** Estimate the equation of interest using the Within estimator on the selected sample ($d_{it} = 1$), including $\hat{\lambda}_{it}$ as an additional regressor:

$$y_{it} = \alpha_i + x'_{it}\beta + \gamma\hat{\lambda}_{it} + e_{it} \quad (11.192)$$

3. **Decision Criterion:** Perform a t -test (with cluster-robust standard errors) on $\hat{\gamma}$. Under $H_0 : \gamma = 0$, the test is asymptotically valid. If we reject H_0 , there is active selectivity bias.

Why the Within Estimator FAILS in Correction (when $\gamma \neq 0$) A chronic methodological error is to use the test equation to correct the bias and report the coefficients β . If H_0 is rejected ($\gamma \neq 0$), the Within estimator with IMR becomes **INCONSISTENT**.

Analytical Justification: The consistency of Within requires strict exogeneity conditioned on selection: $\mathbb{E}[e_{it} \mid x_i, c_i, d_{i1}, \dots, d_{iT}] = 0$. However, if there is contemporaneous selectivity ($\rho \neq 0$), $\mathbb{E}[e_{it} \mid x_i, c_i, d_{it} = 1] = \rho\sigma_e\lambda_{it}$.

When we apply the Within transformation (temporal demeaning):

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)'\beta + \gamma(\hat{\lambda}_{it} - \bar{\lambda}_i) + (e_{it} - \bar{e}_i) \quad (11.193)$$

The transformed error term contains $\bar{e}_i = \frac{1}{T_i} \sum_{s \in S_i} e_{is}$. Since e_{it} is contemporaneously correlated with d_{it} , the error of one period s will be correlated with the selection of other periods. Thus:

$$\mathbb{E}[e_{it} - \bar{e}_i \mid x_i, c_i, s_{i1}, \dots, s_{iT}] \neq 0 \quad (11.194)$$

The Within transformation contaminates the error term of all observed periods with the average of the errors that determined the selection, violating strict exogeneity.

The Consistent Correction Procedure To correct consistently without incurring the contamination bias of the Within transformation, Wooldridge proposes explicitly modeling the fixed effect α_i in levels through the Chamberlain projection:

$$\alpha_i = \phi_0 + \bar{x}'_i\pi + \xi_i \quad (11.195)$$

The specification in levels for the observed sample becomes:

$$\mathbb{E}[y_{it} \mid x_i, d_{it} = 1] = x'_{it}\beta + \bar{x}'_i\pi + \gamma_t\lambda(z'_{it}\alpha_t^s + \bar{z}'_i\phi_t^s) \quad (11.196)$$

Steps of Consistent Estimation:

1. **1st Step (Probit):** Estimate the selection Probit for each period t and calculate $\hat{\lambda}_{it}$.

2. **2nd Step (Pooled OLS in Levels):** Estimate the equation in levels by Pooled OLS including x_{it} , $\bar{x}_i$, and $\hat{\lambda}_{it}$ interacted with time dummies:

$$y_{it} = \theta_t + x'_{it}\beta + \bar{x}'_i\pi + \gamma_t\hat{\lambda}_{it} + v_{it} \quad (11.197)$$

This estimator is consistent, since the fixed effect was controlled via $\bar{x}_i$ and selectivity was purged by $\hat{\lambda}_{it}$.

3. **Inference:** It is mandatory to apply a cluster-robust covariance matrix (at the individual level i).

The Kyriazidou (1997) Semiparametric Estimator

Kyriazidou (1997) proposed a pioneering semiparametric estimator that corrects for selection bias and attrition in panel data without imposing a specific functional form for the distribution of idiosyncratic errors or for the individual effects.

Causal Intuition and Elimination of the Fixed Effect Consider the model with two periods ($T = 2$), where y is observed only if $d_{i1} = d_{i2} = 1$. Applying First Differences:

$$\Delta y_i = \Delta x'_i\beta + \Delta e_i \quad (11.198)$$

Taking the conditional expectation in the doubly selected sample:

$$\mathbb{E}[\Delta y_i \mid x_i, z_i, d_{i1} = d_{i2} = 1] = \Delta x'_i\beta + \mathbb{E}[e_{i2} - e_{i1} \mid x_i, z_i, d_{i1} = d_{i2} = 1] \quad (11.199)$$

The selection bias is $\lambda_{i2} - \lambda_{i1}$.

The Exchangeability Hypothesis Kyriazidou introduces the conditional temporal exchangeability hypothesis: conditioned on the regressors and the fixed effect, the joint distribution of $(e_{i1}, v_{i1}, e_{i2}, v_{i2})$ is identical to that of $(e_{i2}, v_{i2}, e_{i1}, v_{i1})$.

Under this distributional symmetry, if the latent selection indices are identical ($z'_{i1}\alpha = z'_{i2}\alpha$), then $\lambda_{i2} = \lambda_{i1}$ and the bias vanishes:

$$\text{If } z'_{i1}\alpha = z'_{i2}\alpha \implies \lambda_{i2} - \lambda_{i1} = 0 \quad (11.200)$$

In this subset, the First Differences estimator is consistent without parametric modeling:

$$\mathbb{E}[\Delta y_i \mid x_i, z_i, d_{i1} = d_{i2} = 1, z'_{i1}\alpha = z'_{i2}\alpha] = \Delta x'_i\beta \quad (11.201)$$

Operationalization by Kernel-Weighted Least Squares In practice, the probability that $z'_{i1}\alpha = z'_{i2}\alpha$ is zero for continuous variables. Kyriazidou proposes estimating by WLS with Kernel-based weights:

$$\hat{\omega}_i = K \left(\frac{(z_{i2} - z_{i1})'\hat{\alpha}}{h_N} \right) \quad (11.202)$$

Where $K(\cdot)$ is a symmetric Kernel function (e.g., Gaussian) and h_N is the bandwidth that converges to zero. The estimator is:

$$\hat{\beta}_{AP} = \left(\sum_{i=1}^N d_{i1}d_{i2}\hat{\omega}_i(\Delta x_i)(\Delta x_i)' \right)^{-1} \left(\sum_{i=1}^N d_{i1}d_{i2}\hat{\omega}_i(\Delta x_i)(\Delta y_i) \right) \quad (11.203)$$

Advantage: Consistent and asymptotically normal without distributional assumptions.

Disadvantage: Slower convergence rate than $\sqrt{N}$ and requires excluding observations without selection in both periods.

The Sample Attrition Problem

Sample attrition constitutes one of the most severe challenges in longitudinal research. It occurs when individuals who participate in the initial wave systematically leave the panel in subsequent periods. When this process is correlated with unobservable factors that determine the variable of interest, estimation by conventional panel models becomes inconsistent.

Distinction of Scenarios Attrition can be classified under two paradigms:

- **Selection on Unobservables (Endogenous Attrition):** Attrition is contemporaneously correlated with the idiosyncratic errors e_{it} , even after controlling for the history of regressors and fixed effects. It requires parametric Heckman-type corrections.
- **Selection on Observables (Ignorable Attrition):** Conditioned on variables observed in the past, attrition is independent of contemporaneous errors. Ideal for Inverse Probability Weighting (IPW) methods.

The Hausman and Wise (1979) Parametric Model Hausman and Wise (1979) developed the seminal parametric modeling for attrition in two periods ($T = 2$). The decision to remain in period 2 is governed by:

$$s_{i2}^* = \delta y_{i2} + x'_{i2}\theta + w'_{i2}\gamma + v_{i2} \quad (11.204)$$

Substituting $y_{i2} = x'_{i2}\beta + e_{i2} + u_i$:

$$s_{i2}^* = x'_{i2}(\delta\beta + \theta) + w'_{i2}\gamma + \epsilon_{i2} \quad (11.205)$$

Where $\epsilon_{i2} = \delta(u_i + e_{i2}) + v_{i2}$. The covariance between the error of the equation of interest and the selection error is:

$$\text{Cov}(u_i + e_{i2}, \epsilon_{i2}) = \delta(\sigma_u^2 + \sigma_e^2) \neq 0 \quad \text{if } \delta \neq 0 \quad (11.206)$$

This proves that attrition is intrinsically non-random and induces severe selection bias.

Estimation Methods:

- **FIML:** Under joint multivariate normality, the complete log-likelihood integrates the joint density over the observability domains.
- **Two-Stage with Murphy-Topel:** Estimate the selection Probit and correct by OLS with IMR. The Murphy-Topel (1985) correction matrix or complete bootstrap is mandatory for valid inference.

Inverse Probability Weighting (IPW) When the panel has multiple waves and complex non-linear structures, parametric estimation by FIML becomes numerically infeasible. Contemporary econometrics adopts the IPW estimator.

The Ignorability of Attrition Assumption: Conditioned on covariates observed in the initial period (z_{i1}), the attrition indicator is independent of the errors:

$$P(s_{it} = 1 \mid y_{it}, x_{it}, z_{i1}) = P(s_{it} = 1 \mid z_{i1}) \equiv p_t(z_{i1}) \quad (11.207)$$

Two-Stage IPW (Wooldridge, 2002):

1. **1st Step:** For each period t , estimate a Probit for s_{it} against z_{i1} , obtaining $\hat{p}_{it} = \Phi(z'_{i1}\hat{\gamma}_t)$.
2. **2nd Step:** Estimate the equation of interest by weighted M-estimation:

$$\min_{\beta} \sum_{i=1}^N \sum_{t=1}^T \left(\frac{s_{it}}{\hat{p}_{it}} \right) q_t(w_{it}; \beta) \quad (11.208)$$

The weight $\omega_{it} = \frac{s_{it}}{\hat{p}_{it}}$ re-balances the surviving sample. Individuals with low probability of remaining but who in fact remain receive high weights, representing themselves and those who dropped out.

Sequential IPW with Multiplicative Attrition (Robins, Rotnitzky and Zhao, 1995): To allow the attrition probability to depend on contemporary or lagged variables, RRZ (1995) model the cumulative probability as the product of conditional transition probabilities:

$$p_{it}(\theta_t) = p_{i2}(\gamma_2) \cdot p_{i3}(\gamma_3) \cdot \dots \cdot p_{it}(\gamma_t) \quad (11.209)$$

Where each component is $p_{is}(\gamma_s) = P(s_{is} = 1 \mid s_{i,s-1} = 1, z_{i,s-1})$.

Inconsistency Warning: Moffitt, Fitzgerald and Gottschalk (1999) proposed using only the inverse of the contemporary probability ($1/\hat{p}_{it}^*$). Wooldridge (2010) demonstrates that this procedure is inconsistent. Only weighting by the multiplicative cumulative probability guarantees that the expected value of the weighted criterion converges to the unrestricted population expectation.

Valid Inference under IPW Hypothesis testing requires two corrections:

- **Heteroskedasticity and Serial Correlation:** Weighting introduces structural heteroskedasticity and there is serial correlation in the errors. It is mandatory to use a cluster-robust covariance matrix (at the individual level).
- **Generated Regressors:** The weights $\hat{p}_{it}$ are estimated, introducing additional sampling uncertainty. One must use the Wooldridge (2007) correction or complete panel bootstrap (resampling individuals, re-estimating Probits, and recalculating weights in each replication).

Comparative Synthesis: Attrition Correction in Panel Data

Dimension 1: Selection Assumption. Mills Correction (Heckman in FD): Selection on Unobservables (errors e_{it} and v_{it} correlated). IPW: Selection on Observables / Ignorability (attrition determined by z_{it}).

Dimension 2: Distributional Assumption. Mills Correction: Strictly Parametric (joint bivariate normality). IPW: Semiparametric (requires only correct specification of the attrition probability).

Dimension 3: Treatment of Fixed Effect. Mills Correction: Removed via First Differences (FD). IPW: Controlled via Mundlak means or Chamberlain projection in levels.

Dimension 4: Sample Composition. Mills Correction: Restricted to individuals present in consecutive periods. IPW: Uses the entire observed sample via individual re-weighting.

Dimension 5: Use of Exclusion Variable. Mills Correction: Strongly recommended (requires an instrument that affects attrition but not the outcome). IPW: Not necessary (relies on the predictive capacity of $z_{i,t-1}$ on attrition).

Chapter 12

Introduction to Quasiexperimental Methods

12.1 The Potential Outcome Model

The foundation of modern causal inference rests on the Potential Outcomes Framework, formalized by Donald Rubin. Unlike structural approaches that model the data generating process directly, the RCM defines causality through the lens of hypothetical interventions.

12.1.1 Notation and the Switching Equation

Definition 12.1.1 (Potential Outcomes). Let $i \in \{1, \dots, N\}$ denote a unit drawn from a population and let $D_i \in \{0, 1\}$ be the binary treatment indicator, where $D_i = 1$ denotes exposure to treatment and $D_i = 0$ denotes control. For each unit, there exists a pair of *potential outcomes*:

- $Y_i(1) \in \mathbb{R}$: the outcome that would be observed if unit i receives the treatment.
- $Y_i(0) \in \mathbb{R}$: the outcome that would be observed if unit i does not receive the treatment.

The individual causal effect is defined as:

$$\beta_i = Y_i(1) - Y_i(0). \quad (12.1)$$

The relationship between potential outcomes and the realized (observed) outcome is governed by the **Rubin switching equation**:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0). \quad (12.2)$$

Let $P \equiv \Pr(D_i = 1) = \mathbb{E}[D_i]$ denote the unconditional probability of treatment assignment in the population.

Remark 12.1.1 (The Fundamental Problem of Causal Inference). For any given unit i , we observe only one realized outcome:

$$Y_i^{\text{obs}} = \begin{cases} Y_i(1) & \text{if } D_i = 1, \\ Y_i(0) & \text{if } D_i = 0. \end{cases} \quad (12.3)$$

Causal inference is therefore fundamentally a missing data problem: for any treated unit ($D_i = 1$), the counterfactual $Y_i(0)$ remains strictly latent, and vice versa.

Definition 12.1.2 (Structural Conditional Means). To enable rigorous algebraic decomposition of estimators, we define the four population conditional means of the potential outcomes:

$$\begin{aligned} E_{11} &\equiv \mathbb{E}[Y_i(1) \mid D_i = 1] \quad (\text{observed mean of the treated}), \\ E_{00} &\equiv \mathbb{E}[Y_i(0) \mid D_i = 0] \quad (\text{observed mean of the controls}), \\ E_{10} &\equiv \mathbb{E}[Y_i(0) \mid D_i = 1] \quad (\text{latent counterfactual of the treated}), \\ E_{01} &\equiv \mathbb{E}[Y_i(1) \mid D_i = 0] \quad (\text{latent counterfactual of the controls}). \end{aligned} \tag{12.4}$$

12.1.2 Aggregate Causal Parameters

Since individual effects β_i are not identifiable, we focus on population-level causal parameters.

Definition 12.1.3 (ATE, ATT, and ATU).

$$\begin{aligned} \text{ATE} &\equiv \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[\beta_i], \\ \text{ATT} &\equiv \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1] = E_{11} - E_{10}, \\ \text{ATU} &\equiv \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 0] = E_{01} - E_{00}. \end{aligned} \tag{12.5}$$

Remark 12.1.2 (LIE Decomposition of the ATE). Applying the Law of Iterated Expectations (LIE) over the partition generated by $D_i \in \{0, 1\}$:

$$\text{ATE} = \mathbb{E}[\mathbb{E}[Y_i(1) - Y_i(0) \mid D_i]] \quad (\text{by LIE}) \tag{12.6}$$

$$= \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1] \cdot P + \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 0] \cdot (1 - P) \quad (\text{discrete expectation}) \tag{12.7}$$

$$= \text{ATT} \cdot P + \text{ATU} \cdot (1 - P). \quad (\text{by linearity}) \tag{12.8}$$

The ATE is thus a population-weighted average of the ATT and the ATU.

12.1.3 Decomposition of the Naive Estimator

Definition 12.1.4 (Naive Estimator). The naive (simple difference-in-means) estimator compares the observed outcomes of the treated and control groups:

$$R \equiv \mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0]. \tag{12.9}$$

Substituting the switching equation (12.2) into each conditional expectation:

$$\begin{aligned} R &= \mathbb{E}[D_i Y_i(1) + (1 - D_i) Y_i(0) \mid D_i = 1] - \mathbb{E}[D_i Y_i(1) + (1 - D_i) Y_i(0) \mid D_i = 0] \quad (\text{substitution}) \\ &= \mathbb{E}[Y_i(1) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0] \quad (\text{evaluating } D_i \text{ at its conditioned value}) \\ &= E_{11} - E_{00}. \quad (\text{by definition}) \end{aligned} \tag{12.10}$$

Remark 12.1.3 (Decomposition I: ATT and Selection Bias). Adding and subtracting the latent counterfactual $E_{10} = \mathbb{E}[Y_i(0) \mid D_i = 1]$ (neutral additive zero):

$$R = \underbrace{(E_{11} - E_{10})}_{\text{ATT}} + \underbrace{\{E_{10} - E_{00}\}}_{\text{Selection Bias } V}. \quad (12.11)$$

The selection bias $V = \mathbb{E}[Y_i(0) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0]$ captures systematic baseline differences between those who self-select into treatment and those who do not. If more motivated or productive individuals select into treatment, then $V > 0$ and R overstates the ATT.

Remark 12.1.4 (Decomposition II: Heckman's Sorting Gain). Following Heckman (2007), we isolate treatment effect heterogeneity by adding and subtracting the unconditional ATE within Decomposition 12.1.3:

$$R = \underbrace{\text{ATE}}_{\text{Average Return}} + \underbrace{(\text{ATT} - \text{ATE})}_{\text{Sorting Gain}} + \underbrace{(E_{10} - E_{00})}_{\text{Baseline Selection Bias}}. \quad (12.12)$$

Expanding the sorting gain in full:

$$\text{Sorting Gain} = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1] - \mathbb{E}[Y_i(1) - Y_i(0)]. \quad (12.13)$$

This term is nonzero when individuals who self-select into treatment possess systematically higher (or lower) marginal returns than the population average—a phenomenon known as *essential heterogeneity* or *correlated random coefficients*.

12.1.4 The SUTVA Hypothesis

The **Stable Unit Treatment Value Assumption** (SUTVA), formalized by Rubin (1980), constitutes the axiomatic barrier separating classical descriptive statistics from rigorous causal inference. Without SUTVA, the search for counterfactuals becomes mathematically intractable due to the exponential proliferation of latent potential outcomes.

Definition 12.1.5 (SUTVA: Formal Statement). Let $\mathbf{D} = (D_1, \dots, D_N)' \in \{0, 1\}^N$ denote the population treatment assignment vector, describing simultaneously the treatment status of all N units in the sample. The potential outcome of any unit i is, in its most general form, a function of the complete assignment vector: $Y_i(\mathbf{D})$. SUTVA requires that for any two assignment vectors $\mathbf{D}, \mathbf{D}'$:

$$Y_i(\mathbf{D}) = Y_i(\mathbf{D}') \quad \text{if} \quad D_i = D'_i. \quad (12.14)$$

This axiom decomposes into two indispensable sub-assumptions:

- **No Interference (No Spillovers):** Partition the assignment vector as $(d, \mathbf{D}_{-i})$, where $d \in \{0, 1\}$ is the treatment of unit i and $\mathbf{D}_{-i}$ is the $(N - 1) \times 1$ column vector containing the allocation of all other units. No interference postulates:

$$Y_i(d, \mathbf{D}_{-i}) = Y_i(d, \mathbf{D}'_{-i}) \equiv Y_i(d) \quad \forall \mathbf{D}_{-i}, \mathbf{D}'_{-i} \in \{0, 1\}^{N-1}. \quad (12.15)$$

This guarantees that the outcome of unit i is completely invariant to the treatment decisions or assignments made for neighboring or competing units. Historically, in agronomic experiments, this isolation was physically enforced through *guard rows*—unfertilized strips of land preventing nutrient runoff between test plots.

- **No Hidden Variations of Treatment:** For each nominal treatment level $d \in \{0, 1\}$, there exists only one homogeneous version of the intervention available to all units. Formally, if there are different ways to administer the treatment (represented by additional parameters t_1 and t_2), SUTVA requires:

$$Y_i(d, t_1) = Y_i(d, t_2) \equiv Y_i(d). \quad (12.16)$$

This operates as an **exclusion restriction**, preventing unobserved variations in dosage, application rigor, or administration method (e.g., whether an individual received medication impositively or voluntarily) from generating alternative potential outcomes under the same treatment label.

Remark 12.1.5 (The Algebra of Dimensional Collapse: Without SUTVA). To appreciate the mathematical necessity of SUTVA, consider the combinatorial explosion that occurs in its absence. Let the population of interest comprise N individuals, indexed by $i \in \{1, \dots, N\}$. The population treatment assignment vector is:

$$\mathbf{D} = (D_1, D_2, \dots, D_N)' \in \{0, 1\}^N. \quad (12.17)$$

Without SUTVA, the potential outcome of any individual i can be influenced by the treatment assignment of *any other unit* in the sample (presence of spillovers, externalities, or general equilibrium effects). Thus, the potential outcome of individual i must be modeled as a function of the complete vector $\mathbf{D}$:

$$Y_i(\mathbf{D}) = Y_i(D_1, D_2, \dots, D_N). \quad (12.18)$$

The domain of the potential outcome function for individual i is the discrete space $\{0, 1\}^N$, corresponding to an **N -dimensional hypercube**. Since each of the N elements of $\mathbf{D}$ is binary, there are exactly 2^N possible configurations of population treatment assignment (i.e., 2^N possible paths for society). Consequently, for each individual i , nature defines a potential outcome for each of these paths. Each individual possesses 2^N **possible potential outcomes**:

$$\{Y_i(\mathbf{d})\}_{\mathbf{d} \in \{0, 1\}^N}. \quad (12.19)$$

Since we have N individuals in the sample and each reacts potentially uniquely to each configuration of group treatment, the researcher faces a system containing an astronomical number of **latent parameters (potential worlds)**:

$$\text{Total Latent Parameters} = N \times 2^N. \quad (12.20)$$

Numerical Illustration of Mathematical Collapse:

- **Sample of $N = 10$ individuals (a tiny focus group):**

$$\text{Latent Parameters} = 10 \times 2^{10} = 10 \times 1,024 = \mathbf{10,240}. \quad (12.21)$$

For a sample of only 10 people, there are 10,240 possible potential outcomes in the population structure. However, in practice, we observe exactly **1 realized outcome** per individual in the sample. This means we would have only 10 observations to estimate a system with 10,240 latent unknowns. The problem is **strictly underidentified**.

- **Sample of $N = 100$ individuals (a small clinical trial):**

$$\text{Latent Parameters} = 100 \times 2^{100} \approx \mathbf{1.26 \times 10^{32}}. \quad (12.22)$$

The number of parameters to be estimated far exceeds the processing capacity of any machine. Any causal or counterfactual inference becomes mathematically impossible without imposing restrictions.

Remark 12.1.6 (The Algebra of Dimensional Collapse: With SUTVA). The SUTVA hypothesis acts to impose restrictions that force the collapse of this otherwise intractable exponential dimensionality. Under the two sub-assumptions above, the potential outcome function of each unit i simplifies to a **simple binary-domain mapping**:

$$Y_i(\cdot) : \{0, 1\} \rightarrow \mathbb{R}. \quad (12.23)$$

Now, instead of 2^N potential outcomes per individual, each unit possesses **only 2 latent potential outcomes**:

- $Y_i(1)$: The outcome that would be observed if individual i were treated.
- $Y_i(0)$: The outcome that would be observed if individual i were kept in control.

For the complete sample of size N , the total number of latent parameters drops drastically to:

$$\text{Total Latent Parameters} = 2 \times N = 2N. \quad (12.24)$$

Remark 12.1.7 (Direct Comparison of Structures). The dimensional reduction achieved by SUTVA can be summarized across the following metrics. Without SUTVA, the functional form of the potential outcome is $Y_i(D_1, D_2, \dots, D_N)$, with input domain $\{0, 1\}^N$ (the N -dimensional hypercube), yielding 2^N potential outcomes per individual and a total of $N \times 2^N$ parameters in a sample of size N . The observed data in the sample remains N , and the missing data problem involves $N \times (2^N - 1)$ missing data points.

With SUTVA, the functional form collapses to $Y_i(D_i)$, with input domain $\{0, 1\}$ (binary domain), yielding exactly 2 potential outcomes per individual (treated vs. control) and a total of $2N$ parameters in a sample of size N . The observed data in the sample remains N , and the missing data problem reduces to N missing data points (1 counterfactual per unit).

By the Rubin switching equation, under SUTVA, the observed outcome is given by:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0). \quad (12.25)$$

Since we observe exactly Y_i for each individual, we know exactly half of the system's parameters (we know N realized outcome values). Our only causal task becomes **predicting or imputing the N unobserved counterfactuals** (the value of $Y_i(0)$ for those treated and the value of $Y_i(1)$ for those in control). SUTVA therefore **mathematically enables causal inference** by transforming a problem with infinitely negative degrees of freedom into a perfectly tractable missing data problem of dimension N .

Remark 12.1.8 (Violations in Applied Microeconomic Research). In observational or quasi-experimental data, SUTVA is frequently violated by market forces or social interactions:

A. General Equilibrium Effects. Occurs when the scale of program implementation alters relative prices or aggregate market constraints.

Mechanism: Consider evaluating the impact of a professional training program on youth workers' productivity and wages. If the program is evaluated on a very small sample (partial equilibrium), the wage gain reflects only the individual productivity premium. However, if the policy is adopted at scale (universalization), the aggregate supply of skilled labor shifts substantially to the right. By the negatively sloped labor demand curve, equilibrium wages for all skilled workers are forced downward.

Mathematical Violation: The potential wage of individual i under treatment, $Y_i(1, \mathbf{D}_{-i})$, ceases to be invariant. If no other workers are trained ($\mathbf{D}_{-i} = \mathbf{0}$), their potential outcome is elevated. If all are trained ($\mathbf{D}_{-i} = \mathbf{1}$), their potential outcome is depressed by the market price decline, violating the no-interference premise:

$$Y_i(1, \mathbf{D}_{-i} = \mathbf{0}) \neq Y_i(1, \mathbf{D}_{-i} = \mathbf{1}). \quad (12.26)$$

B. Spillovers, Network Effects, and Contamination. Occurs when knowledge diffusion, mutual learning, information exchange, or direct physical contagion happens between treated and control units.

Mechanism: Imagine an agricultural technology adoption or efficient manufacturing incentive program implemented in a factory or community. The observed control group ($D_i = 0$) works alongside or socially interacts with treated units ($D_j = 1$). Through informal conversations or direct observation of others' behavior (knowledge diffusion), control individuals absorb and apply part of the innovative technique, experiencing an indirect productivity gain.

Attenuation Bias on ATT: Let the true average treatment effect on the treated be defined by the counterfactual of absolute isolation, $Y_i(0, \mathbf{0})$:

$$ATT = \mathbb{E}[Y_i(1, \mathbf{D}_{-i}) - Y_i(0, \mathbf{0}) \mid D_i = 1]. \quad (12.27)$$

However, due to network contamination, the untreated control group achieves a realized outcome superior to its pure counterfactual:

$$Y_i^{\text{obs}} \mid (D_i = 0) = Y_i(0, \mathbf{D}_{-i} \neq \mathbf{0}) = Y_i(0, \mathbf{0}) + \delta_i, \quad (12.28)$$

where $\delta_i > 0$ represents the indirect gain absorbed by network contagion.

When calculating the simple difference in post-matching means or regression, the researcher estimates:

$$\begin{aligned} \hat{\tau} &= \mathbb{E}[Y_i^{\text{obs}} \mid D_i = 1] - \mathbb{E}[Y_i^{\text{obs}} \mid D_i = 0] \\ &= \mathbb{E}[Y_i(1, \mathbf{D}_{-i}) \mid D_i = 1] - \mathbb{E}[Y_i(0, \mathbf{0}) + \delta_i \mid D_i = 0]. \end{aligned} \quad (12.29)$$

Under conditional ignorability (CIA), the baseline counterfactual distributions are equivalent between groups, reducing the expression to:

$$\hat{\tau} = ATT - \mathbb{E}[\delta_i \mid D_i = 0]. \quad (12.30)$$

Since $\mathbb{E}[\delta_i \mid D_i = 0] > 0$, the classical estimator suffers **attenuation bias**, systematically underestimating the true causal impact of the program and pushing the estimate toward zero.

C. Signaling (Spence). If education acts as a signal rather than human capital, the social return to educating everyone is zero despite positive private returns. The individual gain from obtaining a degree ($Y_i(1) - Y_i(0) > 0$) stems entirely from differential signaling in a partially educated population. If $\mathbf{D}_{-i} = \mathbf{1}$ (everyone is educated), the signaling value vanishes: $Y_i(1, \mathbf{1}) = Y_i(0, \mathbf{1})$.

Remark 12.1.9 (Methodological Solutions to SUTVA Violations). When interference is expected, researchers frequently redefine the unit of analysis to more aggregate levels (classrooms, schools, or entire communities) where isolation and mutual non-interference are plausible. Alternatively, cluster randomization at higher aggregation levels (villages, municipalities) creates physical or geographic separation between treatment and control clusters, isolating spillovers. For hidden treatment variations, researchers must explicitly redefine treatment categories to include all relevant versions (e.g., “no aspirin,” “aspirin-,” and “aspirin+”) or coarsen the outcome variable (e.g., simplifying health outcome to “alive” or “dead”).

12.2 Randomized Controlled Trials (RCTs)

12.2.1 Identification under Randomization

Remark 12.2.1 (The Identification Bridge). A critical conceptual distinction underpins all of causal inference:

- **Definition** (the theoretical estimand): The ATT is *defined* as $ATT = E_{11} - E_{10}$. The term E_{10} is a population parameter that exists mathematically but is *never directly computable* from observed data.
- **Identification** (the empirical strategy): We seek *statistical assumptions* under which E_{10} can be replaced by a function of *observable* quantities, thereby expressing the causal estimand entirely in terms of the data generating process we can sample from.

Every quasi-experimental method is a different “bridge” that constructs the unobservable counterfactual E_{10} from the observed control group.

Remark 12.2.2 (Identification under Randomization). Random assignment guarantees statistical independence between treatment status and potential outcomes:

$$(Y_i(1), Y_i(0)) \perp\!\!\!\perp D_i. \quad (12.31)$$

This implies:

$$\mathbb{E}[Y_i(0) \mid D_i = 1] = \mathbb{E}[Y_i(0) \mid D_i = 0] = \mathbb{E}[Y_i(0)]. \quad (12.32)$$

Substituting (12.32) into the ATT definition:

$$ATT = E_{11} - \underbrace{\mathbb{E}[Y_i(0) \mid D_i = 1]}_{= E_{00} \text{ by (12.32)}} = \mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0]. \quad (12.33)$$

Moreover, under (12.31), $ATT = ATE = ATU$, and the selection bias vanishes identically: $V = E_{10} - E_{00} = 0$.

Remark 12.2.3 (The OLS Perspective). Consider the linear projection $Y_i = \alpha + \beta D_i + \varepsilon_i$. In observational data, self-selection typically implies $\mathbb{E}[\varepsilon_i | D_i] \neq 0$, violating exogeneity. Under randomization, independence (12.31) forces $\mathbb{E}[\varepsilon_i | D_i] = 0$, so OLS recovers $\beta = \text{ATE}$ without bias.

12.2.2 Internal and External Validity

Definition 12.2.1 (Internal Validity). A study possesses internal validity when its design correctly identifies the causal effect of the treatment for the sample under analysis, free from confounding by observed or unobserved factors. In RCTs, randomization guarantees internal validity by ensuring that treatment and control groups are statistically equivalent *ex ante* in both observable and unobservable characteristics.

Definition 12.2.2 (External Validity). External validity refers to the capacity to generalize or extrapolate the estimated treatment effect beyond the study sample—to other populations, socioeconomic contexts, or time periods. It requires that the experimental sample be representative of the target population of interest.

Remark 12.2.4 (The Double Randomization Principle). The randomization that guarantees external validity does not guarantee internal validity, and vice versa. Achieving both simultaneously requires a **double randomization** procedure:

1. **Random Sampling:** Draw a random subset from the target population. This ensures *external validity* (representativeness).
2. **Random Assignment:** Within the sampled subset, randomly assign treatment and control. This ensures *internal validity* (unbiased identification).

Deviations from this ideal produce asymmetric failures:

- *Representative sample without random assignment:* Potential external validity, but internal validity is compromised by self-selection bias.
- *Random assignment without representative sample:* The Sample Average Treatment Effect (SATE) is consistently estimated (high internal validity), but generalization to the population is invalid due to sample selection bias.

Remark 12.2.5 (Threats to External Validity in RCTs). • **Applicant Bias:** In social experiments, randomization operates only over volunteers (*randomized in/out*). Volunteers differ systematically from the target population in motivation, ability, or support networks, so the SATE is not a consistent estimator of the population ATE.

- **Context Specificity:** An evaluation conducted in a single municipality reflects local socioeconomic and cultural interactions. Extrapolation to a national scale is methodologically unjustified without evidence of effect homogeneity.
- **General Equilibrium and Scale Effects:** Small-scale experiments isolate partial equilibrium impacts. At scale, market forces react (e.g., wages adjust via labor supply shifts), potentially reversing or amplifying the estimated effect.

12.2.3 Practical Challenges in Randomized Experiments

Even when randomization guarantees internal validity *ex ante*, post-randomization behavioral and institutional dynamics can reintroduce selection bias *ex post*.

12.2.3.1 Non-Compliance: Dropouts and Substitution

Definition 12.2.3 (Intention-to-Treat Effect (ITT)). When treatment assignment $Z_i \in \{0, 1\}$ (the randomization lottery) diverges from treatment actually received $D_i \in \{0, 1\}$, the ITT estimates the effect of *being offered* the treatment:

$$\text{ITT} = \mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]. \quad (12.34)$$

Since Z_i is perfectly random, the ITT retains internal validity and measures the policy-relevant impact of offering (not forcing) the program.

Remark 12.2.6 (Dropouts (Non-Compliance of the Treated)). Individuals assigned to treatment ($Z_i = 1$) who choose not to participate ($D_i = 0$) alter the composition of the effectively treated group. If dropout is correlated with unobserved ability or motivation, comparing effective participants to controls produces selection bias.

Remark 12.2.7 (Substitution (Non-Compliance of the Control)). Individuals assigned to control ($Z_i = 0$) who obtain the treatment or a close substitute ($D_i = 1$) contaminate the control group mean: $\mathbb{E}[Y_i \mid Z_i = 0]$ incorporates treatment outcomes for some members, attenuating the naive difference toward zero.

Theorem 12.2.1 (LATE via Wald Estimator (Imbens and Angrist, 1994)). *Under the assumptions of **relevance** ($\Pr(D_i = 1 \mid Z_i = 1) \neq \Pr(D_i = 1 \mid Z_i = 0)$), **exogeneity** ($Z_i \perp\!\!\!\perp (Y_i(1), Y_i(0), D_i(1), D_i(0))$), **exclusion restriction** ($Y_i(z, d) = Y_i(d)$ for $z \in \{0, 1\}$), and **monotonicity** ($D_i(1) \geq D_i(0)$ for all i , ruling out defiers), the Wald estimator identifies the Local Average Treatment Effect:*

$$\text{LATE} = \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\Pr(D_i = 1 \mid Z_i = 1) - \Pr(D_i = 1 \mid Z_i = 0)} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{Complier}]. \quad (12.35)$$

12.2.3.2 Attrition

Remark 12.2.8 (Attrition as Dynamic Selection Bias). In longitudinal experiments, loss of observations between baseline and follow-up (*attrition*) ceases to be a mere sample-size problem when it is *non-random* (Missing Not at Random). Attrition is typically asymmetric: controls, frustrated by lottery rejection, are more likely to refuse follow-up surveys or relocate. If attrition correlates with potential outcomes, the surviving sample breaks the initial randomization balance.

Diagnostics and corrections:

- Test balance of observable characteristics among survivors using normalized differences.
- Model the attrition selection equation jointly with the outcome equation (Heckman-type correction) or use instrumental variables for attrition.

12.2.3.3 Spillovers and SUTVA Violations in the Field

Remark 12.2.9 (Contamination via Externalities). When treatment of unit i generates indirect effects on unit j (e.g., deworming reducing parasite transmission among untreated classmates), SUTVA is violated. The observed control mean absorbs part of the treatment effect: $\mathbb{E}[Y_i(0) \mid D_i = 0, \text{spillover}] > \mathbb{E}[Y_i(0) \mid \text{no spillover}]$, severely underestimating the true impact.

Design solution: Cluster randomization at a higher aggregation level (schools, villages) creates physical or geographic separation between treatment and control clusters, isolating spillovers.

12.2.3.4 The Compliance Typology

Remark 12.2.10 (Latent Compliance Groups). Under the LATE framework, the population is partitioned into four latent behavioral types based on the joint response $(D_i(0), D_i(1))$. The **Compliers** have $D_i = 1$ if $Z_i = 1$ and $D_i = 0$ if $Z_i = 0$; they identify the LATE. The **Always-takers** have $D_i = 1$ regardless of Z_i ; they generate substitution. The **Never-takers** have $D_i = 0$ regardless of Z_i ; they generate dropouts. The **Defiers** have $D_i = 0$ if $Z_i = 1$ and $D_i = 1$ if $Z_i = 0$; they are ruled out by monotonicity.

Only the Compliers contribute to the identification of the LATE. The proportions of each type are not individually point-identified, but the share of Compliers equals the first-stage coefficient: $\Pr(\text{Complier}) = \Pr(D_i = 1 \mid Z_i = 1) - \Pr(D_i = 1 \mid Z_i = 0)$.

12.2.4 Statistical Power and Experimental Design

Definition 12.2.4 (Statistical Power). The power of a hypothesis test is the probability of rejecting H_0 when H_1 is true:

$$\text{Power} = 1 - \beta = \Pr(\text{Reject } H_0 \mid H_1 \text{ is true}), \quad (12.36)$$

where β is the Type II error rate. In RCTs, power is the probability of detecting a nonzero treatment effect when one truly exists.

12.2.4.1 Formal Power Function under Normality

Consider testing $H_0 : \theta = 0$ vs. $H_1 : \theta > 0$ with test statistic $T(0) = \bar{Y}\sqrt{n}/\sigma \sim N(0, 1)$ under H_0 . At significance level α , the critical value is $c = z_{1-\alpha}$. Under the alternative, $T(0) = T(\theta) + \theta\sqrt{n}/\sigma$ where $T(\theta) \sim N(0, 1)$. The power function is:

$$\pi(\theta) = \Pr\left(Z > z_{1-\alpha} - \frac{\theta\sqrt{n}}{\sigma}\right) = 1 - \Phi\left(z_{1-\alpha} - \frac{\theta\sqrt{n}}{\sigma}\right), \quad (12.37)$$

where $\Phi(\cdot)$ is the standard normal CDF. Power is strictly increasing in $|\theta|$ and n , and strictly decreasing in σ and $z_{1-\alpha}$.

12.2.4.2 Determinants of Power in RCTs

Remark 12.2.11 (Key Design Parameters). 1. **Minimum Detectable Effect (MDE):** The smallest θ for which the experiment achieves a target power (conventionally 80%). Smaller MDEs require larger samples.

2. **Significance level (α):** Tightening α (e.g., from 5% to 0.5%) increases $z_{1-\alpha}$ and reduces power. The standard convention is $\alpha = 0.05$.
3. **Outcome variance (σ^2):** Higher idiosyncratic dispersion reduces precision and power.
4. **Allocation ratio (P):** Let $N = N_1 + N_0$ and $P = N_1/N$. Under homoscedasticity:

$$\text{Var}(\bar{Y}_1 - \bar{Y}_0) = \frac{\sigma^2}{N \cdot P(1 - P)}. \quad (12.38)$$

Variance is minimized at $P = 0.5$ (balanced allocation). Asymmetric allocation ($P < 0.5$) may be optimal when per-unit treatment costs exceed control costs.

12.2.4.3 Cluster Randomization and Design Effects

Remark 12.2.12 (Intraclass Correlation and Effective Sample Size). When randomization occurs at the cluster level (schools, municipalities), observations within clusters share common shocks, violating i.i.d. sampling. The intraclass correlation coefficient (ICC, ρ) inflates the variance of the estimator by the **design effect**:

$$\text{DE} = 1 + (m - 1)\rho, \quad (12.39)$$

where m is the number of individuals per cluster. When ρ is substantial, increasing m yields negligible precision gains; power is governed primarily by the **number of clusters**, not the number of individuals.

- Remark 12.2.13* (Stratification, Non-Compliance, and Attrition). • **Stratified (blocked) randomization:** Grouping units into homogeneous strata before randomization removes between-stratum variance from the residual, increasing power relative to simple randomization.
- **Non-compliance:** Partial take-up dilutes the first stage, reducing the effective treatment contrast. The required sample size must be inflated by a factor of $1/\text{FS}^2$, where FS is the first-stage compliance rate.
 - **Attrition:** Anticipated sample loss between baseline and follow-up requires ex ante oversampling to preserve the effective N at endline.

12.3 Matching and Propensity Score Methods

12.3.1 Conditional Ignorability and the Matching Estimator

Remark 12.3.1 (Matching Does Not Resolve SUTVA Violations). A critical conceptual boundary must be established before proceeding: matching methods (whether by covariates or propensity score) do **not** resolve SUTVA violations. SUTVA is an *axiomatic prerequisite* for matching to be theoretically viable—it guarantees that exactly two well-defined potential outcomes $Y_i(1)$ and $Y_i(0)$ exist for each individual. If spillovers or general equilibrium effects are present, SUTVA is violated and matching loses its ontological foundation. Given SUTVA, matching enters with a strict statistical objective: eliminate selection bias V under the hypothesis of Selection on Observables by constructing the counterfactual from control observations sharing the same observed characteristics X_i .

Definition 12.3.1 (Conditional Ignorability (CIA / Unconfoundedness)). Let $X_i \in \mathcal{X} \subset \mathbb{R}^K$ be a vector of pre-treatment observable covariates of dimension $K \times 1$. The ignorability hypothesis postulates that, once we flexibly control for X_i , treatment assignment D_i becomes statistically independent of potential outcomes:

$$(Y_i(1), Y_i(0)) \perp\!\!\!\perp D_i \mid X_i. \quad (12.40)$$

For ATT identification, the weaker condition suffices:

$$\mathbb{E}[Y_i(0) \mid D_i = 1, X_i] = \mathbb{E}[Y_i(0) \mid D_i = 0, X_i] \equiv \mu_0(X_i). \quad (12.41)$$

Definition 12.3.2 (Common Support (Overlap)). For any covariate vector $X_i = x$ observed in the sample, there exists a strictly positive and less-than-one probability of treatment:

$$0 < \Pr(D_i = 1 \mid X_i = x) < 1 \quad \forall x \in \mathcal{X}. \quad (12.42)$$

For the ATE, the strict bi-lateral condition (12.42) is required. For the ATT, only the weaker unilateral condition $\Pr(D_i = 1 \mid X_i = x) < 1$ is needed, ensuring at least one potential control match exists for each treated unit.

Under (12.40) and (12.42), the ATT is identified by integrating local effects over the covariate distribution of the treated:

$$\text{ATT} = \int \left(\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x] \right) dF_{X|D=1}(x). \quad (12.43)$$

12.3.1.1 The Mechanics of M -Nearest Neighbor Matching

We wish to estimate the counterfactual $Y_i(0)$ for each treated unit i (where $D_i = 1$) using the M control observations whose characteristics X_j are closest to X_i .

Distance Metrics. Define a distance metric $\|X_i - X_j\|_V$ in the Euclidean space $\mathbb{R}^K$:

- **Standard Euclidean:** Assumes covariates share the same scale and are uncorrelated:
 $\|X_i - X_j\| = \sqrt{(X_i - X_j)'(X_i - X_j)}.$
- **Mahalanobis:** Corrects for scale and covariance to prevent high-variance regressors from dominating:

$$d_{\text{Maha}}(X_i, X_j) = \sqrt{(X_i - X_j)' \hat{\Sigma}_X^{-1} (X_i - X_j)}, \quad (12.44)$$

where $\hat{\Sigma}_X$ is the sample variance-covariance matrix of covariates computed over the control group.

Matching Set. Let $J_M(i)$ denote the set of indices of the M control observations minimizing distance to X_i :

$$J_M(i) = \left\{ j \in \{1, \dots, N_0\} \left| \sum_{l: D_l=0} \mathbb{I}\{\|X_l - X_i\|_V \leq \|X_j - X_i\|_V\} \leq M \right. \right\}. \quad (12.45)$$

Imputation Estimator. The imputed counterfactual $\hat{Y}_i(0)$ is the simple arithmetic mean of observed outcomes of the M control neighbors:

$$\hat{Y}_i(0) = \frac{1}{M} \sum_{j \in J_M(i)} Y_j. \quad (12.46)$$

Substituting into the sample ATT estimator yields the M -Nearest Neighbor Matching Estimator:

$$\widehat{\text{ATT}}_{\text{match}} = \frac{1}{N_1} \sum_{i:D_i=1} \left(Y_i - \frac{1}{M} \sum_{j \in J_M(i)} Y_j \right). \quad (12.47)$$

Remark 12.3.2 (Methodological Trade-offs). **A. With vs. Without Replacement.**

- *With replacement:* A single control can serve as the “ideal match” for multiple treated units. **Bias:** Low (each treated is compared to the best available control). **Variance:** High (reduces the number of unique control observations effectively used).
- *Without replacement:* Each control matches at most one treated. **Order problem:** Results depend strictly on the sequential order of matching (greedy algorithm); matching the most difficult units first is recommended. **Bias:** Higher (as the eligible control pool depletes, late-treated units are forced to match with progressively distant controls, generating *matching discrepancy bias*).

B. Number of Neighbors (M).

- $M = 1$: Minimizes discrepancy bias but maximizes asymptotic standard error variance.
- $M > 1$: Smooths the counterfactual estimate (reducing variance) but introduces specification bias by including progressively worse controls in the local average.

12.3.1.2 The Curse of Dimensionality: Abadie and Imbens (2006)

Classical covariate matching works perfectly when the dimension K is small. However, when we condition rigorously on a broad vector of characteristics to make ignorability plausible, we encounter the **Curse of Dimensionality**.

Remark 12.3.3 (Asymptotic Bias Decomposition). Abadie and Imbens (2006) analytically demonstrated that the classical matching estimator with M fixed neighbors exhibits the following asymptotic convergence rate:

$$\hat{\theta}_{\text{match}} - \theta_0 = O_p(N^{-1/2} + N^{-1/K}), \quad (12.48)$$

where K is the number of continuous covariates in X .

- If $K = 1$: convergence rate is $O_p(N^{-1/2})$ (standard asymptotic normality and efficiency).
- If $K \geq 2$: the bias convergence rate becomes slower than the square root of sample size ($N^{-1/K}$). The classical matching estimator suffers from a **non-negligible asymptotic bias** that prevents direct application of conventional bootstrap for standard errors, unless a parametric bias correction via local regression in the second stage is applied.

Since the volume of a K -dimensional hypercube grows exponentially with K , the sample density of data in any sub-region of $\mathcal{X}$ collapses to zero. Finding a “close” control in multidimensional space becomes impossible, violating common support in finite samples. This mathematical barrier necessitated finding a one-dimensional balancing function that

resolves dimensionality without violating ignorability—the Propensity Score of Rosenbaum and Rubin (1983).

12.3.2 The Propensity Score Theorem

Definition 12.3.3 (Propensity Score). The propensity score is the scalar function:

$$e(X_i) \equiv \Pr(D_i = 1 \mid X_i) = \mathbb{E}[D_i \mid X_i], \quad (12.49)$$

mapping $\mathbb{R}^K \rightarrow (0, 1)$.

Lemma 12.3.1 (Balancing Property). *Under the definition of $e(X_i)$, conditioned on $e(X_i)$, the covariate vector X_i becomes orthogonal to treatment status:*

$$D_i \perp\!\!\!\perp X_i \mid e(X_i). \quad (12.50)$$

Proof. We must show $\Pr(D_i = 1 \mid X_i, e(X_i)) = \Pr(D_i = 1 \mid e(X_i))$.

Left-hand side: Since $e(X_i)$ is a deterministic function of X_i , conditioning on both is redundant:

$$\Pr(D_i = 1 \mid X_i, e(X_i)) = \Pr(D_i = 1 \mid X_i) = e(X_i). \quad (12.51)$$

Right-hand side: By the Law of Iterated Expectations (LIE), evaluated at a generic support point where $e(X_i) = e$:

$$\begin{aligned} \Pr(D_i = 1 \mid e(X_i) = e) &= \mathbb{E}[D_i \mid e(X_i) = e] \\ &= \mathbb{E}\left[\mathbb{E}[D_i \mid X_i, e(X_i) = e] \mid e(X_i) = e\right] \quad (\text{LIE}) \\ &= \mathbb{E}\left[\mathbb{E}[D_i \mid X_i] \mid e(X_i) = e\right] \quad (\text{redundancy of } e(X_i)) \\ &= \mathbb{E}\left[e(X_i) \mid e(X_i) = e\right] \quad (\text{definition}) \\ &= e. (\text{projection property}) \end{aligned} \quad (12.52)$$

By transitivity, both sides equal $e(X_i)$, proving $D_i \perp\!\!\!\perp X_i \mid e(X_i)$.

Economic intuition: If we group individuals sharing the same propensity score $e(X_i) = 0.35$, the distribution of the complex covariate vector X_i (gender, age, income, capital endowment) will be statistically identical between treated and controls. The score reconstructs the observable endowment balance characteristic of an RCT for that subgroup. $\square$

Theorem 12.3.1 (Rosenbaum and Rubin, 1983). *If conditional ignorability (12.40) holds given X_i , then it remains strictly valid conditional only on the one-dimensional scalar projection $e(X_i)$:*

$$(Y_i(1), Y_i(0)) \perp\!\!\!\perp D_i \mid e(X_i). \quad (12.53)$$

Proof. We show $\mathbb{E}[D_i \mid Y_i(d), e(X_i)] = \mathbb{E}[D_i \mid e(X_i)]$ for $d \in \{0, 1\}$.

By LIE, conditioning on the larger information set $\{Y_i(d), X_i, e(X_i)\}$:

$$\begin{aligned} \mathbb{E}[D_i \mid Y_i(d), e(X_i)] &= \mathbb{E}\left[\mathbb{E}[D_i \mid Y_i(d), X_i, e(X_i)] \mid Y_i(d), e(X_i)\right] \\ &= \mathbb{E}\left[\mathbb{E}[D_i \mid Y_i(d), X_i] \mid Y_i(d), e(X_i)\right]. \quad (\text{redundancy of } e(X_i)) \end{aligned} \quad (12.54)$$

By CIA (12.40), $(Y_i(1), Y_i(0)) \perp\!\!\!\perp D_i \mid X_i$, so the potential outcome $Y_i(d)$ provides no additional information about treatment probability once X_i is controlled:

$$\mathbb{E}[D_i \mid Y_i(d), X_i] = \mathbb{E}[D_i \mid X_i] = e(X_i). \quad (12.55)$$

Therefore:

$$\mathbb{E}[D_i \mid Y_i(d), e(X_i)] = \mathbb{E}[e(X_i) \mid Y_i(d), e(X_i)] = e(X_i). \quad (12.56)$$

Since $\Pr(D_i = 1 \mid e(X_i)) = e(X_i)$ (from the Balancing Lemma), we conclude:

$$\Pr(D_i = 1 \mid Y_i(d), e(X_i)) = e(X_i) = \Pr(D_i = 1 \mid e(X_i)), \quad (12.57)$$

proving $(Y_i(1), Y_i(0)) \perp\!\!\!\perp D_i \mid e(X_i)$ almost surely. $\square$

Remark 12.3.4 (Coarseness Property and Dimensionality Resolution). The propensity score is the *coarsest* balancing score: any other balancing score $b(X_i)$ such that $D_i \perp\!\!\!\perp X_i \mid b(X_i)$ must be a finer partition than $e(X_i)$, meaning $e(X_i)$ is a deterministic function of $b(X_i)$. This makes $e(X_i)$ the optimal dimensionality reduction for removing selection bias.

This theorem brilliantly resolves the nonparametric estimation problem identified by Abadie and Imbens (2006). If the researcher has $K = 30$ continuous covariates, classical cell-based or Euclidean-distance matching becomes numerically impossible due to the absence of common support in finite samples. Rosenbaum and Rubin (1983) prove there is **no loss of causal information** in collapsing K dimensions into a single dimension $e(X_i)$. If selection on observables is valid controlling for X_i , matching or reweighting based strictly on the scalar $e(X_i)$ guarantees consistent and asymptotically unbiased estimation of ATT and ATE.

12.3.3 First Stage: Parametric Modeling of the Propensity Score

In the first stage of a selection-on-observables design, the researcher estimates the conditional probability of treatment given pre-treatment covariates:

$$e(X_i) \equiv \Pr(D_i = 1 \mid X_i) = \mathbb{E}[D_i \mid X_i]. \quad (12.58)$$

Since $D_i \in \{0, 1\}$ is strictly binary, we model the conditional probability through a nonlinear cumulative distribution function (CDF) $F(\cdot)$, ensuring estimates remain in the unit probability interval. Let X_i be the $K \times 1$ covariate vector and γ the coefficient vector:

$$e(X_i) = F(X_i' \gamma). \quad (12.59)$$

- **Probit:** $F(\cdot) = \Phi(\cdot)$, the standard normal CDF: $\Phi(X_i' \gamma) = \int_{-\infty}^{X_i' \gamma} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$.
- **Logit:** $F(\cdot) = \Lambda(\cdot)$, the logistic CDF: $\Lambda(X_i' \gamma) = \frac{\exp(X_i' \gamma)}{1 + \exp(X_i' \gamma)}$.

Remark 12.3.5 (Maximum Likelihood Estimation). For an i.i.d. sample of size N , the log-likelihood function under independent Bernoulli trials is:

$$\ln L(\gamma) = \sum_{i=1}^N \left\{ D_i \ln F(X_i' \gamma) + (1 - D_i) \ln [1 - F(X_i' \gamma)] \right\}. \quad (12.60)$$

The First-Order Conditions (score equations) for $\hat{\gamma}_{\text{MLE}}$ are:

$$s(\gamma) \equiv \frac{\partial \ln L(\gamma)}{\partial \gamma} = \sum_{i=1}^N \left\{ \frac{D_i - F(X_i' \gamma)}{F(X_i' \gamma)[1 - F(X_i' \gamma)]} f(X_i' \gamma) X_i \right\} = \mathbf{0}, \quad (12.61)$$

where $f(z) \equiv dF(z)/dz$ is the associated probability density function (standard normal $\phi(z)$ for Probit, logistic $\lambda(z) = e^z/(1 + e^z)^2$ for Logit).

Geometric intuition: Under the consistent estimator $\hat{\gamma}_{\text{MLE}}$, the treatment prediction residual $e_i \equiv D_i - \hat{e}(X_i)$, weighted by the inverse of the conditional Bernoulli variance and the local density $f(X_i' \hat{\gamma})$, is orthogonal to the space spanned by covariates X_i .

Remark 12.3.6 (Why the Linear Probability Model (LPM) Fails for Stage I). Although the LPM ($D_i = X_i' \gamma + v_i$) offers easy coefficient interpretation, it systematically fails as a propensity score generator due to three mathematical pathologies:

1. **Common support violation:** The LPM does not restrict predictions to $[0, 1]$. Obtaining $\hat{e}(X_i) < 0$ or $\hat{e}(X_i) > 1$ renders IPW estimators mathematically invalid (negative or undefined weights).
2. **Intrinsic heteroskedasticity:** The error variance is structurally dependent on X_i : $\text{Var}(v_i | X_i) = e(X_i)(1 - e(X_i))$. Although correctable by WLS, variance approaches zero near the tails, invalidating conventional OLS inference.
3. **Linear marginal effects at tails:** The LPM forces constant marginal effects for any X_k transition. In Logit/Probit, as $X_i' \gamma \rightarrow \infty$, the marginal effect $\partial e(X_i)/\partial X_k = f(X_i' \gamma) \gamma_k$ converges smoothly to zero, realistically mimicking discrete choice behavior under budget or eligibility constraints.

12.3.4 Diagnostics: Balance and Common Support

Once $\hat{e}(X_i)$ is estimated and matching (or reweighting) performed, the researcher must verify whether the algorithm actually restored pre-treatment balance between treatment and control groups.

Remark 12.3.7 (The Paradox of Classical t -tests). Historically, researchers reported t -tests of mean differences for each covariate pre- and post-matching to “prove” balance:

$$t_k = \frac{\bar{X}_{1,k} - \bar{X}_{0,k}}{\sqrt{\frac{s_{1,k}^2}{N_1} + \frac{s_{0,k}^2}{N_0}}}. \quad (12.62)$$

The critical methodological error: the statistic t_k is a direct inverse function of $\sqrt{N}$, generating a dual statistical paradox:

- **Large samples:** Even with near-perfect overlap (irrelevant mean differences), high N drives the t -statistic denominator toward zero, rejecting the null of equal means ($p < 0.05$), falsely suggesting balance failure.
- **Small samples:** The t -test has low power. It fails to reject the null even with severe distributional imbalances, inducing a false sense of security.

Definition 12.3.4 (Normalized Differences (Imbens and Rubin, 2015)). To eliminate the spurious interference of sample size, the normalized difference for each covariate k is:

$$\Delta_{X_k} = \frac{\bar{X}_{1,k} - \bar{X}_{0,k}}{\sqrt{\frac{s_{1,k}^2 + s_{0,k}^2}{2}}}, \quad (12.63)$$

where $\bar{X}_{1,k}$, $\bar{X}_{0,k}$ are sample means and $s_{1,k}^2$, $s_{0,k}^2$ are sample variances estimated independently for treated and controls.

Mathematical properties:

- **Scale and sample invariance:** Δ_k quantifies mean discrepancy in units of average standard deviation. Since N does not enter the equation, Δ_k remains constant even as $N \rightarrow \infty$.
- **Rule of thumb:** Balance is considered statistically adequate if and only if $|\Delta_k| \leq 0.25$. Values exceeding 0.25 indicate excessively severe linear discrepancy; subsequent second-stage estimates will be dangerously sensitive to parametric specification choices.

Remark 12.3.8 (Common Support Diagnostics and Trimming). Balance validation is mathematically consistent only if strict common support holds in the sample.

Visual diagnostics: Plot mirrored Kernel density histograms of $\hat{e}(X_i)$ between treated and controls, identifying failure zones where treated units have $\hat{e}(X_i) \approx 1$ (no comparable controls) and controls have $\hat{e}(X_i) \approx 0$ (no comparable treated).

Crump, Hotz, Imbens, and Mitnik (2009) Trimming Rule: When overlap is weak at boundaries, classical IPW estimators face numerical instability (infinite variance) due to weights collapsing from near-zero divisions. The authors propose systematic discarding of observations outside the optimal support interval $\mathcal{A}$:

$$\mathcal{A} = \{x \in \mathcal{X} \mid \alpha \leq e(x) \leq 1 - \alpha\}. \quad (12.64)$$

Under arbitrary covariate distributions, Crump et al. (2009) prove the cutoff α minimizing the asymptotic variance of the treatment effect estimator is:

$$\alpha \approx 0.10. \quad (12.65)$$

This formally restricts empirical analysis to the propensity score interval $\hat{e}(X_i) \in [0.10, 0.90]$. Units outside this interval are discarded, and the treatment estimator is interpreted as the causal effect on the subpopulation with stable common support.

12.3.5 Inverse Probability Weighting (IPW)

IPW is a semiparametric method using the Stage I propensity score to reweight observations and mimic an experimental scenario. Under CIA and common support, IPW corrects selection bias by reweighting each unit by the inverse of the probability of actually receiving the treatment it received.

12.3.5.1 Identification of ATE via IPW

Theorem 12.3.2 (IPW-ATE Identification). *Under CIA (12.40) and strict common support, the Horvitz-Thompson IPW estimator identifies the ATE:*

$$ATE = \mathbb{E} \left[\frac{D_i Y_i}{e(X_i)} - \frac{(1 - D_i) Y_i}{1 - e(X_i)} \right]. \quad (12.66)$$

Proof. Part A: Identifying $\mathbb{E}[Y_i(1)]$. Consider the observable weighted expectation:

$$\mathbb{E} \left[\frac{D_i Y_i}{e(X_i)} \right]. \quad (12.67)$$

Substituting the Rubin switching equation $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$ and applying binary algebra ($D_i^2 = D_i$, $D_i(1 - D_i) = 0$):

$$\mathbb{E} \left[\frac{D_i Y_i(1)}{e(X_i)} \right]. \quad (12.68)$$

By LIE conditioning on X_i , and since $e(X_i)$ is deterministic in X_i :

$$= \mathbb{E} \left[\frac{1}{e(X_i)} \mathbb{E}[D_i Y_i(1) \mid X_i] \right]. \quad (12.69)$$

By CIA, $Y_i(1) \perp\!\!\!\perp D_i \mid X_i$, so the conditional expectation factors:

$$\mathbb{E}[D_i Y_i(1) \mid X_i] = \mathbb{E}[D_i \mid X_i] \cdot \mathbb{E}[Y_i(1) \mid X_i] = e(X_i) \cdot \mathbb{E}[Y_i(1) \mid X_i]. \quad (12.70)$$

Substituting back and canceling $e(X_i)$ under strict support ($e(X_i) > 0$):

$$= \mathbb{E} \left[\mathbb{E}[Y_i(1) \mid X_i] \right] = \mathbb{E}[Y_i(1)]. \quad (12.71)$$

Part B: Identifying $\mathbb{E}[Y_i(0)]$. By exact symmetry:

$$\mathbb{E} \left[\frac{(1 - D_i) Y_i}{1 - e(X_i)} \right] = \mathbb{E}[Y_i(0)]. \quad (12.72)$$

By linearity of expectation, combining Parts A and B yields the ATE identification. $\square$

12.3.5.2 Identification of ATT via IPW

Theorem 12.3.3 (IPW-ATT Identification). *Under CIA and common support, the ATT is identified by:*

$$ATT = \mathbb{E} \left[\frac{(D_i - e(X_i)) Y_i}{(1 - e(X_i)) \cdot \mathbb{E}[D_i]} \right]. \quad (12.73)$$

Proof. For ATT, we need $\mathbb{E}[Y_i(0) \mid D_i = 1]$. Consider the reweighted observable:

$$\mathbb{E} \left[\frac{e(X_i)(1 - D_i) Y_i}{1 - e(X_i)} \right]. \quad (12.74)$$

Substituting the switching equation under $D_i = 0$ and applying LIE conditioning on X_i :

$$= \mathbb{E} \left[\frac{e(X_i)}{1 - e(X_i)} \mathbb{E}[(1 - D_i)Y_i(0) \mid X_i] \right]. \quad (12.75)$$

By CIA, $Y_i(0) \perp\!\!\!\perp D_i \mid X_i$:

$$= \mathbb{E} \left[\frac{e(X_i)}{1 - e(X_i)} (1 - e(X_i)) \mathbb{E}[Y_i(0) \mid X_i] \right] = \mathbb{E}[e(X_i) \mathbb{E}[Y_i(0) \mid X_i]]. \quad (12.76)$$

Substituting $e(X_i) = \mathbb{E}[D_i \mid X_i]$ and reconverting by CIA:

$$= \mathbb{E}[\mathbb{E}[D_i Y_i(0) \mid X_i]] = \mathbb{E}[D_i Y_i(0)]. \quad (12.77)$$

By LIE over the discrete partition of D_i :

$$\mathbb{E}[D_i Y_i(0)] = \mathbb{E}[Y_i(0) \mid D_i = 1] \cdot P, \quad (12.78)$$

where $P \equiv \Pr(D_i = 1) = \mathbb{E}[D_i]$. Isolating the counterfactual:

$$\mathbb{E}[Y_i(0) \mid D_i = 1] = \mathbb{E} \left[\frac{e(X_i)(1 - D_i)Y_i}{(1 - e(X_i))P} \right]. \quad (12.79)$$

Combining with $\mathbb{E}[Y_i \mid D_i = 1] = \mathbb{E}[D_i Y_i]/P$ and simplifying the numerator algebraically to $(D_i - e(X_i))/(1 - e(X_i))$ yields the result. $\square$

Remark 12.3.9 (The Hájek Normalized Estimator). The Horvitz-Thompson sample analog presents numerical instability in finite samples: units with $\hat{e}(X_i)$ near 0 or 1 generate excessively large weights, dramatically inflating variance. Additionally, nothing guarantees that $\sum_i D_i/\hat{e}(X_i) = N$.

The **Hájek estimator** forces weight sums to collapse strictly to unity within each treatment group through internal reweighting:

$$\widehat{\text{ATE}}_{\text{Hájek}} = \frac{\sum_i D_i Y_i / \hat{e}(X_i)}{\sum_i D_i / \hat{e}(X_i)} - \frac{\sum_i (1 - D_i) Y_i / (1 - \hat{e}(X_i))}{\sum_i (1 - D_i) / (1 - \hat{e}(X_i))}. \quad (12.80)$$

By normalizing weights by their effective sample sums, the Hájek estimator considerably smooths the impact of extreme propensity scores in distributional tails, substantially reducing finite-sample mean squared error (MSE).

12.3.6 The Doubly Robust Estimator (AIPW)

The Augmented Inverse Probability Weighting (AIPW) estimator, proposed by Robins, Rotnitzky, and Zhao (1995), resolves key limitations of purely regression-based or purely IPW-based estimators. It combines both strategies in a unified moment condition, guaranteeing consistency and asymptotic unbiasedness if *at least one* of the two models is correctly specified.

Definition 12.3.5 (AIPW Estimator). Let $e^*(X_i)$ be the true propensity score and $\mu_1^*(X_i) \equiv \mathbb{E}[Y_i(1) \mid X_i]$ the true outcome regression under treatment. The AIPW estimator for $\mu_1 = \mathbb{E}[Y_i(1)]$ is:

$$\hat{\mu}_1^{\text{DR}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{D_i Y_i}{\hat{e}(X_i)} - \frac{D_i - \hat{e}(X_i)}{\hat{e}(X_i)} \hat{\mu}_1(X_i) \right]. \quad (12.81)$$

Equivalently, the AIPW estimator for the ATE unifies treatment and control components:

$$\widehat{\text{ATE}}^{\text{DR}} = \frac{1}{N} \sum_{i=1}^N \left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{D_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - D_i)(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{e}(X_i)} \right]. \quad (12.82)$$

Theorem 12.3.4 (Double Robustness). *The estimator $\hat{\mu}_1^{\text{DR}}$ is consistent for $\mu_1 = \mathbb{E}[Y_i(1)]$ if either:*

- *The propensity score model is correctly specified: $\text{plim } \hat{e}(X_i) = e^*(X_i)$, **or***
- *The outcome regression model is correctly specified: $\text{plim } \hat{\mu}_1(X_i) = \mu_1^*(X_i)$.*

Proof. Let $e^*(X_i) = \text{plim } \hat{e}(X_i)$ and $\mu_1^*(X_i) = \text{plim } \hat{\mu}_1(X_i)$. The probability limit is:

$$\mu_1^* = \mathbb{E} \left[\frac{D_i Y_i}{e(X_i)} - \frac{D_i - e(X_i)}{e(X_i)} \mu_1(X_i) \right], \quad (12.83)$$

where $e(X_i)$ and $\mu_1(X_i)$ denote the probability limits of the estimated models (not necessarily the true functions).

Case A: Propensity score correct ($e(X_i) = e^*(X_i)$), **outcome regression incorrect** ($\mu_1(X_i) \neq \mu_1^*(X_i)$). By LIE conditioning on X_i , and noting $D_i Y_i = D_i Y_i(1)$ by the switching equation:

$$\begin{aligned} \mu_1^* &= \mathbb{E} \left[\frac{\mathbb{E}[D_i Y_i(1) | X_i]}{e^*(X_i)} - \frac{\mathbb{E}[D_i | X_i] - e^*(X_i)}{e^*(X_i)} \mu_1(X_i) \right] \\ &= \mathbb{E} \left[\frac{e^*(X_i) \mu_1^*(X_i)}{e^*(X_i)} - \frac{e^*(X_i) - e^*(X_i)}{e^*(X_i)} \mu_1(X_i) \right] \quad (\text{by CIA}) \\ &= \mathbb{E}[\mu_1^*(X_i) - 0] = \mathbb{E}[\mu_1^*(X_i)] = \mathbb{E}[Y_i(1)]. \end{aligned} \quad (12.84)$$

The second term vanishes identically because $e^*(X_i) - e^*(X_i) = 0$, even though $\mu_1(X_i)$ is incorrect.

Case B: Outcome regression correct ($\mu_1(X_i) = \mu_1^*(X_i)$), **propensity score incorrect** ($e(X_i) \neq e^*(X_i)$).

$$\begin{aligned} \mu_1^* &= \mathbb{E} \left[\frac{e^*(X_i) \mu_1^*(X_i)}{e(X_i)} - \frac{e^*(X_i) - e(X_i)}{e(X_i)} \mu_1^*(X_i) \right] \\ &= \mathbb{E} \left[\mu_1^*(X_i) \left(\frac{e^*(X_i) - e^*(X_i) + e(X_i)}{e(X_i)} \right) \right] \\ &= \mathbb{E} \left[\mu_1^*(X_i) \cdot \frac{e(X_i)}{e(X_i)} \right] = \mathbb{E}[\mu_1^*(X_i)] = \mathbb{E}[Y_i(1)]. \end{aligned} \quad (12.85)$$

The algebraic simplification inside the parentheses yields exactly 1, regardless of the incorrect $e(X_i)$, as long as strict common support holds ($e(X_i) > 0$). $\square$

Remark 12.3.10 (Semiparametric Efficiency and Neyman Orthogonality). Hahn (1998) derived the semiparametric efficiency bound for ATE estimation under CIA and overlap:

$$V^{\text{Hahn}} = \mathbb{E} \left[\frac{\sigma_1^2(X_i)}{e^*(X_i)} + \frac{\sigma_0^2(X_i)}{1 - e^*(X_i)} + (\mu_1^*(X_i) - \mu_0^*(X_i) - \tau)^2 \right], \quad (12.86)$$

where $\sigma_d^2(X_i) = \text{Var}(Y_i(d) | X_i)$ and $\tau = \text{ATE}$.

When *both* models are correctly specified, the AIPW estimator achieves this bound (attains the semiparametric Cramér-Rao lower bound). Under double correct specification, the estimator also possesses **Neyman orthogonality**: first-stage approximation errors in generated regressors do not contaminate the $\sqrt{N}$ -convergence rate of the second-stage causal parameter. This property makes AIPW ideal for joint application with high-dimensional machine learning algorithms (Double/Debiased Machine Learning, Chernozhukov et al., 2018).

12.4 Difference-in-Differences

12.4.1 The Classical Two-Period DID

Definition 12.4.1 (Setup and Notation). Let $t \in \{0, 1\}$ denote pre- and post-treatment periods. Let $D_i \in \{0, 1\}$ be the time-invariant treatment group indicator. The observed outcome is:

$$Y_{it} = d_{it}Y_{it}(1) + (1 - d_{it})Y_{it}(0) \quad (12.87)$$

where $d_{it} = D_i \times \mathbb{I}\{t = 1\}$ is the contemporaneous treatment indicator.

Definition 12.4.2 (Parallel Trends Assumption). In the absence of treatment, the average change in the outcome would be identical between treated and control groups:

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) | D_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) | D_i = 0] \quad (12.88)$$

Theorem 12.4.1 (Identification of ATT). *Under parallel trends and SUTVA:*

$$ATT = (\mathbb{E}[Y_{i1} | D_i = 1] - \mathbb{E}[Y_{i0} | D_i = 1]) - (\mathbb{E}[Y_{i1} | D_i = 0] - \mathbb{E}[Y_{i0} | D_i = 0]) \quad (12.89)$$

Proof. The DID estimand equals:

$$\begin{aligned} \text{DID} &= \mathbb{E}[\Delta Y_i | D_i = 1] - \mathbb{E}[\Delta Y_i | D_i = 0] \\ &= \mathbb{E}[Y_{i1}(1) - Y_{i0}(0) | D_i = 1] - \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) | D_i = 0] \end{aligned} \quad (12.90)$$

By parallel trends, $\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) | D_i = 0] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) | D_i = 1]$. Substituting:

$$\begin{aligned} \text{DID} &= \mathbb{E}[Y_{i1}(1) - Y_{i0}(0) | D_i = 1] - \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) | D_i = 1] \\ &= \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) | D_i = 1] = \text{ATT} \end{aligned} \quad (12.91)$$

□

Remark 12.4.1 (TWFE Representation). The DID estimator is equivalent to the coefficient θ in the Two-Way Fixed Effects regression:

$$Y_{it} = \alpha_i + \lambda_t + \theta d_{it} + \varepsilon_{it} \quad (12.92)$$

where α_i absorbs time-invariant unit heterogeneity and λ_t absorbs common time shocks.

12.4.2 Covariates in DID: Conditional Parallel Trends

Definition 12.4.3 (Conditional Parallel Trends). When unconditional parallel trends fails, we assume that trends are parallel conditional on a vector of pre-treatment covariates X_i :

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1, X_i] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0, X_i] \quad \text{q.c.} \quad (12.93)$$

Remark 12.4.2 (Why NOT to Include Time-Varying Covariates X_{it}). Although controlling for time-varying covariates may seem intuitive to rescue conditional parallel trends, including variables that change over time generates severe structural problems in DiD designs:

A. Bad Controls (Channel Blocking and Active Stratification). The golden rule of causal specification is that one must **never include variables that could have been affected by the treatment itself**. If a time-varying covariate (e.g., income, employment status) changes as a consequence of the policy under study, including it as a control removes the treatment effect that operates through that channel. Moreover, conditioning on a post-treatment realized variable forces comparison between treated and control individuals who reached the same level of that variable—pairing units with intrinsically different baseline characteristics, reintroducing selection bias and breaking common support.

B. Violation of Strict Exogeneity. For traditional panel estimators (fixed effects or first-differences) to be consistent, explanatory variables must satisfy **strict exogeneity**: $\mathbb{E}[\varepsilon_{it} \mid X_{i1}, \dots, X_{iT}] = 0$. If any temporal feedback exists—where today’s unobserved outcome shock (ε_{it}) affects tomorrow’s covariate value ($X_{i,t+1}$)—strict exogeneity is violated. When the time-varying covariate is not strictly exogenous, first-difference or fixed-effects estimation becomes **inconsistent**, and this inconsistency is not mitigated by increasing the number of time periods.

C. Sample Composition Changes. In repeated cross-section designs, one assumes that the composition of observable and unobservable characteristics remains stable over time. Systematic changes in time-varying covariates post-treatment indicate that **sample composition is dynamically shifting** (e.g., selective attrition or policy-induced migration). If this instability occurs, the DiD estimator captures composition differences between periods as if they were the causal impact of treatment, biasing results.

D. Anticipation Behavior. Time-varying covariates may suffer from endogeneity due to **treatment anticipation**. If individuals know they will be treated in the future and make decisions that alter their covariates before policy implementation, the counterfactual parallel relationship is destroyed, invalidating identification.

Methodological Recommendation: Use strictly **baseline covariates** (X_i measured at $t = 0$ or $g - 1$) or characteristics inherently invariant to treatment (e.g., gender, race), interacting them with time trends to capture differentiated trajectories without generating dynamic endogeneity.

Remark 12.4.3 (The Naive Regression Problem). Including pre-treatment covariates X_i linearly in TWFE:

$$Y_{it} = \alpha_i + \lambda_t + \theta d_{it} + X_i' \gamma + \varepsilon_{it} \quad (12.94)$$

fails because first-differencing eliminates $X_i'\gamma$ entirely:

$$\begin{aligned} Y_{i1} - Y_{i0} &= (\alpha_i + \lambda_1 + \theta D_i + X_i'\gamma + \varepsilon_{i1}) - (\alpha_i + \lambda_0 + 0 + X_i'\gamma + \varepsilon_{i0}) \\ \Delta Y_i &= \lambda^* + \theta D_i + \Delta \varepsilon_i \end{aligned} \quad (12.95)$$

The covariates only control for *level* differences (already absorbed by α_i), not for *trend* differences. Equation (12.95) shows that $X_i'\gamma$ cancels out completely, leaving the estimator identical to the unconditional DiD.

Remark 12.4.4 (Solution 1: Interacting Time-Invariant Covariates with Time). To allow X_i to determine the slope of the trend, we interact the baseline covariate with time dummies or linear trends:

$$Y_{it} = \alpha_i + \lambda_t + \theta d_{it} + (X_i \times \mathbb{I}\{t = 1\})'\gamma + \varepsilon_{it} \quad (12.96)$$

Why this works: Although X_i is constant for each individual (and would be perfectly collinear with α_i), the interaction term $X_i \times \mathbb{I}\{t = 1\}$ *varies over time* for each unit. This allows the fixed-effects model to estimate how the outcome trajectory differs over time according to that invariant characteristic (e.g., permitting different growth trajectories for men and women, or for different regions).

First-differencing yields:

$$\Delta Y_i = \lambda^* + \theta D_i + X_i'\gamma + \Delta \varepsilon_i \quad (12.97)$$

Now X_i survives the transformation and controls for differential trends across groups with different baseline characteristics.

Remark 12.4.5 (Solution 2: Freezing Time-Varying Covariates at Baseline). When the researcher possesses time-varying covariates X_{it} that could serve as useful controls but risk being bad controls if used in their time-varying form, the solution is to **transform them into baseline-stratified variables**:

1. **Isolate the pre-treatment value:** For each individual or cohort g , take the covariate value measured **strictly in the period immediately prior to treatment adoption** ($g - 1$): $X_{i,g-1}$.
2. **Freeze the value:** This value $X_{i,g-1}$ is treated as a **constant individual characteristic** for all subsequent panel periods (it no longer varies over time after freezing).
3. **Apply Solution 1:** Interact this “frozen” pre-treatment value with time dummies: $X_{i,g-1} \times \mathbb{I}\{t\}$.

Why this works: Since the covariate was measured *before* treatment began, it is by definition **immune to anticipation effects and free from treatment contamination**. At the same time, by interacting its baseline value with time, the researcher allows the control group to be reweighted or adjusted to dynamically mimic the trajectories of treated units, cleanly satisfying conditional parallel trends (12.93).

Example: In staggered adoption designs (e.g., Callaway and Sant’Anna, 2021), the propensity score $\hat{p}_g(X) = \Pr(G_g = 1 \mid X, G_g + C = 1)$ uses covariates measured at $g - 1$ precisely to avoid bad controls while allowing conditional parallel trends to hold.

Remark 12.4.6 (Limitations of Linear Interaction). Despite its elegance, the linear interaction approach (12.96) has three fundamental limitations:

- **Functional form rigidity:** Assumes X_i affects trends linearly and homogeneously. If the true relationship is nonlinear (e.g., quadratic in income), the model misspecifies the counterfactual trend.
- **Extrapolation outside support:** If treated and control occupy disjoint regions of X , the model extrapolates counterfactual trends beyond the data support, generating unreliable estimates.
- **Variance weighting distortion:** OLS weights units by the conditional variance of treatment, distorting ATT estimation under heterogeneous effects. This is precisely the pathology that motivated semiparametric alternatives like IPW-DID (Abadie, 2005) and doubly robust estimators.

12.4.3 The IPW-DID Estimator (Abadie, 2005)

Theorem 12.4.2 (Abadie, 2005). *Under conditional parallel trends and strict support ($p(X_i) < 1$ for treated units), the ATT is identified by:*

$$ATT = \mathbb{E} \left[\frac{D_i - p(X_i)}{P(D_i = 1) \cdot (1 - p(X_i))} \cdot \Delta Y_i \right] \quad (12.98)$$

Proof. Decompose the numerator: $D_i - p(X_i) \equiv D_i(1 - p(X_i)) - (1 - D_i)p(X_i)$. The moment condition becomes:

$$\frac{1}{P(D_i = 1)} \left(\underbrace{\mathbb{E}[D_i \Delta Y_i]}_{\text{Term A}} - \underbrace{\mathbb{E} \left[\frac{p(X_i)(1 - D_i)}{1 - p(X_i)} \Delta Y_i \right]}_{\text{Term B}} \right) \quad (12.99)$$

Term A: Since $D_i = 1 \implies Y_{it} = Y_{it}(1)$:

$$\mathbb{E}[D_i \Delta Y_i] = P(D_i = 1) \cdot \mathbb{E}[Y_{i1}(1) - Y_{i0}(0) \mid D_i = 1] \quad (12.100)$$

Term B: By LIE conditioning on X_i , then on D_i :

$$\begin{aligned} \mathbb{E} \left[\frac{p(X_i)(1 - D_i)}{1 - p(X_i)} \Delta Y_i \right] &= \mathbb{E} \left[\frac{p(X_i)}{1 - p(X_i)} (1 - p(X_i)) \mathbb{E}[\Delta Y_i(0) \mid D_i = 0, X_i] \right] \\ &= \mathbb{E}[p(X_i) \mathbb{E}[\Delta Y_i(0) \mid D_i = 1, X_i]] \quad (\text{by conditional parallel trends}) \\ &= \mathbb{E}[D_i \Delta Y_i(0)] \quad (\text{since } p(X_i) = \mathbb{E}[D_i \mid X_i]) \\ &= P(D_i = 1) \cdot \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] \end{aligned} \quad (12.101)$$

Substituting and canceling $P(D_i = 1)$:

$$\begin{aligned} ATT &= \mathbb{E}[Y_{i1}(1) - Y_{i0}(0) \mid D_i = 1] - \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] \\ &= \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1] \end{aligned} \quad (12.102)$$

□

Remark 12.4.7 (Implementation). The sample analog is:

$$\hat{\theta}_{IPW-DID} = \frac{1}{N_1} \sum_{i=1}^N \left(D_i - \frac{\hat{p}(X_i)(1 - D_i)}{1 - \hat{p}(X_i)} \right) \Delta Y_i \quad (12.103)$$

Since $\hat{p}(X_i)$ is a generated regressor, standard errors must account for first-stage estimation uncertainty via Abadie's corrected variance formula or a full two-stage block bootstrap.

12.4.4 Event Studies and Dynamic Effects

Definition 12.4.4 (Event Study Specification). Let E_i be the treatment onset period for unit i . Define relative time dummies $D_{it}^k = \mathbb{I}\{t - E_i = k\}$. The dynamic model is:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \in \mathcal{K}, k \neq -1} \beta_k D_{it}^k + \varepsilon_{it} \quad (12.104)$$

where $k = -1$ is omitted for identification.

Theorem 12.4.3 (Event Study Identification). *Under parallel trends and no anticipation ($Y_{it}(1) = Y_{it}(0)$ for $t < E_i$):*

$$\begin{aligned} \beta_k &= 0 \quad \forall k < 0 \quad (\text{placebo validation}) \\ \beta_k &= ATT_k \equiv \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid t - E_i = k] \quad \forall k \geq 0 \end{aligned} \quad (12.105)$$

Proof. For $k < 0$, no anticipation implies $Y_{it} = Y_{it}(0)$. Under parallel trends:

$$\mathbb{E}[Y_{it} \mid E_i, t] = \alpha_i + \lambda_t \quad \forall t < E_i \quad (12.106)$$

Comparing with the regression specification $\mathbb{E}[Y_{it} \mid E_i, t] = \alpha_i + \lambda_t + \beta_k$ for $k < 0$, we obtain $\beta_k = 0$. $\square$

Remark 12.4.8 (Necessity of Normalization). Omitting $k = -1$ is required to avoid perfect multicollinearity. The sum of all event dummies equals one for each (i, t) , creating a linear dependency with the unit fixed effects α_i (which sum to one across units). This implies $\det(X'X) = 0$, preventing inversion. Normalizing $\beta_{-1} \equiv 0$ restores full rank.

Remark 12.4.9 (Long Differences vs. Short Gaps in Event Studies). The choice of baseline period fundamentally affects the interpretation of pre-treatment coefficients used to validate parallel trends.

Long Differences (Fixed Baseline): All coefficients—both pre- and post-treatment—are computed relative to a single fixed reference period, typically $t = -1$. For any period τ :

$$\delta_\tau^{\text{Long}} = \left(\mathbb{E}[Y \mid D = 1, \tau] - \mathbb{E}[Y \mid D = 1, -1] \right) - \left(\mathbb{E}[Y \mid D = 0, \tau] - \mathbb{E}[Y \mid D = 0, -1] \right) \quad (12.107)$$

Short Gaps (Rolling Baseline): Each pre-treatment period is compared only to its immediate predecessor:

$$\delta_\tau^{\text{Short}} = \left(\mathbb{E}[Y \mid D = 1, \tau] - \mathbb{E}[Y \mid D = 1, \tau - 1] \right) - \left(\mathbb{E}[Y \mid D = 0, \tau] - \mathbb{E}[Y \mid D = 0, \tau - 1] \right) \quad (12.108)$$

Diagnostic Rule: Classical TWFE/OLS *only* estimates Long Differences (it requires omitting exactly one period dummy to avoid multicollinearity). Modern robust estimators (CS, dCDH) can compute both. If the post-treatment estimand accumulates effects relative to a fixed baseline, the pre-treatment falsification test *must* adopt the same Long Differences design for logical consistency. Short Gaps can mask cumulative pre-trends by displaying only marginal period-to-period growth rates, potentially showing “stable” coefficients even when baseline levels are severely diverging.

Implementation: In Stata `csdid`, use option `long2`; in R `did`, set `base_period = "universal"` (not `"rolling"`).

12.4.5 Serial Correlation and Valid Inference

Remark 12.4.10 (Bertrand, Duflo, and Mullainathan (2004)). Standard OLS standard errors assume ε_{it} are i.i.d. In applied DID, errors follow persistent processes (e.g., AR(1)):

$$\varepsilon_{igt} = \rho\varepsilon_{ig,t-1} + u_{igt} \quad \text{with} \quad |\rho| < 1 \quad (12.109)$$

This implies $\text{Cov}(\varepsilon_{igt}, \varepsilon_{igs}) = \rho^{|t-s|}\sigma_\varepsilon^2 \neq 0$. Since the treatment dummy D_{it} is highly persistent (once treated, always treated), the true variance $X'\Omega X \gg \sigma^2 X'X$, causing conventional standard errors to severely underestimate sampling variability. This leads to over-rejection rates up to 45% instead of the nominal 5%.

Definition 12.4.5 (Cluster-Robust Variance Estimator (CRVE)). Group residuals by cluster g (e.g., state):

$$\widehat{\text{Var}}_{CRVE}(\hat{\beta}) = (X'X)^{-1} \left(\sum_{g=1}^{\mathcal{G}} X'_g \hat{e}_g \hat{e}'_g X_g \right) (X'X)^{-1} \quad (12.110)$$

This admits arbitrary autocorrelation and heteroskedasticity within clusters.

Remark 12.4.11 (Small Number of Clusters). CRVE consistency requires $\mathcal{G} \rightarrow \infty$. With few clusters ($\mathcal{G} < 50$), residuals are compressed toward zero due to fixed effects overfitting, causing downward bias. The solution is the **Wild Cluster Bootstrap** (Cameron, Gelbach, and Miller, 2008):

1. Estimate the restricted model under $H_0 : \beta_k = 0$, obtaining restricted residuals $\tilde{e}_g$.
2. Generate bootstrap data: $Y_{igt}^* = X'_{igt} \tilde{\beta} + \tilde{e}_{igt} \times \xi_g$, where $\xi_g \in \{-1, +1\}$ with equal probability (Rademacher distribution).
3. Re-estimate and construct the empirical distribution of t -statistics.

12.4.6 The TWFE Problem Under Staggered Adoption

The discovery that the classical **Two-Way Fixed Effects (TWFE)** estimator can exhibit severe failures under **staggered adoption** designs revolutionized contemporary microeconometrics between the late 2010s and early 2020s. The work of Goodman-Bacon (2021) was a landmark in this movement, opening the “black box” of the static TWFE estimator and analytically demonstrating what it actually computes under heterogeneous treatment timing.

12.4.6.1 The Frisch-Waugh-Lovell Decomposition

Historically, researchers assumed that estimating the classical multi-period TWFE regression:

$$Y_{it} = \alpha_i + \lambda_t + \delta^{\text{TWFE}} D_{it} + \varepsilon_{it} \quad (12.111)$$

would provide a reasonable average of treatment effects, even if the policy were adopted at different times across units.

Applying the **Frisch-Waugh-Lovell (FWL) Theorem**, Goodman-Bacon removed unit (α_i) and time (λ_t) fixed effects through **double demeaning**, projecting the estimator as a simple regression between residualized outcome ($\tilde{Y}_{it}$) and residualized treatment ($\tilde{D}_{it}$). The fundamental result of this mechanical decomposition revealed:

Remark 12.4.12 (The TWFE Decomposition Theorem). The estimator $\hat{\delta}^{\text{TWFE}}$ does not compute a simple, egalitarian average of impacts. It is numerically equivalent to a weighted average of all possible simple 2×2 sub-experiments within the sample, where weights are determined by each subgroup's size and the variance of treatment status over time.

12.4.6.2 The Four 2×2 Sub-Experiments

Consider a sample with three group types:

- **Group k (Early):** Receives treatment at $t = k$.
- **Group l (Late):** Receives treatment at later $t = l$.
- **Group U (Untreated):** Remains untreated throughout.

Goodman-Bacon (2021) proved that TWFE aggregates **four basic 2×2 comparisons**:

1. **Early vs. Untreated ($\hat{\delta}_{kU}^{2 \times 2}$):** Standard comparison where U serves as clean control. **(Valid)**
2. **Late vs. Untreated ($\hat{\delta}_{lU}^{2 \times 2}$):** Standard comparison evaluated at cutoff $t = l$. **(Valid)**
3. **Early vs. Late before Late's treatment ($\hat{\delta}_{kl}^{2 \times 2}$):** Evaluates k 's effect using l (not yet treated) as temporary control. **(Valid)**
4. **Late vs. Early after Early's treatment ($\hat{\delta}_{lk}^{2 \times 2}$):** Evaluates l 's effect using k (already treated) as control. **The “Forbidden Comparison.”**

Remark 12.4.13 (The Mechanical Decomposition Formula).

$$\hat{\delta}^{\text{TWFE}} = \sum_{k \neq U} s_{kU} \hat{\delta}_{kU}^{2 \times 2} + \sum_{k \neq U} \sum_{l > k} \left[s_{klk} \hat{\delta}_{kl}^{2 \times 2, k} + s_{lkl} \hat{\delta}_{lk}^{2 \times 2, l} \right] \quad (12.112)$$

where weights s_{kU} , s_{klk} , and s_{lkl} are all strictly **positive** and sum to 1.

12.4.6.3 The Causal Decomposition: Why “Already-Treated” Generates Bias

While the mechanical decomposition (12.112) is an algebraic identity that always holds on observed data, the **theoretical or causal decomposition** substitutes realized means with potential outcome expectations.

Theorem 12.4.4 (Goodman-Bacon Causal Decomposition). *Under parallel trends, the probability limit of the TWFE estimator is:*

$$\text{plim } \hat{\delta}_{n \rightarrow \infty}^{\text{TWFE}} = VW_{ATT} + VW_{PT} - \Delta_{ATT} \quad (12.113)$$

where:

- VW_{ATT} : Variance-weighted average of the true treatment effect on the treated.
- VW_{PT} : Term associated with parallel trends (collapses to 0 if the classical hypothesis holds conditionally).
- Δ_{ATT} : The **Dynamic Treatment Effect Bias**.

Remark 12.4.14 (The Mechanism of Forbidden Comparison Failure). When estimating the forbidden comparison, the model subtracts the temporal variation of the (already-treated) control group k from the temporal variation of the group currently receiving treatment l .

If the treatment effect is **dynamic** (i.e., impact grows or decays over time since adoption), the observed trajectory of group k in the post-treatment period incorporates both the natural time trend and the **dynamic growth of its own treatment effect**.

- Using k as control, TWFE erroneously assumes k 's dynamic treatment growth is the common baseline trend.
- This causes k 's accumulated dynamic effect to be subtracted from the final estimator, generating contamination that pulls the estimate in the opposite direction.
- If dynamic effects are sufficiently large, Δ_{ATT} can become so severe that **the TWFE estimator's sign inverts completely**, generating false causal conclusions and negative estimates even when the true effect is strictly positive for all units.

Additionally, OLS variance weights maximize the importance of groups treated exactly at the **midpoint of the panel period** (where treatment indicator variance is maximized), at the expense of groups treated at the beginning or end, which could be more representative.

12.4.6.4 The Negative Weights Problem: de Chaisemartin and D'Haultfœuille (2020)

Remark 12.4.15 (Reconciling Bacon and dCDH). The apparent conflict between the **Bacon (2021) Decomposition** and the **de Chaisemartin & D'Haultfœuille (2020, dCDH) Decomposition** is one of the richest debates in contemporary DiD literature. Bacon shows TWFE is a weighted average of 2×2 comparisons with strictly **positive** weights; dCDH proves TWFE can assign **negative** weights.

The reconciliation lies in the fact that both methods **operate at completely different aggregation levels**:

Bacon (Cohort-Time Comparison Level): Decomposes the unified TWFE coefficient into a linear combination of **aggregated 2×2 comparisons**. The weights (s) applied to these equations depend on cohort sizes and treatment variance. Since they represent sample shares and population variances, **these cohort weights are structurally always positive**.

dCDH (Individual Unit-Time Level): Operates at a much more granular perspective. Instead of reweighting aggregated sub-experiments, they decompose TWFE directly over the **individual treatment effects of each treated observation (at unit-time level, (i, t))**. Under this microscopic perspective, the unit weights $w_{i,t}$ applied to each treated observation depend on residuals from projecting treatment onto unit and time fixed effects:

$$w_{i,t} = \frac{\varepsilon_{i,t}}{\frac{1}{NT} \sum_{i,t:D_{i,t}=1} \varepsilon_{i,t}} \quad (12.114)$$

Since these residuals ($\varepsilon_{i,t}$) can take negative values (especially for units treated for many periods or in periods where most of the sample is already treated), **the unit weights $w_{i,t}$ can be strictly negative**.

Theorem 12.4.5 (dCDH Unit-Level Decomposition). *The TWFE estimator can be rewritten directly as a linear combination of treatment effects from each treated unit-time ($D_{i,t} = 1$), weighted by weights $w_{i,t}$:*

$$\hat{\delta}^{TWFE} = \mathbb{E} \left[\sum_{i,t:D_{i,t}=1} w_{i,t} \cdot ATT_{i,t} \right] \quad (12.115)$$

In staggered designs, **several weights $w_{i,t}$ take strictly negative values**. Unlike a conventional weighted average (where all weights are positive and sum to 1), the presence of negative weights causes TWFE to lose the property of being a coherent averaging operator. If treatment effects are heterogeneous across units or over time, negative weights multiplied by these heterogeneous effects can cause TWFE to collapse.

12.4.6.5 Formal Proof of Dynamic Contamination

Theorem 12.4.6 (Analytical Derivation of Dynamic Bias). *Consider a two-period design $t \in \{Pre, Post\}$ with two groups:*

- **Group k (Late Treated):** *Receives treatment only in Post.*
- **Group U (Already Treated):** *Serves as control for k , but was already treated in both Pre and Post.*

The standard DiD estimator is:

$$\hat{\delta}^{DiD} = \left(\mathbb{E}[Y_k | Post] - \mathbb{E}[Y_k | Pre] \right) - \left(\mathbb{E}[Y_U | Post] - \mathbb{E}[Y_U | Pre] \right) \quad (12.116)$$

Decomposing group k 's temporal variation by adding and subtracting $\mathbb{E}[Y_{k0} | Post]$:

$$\mathbb{E}[Y_{k,Post}] - \mathbb{E}[Y_{k,Pre}] = ATT_k + \Delta Y_{k0} \quad (12.117)$$

where $ATT_k \equiv \mathbb{E}[Y_{k1} | Post] - \mathbb{E}[Y_{k0} | Post]$ and $\Delta Y_{k0} \equiv \mathbb{E}[Y_{k0} | Post] - \mathbb{E}[Y_{k0} | Pre]$.

Decomposing group U 's temporal variation by adding and subtracting counterfactuals $\mathbb{E}[Y_{U0} | Post]$ and $\mathbb{E}[Y_{U0} | Pre]$:

$$\mathbb{E}[Y_{U,Post}] - \mathbb{E}[Y_{U,Pre}] = \underbrace{\left(ATT_U(Post) - ATT_U(Pre) \right)}_{\Delta ATT_U} + \Delta Y_{U0} \quad (12.118)$$

Substituting into DiD and reorganizing:

$$\hat{\delta}^{DiD} = ATT_k + \underbrace{\left(\Delta Y_{k0} - \Delta Y_{U0} \right)}_{\text{Parallel Trends Bias (PT)}} - \Delta ATT_U \quad (12.119)$$

Under parallel trends ($PT = 0$):

$$\boxed{\hat{\delta}^{DiD} = ATT_k - \Delta ATT_U} \quad (12.120)$$

Implications:

- **Constant Effect (Homogeneous):** If $ATT_U(Post) = ATT_U(Pre)$, then $\Delta ATT_U = 0$ and $\hat{\delta}^{DiD} = ATT_k$.

- **Dynamic Effect (Heterogeneous):** If impact grows with exposure time ($ATT_U(Post) > ATT_U(Pre)$), then $\Delta ATT_U > 0$. Equation (12.120) proves the DiD estimator **subtracts this variation from k 's true effect**, acting as downward contamination bias. If dynamic growth in the initial cohort is large enough, contamination ΔATT_U will exceed the true effect (ATT_k), causing artificial sign inversion and leading to completely erroneous scientific conclusions.

12.4.6.6 The OLS Variance Bias: Debunking Common Myths

Remark 12.4.16 (False Claim 1: “TWFE penalizes tail cohorts because they have smaller sample sizes (n_k).”). **Why this is false:** Tail penalization (units treated very early or very late) is **not a sample size (n_k) problem, but a treatment timing problem**. Mechanically, the weight OLS assigns each 2×2 comparison in Bacon’s decomposition is proportional to the **treatment status variance of the cohort**:

$$\text{Var}(D_{it}) = \bar{D}_k(1 - \bar{D}_k) \quad (12.121)$$

where $\bar{D}_k$ is the fraction of total panel time cohort k remains treated.

- For a cohort treated **very early** (left tail), the group spends almost all time treated ($\bar{D}_k \rightarrow 1$). Its treatment variance collapses: $\bar{D}_k(1 - \bar{D}_k) \rightarrow 0$.
- For a cohort treated **very late** (right tail), the group spends almost all time under control ($\bar{D}_k \rightarrow 0$). Its treatment variance also collapses.

Even if the late-treated cohort contains the vast majority of sample individuals (high n_k), OLS applies an **extremely low weight** to it in calculating the unified second-stage coefficient. Penalization occurs due to scarcity of temporal treatment variation within the group, not lack of observations.

Remark 12.4.17 (False Claim 2: “TWFE weights are optimal for causal identification because they prioritize groups with better counterfactuals.”). **Why this is false:** TWFE weights are purely mechanical byproducts of OLS mathematical minimization, without any criterion or concern for causal identification or population representativeness. As demonstrated by Słoczyński (2022), OLS was designed to maximize predictive fit of realized outcomes (Y_{it}), not to impute unobserved counterfactuals ($Y_{it}(0)$).

TWFE gives more weight to cohorts at the **panel midpoint** (where $\bar{D}_k \approx 0.50$, maximizing variance at 0.25) simply because these observations provide greater statistical leverage for the regression slope. Econometrically, prioritizing the “panel middle” at the expense of tails **has no justification in internal or external validity**; it is a purely arbitrary bias of OLS projection linear algebra.

Remark 12.4.18 (False Claim 3: “Extending panel length by adding pure pre-treatment periods is innocuous and only increases precision.”). **Why this is false:** In conventional cross-section linear regressions, adding historically clean observations usually increases precision without altering consistency. In DiD with static TWFE, **extending the panel artificially alters the estimated treatment effect**.

By adding, for example, 10 years of pre-treatment data before any unit is treated:

1. You identically alter the treated time fraction ($\bar{D}_k$) of **all** panel cohorts.
2. This mechanically shifts each group’s treatment variance.

3. Cohorts previously at the variance distribution center (maximum weight) are pushed to the tails (penalized), and vice versa.

As Goodman-Bacon weights are recalculated over the new temporal support, the estimator $\hat{\delta}^{\text{TWFE}}$ undergoes **mechanical weight redistribution**. If treatment effects are heterogeneous across cohorts, this simple addition of baseline periods can alter estimate magnitude and even **cause sign inversion**, even if original impact data remain untouched.

Remark 12.4.19 (False Claim 4: “Tail penalization prevents TWFE from suffering the negative unit weights problem.”). **Why this is false:** Tail penalization (low variance weights in Bacon’s decomposition) **does not eliminate the negative unit weights problem** mapped by de Chaisemartin and D’Haultfoeuille (2020).

In fact, very early-treated cohorts (left tail), even receiving low variance weights in the overall estimator, continue perniciously acting as “control group” for late-entering cohorts (the forbidden comparison). Since these early cohorts have been treated for many periods, their dynamic treatment effects continue propagating over time.

OLS subtracts this growth trajectory from the global estimator, which translates directly into **negative weights applied to individual unit-time treatment effects**. OLS’s geometric tail penalization does not protect the model from obtaining a coefficient with incorrect sign under heterogeneous treatment dynamics.

12.4.7 Modern Robust Estimators for Staggered Adoption

The identification of these pathologies motivated the creation of new estimators based on **imputation** or **semiparametric reweighting** strategies that explicitly avoid using already-treated units as controls.

12.4.7.1 Callaway and Sant’Anna (2021): The Group-Time ATT Framework

The Callaway and Sant’Anna (2021, CS) estimator was developed to fill a critical methodological gap identified in the quasi-experimental literature: the inability of the classical TWFE estimator to robustly handle **heterogeneous treatment effects** in **staggered adoption** scenarios.

Instead of trying to recover a single global average parameter under restrictive homogeneous effect assumptions (as static TWFE does), the CS method adopts a cellular decomposition strategy. It estimates the smallest clinically clean fundamental building blocks—known as **group-time effects** ($\text{ATT}(g, t)$)—and then aggregates them in a weighted and transparent manner to answer the researcher’s questions of interest.

Definition 12.4.6 (CS Data Structure and Notation). • **g (Group or Cohort):**

Identifies the exact period when a given group of units first received treatment (e.g., $g = 1986$).

- **t (Time):** The calendar period when the outcome is evaluated.
- **G_g :** Binary indicator equal to 1 if the unit belongs to the group treated in period g , 0 otherwise.
- **C :** Control group indicator. The researcher can choose whether the control pool consists strictly of **never-treated** units or **not-yet-treated** units up to the period

of interest. For units already treated in the past, C is considered undefined, ensuring they are discarded from the control group.

Definition 12.4.7 (The Four Identifying Assumptions of CS). 1. **Irreversible**

Treatment: Once a unit enters treatment, it remains treated until the end of the panel (treatment cannot be “turned off”).

2. **No Anticipation:** Treatment generates no effects in periods prior to its physical implementation ($ATT(g, t) = 0$ for all $t < g$).
3. **Conditional (or Unconditional) Parallel Trends:** In the absence of treatment, the average temporal evolution of treated units’ counterfactual outcome is identical to the average evolution of control units, conditional on a vector of pre-treatment covariates X :

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, G_g = 1] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, C = 1] \quad (12.122)$$

4. **Weak Common Support:** Ensures that for each treatment group, sufficient overlap exists in the propensity score distribution of control units, allowing adequate matching or reweighting.

Definition 12.4.8 (The Group-Time Propensity Score). When the researcher assumes parallel trends hold only conditionally on a vector of exogenous characteristics X , it is necessary to model the selection mechanism. The propensity score in CS is defined specifically for each treated group g :

$$\hat{p}_g(X) \equiv \Pr(G_g = 1 \mid X, G_g + C = 1) \quad (12.123)$$

Dynamic nature of the score: Since the eligible control group ($C = 1$) changes from cohort to cohort (because units being treated are eliminated from the control pool), **the propensity score is estimated independently for each group g .**

Bad controls: To prevent covariates from being contaminated by the dynamic effects of treatment itself, vector X must be measured strictly in the period immediately prior to that group’s treatment, i.e., at $g - 1$ (the cohort’s baseline endowment).

Theorem 12.4.7 (The IPW Estimator of Callaway and Sant’Anna). *The classical version of the CS estimator incorporates baseline covariates X by applying the following **Inverse Probability Weighting (IPW)** scheme to obtain the effect of group g in period t :*

$$ATT(g, t) = \mathbb{E} \left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{\hat{p}_g(X) \cdot C}{1 - \hat{p}_g(X)}}{\mathbb{E} \left[\frac{\hat{p}_g(X) \cdot C}{1 - \hat{p}_g(X)} \right]} \right) (Y_t - Y_{g-1}) \right] \quad (12.124)$$

Deconstructing the IPW Formula: Equation (12.124) can be thought of as a structure of “four weighted means and three subtractions” applied to temporal variation:

- **Treatment Term (Left):** $\frac{G_g}{\mathbb{E}[G_g]}$ isolates and equally weights observations belonging to treated cohort g .
- **Rewighted Control Term (Right):** $\frac{\frac{\hat{p}_g(X) \cdot C}{1 - \hat{p}_g(X)}}{\mathbb{E} \left[\frac{\hat{p}_g(X) \cdot C}{1 - \hat{p}_g(X)} \right]}$ acts on control units ($C = 1$), assigning higher weights to those with observable characteristics X most similar to the treated group.

- **Temporal Difference Operator:** $(Y_t - Y_{g-1})$ calculates each unit's outcome growth or evolution between the period under analysis (t) and the period immediately prior to the group's treatment ($g-1$), establishing the counterfactual baseline cleanly.

12.4.7.2 The Double Robustness Property (Sant'Anna and Zhao, 2020)

Definition 12.4.9 (The DR Estimator for $ATT(g, t)$). Let $G_g \in \{0, 1\}$ indicate cohort membership, $C \in \{0, 1\}$ indicate eligible control group membership, and $S_g \equiv G_g + C$ restrict analysis to the subpopulation of interest. Define $\Delta Y_t \equiv Y_t - Y_{g-1}$ as temporal outcome variation.

The first-stage models are:

- **Propensity Score Model** ($p_g(X)$): $\mathbb{E}[G_g \mid X, S_g = 1] = p_g^*(X)$.
- **Outcome Regression Model** ($\mu_{g,t}(X)$): $\mathbb{E}[\Delta Y_t \mid X, C = 1] = \mu_{g,t}^*(X) \equiv \mathbb{E}[\Delta Y_t(0) \mid X, C = 1]$.

The doubly robust estimator is:

$$\theta^{\text{DR}}(g, t) = \mathbb{E} \left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_g(X) \cdot C}{1 - p_g(X)}}{\mathbb{E} \left[\frac{p_g(X) \cdot C}{1 - p_g(X)} \right]} \right) (\Delta Y_t - \mu_{g,t}(X)) \mid S_g = 1 \right] \quad (12.125)$$

Theorem 12.4.8 (Double Robustness of Sant'Anna and Zhao (2020)). *The estimator $\theta^{\text{DR}}(g, t)$ consistently identifies the true $ATT(g, t)$ if either:*

- *The propensity score model is correctly specified: $p_g(X) = p_g^*(X)$, **or***
- *The outcome regression model is correctly specified: $\mu_{g,t}(X) = \mu_{g,t}^*(X)$.*

Proof. Case A: Propensity Score Correct, Outcome Regression Incorrect.
Premises: $p(X) = p_g^*(X)$ and $\mu_{g,t}(X) \neq \mu_{g,t}^*(X)$.

Applying LIE conditioning internally on X :

$$\mathbb{E}[\theta^{\text{DR}}(g, t) \mid S_g = 1] = \mathbb{E} \left[\mathbb{E} \left[\frac{G_g(\Delta Y_t - \mu_{g,t}(X))}{\mathbb{E}[G_g]} - \frac{p_g^*(X)C(\Delta Y_t - \mu_{g,t}(X))}{(1 - p_g^*(X))\mathbb{E}[G_g]} \mid X, S_g = 1 \right] \mid S_g = 1 \right] \quad (12.126)$$

Analyzing terms containing the incorrect function $\mu_{g,t}(X)$:

$$\begin{aligned} & \mathbb{E} \left[-\frac{G_g \mu_{g,t}(X)}{\mathbb{E}[G_g]} + \frac{p_g^*(X)C \mu_{g,t}(X)}{(1 - p_g^*(X))\mathbb{E}[G_g]} \mid X, S_g = 1 \right] \\ &= \frac{\mu_{g,t}(X)}{\mathbb{E}[G_g]} \left(-\mathbb{E}[G_g \mid X, S_g = 1] + \frac{p_g^*(X)}{1 - p_g^*(X)} \mathbb{E}[C \mid X, S_g = 1] \right) \\ &= \frac{\mu_{g,t}(X)}{\mathbb{E}[G_g]} \left(-p_g^*(X) + \frac{p_g^*(X)}{1 - p_g^*(X)} (1 - p_g^*(X)) \right) = \mathbf{0} \end{aligned} \quad (12.127)$$

The term containing the incorrect regression function collapses to **zero** almost surely.

Remaining terms associated with outcome ΔY_t :

$$\begin{aligned} & \mathbb{E} \left[\frac{G_g \Delta Y_t}{\mathbb{E}[G_g]} - \frac{p_g^*(X) C \Delta Y_t}{(1 - p_g^*(X)) \mathbb{E}[G_g]} \mid X, S_g = 1 \right] \\ &= \frac{p_g^*(X)}{\mathbb{E}[G_g]} \left(\mathbb{E}[\Delta Y_t(1) \mid G_g = 1, X] - \mathbb{E}[\Delta Y_t(0) \mid C = 1, X] \right) \end{aligned} \quad (12.128)$$

Invoking Conditional Parallel Trends: $\mathbb{E}[\Delta Y_t(0) \mid C = 1, X] = \mathbb{E}[\Delta Y_t(0) \mid G_g = 1, X]$:

$$= \frac{p_g^*(X)}{\mathbb{E}[G_g]} \text{ATT}(g, t \mid X) \quad (12.129)$$

Integrating over X distribution via external LIE and applying Bayes' Theorem:

$$\mathbb{E}[\theta^{\text{DR}}(g, t) \mid S_g = 1] = \int \text{ATT}(g, t \mid X) f(X \mid G_g = 1) dX = \mathbf{ATT}(\mathbf{g}, \mathbf{t}) \quad (12.130)$$

Case B: Outcome Regression Correct, Propensity Score Incorrect. Premises: $\mu_{g,t}(X) = \mu_{g,t}^*(X)$ and $p_g(X) \neq p_g^*(X)$.

Analyzing the second term carrying the incorrect propensity score:

$$\begin{aligned} & \mathbb{E} \left[\frac{p_g(X) C (\Delta Y_t - \mu_{g,t}^*(X))}{(1 - p_g(X)) \mathbb{E}[G_g]} \mid X, S_g = 1 \right] \\ &= \frac{p_g(X)}{(1 - p_g(X)) \mathbb{E}[G_g]} \mathbb{E}[C \mid X, S_g = 1] \cdot \mathbb{E}[\Delta Y_t(0) - \mu_{g,t}^*(X) \mid C = 1, X] \\ &= \frac{p_g(X)}{(1 - p_g(X)) \mathbb{E}[G_g]} (1 - p_g^*(X)) \cdot \underbrace{(\mu_{g,t}^*(X) - \mu_{g,t}^*(X))}_{=0} = \mathbf{0} \end{aligned} \quad (12.131)$$

The complete term containing the incorrect propensity specification collapses to **zero**.

Evaluating the remaining first term:

$$\begin{aligned} \mathbb{E} \left[\frac{G_g (\Delta Y_t - \mu_{g,t}^*(X))}{\mathbb{E}[G_g]} \mid X, S_g = 1 \right] &= \frac{p_g^*(X) \mathbb{E}[\Delta Y_t(1) \mid G_g = 1, X] - p_g^*(X) \mathbb{E}[\Delta Y_t(0) \mid C = 1, X]}{\mathbb{E}[G_g]} \\ &= \frac{p_g^*(X)}{\mathbb{E}[G_g]} \text{ATT}(g, t \mid X) \end{aligned} \quad (12.132)$$

Integrating over X support via LIE:

$$\mathbb{E}[\theta^{\text{DR}}(g, t) \mid S_g = 1] = \mathbf{ATT}(\mathbf{g}, \mathbf{t}) \quad (12.133)$$

□

12.4.7.3 Aggregation Schemes: CS vs. TWFE Weights

Remark 12.4.20 (The Three Classical Aggregation Schemes). Once the complete matrix of fundamental building blocks $\text{ATT}(g, t)$ is estimated for all cohorts and viable post-treatment periods, CS (2021) offers three main aggregation schemes:

A. Simple ATT: Calculates the simple average of all individual $\text{ATT}(g, t)$ across cohorts g in each post-treatment period $t \geq g$, weighted by the proportion of treated units in each group:

$$\theta_{\text{simple}} = \sum_g \sum_{t \geq g} w(g, t) \cdot \text{ATT}(g, t) \quad (12.134)$$

where weight $w(g, t)$ is proportional to the fraction of observations belonging to group g in period t .

B. Group/Cohort ATT: Aggregates effects at the entry cohort level, revealing whether units receiving treatment early respond differently from those treated late:

$$\text{ATT}(g) = \frac{1}{\mathcal{T}_g} \sum_{t \geq g} \text{ATT}(g, t), \quad \theta_{\text{group}} = \sum_g p_g \cdot \text{ATT}(g) \quad (12.135)$$

where $\mathcal{T}_g$ is the number of viable post-treatment periods for group g and $p_g = N_g / \sum_{g'} N_{g'}$.

C. Event Study (Dynamic Effects): Organizes $\text{ATT}(g, t)$ according to **relative time of exposure** to policy: $e = t - g$:

$$\text{ATT}(e) = \sum_{g: (g+e) \leq T} w_e(g) \cdot \text{ATT}(g, g+e) \quad (12.136)$$

This aggregation definitively solves the problem of TWFE dynamic estimates, which suffer from “tail contamination” and weight spreading across adjacent periods.

Remark 12.4.21 (The Crucial Choice of Weights: CS vs. TWFE). The contrast between Static TWFE Weights (OLS) and CS (2021) Estimator Weights can be summarized as follows. In terms of **Weight Origin**, TWFE weights are **Mechanical and Implicit**, determined strictly by OLS projection algebra (FWL Theorem); CS weights are **Explicit and Postulated**, defined directly by the researcher based on the estimand of interest. In terms of **Weighting Criterion**, TWFE performs **Variance Maximization**, assigning disproportionately higher weights to groups whose treatment occurs at the **panel midpoint** (where D_{it} dummy variance is highest); CS performs **Population Representativeness**, with weights based strictly on the relative sample size (fraction of units) of each treated cohort. In terms of **Weight Sign**, TWFE is **Potentially Negative**, assigning negative weights to several unit-times and generating false sign inversions due to heterogeneity and dynamic effects; CS is **Strictly Non-Negative**, with weights structurally restricted to be positive or null, guaranteeing a coherent averaging operator. Regarding **Control Treatment**, TWFE uses already-treated cohorts as control (forbidden comparisons), generating contamination and bias, while CS excludes already-treated groups from the control pool, keeping control weights restricted to untreated or not-yet-treated units.

Epistemological Implication: In classical OLS, there is no guarantee the estimated coefficient represents impact for the “average individual” in the sample, since OLS prioritizes statistical variance endowment. In CS, if you want to estimate the effect on the average Brazilian municipality, aggregation weights exactly reproduce the density of treated municipalities in the sample.

12.4.7.4 The Treatment Irreversibility Restriction

Remark 12.4.22 (Why CS Requires Irreversibility). The **treatment irreversibility** restriction is one of the most important—and simultaneously most limiting—simplifying

assumptions of modern DiD models under staggered adoption, such as CS (2021).

Mathematically, if the treatment indicator is $D_{it} \in \{0, 1\}$:

$$D_{it} = 1 \implies D_{is} = 1 \quad \forall s > t \quad (12.137)$$

This configures what the microeconometrics literature calls a **stopping time model**. Treatment is non-recurrent: units are not allowed to enter, exit, and reactivate the policy dynamically over time.

A. Unambiguous Cohort Definition: The CS estimator is based on calculating group-time effects $ATT(g, t)$, where g is the cohort defined by the **exact period when treatment began**. Under irreversibility, group g is a mutable but stable partition: we know exactly when treatment started and that it will remain active.

If units could turn treatment on and off, the definition of “group” would collapse. A unit treated at $t = 3$, turned off at $t = 4$, and reactivated at $t = 6$ would belong to which cohort? Group taxonomy would cease to be one-dimensional (based only on entry time g) and would depend on **complex treatment trajectories** (*treatment history sequences*).

B. Common Support Collapse: If we allow treatment to switch on and off, the number of possible treatment trajectories in a τ -period panel becomes 2^τ . With so many possible sequences, the **common support** (overlap) hypothesis fails drastically. It would be virtually impossible to find control units in the donor pool sharing exactly the same history of observable endowments X_i and prior treatment history, invalidating any propensity score matching or reweighting.

Remark 12.4.23 (The dCDH Solution and the “No Carryover” Cost). To fill this gap and allow estimating causal effects in environments where units freely enter and exit treatment, de Chaisemartin and D’Haultfoeulle (dCDH) proposed flexible estimators (such as DID_M).

However, to relinquish the irreversibility restriction, the dCDH model requires a heavy theoretical counterpart: the **no carryover assumption**.

What No Carryover Assumes: Postulates that the unit’s current outcome depends **exclusively on its current treatment status** (whether on or off in period t), being completely independent of past treatment history. This allows interpreting the causal effect as an instantaneous change.

The Econometric Trade-off: This hypothesis is extremely restrictive and unrealistic for most economic applications. If a school reinforcement policy or technological innovation subsidy is turned off, it is natural to expect treatment effects have accumulated in the form of human capital or learning (*carryover* or *lagged effects*). In the presence of cumulative effects, the no carryover premise is violated, and the reversibility estimator loses its causal validity.

Researcher’s Choice:

- If treatment is **irreversible** (or can be reasonably approximated as such), **Callaway and Sant’Anna (2021)** is the gold standard, as it freely tolerates heterogeneity and dynamic effects accumulated over time.
- If treatment **dynamically switches on and off**, **de Chaisemartin and D’Haultfoeulle (2020)** estimators are the only ones capable of calculating impact,

but at the cost of assuming the policy's past leaves no traces on individuals' present behavior (no carryover hypothesis).

12.4.8 Advanced Extensions

12.4.8.1 Difference-in-Difference-in-Differences (DDD)

When group-specific shocks violate parallel trends, introduce a second control group. For example, if a state policy targets elderly ($A_i = 1$) in treated states ($G_i = 1$):

$$Y_{igt} = \theta(G_i \times A_g \times T_t) + \text{lower-order terms} + \varepsilon_{igt} \quad (12.138)$$

DDD requires *parallel differences in differences*: the gap between elderly and young evolves identically in treated and control states.

12.4.8.2 Changes-in-Changes (Athey and Imbens, 2006)

The CiC model is scale-invariant and identifies distributional effects. The counterfactual is:

$$Y_{it}(0) = h(U_i, t) \quad (12.139)$$

where U_i is unobserved heterogeneity and $h(\cdot, t)$ is strictly increasing. This allows identification of quantile treatment effects without distributional assumptions.

12.4.8.3 Synthetic DID (Arkhangelsky et al., 2021)

SDID combines DID with synthetic control by estimating unit weights w_i and time weights λ_t :

$$\min_{\alpha, \beta, \theta} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \alpha_i - \beta_t - \theta D_{it})^2 w_i^{\text{SDID}} \lambda_t^{\text{SDID}} \quad (12.140)$$

Unit weights align pre-treatment levels and trends; time weights emphasize periods most representative of the post-treatment trajectory.

Remark 12.4.24 (Summary of Key Diagnostics). • **Pre-trends test:** All $\beta_k = 0$ for $k < 0$ in event studies (using Long Differences design).

- **Common support:** Normalized differences $|\Delta_{X_k}| < 0.25$; histogram overlap of propensity scores.
- **Inference:** Cluster-robust standard errors at the level of treatment assignment; wild bootstrap if $\mathcal{G} < 50$.
- **No anticipation:** Verify β_{-2} is not significantly different from zero; if violated, redefine treatment onset.
- **Negative weights diagnostic:** Under staggered adoption, report the fraction of observations receiving negative weights (dCDH algorithm) before interpreting static TWFE estimates.
- **Estimator choice:** If treatment is irreversible and effects are dynamic, prefer CS (2021); if treatment switches on/off and carryover effects are negligible, prefer dCDH (2020).

12.5 Regression Discontinuity Design (RDD)

The Regression Discontinuity Design (RDD) is widely regarded as the quasi-experimental method closest to a randomized controlled trial. Unlike selection-on-observables approaches that require global ignorability, RDD exploits an institutional eligibility rule based on a continuous *running variable* to identify causal effects through a *local* comparison at a known cutoff.

12.5.1 Sharp RDD: Setup and Identification

Definition 12.5.1 (Assignment Rule and Running Variable). Let $Z_i \in \mathbb{R}$ denote the **running variable** (also called forcing variable or assignment score)—a continuous observable that governs treatment eligibility through a known institutional threshold $c \in \mathbb{R}$. In the **Sharp RDD**, treatment assignment is a deterministic function of Z_i :

$$D_i = \mathbb{I}\{Z_i \geq c\}. \quad (12.141)$$

The population is partitioned into:

- **Treated group** ($D_i = 1$): All units with $Z_i \geq c$, obligatorily receiving treatment.
- **Control group** ($D_i = 0$): All units with $Z_i < c$, strictly ineligible.

Remark 12.5.1 (The Dual Role of the Running Variable). Structurally, Z_i plays a dual causal role:

1. **Smooth direct effect:** Z_i exerts a continuous, smooth influence on the outcome Y_i through underlying ability, socioeconomic status, or other confounding channels.
2. **Discontinuous indirect effect:** Z_i induces an abrupt change in treatment status D_i exactly at c , creating a vertical jump in the observed outcome.

The identification strategy consists in analytically separating the smooth trajectory of Z_i from the discrete jump provoked by treatment at the cutoff.

Remark 12.5.2 (The Pioneer Example: Thistlethwaite and Campbell (1960)). The first formal application of RDD in the scientific literature evaluated the impact of receiving a prestigious *National Merit Scholarship* on high-school students' career plans and graduate school intentions. The scholarship was assigned strictly based on PSAT exam scores (the running variable Z_i) relative to a fixed cutoff c . Students scoring above the threshold received the award ($D_i = 1$); those below were disqualified ($D_i = 0$).

Although students with higher academic performance naturally exhibit greater propensity to pursue graduate studies, this continuous relationship can be controlled through smooth regression adjustment. Since students located immediately above and below the eligibility threshold differ only by infinitesimal score fractions—equivalent to pure random fluctuations or “luck” on exam day—they constitute perfect comparison groups. Any vertical discontinuity (jump) observed in post-treatment success rates at c is cleanly interpreted as the causal effect of treatment for individuals at the margin.

12.5.2 The Continuity Hypothesis

Definition 12.5.2 (Conditional Expectation Functions). Let $Y_i(1)$ and $Y_i(0)$ denote potential outcomes under treatment and control. Define the conditional expectation

functions of potential outcomes given the running variable:

$$\begin{aligned} m_1(z) &\equiv \mathbb{E}[Y_i(1) \mid Z_i = z], \\ m_0(z) &\equiv \mathbb{E}[Y_i(0) \mid Z_i = z]. \end{aligned} \quad (12.142)$$

Definition 12.5.3 (Continuity Hypothesis (H1)). The **continuity hypothesis** postulates that the conditional expectation functions $m_1(z)$ and $m_0(z)$ are **strictly continuous** in z at the cutoff point $z = c$:

$$\lim_{z \downarrow c} m_1(z) = m_1(c) \quad \text{and} \quad \lim_{z \uparrow c} m_0(z) = m_0(c). \quad (12.143)$$

Remark 12.5.3 (Economic Intuition and Identification Role). The continuity hypothesis guarantees that, in the absence of treatment, individuals located immediately above the cutoff would have, on average, the same behavior as those immediately below.

In traditional selection-on-observables designs, inference requires the **overlap hypothesis**: for any covariate vector X , there exists a strictly positive probability of observing both treated and control units ($0 < \Pr(D_i = 1 \mid X_i) < 1$). In Sharp RDD, the assignment rule is deterministic ($D_i = \mathbb{I}\{Z_i \geq c\}$), which causes an **absolute violation of common support**:

- For all $z \geq c$: $\Pr(D_i = 1 \mid Z_i = z) = 1$.
- For all $z < c$: $\Pr(D_i = 1 \mid Z_i = z) = 0$.

Consequently, we never observe the control outcome $Y_i(0)$ for values to the right of c , nor the treatment outcome $Y_i(1)$ for values to the left.

Continuity acts as the **mathematical substitute for common support**. Under the premise that curves $m_1(z)$ and $m_0(z)$ evolve smoothly and without abrupt jumps around c , it is statistically legitimate to **extrapolate the limiting behavior** of groups from each side to obtain the unobserved counterfactual exactly at the cutoff. Thus, the one-sided limits of the realized outcome Y_i reconstruct the potential outcome expectations at the threshold:

$$\begin{aligned} Y^+ &\equiv \lim_{\varepsilon \downarrow 0} \mathbb{E}[Y_i \mid Z_i = c + \varepsilon] = \mathbb{E}[Y_i(1) \mid Z_i = c], \\ Y^- &\equiv \lim_{\varepsilon \downarrow 0} \mathbb{E}[Y_i \mid Z_i = c - \varepsilon] = \mathbb{E}[Y_i(0) \mid Z_i = c]. \end{aligned} \quad (12.144)$$

The vertical difference between these two limits identifies the **Local Average Treatment Effect (LATE)** in the neighborhood of c :

$$\tau_{\text{RDD}} = Y^+ - Y^- = \mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = c]. \quad (12.145)$$

Remark 12.5.4 (The Danger of Counterfactual Discontinuity). If the continuity hypothesis is violated, Sharp RDD collapses, as it becomes impossible to distinguish the true policy effect from other confounding factors.

Mathematically, if we allow the counterfactual control function $m_0(z)$ to suffer a discrete jump of magnitude Δ_0 at $z = c$, the observed discontinuity in the realized outcome becomes:

$$Y^+ - Y^- = \tau_{\text{RDD}} + \Delta_0. \quad (12.146)$$

The RDD estimator inherits a **counterfactual discontinuity bias** (Δ_0), invalidating causal identification. In applied practice, this violation occurs primarily through two channels:

A. Coexistence of Multiple Rules (Competing Policies). If other public policies, tax criteria, or institutional benefits use the **same cutoff** c on the running variable, units suffer simultaneous jumps in multiple potential outcomes. For example, if eligibility for an educational subsidy and allocation of funds for school infrastructure both occur under the same poverty threshold of 25%, the observed jump in student scores reflects the joint impact of both interventions, violating the continuity of the isolated control potential outcome.

B. Systematic Manipulation (Perfect Self-Selection). Continuity is violated if agents possess **perfect control over the running variable** Z_i and use this control to intentionally position themselves on the more advantageous side of threshold c . If only the most motivated, wealthy, or well-informed individuals manage to manipulate their scores to surpass the cutoff, there will be an abrupt change in the composition of observable and unobservable characteristics of samples around c . This generates an artificial jump in $m_0(z)$ at point c , breaking the local quasi-randomization premise.

12.5.3 Local Ignorability and Lee (2008)

Remark 12.5.5 (From Global to Local Ignorability). In classical selection-on-observables methods (linear regression, matching), one assumes the **Global Ignorability Hypothesis (Unconfoundedness)**: conditioning on a covariate vector X_i , treatment assignment D_i is independent of potential outcomes:

$$(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i \mid X_i. \quad (12.147)$$

In **Sharp RDD**, since treatment is a deterministic function of Z_i (i.e., $D_i = \mathbb{I}\{Z_i \geq c\}$), the ignorability hypothesis conditional on Z_i is **trivially satisfied**: there is no treatment variation for the same value of Z_i . For any value z , treatment status is perfectly known (1 or 0).

However, Sharp RDD absolutely violates the overlap hypothesis, as it is impossible to observe treated and control units with the same value of Z_i . Here enters **local ignorability**.

Instead of requiring overlap over the entire support, we restrict analysis to an **infinitesimal neighborhood** around the cutoff c . The local ignorability hypothesis postulates that, around $Z_i = c$, treatment assignment is statistically independent of potential outcomes:

$$(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i \mid Z_i \in [c - \varepsilon, c + \varepsilon] \quad \text{as } \varepsilon \rightarrow 0. \quad (12.148)$$

Under this premise, individuals whose running variable scores place them immediately above or below the threshold constitute equivalent comparison groups. The treatment status of an individual at the margin of the cutoff is determined in a **quasi-random** manner, mimicking a real experiment.

Theorem 12.5.1 (Lee (2008): Imperfect Sorting and Local Randomization). *The theoretical justification for local ignorability was formalized by David Lee (2008) and is based on the **impossibility of perfect manipulation or sorting** around the cutoff.*

*The premise is intuitive: economic agents may have incentives and even capacity to influence or manipulate their running variable Z_i (e.g., students who study harder to pass an exam, or municipalities that alter administrative data to receive funds). However, causal identification of RDD remains perfectly valid as long as **this manipulation is imperfect**.*

*Manipulation is considered imperfect when, even exerting maximum possible effort, the agent **lacks absolute and millimetric control over the final realized value of their score Z_i** . Mathematically, each individual's observed score is composed of a deterministic effort/choice component plus an exogenous and unpredictable random shock (measurement noise, luck on exam day, punctual weather variation, etc.):*

$$Z_i = g(W_i) + e_i, \quad (12.149)$$

where W_i represents observed and unobserved characteristics determining effort, and e_i is the stochastic error.

Since agents do not control the stochastic error e_i , the probability distribution of Z_i conditional on their characteristics is continuous. Consequently, for a student who scored $c + \varepsilon$ (approved by one point) and another who scored $c - \varepsilon$ (failed by one point), treatment assignment in the immediate neighborhood of c is governed strictly by chance.

Remark 12.5.6 (Implications for Internal and External Validity). A. Maximum Internal Validity (Balance of Unobservables). The great advantage of RDD over other selection-on-observables designs (classical regressions, matching) is its impact on **unobservable characteristics** (such as motivation, resilience, or intrinsic ability).

In an RCT, randomization balances both observed and unobserved variables between treatment and control groups. In Sharp RDD, local ignorability guarantees exactly the same property asymptotically in the neighborhood of the cutoff. Since treatment assignment at the margin is random, **the distribution of all pre-treatment characteristics (observable and unobservable) is perfectly continuous and balanced at point c** . There are no confounding variables systematically jumping at the cutoff.

B. The External Validity Trade-off. The mathematical price paid for this internal robustness is the loss of generalizability (external validity):

- Sharp RDD identifies the average treatment effect (ATE) **exclusively for the subpopulation located exactly at the cutoff** ($Z_i = c$).
- Attempting to extrapolate the estimated effect to individuals far from the cutoff constitutes a risky functional specification exercise. RDD estimates the marginal impact of policy at the threshold, which may be irrelevant for individuals in the tails of the running variable distribution.

12.5.4 RDD as the Limit Form of Matching

Remark 12.5.7 (The Overlap Paradox). In classical conditional matching models on observables, identification of ATE or ATT requires two fundamental hypotheses: ignorability (unconfoundedness) and the **overlap hypothesis (common support)**.

The overlap hypothesis postulates that the probability of receiving treatment (the propensity score), conditional on covariates Z_i , must be strictly contained in the open interval

between 0 and 1:

$$0 < \Pr(D_i = 1 \mid Z_i = z) < 1. \quad (12.150)$$

This premise guarantees that, for any observed value of the characteristic vector Z_i , there exist both treated and control units in the sample, enabling direct comparison (matching) between individuals with the same covariate values.

In Sharp RDD, the overlap hypothesis is absolutely violated by design. Since the treatment allocation rule is a deterministic and discontinuous function of the running variable ($D_i = \mathbb{I}\{Z_i \geq c\}$), the propensity score conditional on Z_i assumes only polar values:

$$\begin{aligned} \Pr(D_i = 1 \mid Z_i = z) &= 1 \quad \forall z \geq c, \\ \Pr(D_i = 1 \mid Z_i = z) &= 0 \quad \forall z < c. \end{aligned} \quad (12.151)$$

Consequently, **there exists no value of Z_i at which it is possible to simultaneously observe treated and control units.** Treated units are all concentrated to the right of cutoff c and controls to the left. From the perspective of traditional global matching, causal identification would be completely impossible due to the absence of common support.

Remark 12.5.8 (Mathematical Resolution: The Limit Around a Single Point). Sharp RDD resolves this common support collapse by substituting the requirement of global overlap with the **hypothesis of continuity of conditional expectation functions of potential outcomes** at point c .

Instead of trying to find perfect controls for treated units across the entire support of the running variable, we restrict analysis to an **infinitesimal neighborhood** around the cutoff.

Let $\varepsilon > 0$ be a distance parameter. We can analyze the observed expectations of outcome Y_i for individuals located just above and just below cutoff c :

$$\begin{aligned} Y^+ &\equiv \lim_{\varepsilon \downarrow 0} \mathbb{E}[Y_i \mid Z_i = c + \varepsilon], \\ Y^- &\equiv \lim_{\varepsilon \downarrow 0} \mathbb{E}[Y_i \mid Z_i = c - \varepsilon]. \end{aligned} \quad (12.152)$$

Under the hypothesis that counterfactual outcome functions $\mathbb{E}[Y_i(1) \mid Z_i = z]$ and $\mathbb{E}[Y_i(0) \mid Z_i = z]$ extend smoothly and continuously on each side of the cutoff, the one-sided limits above exactly reconstruct the potential outcome expectations at the cutoff:

$$Y^+ = \mathbb{E}[Y_i(1) \mid Z_i = c] \quad \text{and} \quad Y^- = \mathbb{E}[Y_i(0) \mid Z_i = c]. \quad (12.153)$$

This formulation reveals that Sharp RDD operates as the **ideal limit form of matching that occurs around a single point** ($Z_i = c$). As the observation window around c approaches zero (i.e., $\varepsilon \rightarrow 0$), the difference in the running variable between treated individuals at $c + \varepsilon$ and controls at $c - \varepsilon$ becomes mathematically negligible. At the margin, observable and unobservable characteristics of individuals on both sides are perfectly balanced (perfect local common support) due to local ignorability.

Remark 12.5.9 (The Methodological Cost: The Validity Trade-off). Interpreting Sharp RDD as the limit form of matching evidences the fundamental validity trade-off governing this research design:

- **Maximum Internal Validity (Advantage):** Unlike global matching, which frequently suffers from dimensionality bias when we add many variables to rescue

ignorability, Sharp RDD achieves local consistency without needing to model a high-dimensional regression function. In the immediate neighborhood of c , treatment status simulates a purely random allocation, functioning as a perfect quasi-experiment.

- **Minimum External Validity (Disadvantage):** In conventional matching with broad common support, we can estimate the treatment effect for a representative portion of the population. In Sharp RDD, since matching occurs strictly at the limit boundary of a single point, the estimated effect ($\tau = Y^+ - Y^-$) identifies the causal impact **exclusively for the marginal subpopulation located exactly at cutoff c** . Any generalization of this impact to units distant from the threshold constitutes arbitrary functional extrapolation.

12.5.5 Empirical Validation of Continuity

Remark 12.5.10 (Indirect Diagnostic Tests). Although continuity of potential outcomes at the cutoff is a **directly untestable hypothesis** (since counterfactuals are unobservable), applied econometrics has developed indirect robustness diagnostic tests to evaluate its plausibility:

1. Baseline Covariate Continuity Test (Balance Tests). Estimate the RDD model substituting the outcome variable Y_i with baseline covariates (X_i) that were not affected by treatment. If the design is valid, one expects **no statistically significant jumps** in these characteristics at the cutoff ($\mathbb{E}[X | Z_i = c^+] = \mathbb{E}[X | Z_i = c^-]$), confirming that groups are locally balanced.

2. McCrary Density Test (2008). Evaluate the continuity of the probability density function of Z_i around c . A disproportionate accumulation or “step” in density just above (or below) the cutoff constitutes strong evidence of active manipulation and breaks the premise of continuity of potential outcomes.

12.5.6 Fuzzy RDD and Equivalence with Instrumental Variables

The equivalence between the **Fuzzy Regression Discontinuity Design (Fuzzy RDD)** and the classical **Instrumental Variables (IV) model of Imbens and Angrist (1994)** represents one of the most important milestones in applied microeconomics, formalized by Hahn, Todd, and Van der Klaauw (2001).

Under this perspective, Fuzzy discontinuity ceases to be viewed merely as an adjustment of smoothed curves and begins to be treated as an **incentive experiment with partial compliance**, where the eligibility threshold acts as a local exogenous instrument.

12.5.6.1 The Causal Analogy: Cutoff as a Wald Instrument

Definition 12.5.4 (Fuzzy Assignment Rule). In the Fuzzy RDD, the realized treatment status ($D_i \in \{0, 1\}$) is endogenous because individuals near the cutoff may exercise active selection over participation. The institutional eligibility indicator, defined by:

$$T_i = \mathbb{I}\{Z_i \geq c\}, \quad (12.154)$$

functions analytically as a **binary exogenous instrument**.

Since instrument T_i is determined strictly by the running variable Z_i relative to cutoff c , in the immediate neighborhood of c the allocation of T_i is governed by chance (local ignorability), satisfying the exogenous independence condition required of a good instrument.

12.5.6.2 Local Compliance Structure

Remark 12.5.11 (Three Latent Compliance Strata). Following the Imbens and Angrist (1994) framework based on potential outcomes and treatments, define $D_i(1)$ as the treatment status if the individual is allocated above the cutoff ($Z_i \geq c$) and $D_i(0)$ if allocated below ($Z_i < c$). At the margin of discontinuity ($Z_i \approx c$), the population is divided into three latent compliance strata:

1. **Local Compliers** (π_{co}): Individuals who strictly comply with the cutoff guideline ($D_i(1) = 1$ and $D_i(0) = 0$). They **only receive treatment if they cross the eligibility threshold**.
2. **Local Always-Takers** (π_{at}): Individuals who obtain treatment even below the cutoff ($D_i(1) = 1$ and $D_i(0) = 1$), bypassing initial eligibility.
3. **Local Never-Takers** (π_{nt}): Individuals who refuse or lack access to treatment even above the cutoff ($D_i(1) = 0$ and $D_i(0) = 0$).

Under the hypothesis of **local monotonicity**, one assumes the non-existence of **Local Defiers** ($D_i(1) = 0$ and $D_i(0) = 1$), i.e., no one does the exact opposite of the marginal cutoff incentive.

12.5.6.3 Identifying Hypotheses of LATE in Fuzzy RDD

Definition 12.5.5 (Local IV Hypotheses at c). For the discontinuity to identify the **Local Average Treatment Effect (LATE)** for local compliers, the classical IV hypotheses must be satisfied locally at c :

- **Local Relevance (Discontinuous First Stage)**: The probability of realized treatment must give an abrupt vertical jump at the cutoff:

$$\lim_{\varepsilon \downarrow 0} \Pr(D_i = 1 \mid Z_i = c + \varepsilon) \neq \lim_{\varepsilon \downarrow 0} \Pr(D_i = 1 \mid Z_i = c - \varepsilon). \quad (12.155)$$

- **Local Independence and Exclusion (Exclusion Restriction)**: The cutoff T_i does not directly affect outcome Y_i , except through the channel of effective treatment D_i . This is mathematically guaranteed by the **continuity of potential outcome expectations** $\mathbb{E}[Y_i(0) \mid Z_i]$ and $\mathbb{E}[Y_i(1) \mid Z_i]$ around point c . Without counterfactual jumps, the exclusion restriction is preserved locally.
- **Local Monotonicity**: Excludes the presence of local defiers in the neighborhood of c , ensuring that crossing the eligibility threshold does not decrease the probability of treatment for any individual.

12.5.6.4 The Local Wald Estimator

Theorem 12.5.2 (Fuzzy RDD Identification (Hahn, Todd, and Van der Klaauw, 2001)). *When the local IV hypotheses are satisfied, the Fuzzy RDD estimator reduces analytically*

to the **local Wald ratio**:

$$\tau_{FRD} = \frac{Y^+ - Y^-}{P^+ - P^-} = \frac{\lim_{z \downarrow c} \mathbb{E}[Y_i | Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i | Z_i = z]}{\lim_{z \downarrow c} \mathbb{E}[D_i | Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[D_i | Z_i = z]}. \quad (12.156)$$

According to the LATE Theorem, this estimator identifies the causal effect of the policy **exclusively for local compliers located exactly at the boundary**:

$$\tau_{FRD} = \mathbb{E}[Y_i(1) - Y_i(0) | D_i(1) > D_i(0), Z_i = c]. \quad (12.157)$$

This parameter is considered “local twice”:

1. **Local in the running variable support:** Refers only to individuals whose scores are restricted to the immediate neighborhood of c .
2. **Local in compliance:** Identifies the effect only for the subpopulation that actively reacts to the cutoff incentive. It provides no information about the average gain of always-takers or never-takers.

12.5.6.5 Local Monotonicity and Exclusion of Defiers

Definition 12.5.6 (Local Monotonicity Hypothesis). In the potential outcomes framework adapted to local instrumental variables, the **Local Monotonicity Hypothesis** postulates that, for all units i whose running variable Z_i is located in an infinitesimal neighborhood around cutoff c , the instrument affects treatment status **strictly in the same direction**:

$$\lim_{\varepsilon \rightarrow 0} D_i(c + \varepsilon) \geq D_i(c - \varepsilon) \quad \forall i \quad \text{or, equivalently,} \quad D_i(1) \geq D_i(0) \quad \forall i. \quad (12.158)$$

This premise guarantees that crossing the eligibility threshold from left to right can induce individuals to initiate treatment (or maintain it), but **will never cause any unit to interrupt or refuse treatment** as a direct consequence of having become eligible.

Remark 12.5.12 (Compliance Structure and Defiers Exclusion). Under the monotonicity hypothesis, the population located at the boundary around c is cleanly segmented into three latent compliance subgroups:

- **Local Compliers** (π_{co}): Units that strictly respond to the cutoff incentive, so that $D_i(1) = 1$ and $D_i(0) = 0$. These are individuals who only take treatment if they cross the threshold.
- **Local Always-Takers** (π_{at}): Units that take treatment regardless of which side of the cutoff they are on, exhibiting $D_i(1) = 1$ and $D_i(0) = 1$.
- **Local Never-Takers** (π_{nt}): Units that refuse or do not access treatment under any circumstances, maintaining $D_i(1) = 0$ and $D_i(0) = 0$.

The fourth theoretically possible group is that of **Local Defiers**, characterized by doing the exact opposite of the institutional rule: $D_i(1) = 0$ and $D_i(0) = 1$ (they would participate if below the cutoff, but refuse if they cross the threshold).

The monotonicity hypothesis **eliminates by design the existence of defiers** in the neighborhood of c , establishing that their probability of occurrence in the marginal population is zero:

$$\Pr(D_i(1) < D_i(0) | Z_i = c) = 0. \quad (12.159)$$

In public policy evaluation, the elimination of defiers is highly plausible. In programs such as government subsidies or school scholarships based on cutoff scores, the cutoff functions purely as a facilitator of access or cost reduction, making it implausible that granting a benefit acts as a perverse disincentive leading an agent to abandon the program.

Theorem 12.5.3 (Mathematical Destructiveness of Monotonicity Violation). *Without the monotonicity hypothesis, the causal interpretation of the local Wald ratio collapses severely. To demonstrate this algebraic pathology, we analyze the mathematical decomposition of the two limiting discontinuities that compose the Fuzzy RDD estimator.*

A. The First Stage (Denominator). *The jump in treatment probability at the cutoff ($P^+ - P^-$) is the denominator of the estimator. Expanding this difference in terms of latent compliance subgroups:*

$$\begin{aligned} \Pr(D_i = 1 \mid Z_i = c^+) &= \pi_{co} + \pi_{at} && \text{(at the right limit, compliers and always-takers are treated),} \\ \Pr(D_i = 1 \mid Z_i = c^-) &= \pi_{df} + \pi_{at} && \text{(at the left limit, defiers and always-takers are treated).} \end{aligned} \quad (12.160)$$

Subtracting the two limits:

$$P^+ - P^- = (\pi_{co} + \pi_{at}) - (\pi_{df} + \pi_{at}) = \pi_{co} - \pi_{df}. \quad (12.161)$$

*Thus, in the presence of defiers ($\pi_{df} > 0$), the magnitude of the first-stage jump ceases to simply reflect the proportion of local adherents, instead representing the **net difference between the proportions of compliers and defiers**. If the two forces equalize ($\pi_{co} = \pi_{df}$), the first stage will be null, even with strong local adherence to the rule.*

B. The Intent-to-Treat Effect (Numerator). *Analogously, under the hypotheses of local exclusion and independence, the observed discontinuity in the realized outcome ($Y^+ - Y^-$) expands algebraically as the weighted sum of treatment effects on responsive groups:*

$$Y^+ - Y^- = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{Compliers}, Z_i = c] \pi_{co} - \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{Defiers}, Z_i = c] \pi_{df}. \quad (12.162)$$

C. The Collapse of the Wald Ratio. *Dividing the numerator by the denominator under monotonicity violation, the estimator recovers the following non-identifiable expression:*

$$\tau_{FRD} = \frac{\mathbb{E}[Y_i(1) - Y_i(0) \mid \text{Compliers}, Z_i = c] \pi_{co} - \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{Defiers}, Z_i = c] \pi_{df}}{\pi_{co} - \pi_{df}}. \quad (12.163)$$

This equation reveals two serious pathologies:

1. **Sign and Magnitude Bias:** *The negative sign in the numerator and denominator destroys the LATE identification theorem. It is mathematically possible that the treatment effect is strictly positive for all individuals in the population and yet the Fuzzy RDD estimator reports a **null or negative effect** due to the algebraic cancellation provoked by opposite flows of compliers and defiers.*
2. **Weights Outside Support:** *Since the weights associated with the two treatment effects do not sum to 1 with strictly positive signs (the defier weight enters subtracting), the Wald estimator ceases to be a conventional weighted average of local treatment effects. It can generate estimates that lie completely outside the support of real causal effects of individuals.*

By axiomatically imposing that $\pi_{df} = 0$, the monotonicity hypothesis restores the clean interpretability of the estimator. The defier term disappears from both components of the fraction, and the local Wald ratio collapses exactly onto the **Local Average Treatment Effect (LATE)** for local compliers at the margin of discontinuity:

$$\tau_{FRD} = \frac{\mathbb{E}[Y_i(1) - Y_i(0) \mid \text{Compliers}, Z_i = c] \pi_{co}}{\pi_{co}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{Compliers}, Z_i = c]. \quad (12.164)$$

12.5.7 Implementation: Local Linear Regression

Remark 12.5.13 (Centering at the Cutoff). In applied econometric practice, RDD estimation relies on approximating the conditional expectation functions of potential outcomes, $\mathbb{E}[Y_i(0) \mid Z_i]$ and $\mathbb{E}[Y_i(1) \mid Z_i]$, in a restricted neighborhood around the cutoff c . Although the classical Nadaraya-Watson estimator (local constant) is conceptually simple, the statistical and econometric literature demonstrates that it suffers from serious **boundary bias** when attempting to estimate one-sided limits at extreme support points. For this reason, **Local Linear Regression (LLR)** has become the standard and preferred method for RDD estimation.

The algorithmic implementation of LLR requires a crucial transformation of the running variable: **centering at the cutoff** ($Z_i - c$).

In a conventional linear regression of Y_i against Z_i , the estimated intercept analytically represents the expected value of the outcome when the explanatory variable equals zero ($\mathbb{E}[Y_i \mid Z_i = 0]$). However, in almost all practical RDD applications, the institutional cutoff c differs from zero (e.g., a cutoff score of 50 on an exam, or a family income eligibility limit of R\$ 1,000).

If we estimate the model without adjusting the scale of Z_i , the intercept projections will estimate the outcome behavior at a point distant from the cutoff (where $Z_i = 0$), requiring a long and severely biased linear extrapolation.

By systematically subtracting the cutoff point from each observation, defining the centered variable as:

$$\tilde{Z}_i \equiv Z_i - c, \quad (12.165)$$

we shift the origin of the abscissa axis exactly to the eligibility threshold. Under this new scale:

- Observations located exactly at the cutoff have $\tilde{Z}_i = 0$.
- Observations to the left (control group) have negative values ($\tilde{Z}_i < 0$).
- Observations to the right (treatment group) have positive values ($\tilde{Z}_i \geq 0$).

As a direct consequence of this geometric translation, the estimates of the left and right intercepts automatically and exactly represent the **mathematical one-sided limits of the conditional regression curves exactly at the cutoff boundary**.

Remark 12.5.14 (Separate Local Regressions). An intuitive way to operationalize LLR consists of estimating two independent ordinary least squares linear regressions, using only observations contained within a window of width h (where $h > 0$ is the bandwidth):

A. Left Side (Control): Fit a line for observations below the cutoff within the neighborhood $c - h$:

$$Y_i = \alpha_l + \beta_l \cdot (Z_i - c) + \varepsilon_i \quad \text{if } c - h \leq Z_i < c. \quad (12.166)$$

When the running variable approaches the cutoff from the left ($Z_i \uparrow c$), the conditional term $(Z_i - c)$ converges to zero. Thus, the left limit of the realized outcome (Y^-) is captured directly by the estimated intercept:

$$\lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z] = \alpha_l. \quad (12.167)$$

B. Right Side (Treatment): Fit a line for eligible observations located within the post-cutoff neighborhood $c + h$:

$$Y_i = \alpha_r + \beta_r \cdot (Z_i - c) + \varepsilon_i \quad \text{if } c \leq Z_i < c + h. \quad (12.168)$$

Analogously, when approaching the cutoff from the right ($Z_i \downarrow c$), the term $(Z_i - c)$ zeroes out, so the intercept estimates the right limit (Y^+):

$$\lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] = \alpha_r. \quad (12.169)$$

In Sharp RDD, the local average treatment effect (τ) is obtained directly by the simple arithmetic difference between the two estimated intercepts:

$$\tau = \alpha_r - \alpha_l. \quad (12.170)$$

Definition 12.5.7 (Unified Regression Specification). For statistical inference purposes (calculation of robust standard errors, hypothesis tests, and covariance matrices), the applied literature has consolidated LLR estimation into a **single integrated regression equation**. This is done by combining both sides through interactive terms with the treatment indicator $T_i \equiv \mathbb{I}\{Z_i \geq c\}$:

$$Y_i = \alpha_l + \tau \cdot T_i + \beta_l \cdot (Z_i - c) + \gamma \cdot T_i \cdot (Z_i - c) + \varepsilon_i \quad \text{if } c - h \leq Z_i < c + h. \quad (12.171)$$

In this unified specification:

- α_l is the intercept estimating the counterfactual control value at the left boundary (Y^-).
- τ is the coefficient associated with the eligibility dummy T_i , capturing the **vertical jump (discontinuity) at the cutoff**, which directly represents the treatment effect.
- β_l is the local slope of the regression line estimated for the control group.
- $\gamma \equiv (\beta_r - \beta_l)$ is the interaction term capturing the **change in slope** of the line when crossing the cutoff, allowing the trajectory of the conditional outcome to behave asymmetrically on each side of the threshold.

The main analytical advantage of this unified specification is that **the local average treatment effect (τ) and its associated robust standard error are estimated directly in a single regression**, without the need for algebraic post-processing or Delta Method for the difference of intercepts from separate regressions.

The flexibility provided by the interactive term $\gamma \cdot T_i \cdot (Z_i - c)$ is of vital econometric importance: if the model omitted this interaction (forcing $\gamma = 0$), any change in the actual slope of the function $\mathbb{E}[Y_i | Z_i]$ around c would be incorrectly projected as a vertical discontinuity τ , introducing severe functional specification bias.

Remark 12.5.15 (Estimation via Weighted Least Squares and Kernels). The estimation of the unified model above is operationalized by **Weighted Least Squares (WLS)**, where the weight assigned to each observation is determined by a Kernel function $K(\cdot)$ that penalizes observations distant from c :

$$w_i = K\left(\frac{Z_i - c}{h}\right). \quad (12.172)$$

If we use the **Rectangular Kernel** (uniform weight $w_i = 1$ for all observations within the window h), WLS estimation is equivalent to estimating a regression by conventional Ordinary Least Squares (OLS) strictly restricted to the subsample limited by the interval $[c - h, c + h]$.

In summary, centering at the cutoff ($Z_i - c$) is an indispensable mathematical transformation that simplifies the local identification problem of RDD, allowing estimation of the causal impact of the program and obtaining its statistical significance directly, without the need for additional calculations of extrapolated limits of curves.

12.5.7.1 Fuzzy RDD via Two-Stage Least Squares (2SLS)

Remark 12.5.16 (2SLS Implementation for Fuzzy RDD). In the **Fuzzy Regression Discontinuity Design (Fuzzy RDD)**, the presence of non-compliance causes the effectively received treatment status (denoted by T_i) to differ from the eligibility indicator (denoted by D_i , where $D_i = \mathbb{I}\{Z_i \geq c\}$). Since individuals self-select into or out of the program based on unobservable characteristics, **the realized treatment T_i becomes endogenous**, invalidating estimation by direct Ordinary Least Squares (OLS).

To overcome this local endogeneity problem, Fuzzy RDD is estimated equivalently as a **Two-Stage Least Squares (2SLS)** model, using the eligibility dummy D_i as a local exogenous instrument for treatment T_i within a window h around the cutoff:

1. The First Stage. Model the conditional probability of receiving effective treatment as a function of the exogenous cutoff indicator D_i , flexibly controlling for the running variable on both sides of the threshold:

$$T_i = \gamma_0 + \delta \cdot D_i + \delta_1 \cdot (Z_i - c) + \delta_2 \cdot D_i \cdot (Z_i - c) + u_i. \quad (12.173)$$

where parameter δ identifies the **jump in the probability of realized treatment** exactly at the cutoff. From this model, obtain the projected (predicted) value of realized treatment: $\hat{T}_i$.

2. The Second Stage. Estimate the structural outcome equation substituting the endogenous realized treatment T_i with the exogenous predicted value $\hat{T}_i$ obtained in the first stage:

$$Y_i = \alpha_0 + \tau_{\text{FD}} \cdot \hat{T}_i + \beta_0 \cdot (Z_i - c) + \beta_1 \cdot \hat{T}_i \cdot (Z_i - c) + \varepsilon_i. \quad (12.174)$$

where τ_{FD} consistently identifies the **Local Average Treatment Effect (LATE) exclusively for the subpopulation of local compliers** located at the margin of the cutoff.

Wald Equivalence. If the researcher uses **the same window width** h to estimate the first and second stage equations, the 2SLS estimator for τ_{FD} is **numerically identical to the local Wald ratio**, which divides the observed jump in the outcome by the observed jump in treatment probability at the cutoff:

$$\hat{\tau}_{\text{FD}} = \frac{Y^+ - Y^-}{T^+ - T^-} = \frac{\text{Discontinuity of } Y \text{ at } c}{\text{Discontinuity of } T \text{ at } c}. \quad (12.175)$$

Operationalization under the 2SLS framework considerably simplifies statistical inference, allowing standard econometric software to calculate robust standard errors and test statistics directly and asymptotically correctly in the presence of the estimated first stage.

12.5.7.2 Rejection of High-Order Global Polynomials

Remark 12.5.17 (Econometric Rejection of Global Polynomials). The estimation of the treatment effect in an RDD constitutes, by definition, a **boundary estimation** problem of the running variable support. In traditional applied econometrics, it was common to use high-order global polynomials (such as quadratic, cubic, or higher-degree terms fitted to the entire sample) under the argument of flexibly capturing non-linearities and ensuring robustness of estimates. However, recent theoretical developments in boundary econometrics—strongly synthesized by Gelman and Imbens (2019)—strongly discourage this practice.

The main econometric reasons for rejecting the use of high-order global polynomials in RDD include:

1. **Extreme Variance at the Boundary:** Series and global polynomial estimators exhibit **excessively high variance in the vicinity of boundary points** (support limits). Since RDD is precisely based on estimating the jump at the cutoff (which functions as the boundary for the one-sided limits on the left and right), this high sample variability compromises the precision and reliability of the estimator in the neighborhood of the cutoff. In contrast, the **Local Linear Regression (LLR)** estimator presents much superior statistical properties for boundary estimation, being the standard and recommended technique in the literature.
2. **Numerical Instability and Quasi-Collinearity:** As the degree p of the global polynomial increases, OLS estimation faces severe numerical instability problems. This occurs because powers of the running variable $(X_i, X_i^2, X_i^3, \dots, X_i^p)$ become highly correlated with each other, causing the moment matrix $X'X$ to approach singularity. In computational practice, this frequently results in rounding errors and matrix inversion failure messages.
3. **Erratic Behavior and Sample Sensitivity:** High-order polynomials produce extremely erratic estimates in the tails, behavior that accentuates as the polynomial degree increases. The variance of estimation grows very rapidly with the number of polynomial terms (K), requiring that model complexity be strictly controlled and limited. Attempting to compensate for non-linearities by arbitrarily raising the order of the global polynomial introduces artificial fluctuations (spurious jumps and oscillations) that distort the sign and magnitude of the estimated treatment effect at the cutoff.

For these reasons, while alternative estimators such as Nadaraya-Watson (local constant) suffer from systematic boundary bias, and global polynomials suffer from prohibitive variance, **Local Linear Regression consolidates as the preferred methodological approach** widely used to ensure the internal validity of RDD.

12.5.8 Bandwidth Selection and the Bias-Variance Trade-off

Remark 12.5.18 (Optimal Bandwidth Selection). The choice of bandwidth h embodies the fundamental bias-variance trade-off of RDD:

- **Small h :** Reduces bias (units are closer to c , approximation to the limit is better) but increases variance (fewer observations).
- **Large h :** Reduces variance (more observations) but increases bias (functional form approximation deteriorates away from c).

Imbens and Kalyanaraman (2012) derive the MSE-optimal bandwidth that balances these forces. **Calónico, Cattaneo, and Titiunik (2014)** propose bias-corrected confidence intervals that account for the estimation error in bandwidth selection, ensuring valid inference even when the bandwidth is data-driven.

12.5.8.1 The Bias-Variance Trade-off: Formal Decomposition

Remark 12.5.19 (Geometry of Window Margins). The behavior of the local estimator with respect to the magnitude of h can be decomposed into two operational extremes:

A. Excessively Wide Window (Oversmoothing). When the researcher chooses an arbitrarily large h , the local regression begins to incorporate observations located geographically very far from the cutoff point c .

- **Advantage (Low Variance):** By including a substantial volume of data in the least squares estimation, the effective sample size (N_h) increases significantly. This drastically reduces the sample variability of the estimator, resulting in very narrow standard errors and highly stable estimates (high precision).
- **Disadvantage (High Bias):** The assumption that the conditional expectation function (CEF) $\mathbb{E}[Y_i | Z_i]$ behaves linearly ceases to be a reasonable approximation as we move away from the threshold c . If the true structural relationship is curvilinear (non-linear), imposing a linear adjustment over a wide window will generate severe **specification bias**. The global linear model will fail to track the actual curvature of the data, projecting the distant non-linear slope as if it were a spurious vertical jump (bias) exactly at the cutoff point.

B. Excessively Narrow Window (Undersmoothing). At the other extreme, restricting estimation to a very small interval h forces the estimator to compute local means almost exclusively over observations glued to the boundary of c .

- **Advantage (Low Bias):** In an infinitesimal neighborhood around the cutoff, any smooth continuous function behaves in an approximately linear manner. Thus, approximating local behavior by straight lines becomes a representation almost free of functional specification bias. This scenario approaches the theoretical ideal of the

quasi-experiment, where treated and controls at the margin differ only by stochastic luck.

- **Disadvantage (High Variance):** By drastically reducing the active sample size (N_h), the standard error of the RDD estimator explodes. Few data points in the neighborhood generate highly noisy and erratic estimates, making the causal coefficient extremely sensitive to fluctuations and random noise from small samples (high imprecision).

Theorem 12.5.4 (Asymptotic Integrated Mean Squared Error (AIMSE)). *To formalize this balance, we analyze the behavior of the **Asymptotic Integrated Mean Squared Error (AIMSE)** of the local RDD estimator ($\hat{\theta}_{RDD}$):*

$$AIMSE(h) = \text{Asymptotic Variance} + \text{Asymptotic Bias}^2. \quad (12.176)$$

Under regularity hypotheses and using a local polynomial of order $p = 1$ (Local Linear Regression) with a symmetric Kernel function, the asymptotic decomposition of the error is expressed by:

$$AIMSE(h) \approx \underbrace{\frac{1}{Nh} \cdot \frac{\mathcal{R}_K}{f(c)} \left(\frac{\sigma^2(c^+)}{f(c^+)} + \frac{\sigma^2(c^-)}{f(c^-)} \right)}_{\text{Asymptotic Variance} \propto O((Nh)^{-1})} + \underbrace{h^4 \cdot \left[\frac{\kappa_{2,K}}{2} \cdot (m''(c^+) - m''(c^-)) \right]^2}_{\text{Quadratic Asymptotic Bias} \propto O(h^4)}, \quad (12.177)$$

where:

- N is the total sample size, and $f(c)$ is the marginal density of the running variable at the cutoff point.
- $\sigma^2(c^+)$ and $\sigma^2(c^-)$ are the conditional variances of the outcome to the right and left of the threshold, respectively.
- $\mathcal{R}_K = \int K^2(u)du$ and $\kappa_{2,K} = \int u^2 K(u)du$ are geometric penalization constants associated with the Kernel function used.
- $m''(c^+)$ and $m''(c^-)$ represent the **second derivatives** (curvatures) of the true conditional regression function of potential outcomes with respect to the running variable at the margin of the cutoff.

Analysis of limiting terms:

1. **If $h \rightarrow 0$:** The quadratic bias term, governed by h^4 , converges rapidly to zero. However, the variance term explodes at a rate of $1/(Nh)$, since the local sample denominator collapses.
2. **If $h \rightarrow \infty$:** The sample variance error is minimized at the rate of $1/N$, but the quadratic bias grows exponentially under the force of h^4 , multiplying the approximation error if there are sharp curvatures in the counterfactual functions ($m''(c) \neq 0$).

The theoretical optimal bandwidth (h^) is obtained by deriving the AIMSE loss function with respect to h and setting it equal to zero, resulting in an optimal window proportional to the sample rate of $N^{-1/5}$:*

$$h^* \propto \left(\frac{\text{Residual Variability}}{\text{Curvature } [m''(c)]^2 \cdot N} \right)^{1/5} \propto O(N^{-1/5}). \quad (12.178)$$

This relationship formalizes that the ideal window should be narrower in large samples (where variance can be sacrificed in favor of bias reduction) and when the true regression function exhibits strong curvature (high second derivatives) near the cutoff.

12.5.8.2 Cross-Validation Bandwidth Selection

Remark 12.5.20 (Leave-One-Out Cross-Validation (LOO-CV)). The determination of the optimal window (h) in RDD requires data-driven methods to avoid arbitrary choices by the researcher. The **Leave-One-Out (LOO) Cross-Validation** algorithm, formalized for the RDD context by Imbens and Lemieux (2008), stands out as one of the main non-parametric solutions to optimize this trade-off.

Algorithmic Mechanics. The essence of LOO cross-validation consists of measuring the *out-of-sample* predictive capacity of different windows h , estimating successive local regressions where each observation i is systematically excluded from the estimation step.

For each observation i in the panel of size N , the algorithm executes the following steps:

1. **Unit Exclusion:** Remove the i -th observation (Y_i, Z_i) from the dataset.
2. **Definition of Local Neighborhood:** For a window of width h , select only observations $j \neq i$ that are within the local neighborhood of Z_i , i.e., $Z_j \in [Z_i - h, Z_i + h]$.
3. **Local Estimation and Prediction:** Fit the local regression model (linear or local constant) using this subset of data of size $N - 1$. From the estimated coefficients, compute the predicted value for the omitted observation, denoted by $\hat{Y}_i^{-i}(Z_i)$.
4. **Calculation of Prediction Error:** Define the out-of-sample prediction error as:

$$\tilde{e}_i(h) = Y_i - \hat{Y}_i^{-i}(Z_i). \quad (12.179)$$

Repeating the procedure iteratively for all N observations in the sample, we define the **outcome cross-validation criterion function**, represented by the Predictive Mean Squared Error:

$$CV_Y(h) = \frac{1}{N} \sum_{i=1}^N \left(Y_i - \hat{Y}_i^{-i}(Z_i) \right)^2. \quad (12.180)$$

In RDD practice, the applied literature frequently restricts the summation in $CV_Y(h)$ to observations located reasonably close to the cutoff point c (e.g., within a predefined maximum window), ensuring that the selected bandwidth optimizes local predictive capacity precisely where the treatment effect will be identified (at the cutoff boundary).

The **optimal cross-validation window** (h_Y^*) is defined as the argument that minimizes this predictive loss function:

$$h_Y^* = \arg \min_{h \geq h_l} CV_Y(h), \quad (12.181)$$

where $h_l > 0$ represents a lower limit (minimum window) imposed by the econometrician to prevent the algorithm from evaluating excessively narrow windows, which would suffer from local sample degeneration and infinite variance due to lack of sufficient degrees of freedom to estimate the intercepts.

Optimization and Numerical Solution. Since the function $CV_Y(h)$ is typically non-linear and does not exhibit a closed analytical solution, the global minimum point h_Y^* must be computed through **numerical procedures**:

- **Grid Search:** Create a discrete grid of potential values for the window, $H = [h_1, h_2, \dots, h_J]$. The criterion $CV_Y(h_j)$ is evaluated at each point j , and the lowest value is selected. The graphical behavior of $CV_Y(h)$ against h is plotted as an essential diagnostic to verify convexity and the existence of a unique global minimum.
- **Golden-Section Search:** An asymptotically faster and computationally more efficient two-dimensional numerical search algorithm than traditional grid search.

In the asymptotic limit, the cross-validation estimator is an unbiased approximator of the **Integrated Mean Squared Error (IMSE)**, guaranteeing consistency properties in model selection without the need to impose restrictive assumptions such as conditional homoscedasticity.

12.5.8.3 Cross-Validation in Fuzzy RDD

Remark 12.5.21 (Independent Windows for Outcome and Treatment). In the **Fuzzy Regression Discontinuity Design (Fuzzy RDD)**, identification of the local average causal effect (LATE) ceases to be computed by a single discontinuity in the outcome and becomes defined as the **ratio between two distinct vertical jumps at the cutoff point** c :

1. **The Outcome Discontinuity (Reduced Form):** The vertical jump in the conditional expectation of the realized outcome, $\mathbb{E}[Y_i | Z_i]$, at the cutoff boundary.
2. **The Treatment Discontinuity (First Stage):** The vertical jump in the conditional expectation of the effectively received treatment (i.e., the probability of treatment), $\mathbb{E}[T_i | Z_i]$, when crossing the eligibility threshold.

From the perspective of non-parametric estimation, the researcher needs to locally approximate two fundamentally different regression functions with respect to the running variable Z_i . Since each of these conditional relations has its own theoretical curvature (second derivatives), residual variance, and data density, isolated application of the **Leave-One-Out Cross-Validation (LOO-CV)** algorithm results in two potentially very distinct optimal data windows.

Mathematical Formulation of Independent Windows. To determine the bandwidths in a purely data-driven manner, independently estimate the predictive loss functions for the realized outcome (Y_i) and the realized treatment status (T_i):

A. Optimal Window for the Outcome (h_Y^*):

$$CV_Y(h) = \frac{1}{N} \sum_{i=1}^N \left(Y_i - \hat{Y}_i^{-i}(Z_i) \right)^2 \implies h_Y^* = \arg \min_{h \geq h_l} CV_Y(h). \quad (12.182)$$

B. Optimal Window for Treatment Probability (h_T^*):

$$CV_T(h) = \frac{1}{N} \sum_{i=1}^N \left(T_i - \hat{T}_i^{-i}(Z_i) \right)^2 \implies h_T^* = \arg \min_{h \geq h_l} CV_T(h). \quad (12.183)$$

The Dilemma of Inference with Asymmetric Windows. The existence of two independent optimal windows, h_Y^* and h_T^* , creates a methodological impasse in the estimation of the local Wald ratio:

$$\hat{\tau}_{\text{FRD}} = \frac{\hat{\alpha}_{r,Y}(h_Y^*) - \hat{\alpha}_{l,Y}(h_Y^*)}{\hat{\gamma}_{r,T}(h_T^*) - \hat{\gamma}_{l,T}(h_T^*)}. \quad (12.184)$$

Although using distinct windows for the numerator and denominator is theoretically conceivable, the applied econometric literature discourages this practice due to two severe complications:

- **Complexity of Asymptotic Theory:** The combination of two boundary local estimators based on different convergence rates and data windows destroys the simplicity of the Wald ratio limit distribution, making asymptotic variance formulas extremely complex.
- **Local Sample Inconsistency:** By estimating the numerator and denominator in geometrically misaligned local subsamples, the property that two-stage least squares (2SLS) coefficients exactly coincide with the local Wald ratio is lost, impairing the interpretability of classical instrumentation.

The Imbens and Lemieux (2008) Minimum Rule. To circumvent these operational obstacles and ensure the statistical robustness of the estimator, **Imbens and Lemieux (2008)** suggest specifying a **single common window** applied simultaneously to both the outcome regression and the treatment regression.

To define this common smoothing parameter under a conservative perspective of error control, the authors propose selecting the **smallest bandwidth window** among the two independent windows calculated by cross-validation:

$$h_{\text{Fuzzy}}^* = \min(h_Y^*, h_T^*). \quad (12.185)$$

Why Prioritize the Smaller Window (Undersmoothing)? The econometric intuition behind this choice asymmetrically relies on the consequences of deviations in the bias-variance trade-off:

1. **Protection Against Asymptotic Bias (Primary Risk):** Local specification bias arises when the window is wide enough that first-degree polynomials cannot track sharp curvatures in the conditional functions. Since RDD identifies the causal effect strictly at the boundary limit, **asymptotic bias constitutes the most severe threat to estimator consistency**. Choosing the smallest window (min) forces both regressions to operate under the most restrictive window, minimizing biased approximation in the component that exhibits greater non-linearity near the cutoff.
2. **Controlled Sacrifice of Variance (Secondary Risk):** The cost of adopting the smaller window is a marginal loss of precision (increase in sample variance), due to the smaller number of local observations incorporated in the estimation. However, from the perspective of causal identification, the literature prioritizes asymptotic consistency (internal validity free of bias) over small-sample statistical efficiency.

By adopting h_{Fuzzy}^* , the researcher reestablishes the exact mathematical equivalence between the Fuzzy RDD estimator and the local Two-Stage Least Squares (2SLS) model,

enabling the calculation of asymptotically consistent standard errors and robust to heteroscedasticity directly.

12.5.8.4 Asymptotic Optimal Selectors: IK and CCT

Remark 12.5.22 (Imbens-Kalyanaraman (2012) Plug-in Algorithm). Although the Cross-Validation algorithm is a robust and non-parametric data-driven tool, modern econometric literature has advanced in developing **window selectors based on direct analytical minimization of Mean Squared Error (MSE) or Coverage Error**. Among the frontier asymptotic approaches, the selectors developed by **Imbens and Kalyanaraman (2012 — IK)** and by **Calonico, Cattaneo and Titiunik (2014 — CCT)** stand out, consolidated as the most applied analytical and computational standards in empirical research.

The Minimization Target: Combined MSE of Both Regressions. To deduce an optimal selector based on *plug-in*, the first step consists of defining a mathematical loss function to be minimized. Instead of focusing only on the jump effect τ_{RDD} , the **Imbens-Kalyanaraman (IK)** estimator seeks to choose a bandwidth window h that minimizes the **combined Mean Squared Error** of the estimates of the two conditional regression functions, $\hat{m}_1(c)$ and $\hat{m}_0(c)$, estimated at the boundary of the cutoff point c :

$$\text{MSE}(h) = \mathbb{E} \left[(\hat{m}_1(c) - m_1(c))^2 + (\hat{m}_0(c) - m_0(c))^2 \mid \mathbf{Z} \right]. \quad (12.186)$$

Under the local regularity hypotheses of Sharp RDD, the second-order asymptotic approximation for the bias and variance of each Local Linear Regression (LLR) estimator with symmetric Kernel function $K(\cdot)$ provides:

- **Quadratic Asymptotic Bias (squared):** $\text{Bias}^2(\hat{m}_t(c)) \approx h^4 \cdot \left[\frac{\kappa_{2,K}}{2} \cdot m_t''(c) \right]^2$ for $t \in \{0, 1\}$.
- **Asymptotic Variance:** $\text{Var}(\hat{m}_t(c)) \approx \frac{1}{Nh} \cdot \frac{\mathcal{R}_K}{f(c)} \cdot \sigma_t^2(c)$ for $t \in \{0, 1\}$.

Summing the components, the theoretical optimal bandwidth h_{IK}^* that minimizes MSE_{IK} is analytically expressed by:

$$h_{\text{IK}}^* = C_K \cdot \left(\frac{\sigma_1^2(c) + \sigma_0^2(c)}{f(c) \cdot [(m_1''(c))^2 + (m_0''(c))^2]} \right)^{1/5} \cdot N^{-1/5}, \quad (12.187)$$

where $C_K = \left(\frac{\mathcal{R}_K}{\kappa_{2,K}^2} \right)^{1/5}$ is a known constant dependent only on the chosen Kernel (e.g., $C_K \approx 3.437$ for the Triangular Kernel).

The IK Plug-in Algorithm. Since the structural parameters of the local population ($f(c)$, $\sigma_t^2(c)$, and curvatures $m_t''(c)$) are unknown, the **IK** algorithm operationalizes an estimation routine in **three plug-in stages** to consistently estimate each term of the theoretical formula:

- **Stage 1: Estimation of Density and Local Variances.** The density $f(c)$ is estimated at the cutoff point through a simple Kernel estimator with a global pilot window (Silverman's rule of thumb). The conditional variances to the left and right of the cutoff, $\hat{\sigma}_0^2(c)$ and $\hat{\sigma}_1^2(c)$, are computed from the residual variance of local linear regressions in a preliminary neighborhood.

- **Stage 2: Estimation of High-Order Curvatures (The Pivot).** The estimation of second derivatives $m_t''(c)$ is the most sensitive phase of the algorithm.
 1. To avoid specification bias, first estimate the **third derivatives** ($m_t'''(c)$) by fitting a global cubic polynomial over the entire sample.
 2. With these estimates, calculate an **ideal pilot window for the second derivative** (h_2), based on the formulas of Fan and Gijbels (1996).
 3. Using the window h_2 , estimate the local second derivatives $\hat{m}_1''(c)$ and $\hat{m}_0''(c)$ through **local quadratic regressions** fitted on each side of the cutoff.
- **Stage 3: Calculation of Final Optimal Window.** All estimated parameters are inserted into the equation for h_{IK}^* , generating the final optimal bandwidth $\hat{h}_{IK}$, which converges asymptotically to the MSE minimum at a sample rate of $O(N^{-1/5})$.

Remark 12.5.23 (Calonico, Cattaneo, and Titiunik (2014): Robust Bias Correction). Despite the asymptotic optimality of the IK algorithm for point estimation, **Calonico, Cattaneo, and Titiunik (2014)** identified a serious statistical limitation regarding **inference (hypothesis tests and confidence intervals)** under MSE-optimal windows.

The Problem of Non-Negligible Bias. When using the optimal window $h^* \propto N^{-1/5}$, the bias and standard error of the RDD jump estimator converge to zero at exactly the same asymptotic rate ($O(N^{-2/5})$). Consequently, the distribution of the conventional standardized estimator is not centered at zero:

$$\frac{\hat{\tau}_{RDD}(h^*) - \tau}{SE(\hat{\tau}_{RDD}(h^*))} \xrightarrow{d} \mathcal{N}(B, 1) \quad \text{where } B \neq 0. \quad (12.188)$$

This causes conventional confidence intervals to suffer from **severe undercoverage** (empirical rejection rates much higher than the nominal significance level α).

The CCT Solution: Robust Bias Correction (RBC). To solve this pathology, CCT developed the **Robust Bias-Correction (RBC)** methodology, structured under two pillars:

1. **Bias Estimation with Differentiated Pilot Window (b):** The local RDD estimator is corrected by explicitly subtracting an estimate of the second-order bias, computed using a pilot window b (where the ratio b/h is optimized to minimize approximation error).
2. **Variance Adjustment of the Correction:** Since the bias correction term itself is a sample estimate, it has its own stochastic variability. CCT's crucial innovation was **incorporating the variance and covariance of the estimated bias term directly into the final standard error formula**.

The resulting robust test statistic, defined by:

$$T_{RBC} = \frac{\hat{\tau}_{BC}(h, b) - \tau}{SE_{RBC}(h, b)} \xrightarrow{d} \mathcal{N}(0, 1), \quad (12.189)$$

corrects the asymptotic displacement without uncontrollably inflating the variance in finite samples. The CCT selectors (available in the econometric package **rdrobust**) simultaneously provide the optimal estimation window h and the bias pilot window b , consistently minimizing the coverage error of confidence intervals.

12.5.9 RDD with Covariates

12.5.9.1 Motivations and Control Mechanics

Remark 12.5.24 (Precision Gains and Variance Reduction). One of the most elegant methodological properties of the Regression Discontinuity Design (RDD) is that **the inclusion of baseline covariates is not necessary for identification** of the conditional average treatment effect at the cutoff point. If the hypotheses of continuity of conditional regression functions of potential outcomes at c are satisfied, the observed discontinuity in the realized outcome perfectly identifies the causal impact non-parametrically.

However, in applied econometrics, control for pre-treatment observed characteristics (X_i) is widely employed due to two fundamental reasons: **statistical precision gains** and **robustness validation**.

The Mechanism of Residual Variance Reduction. In a local RDD specification, the precision of the discontinuity parameter estimate ($\hat{\tau}$) is directly linked to the **conditional variance of the error** (residual noise of the equation) in the neighborhood of the cutoff point, denoted by $\sigma^2(c)$ or s^2 .

When we estimate the vertical jump without incorporating additional controls, all the variability of the outcome Y_i that is not explained by the running variable Z_i is thrown directly onto the residual error term u_i . If the outcome presents high intrinsic variability, the residual variance will be elevated.

By introducing into the model a vector of pre-treatment covariates X_i that are **strong predictors of the outcome** (such as the history of grades from previous years in educational impact studies), these variables systematically capture part of this unexplained variability. The mathematical impact of this adjustment is direct:

- The **Sum of Squared Residuals (SSR)** of the local regression is drastically reduced.
- The standard error of the estimator ($SE(\hat{\tau})$), which is proportional to the estimated residual variance s^2 , decreases proportionally.
- **Confidence intervals become narrower** and the **statistical power of hypothesis tests is expanded**, allowing detection of statistically significant effects even in local windows with restricted samples.

It is crucial to highlight that this efficiency gain strictly depends on the **real predictive power of the covariates**. As demonstrated by multiple regression theory, the inclusion of weak or irrelevant covariates that do not explain the outcome does not reduce the residual variance sufficiently to compensate for the loss of degrees of freedom, possibly, on the contrary, inflating the estimated standard errors.

Point Estimate Behavior under Large Samples. In terms of point estimation, RDD with covariates behaves analogously to a randomized controlled trial (RCT). Due to the premise of local ignorability, individuals located immediately above and below the eligibility threshold are identical in all observable and unobservable characteristics on average.

Since covariates X_i are locally balanced at the cutoff point, the vector X_i is orthogonal to the local eligibility status. Thus, **in sufficiently large samples, the inclusion of**

X_i **does not alter the point estimate of $\hat{\tau}$** . The inclusion of variables cleans the background noise of the sample without introducing bias, operating in a strictly beneficial manner for the asymptotic efficiency of the estimator.

Remark 12.5.25 (Correction of Local Imbalances in Finite Samples). Although the asymptotic identification of the treatment effect in a Sharp RDD is based on the premise that individuals in the neighborhood of the cutoff point are, on average, identical in all their characteristics, the reality of empirical applications imposes the challenge of working with **finite samples**. Under samples of limited size, local quasi-randomization at the cutoff boundary may fail to produce perfect balancing, allowing the emergence of **small random imbalances in baseline observable characteristics** between groups immediately above and below the threshold c . The introduction of exogenous covariates in the local regression model acts directly to correct these sample fluctuations, shielding the estimator from stochastic noise and stabilizing local consistency.

The Origin of Imbalances in Finite Samples. Theoretically, under the hypothesis of local ignorability, treatment assignment at the margin of the cutoff point is governed by chance, mimicking the internal validity properties of a randomized controlled trial (RCT). In asymptotic terms (as $N \rightarrow \infty$), the conditional expectation of any pre-treatment covariate X_i is perfectly continuous around the cutoff:

$$\lim_{z \downarrow c} \mathbb{E}[X_i \mid Z_i = z] = \lim_{z \uparrow c} \mathbb{E}[X_i \mid Z_i = z]. \quad (12.190)$$

However, in finite samples, the empirical distribution of control variables may present small asymmetries and purely fortuitous mean deviations on each side of the threshold. If a baseline covariate X_{ki} that exhibits a fortuitous sample imbalance is correlated with the outcome Y_i , the simple comparison of local means will inherit a small-sample bias, distorting the true vertical jump τ .

By incorporating the vector of covariates X_i directly into the local regression equation, the econometrician **purges the discontinuity estimator of this stochastic noise**. The covariates absorb the marginal variations of the outcome that are explained by punctual differences in baseline attributes, neutralizing the impact of casual imbalances on the treatment effect estimate.

Mathematical Formulation with Covariate Adjustment. The integration of baseline covariates X_i into the unified Local Linear Regression (LLR) model is modeled by the following weighted specification:

$$Y_i = \alpha_l + \tau \cdot T_i + \beta_l \cdot (Z_i - c) + \gamma \cdot T_i \cdot (Z_i - c) + X_i' \theta + \varepsilon_i \quad \text{if } c - h \leq Z_i < c + h. \quad (12.191)$$

where X_i is a vector of pre-treatment covariates centered in relation to their local sample mean, and θ represents the vector of linear association coefficients with the outcome.

Under this specification, the point estimator $\hat{\tau}$ remains asymptotically equivalent to the estimator without covariates, since the vector X_i is locally orthogonal to the treatment variable T_i due to the continuity of potential outcomes and baseline characteristics at the cutoff. However, in finite samples:

- The term $X_i' \theta$ actively corrects any non-null residual covariance between pre-treatment characteristics and the eligibility variable generated by the limited sample size.

- The point estimator of the causal jump becomes **statistically more stable and robust** to marginal variations in the choice of the bandwidth window h .

12.5.9.2 The Danger of Bad Controls

Remark 12.5.26 (Bad Controls and Selection Bias). Although the inclusion of exogenous and pre-treatment baseline covariates is beneficial for increasing statistical precision and mitigating imbalances in finite samples, the introduction of variables measured after policy implementation requires extreme caution. In the microeconomic literature, variables that are measured or determined **concomitantly with or after treatment allocation** are called **bad controls**. The inclusion of these variables in the local regression model of an RDD destroys the internal validity of the design, reintroducing the selection bias that the quasi-experiment had originally eliminated.

What Defines a “Bad Control”? The fundamental distinction between a valid control and a bad control does not reside in its correlation with the outcome, but rather in its **causal precedence relationship with the treatment**:

- **Valid Controls (Good Controls):** Are baseline covariates or fixed demographic characteristics (such as gender, race, or age) that **cannot have been influenced by the treatment**, since their determination occurred at a chronologically prior moment to the start of the program.
- **Bad Controls:** Are intermediate outcome variables that **were or may have been altered directly or indirectly by the intervention**.

The Intuition of Artificial Selection Bias (Angrist-Pischke Decomposition). To mathematically understand the collapse of identification, consider evaluating the impact of a professional training program on individuals’ income (Y_i). The treatment T_i (eligibility around a cutoff point) affects income, but also increases the probability of the individual obtaining formal employment (denoted by the binary variable $S_i \in \{0, 1\}$, which represents post-treatment employment status).

If the researcher, in an attempt to analyze the “direct effect on the salary of those employed,” decides to control for the variable S_i in the regression, they will be incurring the bad control problem.

Following the potential outcomes framework, each individual has potential employment states: $S_i(1)$ under treatment and $S_i(0)$ under control. By conditioning the analysis on the subsample of individuals effectively employed ($S_i = 1$) on both sides of the cutoff, the observed difference in means will be expressed by:

$$\mathbb{E}[Y_i \mid T_i = 1, S_i = 1] - \mathbb{E}[Y_i \mid T_i = 0, S_i = 1]. \quad (12.192)$$

Substituting the realized outcomes with the corresponding potential outcomes, this expression is equivalent to:

$$\underbrace{\mathbb{E}[Y_i(1) \mid T_i = 1, S_i(1) = 1]}_{\text{Treated Employed}} - \underbrace{\mathbb{E}[Y_i(0) \mid T_i = 0, S_i(0) = 1]}_{\text{Controls Employed}}. \quad (12.193)$$

Even if RDD guarantees local quasi-randomization of treatment at the boundary of c (so that global potential expectations are identical, $\mathbb{E}[Y_i(d) \mid T_i = 1] = \mathbb{E}[Y_i(d) \mid T_i = 0]$), the

equality **does not hold** when we condition on S_i :

$$\mathbb{E}[Y_i(0) \mid T_i = 1, S_i(1) = 1] \neq \mathbb{E}[Y_i(0) \mid T_i = 0, S_i(0) = 1]. \quad (12.194)$$

Why Does This Inequality Occur? The composition of employed groups on each side of the cutoff is latently different:

- **Employed Control Group** ($S_i(0) = 1$): Is composed strictly of individuals who would find employment even without the program's help (individuals with high motivation, ability, or exceptional contact networks—the so-called *always-takers* of employment).
- **Employed Treatment Group** ($S_i(1) = 1$): Is composed of a mixture of two types of people:
 - Those who would get a job anyway (the *always-takers*).
 - Those who only got a job because they received the professional training provided by the program (the *compliers* of employment).

Since training expanded employment, the group of treated employed begins to include individuals of lower relative ability or motivation (who needed the program's incentive to get employed) compared to the employed control group. Consequently, by conditioning on $S_i = 1$, we are comparing a highly selected control group with a treatment group of inferior average composition. This generates an **artificial negative selection bias**, severely underestimating the true wage impact of the policy.

Implications for the RDD Specification Rule. In the Regression Discontinuity Design, the specification rule for covariates must be rigorous:

- **Golden Rule:** The vector of control characteristics X_i must be composed **strictly of exogenous pre-treatment variables**.
- If there is suspicion that a covariate was affected by the intervention, it should be excluded from the main outcome regression and treated, if necessary, as a secondary outcome variable to evaluate causal transmission mechanisms (mediation analysis).

12.5.9.3 Covariates as a Robustness Test

Remark 12.5.27 (The Point Estimate Stability Theorem). Under the null hypothesis that RDD is a valid research design, individuals located immediately above and immediately below the eligibility threshold c are statistically equivalent, differing only by marginal stochastic chance (local common support and local ignorability). This local quasi-randomization implies that the vector of baseline characteristics X_i must be perfectly balanced in the neighborhood of the cutoff point.

From an asymptotic perspective, the local orthogonality between exogenous covariates and eligibility status T_i dictates that:

$$\text{plim} (\hat{\tau}_{\text{with covariates}} - \hat{\tau}_{\text{without covariates}}) = 0. \quad (12.195)$$

That is, **in sufficiently large samples, the inclusion of the vector X_i in the RDD local linear model should not statistically significantly alter the point estimate**

of the causal effect ($\hat{\tau}$). The coefficient should remain stable, ideally recording only a reduction in its associated standard error and the consequent narrowing of its confidence intervals due to purging the residual variability of the outcome.

Symptoms of Violation of Identification Premises. If, when estimating the unified equation with exogenous covariates, the researcher observes a substantial displacement in the estimated causal coefficient ($\hat{\tau}$)—such as sign inversion, severe loss of magnitude, or erratic fluctuations under different sets of controls—there are **strong indications that the fundamental premises of continuity or non-manipulation have been violated**. Econometrically, this displacement points to two serious structural pathologies:

A. Coexistence of Competing Policies (Lack of Local Exclusion). If the point estimate of the vertical jump systematically changes with the inclusion of baseline covariates, it is likely that other institutional rules, tax benefits, or concurrent assistance programs operate under the same cutoff point c . Under this specification violation, the discontinuities of other outcomes and covariates are confused with the effect of the intervention analyzed, revealing that **the observed vertical jump in Y_i at the cutoff cannot be isolatedly attributed to the treatment of interest**.

B. Systematic Manipulation and Self-Selection (Sorting). When individuals have the capacity to predict the eligibility threshold c and hold control (even if partial) over their running variable Z_i , they tend to actively manipulate their scores to deliberately position themselves on the most advantageous side of the cutoff.

This systematic manipulation induces an **artificial discontinuity in the density of the assignment variable** and, simultaneously, breaks the premise that individuals at the margin of c are equivalent in unobservable and observable variables. The extreme sensitivity of $\hat{\tau}$ to the insertion of X_i reflects, in finite samples, this imbalance and local selection bias.

Graphical and Statistical Operationalization of the Robustness Test. To formalize the validation of specification robustness, the applied literature recommends that the researcher adopt two rigorous statistical routines:

1. **Tabular Sensitivity to Control Blocks:** Present a regression table where the LLR estimator is computed iteratively, adding progressive blocks of exogenous baseline covariates (such as demographic controls, socioeconomic controls, and lagged outcomes). The design is considered robust if variations in $\hat{\tau}$ are statistically null.
2. **RDD Regression on Baseline Covariates (Balance Test):** Estimate the same unified LLR equation, but substituting the outcome variable Y_i with each of the baseline covariates X_{ki} as the dependent variable:

$$X_{ki} = \alpha_{l,k} + \tau_k \cdot T_i + \beta_{l,k} \cdot (Z_i - c) + \gamma_k \cdot T_i \cdot (Z_i - c) + u_{ki}. \quad (12.196)$$

Under the hypothesis of RDD validity, one expects that $\hat{\tau}_k$ **is not statistically different from zero for any covariate k** ($\hat{\tau}_k \approx 0$), graphically and statistically proving that baseline characteristics pass continuously through the cutoff point, without registering spurious vertical discontinuities at the boundary.

12.5.9.4 Nonparametric Approach with Covariates

Definition 12.5.8 (Local Common Support of Covariates at the Threshold). In applied econometric practice, the introduction of covariates in Regression Discontinuity Designs (RDD) is traditionally conducted in an ad-hoc manner, through simple additive linear specifications or low-order polynomials in the assignment variable. However, when formalizing **fully non-parametric modeling with covariates**, the impact evaluation literature requires the rigorous definition of local distributional assumptions to ensure consistent identification of the local average causal effect (LATE). The central premise of this approach is the **hypothesis of local common support** of covariates in the neighborhood of the cutoff point.

Mathematically, the support of a random variable corresponds to the smallest closed set of values that concentrates probability of occurrence equal to 1. In the context of global matching on observables, the **overlap** hypothesis (common support) establishes that for each value of exogenous covariate X , the probability of treatment must be strictly positive and less than 1 (i.e., $0 < \Pr(T_i = 1 | X_i) < 1$). This guarantees the coexistence of treated and control units over the entire support of X . In Sharp RDD, since the global probability of treatment jumps deterministically from exactly 0 to 1 at the cutoff point, this global overlap premise is violated by design.

Non-parametric modeling of RDD with covariates circumvents this violation by limiting the overlap premise to an **infinitesimal local boundary**. To formalize this assumption, let N_ε be a symmetric neighborhood of radius $\varepsilon > 0$ around the eligibility threshold z_0 . This neighborhood is partitioned into two subsets on each side of the cutoff:

- $N_\varepsilon^+ = \{z : z \geq z_0, z \in N_\varepsilon\}$, representing the right side (eligibility/treatment group at the margin);
- $N_\varepsilon^- = \{z : z < z_0, z \in N_\varepsilon\}$, representing the left side (control group at the margin).

The **Local Common Support Hypothesis** (also denoted as RDD-4 (i) hypothesis in the non-parametric approach) postulates that the supports of baseline covariates X must coincide perfectly when the width of this limit data window converges to zero:

$$\lim_{\varepsilon \rightarrow 0} \text{Supp}(X | Z \in N_\varepsilon^+) = \lim_{\varepsilon \rightarrow 0} \text{Supp}(X | Z \in N_\varepsilon^-). \quad (12.197)$$

This hypothesis guarantees that, in the limit of approximation to the cutoff point, the distribution of observable characteristics of individuals located immediately above and immediately below the threshold shares exactly the **same population support**.

The methodological necessity of this premise derives directly from the mechanics of non-parametric identification of RDD:

1. **Recovery of Unconditional LATE:** RDD initially estimates effects conditional on the vector of characteristics, i.e., $\mathbb{E}[Y_i(1) - Y_i(0) | X_i = x, Z_i = z_0]$. To recover the unconditional LATE at the boundary, the estimator requires integrating these conditional expectations with respect to the marginal distribution of covariates.
2. **The Collapse of the Local Counterfactual:** If the support of X differs between the two sides of the threshold at the margin, mathematical integration becomes unfeasible for regions without common support, since there will be some subpopulation with specific observable characteristics present on one side of the cutoff for which **there is no equivalent local counterfactual** on the other side.

3. **Redefinition of the Estimand:** Under the failure of local common support, the researcher is mathematically forced to abandon identification of the global unconditional effect over the entire neighborhood and **redefine their estimand**, restricting identification of the LATE to the common intersection of the supports of local covariates.

Finally, the non-parametric literature observes that, although the support of covariates needs to be identical on both sides, the **conditional density functions of covariates** in the neighborhood of z_0 may be discontinuous, so that the limiting densities to the right and left at the threshold differ ($f_+(x | z_0) \neq f_-(x | z_0)$). Under the presence of such discontinuity, the common support premise guarantees that non-parametric control by X is feasible and indispensable to purge the identification bias that would occur even in locally robust RDD designs.

Theorem 12.5.5 (Frölich and Huber (2018) Nonparametric Integration Estimator). *Conventional econometric practice frequently incorporates baseline covariates X_i in Regression Discontinuity (RDD) models through ad-hoc additive linear specifications, typically expressed as:*

$$Y_i = \alpha_l + \tau \cdot T_i + \beta_l \cdot (Z_i - c) + \gamma \cdot T_i \cdot (Z_i - c) + X_i' \theta + \varepsilon_i. \quad (12.198)$$

*Although this parametric formulation is convenient for estimation by Ordinary Least Squares (OLS) and provides precision gains in finite samples, it rests on extremely restrictive premises: **additive separability** between covariates and the running variable, and **homogeneity of treatment effects** over the support of X_i . If the true treatment effect interacts non-linearly with baseline characteristics, or if counterfactual outcome trajectories $\mathbb{E}[Y_i(d) | Z_i, X_i]$ present distinct curvatures for different demographic profiles, the ad-hoc linear estimator will fail to recover the unconditional average causal effect at the boundary, generating an estimand weighted in a difficult-to-interpret or severely biased manner.*

*To circumvent this specification pathology, Frölich (2007) and Frölich and Huber (2018) formalized an approach of **fully non-parametric identification and estimation with covariates based on local integration (or reweighting)**.*

The Theoretical Identification Mechanism. *Let Z_i be the running variable centered at the cutoff c , X_i the vector of baseline covariates, and $Y_i(1), Y_i(0)$ the potential outcomes under treatment and control, respectively. Define the non-parametric conditional expectation functions of the realized outcome Y_i conditional on the entire vector of observables (Z_i, X_i) on each side of the boundary:*

$$\begin{aligned} m_Y(z, x) &\equiv \mathbb{E}[Y_i | Z_i = z, X_i = x], \\ m_D(z, x) &\equiv \mathbb{E}[D_i | Z_i = z, X_i = x] \quad (\text{for Fuzzy RDD}). \end{aligned} \quad (12.199)$$

The limiting expectations at the right margin (c^+) and left (c^-) of the cutoff point are mapped non-parametrically for each specific value of covariate $X_i = x$:

$$\begin{aligned} m_Y(c^+, x) &= \lim_{z \downarrow c} \mathbb{E}[Y_i | Z_i = z, X_i = x], \\ m_Y(c^-, x) &= \lim_{z \uparrow c} \mathbb{E}[Y_i | Z_i = z, X_i = x]. \end{aligned} \quad (12.200)$$

Different from traditional multiple linear regression, which projects a global average slope for covariates, the non-parametric integration estimator estimates these limits flexibly for

each point of the support of X_i . The unconditional effect at the boundary is reestablished by integrating (or calculating the weighted average) of these limits with respect to the **conditional cumulative distribution of covariates exactly at the threshold of the cutoff point**, denoted by $F_{X|Z=c}(x)$:

A. Sharp RDD Case (Direct Integration):

$$\tau_{SRD}^{NP} = \int_{\mathcal{X}} [m_Y(c^+, x) - m_Y(c^-, x)] dF_{X|Z=c}(x). \quad (12.201)$$

B. Fuzzy RDD Case (Ratio of Integrals):

$$\tau_{FRD}^{NP} = \frac{\int_{\mathcal{X}} [m_Y(c^+, x) - m_Y(c^-, x)] dF_{X|Z=c}(x)}{\int_{\mathcal{X}} [m_D(c^+, x) - m_D(c^-, x)] dF_{X|Z=c}(x)}. \quad (12.202)$$

Under the premises of local continuity and common support of covariates at the cutoff (established in the previous definition), the integration perfectly robustly reconstructs the **unconditional Local Average Treatment Effect (LATE)** at the boundary. This estimator is immune to functional specification biases because it imposes no restriction on how covariates X_i affect the outcome or interact with treatment status.

Algorithmic Implementation and Empirical Distribution. Since the true conditional distribution function $F_{X|Z=c}(x)$ is unknown in practice, the Frölich and Huber estimator substitutes the theoretical integral with a **weighted sample mean** over the observations in the sample. The two-stage structured estimation mechanics proceeds as follows:

1. **Non-Parametric Estimation of Boundary Curves:** For each individual i in the sample, estimate the predicted values $\hat{m}_Y(c^+, X_i)$ and $\hat{m}_Y(c^-, X_i)$. This is commonly operationalized by fitting **Local Linear Regressions (LLR)** or Kernel estimators for the outcome Y against the running variable Z , conditioning locally on the vector of baseline covariates $X_i = x$.
2. **Reweighting and Local Integration at the Threshold:** To collapse individual conditional predictions into the unconditional estimand, calculate the weighted average of predicted differences. The weights w_i are determined by a Kernel function that punishes observations whose running variable Z_i is distant from the cutoff point c , isolating the empirical distribution of covariates exactly in the neighborhood of the cutoff:

$$\hat{\tau}_{SRD}^{NP} = \sum_{i=1}^N w_i [\hat{m}_Y(c^+, X_i) - \hat{m}_Y(c^-, X_i)], \quad (12.203)$$

where the normalized weights are given by:

$$w_i = \frac{K\left(\frac{Z_i - c}{h}\right)}{\sum_{j=1}^N K\left(\frac{Z_j - c}{h}\right)}. \quad (12.204)$$

For Fuzzy RDD, the denominator is computed in an entirely analogous manner, reweighting the estimated local discontinuities for the probability of realized treatment $\hat{m}_D(c^+, X_i) - \hat{m}_D(c^-, X_i)$.

Statistical Properties and Inference.

- **Convergence Rates and Curse of Dimensionality:** Although non-parametric estimation of curves $m_Y(z, x)$ suffers from the slow asymptotic convergence typical of high-dimensional estimators (the curse of dimensionality), Frölich (2007) demonstrates that, under regularity conditions, the integration step over $dF_{X|Z=c}$ acts as an average smoother, allowing the final estimator $\hat{\tau}^{NP}$ to converge at reasonable asymptotic rates and exhibit a normal limit distribution.
- **Consistent Inference via Bootstrap:** The analytical calculation of asymptotic standard errors for this class of two-stage integrated estimators is extremely complex due to the propagation of first-stage non-parametric estimation errors to the weighted empirical distribution. Given this, Frölich and Huber (2018) propose the application of **non-parametric bootstrap routines with cluster-level resampling** to obtain statistically robust confidence intervals and with consistent empirical coverage rates in finite samples.

Remark 12.5.28 (Summary of Key RDD Diagnostics). • **Density test (McCrary):** Verify continuity of the running variable density at c ; detect manipulation.

- **Covariate balance tests:** Estimate RDD on pre-treatment covariates; expect no significant jumps.
- **Sensitivity to bandwidth:** Report estimates across multiple bandwidths; results should be stable.
- **Placebo cutoffs:** Estimate RDD at false cutoffs where no policy change occurred; expect null effects.
- **First-stage discontinuity (Fuzzy RDD):** Verify a significant jump in treatment probability at c .
- **Exclusion restriction:** Argue that no other policies use the same cutoff; test for jumps in alternative outcomes.
- **Covariate stability:** Estimate RDD with and without baseline covariates; point estimate $\hat{\tau}$ should remain stable.
- **Optimal bandwidth:** Use IK (2012) or CCT (2014) selectors; report robust bias-corrected confidence intervals.
- **Avoid global polynomials:** Use local linear regression (LLR) with triangular or uniform kernel; do not fit high-order polynomials to the entire sample.

Chapter 13

Comprehensive Review: Estimators and Models

13.1 Cross-Section and System Estimators: OLS, SOLS, SUR

13.1.1 OLS (Ordinary Least Squares)

13.1.1.1 Motivation and Intuition

The OLS estimator is the fundamental pillar of econometrics. In applied microeconomic research (e.g., estimating the return to schooling on wages or the production function of firms), OLS seeks to project the dependent variable y onto the column space generated by regressors X , minimizing the sum of squared vertical distances from the points to the regression hyperplane.

Remark 13.1.1 (Economic Intuition). OLS provides the **Best Linear Unbiased Estimator (BLUE)** under the Gauss-Markov assumptions. It is the workhorse of empirical research because it is:

- **Simple to compute and interpret**
- **Optimal** among linear unbiased estimators under homoscedasticity
- **Consistent** under weak exogeneity conditions

However, OLS fails when regressors are endogenous ($\mathbb{E}[X_i u_i] \neq 0$), requiring instrumental variables or other correction methods.

13.1.1.2 Notation and Classical Linear Regression Model Hypotheses

Definition 13.1.1 (Operational Notation). • Let $i \in \{1, 2, \dots, N\}$ index observations in a cross-sectional sample.

- Let $y_i \in \mathbb{R}$ be the observed dependent variable for unit i .
- Let $x_i \in \mathbb{R}^{K \times 1}$ be the column vector of K regressors for unit i , where the first element is typically 1 (constant).

- Let $\beta \in \mathbb{R}^{K \times 1}$ be the vector of unknown population parameters to be estimated.
- Let $u_i \in \mathbb{R}$ be the unobserved structural/idiosyncratic error term for unit i .

Stacking the sample into vectors and matrices:

$$\begin{aligned} y &= [y_1, y_2, \dots, y_N]' \in \mathbb{R}^{N \times 1} \\ X &= [x_1, x_2, \dots, x_N]' \in \mathbb{R}^{N \times K} \quad (\text{the design matrix}) \\ u &= [u_1, u_2, \dots, u_N]' \in \mathbb{R}^{N \times 1} \end{aligned} \quad (13.1)$$

The population model in matrix form is given by:

$$y = X\beta + u \quad (13.2)$$

Definition 13.1.2 (Formal Hypotheses). **OLS.1 (Linearity in Parameters):** The data generating process (DGP) obeys the linear structure described in Equation (13.2).

OLS.2 (Strict Exogeneity / Zero Conditional Mean):

$$\mathbb{E}[u_i \mid X] = 0, \quad \forall i = 1, \dots, N \quad (13.3)$$

Role: Guarantees unbiasedness of $\hat{\beta}_{OLS}$.

Practical translation: Requires that no omitted variable determining y_i is correlated with regressors in X , and that there is no measurement error in X or simultaneity.

Counterexample without it: If y_i is wage and x_i is schooling, omitting “innate ability” A_i (contained in u_i) correlated with x_i makes $\mathbb{E}[u_i \mid x_i] \neq 0$, generating omitted variable bias and rendering OLS inconsistent.

OLS.3 (Full Rank / No Perfect Multicollinearity):

$$\text{rank}(X) = K \leq N \quad (13.4)$$

Role: Guarantees that the $K \times K$ matrix $X'X$ is strictly positive definite and therefore invertible.

Counterexample without it: If we include age in years and age in months in the same model, one column will be a perfect linear combination of the other; $\det(X'X) = 0$, making it impossible to invert $X'X$.

OLS.4 (Sphericity of Errors / Gauss-Markov):

$$\text{Var}(u \mid X) = \mathbb{E}[uu' \mid X] = \sigma^2 I_N \quad (13.5)$$

where $\sigma^2 > 0$ is the constant scalar variance and I_N is the $N \times N$ identity matrix.

Role: Guarantees that errors are conditionally homoscedastic ($\text{Var}(u_i \mid X) = \sigma^2$) and not autocorrelated ($\text{Cov}(u_i, u_j \mid X) = 0, \forall i \neq j$). Under OLS.1-OLS.4, the Gauss-Markov Theorem guarantees that OLS is the Best Linear Unbiased Estimator (BLUE).

13.1.1.3 When to Use vs. When NOT to Use

Remark 13.1.2 (Decision Guide for OLS). **WHEN TO USE:**

- When all Gauss-Markov assumptions hold (OLS.1-OLS.4)
- When regressors are exogenous ($\mathbb{E}[X_i u_i] = 0$)
- As a baseline estimator before testing for violations

WHEN NOT TO USE:

- **DO NOT use** if there is endogeneity ($\mathbb{E}[X_i u_i] \neq 0$) due to omitted variables, measurement error, or simultaneity. Use **IV/2SLS** instead.
- **DO NOT use** if errors are heteroscedastic ($\text{Var}(u_i | X_i) = \sigma_i^2$). Use **White's robust standard errors** or **WLS**.
- **DO NOT use** if errors are autocorrelated (time series or panel data). Use **Newey-West** or **cluster-robust standard errors**.

13.1.1.4 Granular Derivation of the OLS Estimator

Define the Sum of Squared Residuals function $S(b)$ for a candidate vector $b \in \mathbb{R}^{K \times 1}$:

$$S(b) = (y - Xb)'(y - Xb) \quad (13.6)$$

Expanding the scalar product (by the distributive property of transposition):

$$S(b) = (y' - b'X')(y - Xb) = y'y - y'Xb - b'X'y + b'X'Xb \quad (13.7)$$

Since the term $y'Xb$ is a 1×1 scalar, its transpose is identical to itself ($(y'Xb)' = b'X'y$). Thus:

$$S(b) = y'y - 2b'X'y + b'X'Xb \quad (13.8)$$

To find the minimizer $\hat{\beta}_{\text{OLS}}$, we derive $S(b)$ with respect to vector b and set equal to the null vector $0_{K \times 1}$:

$$\frac{\partial S(b)}{\partial b} = -2X'y + 2X'Xb = 0 \quad (13.9)$$

Evaluating at the solution $\hat{\beta}_{\text{OLS}}$:

$$2X'X\hat{\beta}_{\text{OLS}} = 2X'y \quad (13.10)$$

Dividing both sides by scalar 2:

$$X'X\hat{\beta}_{\text{OLS}} = X'y \quad (13.11)$$

This is the **OLS Normal Equation**.

By Hypothesis OLS.3 ($\text{rank}(X) = K$), matrix $X'X$ has inverse $(X'X)^{-1}$. Multiplying Equation (13.11) on the left by $(X'X)^{-1}$:

$$(X'X)^{-1}X'X\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'y \quad (13.12)$$

Since $(X'X)^{-1}(X'X) = I_K$ (by definition of inverse matrix):

$$\boxed{\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'y} \quad (13.13)$$

13.1.1.5 Formal Properties of the OLS Estimator

Theorem 13.1.1 (Unbiasedness). *Substituting the structural DGP equation $y = X\beta + u$ into Equation (13.13):*

$$\hat{\beta}_{OLS} = (X'X)^{-1}X'(X\beta + u) = (X'X)^{-1}X'X\beta + (X'X)^{-1}X'u = \beta + (X'X)^{-1}X'u \quad (13.14)$$

Taking the conditional mathematical expectation on X :

$$\mathbb{E}[\hat{\beta}_{OLS} | X] = \mathbb{E}[\beta | X] + \mathbb{E}[(X'X)^{-1}X'u | X] \quad (13.15)$$

Since β and X are deterministic under conditioning:

$$\mathbb{E}[\hat{\beta}_{OLS} | X] = \beta + (X'X)^{-1}X'\mathbb{E}[u | X] \quad (13.16)$$

Applying Hypothesis OLS.2 ($\mathbb{E}[u | X] = 0$):

$$\mathbb{E}[\hat{\beta}_{OLS} | X] = \beta + (X'X)^{-1}X' \cdot 0 = \beta \quad (13.17)$$

By the Law of Iterated Expectations, $\mathbb{E}[\hat{\beta}_{OLS}] = \mathbb{E}[\mathbb{E}[\hat{\beta}_{OLS} | X]] = \mathbb{E}[\beta] = \beta$.

Theorem 13.1.2 (Conditional Variance-Covariance Matrix).

$$\text{Var}(\hat{\beta}_{OLS} | X) = \mathbb{E}[(\hat{\beta}_{OLS} - \beta)(\hat{\beta}_{OLS} - \beta)' | X] \quad (13.18)$$

Substituting $(\hat{\beta}_{OLS} - \beta) = (X'X)^{-1}X'u$:

$$\text{Var}(\hat{\beta}_{OLS} | X) = \mathbb{E} \left[\left((X'X)^{-1}X'u \right) \left((X'X)^{-1}X'u \right)' \middle| X \right] \quad (13.19)$$

Applying the transposition rule $(AB)' = B'A'$ and symmetry of $X'X$:

$$\text{Var}(\hat{\beta}_{OLS} | X) = \mathbb{E} \left[(X'X)^{-1}X'uu'X(X'X)^{-1} \middle| X \right] \quad (13.20)$$

Factoring terms in X outside the conditional expectation:

$$\text{Var}(\hat{\beta}_{OLS} | X) = (X'X)^{-1}X'\mathbb{E}[uu' | X]X(X'X)^{-1} \quad (13.21)$$

Applying Hypothesis OLS.4 ($\mathbb{E}[uu' | X] = \sigma^2 I_N$):

$$\text{Var}(\hat{\beta}_{OLS} | X) = (X'X)^{-1}X'(\sigma^2 I_N)X(X'X)^{-1} = \sigma^2(X'X)^{-1}(X'X)(X'X)^{-1} \quad (13.22)$$

Since $(X'X)(X'X)^{-1} = I_K$:

$$\boxed{\text{Var}(\hat{\beta}_{OLS} | X) = \sigma^2(X'X)^{-1}} \quad (13.23)$$

13.1.1.6 Main Tests and Diagnostics in OLS

Remark 13.1.3 (White Test for Heteroscedasticity). • $H_0 : \text{Var}(u_i | x_i) = \sigma^2$ (Homoscedasticity) vs. $H_1 : \text{Var}(u_i | x_i) = g(x_i)$.

- **Mechanics:** Regress squared OLS residuals ($\hat{e}_i^2$) against all regressors in X , their squares, and cross products.
- **Test Statistic:** $LM = N \cdot R_{\text{auxiliary}}^2 \xrightarrow{d} \chi_P^2$, where P is the number of regressors in the auxiliary regression.
- **Decision Rule:** Reject H_0 if $LM > \chi_{\alpha, P}^2$. If rejected, conventional standard errors given by Equation (13.23) are invalid, requiring use of White's Heteroscedasticity-Robust Covariance Matrix (HC0-HC3).

Remark 13.1.4 (Advantages and Limitations of OLS). **Advantages:**

- **BLUE** under Gauss-Markov assumptions (minimum variance among linear unbiased estimators)
- Simple computation and interpretation
- Consistent under weak exogeneity

Limitations:

- **Inconsistent under endogeneity** ($\mathbb{E}[X_i u_i] \neq 0$)
- **Inefficient under heteroscedasticity** (though still consistent)
- Cannot estimate coefficients of time-invariant variables in panel fixed effects models

13.1.2 SOLS (System Ordinary Least Squares)

13.1.2.1 Motivation and Economic Intuition

In many microeconomic applications, we study systems of M related behavioral equations (e.g., demand systems for multiple goods, production functions across sectors). The SOLS estimator applies OLS to each equation separately, ignoring potential cross-equation correlation in the errors.

Remark 13.1.5 (When to Use SOLS). SOLS is appropriate when:

- Equations are **truly unrelated** (errors are uncorrelated across equations: $\sigma_{mj} = 0$)
- You want to **isolate specification errors** (if equation 2 is misspecified, SOLS keeps equation 1 consistent)
- Computational simplicity is prioritized over efficiency

13.1.2.2 Matrix Notation of the Stacked System

Let the system of M equations for N sample units be:

$$y_{im} = x'_{im} \beta_m + u_{im}, \quad m = 1, \dots, M, \quad i = 1, \dots, N \quad (13.24)$$

For each equation m , define vectors and matrices of individual dimension:

- $y_m \in \mathbb{R}^{N \times 1}$
- $X_m \in \mathbb{R}^{N \times K_m}$
- $\beta_m \in \mathbb{R}^{K_m \times 1}$
- $u_m \in \mathbb{R}^{N \times 1}$

Stacking the M equations vertically:

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_M \end{bmatrix}_{MN \times 1}, \quad X = \begin{bmatrix} X_1 & 0 & \dots & 0 \\ 0 & X_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & X_M \end{bmatrix}_{MN \times K_{\text{tot}}}, \quad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_M \end{bmatrix}_{K_{\text{tot}} \times 1}, \quad u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_M \end{bmatrix}_{MN \times 1} \quad (13.25)$$

where $K_{\text{tot}} = \sum_{m=1}^M K_m$.

The SOLS estimator for the complete system is defined by:

$$\hat{\beta}_{\text{SOLS}} = (X'X)^{-1}X'y \quad (13.26)$$

13.1.2.3 Identification Hypotheses

Definition 13.1.3 (SOLS Hypotheses). 1. **Local Exogeneity (Within Each Equation):**

$$\mathbb{E}[u_{im} \mid X_{im}] = 0 \quad \forall m = 1, \dots, M \quad (13.27)$$

Economic Intuition: Regressors in equation m must be orthogonal only to the error in equation m . SOLS does **not** require orthogonality across equations.

2. **No Perfect Multicollinearity Within Each Equation:**

$$\text{rank}(X_m) = K_m \leq N \quad \forall m \quad (13.28)$$

3. **Contemporaneous Error Correlation (Optional):**

$$\mathbb{E}[u_i u_i' \mid X] = \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1M} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{M1} & \sigma_{M2} & \dots & \sigma_{MM} \end{bmatrix}_{M \times M} \quad (13.29)$$

where $\sigma_{mj} = \text{Cov}(u_{im}, u_{ij} \mid X)$ can be non-zero. SOLS remains consistent even if $\sigma_{mj} \neq 0$, but becomes **inefficient**.

13.1.2.4 Granular Proof of Numerical Equivalence between SOLS and Equation-by-Equation OLS

Theorem 13.1.3 (SOLS Equivalence Theorem). *If all M equations in the system share exactly the same set of regressors ($X_1 = X_2 = \dots = X_M = X_0 \in \mathbb{R}^{N \times K_0}$), the SOLS estimator applied to the stacked system generates point estimates numerically identical to applying isolated OLS equation by equation.*

Proof. If $X_m = X_0$ for all $m = 1, \dots, M$, the matrix X of the stacked system can be rewritten using the Kronecker product ($\otimes$):

$$X = I_M \otimes X_0 \quad (13.30)$$

where I_M is the $M \times M$ identity matrix.

Calculating matrix $X'X$:

$$X'X = (I_M \otimes X_0)'(I_M \otimes X_0) \quad (13.31)$$

Applying the Kronecker product transpose property $((A \otimes B)' = A' \otimes B')$:

$$X'X = (I_M' \otimes X_0')(I_M \otimes X_0) \quad (13.32)$$

Applying the Kronecker product property $((A \otimes B)(C \otimes D) = (AC) \otimes (BD))$:

$$X'X = (I_M I_M) \otimes (X_0' X_0) = I_M \otimes (X_0' X_0) \quad (13.33)$$

(since $I_M' I_M = I_M$).

Inverting matrix $X'X$:

$$(X'X)^{-1} = (I_M \otimes (X_0' X_0))^{-1} \quad (13.34)$$

Applying the Kronecker product inverse property $((A \otimes B)^{-1} = A^{-1} \otimes B^{-1})$:

$$(X'X)^{-1} = I_M^{-1} \otimes (X_0' X_0)^{-1} = I_M \otimes (X_0' X_0)^{-1} \quad (13.35)$$

Now, calculating vector $X'y$:

$$X'y = (I_M \otimes X_0)'y = (I_M \otimes X_0') \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_M \end{bmatrix} = \begin{bmatrix} X_0' y_1 \\ X_0' y_2 \\ \vdots \\ X_0' y_M \end{bmatrix} \quad (13.36)$$

Multiplying to obtain $\hat{\beta}_{\text{SOLS}}$:

$$\hat{\beta}_{\text{SOLS}} = (I_M \otimes (X_0' X_0)^{-1}) \begin{bmatrix} X_0' y_1 \\ X_0' y_2 \\ \vdots \\ X_0' y_M \end{bmatrix} = \begin{bmatrix} (X_0' X_0)^{-1} X_0' y_1 \\ (X_0' X_0)^{-1} X_0' y_2 \\ \vdots \\ (X_0' X_0)^{-1} X_0' y_M \end{bmatrix} = \begin{bmatrix} \hat{\beta}_{1,\text{OLS}} \\ \hat{\beta}_{2,\text{OLS}} \\ \vdots \\ \hat{\beta}_{M,\text{OLS}} \end{bmatrix} \quad (13.37)$$

The SOLS estimate for the parameter vector of equation m is strictly $(X_0' X_0)^{-1} X_0' y_m$, which is the exact single-equation OLS formula. $\square$

13.1.2.5 SOLS Limitations and Cross-Covariance Calculation

Although point estimates coincide when $X_m = X_0$, SOLS offers a crucial advantage over isolated OLS: calculation of the **Cross-Covariance Matrix between Estimators of Different Equations**.

Consider the contemporaneous error structure of the system. For unit i , let $u_i = [u_{i1}, u_{i2}, \dots, u_{iM}]'$ be the vector of errors from the M equations. Assume:

$$\mathbb{E}[u_i u_i' | X] = \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1M} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{M1} & \sigma_{M2} & \dots & \sigma_{MM} \end{bmatrix}_{M \times M} \quad (13.38)$$

where $\sigma_{mj} = \text{Cov}(u_{im}, u_{ij} | X) \neq 0$ represents contemporaneous error correlation between equations m and j .

The variance matrix of the stacked system error u is given by:

$$\text{Var}(u | X) = \Omega = \Sigma \otimes I_N \quad (13.39)$$

Calculating the true variance of the SOLS estimator under this cross-correlation structure:

$$\text{Var}(\hat{\beta}_{\text{SOLS}} | X) = (X'X)^{-1} X'(\Sigma \otimes I_N) X (X'X)^{-1} \quad (13.40)$$

When $X = I_M \otimes X_0$:

$$\text{Var}(\hat{\beta}_{\text{SOLS}} | X) = \Sigma \otimes (X_0' X_0)^{-1} \quad (13.41)$$

Thus, the generic block of the covariance matrix between the estimator of equation m and equation j is given by:

$$\text{Cov}(\hat{\beta}_m, \hat{\beta}_j | X) = \sigma_{mj} (X_0' X_0)^{-1} \quad (13.42)$$

Remark 13.1.6 (SOLS Limitation). SOLS is inefficient when Σ is not diagonal ($\sigma_{mj} \neq 0$) and regressors differ between equations, because it does not use the weight matrix Σ^{-1} to reweight observations. This motivates the SUR estimator.

13.1.3 SUR (Seemingly Unrelated Regressions) / FGLS in System

13.1.3.1 Motivation and Economic Intuition

Developed by Arnold Zellner (1962), the SUR model recognizes that equations seemingly unrelated on the regressor side may be linked through contemporaneous unobserved shocks contained in the error terms.

Remark 13.1.7 (Economic Example). Consider a demand system for meats (beef, chicken, fish). Each equation has different regressors (prices of each meat, income, demographics). However, unobserved shocks (health trends, cultural preferences, seasonal effects) affect all demand equations simultaneously. SUR exploits this cross-equation correlation to achieve **asymptotic efficiency gains** over equation-by-equation OLS.

13.1.3.2 Identification Hypotheses

Definition 13.1.4 (SUR Hypotheses). 1. **Strict Cross-Exogeneity:**

$$\mathbb{E}[u_{im} \mid X_{i1}, X_{i2}, \dots, X_{iM}] = 0 \quad \forall m \quad (13.43)$$

Economic Intuition: **All** regressors from **all** equations must be orthogonal to **all** errors. This is a stronger assumption than SOLS.

2. **Contemporaneous Error Correlation:**

$$\text{Cov}(u_{im}, u_{ij} \mid X) = \sigma_{mj} \neq 0 \quad \text{for some } m \neq j \quad (13.44)$$

If $\sigma_{mj} = 0$ for all $m \neq j$, SUR collapses to SOLS (no efficiency gain).

3. **Different Regressors Across Equations:**

$$X_m \neq X_j \quad \text{for some } m \neq j \quad (13.45)$$

If $X_1 = X_2 = \dots = X_M$, SUR collapses to SOLS (Zellner's Theorem).

13.1.3.3 When to Use vs. When NOT to Use

Remark 13.1.8 (Decision Guide for SUR). **WHEN TO USE:**

- When equations have **different regressors** ($X_m \neq X_j$)
- When errors are **contemporaneously correlated** ($\sigma_{mj} \neq 0$)
- When you want to test **cross-equation restrictions** (e.g., $\beta_{1m} = \beta_{1j}$)

WHEN NOT TO USE:

- **DO NOT use** if $X_1 = X_2 = \dots = X_M$ (SUR = SOLS, no gain)
- **DO NOT use** if $\sigma_{mj} = 0$ for all $m \neq j$ (SUR = SOLS, no gain)
- **DO NOT use** if strict cross-exogeneity fails (SUR becomes inconsistent)

13.1.3.4 Zellner's GLS and FGLS Estimator

If the contemporaneous covariance matrix Σ is known, the Generalized Least Squares (GLS) estimator of the stacked system $y = X\beta + u$ is given by:

$$\hat{\beta}_{\text{GLS}} = (X'\Omega^{-1}X)^{-1} X'\Omega^{-1}y = [X'(\Sigma^{-1} \otimes I_N)X]^{-1} X'(\Sigma^{-1} \otimes I_N)y \quad (13.46)$$

Since Σ is unknown in practice, we use the **Feasible GLS (FGLS / 2-Stage SUR)** estimator:

Remark 13.1.9 (Two-Stage FGLS Procedure). 1. **Step 1 (Initial Stage via OLS):**

Estimate each equation isolatedly by OLS and extract the residual vector $\hat{u}_m = y_m - X_m\hat{\beta}_{m,\text{OLS}}$.

2. **Step 2 (Consistent Estimate of Σ):** Calculate element (m, j) of matrix $\hat{\Sigma}$:

$$\hat{\sigma}_{mj} = \frac{1}{N} \sum_{i=1}^N \hat{u}_{im} \hat{u}_{ij} \quad \text{or} \quad \hat{\sigma}_{mj} = \frac{\hat{u}'_m \hat{u}_j}{\sqrt{(N - K_m)(N - K_j)}} \quad (13.47)$$

3. Step 3 (FGLS/SUR Estimator): Substitute $\hat{\Sigma}$ into the GLS formula:

$$\hat{\beta}_{\text{SUR}} = \left[X'(\hat{\Sigma}^{-1} \otimes I_N)X \right]^{-1} X'(\hat{\Sigma}^{-1} \otimes I_N)y \quad (13.48)$$

13.1.3.5 Granular Proof of Zellner's Equivalence Theorems (SUR $\equiv$ OLS/SOLS)

Theorem 13.1.4 (Zellner (1962)). *The SUR estimator ($\hat{\beta}_{\text{GLS}}$) reduces exactly to the OLS/SOLS estimator in two sufficient cases:*

1. *When the contemporaneous error covariance matrix is diagonal ($\sigma_{mj} = 0, \forall m \neq j$).*
2. *When regressors are identical in all equations ($X_1 = X_2 = \dots = X_M = X_0$).*

Case 1: Σ is diagonal. If $\sigma_{mj} = 0$ for all $m \neq j$, matrix Σ is diagonal:

$$\Sigma = \text{diag}(\sigma_{11}, \sigma_{22}, \dots, \sigma_{MM}) \implies \Sigma^{-1} = \text{diag}\left(\frac{1}{\sigma_{11}}, \frac{1}{\sigma_{22}}, \dots, \frac{1}{\sigma_{MM}}\right) \quad (13.49)$$

Expanding the Kronecker product $\Sigma^{-1} \otimes I_N$:

$$\Sigma^{-1} \otimes I_N = \begin{bmatrix} \frac{1}{\sigma_{11}}I_N & 0 & \dots & 0 \\ 0 & \frac{1}{\sigma_{22}}I_N & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\sigma_{MM}}I_N \end{bmatrix}_{MN \times MN} \quad (13.50)$$

Since $X = \text{diag}(X_1, X_2, \dots, X_M)$, we calculate matrix $X'(\Sigma^{-1} \otimes I_N)X$:

$$X'(\Sigma^{-1} \otimes I_N)X = \begin{bmatrix} \frac{1}{\sigma_{11}}X_1'X_1 & 0 & \dots & 0 \\ 0 & \frac{1}{\sigma_{22}}X_2'X_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\sigma_{MM}}X_M'X_M \end{bmatrix} \quad (13.51)$$

The inverse of a block diagonal matrix is the matrix of block inverses:

$$\left[X'(\Sigma^{-1} \otimes I_N)X \right]^{-1} = \begin{bmatrix} \sigma_{11}(X_1'X_1)^{-1} & 0 & \dots & 0 \\ 0 & \sigma_{22}(X_2'X_2)^{-1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{MM}(X_M'X_M)^{-1} \end{bmatrix} \quad (13.52)$$

Multiplying to obtain $\hat{\beta}_{\text{GLS}}$:

$$\hat{\beta}_{\text{GLS}} = \begin{bmatrix} (X_1'X_1)^{-1}X_1'y_1 \\ (X_2'X_2)^{-1}X_2'y_2 \\ \vdots \\ (X_M'X_M)^{-1}X_M'y_M \end{bmatrix} = \begin{bmatrix} \hat{\beta}_{1,\text{OLS}} \\ \hat{\beta}_{2,\text{OLS}} \\ \vdots \\ \hat{\beta}_{M,\text{OLS}} \end{bmatrix} \quad (13.53)$$

(by cancellation of variance scalars $\sigma_{mm}/\sigma_{mm} = 1$). □

Case 2: $X_1 = X_2 = \cdots = X_M = X_0$. If $X_m = X_0$ for all m , we have $X = I_M \otimes X_0$.

Calculating $X'(\Sigma^{-1} \otimes I_N)X$:

$$X'(\Sigma^{-1} \otimes I_N)X = (I_M \otimes X_0)'(\Sigma^{-1} \otimes I_N)(I_M \otimes X_0) = \Sigma^{-1} \otimes (X_0'X_0) \quad (13.54)$$

Inverting the resulting matrix:

$$\left[X'(\Sigma^{-1} \otimes I_N)X\right]^{-1} = \Sigma \otimes (X_0'X_0)^{-1} \quad (13.55)$$

Multiplying to obtain $\hat{\beta}_{\text{GLS}}$:

$$\hat{\beta}_{\text{GLS}} = \left(\Sigma \otimes (X_0'X_0)^{-1}\right) (\Sigma^{-1} \otimes X_0')y = I_M \otimes \left((X_0'X_0)^{-1}X_0'\right)y = \hat{\beta}_{\text{SOLS}} \quad (13.56)$$

(since $\Sigma\Sigma^{-1} = I_M$). □

13.1.3.6 Breusch-Pagan Test of Contemporaneous Independence

To test whether there is efficiency gain from using SUR instead of OLS/SOLS:

- **Null Hypothesis:** $H_0 : \sigma_{mj} = 0 \quad \forall m \neq j$ (Matrix Σ is diagonal)
- **Alternative Hypothesis:** $H_1 : \exists m \neq j$ such that $\sigma_{mj} \neq 0$

Test Statistic:

$$LM = N \sum_{m=1}^{M-1} \sum_{j=m+1}^M \hat{r}_{mj}^2 \quad (13.57)$$

where $\hat{r}_{mj} = \frac{\hat{\sigma}_{mj}}{\sqrt{\hat{\sigma}_{mm}\hat{\sigma}_{jj}}}$ is the sample correlation between OLS residuals from equations m and j .

Asymptotic Distribution and Decision Rule: Under H_0 , $LM \xrightarrow{d} \chi_{M(M-1)/2}^2$. Reject H_0 at significance level α if $LM > \chi_{\alpha, M(M-1)/2}^2$. Rejection confirms that errors are contemporaneously correlated and that the SUR estimator is strictly more efficient than OLS/SOLS.

Remark 13.1.10 (Advantages and Limitations of SUR). **Advantages:**

- **Asymptotic efficiency gains** when $X_m \neq X_j$ and $\sigma_{mj} \neq 0$
- Allows testing **cross-equation restrictions** with valid standard errors
- Provides consistent estimates of **cross-equation covariances**

Limitations:

- **Cross-contamination risk:** If one equation is misspecified, the entire system becomes inconsistent
- Requires **strict cross-exogeneity** (stronger than SOLS)
- No efficiency gain if $X_1 = X_2 = \cdots = X_M$ or if $\sigma_{mj} = 0$

13.1.4 Comparative Summary and Research Decision Guide

Remark 13.1.11 (Comparative Summary: OLS, SOLS, and SUR). **OLS:** Data structure is a single cross-section (N); key exogeneity hypothesis is $\mathbb{E}[u_i | x_i] = 0$; error covariance matrix is $\sigma^2 I_N$ (homoscedastic and i.i.d.); behavior when $X_1 = X_2 = \dots = X_M$ is not applicable (single equation).

SOLS: Data structure is a system of equations (M equations, N observations); key exogeneity hypothesis is $\mathbb{E}[u_{im} | X_{im}] = 0$ (local); error covariance matrix is treated as $I_M \otimes \sigma^2 I_N$, but cross-covariances $\text{Cov}(\hat{\beta}_m, \hat{\beta}_j)$ are calculated; behavior when $X_1 = X_2 = \dots = X_M$ is **numerically identical** to equation-by-equation OLS.

SUR (FGLS): Data structure is a system of equations (M equations, N observations); key exogeneity hypothesis is $\mathbb{E}[u_{im} | X_1, \dots, X_M] = 0$ (strict cross); error covariance matrix is $\Sigma_{M \times M} \otimes I_N$ (contemporaneous correlation); behavior when $X_1 = X_2 = \dots = X_M$ is **reduces to SOLS/OLS** (Zellner's Theorem). No efficiency gain.

Remark 13.1.12 (Decision Framework: Choosing Between OLS, SOLS, and SUR). 1.

Single equation, cross-sectional data? $\implies$ Use **OLS**

2. **Multiple equations, same regressors** ($X_1 = X_2 = \dots = X_M$)? $\implies$ Use **SOLS** (SUR provides no efficiency gain)

3. **Multiple equations, different regressors** ($X_m \neq X_j$), **correlated errors** ($\sigma_{mj} \neq 0$)? $\implies$ Use **SUR** for efficiency gains

4. **Panel data, unobserved heterogeneity correlated with regressors** ($\mathbb{E}[c_i | X_i] \neq 0$)? $\implies$ Use **Fixed Effects (FE)** or **First Differences (FD)**

13.2 Instrumental Variables: IV, 2SLS, and GMM

13.2.1 IV (Instrumental Variables — Exactly Identified Case)

13.2.1.1 Motivation and Economic Intuition

The Instrumental Variables (IV) estimator was developed to solve the failure of regressor exogeneity in the Ordinary Least Squares (OLS) model. In applied microeconometrics, regressor endogeneity ($\mathbb{E}[x_i u_i] \neq 0$) arises from three main reasons:

1. **Omitted Variable:** Absence of control for unobserved factors correlated with x_i and y_i (e.g., innate ability in the wage equation).
2. **Measurement Error in Regressors:** Coefficient attenuation due to random noise in the explanatory variable.
3. **Simultaneity / Reverse Causality:** Joint determination between the dependent and explanatory variable (e.g., market supply and demand).

The IV method uses an external variable z_i (the instrument) to isolate the variation in x_i that is exogenous to the error term u_i , allowing recovery of the consistent estimate of the structural parameter.

Remark 13.2.1 (Economic Intuition: The “Clean Variation” Principle). The core idea of IV is to find an external source of variation in the endogenous regressor x_i that is **orthogonal**

to the structural error u_i . The instrument z_i acts as a “natural experiment” that shifts x_i for reasons unrelated to the unobserved determinants of y_i . By using only this “clean” variation, we can consistently estimate the causal effect β even when OLS fails.

13.2.1.2 Population Model and Notation

Consider the simple linear regression model with one endogenous scalar regressor x_i :

$$y_i = \beta_0 + \beta_1 x_i + u_i, \quad \text{with } \text{Cov}(x_i, u_i) \neq 0 \quad (13.58)$$

Let z_i be a scalar instrument. In matrix notation for the general exactly identified case with K regressors and $L = K$ instruments:

$$y = X\beta + u, \quad \text{where } y \in \mathbb{R}^{N \times 1}, X \in \mathbb{R}^{N \times K}, Z \in \mathbb{R}^{N \times K} \quad (13.59)$$

13.2.1.3 Identification Hypotheses

Definition 13.2.1 (IV Hypotheses). 1. **Instrument Relevance:**

- *Scalar Formulation:* $\text{Cov}(z_i, x_i) \neq 0$. In the first-stage regression $x_i = \pi_0 + \pi_1 z_i + v_i$, we require $\pi_1 \neq 0$.
- *Matrix Formulation:* The cross-moments matrix $Q_{ZX} = \mathbb{E}[Z_i X_i']$ has full rank K .
- *Economic Intuition:* The instrument must have direct explanatory capacity over the endogenous regressor. Without relevance, the IV estimator is undefined (division by zero in the Wald formula).

2. **Instrument Exogeneity / Exclusion Restriction:**

- *Scalar Formulation:* $\text{Cov}(z_i, u_i) = 0$ (and $\mathbb{E}[u_i | z_i] = 0$).
- *Matrix Formulation:* $\mathbb{E}[Z_i u_i] = 0_{L \times 1}$.
- *Economic Intuition:* The instrument affects the dependent variable y_i **exclusively through the regressor** x_i , having no direct effect on y_i nor being correlated with unobserved determinants in u_i . This is the critical assumption that makes IV work—it must be justified by economic theory and institutional knowledge, not by statistical tests.

13.2.1.4 When to Use vs. When NOT to Use

Remark 13.2.2 (Decision Guide for IV). **WHEN TO USE:**

- When there is statistically confirmed endogeneity ($\mathbb{E}[X_i u_i] \neq 0$).
- When a valid instrument is available (relevant and exogenous).
- When the exclusion restriction can be credibly defended on economic/institutional grounds.

WHEN NOT TO USE:

- **DO NOT use** when X is strictly exogenous ($\mathbb{E}[X_i u_i] = 0$). OLS is asymptotically more efficient (lower variance) than IV under exogeneity; unnecessary use of IV reduces estimate precision.
- **DO NOT use** when instruments are weak ($\text{Cov}(z_i, x_i) \approx 0$), which invalidates hypothesis tests and inflates finite-sample bias.
- **DO NOT use** when the exclusion restriction is dubious. A slightly invalid instrument can generate estimates more biased than OLS.

13.2.1.5 Mathematical Derivation of the IV Estimator

Starting from the population equation $y_i = \beta_0 + \beta_1 x_i + u_i$, we take the population covariance of both sides with respect to z_i :

$$\text{Cov}(z_i, y_i) = \text{Cov}(z_i, \beta_0 + \beta_1 x_i + u_i) = \beta_1 \text{Cov}(z_i, x_i) + \text{Cov}(z_i, u_i) \quad (13.60)$$

By the instrument exogeneity hypothesis ($\text{Cov}(z_i, u_i) = 0$):

$$\text{Cov}(z_i, y_i) = \beta_1 \text{Cov}(z_i, x_i) \implies \beta_1 = \frac{\text{Cov}(z_i, y_i)}{\text{Cov}(z_i, x_i)} \quad (13.61)$$

Substituting with sample covariances:

$$\hat{\beta}_{1,IV} = \frac{\sum_{i=1}^N (z_i - \bar{z})(y_i - \bar{y})}{\sum_{i=1}^N (z_i - \bar{z})(x_i - \bar{x})} = \frac{\hat{\pi}_{21}}{\hat{\pi}_{11}} \quad (13.62)$$

where $\hat{\pi}_{21}$ is the reduced-form regression coefficient of y_i on z_i , and $\hat{\pi}_{11}$ is the first-stage regression coefficient of x_i on z_i (Indirect Least Squares ratio / ILS).

Remark 13.2.3 (Binary Instrument Case (Wald Estimator)). If $z_i \in \{0, 1\}$, the formula reduces to the ratio of mean differences:

$$\hat{\beta}_{\text{Wald}} = \frac{\bar{y}_1 - \bar{y}_0}{\bar{x}_1 - \bar{x}_0} = \frac{\mathbb{E}[y_i | z_i = 1] - \mathbb{E}[y_i | z_i = 0]}{\mathbb{E}[x_i | z_i = 1] - \mathbb{E}[x_i | z_i = 0]} \quad (13.63)$$

Remark 13.2.4 (Matrix Derivation for the Just-Identified Case ($L = K$)). Multiplying the matrix equation $y = X\beta + u$ on the left by Z' :

$$Z'y = Z'X\beta + Z'u \quad (13.64)$$

Since $Z'X$ is a $K \times K$ square matrix of full rank K , we multiply on the left by $(Z'X)^{-1}$:

$$(Z'X)^{-1}Z'y = (Z'X)^{-1}Z'X\beta + (Z'X)^{-1}Z'u \implies \hat{\beta}_{IV} = (Z'X)^{-1}Z'y \quad (13.65)$$

13.2.1.6 IV Estimator Consistency Proof

Theorem 13.2.1 (IV Consistency). *Substituting $y = X\beta + u$ into the IV estimator:*

$$\hat{\beta}_{IV} = (Z'X)^{-1}Z'(X\beta + u) = \beta + (Z'X)^{-1}Z'u = \beta + \left(\frac{1}{N}Z'X\right)^{-1} \left(\frac{1}{N}Z'u\right) \quad (13.66)$$

Taking the probability limit when $N \rightarrow \infty$:

$$\text{plim}_{N \rightarrow \infty} \hat{\beta}_{IV} = \beta + \left(\text{plim} \frac{1}{N} Z'X \right)^{-1} \left(\text{plim} \frac{1}{N} Z'u \right) \quad (13.67)$$

By the Weak Law of Large Numbers (WLLN):

1. $\text{plim} \frac{1}{N} Z'X = \mathbb{E}[Z_i X_i'] = Q_{ZX}$, non-singular by Relevance.
2. $\text{plim} \frac{1}{N} Z'u = \mathbb{E}[Z_i u_i] = 0_{K \times 1}$, by Exogeneity.

Thus:

$$\text{plim}_{N \rightarrow \infty} \hat{\beta}_{IV} = \beta + Q_{ZX}^{-1} \cdot 0 = \beta \quad (\text{The IV estimator is consistent}) \quad (13.68)$$

13.2.1.7 IV Asymptotic Variance

Under the conditional homoscedasticity hypothesis ($\text{Var}(u_i | z_i) = \sigma^2$), the asymptotic covariance matrix of IV is:

$$\text{Var}(\hat{\beta}_{IV} | Z) = \sigma^2 (Z'X)^{-1} (Z'Z) (X'Z)^{-1} \quad (13.69)$$

In the scalar case of a single regressor x_i and instrument z_i :

$$\text{Var}(\hat{\beta}_{1,IV}) = \frac{\sigma^2}{N \cdot \text{Var}(x) \cdot R_{x,z}^2} \quad (13.70)$$

where $R_{x,z}^2$ is the coefficient of determination from the first-stage regression of x_i on z_i .

Remark 13.2.5 (Analysis of IV Efficiency Loss). Since $R_{x,z}^2 \leq 1$, $\text{Var}(\hat{\beta}_{1,IV}) \geq \text{Var}(\hat{\beta}_{1,OLS}) = \frac{\sigma^2}{N \cdot \text{Var}(x)}$. When the instrument is weak ($R_{x,z}^2 \rightarrow 0$), the IV estimator variance explodes. This is the **price of correcting endogeneity**: IV is always less efficient than OLS when OLS is consistent, and dramatically so when instruments are weak.

13.2.2 2SLS (Two-Stage Least Squares — Overidentified Case)

13.2.2.1 Motivation and Economic Intuition

When we have more valid exogenous instruments than endogenous regressors ($L > K$), the system is **overidentified**. It is not possible to apply the traditional IV $(Z'X)^{-1}Z'Y$, since the $L \times K$ matrix $Z'X$ is not square.

The Two-Stage Least Squares estimator (**2SLS** or **TSLS**), developed by Basmann (1957) and Theil (1953), constructs the optimal linear combination of all L exogenous moment conditions that minimizes the estimator variance in the instrumental variables framework.

Remark 13.2.6 (Economic Intuition: Combining Multiple Instruments). When multiple instruments are available, 2SLS finds the “best” way to combine them into a single synthetic instrument $\hat{X}$ that captures all the exogenous variation in X . This is analogous to constructing a portfolio of assets that maximizes the signal-to-noise ratio. The first stage projects X onto the space spanned by Z , extracting only the variation in X that is orthogonal to the error u .

13.2.2.2 Equivalent Representations of 2SLS

The 2SLS estimator has 4 algebraically equivalent formulations:

1. **Orthogonal Projection of X onto Z :** Define the symmetric and idempotent projection matrix of Z :

$$P_Z = Z(Z'Z)^{-1}Z' \quad (P_Z' = P_Z \text{ and } P_Z P_Z = P_Z) \quad (13.71)$$

Calculate predicted values of regressors projected onto the space of Z :

$$\hat{X} = P_Z X = Z(Z'Z)^{-1}Z'X \quad (13.72)$$

2. **Second Stage (Regression of Y on $\hat{X}$):**

$$\hat{\beta}_{2SLS} = (\hat{X}'\hat{X})^{-1}\hat{X}'Y = (X'P_Z X)^{-1}X'P_Z Y = [X'Z(Z'Z)^{-1}Z'X]^{-1}X'Z(Z'Z)^{-1}Z'Y \quad (13.73)$$

3. **Generated Instruments in IV Framework:** Using $\hat{X}$ as the instrument matrix for X : $\hat{\beta}_{2SLS} = (\hat{X}'X)^{-1}\hat{X}'Y$.
4. **GMM Quadratic Form:** It is the GMM estimator with weight matrix $W = (Z'Z)^{-1}$ or $(Z'Z/N)^{-1}$.

13.2.2.3 Critical Alert: Invalidation of Standard Errors in Manual 2nd Stage Estimation

When calculating 2SLS by executing the two stages manually via two consecutive OLS commands:

1. *1st Stage:* Regress each column of X on Z to obtain $\hat{X}$.
2. *2nd Stage:* Regress Y on $\hat{X}$ by OLS.

The generated point estimator $\hat{\beta}_{2SLS}$ is perfectly correct, but **the standard errors, t -statistics, and F -tests reported by OLS in the 2nd stage are completely invalid.**

Remark 13.2.7 (The Algebraic Reason). The second-stage OLS calculates residual variance based on residuals from the transformed model:

$$\hat{e}^* = Y - \hat{X}\hat{\beta}_{2SLS} \quad (13.74)$$

However, the variance of the structural model equation depends on the true equation error:

$$u = Y - X\beta \implies \hat{u}_{2SLS} = Y - X\hat{\beta}_{2SLS} \quad (13.75)$$

Since $\hat{X} \neq X$, $\hat{e}^* \neq \hat{u}_{2SLS}$, causing the estimate $\hat{\sigma}_{\text{manual}}^2 = \frac{\hat{e}^{*'}\hat{e}^*}{N-K}$ to be a biased estimator of the structural error variance. Econometric programs correct this automatically by calculating $\hat{\sigma}^2$ with the original structural residuals $\hat{u}_{2SLS}$.

13.2.2.4 Fundamental Tests in IV / 2SLS

Remark 13.2.8 (a) Durbin-Wu-Hausman Endogeneity Test). • **Objective:** Evaluate whether suspected regressors X_2 are actually endogenous.

- **Null Hypothesis (H_0):** $\mathbb{E}[X_i u_i] = 0$ (Regressors are exogenous; OLS is unbiased, consistent, and efficient).
- **Alternative Hypothesis (H_1):** $\mathbb{E}[X_i u_i] \neq 0$ (Regressors are endogenous; OLS is inconsistent, 2SLS is necessary).
- **Wu Auxiliary Regression Procedure:**
 1. Estimate the first stage of the endogenous regressor by OLS: $X_{2i} = Z_i \Pi + v_i$, extracting residuals $\hat{v}_i$.
 2. Add the residual vector $\hat{v}_i$ as an additional explanatory regressor in the original structural equation and estimate by OLS: $Y_i = X_{1i} \beta_1 + X_{2i} \beta_2 + \rho \hat{v}_i + e_i$.
- **Decision Rule:** Perform a traditional t -test on coefficient $\hat{\rho}$. Reject H_0 if p -value $\leq \alpha$. Rejection confirms that X_2 is endogenous and validates the need for the 2SLS estimator.

Remark 13.2.9 (b) Sargan Overidentification Test (under Homoscedasticity)). • **Requirement:** Overidentified model ($L > K$, where $L - K$ are the test degrees of freedom).

- **Null Hypothesis (H_0):** $\mathbb{E}[Z_i u_i] = 0_{L \times 1}$ (All L instruments are exogenous and valid).
- **Alternative Hypothesis (H_1):** At least one instrument is invalid ($\text{Cov}(z_{li}, u_i) \neq 0$) or the structural equation is misspecified.
- **Sargan Test Mechanics:**
 1. Extract structural residuals from 2SLS: $\hat{u}_i = Y_i - X_i' \hat{\beta}_{2\text{SLS}}$.
 2. Regress $\hat{u}_i$ on the complete matrix of L instruments Z : $\hat{u}_i = Z_i \gamma + e_i$.
- **Statistic and Decision:**

$$\text{Sargan} = N \cdot R_{\text{aux}}^2 \xrightarrow{d} \chi_{L-K}^2 \quad (13.76)$$

Reject H_0 at significance level α if $\text{Sargan} > \chi_{\alpha, L-K}^2$. Rejection invalidates the instrument set.

Remark 13.2.10 (c) Weak Instrument Test (1st Stage F -Statistic and Stock-Yogo)). • **Objective:** Diagnose whether the correlation between excluded instruments Z_2 and the endogenous regressor X_2 is too weak.

- **Null Hypothesis (H_0):** Instruments are weak (2SLS bias exceeds $b\%$ of OLS bias, or the Wald test rejection rate exceeds 10%).
- **Statistic:** F -statistic of joint significance of excluded instruments in the first-stage regression.
- **Decision Rules:**

- *Traditional Rule of Thumb (Staiger-Stock)*: Reject H_0 if $F > 10$.
- *Stock-Yogo Rule*: Compare the calculated F -statistic with critical values tabulated by Stock and Yogo based on bias limits and size distortion.
- *Consequence of Non-Rejection*: If $F < 10$, instruments are weak; 2SLS suffers from severe bias in the direction of OLS and traditional t -statistics lose the Gaussian asymptotic distribution.

13.2.3 GMM (Generalized Method of Moments) — General Framework

13.2.3.1 Motivation and Econometric Unification

Formulated in a paper by Lars Peter Hansen (1982), the Generalized Method of Moments (GMM) is the semi-parametric unifying framework of modern econometrics. GMM encompasses OLS, 2SLS, SUR, and 3SLS as special cases.

Its main motivation relative to 2SLS is to provide an estimator that achieves **optimal asymptotic efficiency under conditional heteroscedasticity of unknown form** or error autocorrelation.

Remark 13.2.11 (Economic Intuition: Moment Conditions as Economic Theory). GMM translates economic theory directly into estimation. Each theoretical restriction (e.g., “the instrument is uncorrelated with the error”) becomes a **moment condition** $E[g(w_i, \theta)] = 0$. The estimator finds the parameter values that make the sample moments as close to zero as possible. This is more flexible than MLE because it does not require specifying the full distribution of the data—only the moments that economic theory predicts.

13.2.3.2 Notation and Formal Moment Structure

Let $w_i = (y_i, x_i, z_i)$ be the vector of observed variables for unit i . The population model imposes a set of L orthogonal moment conditions indexed by the parameter vector of interest $\beta_0 \in \mathbb{R}^{K \times 1}$:

$$g(\beta_0) = \mathbb{E}[g_i(\beta_0)] = \mathbb{E}[m(w_i, \beta_0)] = 0_{L \times 1} \quad (13.77)$$

In the linear regression model with instruments $Z_i \in \mathbb{R}^{L \times 1}$, the moment function is the product of the instrument vector by the structural error $u_i(\beta) = Y_i - X_i' \beta$:

$$m(w_i, \beta) = Z_i(Y_i - X_i' \beta) \implies \mathbb{E}[Z_i(Y_i - X_i' \beta_0)] = 0_{L \times 1} \quad (13.78)$$

Definition 13.2.2 (Formal GMM Hypotheses). 1. **Identification:** $\mathbb{E}[m(w_i, \beta)] = 0 \iff \beta = \beta_0$.

2. **Rank Condition:** $\text{rank}(G) = K \leq L$, where $G = \mathbb{E} \left[\frac{\partial m_i(\beta_0)}{\partial \beta'} \right] = -\mathbb{E}[Z_i X_i']$ is the $L \times K$ Jacobian matrix.

3. **Existence of S :** The asymptotic covariance matrix of moments $S = \text{Avar}(\sqrt{N} \bar{g}(\beta_0)) = \mathbb{E}[m(w_i, \beta_0) m(w_i, \beta_0)'] = \mathbb{E}[u_i^2 Z_i Z_i']$ is positive definite and invertible.

13.2.3.3 Objective Function Derivation and GMM FOC

The sample average of moment conditions is given by:

$$g_N(\beta) = \frac{1}{N} \sum_{i=1}^N m(w_i, \beta) = \frac{1}{N} Z'(Y - X\beta) \quad (13.79)$$

If the system is overidentified ($L > K$), it is not possible to find a β that zeros all L equations simultaneously. The GMM estimator minimizes the quadratic norm of the weighted distance of moments from the origin:

$$Q_N(\beta) = g_N(\beta)' W_N g_N(\beta) \quad (13.80)$$

where W_N is an $L \times L$ symmetric and positive definite weight matrix.

Substituting $g_N(\beta)$:

$$Q_N(\beta) = \frac{1}{N^2} (Y - X\beta)' Z W_N Z' (Y - X\beta) \quad (13.81)$$

Deriving $Q_N(\beta)$ with respect to vector β and setting equal to the null vector:

$$\frac{\partial Q_N(\beta)}{\partial \beta} = -\frac{2}{N^2} X' Z W_N Z' (Y - X\beta) = 0_{K \times 1} \quad (13.82)$$

$$X' Z W_N Z' X \hat{\beta}_{\text{GMM}} = X' Z W_N Z' Y \quad (13.83)$$

Inverting the $K \times K$ matrix $X' Z W_N Z' X$:

$$\boxed{\hat{\beta}_{\text{GMM}}(W_N) = (X' Z W_N Z' X)^{-1} X' Z W_N Z' Y} \quad (13.84)$$

13.2.3.4 Optimal Weight Matrix and Efficient 2-Stage GMM Estimator

The asymptotic variance of $\hat{\beta}_{\text{GMM}}(W_N)$ depends on the choice of matrix W_N :

$$\text{Avar}(\hat{\beta}_{\text{GMM}}) = \frac{1}{N} (G' W G)^{-1} (G' W S W G) (G' W G)^{-1} \quad (13.85)$$

Theorem 13.2.2 (Optimal Weight Matrix Theorem (Hansen, 1982)). *The weight matrix W^* that minimizes the asymptotic covariance matrix of the GMM estimator in the positive definite matrix sense is the inverse of the moment conditions covariance matrix:*

$$W^* = S^{-1} \quad \text{where } S = \mathbb{E}[u_i^2 Z_i Z_i'] \quad (13.86)$$

Substituting $W_N = S^{-1}$, the asymptotic variance reaches the semi-parametric efficiency lower bound:

$$\text{Avar}(\hat{\beta}_{\text{GMM}}^*) = \frac{1}{N} (G' S^{-1} G)^{-1} = \frac{1}{N} (X' Z S^{-1} Z' X)^{-1} \quad (13.87)$$

Remark 13.2.12 (Two-Step Estimation Procedure (Two-Step Efficient GMM)). 1.

Step 1 (Initial Stage): Obtain a consistent preliminary estimator $\hat{\beta}_1$ using a suboptimal non-stochastic weight matrix, typically $W_1 = (Z'Z/N)^{-1}$ (which corresponds to the 2SLS estimator).

2. **Step 2 (Calculation of Matrix $\hat{S}$):** Extract first-stage residuals $\hat{u}_i = Y_i - X_i'\hat{\beta}_1$ and construct the consistent estimate of the White/Eicker moment covariance matrix:

$$\hat{S} = \frac{1}{N} \sum_{i=1}^N \hat{u}_i^2 Z_i Z_i' \quad (13.88)$$

3. **Step 3 (Efficient GMM Estimator):** Invert $\hat{S}$ to form $W^* = \hat{S}^{-1}$ and re-estimate the model:

$$\hat{\beta}_{\text{GMM}}^* = \left(X'Z\hat{S}^{-1}Z'X \right)^{-1} X'Z\hat{S}^{-1}Z'Y \quad (13.89)$$

13.2.3.5 Iterated GMM and Continuous Updating GMM (CU-GMM)

- **Iterated GMM:** Recalculate residuals $\hat{u}_i$ with the Step 3 estimate $\hat{\beta}_{\text{GMM}}^*$, update $\hat{S}$, and re-optimize the parameter vector successively until convergence of $\hat{\beta}$. It removes estimator dependence on the initial weight matrix choice W_1 .
- **Continuous Updating GMM (CU-GMM / Hansen-Heaton-Yaron):** Instead of keeping $\hat{S}$ fixed during second-stage minimization, minimize the quadratic objective function by continuously updating the covariance matrix $S(\beta)$ for each candidate value of β :

$$\hat{\beta}_{\text{CU-GMM}} = \arg \min_{\beta} g_N(\beta)' \left(\frac{1}{N} \sum_{i=1}^N u_i(\beta)^2 Z_i Z_i' \right)^{-1} g_N(\beta) \quad (13.90)$$

Advantage: CU-GMM is invariant to normalizations/reparameterizations and has substantially lower finite-sample bias than 2-step GMM in models with weak or numerous instruments.

13.2.3.6 GMM Collapse and Equivalence Theorems

Theorem 13.2.3 (Theorem 1: Efficient GMM Collapse to 2SLS under Conditional Homoscedasticity). *If errors are conditionally homoscedastic ($\mathbb{E}[u_i^2 | Z_i] = \sigma^2$), the efficient 2-step GMM estimator collapses and reduces exactly to the classical 2SLS estimator.*

Proof. Under conditional homoscedasticity, matrix S simplifies to:

$$S = \mathbb{E}[u_i^2 Z_i Z_i'] = \mathbb{E}[\mathbb{E}[u_i^2 | Z_i] Z_i Z_i'] = \sigma^2 \mathbb{E}[Z_i Z_i'] \quad (13.91)$$

The consistent estimate of S is $\hat{S} = \hat{\sigma}^2 \left(\frac{Z'Z}{N} \right)$. Its inverse is $\hat{S}^{-1} = \frac{1}{\hat{\sigma}^2} \cdot N(Z'Z)^{-1}$.

Substituting $\hat{S}^{-1}$ into the Efficient GMM formula:

$$\hat{\beta}_{\text{GMM}}^* = \left(X'Z \left[\frac{N}{\hat{\sigma}^2} (Z'Z)^{-1} \right] Z'X \right)^{-1} X'Z \left[\frac{N}{\hat{\sigma}^2} (Z'Z)^{-1} \right] Z'Y \quad (13.92)$$

Factoring the scalar $\frac{N}{\sigma^2}$ outside parentheses and inversions:

$$\hat{\beta}_{\text{GMM}}^* = \left(X'Z(Z'Z)^{-1}Z'X \right)^{-1} X'Z(Z'Z)^{-1}Z'Y \equiv \hat{\beta}_{2\text{SLS}} \quad (13.93)$$

□

Theorem 13.2.4 (Theorem 2: Equivalence in Just-Identified Models ($L = K$)). *If the number of moment conditions exactly equals the number of parameters ($L = K$), the GMM estimator is numerically identical to the traditional IV estimator, regardless of the weight matrix W_N used.*

Proof. If $L = K$, matrix $Z'X$ has dimension $K \times K$ and is invertible. Decomposing the inverse of the product in $\hat{\beta}_{\text{GMM}}$:

$$\hat{\beta}_{\text{GMM}} = (X'ZW_NZ'X)^{-1} X'ZW_NZ'Y = (Z'X)^{-1}W_N^{-1}(X'Z)^{-1}X'ZW_NZ'Y \quad (13.94)$$

Since $(X'Z)^{-1}X'Z = I_K$ and $W_N^{-1}W_N = I_K$:

$$\hat{\beta}_{\text{GMM}} = (Z'X)^{-1}I_KI_KZ'Y = (Z'X)^{-1}Z'Y \equiv \hat{\beta}_{\text{IV}} \quad (13.95)$$

□

13.2.3.7 Hansen J -Test (Overidentification Restrictions in GMM)

- **Requirement:** Overidentified model ($L > K$) and use of optimal weight matrix $\hat{S}^{-1}$.
- **Null Hypothesis (H_0):** $\mathbb{E}[m(w_i, \beta_0)] = 0_{L \times 1}$ (All L moment conditions are statistically valid and orthogonal to the error).
- **Alternative Hypothesis (H_1):** At least one moment condition is invalid ($\mathbb{E}[Z_i u_i] \neq 0$) or the structural model is incorrectly specified.

J -Test Statistic:

$$J = N \cdot g_N(\hat{\beta}_{\text{GMM}}^*)' \hat{S}^{-1} g_N(\hat{\beta}_{\text{GMM}}^*) \xrightarrow{d} \chi_{L-K}^2 \quad (13.96)$$

where $g_N(\hat{\beta}_{\text{GMM}}^*) = \frac{1}{N}Z'(Y - X\hat{\beta}_{\text{GMM}}^*)$ is the vector of sample moments evaluated at the efficient GMM estimator.

Decision Rule: Reject H_0 at significance level α if $J > \chi_{\alpha, L-K}^2$. Rejection indicates that instruments fail the exclusion restriction or that the structural model suffers from specification error.

Remark 13.2.13 (Note). Under conditional homoscedasticity, Hansen's J -statistic reduces exactly to the 2SLS Sargan overidentification statistic $N \cdot R_{\text{aux}}^2$.

13.2.4 Synthetic Comparative Framework (IV vs. 2SLS vs. GMM)

Remark 13.2.14 (Comparative Summary: IV, 2SLS, and Efficient GMM). **IV (Just-Identified):** Identification condition is exact ($L = K$); weight matrix is irrelevant (cancels

out); hypothesis about errors is homoscedastic or heteroscedastic; asymptotic efficiency is less than 2SLS under homoscedasticity; overidentification test is not applicable ($df = 0$).

2SLS (Overidentified): Identification condition is overidentified ($L > K$); weight matrix is $W = (Z'Z)^{-1}$; hypothesis about errors is homoscedastic ($\mathbb{E}[u_i^2 | Z_i] = \sigma^2$); asymptotic efficiency is optimal under homoscedasticity; overidentification test is Sargan / Basman (requires homoscedasticity).

Efficient GMM (2-Step): Identification condition is overidentified ($L > K$); weight matrix is $W^* = \hat{S}^{-1} = \left(\frac{1}{N} \sum \hat{u}_i^2 Z_i Z_i'\right)^{-1}$; hypothesis about errors is heteroscedastic of unknown form ($\mathbb{E}[u_i^2 | Z_i] = \sigma^2(Z_i)$); asymptotic efficiency is **optimal under heteroscedasticity** (attains semi-parametric efficiency bound); overidentification test is Hansen's J -test (robust to heteroscedasticity).

Remark 13.2.15 (Decision Framework: IV vs. 2SLS vs. GMM). 1. **Exactly identified ($L = K$)?** $\implies$ Use **IV**. 2SLS and GMM collapse to IV.

2. **Overidentified ($L > K$) with homoscedastic errors?** $\implies$ Use **2SLS**. It is optimal under homoscedasticity and simpler to compute.

3. **Overidentified ($L > K$) with heteroscedasticity of unknown form?** $\implies$ Use **Efficient 2-Step GMM**. It attains the semi-parametric efficiency bound.

4. **Concerned about finite-sample bias with many instruments?** $\implies$ Use **CU-GMM** or **Iterated GMM** to reduce dependence on initial weight matrix.

5. **Always test:** First-stage $F > 10$ (weak instruments), Hansen J -test (overidentification), Durbin-Wu-Hausman (endogeneity).

Remark 13.2.16 (Final Remarks on IV/2SLS/GMM). 1. **Exclusion Restriction is King:** No statistical test can validate the exclusion restriction. It must be defended on economic, institutional, and theoretical grounds. A clever instrument with a dubious exclusion restriction is worse than no instrument at all.

2. **Weak Instruments are a Serious Problem:** Even with a valid exclusion restriction, weak instruments generate severe finite-sample bias toward OLS and invalid inference. Always report first-stage F -statistics and consider Stock-Yogo critical values.

3. **LATE Interpretation:** In the presence of heterogeneous treatment effects, IV/2SLS/GMM identify a **Local Average Treatment Effect (LATE)** for the subpopulation of *Compliers*, not the global ATE. Be precise about the interpretation.

4. **GMM is the Modern Standard:** For overidentified models with potential heteroscedasticity, efficient 2-step GMM is the state-of-the-art estimator. It subsumes 2SLS as a special case under homoscedasticity.

5. **Report Everything:** Always present first-stage results, weak instrument tests, overidentification tests, and sensitivity analysis. Transparency is essential for credible IV estimation.

13.3 Panel Data: POLS, FGLS, FE, FD, RE, and Dynamic Panel

13.3.1 POLS (Pooled Ordinary Least Squares)

13.3.1.1 Motivation and Economic Intuition

In panel data, we follow N individual units over T time periods ($i = 1, \dots, N; t = 1, \dots, T$), totaling NT observations. POLS ignores the panel structure and treats all NT observations as independent, applying classical OLS to the stacked data.

Remark 13.3.1 (When to Use POLS). POLS is appropriate when:

- Unobserved individual heterogeneity c_i is **uncorrelated** with regressors ($\mathbb{E}[c_i | X_{it}] = 0$)
- You want to estimate coefficients of **time-invariant variables** (gender, race, initial education)
- You are willing to use **cluster-robust standard errors** to correct inference

13.3.1.2 Panel Data Structure and Composite Error

The static panel model is given by:

$$y_{it} = x'_{it}\beta + v_{it}, \quad \text{where } v_{it} = c_i + u_{it} \quad (13.97)$$

where:

- c_i is the **unobserved fixed effect / individual heterogeneity** (constant over time for unit i).
- u_{it} is the **idiosyncratic error** (varies over time and between units).
- v_{it} is the **composite error term**.

13.3.1.3 Identification Hypotheses

Definition 13.3.1 (POLS Hypotheses). 1. **Orthogonal Mean Exogeneity:**

$$\mathbb{E}[x_{it}v_{it}] = 0 \iff \mathbb{E}[x_{it}c_i] = 0 \quad \text{and} \quad \mathbb{E}[x_{it}u_{it}] = 0 \quad (13.98)$$

Economic Intuition: Unobserved individual heterogeneity c_i (ability, management quality, culture) must be uncorrelated with regressors. If $\mathbb{E}[x_{it}c_i] \neq 0$, POLS suffers from **omitted variable bias** and becomes inconsistent.

2. **No Perfect Multicollinearity in Stacked Sample:**

$$\text{rank} \left(\sum_{i=1}^N \sum_{t=1}^T x_{it}x'_{it} \right) = K \quad (13.99)$$

13.3.1.4 POLS Matrix Notation

The **Pooled OLS (POLS)** estimator ignores the panel structure (the individual dimension i and temporal t) and stacks all NT observations, applying the classical OLS formula:

$$\hat{\beta}_{\text{POLS}} = (X'X)^{-1} X'y = \left(\sum_{i=1}^N \sum_{t=1}^T x_{it}x'_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T x_{it}y_{it} \right) \quad (13.100)$$

13.3.1.5 What Breaks if $\mathbb{E}(x_{it}c_i) \neq 0$ (POLS Failure)

Remark 13.3.2 (Unobserved Heterogeneity Bias). In microeconometrics of firms or individuals, unobserved heterogeneity c_i (ability, management capacity, technology) is almost always correlated with regressors x_{it} (schooling, capital, inputs).

Calculating the probability limit of $\hat{\beta}_{\text{POLS}}$ when $\mathbb{E}[x_{it}c_i] \neq 0$:

$$\text{plim}_{N \rightarrow \infty} \hat{\beta}_{\text{POLS}} = \beta + \left(\text{plim}_{N \rightarrow \infty} \frac{1}{NT} X'X \right)^{-1} \left(\text{plim}_{N \rightarrow \infty} \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T x_{it}c_i \right) \neq \beta \quad (13.101)$$

This results in the **Unobserved Heterogeneity Omitted Variable Bias** (or *Heterogeneity Bias*). In these cases, POLS is unviable, requiring **Fixed Effects (FE / Within)** or **First Differences (FD)** estimators to eliminate c_i .

13.3.1.6 POLS Error Covariance Matrix Structure

Even assuming c_i is uncorrelated with x_{it} ($\mathbb{E}[x_{it}c_i] = 0$), using POLS faces a second serious problem in **inference**.

Assume $c_i \sim \text{i.i.d.}(0, \sigma_c^2)$ and $u_{it} \sim \text{i.i.d.}(0, \sigma_u^2)$, mutually independent. Calculating the variance and covariance of the composite error $v_{it} = c_i + u_{it}$ for the same individual i :

1. Contemporaneous Variance:

$$\text{Var}(v_{it} \mid X_i) = \text{Var}(c_i + u_{it} \mid X_i) = \sigma_c^2 + \sigma_u^2 \quad (13.102)$$

2. Intertemporal Covariance ($t \neq s$) for the same unit i :

$$\text{Cov}(v_{it}, v_{is} \mid X_i) = \mathbb{E}[(c_i + u_{it})(c_i + u_{is}) \mid X_i] = \mathbb{E}[c_i^2 \mid X_i] = \sigma_c^2 \neq 0 \quad (13.103)$$

This implies that **errors from the same unit i in different periods are positively correlated over time** ($\text{Corr}(v_{it}, v_{is}) = \frac{\sigma_c^2}{\sigma_c^2 + \sigma_u^2} > 0$).

The covariance matrix of the error for unit i ($v_i = [v_{i1}, \dots, v_{iT}]' \in \mathbb{R}^{T \times 1}$) has $T \times T$ block structure:

$$\Omega_i = \mathbb{E}[v_i v_i' \mid X_i] = \sigma_u^2 I_T + \sigma_c^2 J_T \quad (13.104)$$

where $J_T = \mathbf{1}_T \mathbf{1}_T'$ is the $T \times T$ matrix of ones.

Stacking for the entire panel, the global covariance matrix is block-diagonal:

$$\Omega = I_N \otimes \Omega_i \neq \sigma^2 I_{NT} \quad (13.105)$$

13.3.1.7 Overwhelming Consequence on Standard POLS Inference

Remark 13.3.3 (Invalidation of Conventional Standard Errors). If the researcher runs POLS and uses the conventional OLS standard error formula pre-installed in software ($\widehat{\text{Var}}_{\text{standard}}(\hat{\beta}_{\text{POLS}}) = s^2(X'X)^{-1}$), the formula incorrectly assumes $\Omega = \sigma^2 I_{NT}$.

The **true asymptotic variance** of POLS under group dependence is given by the “sandwich” structure:

$$\text{Var}_{\text{real}}(\hat{\beta}_{\text{POLS}} | X) = (X'X)^{-1} \left(\sum_{i=1}^N X_i' \Omega_i X_i \right) (X'X)^{-1} \quad (13.106)$$

Since $\sigma_c^2 > 0$, ignoring the intertemporal serial correlation term Ω_i causes the conventional OLS formula to **severely underestimate coefficient standard errors**. As a result:

- Conventionally calculated $t = \frac{\hat{\beta}}{\text{SE}(\hat{\beta})}$ statistics become **substantially inflated**.
- p -values become **falsely tiny**.
- The **Type I Error rate explodes**, leading the researcher to declare highly “significant” effects when it is merely autocorrelated noise.

13.3.1.8 Inferential Solution: Cluster-Robust Standard Errors

If the researcher chooses to use POLS, it is **mandatory** to correct standard errors using the Individual Cluster-Robust Covariance Matrix (*Cluster-Robust Variance Estimator* — *CRVE* by Liang-Zeger / Arellano / Wooldridge):

$$\widehat{\text{Var}}_{\text{cluster}}(\hat{\beta}_{\text{POLS}}) = (X'X)^{-1} \left(\sum_{i=1}^N X_i' \hat{v}_i \hat{v}_i' X_i \right) (X'X)^{-1} \quad (13.107)$$

where $\hat{v}_i = y_i - X_i \hat{\beta}_{\text{POLS}}$ is the $T \times 1$ vector of POLS residuals for unit i .

This formula does not require the researcher to specify the parametric form of Ω_i ; it consistently adjusts variance for any form of heteroscedasticity and intertemporal autocorrelation within unit i when $N \rightarrow \infty$.

Remark 13.3.4 (Advantages and Limitations of POLS). **Advantages:**

- Allows estimation of coefficients for **time-invariant variables** (gender, race, geography)
- Simple to implement and interpret
- Consistent if $\mathbb{E}[x_{it}c_i] = 0$

Limitations:

- **Inconsistent** if $\mathbb{E}[x_{it}c_i] \neq 0$ (omitted variable bias from unobserved heterogeneity)
- **Inefficient** relative to Random Effects (RE) when $\mathbb{E}[c_i | X_i] = 0$
- Requires **cluster-robust standard errors** for valid inference

13.3.2 FGLS (Feasible Generalized Least Squares)

13.3.2.1 Motivation and Intuition in Panel Context and Structured Errors

The Generalized Least Squares (GLS) estimator is the Gauss-Markov solution for linear models where the conditional variance-covariance matrix of errors given regressors, $\text{Var}(u | X) = \Omega$, is not spherical ($\Omega \neq \sigma^2 I$). In the presence of arbitrary heteroscedasticity between individuals or temporal autocorrelation within each individual, OLS/POLS loses asymptotic efficiency and generates incorrect standard errors.

Since the true matrix Ω depends on unknown population scale/variance parameters, **FGLS** replaces Ω with a consistent estimate $\hat{\Omega}$ constructed in two stages.

13.3.2.2 Mathematical Formulation and Estimator Derivation

Consider the linear model $y = X\beta + u$, with $y \in \mathbb{R}^{M \times 1}$, $X \in \mathbb{R}^{M \times K}$, and $\text{Var}(u | X) = \Omega$, where $M = N \cdot T$ in panel data.

Remark 13.3.5 (1. The Infeasible Estimator (GLS)). If Ω were known, applying the Cholesky decomposition $\Omega^{-1} = P'P$ (where P is a non-singular transformation matrix):

$$Py = PX\beta + Pu \implies y^* = X^*\beta + u^* \quad (13.108)$$

It is verified that $\text{Var}(u^* | X) = \text{Var}(Pu | X) = P \text{Var}(u | X) P' = P\Omega P' = P(P'P)^{-1}P' = I_M$.

Applying OLS to the transformed model $y^* = X^*\beta + u^*$:

$$\hat{\beta}_{\text{GLS}} = (X^{*'}X^*)^{-1}X^{*'}y^* = (X'P'PX)^{-1}X'P'Py = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y \quad (13.109)$$

Remark 13.3.6 (2. The Feasible Estimator (2-Stage FGLS)). Since $\Omega = \Omega(\theta)$ depends on a parameter vector θ , the FGLS procedure operates as follows:

1. **1st Stage:** Estimate the model by OLS/POLS, obtaining consistent residuals $\hat{u}_i = y_i - X_i\hat{\beta}_{\text{OLS}}$.
2. **Estimation of Ω :** Use residuals $\hat{u}_i$ to consistently estimate elements of θ , forming $\hat{\Omega} = \Omega(\hat{\theta})$.
3. **2nd Stage (FGLS):**

$$\hat{\beta}_{\text{FGLS}} = (X'\hat{\Omega}^{-1}X)^{-1}X'\hat{\Omega}^{-1}y \quad (13.110)$$

Theorem 13.3.1 (Asymptotic Equivalence Theorem). *If $\text{plim}_{N \rightarrow \infty} \frac{1}{N}X'(\hat{\Omega}^{-1} - \Omega^{-1})u = 0$, then $\sqrt{N}(\hat{\beta}_{\text{FGLS}} - \beta)$ has the same Gaussian asymptotic distribution as $\sqrt{N}(\hat{\beta}_{\text{GLS}} - \beta)$, with asymptotic variance:*

$$\text{Avar}(\hat{\beta}_{\text{FGLS}}) = (X'\Omega^{-1}X)^{-1} \quad (13.111)$$

13.3.2.3 FGLS Identification Hypotheses and Limitations

- **Strict Exogeneity / Orthogonality:** $\mathbb{E}[u_{it} | X_i] = 0$ (or $\mathbb{E}[X_i'\Omega^{-1}u_i] = 0$).
- **Rank Condition:** $\text{rank}(X'\hat{\Omega}^{-1}X) = K$.

- **Inconsistency Risk:** If strict exogeneity fails (e.g., in dynamic models with predetermined or lagged regressors), the transformation by $\hat{\Omega}^{-1}$ mixes past and future errors with regressors, rendering FGLS **inconsistent**. In these cases, prefer POLS/FE with cluster-robust standard errors (*CRVE*).

13.3.3 FE (Fixed Effects — Within Estimator)

13.3.3.1 Motivation and the Unobserved Heterogeneity Problem

Consider the static panel data model:

$$y_{it} = x'_{it}\beta + c_i + u_{it}, \quad i = 1, \dots, N; \quad t = 1, \dots, T \quad (13.112)$$

where c_i represents the **time-invariant unobserved individual heterogeneity** (e.g., individual's innate ability, firm's organizational culture).

If c_i is correlated with regressors x_{it} ($\mathbb{E}[x_{it}c_i] \neq 0$), POLS and RE estimators suffer from **heterogeneity omitted variable bias**. The **Fixed Effects (FE)** estimator solves this problem by applying an orthogonal transformation that algebraically eliminates c_i from the equation before estimation.

13.3.3.2 The Within Transformation (Centering on Individual Mean)

Calculating the temporal mean for each individual i :

$$\bar{y}_i = \bar{x}'_i\beta + c_i + \bar{u}_i \quad \text{where } \bar{y}_i = \frac{1}{T} \sum_{t=1}^T y_{it}, \quad \bar{x}_i = \frac{1}{T} \sum_{t=1}^T x_{it}, \quad \bar{u}_i = \frac{1}{T} \sum_{t=1}^T u_{it} \quad (13.113)$$

Subtracting the temporal mean from the original equation for each period t :

$$(y_{it} - \bar{y}_i) = (x_{it} - \bar{x}_i)'\beta + (c_i - c_i) + (u_{it} - \bar{u}_i) \quad (13.114)$$

$$\ddot{y}_{it} = \ddot{x}'_{it}\beta + \ddot{u}_{it} \quad (13.115)$$

where $\ddot{y}_{it} = y_{it} - \bar{y}_i$, $\ddot{x}_{it} = x_{it} - \bar{x}_i$, and $\ddot{u}_{it} = u_{it} - \bar{u}_i$ are the time-demeaned variables. Note that $c_i - c_i = 0$, completely eliminating unobserved individual heterogeneity.

13.3.3.3 Matrix Derivation of the FE Estimator

Define the Within projection matrix $M_Q = I_T - \frac{1}{T}\mathbf{1}_T\mathbf{1}'_T$, symmetric and idempotent ($M'_Q = M_Q$ and $M_Q M_Q = M_Q$).

For each individual i , $\ddot{y}_i = M_Q y_i$ and $\ddot{X}_i = M_Q X_i$.

The Fixed Effects estimator is obtained by applying OLS to the transformed data:

$$\hat{\beta}_{\text{FE}} = (\ddot{X}'\ddot{X})^{-1}\ddot{X}'\ddot{y} = \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it}\ddot{x}'_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it}\ddot{y}_{it} \right) \quad (13.116)$$

13.3.3.4 FE Estimator Identification Hypotheses

Definition 13.3.2 (FE Hypotheses). 1. **Strict Idiosyncratic Exogeneity:**

$$\mathbb{E}[u_{it} \mid X_i, c_i] = 0, \quad \forall t = 1, \dots, T \quad (13.117)$$

The idiosyncratic error u_{it} must be uncorrelated with regressors from **all** time periods (past, present, and future).

2. **Full Rank of the Transformed Matrix:**

$$\text{rank} \left(\sum_{i=1}^N \ddot{X}_i' \ddot{X}_i \right) = K \quad (13.118)$$

Consequence: If a variable in x_{it} is **time-invariant** (e.g., gender, race, years of schooling of an adult), we have $x_{it} = \bar{x}_i \implies \ddot{x}_{it} = 0$. Time-invariant variables are eliminated by the Within transformation and their coefficients **cannot be identified by the FE estimator**.

13.3.3.5 Advantages, Disadvantages, and Fixed Effects F -Test

- **Advantages:** Allows tying/correlating unobserved heterogeneity c_i with regressors x_{it} in a totally arbitrary way without generating bias.
- **Disadvantages:**
 - Does not estimate the effect of time-invariant regressors.
 - Consumes N degrees of freedom to eliminate c_i .
 - If the strict exogeneity hypothesis fails (e.g., feedback/predetermination or dynamic lag $y_{i,t-1}$), $\hat{\beta}_{\text{FE}}$ suffers from **Nickell Bias** ($O(1/T)$).

Remark 13.3.7 (F -Test for the Presence of Individual Fixed Effects). • **Null Hypothesis:** $H_0 : c_1 = c_2 = \dots = c_N = c$ (All individual effects are equal; POLS is adequate).

- **Test Statistic:**

$$F = \frac{(SSR_{\text{POLS}} - SSR_{\text{FE}})/(N-1)}{SSR_{\text{FE}}/(NT-N-K)} \sim F_{N-1, NT-N-K} \quad (13.119)$$

- **Decision Rule:** Reject H_0 if $F > F_{\alpha, N-1, NT-N-K}$. Rejection confirms the presence of significantly varying individual heterogeneity between individuals.

13.3.4 FD (First Differences Estimator)

13.3.4.1 Motivation and the Difference Operation

Like FE, the **First Differences (FD)** estimator is designed to eliminate unobserved heterogeneity c_i when $\mathbb{E}[x_{it}c_i] \neq 0$. Instead of subtracting the temporal mean, FD subtracts the observation from the immediately preceding period ($t-1$).

13.3.4.2 Formulation and Estimator Derivation

Consider $y_{it} = x'_{it}\beta + c_i + u_{it}$. Subtracting the equation at $t - 1$:

$$(y_{it} - y_{i,t-1}) = (x_{it} - x_{i,t-1})'\beta + (c_i - c_i) + (u_{it} - u_{i,t-1}) \quad (13.120)$$

$$\Delta y_{it} = \Delta x'_{it}\beta + \Delta u_{it}, \quad \text{for } t = 2, \dots, T \quad (13.121)$$

where $\Delta y_{it} = y_{it} - y_{i,t-1}$, $\Delta x_{it} = x_{it} - x_{i,t-1}$ and $\Delta u_{it} = u_{it} - u_{i,t-1}$.

Applying Pooled OLS to the first-differenced equation for the $T - 1$ transformed periods:

$$\hat{\beta}_{\text{FD}} = (\Delta X' \Delta X)^{-1} \Delta X' \Delta y = \left(\sum_{i=1}^N \sum_{t=2}^T \Delta x_{it} \Delta x'_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=2}^T \Delta x_{it} \Delta y_{it} \right) \quad (13.122)$$

13.3.4.3 Critical Comparison: FE vs. FD

Remark 13.3.8 (1. Theoretical Equivalence for $T = 2$). For a panel of exactly two periods ($T = 2$), the Fixed Effects estimator and the First Differences estimator produce **algebraically and numerically identical** point estimates, residuals, and t -statistics ($\hat{\beta}_{\text{FE}} \equiv \hat{\beta}_{\text{FD}}$).

Remark 13.3.9 (2. Efficiency Choice for $T \geq 3$). For $T \geq 3$, both estimators are unbiased and consistent under strict exogeneity ($\mathbb{E}[u_{it} | X_i, c_i] = 0$). The choice between FE and FD depends on the **autocorrelation structure of the idiosyncratic error** u_{it} :

- **If u_{it} is not autocorrelated ($u_{it} \sim \text{i.i.d.}$):** The Within transformation of FE preserves non-autocorrelated errors, making **FE asymptotically more efficient than FD**. Note that, under i.i.d. u_{it} , the differenced error $\Delta u_{it} = u_{it} - u_{i,t-1}$ presents serial correlation of $\text{Cov}(\Delta u_{it}, \Delta u_{i,t-1}) = -\text{Var}(u_{i,t-1}) = -\sigma_u^2$, generating -0.5 autocorrelation in FD.
- **If u_{it} is a Random Walk ($u_{it} = u_{i,t-1} + e_{it}$, where $e_{it} \sim \text{i.i.d.}$):** First difference eliminates the unit root ($\Delta u_{it} = e_{it} \sim \text{i.i.d.}$), eliminating autocorrelation. In this scenario, **FD is strictly more efficient than FE**.

13.3.5 RE (Random Effects — GLS in Panel / Quasi-Demeaning)

13.3.5.1 Motivation and the Orthogonality Hypothesis

The **Random Effects (RE)** estimator is applicable when it is reasonably assumed that unobserved individual heterogeneity c_i **is not correlated with regressors**:

$$\mathbb{E}[c_i | x_{i1}, \dots, x_{iT}] = 0 \iff \text{Cov}(x_{it}, c_i) = 0 \quad (13.123)$$

In this case, c_i does not generate omitted variable bias. Treat $c_i + u_{it} = v_{it}$ as a **composite error term**. The RE estimator uses the structure of the composite error covariance matrix to apply FGLS, being asymptotically more efficient than POLS and allowing inclusion of time-invariant regressors.

13.3.5.2 The Quasi-Time Demeaning Transformation

The covariance matrix of the composite error $v_i = [v_{i1}, \dots, v_{iT}]'$ for individual i is:

$$\Omega_i = \mathbb{E}[v_i v_i' | X_i] = \sigma_u^2 I_T + \sigma_c^2 \mathbf{1}_T \mathbf{1}_T' \quad (13.124)$$

By spectral inversion of Ω_i , it is demonstrated that the FGLS transformation that purifies errors is equivalent to subtracting a **fraction θ of the individual mean**:

$$(y_{it} - \theta \bar{y}_i) = (x_{it} - \theta \bar{x}_i)' \beta + ((1 - \theta)c_i + (u_{it} - \theta \bar{u}_i)) \quad (13.125)$$

$$\tilde{y}_{it} = \tilde{x}_{it}' \beta + \tilde{v}_{it} \quad (13.126)$$

where the weighting parameter θ is given by:

$$\theta = 1 - \sqrt{\frac{\sigma_u^2}{\sigma_u^2 + T\sigma_c^2}} \in [0, 1] \quad (13.127)$$

Remark 13.3.10 (Analysis of Limit Cases of θ). • **If $\sigma_c^2 \rightarrow 0$: $\theta \rightarrow 0 \implies \tilde{y}_{it} = y_{it}$.**
The RE estimator reduces to **Pooled OLS (POLS)**.

- **If $\sigma_c^2 \rightarrow \infty$ or $T \rightarrow \infty$: $\theta \rightarrow 1 \implies \tilde{y}_{it} = y_{it} - \bar{y}_i$.** The RE estimator collapses to **Fixed Effects (FE)**.

13.3.5.3 Hausman Specification Test (RE vs. FE)

Remark 13.3.11 (Hausman Test). • **Objective:** Formally test the exogeneity hypothesis of the unobserved effect c_i .

- **Null Hypothesis (H_0):** $\mathbb{E}[c_i | X_i] = 0$ (RE and FE are both consistent, but RE is asymptotically more efficient).
- **Alternative Hypothesis (H_1):** $\mathbb{E}[c_i | X_i] \neq 0$ (RE is biased and inconsistent; FE is consistent).

Hausman Test Wald Statistic: Calculate the standardized quadratic distance between estimated coefficient vectors (restricting to the subvector of K_1 coefficients of time-varying variables):

$$W = (\hat{\beta}_{FE} - \hat{\beta}_{RE})' [\widehat{\text{Var}}(\hat{\beta}_{FE}) - \widehat{\text{Var}}(\hat{\beta}_{RE})]^{-1} (\hat{\beta}_{FE} - \hat{\beta}_{RE}) \xrightarrow{d} \chi_{K_1}^2 \quad (13.128)$$

Note: The central matrix simplifies to $\text{Var}(\hat{\beta}_{FE}) - \text{Var}(\hat{\beta}_{RE})$ because, under H_0 , the RE estimator reaches the GLS efficiency limit, causing $\text{Cov}(\hat{\beta}_{FE} - \hat{\beta}_{RE}, \hat{\beta}_{RE}) = 0$.

Decision Rule: Reject H_0 if $W > \chi_{\alpha, K_1}^2$. If H_0 is rejected, adopt the **Fixed Effects (FE)** estimator.

13.3.6 Arellano-Bond (First-Differenced GMM) and System GMM

13.3.6.1 Nickell Bias in Dynamic Panels

Consider a first-order autoregressive model with panel data (dynamic AR(1)):

$$y_{it} = \rho y_{i,t-1} + x'_{it}\beta + c_i + u_{it}, \quad |\rho| < 1 \quad (13.129)$$

If we try to apply the Fixed Effects (FE/Within) estimator to eliminate c_i :

$$(y_{it} - \bar{y}_i) = \rho(y_{i,t-1} - \bar{y}_{i,-1}) + (x_{it} - \bar{x}_i)'\beta + (u_{it} - \bar{u}_i) \quad (13.130)$$

By construction, the temporal mean of the error $\bar{u}_i = \frac{1}{T} \sum_{s=1}^T u_{is}$ contains the term $u_{i,t-1}$. Since $y_{i,t-1}$ depends directly on $u_{i,t-1}$, the transformed regressor $(y_{i,t-1} - \bar{y}_{i,-1})$ is **mechanically correlated** with the transformed error $(u_{it} - \bar{u}_i)$.

This violation of strict exogeneity generates **Nickell Bias (1981)**. Even for $N \rightarrow \infty$, the FE estimator is **inconsistent for fixed T** , presenting severe negative bias in the persistence parameter estimate ρ .

13.3.6.2 The Arellano and Bond (1991) Estimator — FD-GMM

Manuel Arellano and Stephen Bond (1991) solved Nickell bias by combining **first differencing** of the equation with the use of **lagged levels as instrumental variables in the GMM framework**.

Remark 13.3.12 (Step 1: Elimination of Unobserved Heterogeneity via First Difference).

$$\Delta y_{it} = \rho \Delta y_{i,t-1} + \Delta x'_{it}\beta + \Delta u_{it}, \quad \text{for } t = 3, \dots, T \quad (13.131)$$

where $\Delta u_{it} = u_{it} - u_{i,t-1}$.

Remark 13.3.13 (Step 2: Construction of the Instrument Set under Sequential Exogeneity). Assume the hypothesis of **Sequential Exogeneity / Predetermination** of the idiosyncratic error (absence of autocorrelation in u_{it}):

$$\mathbb{E}[u_{it} \mid y_{i,t-1}, y_{i,t-2}, \dots, y_{i1}, c_i] = 0 \quad (13.132)$$

Analyzing orthogonality between past levels of y and the first-differenced error $\Delta u_{it} = u_{it} - u_{i,t-1}$:

- For $t = 3$: $\Delta u_{i3} = u_{i3} - u_{i2}$. Level y_{i1} is orthogonal to u_{i3} and u_{i2} . Thus, $\mathbb{E}[y_{i1} \Delta u_{i3}] = 0$. (1 valid instrument).
- For $t = 4$: $\Delta u_{i4} = u_{i4} - u_{i3}$. Levels y_{i1} and y_{i2} are orthogonal to Δu_{i4} . Thus, $\mathbb{E}[y_{i1} \Delta u_{i4}] = 0$ and $\mathbb{E}[y_{i2} \Delta u_{i4}] = 0$. (2 valid instruments).

Population Moment Conditions for y :

$$\mathbb{E}[y_{i,s} \cdot \Delta u_{it}] = 0, \quad \forall s \leq t-2; \quad t = 3, \dots, T \quad (13.133)$$

Block Instrument Matrix Structure (Z_i^{FD}): The Arellano-Bond instrument matrix has dimension $(T - 2) \times L$, expanding the number of instruments as t advances:

$$Z_i^{\text{FD}} = \begin{bmatrix} y_{i1} & 0 & 0 & \dots & 0 & \dots & 0 \\ 0 & y_{i1} & y_{i2} & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & y_{i1} & \dots & y_{i,T-2} \end{bmatrix}_{(T-2) \times \frac{(T-1)(T-2)}{2}} \quad (13.134)$$

13.3.6.3 System GMM (Blundell and Bond, 1998 / Arellano and Bover, 1995)

Remark 13.3.14 (The Arellano-Bond Limitation (Weak Instruments)). When the series y_{it} is **highly persistent** ($\rho \rightarrow 1$) or the variance of c_i is very large relative to u_{it} , lagged levels $y_{i,t-2}$ become **weak instruments** for the first difference $\Delta y_{i,t-1}$, causing severe finite-sample bias and loss of precision in FD-GMM.

Remark 13.3.15 (The System GMM Solution). The **System GMM** estimator stacks a system with two equations:

1. The **first-differenced** equation (Δy_{it}), instrumented by lagged levels ($y_{i,t-2}, \dots$).
2. The **levels** equation (y_{it}), instrumented by **lagged first differences** ($\Delta y_{i,t-1}$).

New Identification Hypothesis (Initial Mean Stationarity): To validate instruments in first difference ($\Delta y_{i,t-1}$) for the levels equation (where the error contains $c_i + u_{it}$), it is required that the correlation between differences of y and the fixed effect c_i be zero:

$$\mathbb{E}[\Delta y_{i,t-1} \cdot c_i] = 0 \implies \mathbb{E}[\Delta y_{i,t-1} (c_i + u_{it})] = 0, \quad \text{for } t = 3, \dots, T \quad (13.135)$$

13.3.6.4 Fundamental Diagnostic Tests in Dynamic Panel

Remark 13.3.16 (1. Arellano-Bond Residual Autocorrelation Test). Since the estimated equation is in first differences ($\Delta u_{it} = u_{it} - u_{i,t-1}$):

- **1st Order Autocorrelation — AR(1):** It is **expected and natural** to have negative correlation between Δu_{it} and $\Delta u_{i,t-1}$ due to the shared term $u_{i,t-1}$ ($\text{Cov}(\Delta u_{it}, \Delta u_{i,t-1}) = -\sigma_u^2$). Reject H_0 of absence of AR(1).
- **2nd Order Autocorrelation — AR(2):** This is the **decisive validation test**. Evaluate $H_0 : \text{Cov}(\Delta u_{it}, \Delta u_{i,t-2}) = 0$.
 - **Decision Rule: Cannot reject H_0 in AR(2)** ($p\text{-value} > \alpha$). If the test rejects H_0 for AR(2), it means that the original idiosyncratic error u_{it} has autocorrelation, **invalidating lags at $t - 2$ as instruments**. In these cases, one must move the instrument matrix back to $t - 3$ or earlier periods.

Remark 13.3.17 (2. Hansen J -Test (Instrument Validity and Proliferation)). Being the model strongly overidentified, apply the statistic $J = N\bar{g}'\hat{S}^{-1}\bar{g} \sim \chi_{L-K}^2$.

- **Rule:** Do not reject H_0 ($p\text{-value} > 0.05$).
- **Instrument Proliferation Warning:** If T is high, the instrument matrix Z_i grows quadratically ($O(T^2)$), reducing the power of the J -test and generating a falsely equal p -value to 1.000. It is required to collapse the instrument matrix (*collapse option*) or limit the depth of lags.

13.3.7 Panel Comparative Framework and Decision Guide

Remark 13.3.18 (Comparative Summary: Panel Estimators). **POLS:** Does not eliminate c_i ; exogeneity hypothesis is $\mathbb{E}[x_{it}c_i] = 0$ and $\mathbb{E}[x_{it}u_{it}] = 0$ (orthogonal mean exogeneity); allows time-invariant regressors; in dynamic panel, biased upward ($\hat{\rho}_{\text{POLS}} > \rho$).

FE (Within): Eliminates c_i (subtracts $\bar{y}_i$); exogeneity hypothesis is strict exogeneity $\mathbb{E}[u_{it} | X_i, c_i] = 0$; does not allow time-invariant regressors (eliminated); in dynamic panel, biased downward — **Nickell Bias** ($\hat{\rho}_{\text{FE}} < \rho$).

FD: Eliminates c_i (subtracts $y_{i,t-1}$); exogeneity hypothesis is strict exogeneity $\mathbb{E}[u_{it} | X_i, c_i] = 0$; does not allow time-invariant regressors (eliminated); in dynamic panel, biased if applied by simple OLS.

RE (GLS): Does not eliminate c_i ; exogeneity hypothesis is orthogonality $\mathbb{E}[c_i | X_i] = 0$ and strict exogeneity; allows time-invariant regressors; in dynamic panel, biased and inconsistent.

Arellano-Bond: Eliminates c_i (via FD); exogeneity hypothesis is sequential exogeneity $\mathbb{E}[u_{it} | y_{i,t-1}, \dots] = 0$; does not allow time-invariant regressors; in dynamic panel, consistent (FD-GMM). Weak if $\rho \rightarrow 1$.

System GMM: Eliminates c_i (FD + Levels); exogeneity hypothesis is sequential exogeneity + mean stationarity; allows time-invariant regressors (in levels equation); in dynamic panel, consistent and efficient for persistent series.

13.4 Discrete and Qualitative Choice Models

13.4.1 Binary Choice Models: LPM, Probit, and Logit

13.4.1.1 Motivation and Linear Probability Model (LPM) Limitations

When the dependent variable is binary (i.e., $y_i \in \{0, 1\}$, indicating decisions such as buy/not buy, work/not work, pay/not pay dividends), direct application of the Ordinary Least Squares Model is called the **Linear Probability Model (LPM)**:

$$y_i = x_i'\beta + u_i \implies P(y_i = 1 | x_i) = \mathbb{E}[y_i | x_i] = x_i'\beta \quad (13.136)$$

Remark 13.4.1 (Why the LPM is Conceptually Flawed). 1. **Predictions Outside the Probability Limit:** Since the function $x_i'\beta$ is an unrestricted linear combination, for extreme values of regressors x_i , the model produces predicted probabilities that are negative ($\hat{y}_i < 0$) or greater than 1 ($\hat{y}_i > 1$).

2. **Intrinsic Heteroscedasticity of the Error Term:** Since y_i assumes only two values (0 or 1), the error $u_i = y_i - x_i'\beta$ has a discrete conditional distribution of two points:

- $u_i = 1 - x_i'\beta$ with probability $x_i'\beta$
- $u_i = -x_i'\beta$ with probability $1 - x_i'\beta$

The conditional variance of the error is a direct function of the regressors:

$$\text{Var}(u_i | x_i) = x_i'\beta(1 - x_i'\beta) \quad (13.137)$$

This violates the Gauss-Markov homoscedasticity hypothesis, rendering OLS inefficient and invalidating conventional t and F statistics.

3. **Unrealistic Constant Marginal Effect:** The LPM assumes that the rate of change of probability with respect to a regressor is constant ($\frac{\partial P}{\partial x_k} = \beta_k$) for any level of x_i . In behavioral reality, the marginal impact tends to decrease near the extremes ($P \rightarrow 0$ or $P \rightarrow 1$), exhibiting an “S-shaped” (sigmoid) curve format.

13.4.1.2 Latent Variable Structure and Index Models

To guarantee that $0 \leq P(y_i = 1 \mid x_i) \leq 1$, we map the linear index $x_i'\beta$ through a **continuous and monotonically increasing Cumulative Distribution Function (CDF)** $G(z) \in (0, 1)$:

$$P(y_i = 1 \mid x_i) = G(x_i'\beta) \quad (13.138)$$

Remark 13.4.2 (Derivation of the Latent Variable Model). Consider an unobserved (latent) variable y_i^* that represents the net utility or unobserved propensity of individual i to make the decision:

$$y_i^* = x_i'\beta + e_i \quad (13.139)$$

The observed decision rule is given by the indicator function:

$$y_i = \mathbb{I}(y_i^* > 0) = \begin{cases} 1 & \text{if } y_i^* > 0 \\ 0 & \text{if } y_i^* \leq 0 \end{cases} \quad (13.140)$$

Calculating the conditional response probability:

$$P(y_i = 1 \mid x_i) = P(y_i^* > 0 \mid x_i) = P(x_i'\beta + e_i > 0 \mid x_i) = P(e_i > -x_i'\beta \mid x_i) \quad (13.141)$$

Assuming that the error distribution e_i is symmetric around zero ($1 - G(-z) = G(z)$):

$$P(y_i = 1 \mid x_i) = 1 - G(-x_i'\beta) = G(x_i'\beta) \quad (13.142)$$

13.4.1.3 Probit and Logit Specifications

The choice of the latent error distribution e_i defines the model:

Definition 13.4.1 (Probit Model). Assumes that $e_i \sim N(0, 1)$ has a Standard Normal distribution:

$$G(x_i'\beta) = \Phi(x_i'\beta) = \int_{-\infty}^{x_i'\beta} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du \quad (13.143)$$

where $\Phi(\cdot)$ is the Standard Normal CDF and $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is its density (pdf).

Definition 13.4.2 (Logit Model). Assumes that e_i has a Standard Logistic distribution:

$$G(x_i'\beta) = \Lambda(x_i'\beta) = \frac{\exp(x_i'\beta)}{1 + \exp(x_i'\beta)} = \frac{1}{1 + \exp(-x_i'\beta)} \quad (13.144)$$

the logistic density is given by $\lambda(z) = \frac{d\Lambda(z)}{dz} = \Lambda(z)(1 - \Lambda(z))$.

Remark 13.4.3 (Comparison between Logit and Probit and Conversion Rules). •

Tail Format: The logistic distribution has slightly heavier tails than the Normal distribution. This means that the predicted probabilities $P(y = 1 | x)$ of the Logit approach 0 or 1 at a slower rate than in the Probit.

- **Unidentified Scale of Variance:** The variance of the latent error is not identifiable in the data.
 - In the Probit, we normalize $\text{Var}(e_i) = 1$.
 - In the Logit, the standard logistic distribution has fixed variance $\text{Var}(e_i) = \frac{\pi^2}{3} \approx 3.29$.
- **Practical Coefficient Conversion Rules:** Since raw coefficients β depend on the model's variance scale, they cannot be directly compared. The following empirical approximations are used:

$$\begin{aligned}\beta_{\text{logit}} &\approx 1.6 \text{ to } 1.8 \times \beta_{\text{probit}} \\ \beta_{\text{probit}} &\approx 0.55 \text{ to } 0.625 \times \beta_{\text{logit}} \\ \beta_{\text{LPM}} &\approx 0.25 \times \beta_{\text{logit}} \quad (\text{for intercept: } \beta_{\text{LPM}} \approx 0.25\beta_{\text{logit}} + 0.5)\end{aligned}\tag{13.145}$$

13.4.1.4 Maximum Likelihood Estimation (MLE)

Since the model is non-linear in parameters, estimation by OLS fails. We use the **Maximum Likelihood Estimator (MLE)**.

For an i.i.d. sample with $y_i \in \{0, 1\}$, the conditional Bernoulli probability mass function is:

$$f(y_i | x_i; \beta) = [G(x_i' \beta)]^{y_i} [1 - G(x_i' \beta)]^{1-y_i}\tag{13.146}$$

Definition 13.4.3 (Log-Likelihood Function).

$$\ln L(\beta) = \sum_{i=1}^n \{y_i \ln G(x_i' \beta) + (1 - y_i) \ln [1 - G(x_i' \beta)]\}\tag{13.147}$$

Remark 13.4.4 (First-Order Condition (Score Equation)). Deriving with respect to vector β and setting equal to zero:

$$S(\beta) = \frac{\partial \ln L(\beta)}{\partial \beta} = \sum_{i=1}^n \left[\frac{y_i - G(x_i' \beta)}{G(x_i' \beta)(1 - G(x_i' \beta))} g(x_i' \beta) x_i \right] = 0_{K \times 1}\tag{13.148}$$

Simplification in the Logit: Since $g(z) = \lambda(z) = \Lambda(z)(1 - \Lambda(z))$, the central term cancels perfectly:

$$\sum_{i=1}^n (y_i - \Lambda(x_i' \beta)) x_i = 0_{K \times 1}\tag{13.149}$$

The Logit FOC imposes that the sum of residuals weighted by regressors is exactly zero (similar to OLS).

13.4.1.5 Parameter Interpretation and Marginal Effects

The raw coefficient β_k in index models **does not indicate the direct variation in probability** (except for its sign, which indicates the direction of the effect).

Definition 13.4.4 (Marginal Effects for Continuous Regressors). Differentiating $P(y_i = 1 | x_i) = G(x_i'\beta)$ with respect to regressor x_k :

$$ME_k(x_i) = \frac{\partial P(y_i = 1 | x_i)}{\partial x_{ik}} = g(x_i'\beta) \cdot \beta_k \quad (13.150)$$

where $g(z) = \frac{dG(z)}{dz}$ is the marginal density (in Probit $g = \phi$; in Logit $g = \lambda$). The marginal effect depends on the value of all regressors contained in $x_i'\beta$.

Remark 13.4.5 (Ways to Report the Marginal Effect). • **APE / AME (Average Marginal Effect)**: Calculate the marginal effect for each individual in the sample and take the average:

$$AME_k = \frac{1}{n} \sum_{i=1}^n g(x_i'\hat{\beta})\hat{\beta}_k \quad (13.151)$$

- **PEA (Partial Effect at the Mean)**: Evaluate the density at the midpoint of regressors $\bar{x}$:

$$PEA_k = g(\bar{x}'\hat{\beta})\hat{\beta}_k \quad (13.152)$$

Remark 13.4.6 (Discrete Effect for Dummy (Binary) Variables). For a binary regressor $d_i \in \{0, 1\}$ (e.g., having a college degree), the discrete effect is calculated by the finite difference of predicted probabilities:

$$\Delta P = G(x'_{\text{others}}\beta_{\text{others}} + \beta_d) - G(x'_{\text{others}}\beta_{\text{others}}) \quad (13.153)$$

Remark 13.4.7 (Interpretation in Logit via Odds Ratio). In the Logit, isolating the odds ratio:

$$\text{Odds} = \frac{P(y_i = 1 | x_i)}{1 - P(y_i = 1 | x_i)} = \exp(x_i'\beta) \quad (13.154)$$

Taking the natural logarithm (the *Log-Odds* or *Logit link*):

$$\ln\left(\frac{P_i}{1 - P_i}\right) = x_i'\beta = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} \quad (13.155)$$

- **Interpretation**: β_k represents the change in log-odds for a unit increase in x_k .
- **Odds Ratio** ($\exp(\beta_k)$): A 1-unit increase in x_k multiplies the odds of success by $\exp(\beta_k)$.

13.4.1.6 Hypothesis Tests and Goodness of Fit

Remark 13.4.8 (Likelihood Ratio (LR) Test). Tests q exclusion restrictions by comparing the log-likelihood of the unrestricted model ($\ln L_{\text{ur}}$) and the restricted model ($\ln L_r$):

$$LR = 2 (\ln L_{\text{ur}} - \ln L_r) \xrightarrow{d} \chi_q^2 \quad (13.156)$$

Reject H_0 if $LR > \chi_{\alpha, q}^2$.

Definition 13.4.5 (McFadden's Pseudo- R^2).

$$\text{Pseudo-}R^2 = 1 - \frac{\ln L_{\text{model}}}{\ln L_0} \quad (13.157)$$

where $\ln L_0$ is the log-likelihood of the model containing only the intercept constant.

13.4.1.7 Binary Models in Panel Data and Endogeneity

Remark 13.4.9 (Panel Data Challenges). • **Infeasibility of Fixed Effects Probit**

(Incidental Parameters Problem): In panels with large N and small T , attempting to estimate the N fixed effects c_i as parameters to be optimized via MLE generates **severely biased and inconsistent** estimators (100% bias when $T = 2$).

- **Conditional Fixed Effects Logit (Chamberlain, 1980):** The Logit allows eliminating c_i from the likelihood by conditioning on the temporal sum of each individual's successes ($S_i = \sum_{t=1}^T y_{it}$). Observations with $S_i = 0$ or $S_i = T$ (no variation over time) do not enter the conditional likelihood.
- **IV Probit (Continuous Endogeneity):** If $y_1^* = x_1'\beta_1 + y_2\beta_2 + e_1$ contains a continuous endogenous regressor y_2 , we use the **Rivers-Vuong Control Function Approach**:
 1. Regress y_2 on all exogenous instruments Z by OLS and extract residuals $\hat{v}_2$.
 2. Include $\hat{v}_2$ as an additional regressor in the Probit. The t -test on the coefficient of $\hat{v}_2$ is a direct test of endogeneity.

13.4.1.8 Chamberlain's Conditional Fixed Effects Logit (1980)

Remark 13.4.10 (The Incidental Parameters Problem). In the static Logit model with individual fixed effects:

$$P(y_{it} = 1 \mid x_{it}, \alpha_i) = \Lambda(x_{it}'\beta + \alpha_i) = \frac{\exp(x_{it}'\beta + \alpha_i)}{1 + \exp(x_{it}'\beta + \alpha_i)} \quad (13.158)$$

When $N \rightarrow \infty$ with T fixed and small (e.g., $T = 2$), the number of individual parameters α_i grows at the same rate as N . Attempting to estimate β and the N parameters α_i simultaneously via unrestricted Maximum Likelihood renders the estimators **inconsistent and severely biased** (in the case of $T = 2$, the estimate of β converges to 2β , presenting a 100% bias).

Remark 13.4.11 (Chamberlain's Solution: Conditional Likelihood). Gary Chamberlain (1980) demonstrated that the temporal sum of each individual's successes, $S_i = \sum_{t=1}^T y_{it}$, is a **sufficient statistic** for the individual effect α_i .

By conditioning the joint probability of the observed sequence of choices $(y_{i1}, \dots, y_{iT})$ on its total sum S_i , the individual parameter α_i is **completely eliminated by algebraic cancellation**.

Theorem 13.4.1 (Mathematical Derivation (Classical Case with $T = 2$)). *Consider a panel of $T = 2$ periods. The observed responses of individual i are (y_{i1}, y_{i2}) .*

The sum of successes $S_i = y_{i1} + y_{i2}$ can assume three values: 0, 1, or 2.

Case 1: If $S_i = 0$: The only possible sequence is $(0, 0)$. The conditional probability $P(y_{i1} = 0, y_{i2} = 0 \mid S_i = 0) = 1$, providing no information about β .

Case 2: If $S_i = 2$: The only possible sequence is $(1, 1)$. The conditional probability $P(y_{i1} = 1, y_{i2} = 1 \mid S_i = 2) = 1$, also providing no information about β .

Case 3: If $S_i = 1$: There are two possible sequences: event $A = (0, 1)$ or event $B = (1, 0)$.

Step 1: Unconditional Probability of Events. Under the hypothesis of temporal independence conditional on (x_i, α_i) , we calculate the probability of sequence $A = (0, 1)$:

$$\begin{aligned} P(0, 1 \mid x_i, \alpha_i) &= P(y_{i1} = 0 \mid x_{i1}, \alpha_i) \cdot P(y_{i2} = 1 \mid x_{i2}, \alpha_i) \\ &= \left(\frac{1}{1 + \exp(x'_{i1}\beta + \alpha_i)} \right) \left(\frac{\exp(x'_{i2}\beta + \alpha_i)}{1 + \exp(x'_{i2}\beta + \alpha_i)} \right) \\ &= \frac{\exp(x'_{i2}\beta + \alpha_i)}{\prod_{t=1}^2 (1 + \exp(x'_{it}\beta + \alpha_i))} \end{aligned} \quad (13.159)$$

Analogously, for sequence $B = (1, 0)$:

$$P(1, 0 \mid x_i, \alpha_i) = \frac{\exp(x'_{i1}\beta + \alpha_i)}{\prod_{t=1}^2 (1 + \exp(x'_{it}\beta + \alpha_i))} \quad (13.160)$$

Step 2: Probability of the Conditioning Event ($S_i = 1$). Since events $(0, 1)$ and $(1, 0)$ are mutually exclusive:

$$P(S_i = 1 \mid x_i, \alpha_i) = P(0, 1 \mid x_i, \alpha_i) + P(1, 0 \mid x_i, \alpha_i) = \frac{\exp(x'_{i1}\beta + \alpha_i) + \exp(x'_{i2}\beta + \alpha_i)}{\prod_{t=1}^2 (1 + \exp(x'_{it}\beta + \alpha_i))} \quad (13.161)$$

Step 3: The Conditional Probability and Elimination of α_i . By the conditional probability rule:

$$P(0, 1 \mid S_i = 1, x_i, \alpha_i) = \frac{P(0, 1 \mid x_i, \alpha_i)}{P(S_i = 1 \mid x_i, \alpha_i)} \quad (13.162)$$

Substituting the obtained expressions:

$$P(0, 1 \mid S_i = 1, x_i, \alpha_i) = \frac{\frac{\exp(x'_{i2}\beta + \alpha_i)}{\prod_{t=1}^2 (1 + \exp(x'_{it}\beta + \alpha_i))}}{\frac{\exp(x'_{i1}\beta + \alpha_i) + \exp(x'_{i2}\beta + \alpha_i)}{\prod_{t=1}^2 (1 + \exp(x'_{it}\beta + \alpha_i))}} \quad (13.163)$$

The common denominator $\prod_{t=1}^2 (1 + \exp(x'_{it}\beta + \alpha_i))$ cancels perfectly:

$$P(0, 1 \mid S_i = 1, x_i, \alpha_i) = \frac{\exp(x'_{i2}\beta + \alpha_i)}{\exp(x'_{i1}\beta + \alpha_i) + \exp(x'_{i2}\beta + \alpha_i)} \quad (13.164)$$

Dividing numerator and denominator by $\exp(x'_{i1}\beta + \alpha_i)$:

$$P(0, 1 \mid S_i = 1, x_i, \alpha_i) = \frac{\frac{\exp(x'_{i2}\beta + \alpha_i)}{\exp(x'_{i1}\beta + \alpha_i)}}{1 + \frac{\exp(x'_{i2}\beta + \alpha_i)}{\exp(x'_{i1}\beta + \alpha_i)}} = \frac{\exp((x_{i2} - x_{i1})'\beta)}{1 + \exp((x_{i2} - x_{i1})'\beta)} \quad (13.165)$$

Remark 13.4.12 (Important Conclusions from the Result). 1. **Independence from α_i** : The resulting conditional probability **does not contain the term α_i** .

2. **Analogous to Within / First Differences Estimator**: The conditional model reduces to a standard binary Logit where the explanatory variable is the first difference of regressors $(x_{i2} - x_{i1})$.
3. **Discarding Observations Without Variation**: Individuals who do not change their state over time ($S_i = 0$ or $S_i = T$) have conditional probability equal to 1 and are discarded from the likelihood.
4. **Time-Invariant Variables**: Variables that do not change over time ($x_{i2} = x_{i1}$) have $(x_{i2} - x_{i1}) = 0$ and their coefficients cannot be identified.

13.4.2 Multinomial Choice Models

When the dependent variable Y_i assumes more than two discrete categories ($Y_i \in \{1, 2, \dots, J\}$), models divide into two fundamental families based on data nature:

1. **Unordered Choices**: Nominal categories without hierarchy (e.g., mode of transport: car, bus, train; occupation choice).
2. **Ordered Choices**: Hierarchical ordinal categories (e.g., credit risk rating AAA, AA, BBB; degree of satisfaction).

13.4.2.1 Random Utility Model (RUM) Framework

Definition 13.4.6 (RUM Structure). Individual i chooses alternative $j \in \{1, \dots, J\}$ that maximizes their latent utility U_{ij}^* :

$$U_{ij}^* = V_{ij} + \varepsilon_{ij} \quad (13.166)$$

where V_{ij} is the deterministic/observable part of utility and ε_{ij} is the unobserved stochastic shock.

The probability that individual i chooses option j is the probability that U_{ij}^* is strictly greater than the utility of all other alternatives:

$$P_{ij} = P(U_{ij}^* > U_{ik}^*, \forall k \neq j) = P(\varepsilon_{ik} - \varepsilon_{ij} < V_{ij} - V_{ik}, \forall k \neq j) \quad (13.167)$$

13.4.2.2 Multinomial Logit (MNL) vs. Conditional Logit (CL)

The classical econometric distinction between MNL and CL lies in the nature of the regressors:

Definition 13.4.7 (Multinomial Logit (MNL — Schmidt & Strauss, 1975)). Regressors w_i are **individual characteristics** (e.g., age, income, schooling) and do not vary between alternatives. To allow individual characteristics to affect choice, parameters β_j must vary by alternative:

$$P(y_i = j \mid w_i) = \frac{\exp(w_i' \beta_j)}{\sum_{m=1}^J \exp(w_i' \beta_m)} \quad (13.168)$$

Identification / Base Category Normalization: Since the sum of probabilities must be 1, parameters β_j are only identified relative to a reference category. We normalize category J as the base ($\beta_J = 0_{K \times 1}$):

$$\begin{aligned} P(y_i = J \mid w_i) &= \frac{1}{1 + \sum_{m=1}^{J-1} \exp(w'_i \beta_m)} \\ P(y_i = j \mid w_i) &= \frac{\exp(w'_i \beta_j)}{1 + \sum_{m=1}^{J-1} \exp(w'_i \beta_m)}, \quad \text{for } j = 1, \dots, J-1 \end{aligned} \quad (13.169)$$

Definition 13.4.8 (Conditional Logit (CL — McFadden, 1974)). Regressors x_{ij} are **attributes of the alternatives** (e.g., cost of each mode of transport, travel time). The parameter vector γ is homogeneous/common for all options:

$$P(y_i = j \mid x_i) = \frac{\exp(x'_{ij} \gamma)}{\sum_{m=1}^J \exp(x'_{im} \gamma)} \quad (13.170)$$

Note: In modern applications, econometric packages estimate the **Mixed/Combined Multinomial Logit**, simultaneously encompassing individual regressors w_i and attributes x_{ij} .

13.4.2.3 Independence of Irrelevant Alternatives (IIA) Hypothesis

Remark 13.4.13 (IIA Consequence). The Multinomial Logit model assumes that errors ε_{ij} are **i.i.d. distributed according to the Type I Extreme Value distribution (Gumbel)**.

Mathematical-Behavioral Consequence: The probability ratio (relative odds) between any two options j and k depends **exclusively on the attributes/characteristics of j and k** , being totally independent of the presence or attributes of a third alternative m :

$$\frac{P_{ij}}{P_{ik}} = \frac{\exp(x'_{ij} \gamma)}{\exp(x'_{ik} \gamma)} = \exp((x_{ij} - x_{ik})' \gamma) \quad (13.171)$$

Remark 13.4.14 (The IIA Paradox (Red Bus / Blue Bus Paradox)). If a commuter chooses between Car ($P = 0.5$) and Red Bus ($P = 0.5$), the odds ratio is 1. If we introduce the Blue Bus (identical to the red one, except for color), intuition indicates that people will divide themselves between Car (50%), Red Bus (25%), and Blue Bus (25%). However, under IIA, the model forces the odds ratio between Red Bus and Blue Bus to be preserved, erroneously predicting 33.3% for each of the three options.

Remark 13.4.15 (Hausman-McFadden IIA Test). Estimate the complete model with J choices ($\hat{\gamma}_f$) and re-estimate excluding one of the options ($\hat{\gamma}_s$):

$$W_{\text{IIA}} = (\hat{\gamma}_s - \hat{\gamma}_f)' \left[\widehat{\text{Var}}(\hat{\gamma}_s) - \widehat{\text{Var}}(\hat{\gamma}_f) \right]^{-1} (\hat{\gamma}_s - \hat{\gamma}_f) \xrightarrow{d} \chi_K^2 \quad (13.172)$$

Rejecting H_0 indicates violation of the IIA hypothesis.

13.4.2.4 Models Relaxing the IIA Hypothesis

To overcome IIA rigidity, three advanced extensions are used:

Remark 13.4.16 (Advanced Multinomial Models). 1. **Nested Logit Model:** Groups substitute alternatives into “branches” or “nests”. Errors are correlated within the same nest, but independent between nests. Introduces the **dissimilarity/inclusion parameter** (t_b) for each branch. If $t_b = 1$, the model collapses to the traditional Conditional Logit.

2. **Mixed Logit / Random Parameters Logit (RPL):** Allows preference parameters to be random across individuals ($\gamma_i \sim N(\gamma, \Sigma)$). Offers total flexibility to substitute preferences and eliminates the IIA restriction.

3. **Multinomial Probit (MNP):** Assumes that the error vector has a **Multivariate Normal** distribution with unrestricted covariance matrix $\varepsilon_i \sim N(0, \Sigma)$.

- **Advantage:** Allows totally arbitrary patterns of correlation and heteroscedasticity between alternatives.
- **Disadvantage:** The choice probability involves a $(J - 1)$ -dimensional integral with no analytical solution, requiring stochastic simulation integration methods (such as the **GHK Algorithm**).

13.4.3 Ordered Choice Models (Ordered Response Models)

13.4.3.1 Formulation via Latent Variable and Cut-Points

When response categories are ordinal ($Y_i \in \{1, 2, \dots, J\}$), using unordered multinomial discards ordering information and is inefficient.

Definition 13.4.9 (Ordered Model Structure). The model is based on a single continuous latent variable y_i^* :

$$y_i^* = x_i' \beta + e_i \quad (13.173)$$

We define a set of $J - 1$ **unknown cut-points / thresholds** ($\alpha_1 < \alpha_2 < \dots < \alpha_{J-1}$) that partition the real line:

$$y_i = \begin{cases} 1 & \text{if } y_i^* \leq \alpha_1 \\ 2 & \text{if } \alpha_1 < y_i^* \leq \alpha_2 \\ \vdots & \vdots \\ J & \text{if } y_i^* > \alpha_{J-1} \end{cases} \quad (13.174)$$

13.4.3.2 Conditional Probabilities in Ordered Probit / Logit

Calculating the probability of occurrence of each category j :

$$\begin{aligned} P(y_i = 1 \mid x_i) &= F(\alpha_1 - x_i' \beta) \\ P(y_i = 2 \mid x_i) &= F(\alpha_2 - x_i' \beta) - F(\alpha_1 - x_i' \beta) \\ &\vdots \\ P(y_i = j \mid x_i) &= F(\alpha_j - x_i' \beta) - F(\alpha_{j-1} - x_i' \beta) \\ &\vdots \\ P(y_i = J \mid x_i) &= 1 - F(\alpha_{J-1} - x_i' \beta) \end{aligned} \quad (13.175)$$

- If $e_i \sim N(0, 1) \implies F = \Phi$ (**Ordered Probit**).
- If $e_i \sim \text{Logistic}(0, \pi^2/3) \implies F = \Lambda$ (**Ordered Logit**).

13.4.3.3 Coefficient Interpretation and Intermediate Sign Ambiguity

Remark 13.4.17 (Interpretation Requires Rigorous Caution). The interpretation of the coefficient vector β requires rigorous caution. Differentiating the probability of category j with respect to regressor x_k :

Lower Extreme Category ($j = 1$):

$$\frac{\partial P(y_i = 1 \mid x_i)}{\partial x_k} = -f(\alpha_1 - x_i' \beta) \cdot \beta_k \quad (13.176)$$

The sign of the marginal effect is **strictly opposite** to the sign of β_k .

Upper Extreme Category ($j = J$):

$$\frac{\partial P(y_i = J \mid x_i)}{\partial x_k} = f(\alpha_{J-1} - x_i' \beta) \cdot \beta_k \quad (13.177)$$

The sign of the marginal effect is **identical** to the sign of β_k .

Intermediate Categories ($1 < j < J$):

$$\frac{\partial P(y_i = j \mid x_i)}{\partial x_k} = [f(\alpha_{j-1} - x_i' \beta) - f(\alpha_j - x_i' \beta)] \cdot \beta_k \quad (13.178)$$

Sign Ambiguity: The sign of the marginal effect for intermediate choices is **not determined by the sign of β_k** . It depends on the relative displacement of densities at the two cut-points. Therefore, it is **mandatory** to calculate and report average marginal effects (AME) for all J categories.

Remark 13.4.18 (Note on *dummy* Variables). For binary variables, the derivative is not appropriate. The correct marginal effect must be calculated as the difference in predicted probabilities: $\Delta P = P(x_k = 1, \mathbf{x}_{i,-k}) - P(x_k = 0, \mathbf{x}_{i,-k})$, evaluated at sample means.

Zero-sum restriction: For any variable x_k , the sum of marginal effects over all J categories must be exactly zero.

13.4.4 Synthetic Decision Framework for Discrete Choice Models

Remark 13.4.19 (Decision Guide for Discrete Choice Models). • **Cross-sectional data:** Logit or Probit. Coefficients not directly comparable.

- **Binary panel, only ratios:** Conditional Logit (Chamberlain). Discards individuals without variation.
- **Binary panel, marginal effects:** Mundlak (CRE) or Random Effects. Allows AME and probability calculation.
- **Multiple choices per individual:** Mixed Logit. Captures preference heterogeneity.

- **Correlated alternatives:** Nested Logit or Mixed Logit. Relax IIA.
- **Stated + revealed choice:** Nested Logit with different scale. Controls statement bias.

Binary Modeling Checklist:

1. **Justification:** Is the dependent variable genuinely binary? If it is truncated continuous, use Tobit. If ordinal, use ordered models.
2. **Specification:** Logit or Probit? In practice, predictions are virtually identical. Logit is slightly more common in panel due to Chamberlain's Conditional Logit.
3. **Diagnosis:** Check complete separation (explosive coefficients) and multicollinearity.
4. **Marginal Effects:** Never interpret β in isolation. Report AME for continuous variables and discrete differences for dummies.
5. **Panel Data:** Avoid unconditional Fixed Effects for small T . Prefer Mundlak or Random Effects, testing the correlation between α_i and $\mathbf{x}_{it}$ via significance of δ .

The central message is that binary choice models are not mechanical extensions of OLS. They require attention to the latent structure, identification normalizations, correct interpretation of marginal effects, and rigorous treatment of unobserved heterogeneity in panel data.

13.5 Limited Dependent Variables: Tobit and Heckman Selection

13.5.1 Fundamental Taxonomy: Censoring, Truncation, and Corner Response

In applied microeconometrics, limited dependent variables arise under different behavioral and sampling mechanisms. The conceptual distinction between **Corner Response**, **Censoring**, and **Truncation** is indispensable for correct econometric specification and parameter interpretation.

13.5.1.1 Corner Solution Response

Definition 13.5.1 (Corner Solution). The dependent variable y is the result of an optimization choice by economic agents in which a significant fraction of the population chooses the value zero (or a limit c), while the remainder chooses strictly positive values continuously.

Examples: Hours worked in labor supply, charitable donations, spending on durable consumer goods, amount invested in research and development (R&D).

Observability: There is no missing data or sampling censoring problem; zero and positive choices are perfectly observed for the entire population.

Latent Variable: The variable y^* represents the agent's unobserved desire or propensity (which may assume negative values in terms of utility).

13.5.1.2 Data Censoring

Definition 13.5.2 (Data Censoring). The population variable of interest y^* is continuous and has no probability accumulation at zero. However, due to the sampling design, institutional rules, or measurement limitations, the observed variable w is restricted above or below a threshold c .

Examples: Wages with a maximum limit (*top-coding*) in sample surveys for confidentiality preservation; time until re-entry into the labor market in studies with a fixed follow-up period end (*right censoring*).

Observability: Sampling is random and the regressor vector x_i is observed for **all** units. For censored units, the exact value of y^* is not observed, but it is known with certainty that $y^* \leq c$ (left censoring) or $y^* \geq c$ (right censoring).

13.5.1.3 Data Truncation

Definition 13.5.3 (Data Truncation). The sampling process entirely excludes from the database units whose outcome y^* does not reach a certain criterion or threshold c .

Examples: Family budget surveys restricted only to families with income below 1.5 times the poverty line; surveys conducted at the service location (*on-site sampling*), such as interviews with park visitors where non-visitors are not recorded.

Observability: Information on y_i is completely lost **and also on regressors** x_i for the entire subpopulation that does not meet the criterion.

13.5.2 Mathematical Preliminaries: Truncated Expectation and the Inverse Mills Ratio

Theorem 13.5.1 (Derivation of the Truncated Conditional Expectation). *Let the latent variable $Y^* \sim N(\mu, \sigma^2)$. Suppose Y^* is left-truncated at threshold c (we observe only $Y = Y^*$ when $Y^* > c$).*

The probability of non-truncation is:

$$\Pr(Y^* > c) = 1 - \Phi\left(\frac{c - \mu}{\sigma}\right) = \Phi\left(\frac{\mu - c}{\sigma}\right) \quad (13.179)$$

The conditional density of Y given $Y > c$ is:

$$f(y | Y^* > c) = \frac{\frac{1}{\sigma}\phi\left(\frac{y - \mu}{\sigma}\right)}{\Phi\left(\frac{\mu - c}{\sigma}\right)} \quad (13.180)$$

Calculating the conditional expectation $\mathbb{E}[Y^ | Y^* > c]$:*

$$\mathbb{E}[Y^* | Y^* > c] = \int_c^\infty y \frac{\frac{1}{\sigma}\phi\left(\frac{y - \mu}{\sigma}\right)}{\Phi\left(\frac{\mu - c}{\sigma}\right)} dy \quad (13.181)$$

Making the change of variable $z = \frac{y - \mu}{\sigma} \implies dy = \sigma dz$. The lower limit becomes $a = \frac{c - \mu}{\sigma}$:

$$\mathbb{E}[Y^* | Y^* > c] = \mu + \frac{\sigma}{\Phi(-a)} \int_a^\infty z\phi(z) dz \quad (13.182)$$

Since the derivative of $\phi(z)$ is $\phi'(z) = -z\phi(z)$, we have $\int_a^\infty z\phi(z) dz = [-\phi(z)]_a^\infty = \phi(a)$. Recalling that by the symmetry of the Normal, $\phi(a) = \phi(-a) = \phi\left(\frac{\mu-c}{\sigma}\right)$ and $\Phi(-a) = \Phi\left(\frac{\mu-c}{\sigma}\right)$:

$$\mathbb{E}[Y^* | Y^* > c] = \mu + \sigma \left[\frac{\phi\left(\frac{\mu-c}{\sigma}\right)}{\Phi\left(\frac{\mu-c}{\sigma}\right)} \right] = \mu + \sigma \lambda\left(\frac{\mu-c}{\sigma}\right) \quad (13.183)$$

where $\lambda(z) = \frac{\phi(z)}{\Phi(z)}$ is the **Inverse Mills Ratio (IMR)**.

Theorem 13.5.2 (Derivation of Heckman's Second Stage). *Consider the two-equation model:*

1. **Selection:** $s_i^* = z_i'\gamma + v_i$, with $s_i = \mathbb{I}(s_i^* > 0)$.

2. **Outcome:** $y_i^* = x_i'\beta + u_i$, observed only if $s_i = 1$.

Assume the Bivariate Normal distribution of errors:

$$\begin{pmatrix} u_i \\ v_i \end{pmatrix} \sim N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{bmatrix} \sigma_u^2 & \sigma_{uv} \\ \sigma_{uv} & 1 \end{bmatrix} \right) \quad (13.184)$$

The conditional expectation of the outcome variable in the selected subsample ($s_i = 1 \iff v_i > -z_i'\gamma$) is given by:

$$\mathbb{E}[y_i | x_i, z_i, s_i = 1] = x_i'\beta + \mathbb{E}[u_i | x_i, z_i, v_i > -z_i'\gamma] \quad (13.185)$$

By the Bivariate Normal property, the conditional expectation of error u_i given v_i is linear: $\mathbb{E}[u_i | v_i] = \frac{\sigma_{uv}}{\sigma_v^2} v_i = \sigma_{uv} v_i$ (since $\sigma_v^2 = 1$). Therefore:

$$\mathbb{E}[u_i | v_i > -z_i'\gamma] = \sigma_{uv} \mathbb{E}[v_i | v_i > -z_i'\gamma] \quad (13.186)$$

Using the truncated expectation result derived above:

$$\mathbb{E}[v_i | v_i > -z_i'\gamma] = \frac{\phi(-z_i'\gamma)}{1 - \Phi(-z_i'\gamma)} = \frac{\phi(z_i'\gamma)}{\Phi(z_i'\gamma)} \equiv \lambda(z_i'\gamma) \quad (13.187)$$

Substituting back into the main equation:

$$\mathbb{E}[y_i | x_i, z_i, s_i = 1] = x_i'\beta + \sigma_{uv} \lambda(z_i'\gamma) = x_i'\beta + \theta_\lambda \lambda(z_i'\gamma) \quad (13.188)$$

where $\theta_\lambda = \sigma_{uv} = \rho\sigma_u$.

This generates **Heckman's second-stage regression equation**:

$$y_i = x_i'\beta + \theta_\lambda \hat{\lambda}_i + e_i \quad (13.189)$$

13.5.3 Truncated Regression

13.5.3.1 Concept and Definition

Definition 13.5.4 (Truncated Regression). **Truncated Regression** occurs when the sampling/data collection process itself **entirely excludes from the database** units whose outcomes Y_i fall outside a pre-determined interval (e.g., $Y_i \leq c$).

Total Loss of Information (Missing Full Real Distribution): For excluded units, we completely lose the value of outcome Y_i and also the regressor vector X_i . They do not exist in the data file.

Unintentional Truncation: Occurs when the researcher applies a sample filter (e.g., analyzes only people with wages below a certain ceiling or students of a certain range) without realizing that they have destroyed the randomness of sampling from the entire population.

13.5.3.2 When to Use vs. When NOT to Use

Remark 13.5.1 (Decision Guide for Truncated Regression). **WHEN TO USE:**

- When the database was intentionally or institutionally collected covering only a subpopulation (e.g., surveys directed exclusively to beneficiaries below the poverty line) and the researcher wishes to make inference about the parameters of the complete population.

WHEN NOT TO USE:

- **NEVER use** if you have the data on X_i available for individuals with $Y_i \leq c$. Voluntarily discarding zeros/censored observations to apply truncated regression means throwing away useful information.
- **NEVER use** in duration analyses, *top-coding*, or zero consumption choices where the regressors of limited observations were recorded (in these cases, use Tobit or Heckit).

13.5.3.3 Identification Hypotheses and Economic Intuition

Definition 13.5.5 (Truncated Regression Identification). 1. **Strict Normality and Homoscedasticity of the Latent Error:** $u_i \mid x_i \sim N(0, \sigma^2)$ in the untruncated model $y_i^* = x_i' \beta + u_i$.

2. **Known Cut-off Point (c):** The truncation threshold c is known and exogenous to the errors.

3. **Population Exogeneity:** $\mathbb{E}[u_i \mid X_i] = 0$ in the untruncated population.

Remark 13.5.2 (Economic Intuition). The sampling process has physically cut one tail of the population's Normal distribution. To identify the original parameters β without bias, the model must “reconstruct” the portion of the curve that was discarded. This requires knowing the exact functional form of the unobserved errors' distribution—hence the strict Normality and Homoscedasticity requirements. By knowing the precise shape of the Normal curve, the selection probability adjusts the slope of the regression line, allowing recovery of the population parameters from the truncated sample.

13.5.3.4 Statistical Structure and Likelihood

Definition 13.5.6 (Conditional Expectation in the Truncated Sample). Since sampling conditions on $Y_i > c$ (equivalently, $u_i > c - X_i' \beta$), the observed conditional expectation

becomes non-linear:

$$\mathbb{E}[Y_i | X_i, Y_i > c] = X_i' \beta + \sigma \cdot \lambda \left(\frac{X_i' \beta - c}{\sigma} \right) \quad (13.190)$$

where $\lambda(z) = \frac{\phi(z)}{\Phi(z)}$ is the **Inverse Mills Ratio (IMR)**.

Definition 13.5.7 (Truncated Regression Log-Likelihood). The conditional density reweights the Normal density by dividing it by the probability of being in the sample:

$$f(y_i | y_i > c, x_i) = \frac{\frac{1}{\sigma} \phi \left(\frac{y_i - x_i' \beta}{\sigma} \right)}{1 - \Phi \left(\frac{c - x_i' \beta}{\sigma} \right)} \quad (13.191)$$

The log-likelihood for a sample left-truncated at c of size n is:

$$\ln L(\beta, \sigma^2) = \sum_{i=1}^n \left\{ -\frac{1}{2} \ln(2\pi\sigma^2) - \frac{(y_i - x_i' \beta)^2}{2\sigma^2} - \ln \left[1 - \Phi \left(\frac{c - x_i' \beta}{\sigma} \right) \right] \right\} \quad (13.192)$$

13.5.3.5 Advantages and Disadvantages

Remark 13.5.3 (Truncated Regression Trade-offs). **Advantages:** Corrects the sample bias of simple OLS that would occur when trying to fit a line in a sample where part of the population has been removed.

Disadvantages:

- **Extreme Fragility to Specification Errors:** If error u_i is heteroscedastic or non-normal, the Maximum Likelihood estimator of Truncated Regression becomes **severely inconsistent**.
- Inefficient if censored data is available in the original sample.

13.5.3.6 Main Diagnostic Tests

Remark 13.5.4 (Truncated Regression Diagnostics). 1. **Likelihood Ratio (LR) Test** for model significance:

$$LR = 2(\ln L_{ur} - \ln L_r) \sim \chi_q^2 \quad (13.193)$$

Reject H_0 if $LR > \chi_{\alpha, q}^2$.

2. **Specification Tests for Non-Normality and Heteroscedasticity:** Conditional moment tests (Pagan-Vella).

13.5.4 The Tobit Model (Type I Tobit / Corner Solution)

13.5.4.1 Concept and Definition

Definition 13.5.8 (Tobit Model). The **Tobit Model (Type I)** addresses situations in which the regressor vector X_i is observed for **all units in the sample**, but the dependent variable Y_i presents probability accumulation (discrete mass) at a limit (usually zero):

1. **Data Censoring (*Data Censoring*):** The variable of interest Y^* is continuous in the population, but the research/measurement rule limits the observed value $Y = \max(c, Y^*)$ (e.g., wage *top-coding* to preserve confidentiality, durability time in studies ending at a fixed date).

2. **Corner Solution Response:** $Y \geq 0$ is the result of an optimization choice by economic agents in which a relevant fraction voluntarily decides to consume/invest zero (e.g., charitable spending, hours worked, stock purchases). There is no missing data or research error.

13.5.4.2 When to Use vs. When NOT to Use

Remark 13.5.5 (Decision Guide for Tobit). **WHEN TO USE:**

- When the sample contains regressors X_i for all units and the dependent variable Y_i has a discrete mass of zeros and continuous positive values.

WHEN NOT TO USE:

- **DO NOT use** if the sample did not record regressors X_i for units with $Y = 0$ (use Truncated Regression).
- **DO NOT use** if the participation decision ($Y > 0$) and the conditional quantity decision ($Y \mid Y > 0$) are governed by **different variables or coefficients** (use Heckit or Hurdle/Two-Part models).

13.5.4.3 Model Structure and Latent Variable

Originally proposed by James Tobin (1958), the Tobit Model (or Type I Tobit) is formulated through a linear regression equation for the latent variable y_i^* with normal and homoscedastic errors:

$$y_i^* = x_i' \beta + u_i, \quad u_i \mid x_i \sim N(0, \sigma^2) \quad (13.194)$$

The observed dependent variable y_i is determined by the left censoring/corner rule at zero:

$$y_i = \max(0, y_i^*) = \begin{cases} y_i^* & \text{if } y_i^* > 0 \\ 0 & \text{if } y_i^* \leq 0 \end{cases} \quad (13.195)$$

13.5.4.4 Identification Hypotheses and Economic Intuition

Definition 13.5.9 (Tobit Identification). 1. **Latent Variable with Normal and Homoscedastic Errors:** $y_i^* = x_i' \beta + u_i$, with $u_i \mid x_i \sim N(0, \sigma^2)$.

2. **Single Mechanism (Single-Index Assumption):** A single parameter vector β governs both $\Pr(Y_i > 0 \mid x_i)$ and $\mathbb{E}[Y_i^* \mid x_i]$.

Remark 13.5.6 (Economic Intuition). The Tobit model assumes that the decision to “participate” (have expenditure greater than zero) and the decision of “how much to spend” are two sides of the same economic coin. For example, the same preference vector (β) determines both whether someone buys a car and the amount spent on the vehicle. This **single-mechanism assumption** is both the Tobit’s strength (parsimony) and its main limitation (structural rigidity).

13.5.4.5 Advantages and Disadvantages

Remark 13.5.7 (Tobit Trade-offs). **Advantages:**

- Resolves the attenuation bias of OLS in the complete sample ($\text{plim } \hat{\beta}_{\text{OLS}} \approx (1 - \pi)\beta_{\text{Tobit}}$, where π is the proportion of zeros).
- Avoids the truncation/omitted variable bias of OLS in the positive subsample.
- Exploits the full population information without discarding the mass of zeros.

Disadvantages:

- **Structural Rigidity:** Imposes that the sign of the marginal effect of x_k on the participation probability ($\frac{\partial \Pr(Y>0)}{\partial x_k}$) is necessarily equal to the sign of the effect on the conditional level ($\frac{\partial \mathbb{E}[Y|Y>0]}{\partial x_k}$).
- **Extreme Sensitivity to Non-Normality and Heteroscedasticity:** If u_i is heteroscedastic or non-normal, Tobit by MLE becomes **totally inconsistent**.

13.5.4.6 Mixed (Continuous/Discrete) Log-Likelihood Function

Since the observed distribution of y_i has a discrete probability mass at point $y_i = 0$ and a continuous density for $y_i > 0$, the likelihood function is of a mixed nature:

- **For zero observations** ($y_i = 0$):

$$\Pr(y_i = 0 \mid x_i) = \Pr(u_i \leq -x_i'\beta) = \Phi\left(\frac{-x_i'\beta}{\sigma}\right) = 1 - \Phi\left(\frac{x_i'\beta}{\sigma}\right) \quad (13.196)$$

where $\Phi(\cdot)$ represents the Standard Normal cumulative distribution function (CDF).

- **For positive observations** ($y_i > 0$): The conditional density of the observed outcome is the density of the Normal distribution with mean $x_i'\beta$ and variance σ^2 :

$$f(y_i \mid x_i, y_i > 0) = \frac{1}{\sigma} \phi\left(\frac{y_i - x_i'\beta}{\sigma}\right) \quad (13.197)$$

where $\phi(\cdot)$ is the Standard Normal probability density function (PDF).

Definition 13.5.10 (Global Log-Likelihood Function). The mixed log-likelihood combines the Probit term for zeros with the Normal density for positives:

$$\ln L(\beta, \sigma^2) = \sum_{i:y_i=0} \ln \left[1 - \Phi\left(\frac{x_i'\beta}{\sigma}\right) \right] + \sum_{i:y_i>0} \left\{ -\frac{1}{2} \left[\ln(2\pi) + \ln(\sigma^2) + \frac{(y_i - x_i'\beta)^2}{\sigma^2} \right] \right\} \quad (13.198)$$

Maximization of $\ln L$ with respect to (β, σ^2) via Maximum Likelihood generates consistent and asymptotically normal non-linear estimators (Amemiya, 1973).

13.5.4.7 Conditional Means and Inverse Mills Ratio (λ)

In the Tobit model, we distinguish three conditional expectations of empirical interest:

Remark 13.5.8 (Three Conditional Expectations). **1. Latent Variable Mean:**

$$\mathbb{E}[y_i^* | x_i] = x_i' \beta \quad (13.199)$$

2. Mean Conditional on Positive Observations (Truncated Expectation):

$$\mathbb{E}[y_i | x_i, y_i > 0] = x_i' \beta + \sigma \lambda \left(\frac{x_i' \beta}{\sigma} \right) \quad (13.200)$$

where $\lambda(z) = \frac{\phi(z)}{\Phi(z)}$ is the **Inverse Mills Ratio**. Since $\lambda(z) > 0$, the mean of positive observations is strictly greater than $x_i' \beta$.

3. Unconditional Mean of the Observed Outcome (Censored Expectation): By the law of iterated expectations:

$$\begin{aligned} \mathbb{E}[y_i | x_i] &= \Pr(y_i > 0 | x_i) \cdot \mathbb{E}[y_i | x_i, y_i > 0] + \Pr(y_i = 0 | x_i) \cdot 0 \\ &= \Phi \left(\frac{x_i' \beta}{\sigma} \right) \left[x_i' \beta + \sigma \lambda \left(\frac{x_i' \beta}{\sigma} \right) \right] \\ &= \Phi \left(\frac{x_i' \beta}{\sigma} \right) x_i' \beta + \sigma \phi \left(\frac{x_i' \beta}{\sigma} \right) \end{aligned} \quad (13.201)$$

13.5.4.8 Classical OLS Estimator Bias

Application of the OLS estimator to the Tobit model is strictly inconsistent in both traditional approaches:

Remark 13.5.9 (Two Types of OLS Bias). **1. OLS on the Complete Sample (including zeros):** Since the true conditional expectation function $\mathbb{E}[y_i | x_i]$ is non-linear (Equation above), the OLS linear regression attempts to fit a line to an “S-shaped” curve attenuated by the mass of zeros. Greene derived the pocket rule of attenuation:

$$\text{plim } \hat{\beta}_{\text{OLS}} \approx (1 - \pi) \beta_{\text{Tobit}} \quad (13.202)$$

where $\pi = \Pr(y_i = 0)$ is the proportion of censored/zero observations in the sample. The higher the proportion of zeros, the greater the attenuation bias toward zero.

2. OLS Only on the Positive Subsample ($y_i > 0$): Regressing y_i on x_i omitting zeros is equivalent to estimating the truncated mean equation:

$$y_i = x_i' \beta + \sigma \lambda \left(\frac{x_i' \beta}{\sigma} \right) + e_i \quad (13.203)$$

Omission of the Inverse Mills Ratio $\sigma \lambda(\cdot)$ generates severe omitted variable bias, since $\lambda(x_i' \beta / \sigma)$ is correlated with x_i .

13.5.4.9 Marginal Effects: McDonald & Moffitt Decomposition

In Tobit models, parameter β_k measures only the variation in the latent variable y^* ($\frac{\partial \mathbb{E}[y^* | x]}{\partial x_k} = \beta_k$). To analyze the effect of a continuous regressor x_k on the observed variable y , we use the **McDonald & Moffitt (1980)** decomposition:

Definition 13.5.11 (Tobit Marginal Effects (McDonald-Moffitt)). **1. Marginal Effect on the Unconditional Mean** $\mathbb{E}[y \mid x]$:

$$\frac{\partial \mathbb{E}[y_i \mid x_i]}{\partial x_{ik}} = \beta_k \cdot \Phi\left(\frac{x'_i \beta}{\sigma}\right) \quad (13.204)$$

Interpretation (Adjustment Rule): To obtain the marginal effect comparable to OLS, multiply $\hat{\beta}_k$ by the proportion of non-censored/positive observations in the sample $\Phi(x'_i \beta / \sigma)$. This is the **McDonald-Moffitt decomposition**: the total effect on the unconditional mean equals the latent effect scaled by the probability of being positive.

2. Marginal Effect on the Mean Conditional on Positives $\mathbb{E}[y \mid x, y > 0]$:

$$\frac{\partial \mathbb{E}[y_i \mid x_i, y_i > 0]}{\partial x_{ik}} = \beta_k \left[1 - \lambda \left(\frac{x'_i \beta}{\sigma} \right) \left(\frac{x'_i \beta}{\sigma} + \lambda \left(\frac{x'_i \beta}{\sigma} \right) \right) \right] \quad (13.205)$$

3. Marginal Effect on the Participation Probability $\Pr(y > 0 \mid x)$:

$$\frac{\partial \Pr(y_i > 0 \mid x_i)}{\partial x_{ik}} = \frac{\beta_k}{\sigma} \phi\left(\frac{x'_i \beta}{\sigma}\right) \quad (13.206)$$

13.5.4.10 Tobit Sensitivity and Robust Estimators (CLAD)

Remark 13.5.10 (Vulnerability to Specification Errors). Unlike classical OLS, the consistency of the Tobit Maximum Likelihood estimator **critically depends on the Normality and Homoscedasticity hypotheses of u_i** . If u_i is heteroscedastic or non-normal, the Tobit-MLE estimator becomes **completely inconsistent**.

Definition 13.5.12 (Powell's CLAD Estimator (Censored Least Absolute Deviations)). To circumvent this fragility, James Powell (1984, 1986) developed the **CLAD** estimator.

Mechanics: Based on the property that the conditional median is preserved by non-decreasing monotonic transformations:

$$\text{Med}(y_i \mid x_i) = \text{Med}(\max(0, y_i^*) \mid x_i) = \max(0, \text{Med}(y_i^* \mid x_i)) = \max(0, x'_i \beta) \quad (13.207)$$

Objective Function: CLAD minimizes the sum of absolute deviations of observations from the censored median:

$$\hat{\beta}_{\text{CLAD}} = \arg \min_{\beta} \sum_{i=1}^n |y_i - \max(0, x'_i \beta)| \quad (13.208)$$

Advantages: The CLAD estimator is totally **robust to heteroscedasticity and non-normality** of the error term u_i .

13.5.4.11 Main Diagnostic Tests

Remark 13.5.11 (Tobit Diagnostics). **1. Chesher-Irish / Pagan-Vella Specification Test:** Evaluates the null hypothesis of homoscedasticity and normality of Tobit residuals.

2. Likelihood Ratio (LR) Test: $LR = 2(\ln L_{\text{ur}} - \ln L_r) \sim \chi_q^2$.

13.5.5 Heckman Selection Model (Heckit / Type II Tobit)

13.5.5.1 Motivation and Incidental Truncation

The Heckman selection model (1976, 1979) addresses the problem of **Incidental Truncation** or **Non-Random Sample Selection Bias**.

This occurs when the variable of interest y_i is observed only for a non-randomly selected subsample of the population due to a behavioral or choice process. The classical example of Gronau (1974) and Heckman (1979) is the women's wage equation: the wage offer y_i is only observed for women who decide to work (whose reservation wage is less than the market offer).

13.5.5.2 When to Use vs. When NOT to Use

Remark 13.5.12 (Decision Guide for Heckman). **WHEN TO USE:**

- When the observability of Y_i depends on a participation or eligibility decision ($S_i = 1$) and the determinants of participation may differ from the determinants of the outcome Y_i .

WHEN NOT TO USE:

- **DO NOT use** if selection is strictly random or exogenous conditional on regressors X_i ($\mathbb{E}[u_i | v_i] = 0 \iff \rho = 0$). Simple OLS in the selected subsample is consistent and more efficient.
- **DO NOT use** for unified corner solutions where a single utility mechanism defines zeros and positives (use Tobit Type I).

13.5.5.3 Two-Equation Structure of the Heckman Model

The model specifies two correlated stochastic processes:

Definition 13.5.13 (Heckman's Two Equations). **1. Selection Equation (Participation / Probit):**

$$s_i^* = z_i' \gamma + v_i, \quad s_i = \mathbb{I}(s_i^* > 0) = \begin{cases} 1 & \text{if } s_i^* > 0 \text{ (Observed)} \\ 0 & \text{if } s_i^* \leq 0 \text{ (Not Observed)} \end{cases} \quad (13.209)$$

2. Outcome Equation (Intensity / Offer):

$$y_i^* = x_i' \beta + u_i, \quad y_i = \begin{cases} y_i^* & \text{if } s_i = 1 \\ \text{Missing} & \text{if } s_i = 0 \end{cases} \quad (13.210)$$

Error Structure: Assume that (u_i, v_i) have a Bivariate Normal distribution independent of z_i and x_i :

$$\begin{pmatrix} u_i \\ v_i \end{pmatrix} \bigg| X_i, Z_i \sim N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{bmatrix} \sigma_u^2 & \sigma_{uv} \\ \sigma_{uv} & 1 \end{bmatrix} \right) \quad (13.211)$$

where the variance of the selection equation error v_i is normalized to 1 (Probit identification), and $\rho = \frac{\sigma_{uv}}{\sigma_u}$ represents the correlation between the unobservables of selection and outcome.

13.5.5.4 Identification Hypotheses and Economic Intuition

Definition 13.5.14 (Heckman Identification). 1. **Bivariate Normality of Errors:**

$$(u_i, v_i) \sim N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{bmatrix} \sigma_u^2 & \sigma_{uv} \\ \sigma_{uv} & 1 \end{bmatrix} \right).$$

2. **Exclusion Restriction** (z_{excl}): The selection equation regressor vector z_i must contain **at least one exclusive variable** (z_{excl}) that significantly affects selection (S_i), but **has no direct effect** on the outcome Y_i .

Remark 13.5.13 (Economic Intuition). The decision to participate (work) and the outcome (wage) are governed by **two separate behavioral processes**. Since the unobserved factors that encourage a person to work (v_i , such as labor market attachment, family support) can be correlated with unobserved productivity (u_i , such as innate ability), the subsample of workers is not random. The exclusion variable Z_{excl} (e.g., number of young children, spouse's income) allows us to estimate the clean selection propensity and isolate the bias—it affects the participation decision without directly affecting the wage offer.

13.5.5.5 Advantages and Disadvantages

Remark 13.5.14 (Heckman Trade-offs). **Advantages:**

- Allows different variables and distinct coefficients to govern the participation decision and the intensity of the outcome.
- Corrects non-random selection bias.

Disadvantages:

- Requires a theoretically well-founded Exclusion Restriction (z_{excl}).
- Highly vulnerable to violations of the Bivariate Normality hypothesis of errors.
- Requires second-stage standard error correction due to the generated regressor $\hat{\lambda}_i$ (Murphy-Topel or bootstrap).

13.5.5.6 Heckman's Two-Stage Procedure (Heckit)

Remark 13.5.15 (Two-Stage Heckit Estimator). **Stage 1: Estimation of the Selection Probit Model (Complete Sample of size N)**

1. Use the **complete sample of N observations** (selected units $s_i = 1$ and non-selected $s_i = 0$).
2. Estimate a **Probit** model for the participation probability $\Pr(s_i = 1 \mid z_i) = \Phi(z_i' \gamma)$.
3. Obtain the consistent parameter vector $\hat{\gamma}$ and calculate the estimated Inverse Mills Ratio for each individual in the selected subsample ($s_i = 1$):

$$\hat{\lambda}_i = \lambda(z_i' \hat{\gamma}) = \frac{\phi(z_i' \hat{\gamma})}{\Phi(z_i' \hat{\gamma})} \quad (13.212)$$

Stage 2: Augmented OLS Regression in the Subsample ($s_i = 1$)

1. Restrict the sample **only to selected observations** ($s_i = 1$).

2. Regress y_i on the original regressors x_i and the generated Inverse Mills Ratio ($\hat{\lambda}_i$) by OLS:

$$y_i = x_i' \beta + \theta_\lambda \hat{\lambda}_i + e_i \quad (13.213)$$

3. **Standard Error Adjustment:** Since $\hat{\lambda}_i$ is a generated regressor from the first stage, OLS standard errors in the second stage are incorrect and must be adjusted via the Murphy-Topel analytical formula or *bootstrap*.

13.5.5.7 Selection Bias Test and Exclusion Restriction

Remark 13.5.16 (Selection Bias Test). Perform a t -test (or Wald test) on the coefficient $\hat{\theta}_\lambda$ of the term $\hat{\lambda}_i$ in the second stage.

- **Null Hypothesis** ($H_0 : \theta_\lambda = 0 \iff \rho = 0$): There is no correlation between the unobservables of selection and outcome. Selection is exogenous/random conditional on x_i , and simple OLS in the selected subsample is consistent.
- **Alternative Hypothesis** ($H_1 : \theta_\lambda \neq 0 \iff \rho \neq 0$): There is significant non-random selection bias; inclusion of the Heckit term $\hat{\lambda}_i$ corrects the omitted variable bias.

Decision Rule: Reject H_0 if p -value $< \alpha$ (e.g., 0.05). If H_0 is rejected, the presence of selection bias is confirmed and the need for the Heckman procedure is validated.

Remark 13.5.17 (Crucial Identification Condition (Exclusion Restriction)). For the Heckman estimator to be statistically robust, the selection equation regressor vector z_i must contain **at least one exclusive variable** (z_{excl}) that significantly affects the selection decision (s_i), but **does not directly affect the outcome equation** (y_i).

Consequence of Absence of Exclusion Restriction ($z_i = x_i$): If z_i contains exactly the same variables as x_i , model identification depends **exclusively on the curvature/non-linearity of the Inverse Mills Ratio** $\lambda(x_i' \gamma)$. Since $\lambda(\cdot)$ is approximately linear over wide probability intervals, **severe multicollinearity** arises between $\hat{\lambda}_i$ and regressors x_i in the second stage, excessively inflating the standard errors of coefficients $\hat{\beta}$ and making estimates extremely unstable.

13.5.6 Comparative Summary of Limited Dependent Variable Models

Remark 13.5.18 (Comparative Summary: Truncated, Tobit, and Heckman). **Truncated Regression:** Origin mechanism is pure truncation (incomplete population sampling); observed sample in file is **incomplete** (X and Y totally lost for $Y \leq c$); equation structure is 1 equation; decision mechanism is a single truncated population process; estimation method is **MLE** of Truncated Normal density; key identification requirement is normality and homoscedasticity of u_i ; main tests are LR likelihood tests; robust alternative is semiparametric truncated estimators.

Tobit (Type I / Corner Solution): Origin mechanism is data censoring / corner response (population zeros); observed sample in file is **complete** (X and Y recorded for all, zeros + positives); equation structure is 1 equation ($y^* = x' \beta + u$); decision mechanism is a **single mechanism** (same β governs $P(Y > 0)$ and $E[Y | Y > 0]$); estimation method is

mixed MLE (Probit term on zeros + Normal on positives); key identification requirement is normality and homoscedasticity of u_i ; main tests are normality and homoscedasticity tests (Pagan-Vella); robust alternative is Powell's **CLAD**.

Heckman (Heckit / Type II): Origin mechanism is incidental truncation / non-random sample selection; observed sample in file is **partial** (X, Z recorded for all N ; Y is *missing* for $S = 0$); equation structure is **2 distinct equations** (Selection S^* and Outcome Y^*); decision mechanism is **two mechanisms** (distinct vectors γ for selection and β for outcome); estimation method is **2-stage (Heckit)** (1st Probit on sample $N \rightarrow$ 2nd OLS with $\hat{\lambda}_i$ on $S = 1$); key identification requirement is **exclusion restriction** (z_{excl}) in 1st stage Probit plus bivariate normality; main test is **selection bias test** ($H_0 : \theta_\lambda = 0, \rho = 0$); robust alternative is semiparametric selection models (Ahn-Powell).